

Intelligent Security API

Developer Guide

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DETEKTOR

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

2.1 Introduction

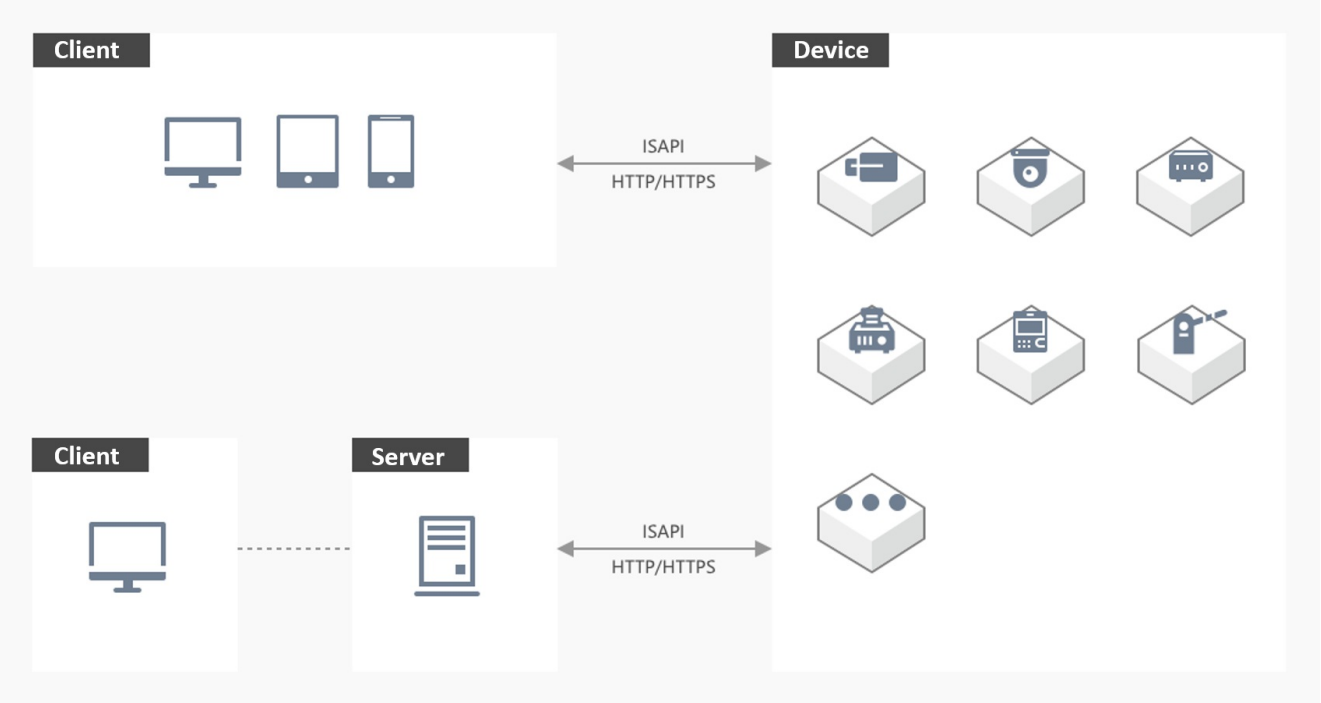
Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on HTTP (Hypertext Transfer Protocol) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

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2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, RTSP (Real-Time Streaming Protocol) based on TCP/UDP for live view and video playback of devices, etc.

Application Layer	Device Discovery SADP	Signaling Interaction ISAPI (HTTP)	Media Stream RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- Network Video Recorders
  - Pro Series

    DS-7604NI-E1, DS-7604NI-E1/4N, DS-7604NI-E1/4P, DS-7604NI-K1, DS-7604NI-K1(B), DS-7604NI-K1/4K, DS-7604NI-K1/4P, DS-7604NI-K1/4P(B), DS-7604NI-K1/4P/4K, DS-7604NI-Q1, DS-7604NI-Q1(E), DS-7604NI-Q1/4P, DS-7604NI-Q1/4P(E), DS-7604NXI-K1, DS-7604NXI-K1(C), DS-7604NXI-K1(D), DS-7604NXI-K1(E), DS-7604NXI-K1/4P, DS-7604NXI-K1/4P(C), DS-7604NXI-K1/4P(D), DS-7604NXI-K1/4P(E), DS-7604NXI-K1/4P/S, DS-7604NXI-K1/4PUHK(D), DS-7604NXI-K1/S, DS-7608NI-E1, DS-7608NI-E2, DS-7608NI-E2/8N, DS-7608NI-E2/8P, DS-7608NI-I2, DS-7608NI-I2/8P, DS-7608NI-K1, DS-7608NI-K1(B), DS-7608NI-K1/4K, DS-7608NI-K1/8P, DS-7608NI-K1/8P(B), DS-7608NI-K1/8P/4K, DS-7608NI-K2, DS-7608NI-K2/4K, DS-7608NI-K2/8P, DS-7608NI-K2/8P/4K, DS-7608NI-Q1, DS-7608NI-Q1(E), DS-7608NI-Q1/8P, DS-7608NI-Q1/8P(E), DS-7608NI-Q2, DS-7608NI-Q2(E), DS-7608NI-Q2/8P, DS-7608NI-Q2/8P(E), DS-7608NXI-I2/4S, DS-7608NXI-I2/8P/4S, DS-7608NXI-I2/8P/S, DS-7608NXI-I2/S, DS-7608NXI-K1, DS-7608NXI-K1(C), DS-7608NXI-K1(D), DS-7608NXI-K1(E), DS-7608NXI-K1/8P, DS-7608NXI-K1/8P(C), DS-7608NXI-K1/8P(D), DS-7608NXI-K1/8P(E), DS-7608NXI-K1/8P/S, DS-7608NXI-K1/8PUHK(D), DS-7608NXI-K1/S, DS-7608NXI-K2, DS-7608NXI-K2(D), DS-7608NXI-K2(E), DS-7608NXI-K2/8P, DS-7608NXI-K2/8P(D), DS-7608NXI-K2/8P(E), DS-7608NXI-Q1/S, DS-7616NI-E1, DS-7616NI-E2, DS-7616NI-E2/16N, DS-7616NI-E2/16P, DS-7616NI-E2/8N, DS-7616NI-E2/8P, DS-7616NI-I2, DS-7616NI-I2/16P, DS-7616NI-K1, DS-7616NI-K1(B), DS-7616NI-K1/4K, DS-7616NI-K2, DS-7616NI-K2/16P, DS-7616NI-K2/16P/4K, DS-7616NI-K2/4K, DS-7616NI-Q1, DS-7616NI-Q1(E), DS-7616NI-Q2, DS-7616NI-Q2(E), DS-7616NI-Q2/16P, DS-7616NI-Q2/16P(E), DS-7616NXI-I2/16P/4S, DS-7616NXI-I2/16P/S, DS-7616NXI-I2/4S, DS-7616NXI-I2/S, DS-7616NXI-K1, DS-7616NXI-K1(C), DS-7616NXI-K1(D), DS-7616NXI-K1(E), DS-7616NXI-K1/S, DS-7616NXI-K2, DS-7616NXI-K2(D), DS-7616NXI-K2(E), DS-7616NXI-K2/16P, DS-7616NXI-K2/16P(D), DS-7616NXI-K2/16P(E), DS-7616NXI-K2/16PUHK(D), DS-7616NXI-Q1/S, DS-7632NI-E2, DS-7632NI-E2/16N, DS-7632NI-E2/16P, DS-7632NI-E2/8N, DS-7632NI-E2/8P, DS-7632NI-I2, DS-7632NI-I2/16P, DS-7632NI-K2, DS-7632NI-K2/16P, DS-7632NI-K2/16P/4K, DS-7632NI-K2/4K, DS-7632NXI-I2/16P/4S, DS-7632NXI-I2/16P/S, DS-7632NXI-I2/4S, DS-7632NXI-I2/S, DS-7632NXI-K2, DS-7632NXI-K2(D), DS-7632NXI-K2(E), DS-7632NXI-K2/16P, DS-7632NXI-K2/16P(D), DS-7632NXI-K2/16P(E), DS-7708NI-E4, DS-7708NI-E4/8P, DS-7708NI-I4, DS-7708NI-I4/8P, DS-7708NI-K4, DS-7708NI-K4/4K, DS-7708NI-K4/8P, DS-7708NI-K4/8P/4K, DS-7708NI-Q4, DS-7708NI-Q4/8P, DS-7708NXI-K4, DS-7708NXI-K4(D), DS-7708NXI-K4(E), DS-7708NXI-K4/8P, DS-7708NXI-K4/8P(D), DS-7708NXI-K4/8P(E), DS-7716NI-E4, DS-7716NI-E4/16P, DS-7716NI-I4, DS-7716NI-I4(B), DS-7716NI-I4/16P, DS-7716NI-I4/16P(B), DS-7716NI-K4, DS-7716NI-K4/16P, DS-7716NI-K4/16P/4K, DS-7716NI-K4/4K, DS-7716NI-Q4, DS-7716NI-Q4/16P, DS-7716NXI-I4/16P/4S, DS-7716NXI-I4/16P/S, DS-7716NXI-I4/4S, DS-7716NXI-I4/S, DS-

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7716NXI-K4, DS-7716NXI-K4(D), DS-7716NXI-K4(E), DS-7716NXI-K4/16P, DS-7716NXI-K4/16P(D), DS-7716NXI-K4/16P(E), DS-7732NI-E4, DS-7732NI-E4/16P, DS-7732NI-I4, DS-7732NI-I4(B), DS-7732NI-I4/16P, DS-7732NI-I4/16P(B), DS-7732NI-I4/24P, DS-7732NI-K4, DS-7732NI-K4/16P, DS-7732NI-K4/16P/4K, DS-7732NI-K4/4K, DS-7732NI-Q4, DS-7732NI-Q4/16P, DS-7732NXI-I4/16P/4S, DS-7732NXI-I4/16P/S, DS-7732NXI-I4/4S, DS-7732NXI-I4/S, DS-7732NXI-K4, DS-7732NXI-K4(D), DS-7732NXI-K4(E), DS-7732NXI-K4/16P, DS-7732NXI-K4/16P(D), DS-7732NXI-K4/16P(E), DS-7764NI-K4, DS-7804NXI-K1, DS-7804NXI-K1/4P, DS-7808NI-I2, DS-7808NI-I2/8P, DS-7808NXI-I2/8P/S, DS-7808NXI-I2/S, DS-7808NXI-K1, DS-7808NXI-K1/8P, DS-7808NXI-K2, DS-7808NXI-K2/8P, DS-7816NI-I2, DS-7816NI-I2/16P, DS-7816NXI-I2/16P/S, DS-7816NXI-I2/S, DS-7816NXI-K1, DS-7816NXI-K2, DS-7816NXI-K2/16P, DS-7832NI-I2, DS-7832NI-I2/16P, DS-7832NXI-K2, DS-7832NXI-K2/16P, DS-7916NI-I4, DS-7916NI-I4/16P, DS-7916NXI-I4/16P/S, DS-7916NXI-I4/S, DS-7916NXI-K4, DS-7932NI-I4, DS-7932NI-I4/16P, DS-7932NI-I4/24P, DS-7932NXI-I4/16P/S, DS-7932NXI-I4/S, DS-8616NXI-I8/S, DS-8616NXI-K8, DS-8616NXI-K8(E), DS-8632NXI-I8/24P/S, DS-8632NXI-I8/S, DS-8632NXI-K8, DS-8632NXI-K8(E), DS-8664NXI-I8/S, DS-9616NXI-I8/S, DS-9632NXI-I8/S, DS-9664NXI-I8/S

2.3 Terms And Definitions

2.3.1 IoTChannel

IoTChannel refers to channels of IoT devices.

2.3.2 Event

Event refers to the information uploaded by the device actively. Event needs to be uploaded by the device in real time for the immediate response from the platform. If the device is offline, the event can be stored in the cache first and then be uploaded again when the connection is restored.

2.3.3 Arming

Arming means that the client establishes connection with the device so that events can be uploaded to the client via the connection. The client can subscribe to some event types, and the device will upload the specified events only, otherwise the device will upload all types of events to the client.

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2.3.4 Motion Detection

It can realize image changes recognition and moving objects detection of a specified area by camera. For example, if a person passes by or a camera is moved, the motion will be recognized and then an alarm will be uploaded for notifications, and the event can trigger recording and alarm output.

2.4 Symbols And Acronyms

IOT: Internet of Things

OSD: On-screen display

ISAPI: Intelligent Security API.

ISUP: Intelligent Security Uplink Protocol.

HTTP: Hypertext Transfer Protocol.

NVR: Network Video Recorder

IPC: IP Camera

VMD: Motion Detection

PTZ: Pan-Tilt-Zoom

FDLib: Face Picture Library

FDID: Face Picture Library ID

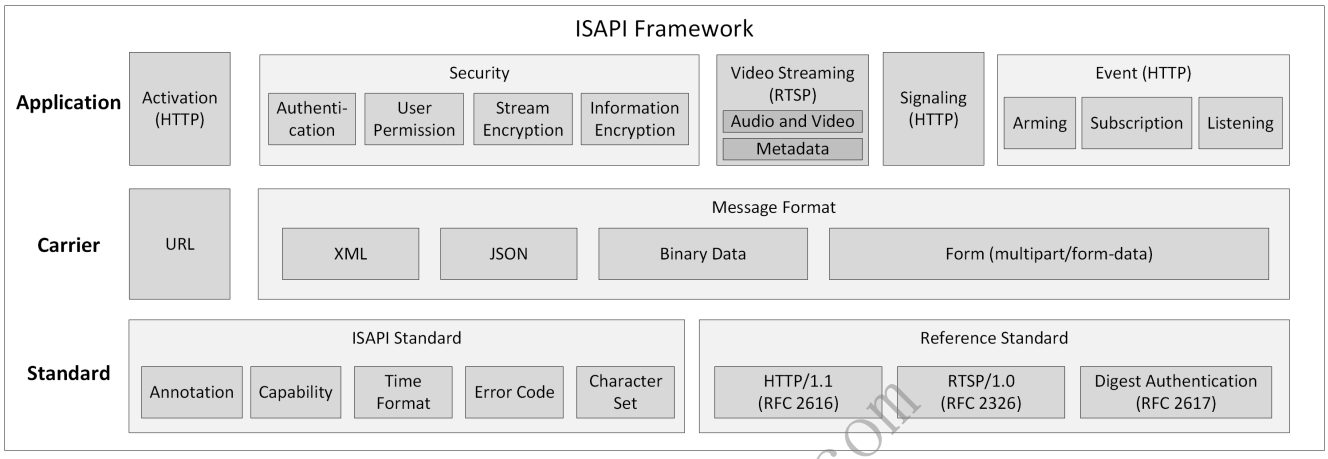
PID: Face Picture ID in the Face Picture Library

2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview



Notes:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

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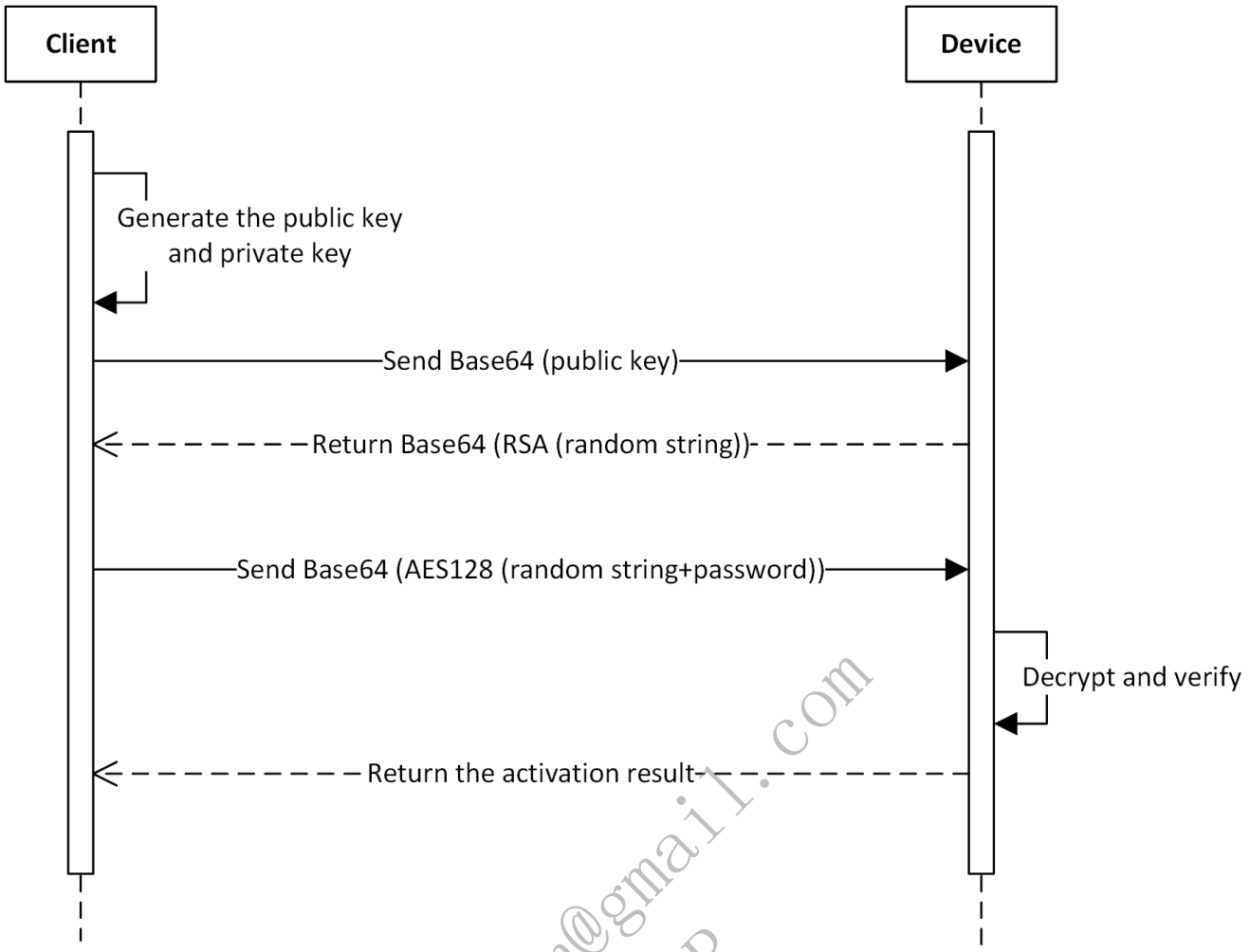
3.2 Activation

The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.



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Firstly, two operations are defined:

- bytesToHexstring: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
- hexStringToBytes: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via bytesToHexstring and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via hexStringToBytes. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via bytesToHexstring, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via hexStringToBytes to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via hexStringToBytes to get a 16-byte AES key. Then the client uses the AES key to encrypt the `"string consisting of the first 16 characters of the random string and the real password"` by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via bytesToHexstring, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`). Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via hexStringToBytes to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.

7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in Authentication of Quick Start Guide.

3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

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3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

3.4 Video Streaming

3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in RFC 7826). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application
reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

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4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)

```java
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

```python
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

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4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the `Content-Type` in the headers of the HTTP request is `application/xml; charset="UTF-8"`.

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g., `http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data, boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to RFC 1867 (Form-Based File Upload in HTML). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

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The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
    "ipAddress": "172.6.64.7",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2009-11-14T15:27+08:00",
    "eventType": "mixedTargetDetection",
    "eventDescription": "Mixed target detection",
    "deviceID": "123456789",
    "CaptureResult": [{
        "targetID": 1,
        "Human": {
            "Rect": {
                "height": 1.0,
                "width": 1.0,
                "x": 0.0,
                "y": 0.0
            },
            "contentID1": "humanImage", /*human body thumbnail*/
            "contentID2": "humanBackgroundImage", /*human body background picture*/
            "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
            "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
        }
    }]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--
```

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4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#Level 1,level2#Level 2,level3#Level 3]-->level1
    </level>
  </Node>
</NodeList>
```

```
{
    "name":  "test",
    /*ro, req, string, name, range:[1,32]*/
    "type":  "type1",
    /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
    "enabled":  true,
    /*ro, opt, bool, enable or not, desc:xxxxxxx*/
    "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
        "scene":  1,
        /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
        "ID":  1
        /*req, int, No., range:[1,8], step:, unit:, unitType:*/
    }
}
```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

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4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupportXxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```
{
    "isSupportMap":  true,
    /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
    "isSupportAlgTrainResourceInfo":  true,
    /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
    "isSupportAlgTrainAuthInfo":  true,
    /*ro, opt, bool, whether it supports ony getting the authorization information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
    "isSupportAlgTrainNodeList":  true,
    /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
format=json*/
    "isSupportNAS":  true
    /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}
```

The capability message example in XML format is shown below:

```
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether it supports searching the network status, desc: related API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>
```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```
{
    "boolType": {
        /*req, object, example of the capability of type bool*/
        "@opt": [true, false]
        /*req, array, options, subType: bool*/
    },
    "integerType": {
        /*req, object, example of the capability of type integer*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
    },
    "stringType": {
        /*req, object, example of the capability of type string*/
        "@min": 0,
        /*ro, req, int, the minimum string size*/
        "@max": 32
        /*ro, req, int, the maximum string size*/
    },
    "enumType": {
        /*req, object, capability example of type enum*/
        "@opt": ["enum1", "enum2", "enum3"]
        /*req, array, options, subType: string*/
    }
}
```

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The example of XML format is shown below:

```
<boolType opt="true,false" def="true">
    <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
    <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
    <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
    <!--ro, opt, string, example of the capability of type enum-->white
</enumType>
```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts ISO 8601 Standard Time Format, which is the same as W3C Standard Date and Time Formats.

Format: `YYYY-MM-DDThh:mm:ss.sTZD`

```
YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)
```

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
    "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
    "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```

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4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to RFC 2044).

No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	'
12	Underscore	_	29	Vertical Bar	|
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	|
17	Octothorpe	#			

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The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in RFC 2616. For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

```
HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
    "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
    "statusCode": 4,
    "statusString": "Invalid Operation",
    "subStatusCode": "notSupport",
    "errorCode": 1073741825,
    "errorMsg": "notSupport"
}
```

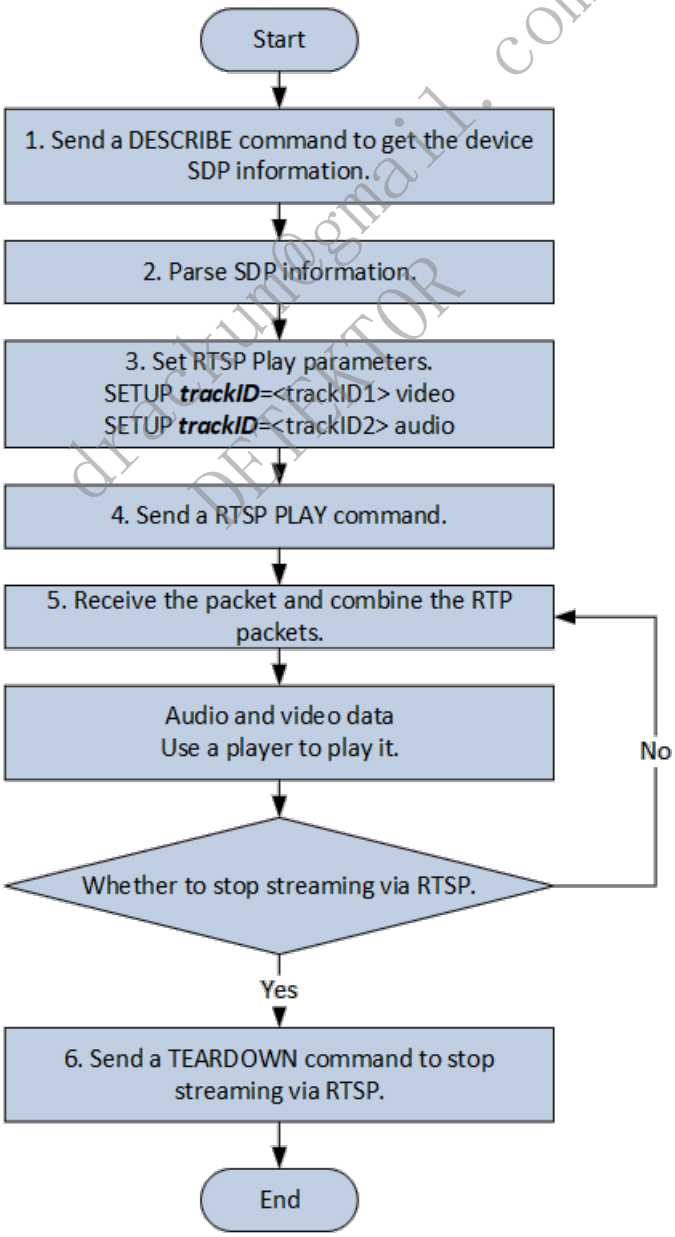
4.3 Real-Time Live View

4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

4.3.2 API Calling Flow



1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`. Digest authentication with devices is required before this step. 2. The client parses the media SDP information returned by the device. 3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For

example, trackID=1 indicates videos while trackID=2 indicates audios. 4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`. 5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed. 6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

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```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.22.126
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

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```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

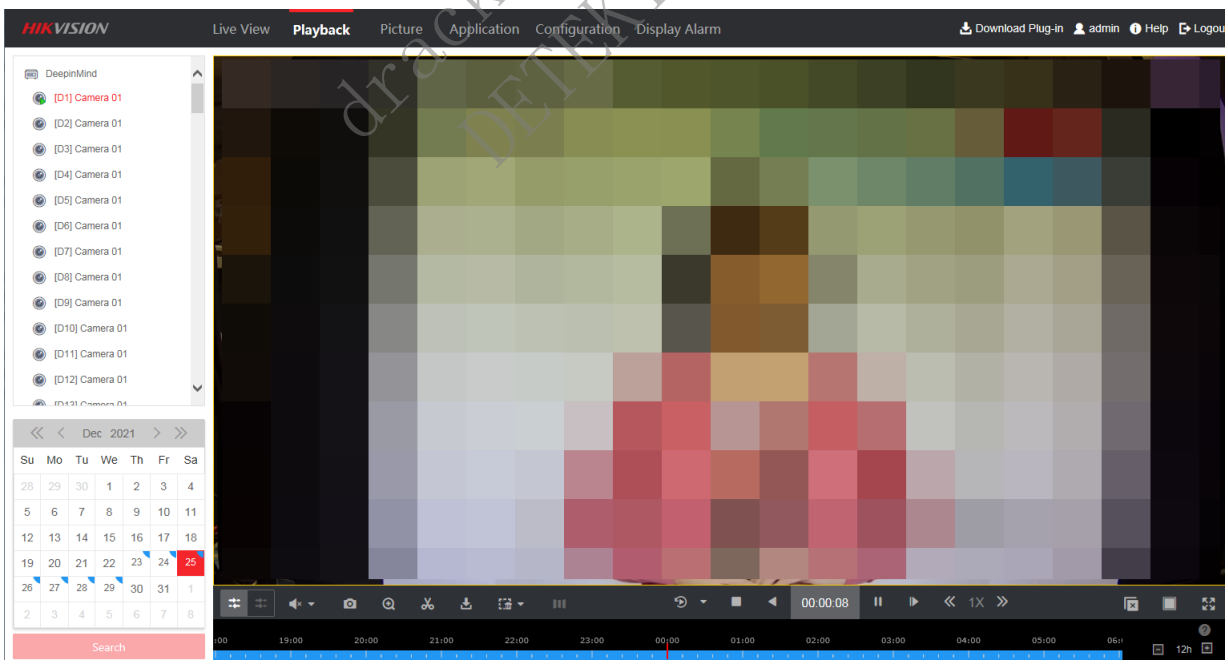
```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

4.4 Playback

4.4.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

Below is an example of playback on a web client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos are shown on the calendar and you can click a date to play back searched videos of the corresponding day.



4.4.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.

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3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. Refer to streaming via RTSP in real-time streaming. See details in RFC 7826 (Real Time Streaming Protocol).

Notes:

- Playback via RTSP needs digest authentication. The method is the same as the one of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in RFC 7826 (Real Time Streaming Protocol).

4.4.3 Example

4.4.3.1 Search for Videos According to Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
    <year>2021</year>
    <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

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```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <dayList>
        <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
        <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
        <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
        <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
        <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
        <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
        <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
        <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
        <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
        <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
        <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
        <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
        <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
        <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
        <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
        <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
        <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
    </dayList>
</trackDailyDistribution>
```

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4.4.3.2 Search for Videos

Request from Client

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
    <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
    <trackList>
        <trackID>101</trackID>
    </trackList>
    <timeSpanList>
        <timeSpan>
            <startTime>2021-08-16T00:00:00Z</startTime>
            <endTime>2021-08-18T23:59:59Z</endTime>
        </timeSpan>
    </timeSpanList>
    <maxResults>100</maxResults>
    <searchResultPostion>0</searchResultPostion>
    <metadataList>
        <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
    </metadataList>
</CMSearchDescription>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
    <responseStatus>true</responseStatus>
    <responseStatusStrg>OK</responseStatusStrg>
    <numOfMatches>1</numOfMatches>
    <matchList>
        <searchMatchItem>
            <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
            <trackID>101</trackID>
            <timeSpan>
                <startTime>2021-08-18T15:18:15Z</startTime>
                <endTime>2021-08-18T15:19:08Z</endTime>
            </timeSpan>
            <mediaSegmentDescriptor>
            <contentType>video</contentType>
            <codecType>H.264-BP</codecType>
            <playbackURI>rtsp://10.14.97.40/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&amp;endtime=20210818T151908Z&amp;name=00000004667000100&amp;size=1400788</playbackURI>
            <lockStatus>unlock</lockStatus>
            <name>00000004667000100</name>
            </mediaSegmentDescriptor>
            <metadataMatches>
                <metadataDescriptor>recordType.meta..com/timing</metadataDescriptor>
            </metadataMatches>
        </searchMatchItem>
    </matchList>
</CMSearchResult>
```

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4.4.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP commands two times, and the server responds to them.

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY request, and the server starts to push streams for playback.

```
PLAY rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

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```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

4.5 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

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4.5.1 Arming

The client establishes a HTTP persistent connection with the device to receive event messages from the device.

There are two methods (arming with subscription and arming without subscription) to receive events from the device. The arming without subscription is to get all event messages from the device via HTTP GET method, while the arming with subscription is to get messages of subscribed events via HTTP POST method.

Notes

- ISAPI arming (with or without subscription) uses the HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send event messages continuously, while it's not supported for clients to send any message to the device via the connection.
- When the heartbeat timed out and no message is received from the device, you should terminate the arming connection and try establishing a new one.

4.5.1.1 Arming without Subscription

1. Establish the connection of arming without subscription: `GET /ISAPI/Event/notification/alertStream` and keep the connection alive via configuring `Connection: keep-alive` in HTTP headers on the client.
2. Receive events sent by the device. The event message will be separated and parsed by boundary. For parsing details, see **Event Message Parsing** below.
3. Terminate the arming connection when no event message needs to be received.

Event Message Parsing:

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

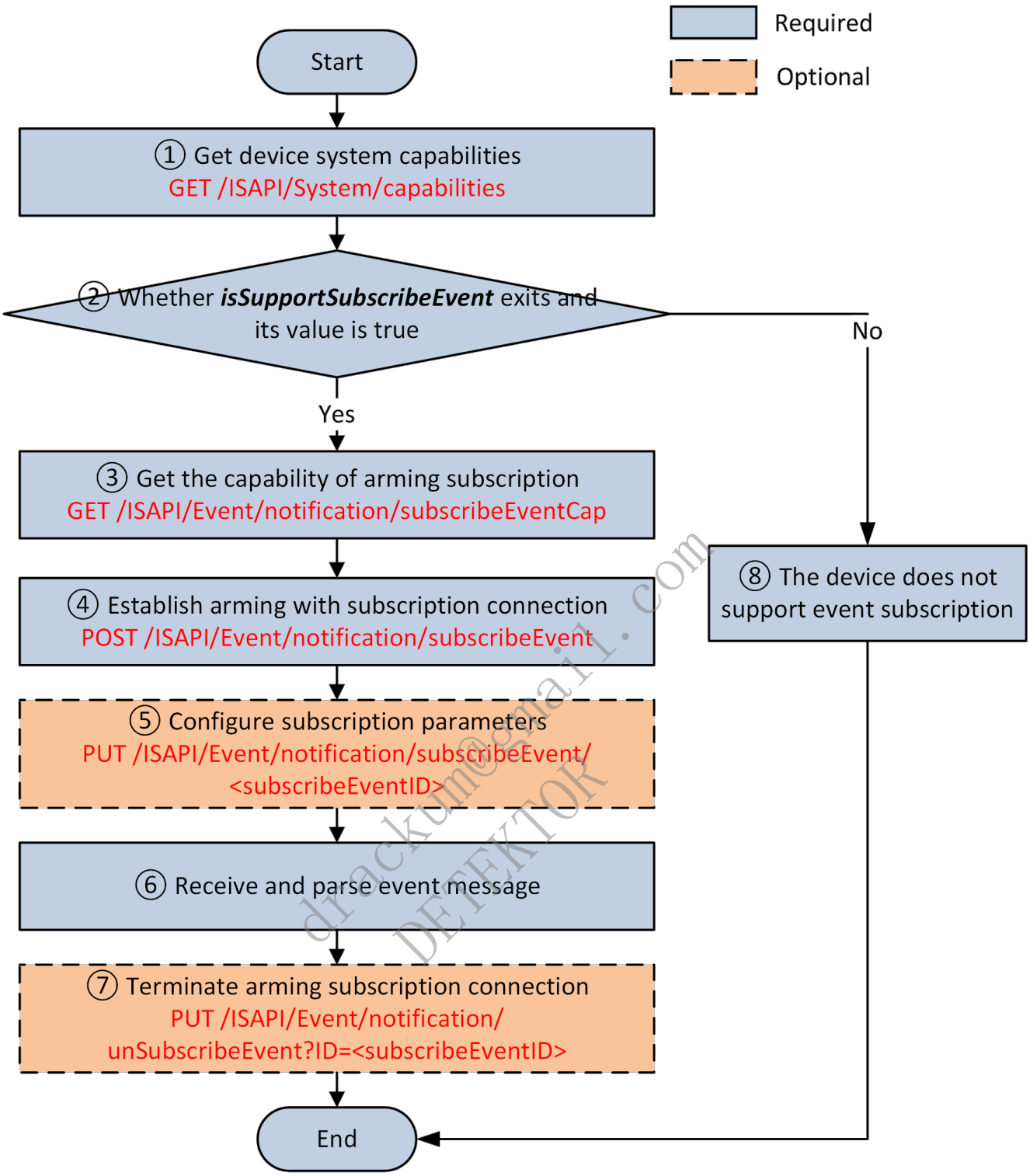
<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

Note: `<data_gateway_ip>` and `<frontier>` are variables, `[Picture Data]` indicates the raw data of a picture.

4.5.1.2 Arming with Subscription

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by the boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

Event Messages Parsing:

Client Creates Arming with Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

Server Responds to Request and Push Event Message

When the client sends a request to the device for establishing an arming connection, the device sends data in HTTP form format (multipart/form-data). In HTTP request headers, the `Content-Type` is `multipart/form-data, boundary=AaB03x`, of which the boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after boundary. In normal conditions, devices will not close the arming link actively, and the end mark `--AaB03x--` will not be sent on the arming link.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Picture Data
--AaB03x--
```

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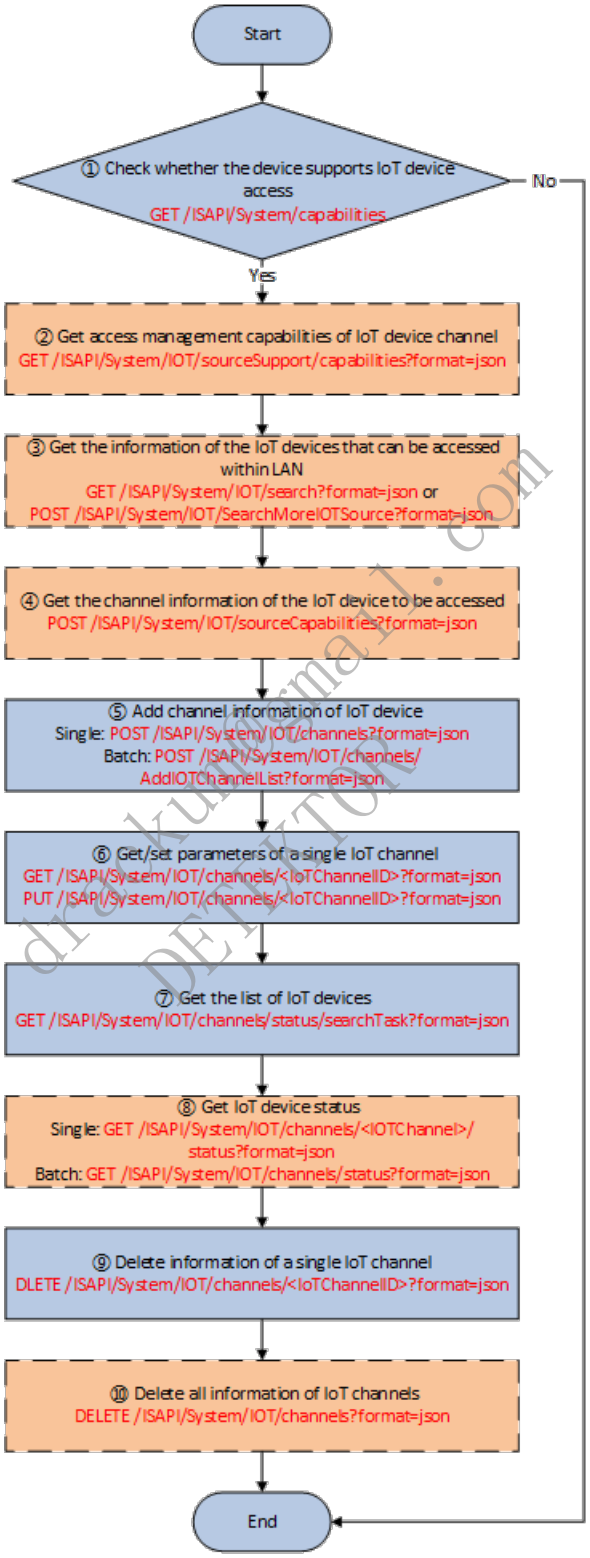
The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description. `form-data` is a piece of form data.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next `\r\n` to the next `--boundary`.

5 Device (General)

5.1 4 IoT Channel Management

5.1.1 IoT Device



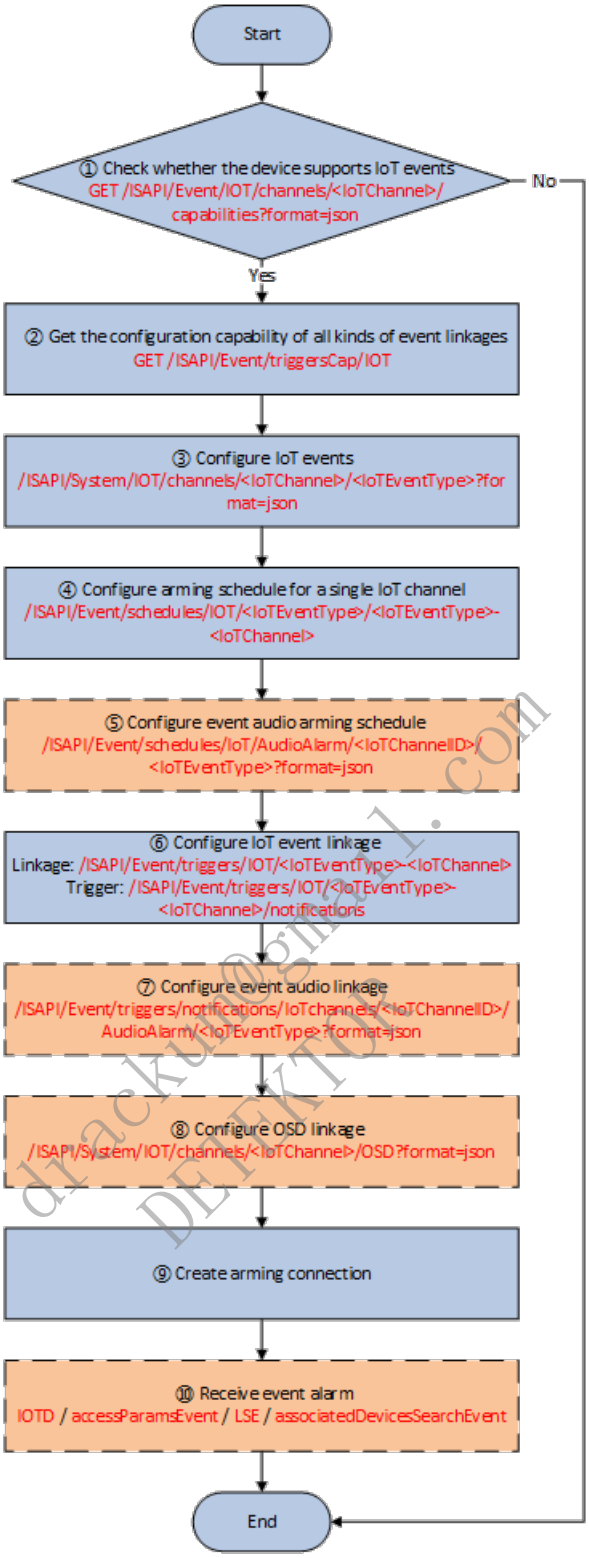
1. Check whether the device supports IoT device access: `GET /ISAPI/System/capabilities`. If the node `IOTCap` exists in the response message, it means that the device supports IoT device access. The content of `IOTCap` is the capability set of IoT device access.
2. (Optional) Get access management capabilities of IoT device channel: `GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json`. Its function is similar to that of the previous step, so it is optional.
3. Get the information of the IoT devices that can be accessed within LAN.

- All results: `GET /ISAPI/System/IOT/search?format=json`
- Results with search conditions: `POST /ISAPI/System/IOT/SearchMoreIOTSource?format=json`

4. Get the channel information of the IoT device to be accessed: `POST /ISAPI/System/IOT/sourceCapabilities?format=json`
5. Add channel information of IoT device.
   - Single: `POST /ISAPI/System/IOT/channels?format=json`
   - Batch: `POST /ISAPI/System/IOT/channels/AddIOTChannelList?format=json`
6. Get/set parameters of a single IoT channel.
   - Get: `GET /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
   - Set: `PUT /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
7. Get the list of IoT devices: `GET /ISAPI/System/IOT/channels/status/searchTask?format=json`
8. (Optional) Get IoT device status. Its function is similar to that of the previous step, so it is optional. It is recommended to get device status via the device list in the previous step.
   - Single: `GET /ISAPI/System/IOT/channels/<IOTChannel>/status?format=json`
   - Batch: `GET /ISAPI/System/IOT/channels/status?format=json`
9. Delete information of a single IoT channel: `DLETE /ISAPI/System/IOT/channels/<IoTChannelID>?format=json`
10. Delete all information of IoT channels: `DELETE /ISAPI/System/IOT/channels?format=json`

5.1.2 IoT Event

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1. Get the event capabilities supported by the IoT channel: `GET /ISAPI/Event/IOT/channels/<IoTChannel>/capabilities?format=json`
2. Get the configuration capability of all kinds of event linkages: `GET /ISAPI/Event/triggersCap/IOT`
3. Configure IoT events.
   - Get IoT event configuration capability: `GET /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>/capabilities?format=json`
   - Set IoT event parameters: `PUT /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>?format=json`
   - Get IoT event parameters: `GET /ISAPI/System/IOT/channels/<IoTChannel>/<IoTEventType>?format=json`
4. Configure arming schedule for a single IoT channel.
   - Get arming schedule of a single IoT channel: `GET /ISAPI/Event/schedules/IOT/<IoTEventType>/<IoTEventType>-<IoTChannel>`
   - Set arming schedule of a single IoT channel: `PUT`

`/ISAPI/Event/schedules/IOT/<IoTEventType>/<IoTEventType>-<IoTChannel>`

5. (Optional) Configure event audio arming schedule. If it is not configured, the device will play audio alarm whenever a linkage alarm is triggered. If it is configured, the device will only play the audio alarm within the arming schedule, and will upload the alarm without playing audio for the rest of the time.
   - Get event audio linkage configuration capability: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>/capabilities?format=json`
   - Set event audio arming schedule parameters: `PUT /ISAPI/Event/schedules/IoT/AudioAlarm/<IoTChannelID>/<IoTEventType>?format=json`
   - Get event audio arming schedule parameters: `GET /ISAPI/Event/schedules/IoT/AudioAlarm/<IoTChannelID>/<IoTEventType>?format=json`
6. Configure IoT event linkage, which consists of two parts: linkage method configuration and trigger method configuration.
   - Set IoT event linkage method: `PUT /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>`
   - Get IoT event linkage method: `GET /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>`
   - Set IoT event trigger method: `PUT /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>/notifications`
   - Get IoT event trigger method: `GET /ISAPI/Event/triggers/IOT/<IoTEventType>-<IoTChannel>/notifications`
7. (Optional) Configure event audio linkage.
   - Get event audio linkage configuration capability: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>/capabilities?format=json`
   - Set event audio linkage parameters: `PUT /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>?format=json`
   - Get event audio linkage parameters: `GET /ISAPI/Event/triggers/notifications/IoTChannels/<IoTChannelID>/AudioAlarm/<IoTEventType>?format=json`
8. Configure IoT channel OSD linkage.
   - Get OSD linkage configuration capability: `GET /ISAPI/System/IOT/channels/<IoTChannel>/OSD/capabilities?format=json`
   - Set OSD linkage parameters: `PUT /ISAPI/System/IOT/channels/<IoTChannel>/OSD?format=json`
   - Get OSD linkage parameters: `GET /ISAPI/System/IOT/channels/<IoTChannel>/OSD?format=json`
9. Upload alarm events. There are four types of IoT events:
   - IoT device events: `IOTD`
   - Getting access parameters: `accessParamsEvent`
   - Logistics scanning event: `LSE`
   - Getting the list of linked devices: `associatedDevicesSearchEvent`

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5.2 Arming and Subscription

5.2.1 Introduction to the Function

With arming and subscription, the client can establish HTTP persistent connection with the device, and continuously receive the event messages from the device.

For ISAPI event arming, the client can receive all types of events by GET method, or receive the subscribed events only by POST method.

5.2.2 API Calling Flow

5.2.2.1 Without Subscription

1. Establish a connection for arming: `GET /ISAPI/Event/notification/alertStream`. You need to set `Connection: keep-alive` in HTTP Headers.
2. When receiving events sent by the device, the event messages can be separated and parsed by boundary. See "Parsing Event Messages" below for details.
3. Disable the arming connection when you do not need to receive event messages.

5.2.2.1.2 Syntax

```
GET /ISAPI/Event/notification/alertStream HTTP/1.1
Host: <data_gateway_ip>
Connection: Keep-Alive

HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

GET /ISAPI/Event/notification/alertStream HTTP/1.1
Authorization: Digest username="admin", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",
uri="/ISAPI/Event/notification/alertStream", cnonce="3d183a245b8729121ae4ca3d41b90f18", nc=00000001, qop="auth", response="f2e0728991bb031f83df557a8f185178"
Host: 10.6.165.192

HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/form-data; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!-- Some alarms are in JSON format, so the upper layer should parse based on the Content-Type field -->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/jpeg
Content-Length: image_length

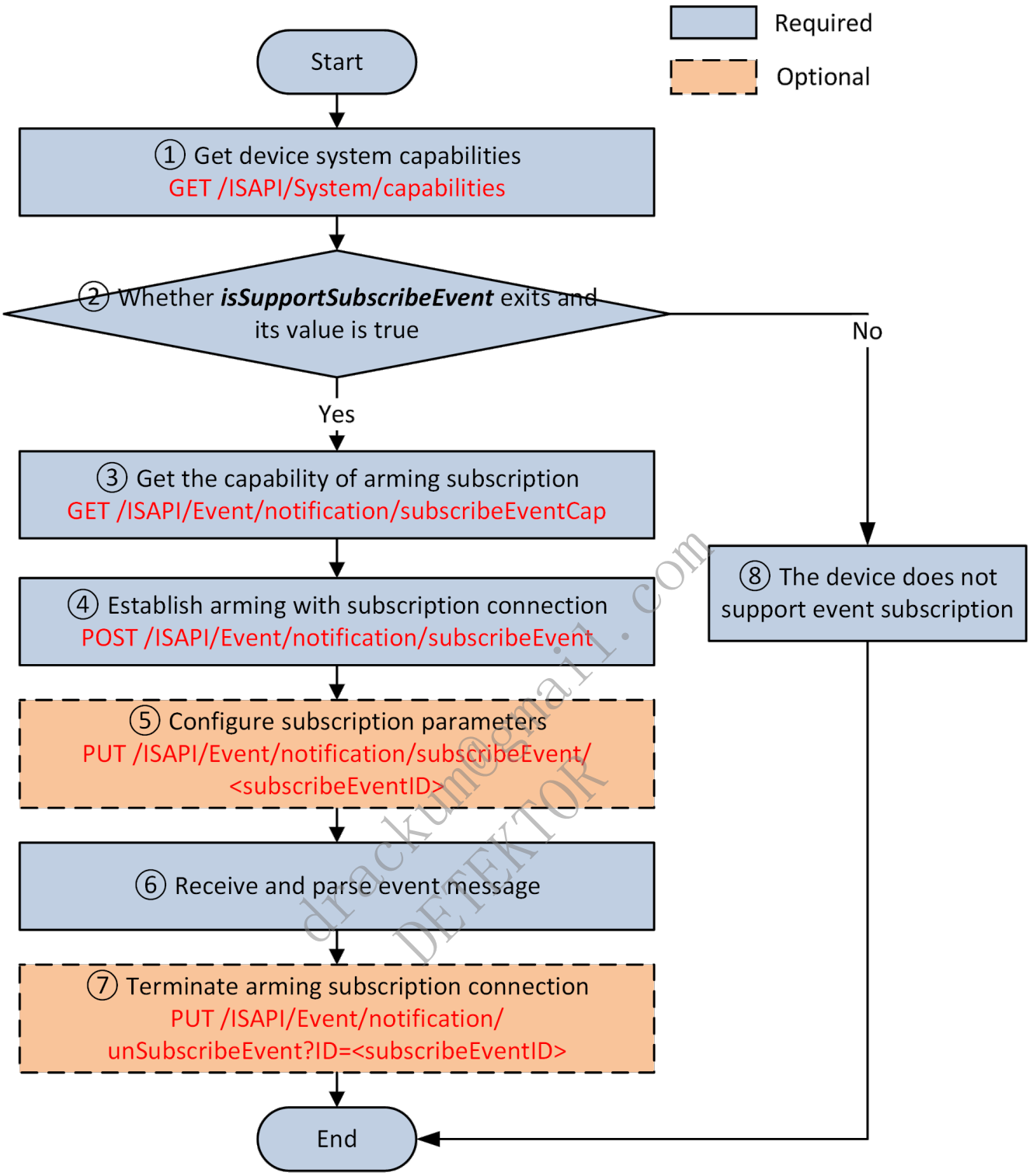
[Image Data]
--<frontier>
```

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Note: `<data_gateway_ip>` and `<frontier>` are variables, and `[Image Data]` is an abbreviated representation indicating the raw data of an image at this location.

5.2.2.2 Subscription

5.2.2.2.1 API Calling Flow



1. Get device system capabilities: `GET /ISAPI/System/capabilities`.
2. Check if event subscription is supported: `isSupportSubscribeEvent` exists and its value is `true`. When `isSupportSubscribeEvent` does not exist or its value is false, the device does not support event subscription.
3. Get the capability of arming with subscription: `GET /ISAPI/Event/notification/subscribeEventCap`.
4. Establish a connection of arming with subscription: `POST /ISAPI/Event/notification/subscribeEvent`. You need to set `Connection: keep-alive` in HTTP Headers.
5. (Optional) Edit parameters of the existing subscription. You need to get the subscription parameters first: `GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`. Then, edit the parameters based on the existing subscription configurations: `PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>`.
6. Receive events sent by the device. The event messages will be separated and parsed by boundary. For parsing description, see **Event Messages Parsing** below.

7. (Optional) Terminate the connection of arming with subscription: `PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>`. When communicating with the device via HTTP directly, there is no need to call this API. You can just terminate the connection.

Note:

Three types of data will be transmitted on the arming link: `<SubscribeEventResponse/>`, `<EventNotificationAlert/>`, and picture data. `<SubscribeEventResponse/>` is the data of first form sent by the device after arming established, see the response parameters of URL (`POST /ISAPI/Event/notification/subscribeEvent`) for details; and `<EventNotificationAlert/>` is the event content or heartbeat, you can identify the event type via field `eventType`, e.g., for heartbeat, the value of `eventType` is `heartBeat`.

5.2.2.2.2 Example

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Host: device_ip
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml;
Content-Length: text_length
Connection: Keep-Alive

<SubscribeEvent/>
HTTP/1.1 401 Unauthorized
Date: Sun, 01 Apr 2018 18:58:53 GMT
Server:
Content-Length: 178
Content-Type: text/html
Connection: keep-alive
Keep-Alive: timeout=10, max=99
WWW-Authenticate: Digest qop="auth", realm="IP Camera(C2183)", nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d", stale="FALSE"

POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message
format accroding to Content-Type when parsing event messages-->
Content-Length: text_length

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length

[Picture Data]
--<frontier>
```

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5.2.2.3 Event Messages Parsing

After the arming connection with the device is established, the data sent by the device is in HTTP form format (multipart/form-data). In an HTTP request, `Content-Type` in Headers is usually `multipart/form-data, boundary=AaB03x`, and boundary is a variable used to divide HTTP Body into multiple units, each being a set of data and has its own Headers and Body. For detailed format description, see RFC 1867 (Form-based File Upload in HTML). An example is shown below. Note the dash `--` before and after boundary. Under normal circumstances, the device will not actively close the arming connection, so the device will not send the form format end symbol `--AaB03x--` on the arming connection.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=AaB03x
Connection: keep-alive
--AaB03x
Content-Disposition: form-data; name="ANPR.xml"; filename="ANPR.xml";
Content-Type: application/xml
Content-Length: 9

<ANPR/>
--AaB03x
Content-Disposition: form-data; name="licensePlatePicture.jpg"; filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 14

Image Data
--AaB03x--
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=AaB03x	Content type. multipart/form-data means the message is in form format.
boundary	AaB03x	Delimiter of the form message. --boundary is the start of a form. --boundary-- is the end of the whole HTTP form message.
Content-Disposition	form-data; name="ANPR.xml"; filename="ANPR.xml";	Content description.
name	"ANPR.xml"	Form name.
filename	"ANPR.xml"	File name of the form.
Content-Length	9	Content length, starting from the next \r\n to the next --boundary.

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5.2.3 Restriction Description

Note that ISAPI arming (with or without subscription) uses HTTP/HTTPS persistent connection. Due to the simplex channel communication mode of HTTP, after establishing the arming connection, the device will send out event messages continuously, while you cannot send any message to the device via the connection.

After the heartbeat time, if you do not receive any message from the device, you should disable the arming connection and try establishing a new one.

5.2.4 Sample Messages

5.2.4.1 Establish Arming Subscription

```
POST /ISAPI/Event/notification/subscribeEvent HTTP/1.1
Authorization: Digest username="admin",realm="IP
Camera(C2183)",nonce="4e5468694e7a42694e7a4d364f4449354d7a6b354d54513d",uri="/ISAPI/Event/notification/alertStream",cnonce="3d183a245b8729121ae4ca3d41b90f18
",nc=00000001,qop="auth",response="f2e0728991bb031f83df557a8f185178"
Host: device_ip

<SubscribeEvent/>
```

5.2.4.2 The Device Responses and Uploads an Event Message

```
HTTP/1.1 200 OK
MIME-Version: 1.0
Connection: close
Content-Type: multipart/mixed; boundary=<frontier>

--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<SubscribeEventResponse/>
--<frontier>
Content-Type: application/xml; charset="UTF-8"  <!--Some alarm messages are in JSON format, so when parsing messages, the upper-layer should distinguish them according to the Content-Type field.-->
Content-Length: text_length

<EventNotificationAlert version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <ipAddress>10.17.133.46</ipAddress>
    <portNo>80</portNo>
    <protocol>HTTP</protocol>
    <macAddress>44:19:b6:6d:24:85</macAddress>
    <channelID>1</channelID>
    <dateTime>2017-05-04T11:20:02+08:00</dateTime>
    <activePostCount>0</activePostCount>
    <eventType>heartBeat</eventType>
    <eventState>active</eventState>
    <eventDescription>heartBeat</eventDescription>
</EventNotificationAlert>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Type: image/pjpeg
Content-Length: image_length
Content-ID: image_ID

[Picture Data]
--<frontier>
```

5.3 Calling Flow of Device Packet Capture

5.3.1 Function Introduction

When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

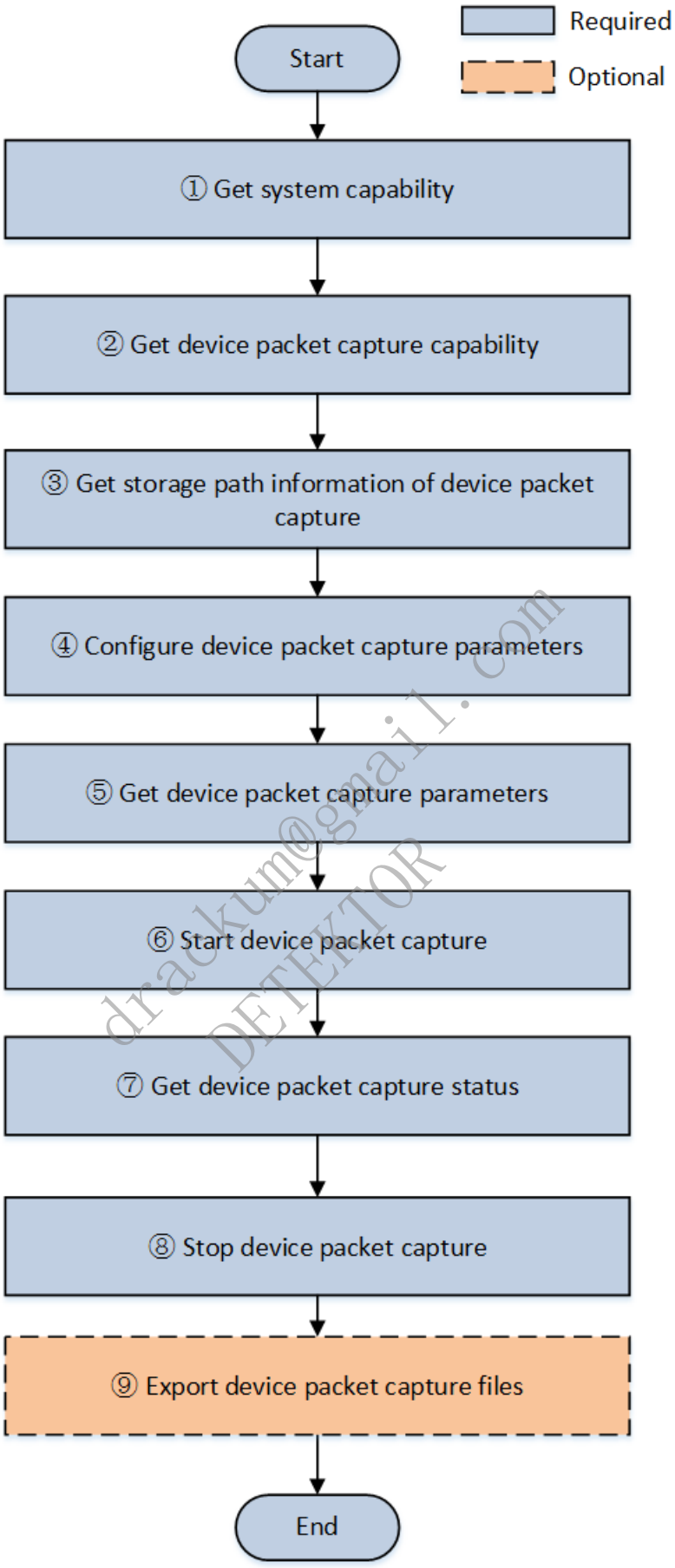
1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.3.2 API Calling Flow

5.3.2.1 Device Packet Capture

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1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: `GET /ISAPI/System/networkCapture/StoragePathInfo?format=json`.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: `PUT /ISAPI/System/networkCapture/captureParams?format=json`.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: `GET /ISAPI/System/networkCapture/captureParams?format=json`.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: `PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>`.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: `GET /ISAPI/System/networkCapture/manualStatus?format=json`.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

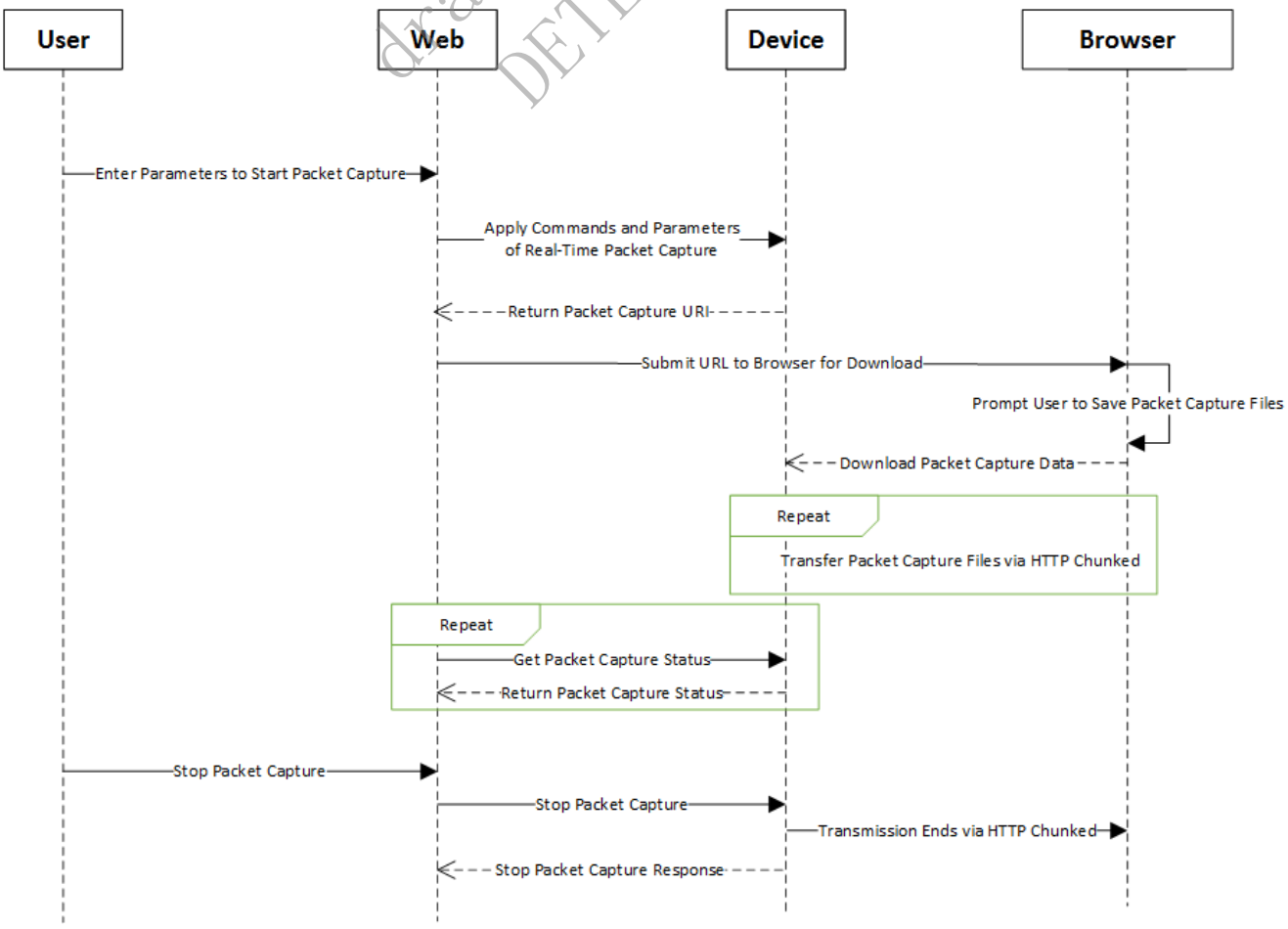
Stop device packet capture: `PUT /ISAPI/System/networkCapture/manualStop?format=json`.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

Export device packet capture files: `GET /ISAPI/System/networkCapture/exportFile?format=json`.

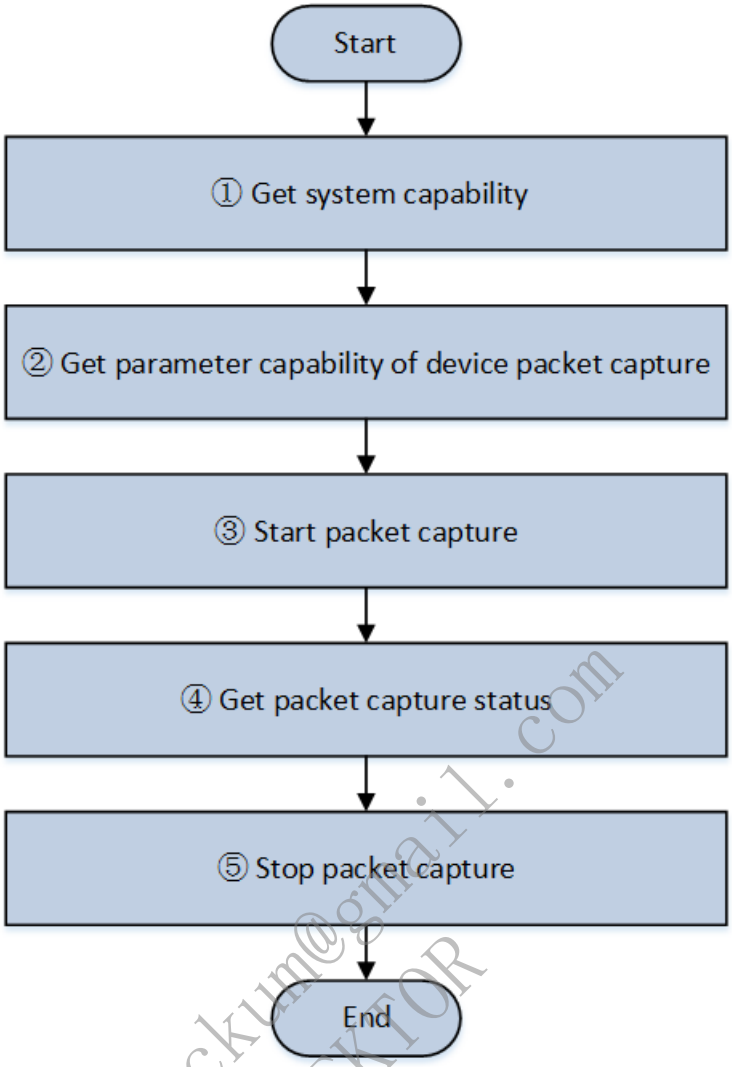
5.3.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



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See the following figure for the calling flow:



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1. Get device system capabilities: `GET /ISAPI/System/capabilities`. `<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

Note:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

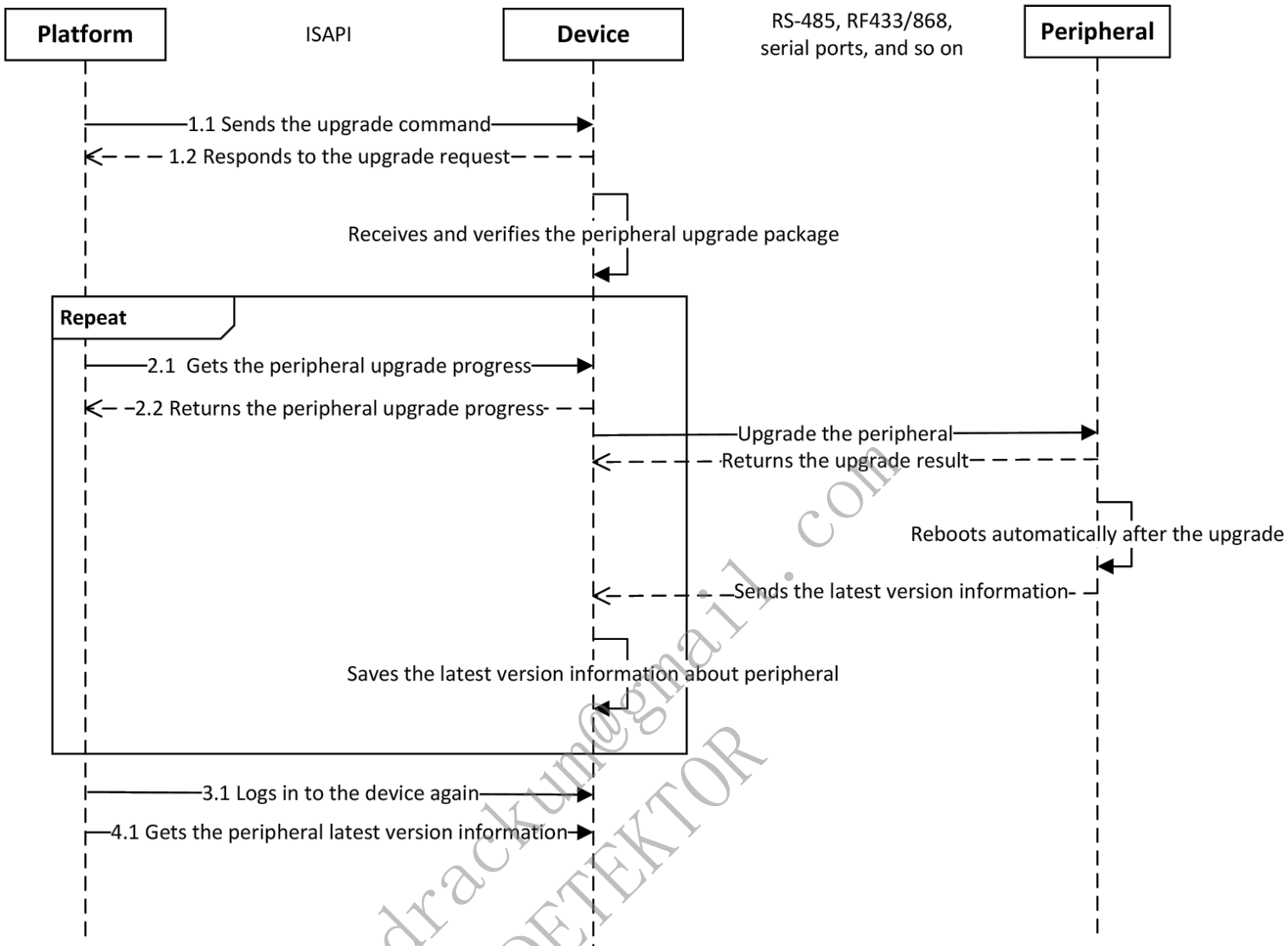
5.4 Device Peripherals Upgrade

5.4.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.4.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



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1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcsUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.
2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.5 Device Time Sync

5.5.1 Introduction to the Function

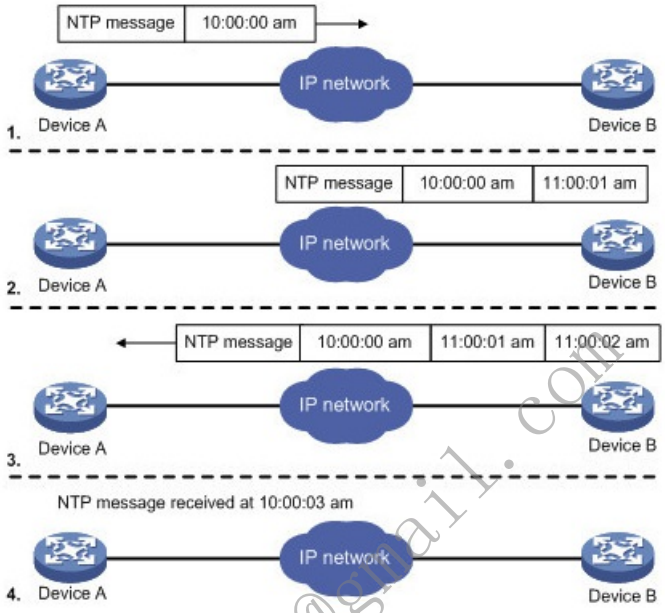
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.5.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B, Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: Delay = (T4-T1) - (T3-T2) = 2 seconds.

Time difference between Device A and Device B: offset = ((T2-T1)+(T3-T4))/2=1 h.

Device A can sync its own time with that of Device B according to calculation results.

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5.5.2 API Calling Flow

5.5.2.1 Time Sync Configuration

1. Get the Capability of Device Time synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows：

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime`、`timeZone`.

Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note： For manual time synchronization (time offset including time zone offset): Set manual time synchronization: localTime refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: localTime refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.5.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

2. Configure time zone parameters

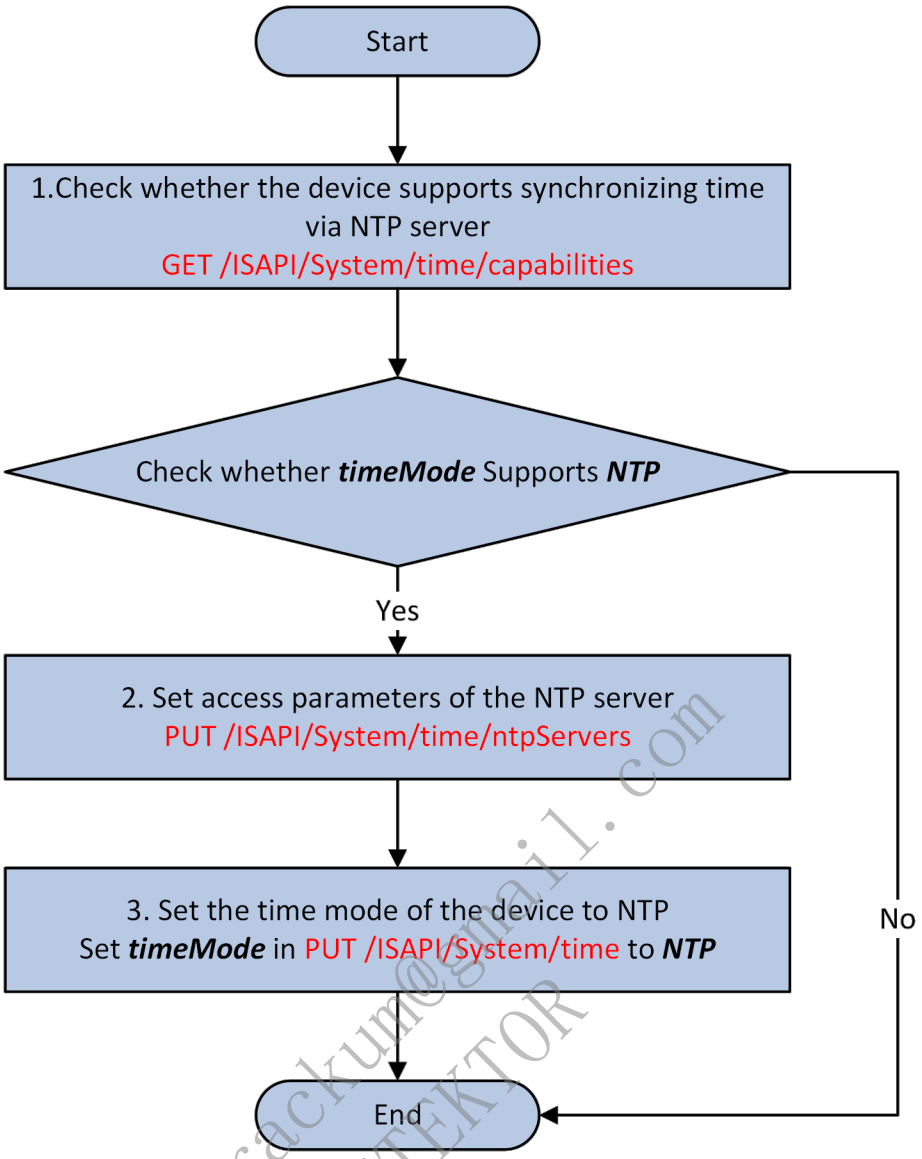
Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.5.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):

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1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: `GET /ISAPI/System/time/capabilities`, and check whether `timeMode` supports `NTP`.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: `GET /ISAPI/System/time/ntpServers/capabilities`

Set access parameters of the NTP server: `PUT /ISAPI/System/time/ntpServers`

Get access parameters of the NTP server: `GET /ISAPI/System/time/ntpServers`

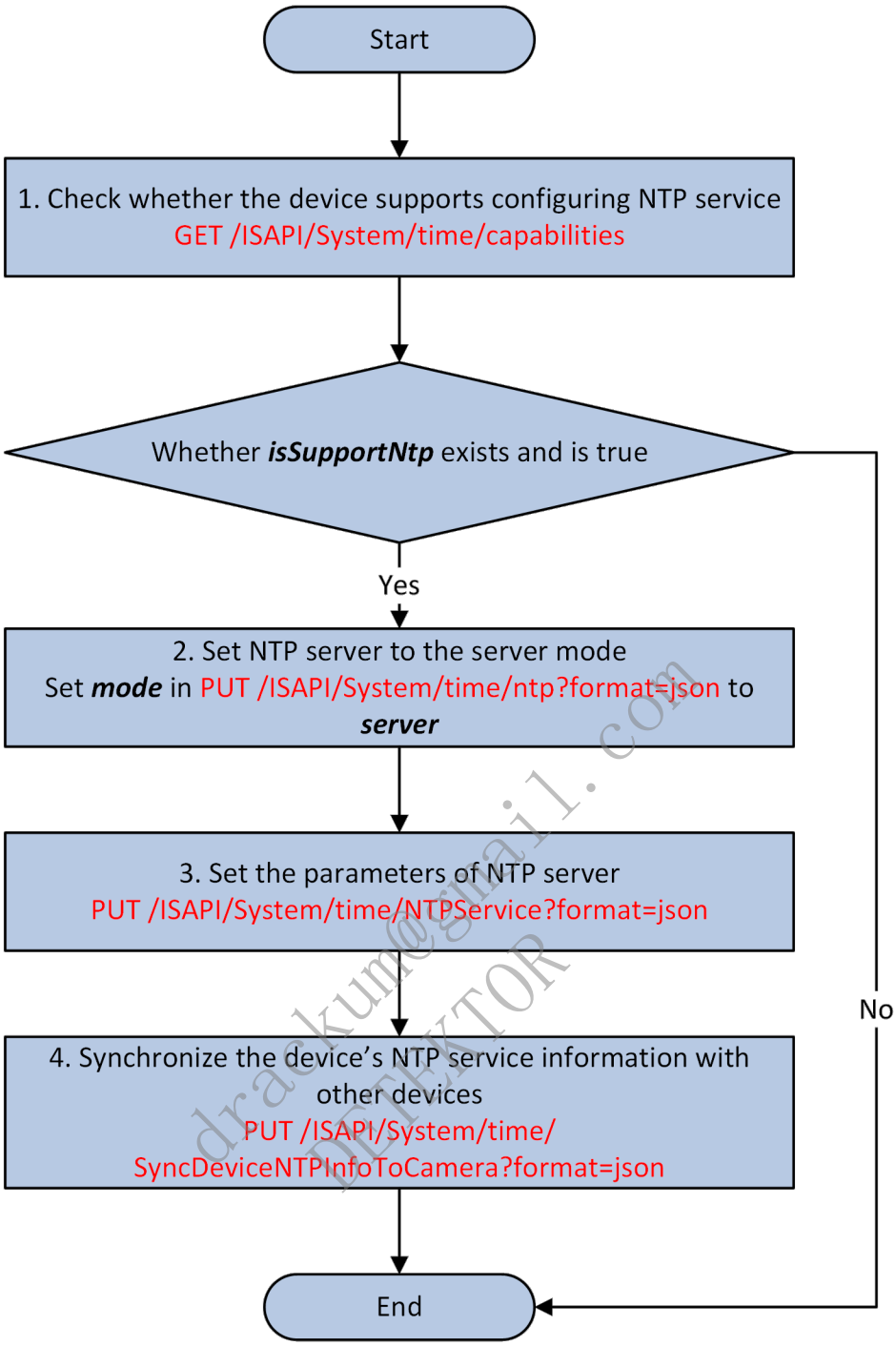
3. Set the time mode of the device to NTP

Supports setting the value of `timeMode` to `NTP`.

Get device time synchronization management parameters: `GET /ISAPI/System/time` Set device time synchronization management parameters: `PUT /ISAPI/System/time`

5.5.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):



1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: `GET /ISAPI/System/time/capabilities`; If `isSupportNtp` is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of `mode` to `server`.

Get the capability of server mode: `GET /ISAPI/System/time/ntp/capabilities?format=json`

Set NTP to server mode: `PUT /ISAPI/System/time/ntp?format=json`

Get parameters of NTP server mode: `GET /ISAPI/System/time/ntp?format=json`

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: `GET /ISAPI/System/time/NTPService/capabilitis?format=json`

Set the NTP server parameters: `PUT /ISAPI/System/time/NTPService?format=json`

Get the parameters of the NTP server: `GET /ISAPI/System/time/NTPService?format=json`

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json`

Synchronize device's NTP service information with the camera: `PUT /ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json`

Get the progress of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json`

Search for the results of synchronizing device's NTP service information with the camera: `POST /ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json`

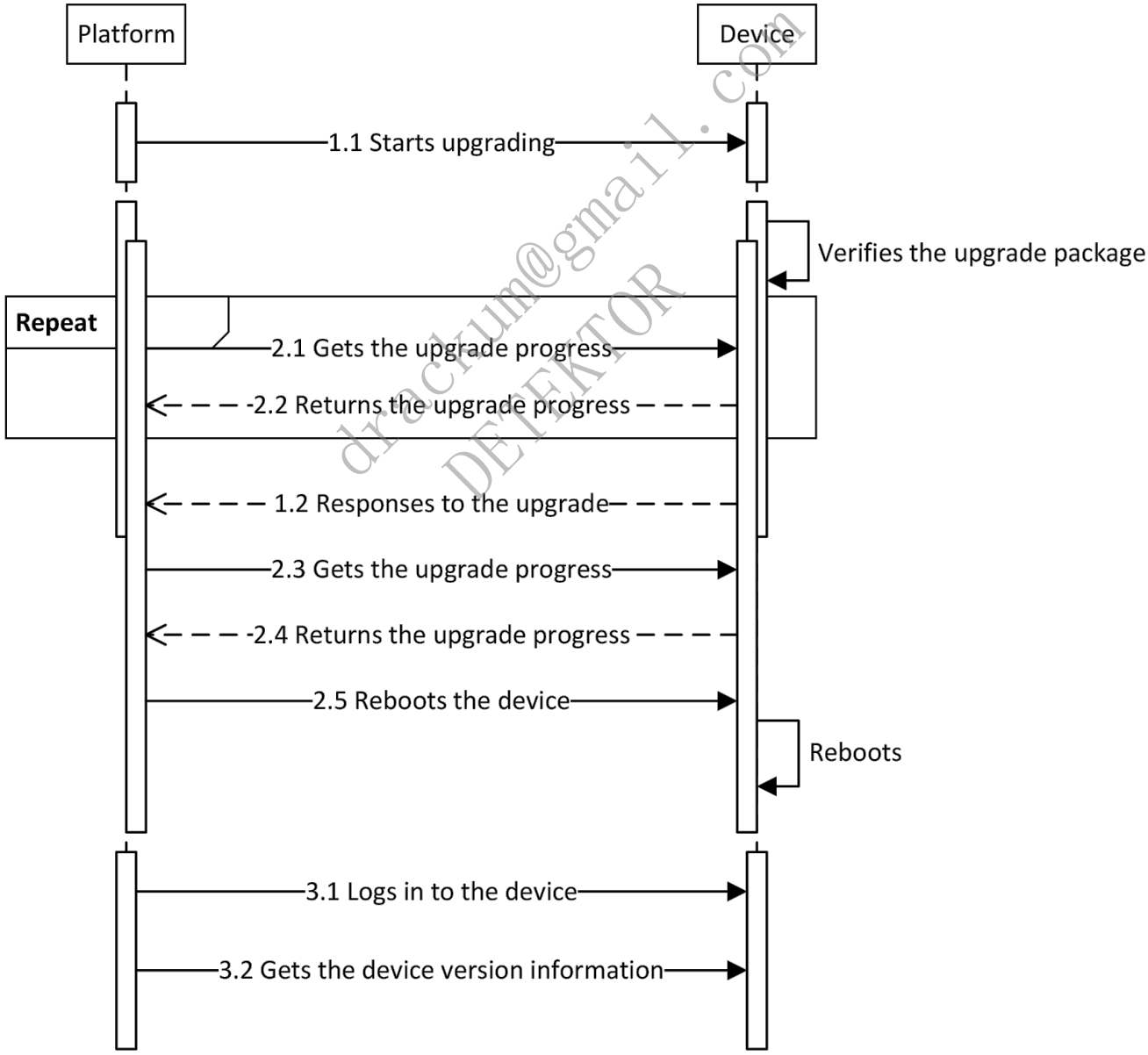
5.6 Device Upgrade

5.6.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.6.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



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1. **Upgrade devices.**

   Upgrade the device firmware: `POST /ISAPI/System/updateFirmware`.

2. **Get the device upgrade progress.**

Get the device upgrade progress: `GET /ISAPI/System/upgradeStatus`.

3. **Reboot devices.**

   Reboot devices: `PUT /ISAPI/System/reboot`.

5.7 Mutually Exclusive Functions

5.7.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

5.7.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: `GET /ISAPI/System/mutexFunction/capabilities?format=json`. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exlusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).
2. Search for the functions that are mutually exclusive with a specified function: `POST /ISAPI/System/mutexFunction?format=json`. Based on the list of mutually exclusive functions returned by `GET /ISAPI/System/mutexFunction/capabilities?format=json`, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disbale the mutually exclusive function.
3. Get the mutual exclusion information when device function exception occurs: `GET /ISAPI/System/mutexFunctionErrorMsg`. After getting the error code, you can call this API to get the current mutually exclusive functions.

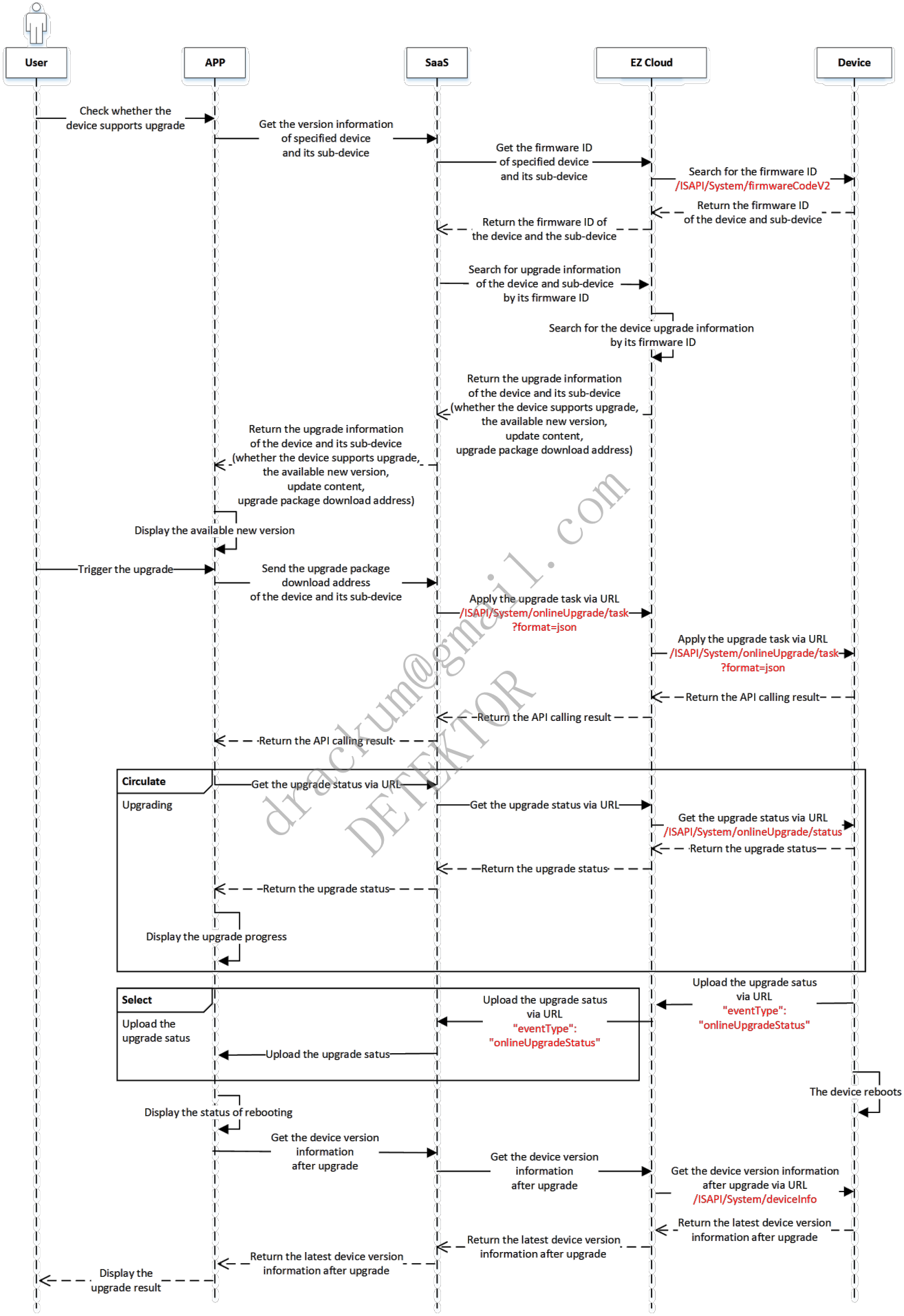
5.8 Remote Online Upgrade Based on The EZ2.0

5.8.1 Introduction to the Function

Considering that the device is connected to EZ via EZ 2.0 Protocol, EZ OTA upgrade solution is not supported. This solution applies to specified version upgrade.

5.8.2 API Calling Flow

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1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by its firmware ID. If there is an available new version, apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade devices (e.g., NVR) or sub-devices (e.g., IPC) via the node

device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.

3. During the device upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` and reboot automatically.
5. After both upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

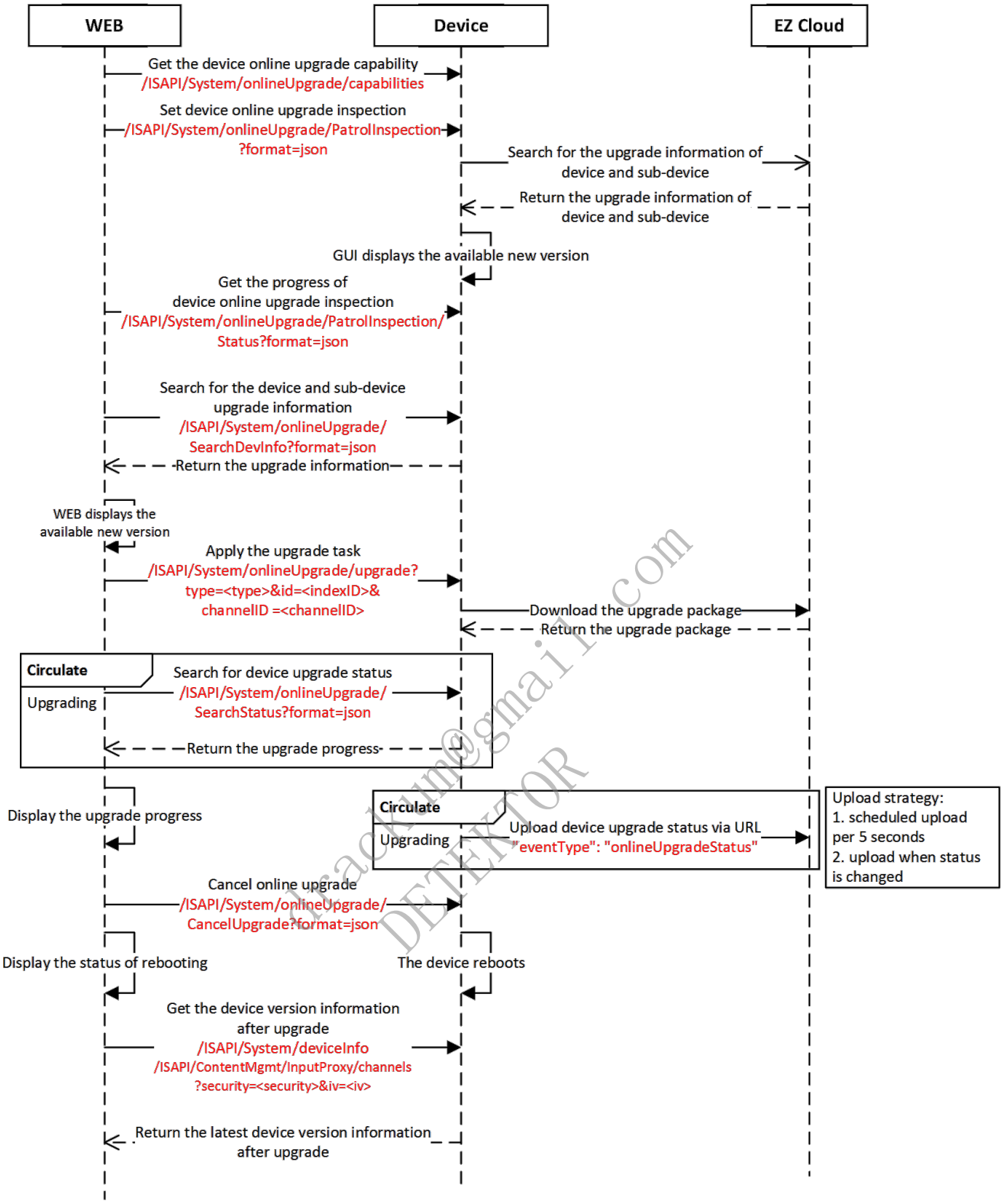
5.9 Remote Online Upgrade Based on the WEB Client

5.9.1 Introduction to the Function

This solution applies to remote online upgrade of the devices and sub-devices connected to WEB client.

5.9.2 API Calling Flow

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1. Get device online upgrade capability: `GET /ISAPI/System/onlineUpgrade/capabilities`.
2. Manually trigger the device online upgrade inspection: `POST /ISAPI/System/onlineUpgrade/PatrolInspection?format=json`. After it is enabled, the device will search for the available new version.
3. During the inspection, the WEB client can get the real-time inspection progress: `GET /ISAPI/System/onlineUpgrade/PatrolInspection/Status?format=json`.
4. After the device inspection is finished, search for the device and sub-device upgrade information: `GET /ISAPI/System/onlineUpgrade/SearchDevInfo?format=json`.
5. Apply the upgrade task: `PUT /ISAPI/System/onlineUpgrade/upgrade?type=<type>&id=<indexID>&channelID =<channelID>`. You can upgrade a specified device via the node `<channelID>` in the URL. Specify sub-device upgrade by configuring `<type>`.

6. You can search for online upgrade status during the upgrade: `GET /ISAPI/System/onlineUpgrade/SearchStatus?format=json`.
7. You can cancel the upgrade during the upgrade: `PUT /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json`.
8. After the upgrade and reboot are finished, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

5.10 Serial Port Accessed External Device Management

5.10.1 Introduction to the Function

Information management of device accessed the serial are as follows: 1. Configure manufacturer, type, and model information of the specific serial port access device. 2. Search for the device type or model supported by the specific serial port.

5.10.2 API Calling Flow

1. Check whether the device supports information management of serial port. Get the capability of the device serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportDeviceInfo>` is returned, the device supports information configuration of devices access the serial port.

2. Set the information of serial port:

- Get the capability of device information parameters of a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Get device information parameters access single serial port: `GET /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.
- Set device information parameter of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>/deviceInfo?format=json`.

3. Check whether the device supports linking information of devices access the serial port: `GET /ISAPI/System/Serial/capabilities`; If `<isSupportSearchDeviceInfoRelations>` is returned, it indicates that the device supports searching for linked information od devices access the serial port.

4. Search for linked information of devices access the serial port.

- Get the capability of searching for linked parameters of information of devices access a single serial port: `GET /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations/capabilities?format=json`.
- Search for linked parameters of information of devices access a single serial port:`POST /ISAPI/System/Serial/ports/<portID>/searchDeviceInfoRelations?format=json`.

5.11 Serial Port Data Transparent Transmission

5.11.1 Introduction to the Function

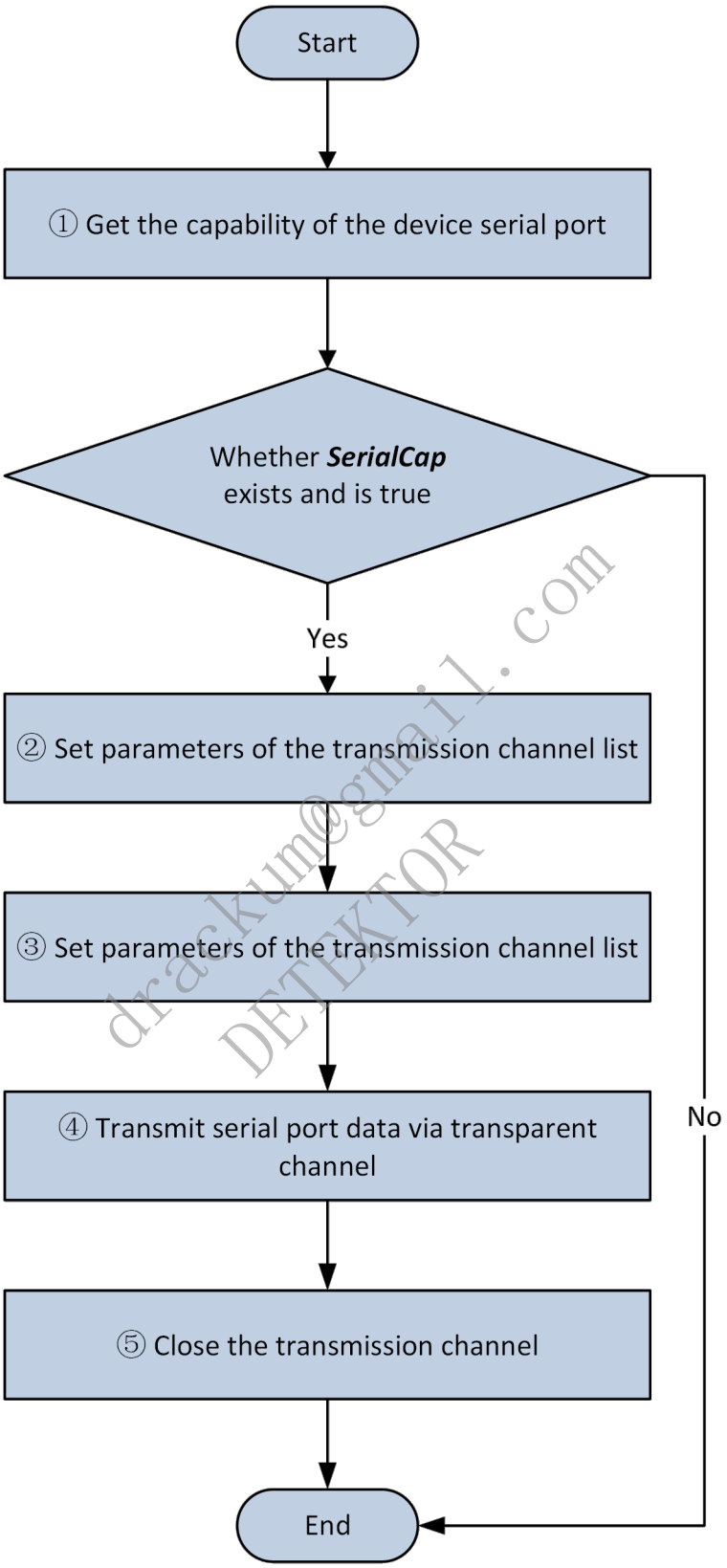
RS485, RS422 and RS232 serial ports external to the device are used as transparent channels to transmit serial port data. Supports the client sending serial data to the device, which then forwards the data to the serial port; conversely, when the serial port sends data to the device, which transparently transmits the data to the client. Note that the device's transparent transmission of serial data is half-duplex communication, meaning that bidirectional communication can be implemented between the device and the client, but the communication data is not necessarily in a question-and-answer format, and the client must match the request and response relationship on its own.

Since most standard HTTP request libraries do not support the persistent connection for receiving and sending data in real time, it is recommended that the client uses two TCP clients to implement the sending and receiving of

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transparently transmitted serial data.

5.11.2 API Calling Flow



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1. Check whether the device supports serial port data transmission.

Get the capability of the device serial port: `GET /ISAPI/System/capabilities`. If `SerialCap` is returned and the value is true, it indicates that the device supports the functions of the serial port.

2. Set parameters of the transmission channel list.

- Get parameters of the specific transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.
- Configure parameters of the specific transmission channel: `GET`

`/ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>`.

3. Open the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/open`;

4. Transmit serial port data via transparent channel.

- Receive data uploaded by device serial port through transmission channel: `GET /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.
- Send data to device serial port through transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/transData`.

5. Close the transmission channel: `PUT /ISAPI/System/Serial/ports/<portID>/Transparent/channels/<channelID>/close`.

5.12 Serial Port Parameter Configuration

5.12.1 Introduction to the Function

Serial port parameter configuration.

5.12.2 API Calling Flow

1. Check whether the device supports configuring serial port parameters.

Get the capability of device serial port: `GET /ISAPI/System/capabilities. If is returned and its value is true, the device supports functions of serial port.

3. Get parameters of all serial ports.

- Get the capability of all serial ports: `GET /ISAPI/System/Serial/capabilities`.
- Get control parameters of all serial ports: `GET /ISAPI/System/Serial/ports?permissionController=<indexID>`.

4. Set control parameters of a single serial port.

- Get control parameters of single serial port: `GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.
- Configure control parameters of single serial port: `PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>`.

5. Get the status of single serial port:`GET /ISAPI/System/Serial/ports/<portID>/status`.

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5.13 Sub-device Batch Upgrade

5.13.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.13.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: `/ISAPI/System/capabilities`.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgraading sub-devices.

Sub-device batch upgrade: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.

Batch sub-device upgrade progress search: `POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json`.

Sub-device Batch Upgrade Task Management

Get device system capabilities `/ISAPI/System/capabilities`, If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that batch upgrade of sub devices is supported,

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node `isSupportSearchBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading devices can be searched,

If the node `isSupportModifyBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node `isSupportDeleteBulkUpgradeChildDeviceTask` exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`.

Batch upgrade sub devices: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.****

Search for tasks of batch sub-device upgrade: `POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`.

Edit tasks of batch sub-device upgrade: **`PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`**.****

Delete tasks of batch sub-device upgrade: `PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json`.

Note: 1. The two methods for batch upgrading child devices are the same API: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`. If the `BulkUpgradeChildDeviceListCap.taskID` is returned in `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API `POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json`. Sub devices with the `tag` `isUpgradable` and `value` `true` indicate that the sub devices support upgrades.

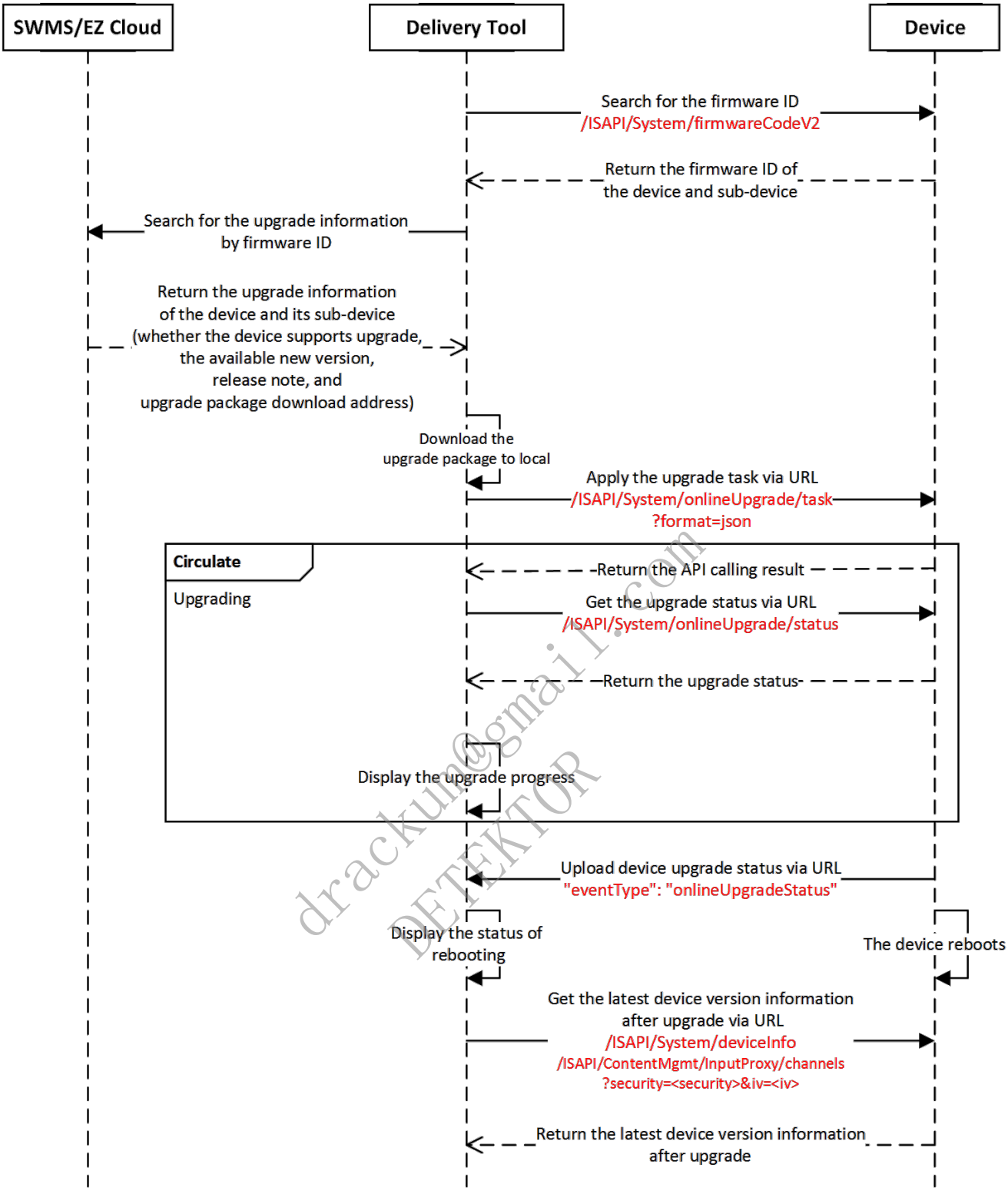
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5.14 Upgrade Based on Interactive Tools Local Deployment

5.14.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI. If the device is connected via ISAPI protocols, it is required to set up HTTP(s) service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK, you can implement the device upgrade via transmitting ISAPI.

5.14.2 API Calling Flow



1. Search for the device firmware ID: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information by firmware ID on the SWMS/EZ Cloud, download upgrade package to local storage, and apply the upgrade task to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., camera) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress: `GET /ISAPI/System/onlineUpgrade/status`.
4. After upgrade is completed, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and reboot automatically.
5. After the device rebooting, you can get the latest device version information.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=`

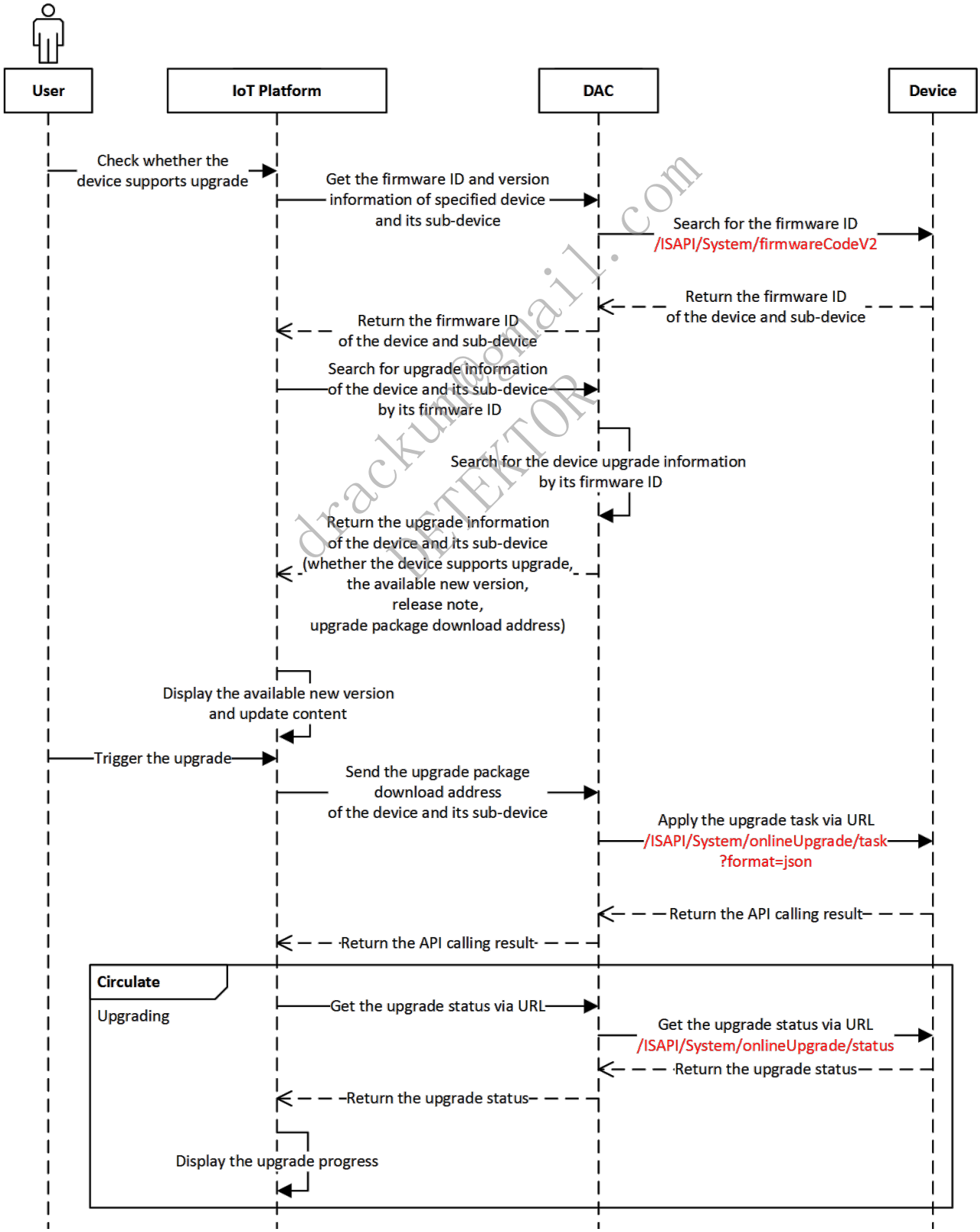
`<security>&iv=<iv>`.

5.15 Upgrade Based on Platform Local Deployment

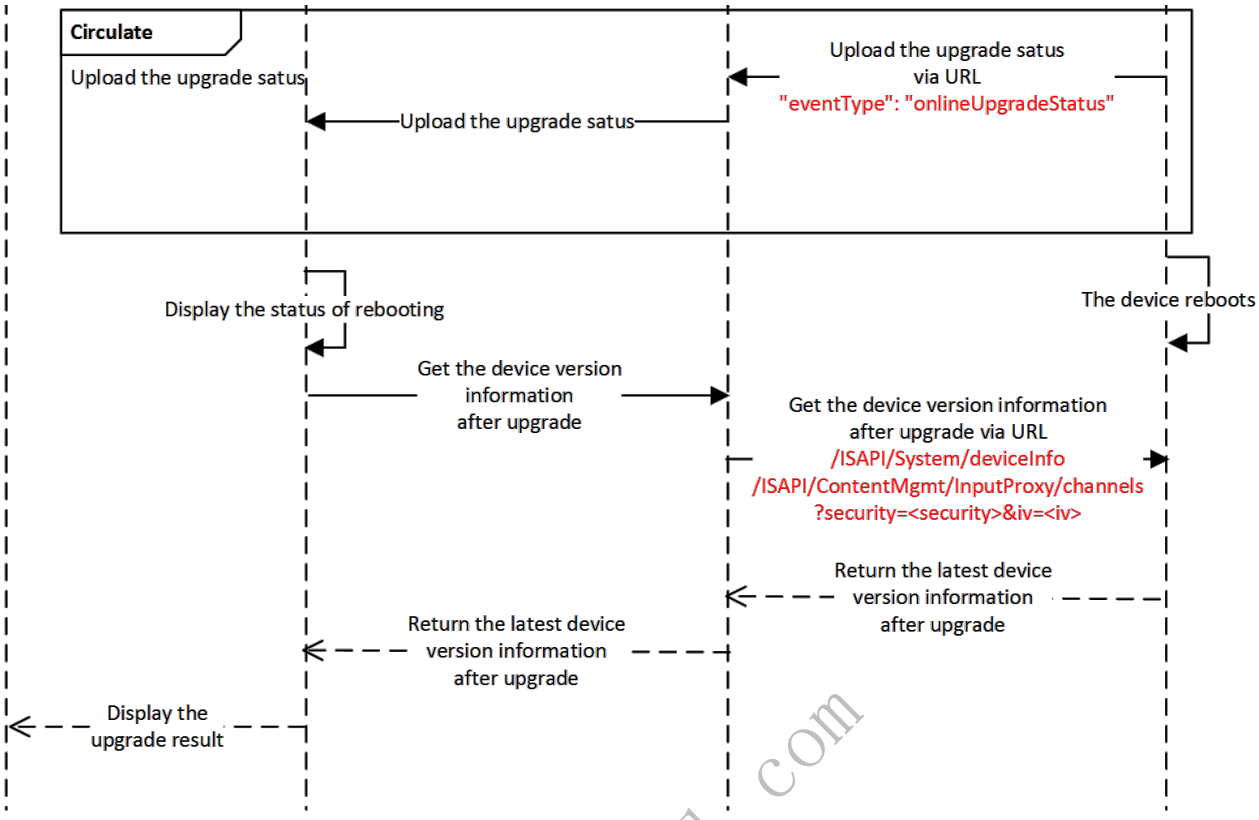
5.15.1 Introduction to the Function

This solution applies to the upgrade of devices and sub-devices connected via ISAPI and ISUP. If via ISAPI, it is required to set up HTTP(s) service or transfer binary upgrade package files so as to store device upgrade package and generate URL for downloading upgrade package. If via ISUP, it is required to set up FTP service to store device upgrade package and generate URL for downloading upgrade package. If the device is connected via HCNetSDK protocols, you can realize device upgrade via transmitting ISAPI.

5.15.2 API Calling Flow



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1. Search for device firmware ID on the platform: `GET /ISAPI/System/firmwareCodeV2`.
2. Search for the device upgrade information in DAC by device firmware ID on the platform. If there is an available new version, DAC will trigger an upgrade task and apply it to the device: `POST /ISAPI/System/onlineUpgrade/task?format=json`. You can specify to upgrade device (e.g., NVR) or sub-device (e.g., IPC) via the node `<channel>`.
3. During the upgrade, you can get the real-time upgrade progress on the platform: `GET /ISAPI/System/onlineUpgrade/status`.
4. After the device upgrade is finished, the device will upload the `"onlineUpgradeStatus"` (upgrade status), and will reboot automatically.
5. After the upgrade and reboot are finished, you can get the latest device version information on the platform.

- Get the device (e.g., NVR) version information: `GET /ISAPI/System/deviceInfo`.
- Get the sub-device (e.g., IPC) version information: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

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6 Video (General)

6.1 . OSD

6.1.1 Introduction to the Function

On-screen display is applied in the display interface of cameras, and is used to display some characters or graphics on the video to provide some information to the audience. After OSD configurations on the client software, the configured information text will be overlaid on the video stream of cameras and displayed on the screen.

The typical application scenarios of dynamic OSD: elevator floor number overlay for elevator monitoring, characters overlay for toll stations on highway/road, notification information overlay during live video programs. The typical application scenarios of static OSD: fixed information (such as camera position, date, time) overlay on the video signal.

6.1.2 API Calling Flow

6.1.2.1 OSD

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`.
2. The client software configures OSD parameters, including whether to enable OSD, custom text overlay, time overlay, and overlay format, for a specified channel.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   Set: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

   Remarks:

   - The configuration URL supports various parameters, including custom text overlay, channel name overlay, time overlay, overlay format, and so on. Some of these parameters can also be configured by other URLs, as shown below: `TextOverlayList` can also be configured via URL of custom text overlay.

     `channelNameOverlay` can also be configured via URL of channel name overlay.

     `DateTimeOverlay` can also be configured via URL of channel time overlay.

   - The overlaid information alignment mode (defined by `alignment`) refers to the alignment mode of custom text:

     `customize`: custom mode. The client software sends the overlay position coordinates of custom text to the device, and the device will overlay the characters on the stream for display according to the position coordinates.

     `alignRight`: right align. The device will display custom text on the right boundary of screen.

     `alignLeft`: left align. The device will display custom text on the left boundary of screen.

     `allRight`: all right align. The device will display all overlay contents, including custom text, the channel name, the time on the right boundary of screen.

     `allLeft`： all left align. The device will display all overlay contents, including custom text, the channel name, the time on the left boundary of screen.

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6.1.2.2 Custom Text Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `TextOverlayList` is returned, it indicates that custom text overlay is supported and `size` defines the maximum number of custom texts.
2. The client software configures custom text OSD. `textID` refers to the custom text No., and the value range is: [1, `size`]:

   Create a custom text for a specified channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Get overlay parameters of a custom text for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Set overlay parameters of a custom text for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Delete a custom text: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Get overlay parameters of all custom texts for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

Set overlay parameters of all custom texts for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

Delete all custom texts for a specified channel: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

6.1.2.3 Channel Name Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `channelNameOverlay` is returned, it indicates that channel name overlay is supported.
2. The client software configures the channel name overlay, including whether to enable channel name overlay, overlay position, and so on.

   Get channel name overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

   Set channel name overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

6.1.2.4 Channel Time Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `DateTimeOverlay` is returned, it indicates that channel time overlay is supported.
2. The client software configures the channel time overlay, including whether to enable channel time overlay, overlay position, date format, and so on.

   Get channel time overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

   Set channel time overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

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6.2 Async Capturing Manually

6.2.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately responds by returning the URL of the storage server where the captured picture is stored. The client can request image data from the storage server based on the received URL. It is suitable for the limited network bandwidth between the client and the device or the client centrally manages a large number of devices. The client can choose to use the image URL address carried in the alarm content to request pictures from the storage server when the network conditions improve or there is no concurrency of a large number of requests, so as to reduce the probability of image download failure and the risk of network congestion.

6.2.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportSnapshotAsync` is returned and its value is `true`, async manual capturing shall be supported.
2. (Optional) The client gets the async manual capturing capability of a specified channel: `GET /ISAPI/Streaming/channels/<channelID>/picture/async/capabilities?format=json`.
3. The client controls the specified channel to manually and asynchronously capture pictures: `GET /ISAPI/Streaming/channels/<channelID>/picture/async?format=json&imageType=<imageType>&URLType=<URLType>`, and the device returns the URL of the storage server for capturing pictures.

Remarks:

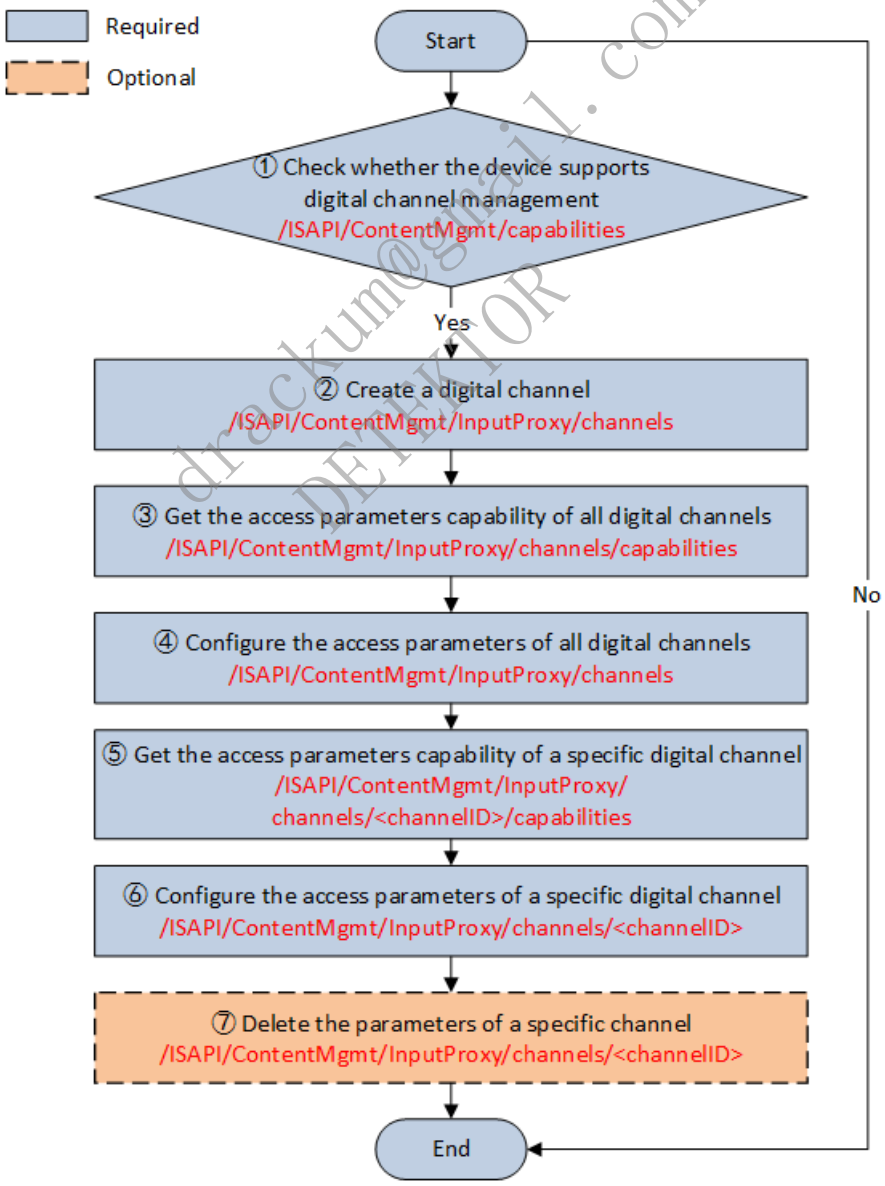
- If the step 2 returns a failure during the integration，it is recommended that you check whether the channel supports async manual capturing according to steps 1 and 3.

6.3 Digital Channel Access Management

6.3.1 Introduction to the Function

- Video channel includes analog channel and digital channel. The service involves channel access protocols, streaming protocols, address information, version No. of channel access firmware, and so on.
- Analog channel refers to the local video channels of the device.
- Digital channel refers to the video channels accessed by the device by IP address or POE port (e.g., NVR accessing the video channel of network camera).
- Analog channel No. and digital channel No. are unique. Devices that do not support digital channels will only return analog channel No. For example, analog channel No. ranges from 1 to 8 (determined by the number of analog channel ports of the device), digital channel No. ranges from 9 to 32 (within the maximum number of digital channels, determined by the number of added network cameras).

6.3.2 API Calling Flow



1. Get the capability node to check whether the device supports digital channel management: `GET /ISAPI/ContentMgmt/capabilities`. If the value of the node `iSptInputProxyChanCap` is true, it indicates that the device supports digital channel management.

2. Add digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
3. Get the access parameters capability of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities?security=<security>&iv=<iv>`.
4. Configure the access parameters of all digital channels:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

   Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
5. Get the capability of configuring the access parameters of a specific digital channel: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/capabilities`.
6. Configure the access parameters of a specific digital channel:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.

   Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.
7. Delete the parameters of a specific digital channel: `DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>`.

6.3.3 Video Channel Application

1.Get the list of video channels, which is the sum of digital channels and analog channels. Digital channels and analog channels need to be obtained separately:

- 1) Call "Get all video input channel parameters" to get all analog channels via GET /ISAPI/System/Video/inputs/channels and VideoInputChannelList parameter is returned. The id represents the analog channel No.
- 2) Call the "Get all digital channel access parameters" to get all digital channels via GET /ISAPI/ContentMgmt/InputProxy/channels?security=&iv= and InputProxyChannelList parameter is returned. The id represents the digital channel No.

2. Video channel function configuration supports unified configuration of analog and digital channels. Both the analog channel interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/` and the digital channel interface starting with `/ISAPI/ContentMgmt/InputProxy/channels/<channelID>/` are supported. **It is recommended to use the channel configuration interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/`, where <channelID> represents all channel No.s supported by the device, including analog channels and digital channels.**

> For example, configure text overlay parameters for a channel: Both `GET/PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays` and `GET/PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/overlays` are supported, and it is recommended to use `/ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

6.4 Live View

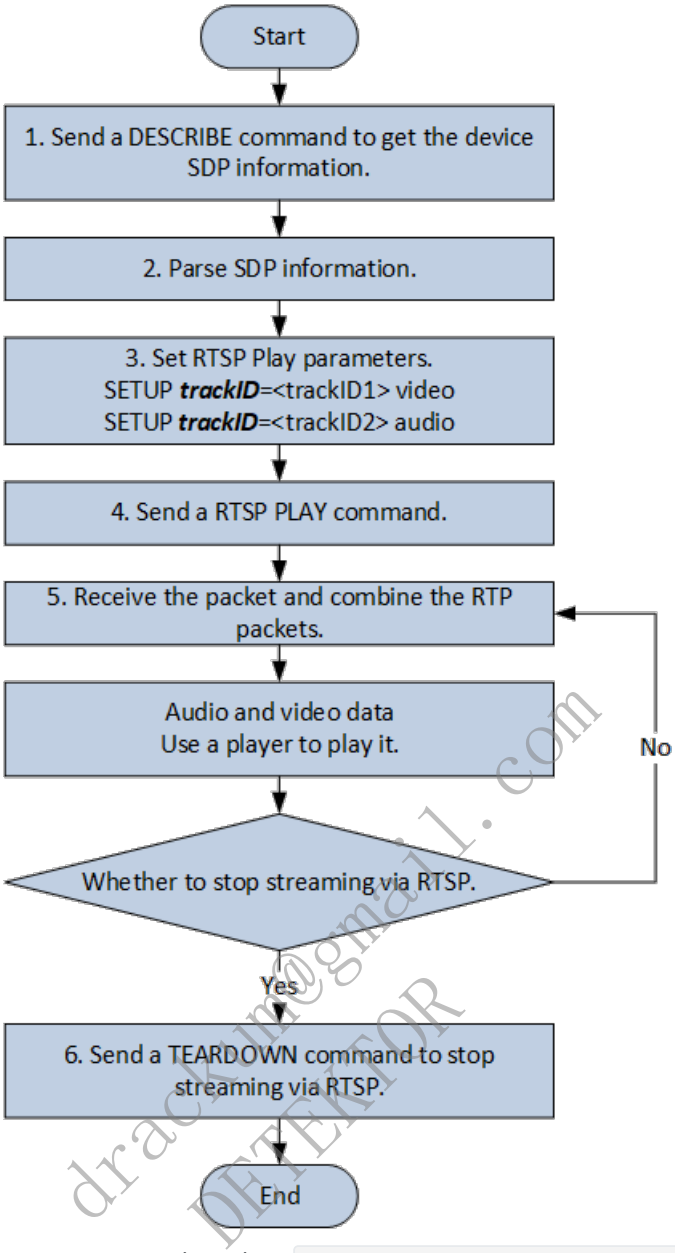
6.4.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

6.4.2 API Calling Flow

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1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`; Digest authentication with devices is required.
2. The client parses the media SDP information returned by the device.
3. Set RTSP play parameters, set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP package sent by device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address, `[:port]` is optional, and 554 by default, `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream); for example, the IP address of the target device is `172.7.203.11`, and if the channel number is 172, the streaming address will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

6.4.3. Example

1. A client sends an RTSP DESCRIBE request.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s=Media Server V4.22.126
i=Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000020000011171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

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5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J.............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

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6.5 Manual Capturing Events

6.5.1 Introduction to the Function

According to the preset information and capture interval configured by the client, the device manually captures images at the specified preset and reports the image information in the form of events. It is used for the client to identify emergency situations and perform an emergency patrol on some presets for analysis. It has a higher priority than the daily scheduled patrol capture. Common usage scenarios: according to the scheduled capturing event (eventType: imageCapture) rules, the store camera regularly audits in a number of key areas. If in the event of an emergency, and a worker wants to immediately re-inspect each area, you can set a manual capture rule to complete an independent audit. After the audit is completed, the camera will continue to audit according to the scheduled capturing rules.

6.5.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualImageCapture` returns `true`, manual capturing shall be supported.
2. The client gets manual capturing capability for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture/capabilities?format=json`.
3. The client sets the parameters of manual capture events for a specified channel: Get parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`. Set parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`.
4. Receive the event uploaded by the device in arming or listening mode: `eventType:manualImageCapture`.

Remarks:

- Each time the client sets manual capture event parameters, the channel immediately takes a patrol capture of the relevant presets. When the last preset capture is finished, the manual capture event ends.

6.6 Motion Detection

6.6.1 Introduction to the Function

Commonly used in unattended camera recording and automatic alarm, motion detection can help to reduce the cost of manual monitoring for public institutions and companies. Besides, it can greatly improve monitoring efficiency and accuracy by avoiding the errors of manual monitoring due to fatigue after long-time work. For example, it can be applied in home and company theft proof, as well as monitoring and alarm in institutions like medical organizations and nursing homes.

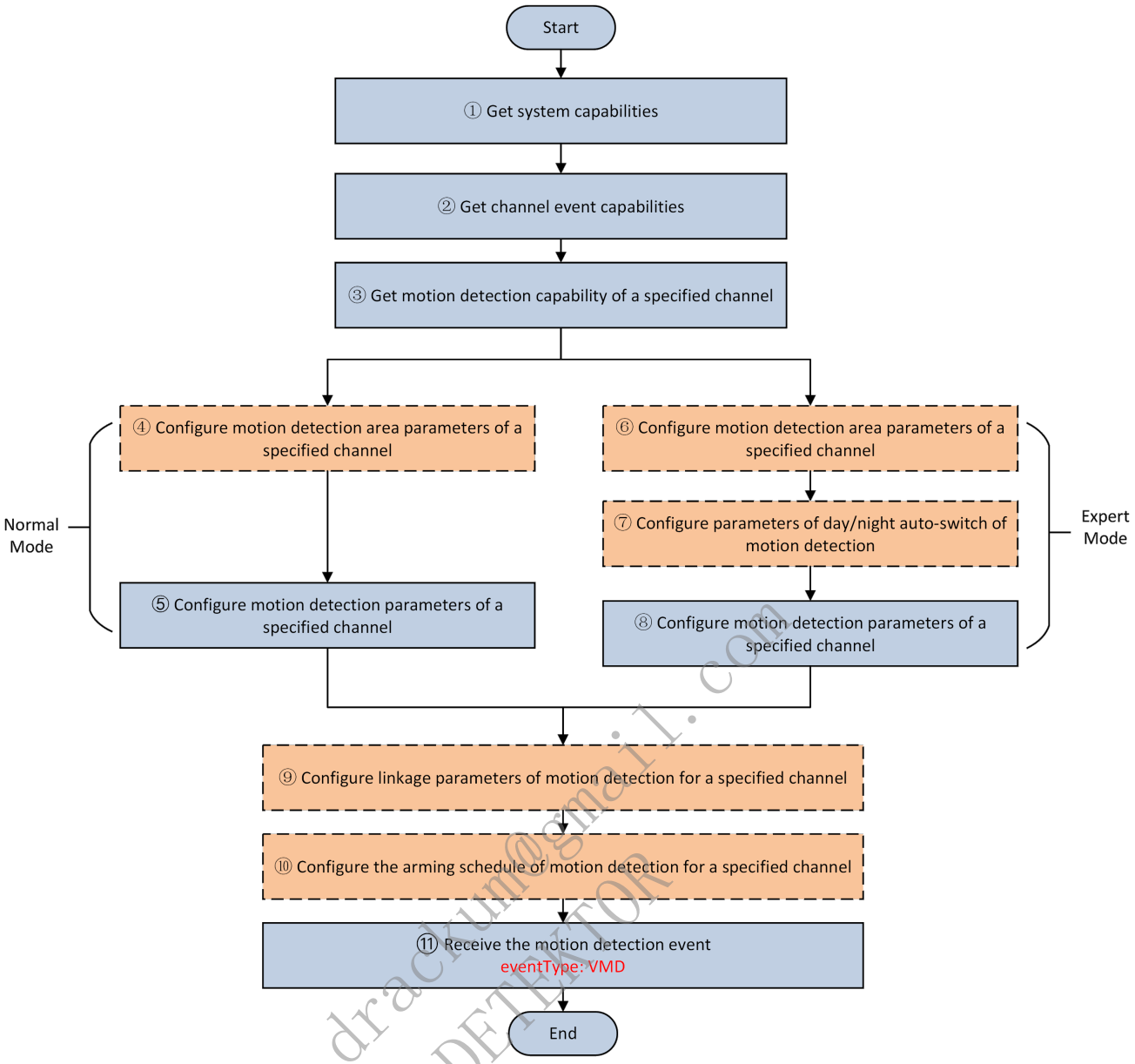
Motion detection has two modes: normal mode and expert mode.

Normal mode: the default mode of motion detection. In this mode, all detected areas only support a same sensitivity.

Expert mode: in this mode, every detected area can have its own detection sensitivity and percentage of objects in area. It supports configuring different sensitivities according to area types. For example, for the motion detection in the company hall, you can set higher detection sensitivity for the gate area to focus on people's entering and exiting.

6.6.2 API Calling Flow

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1. Get the system capability: `GET /ISAPI/System/capabilities`. If the node `isSupportMotionDetection` is returned and its value is `true`, it indicates that the device supports motion detection.

2. Get the device channel event capability. If the returned `eventType` contains `VMD`, it indicates that the corresponding channel supports motion detection.

   Get the event capability of all channels: `GET /ISAPI/Event/channels/capabilities`.

   Get the event capability of a specified channel: `GET /ISAPI/Event/channels/<channelID>/capabilities`.

3. Get the motion detection capability of a specified channel (normal mode): `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/capabilities`.

   Check the motion detection parameters (such as function enabling, detection sensitivity) supported by the device. You can check whether the device supports multiple areas via checking whether the node `regionType` contains `region`.

4. (Optional) Configure parameters of the motion detection area (normal mode).

   Capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout/capabilities`.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

5. Configure the motion detection parameters (such as motion detection enabling, detection sensitivity) (normal mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

6. (Optional) Configure the parameters of motion detection area (expert mode). You can configure different detection conditions such as sensitivity for different areas.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

7. (Optional) Configure day/night auto switch of motion detection (expert mode). You can configure the time switch schedule of `day` and `night`.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

8. Configure motion detection conditions (expert mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

9. (Optional) Configure linkage parameters of the motion detection.

   Get: `GET /ISAPI/Event/triggers/VMD-<channelID>`.

   Configure: `PUT /ISAPI/Event/triggers/VMD-<channelID>`.

   Delete: `DELETE /ISAPI/Event/triggers/VMD-<channelID>`.

10. (Optional) Configure the arming schedule of motion detection.

    Get: `GET /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

    Configure: `PUT /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

11. Receive events reported by device in arming or listening mode: `eventType:VMD`.

Remarks:

1. For motion detection (normal mode), the uploaded `VMD` event includes device basic information only, such as channel No. and event time. For details, see 4.3.1 Motion Detection Event (Normal Mode).

2. For motion detection (expert mode), in addition to the device basic information, the uploaded `VMD` event includes the detection area ID, area coordinates and area detection sensitivity. For details, see 4.3.2 Motion Detection Event (Expert Mode).

3. In practical employment, you can select one mode as needed. There are no logical relations between the two modes.

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6.6.3 Message Example

6.6.3.1 Motion Detection Event (Normal Mode)

```
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T19:42:27+08:00</dateTime>
<activePostCount>4</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```

6.6.3.2 Motion Detection Event (Expert Mode)

```
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T20:36:34+08:00</dateTime>
<activePostCount>11</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<DetectionRegionList>
<DetectionRegionEntry>
<regionID>1</regionID>
<sensitivityLevel>50</sensitivityLevel>
<RegionCoordinatesList>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
</RegionCoordinatesList>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```

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6.7 Preset Dedicated Focusing

6.7.1 Function Introduction

The preset dedicated focusing algorithm is designed for specific scenarios (such as instrument and gauge images) where the image is not clear at a certain position, to achieve clearer and faster focusing. The dedicated focusing algorithm includes foreground focusing, background focusing, and area focusing.

6.7.2 API Calling Flow

1. The client gets the device's PTZ capabilities via URL `GET /ISAPI/PTZCtrl/capabilities`. If `isSupportPresetsProFocus` is returned with `true` value, the device supports preset dedicated focusing.

2. The client gets the capability of preset dedicated focusing for the specified channel via URL `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg/capabilities?format=json`.
3. The client batch configures presets' dedicated focusing parameters for the specified channel.
   - Get parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`
4. The client configures the dedicated focusing parameters of a single preset for the specified channel. If a preset is not clear under the batch configuration in step 3, it can be configured separately.
   - Add dedicated focusing parameters of a preset for a specified channel: `POST /ISAPI/PTZCtrl/channels/<channelID>/AddPresetsProFocusCfg?format=json`
   - Get parameters: `POST /ISAPI/PTZCtrl/channels/<channelID>/GetPresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/ModifyPresetsProFocusCfg?format=json`
   - Delete: `POST /ISAPI/PTZCtrl/channels/<channelID>/DeletePresetsProFocusCfg?format=json`
   - Get parameters of multiple presets: `GET /ISAPI/PTZCtrl/channels/<channelID>/GetMultiPresetsProFocusCfg?format=json`

6.7.3 Exception Handling

Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Chinese Description
4	Invalid Operation	numberReachedLlimit	0x40001119	The number reached limit.	The configured items have reached the limit and cannot be added further. Please delete unnecessary ones and try again.

6.8 Scene Change Detection

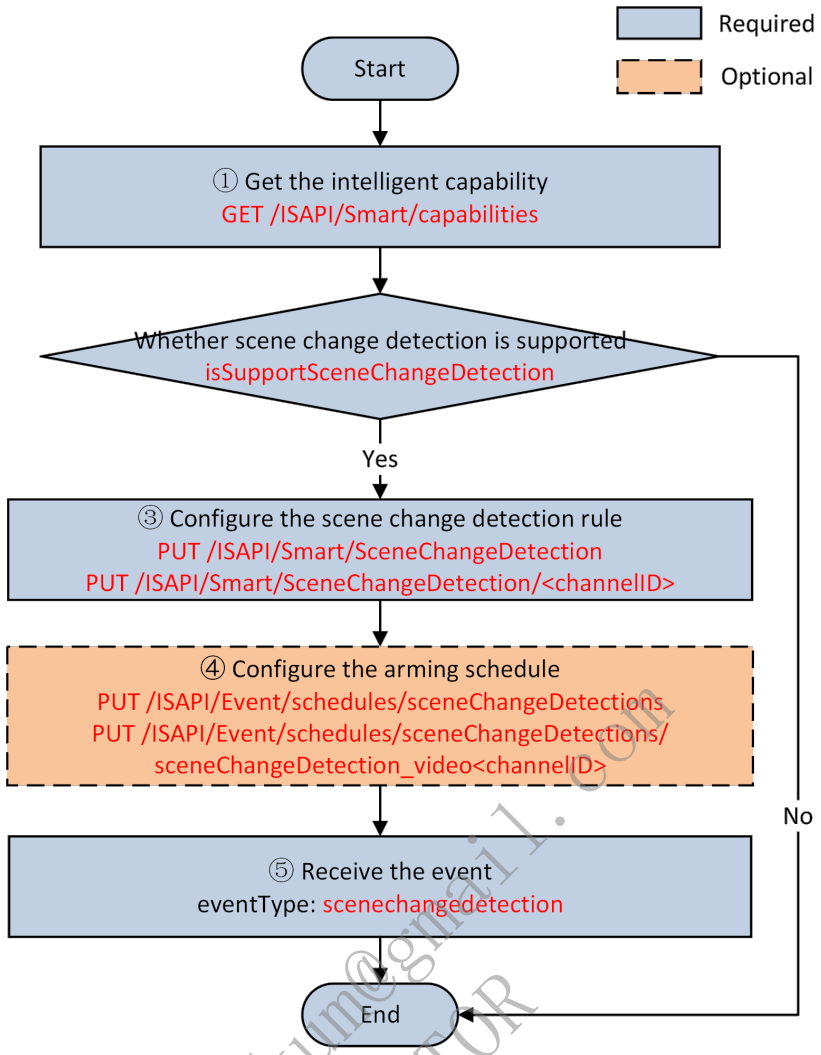
6.8.1 Function Introduction

The scene change detection function identifies whether a scene has changed based on configured rules. Once a scene change is detected, an alarm is triggered.

This function is commonly used to monitor whether external forces have caused changes in camera's viewing angle. When the viewing angle is changed by external interference, or the monitoring scene is tampered, affecting normal monitoring, relevant personnel can be notified promptly to take actions.

6.8.2 API Calling Flow

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1. Get the device's intelligent capability.

   `GET /ISAPI/Smart/capabilities`. If the response returns `isSupportSceneChangeDetection` as `true`, it indicates that the device supports the scene change detection.

2. Configure the detection rules.

   - Get the scene change detection capability: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>/capabilities`.
   - Get scene change detection parameters: `GET /ISAPI/Smart/SceneChangeDetection`.
   - Set scene change detection parameters: `PUT /ISAPI/Smart/SceneChangeDetection`.
   - Get scene change detection parameters for a specific channel: `GET /ISAPI/Smart/SceneChangeDetection/<channelID>`.
   - Set scene change detection parameters for a specific channel: `PUT /ISAPI/Smart/SceneChangeDetection/<channelID>`.

   Note: A higher **sensitivityLevel** makes the camera more sensitive to scene changes. Adjust it based on actual environment.

3. (Optional) Configure the arming schedule.

   - Get schedule parameters of scene change detection for all channels: `GET /ISAPI/Event/schedules/sceneChangeDetections`.
   - Set schedule parameters of scene change detection for all channels: `PUT /ISAPI/Event/schedules/sceneChangeDetections`.

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- Get schedule parameters of scene change detection for a specific channel: `GET /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.
- Set schedule parameters of scene change detection for a specific channel: `PUT /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>`.

Note:

- Up to 8 time periods can be configured per day.
- Time periods must not overlap.

4. Receive the event (`"eventType":"scenechangedetection"`).

6.9 Sub-device's Transparent Transmission

6.9.1 Introduction to the Function

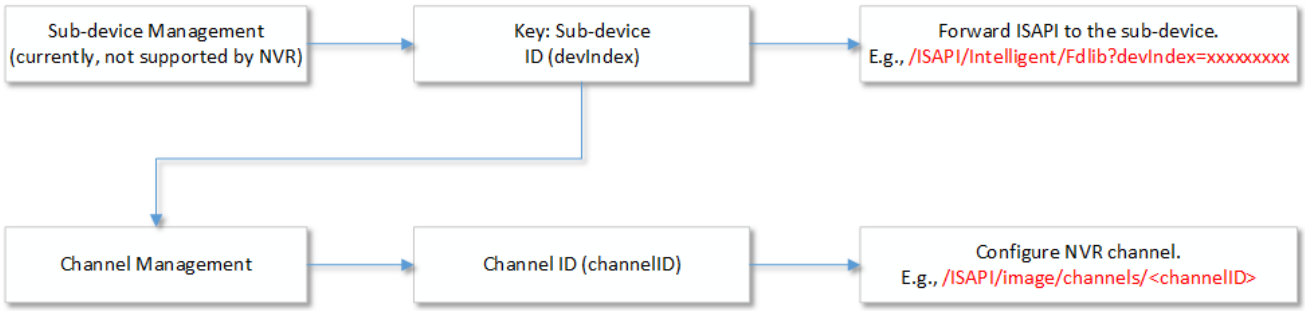
In actual application scenarios, the NVR being used is generally not the latest version, if an IPC with latest functions is added to the system, the NVR will fail to recognize the IPC's new functions due to version incompatibility, so the platform (including device access platforms, client softwares, device's web plug-ins, etc, hereinafter referred to as the platform) will fail to operate new functions of the IPC.

Besides, in video channel resource management, the platform can only get information about digital channels (also known as IP channels) added to the NVR, but not the device that the channels belong to. For example, two video channels (channel 13 and channel 18) of a dual-lens people counting camera are added to the NVR. The platform can only get that digital channel 13 and 18 are added to the NVR, but not that the two channels belong to one camera. Such information loss may lead to repeated operations, which will further cause problems in actual practices. For example, users may fail to upgrade IPC via digital channel No. if not knowing the relationship between camera and digital channel.

To settle above problems, the sub-device's transmission function is introduced to the video channel resources management. In this way, users can directly access to the device of the digital channels via the devIndex (sub-device index) allocated to the device by NVR.

Notably, despite the introduction of the sub-device's transparent transmission function, the NVR does not function as an access gateway, and so it does not function with sub-device management on gateway. Refer to the Sub-device Management of Device Management for details.

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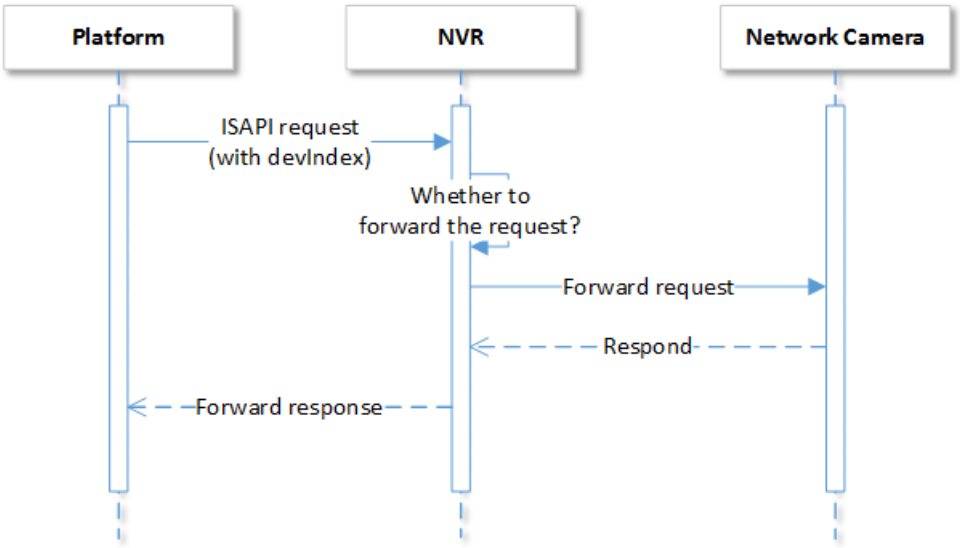


Steps are as follows:

1. First, the platform will check whether the NVR supports sub-device's transparent transmission. If it is supported, follow the next step. Second, add devIndex to the request URL.
2. For the ISAPI request sent to the NVR by the platform, if it contains devIndex, it will be forwarded to the specific sub-device; otherwise, the NVR will respond to the request.
3. When the request is forwarded to the sub-device by the NVR, delete the devIndex in the request URL so as to realize ISAPI interaction.

For example:

1. Send the request URL (`GET /ISAPI/System/deviceInfo`) to get the information of the platform's NVR device.
2. Send the request URL (`GET /ISAPI/System/deviceInfo?devIndex=<devIndex>`) to get the information of a specific sub-device of the NVR.

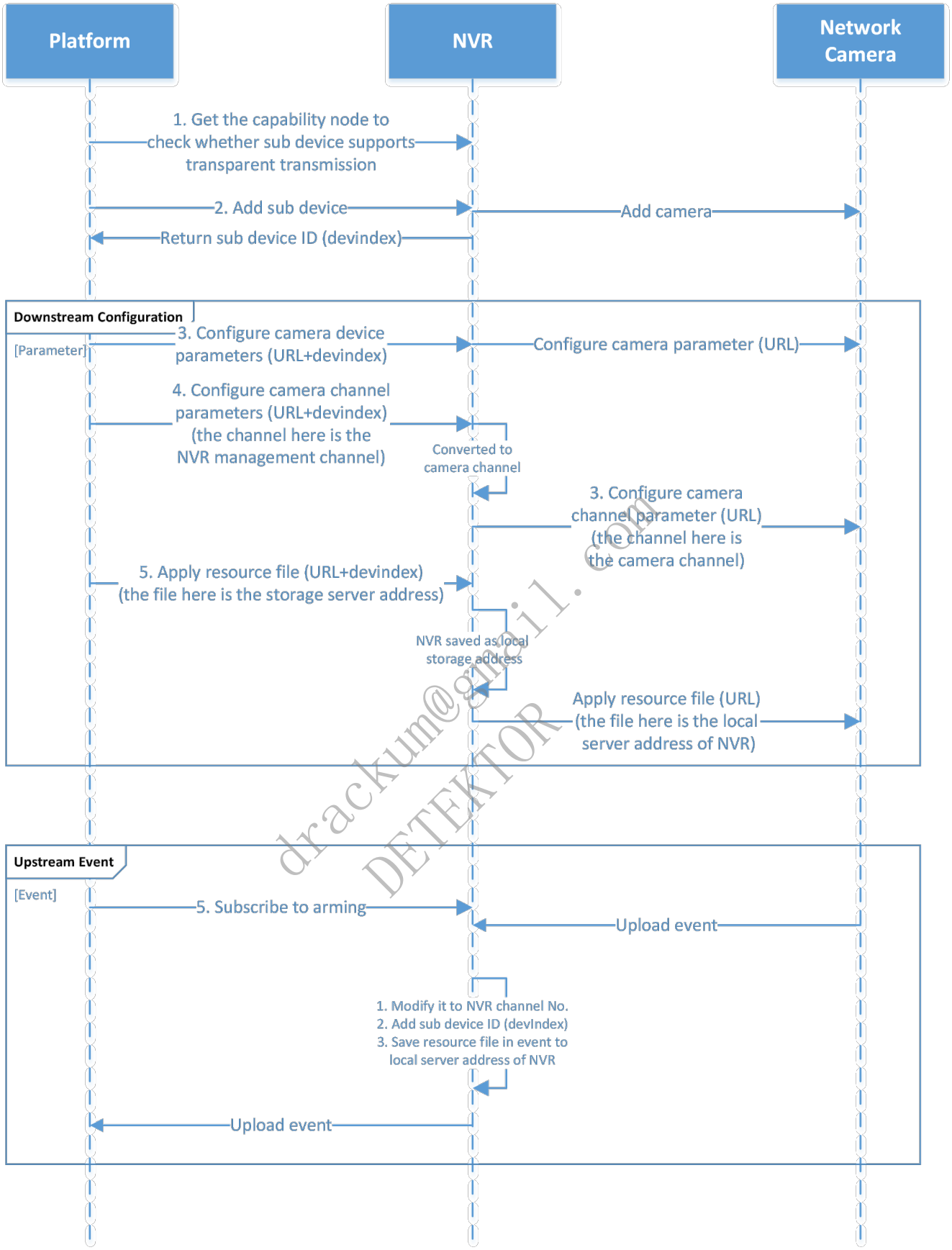


Note:

1. This function is applicable to transitory connections, while inapplicable to persistent connections (e.g., arming subscription, two-way audio).
2. It is required to delete the devIndex in the URL before NVR forwards the ISAPI request.
3. For the VCA function of NVR, the transmission solution of sub-devices is not adopted. You can switch the analysis unit by the following URI: /ISAPI/Smart/analysisUnitSwitch/channels/`<channelID>`/event/`<EventType>`.

6.9.2 API Calling Flow

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1. Get the capability node to check whether the device supports transparent transmission. If the value of the node `isSupportDevIndex` is true, it indicates that the device supports transparent transmission.
2. Create digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`. The sub device ID `devindex` will be returned.
3. Get access parameters of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
   Each digital channel information contains one parameter `devIndex`, which is the same for digital channels of a same physical device.

4. When splicing ISAPI request, if the URL contains `devIndex=<devIndex>`, it represents a request for the sub-device. For example, you can send `12345678` (the request URL) to get the information of the sub-device `GET /ISAPI/System/deviceInfo?devIndex=12345678` of the target device. Note: 1) For the downstream configuration of a video channel, the platform will apply the channel managed by NVR, and before applying, the NVR will transform channel into the video channel. 2) When the platform applies the storage address of resource files, the camera cannot access the storage service to get resource files because of the network isolation between the camera and platform, so the storage address must be saved as a local storage address by the NVR before being applied.

6.10 Sync Capturing Manually

6.10.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately returns the captured picture data in respond. It is often used in scenarios with high network bandwidth between clients and devices to facilitate the rapid transmission of image data, such as a LAN built on a company campus.

6.10.2 API Calling Flow

1. (Optional) The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualSnapPicture` returns `true`, sync manual capturing shall be supported.
2. (Optional) The client gets sync manual capturing capability for a specified channel: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json`.
3. The client controls the specified channel to perform sync manual capturing: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>`, and the device returns the captured binary picture data.

Remarks:

- If you only need to capture a JEPG picture, you can jump to step 3.
- If you need to capture pictures other than JEPG or capture in the designated area of the video, you need to perform steps 1 and 2 to determine whether the device supports it. If it does, go to step 3.
- In step 3, the value ranges of `videoResolutionWidth` and `videoResolutionHeight` are the same as `<videoResolutionWidth> and <videoResolutionHeight>` returned by `GET /ISAPI/Streaming/channels/<trackStreamID>/capabilities`.

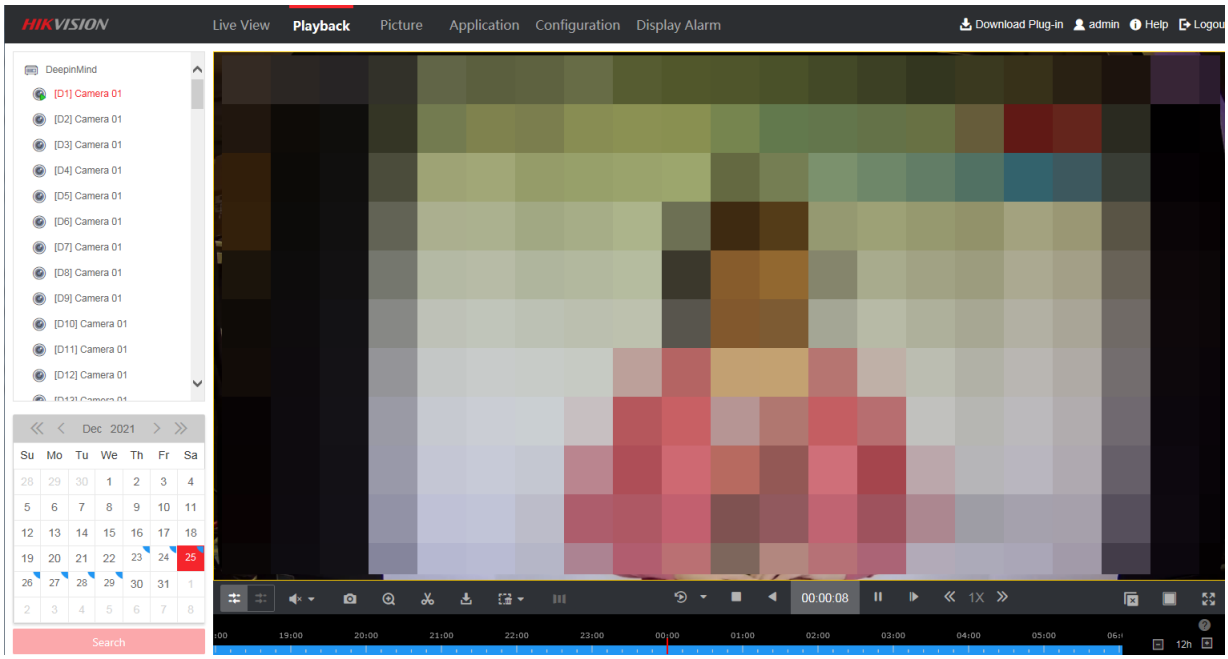
6.11 Video Playback

6.11.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

The below is an example of playback on a Web Client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos can be shown on the calendar and you can click a date to view videos.

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6.11.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.
3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. See details in RFC 7826 (Real Time Streaming Protocol).

Remarks:

- Playback via RTSP needs digest authentication. The method is the same as that of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set the Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in RFC 7826 (Real Time Streaming Protocol).

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6.11.3 Example

6.11.3.1 Search for Video on Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
    <year>2021</year>
    <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <dayList>
        <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
        <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
        <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
        <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
        <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
        <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
        <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
        <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
        <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
        <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
        <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
        <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
        <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
        <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
        <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
        <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
        <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
    </dayList>
</trackDailyDistribution>
```

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6.11.3.2 Search for Video

Request from Client

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
    <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
    <trackList>
        <trackID>101</trackID>
    </trackList>
    <timeSpanList>
        <timeSpan>
            <startTime>2021-08-16T00:00:00Z</startTime>
            <endTime>2021-08-18T23:59:59Z</endTime>
        </timeSpan>
    </timeSpanList>
    <maxResults>100</maxResults>
    <searchResultPostion>0</searchResultPostion>
    <metadataList>
        <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
    </metadataList>
</CMSearchDescription>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
    <responseStatus>true</responseStatus>
    <responseStatusStrg>OK</responseStatusStrg>
    <numOfMatches>1</numOfMatches>
    <matchList>
        <searchMatchItem>
            <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
            <trackID>101</trackID>
            <timeSpan>
                <startTime>2021-08-18T15:18:15Z</startTime>
                <endTime>2021-08-18T15:19:08Z</endTime>
            </timeSpan>
            <mediaSegmentDescriptor>
            <contentType>video</contentType>
            <codecType>H.264-BP</codecType>
            <playbackURI>rtsp://10.14.97.40/Streaming/tracks/101/?
starttime=20210818T151815Z&amp;endtime=20210818T151908Z&amp;name=00000004667000100&amp;size=1400788</playbackURI>
            <lockStatus>unlock</lockStatus>
            <name>00000004667000100</name>
            </mediaSegmentDescriptor>
            <metadataMatches>
                <metadataDescriptor>recordType.meta.com/timing</metadataDescriptor>
            </metadataMatches>
        </searchMatchItem>
    </matchList>
</CMSearchResult>
```

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6.11.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP commands two times, and the server responds to them.

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000000046670001 00&size=1400788/trackID=video
RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000000046670001 00&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000000046670001 00&size=1400788/trackID=audio
RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000000046670001 00&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY request, and the server starts to push streams for playback.

```
PLAY rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000000046670001 00&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=0000000046670001 00&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

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```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

6.12 Video Privacy Mask

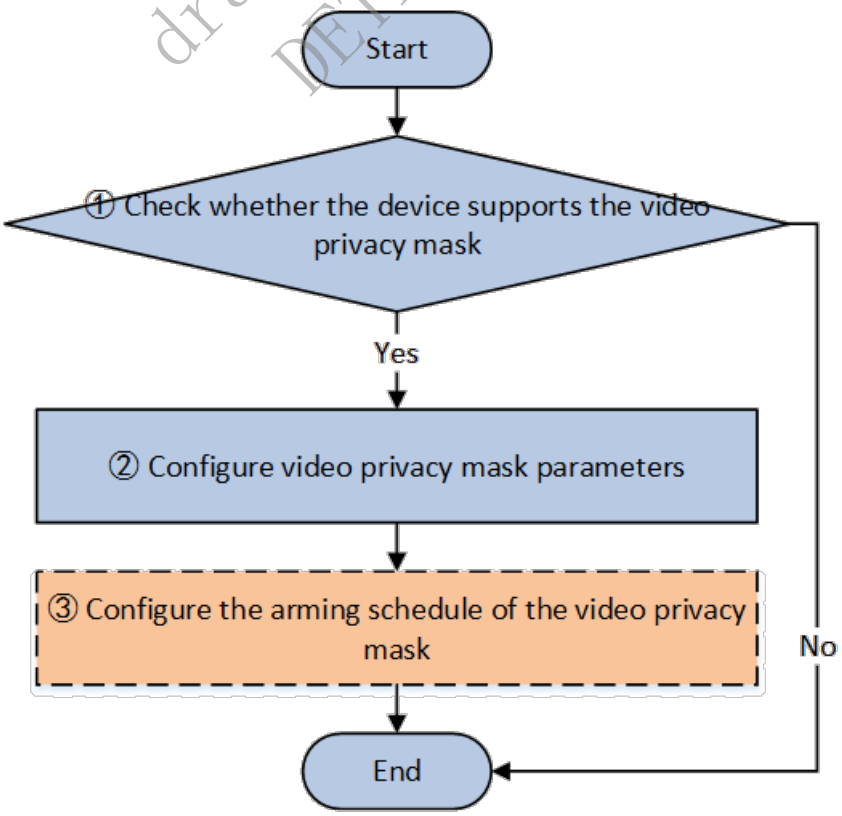
6.12.1 Introduction to the Function

Video privacy mask is to cover certain areas on the live image to protect personal privacy from being live viewed and recorded. You can use the masking method such as a mosaic or black frame as needed to make the area invisible. The device supporting privacy mask can specify the area(s) to be masked permanently on the live image, as a result of which the live view and recorded video during the live view will be influenced and the image cannot be restored. The following image shows the effect of masking an area with the mosaic.

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6.12.2 API Calling Flow

6.12.2.1 Video Privacy Mask



1. Check whether the device supports the video privacy mask. Get the device's system capability: `GET /ISAPI/System/capabilities`. If `isSupportPrivacyMask` is returned and its value is true, it indicates that the video privacy mask is supported by the device.

2. Configure the video privacy mask parameters. You can specify sensitive area(s) to be masked on the live image (related field: `PrivacyMaskRegionList`), and up to 8 areas are supported. You can also configure the masking type, such as mosaic, black frame, and red frame (related field: `maskType`). Get the video privacy mask capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/privacyMaskCap`. Configure the video privacy mask parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`. Get the video privacy mask parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask`.

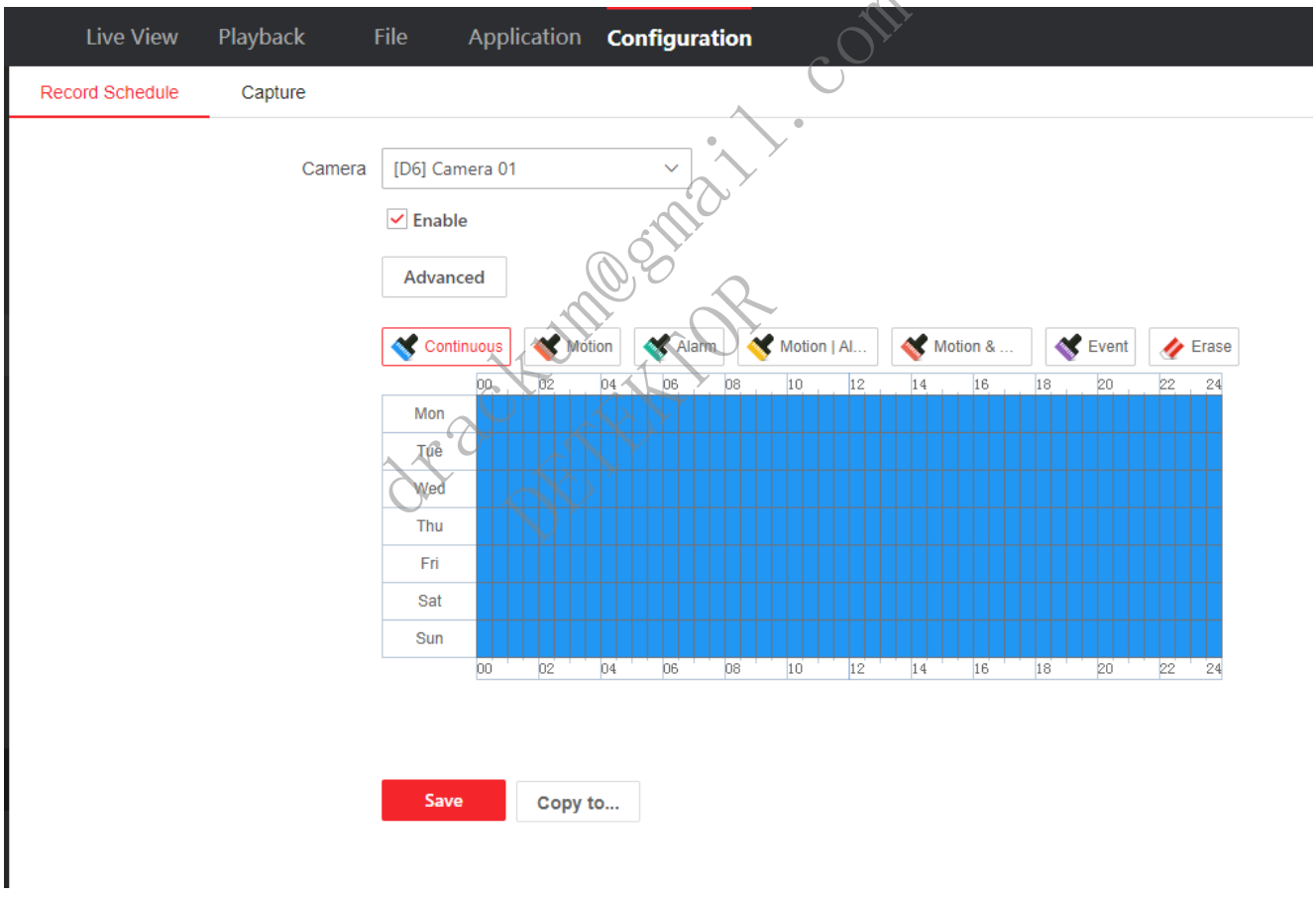
3. (Optional) Configure the arming schedule of the video privacy mask. You can configure the privacy mask schedule as needed. If not configured, the privacy mask is performed 24 hours a day by default. Get the video privacy mask schedule: `GET /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`. Set the video privacy mask schedule:`PUT /ISAPI/Event/schedules/privacyMask/privacyMask-<channelID>`.

7 Video Management

7.1 . Scheduled Recording

7.1.1 Introduction to the Function

The web interface for reference is shown below.



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7.1.2 API Calling Flow

1. The client software gets the device system capability: `GET /ISAPI/System/capabilities`. When `isSupportScheduledRecordUpload` is returned and its value is `true`, it indicates that scheduled recording is supported.

2. The client software configure schedule parameters of scheduled recording. You can configure recording schedules for all channels or configure the recording schedule for a specified channel.

- Recording schedule management for all channels: Add a recording schedule: `POST /ISAPI/ContentMgmt/record/tracks`. Get all recording schedules parameters: `GET /ISAPI/ContentMgmt/record/tracks`. Set all recording schedules: `PUT /ISAPI/ContentMgmt/record/tracks`.
- Recording schedule management for a specified channel: Get the capability of configuring the recording schedule for a specified channel: `GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/capabilities?security=`

`<security>&iv=<iv>`. Get the recording schedule parameters for a specified channel: `GET /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`. Set the recording schedule for a specified channel: `PUT /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`. Delete the recording schedule for a specified channel: `DELETE /ISAPI/ContentMgmt/record/tracks/<trackStreamID>?security=<security>&iv=<iv>`.

7.2 Search for Event Video

7.2.1 Introduction to the Function

This function supports searching for the videos of linked cameras when an event occurs.

7.2.2 API Calling Flow

1. Check whether the device supports searching for event videos.

Get the device sysgem capability: `GET /ISAPI/System/capabilities`. If the node `<SerialCap>` is returned and its value is true, the device supports searching for event videos.

2. Search for event videos: `POST /ISAPI/ContentMgmt/eventRecordSearch?format=json`.

7.3 Search for Videos and Pictures

7.3.1 Introduction to the Function

This function supports searching for the video/picture data stored in the device. Video data source includes scheduled recording, manual recording and event recording, and picture data source includes scheduled capture and event linkage capture.

7.3.2 API Calling Flow

1. Check the device's supported time format. Get device's system capabilities: `GET /ISAPI/System/capabilities`.
   - If node `isSupportTimeTypeSTD` is returned with value `true`, it indicates that the device supports the ISO 8601 standard time, and URLs of functions related to ISO 8601 time parameters support additional query parameter `timeType=STD`.
   - If node `isSupportTimeTypeSTD` is not returned. Call `GET /ISAPI/System/time/timeType?format=json` to get the time type. If the `type` field returns `local`, it indicates that the device video search and playback use local time. If the `type` field returns `UTC`, it indicates that the device video search and playback use ISO 8601 time format.
   - If the device does not support API `GET /ISAPI/System/time/timeType?format=json`. Call `GET /ISAPI/System/deviceInfo` and check the time type of device via field `deviceType`. If `deviceType` (device type) is CVR, NVR, DVR, or ITCCAM, the device video search and playback use local time. Otherwise, follow the protocol specifications and use the TZ format, for example, `2024-05-13T16:00:00Z`.

2. Check whether the device supports video/picture search. Get the device's video search capability via `GET /ISAPI/ContentMgmt/capabilities`. If `<isSupportRecordSearch>` is returned, it indicates that the device supports video/picture search.

3. Search for videos and pictures:
   i. Get the capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/capabilities`. Get the picture search types that the device supports searching via the node `<pictureSearchType>` returned in the message, including pictures of event capture, scheduled capture and manual capture. Get the video types that the device supports searching for via the node `<recordSearchType>` returned in the message.
   ii. Get the parameters capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/search/capabilities`. Get the periods when the device supports searching for videos and pictures via the node `<timeSpanList>` returned in the message. Get the file tags that the device supports searching via the node `<labelSearch>` returned in the message.
   iii. Search for videos and pictures: `POST /ISAPI/ContentMgmt/search?security=<security>&iv=<iv>`.

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- The URL containing `timeType=STD` indicates that the time in the input and output messages are according to the ISO 8601 time format. The URL without `timeType=STD` indicates to use device's original time.
- Check whether it is picture or video to be searched via the node `<trackID>` in the request message `<CMSearchDescription>`. "101" (search for the main stream videos of channel 1), "103" (search for the pictures of channel 1), and it is channel 1 by default .

Note:

- The value of `CMSearchDescription`.`trackIDList`.`trackID` is obtained from the `id` value returned in the response message of `GET /ISAPI/ContentMgmt/record/tracks`. The meaning of `id`: `101` is the main stream of channel 1, `102` is the sub-stream of channel 1, `103` is the capture schedule of channel 1, `201` is the main stream of channel 2, and so on. When the device saves only one copy of channel videos, it does not distinguish between the main and sub stream, and only supports setting `trackID` to `xx1`.
- If `<trackIDList>` is returned in response message of `GET /ISAPI/ContentMgmt/search/capabilities`, it indicates that the device supports accurate search, and the returned value in `<trackIDList>` indicates the channels that support independent video search.

7.4 Search for Videos by Target Type

7.4.1 Introduction to the Function

This function realizes persons, vehicles or mixed targets distinguishing when searching for motion detection event videos. Searching for videos by target type is newly added. The search conditions include target type information, and the returned result of the search contains the start and end time as well as the file name of the video.

7.4.2 API Calling Flow

1. Check whether the device supports searching by target type:

Get the device capability: `GET /ISAPI/System/capabilities`. If the node is returned and its value is true, it indicates that the device supports searching by target type. Currently, the device supports searching by person and by vehicle.

2. Get the capability of searching by target type: `GET /ISAPI/ContentMgmt/SearchByTargetType/capabilities?format=json`.

3. Search for videos by target type: `POST /ISAPI/ContentMgmt/SearchByTargetType?format=json`.

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7.5 Video Download

7.5.1 Introduction to the Function

After completing searching for videos (refer to the calling flow of video search), users can download the searched videos by time or file. When downloading the videos by time or file via web browser, users should enable browser plug-ins.

7.5.2 API Calling Flow

1. Check whether the device supports video download: `GET /ISAPI/System/capabilities`. If the node is returned and its value is true, it indicates that the device supports downloading videos by time.

2. Download videos (plug-in-free):

   Get the capability of video download: `GET /ISAPI/ContentMgmt/download/capabilities>`.

   Download the videos: `GET /ISAPI/ContentMgmt/download?playbackURI=<playbackURI>&mediaID=<recordID>&downType=<downType>&encodeType=<encodeType>`.

   You can download videos by time or file via this protocol, and you can specify download method via the in the message. Details are as follows:

   - Download by file: the playbackURI uploads the start and end time, file name and size of the current file; `rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18T00:00:00Z;endtime=2016-07-`

`18T00:25:04Z;name=00000000721000000&size=1065437356`.

- Download by time: the playbackURI only uploads the start and end time; the device only parses time; `rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18T00:00:00Z;endtime=2016-07-18T00:25:04Z`.

Note: The time in the URL should be in accordance with the ISO 8601 time format.

3. Download videos (with plug-in):

   Get the capability of video download: `GET /ISAPI/ContentMgmt/download/capabilities>`.

   Download the videos: `POST /ISAPI/ContentMgmt/download`.

   You can download videos by time or file via this protocol, and you can specify download method via the in the message. Details are as follows:

   - Download by file: the playbackURI uploads the start and end time, file name and size of the current video; `rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18 00:00:00&endtime=2016-07-18 00:25:04Z&name=00000000721000000&size=1065437356`.
   - Download by time: the playbackURI only uploads the start and end time; the device only parses time; `rtsp://IpAddress/Streaming/tracks/101?starttime=2016-07-18 00:00:00&endtime=2016-07-18 00:25:04Z`.

   Note: For website integrations, spaces in URL are not allowed, you can use &amp to escape the space.

8 PTZ

8.1 . PTZ Lock

8.1.1 Introduction to the Function

The client can control a network camera's PTZ lock. During the locked period, the PTZ linked with alarm events will not work, but manual rotation is available until the client unlocks the PTZ. This function can be used for long-term monitoring of key locations to prevent the lens from being moved. When configuring or modifying intelligent applications that involve patrol, the PTZ should be locked first and then unlocked after the configuration is completed to achieve internal control of the intelligent application's operation, while the actual PTZ operation remains functional.

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8.1.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `ManualPTZLockCap` indicates the support for manual `PTZ` control locking;

2. (Client) Control PTZ manually:

   Manually lock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLock?format=json`;

   Manually unlock the PTZ: `PUT /ISAPI/PTZCtrl/channels/<channelID>/manualPTZUnlock?format=json`.

3. (Optional) The client configures the PTZ lock parameters to specify the lock duration:

   Capability: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `lockTime` indicates the maximum lock duration.

   Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.

   Get: `GET /ISAPI/PTZCtrl/channels/<channelID>/lockPTZ`.

4. The client gets the remaining time for PTZ unlock: `GET /ISAPI/PTZCtrl/channels/<channelID>/manualPTZLockCfg?format=json`.

8.2 Auxiliary Device Control

8.2.1 Introduction to the Function

The PTZ auxiliary devices help the device better display video content. The auxiliary devices includes lights, wipers, fans, and heaters. This function is used to control PTZ auxiliary devices.

8.2.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `PTZAuxType` indicates the support for PTZ auxiliary device control;
2. The client configures the PTZ auxiliary device parameters to control the auxiliary devices. It is recommended to first get all auxiliary device parameters before modifying parameter configurations:

   (Optional) Get all auxiliary device parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

   (Optional) Get a single auxiliary device parameter: `GET /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`;

   (Optional) Configure auxiliary device parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols`;

   Configure a single auxiliary device parameter: `PUT /ISAPI/PTZCtrl/channels/<channelID>/auxcontrols/<auxCtrlID>`.

Notes:

The `id` in the parameter configuration message corresponds to the `auxCtrlID` in the `url`, which is the auxiliary device index and starts from 1 by default.

8.3 PTZ 3D Positioning Control

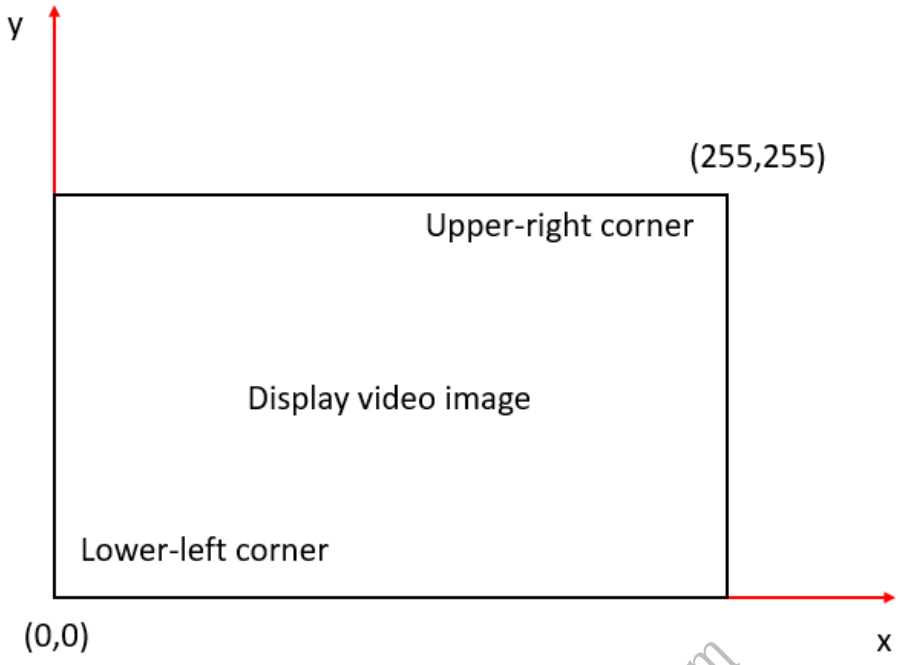
8.3.1 Introduction to the Function

Supports centering a point in the video image, and supports centering and zooming a rectangular area in the video image.

8.3.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPosition3D` indicates the support for 3D positioning control;
2. (Client) Send a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/position3D`:
   - The point in the video image is centered by the device, and the coordinates of `StartPoint` and `EndPoint` are the same;
   - The device zooms in and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is smaller than that of `EndPoint`;
   - The device zooms out and centers a rectangular area in the video image, and the X coordinate value of `StartPoint` is greater than that of `EndPoint`.

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8.4 PTZ Absolute Position

8.4.1 Introduction to the Function

The PTZ absolute position is unique and remains the same. The client can record certain PTZ absolute positions in actual application scenarios to repeatedly inspect at specified positions. For example, in a supermarket, record the PTZ absolute position of the cash register and inspect the position periodically.

8.4.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. the returned `isSupportAbsoluteEx` indicates the support for high-precision absolute `PTZ` movement control;
2. (Client) Get the high-precision absolute `PTZ` movement control capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/RailwayRobot/capabilities?format=json`. The value range of control parameters will be returned;
3. The client specifies the PTZ to move to an absolute position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`. The `elevation` and `azimuth` represent the angles in the vertical and horizontal directions, `absoluteZoom` represents the zoom range, `focus` represents the focus range, and `absoluteRotate` represents the absolute rotation angle.
4. (Client) Get the PTZ absolute position: `GET /ISAPI/PTZCtrl/channels/<channelID>/absoluteEx`.

Notes:

- The zero coordinate position of `elevation` and `azimuth`, i.e., PT(0,0), can be specified by `PUT /ISAPI/PTZCtrl/channels/<channelID>/homeposition`.

8.5 PTZ Manual Control

8.5.1 Introduction to the Function

Adjust the horizontal and vertical positions of the PTZ, and control the zoom and focus of the lens at specified rotation/changing speeds. The client adjusts PTZ continuously through applying control commands, which is usually used for manual direction adjustment in remote live view.

8.5.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned

`ContinuousPanTiltSpace` in the message indicates the support for PTZ manual control;

2. The client sends a control command to the device: `PUT /ISAPI/PTZCtrl/channels/<channelID>/continuous`. The values of `pan`, `tilt`, `zoom`, and `rotate` represent the speed of rotation/change, and the characteristics (positive and negative) of the values represent the direction of rotation/change. If the control needs to be stopped, another command should be sent and values should be set as 0.

8.6 PTZ One-Touch Focus

8.6.1 Introduction to the Function

Help PTZ devices to focus. When defocus (blurry image) occurs , one-touch focus can make the image clear.

8.6.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsStart` indicates the support for one-touch focus;
2. (Client) Configure PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/start`;
3. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportOnepushfoucsReset` indicates the support for resetting focus parameters;
4. (Client) Reset the PTZ one-touch focus parameters: `PUT /ISAPI/PTZCtrl/channels/<channelID>/onepushfoucs/reset`. The PTZ focus parameter setting will be restored.

8.7 Save PTZ Position

8.7.1 Introduction to the Function

The client controls the device to save the current PTZ position information. After the PTZ leaves the current position, the client can control the PTZ to return to the saved position, but only one position can be saved for each PTZ.

8.7.2 API Calling Flow

1. (Client) Get the PTZ capabilities: `GET /ISAPI/PTZCtrl/channels/<channelID>/capabilities`. The returned `isSupportPTZSave` and `isSupportPTZSaveGet` indicates the support for keeping and getting `PTZ` parameters configurations;
2. (Client) Configure the device to save `PTZ` information: `PUT /ISAPI/PTZCtrl/channels/<channelID>/save?format=json`;
3. The client controls the PTZ to move to the saved position: `PUT /ISAPI/PTZCtrl/channels/<channelID>/gotoPosition?format=json`. The PTZ will rotates to the saved position;
4. The client gets the saved PTZ position information: `GET /ISAPI/PTZCtrl/channels/<channelID>/save`. The PTZ position information will be returned and PTZ does not move.

9 People Counting

9.1 People Counting Statistics Search

9.1.1 Introduction to the Function

For devices with local storage function, the people counting data will be stored in the local storage after the people counting function is enabled. You can search the local data to get multiple types of reports by setting different conditions.

9.1.2 API Calling Flow

9.1.2.1 Single Channel

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1. Get the capability of searching for people counting data of a single channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/search/capabilities`.
2. Search for people counting data of a single channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/counting/search`.

9.1.2.2 Multiple Channels

1. Get video capability: `GET /ISAPI/System/Video/capabilities`. If the node `isSupportMultiChannelCounting` is returend and its value is `true`, it indicates that searching for people counting data of multiple channels is supported.
2. Get the capability of searching for people counting data of multiple channels: `GET /ISAPI/System/Video/inputs/channels/counting/search/capabilities`.
3. Search for the people counting data of multiple channels: `POST /ISAPI/System/Video/inputs/channels/counting/search`.

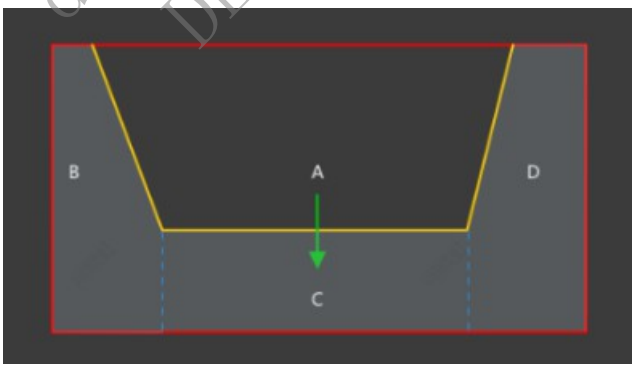
9.2 People Flow Direction Analysis

9.2.1 Introduction to the Function

People flow direction analysis is used for counting the people who crosses detection lines and enters the detection area. The flow analysis is valid only when the detection area is quadrilateral and the detection line and the detection area are overlapped only in the start and end points.

The requirements for drawing the rule/detection area of flow direction analysis are as follows:

1. The detection line (in yellow) and the detection area (in red) should have but two overlapped points.
2. The detection line should be perpendicular to the detection direction (along the direction of green arrow).
3. The four points on the detection line respectively are: point 1, 2, 3, 4.
4. Along the detection direction (along the green arrow), draw two parallel lines from point 2 and 3 to intersect the detection area (in red). The area between the two parallel lines is the counting region of area 2 (area C in the figure), and the rest two are area 1 (area B in the figure) and area 3 (area D in the figure).



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9.2.2 API Calling Flow

The process of people flow direction analysis is the same with that of the real-time people counting. You need to enable the flow direction analysis to receive and parse the flow statistics data.

1. Get the capability of people counting: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`. If `flowDirectionAnalysis` is returned, it indicates that the device supports flow analysis.
2. Configure people counting parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting`. Configure detection lines parameters (`CountingLineItemList`) and rule parameters (`CountingRegionList.RuleRegionCoordinatesList`), select scene mode (`countingSceneMode`), enable the flow direction analysis (`flowDirectionAnalysis` is `true`).
3. Receive the event uploaded by devices in arming or listening mode: `PeopleCounting`. Parse

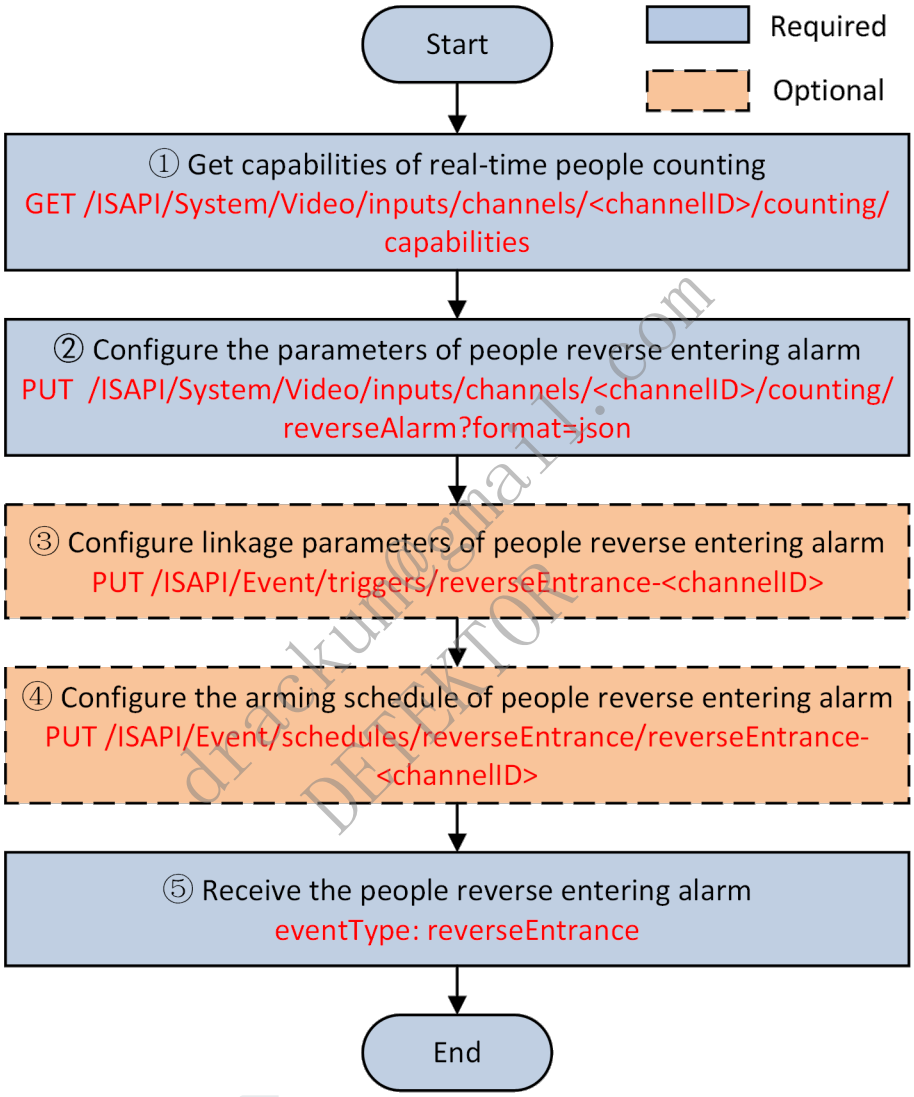
`peopleCounting.FlowStatisticsList` to get the flow data.

9.3 People Reverse Entering Alarm

9.3.1 Introduction to the Function

The people reverse entering alarm will be triggered if a person crosses the detection line from the opposite direction within a specified period. It applies to the circumstance when only single direction passing is allowed within a specified period. For example, if someone exits the workshop within working time, an alarm will be uploaded for notification.

9.3.2 API Calling Flow



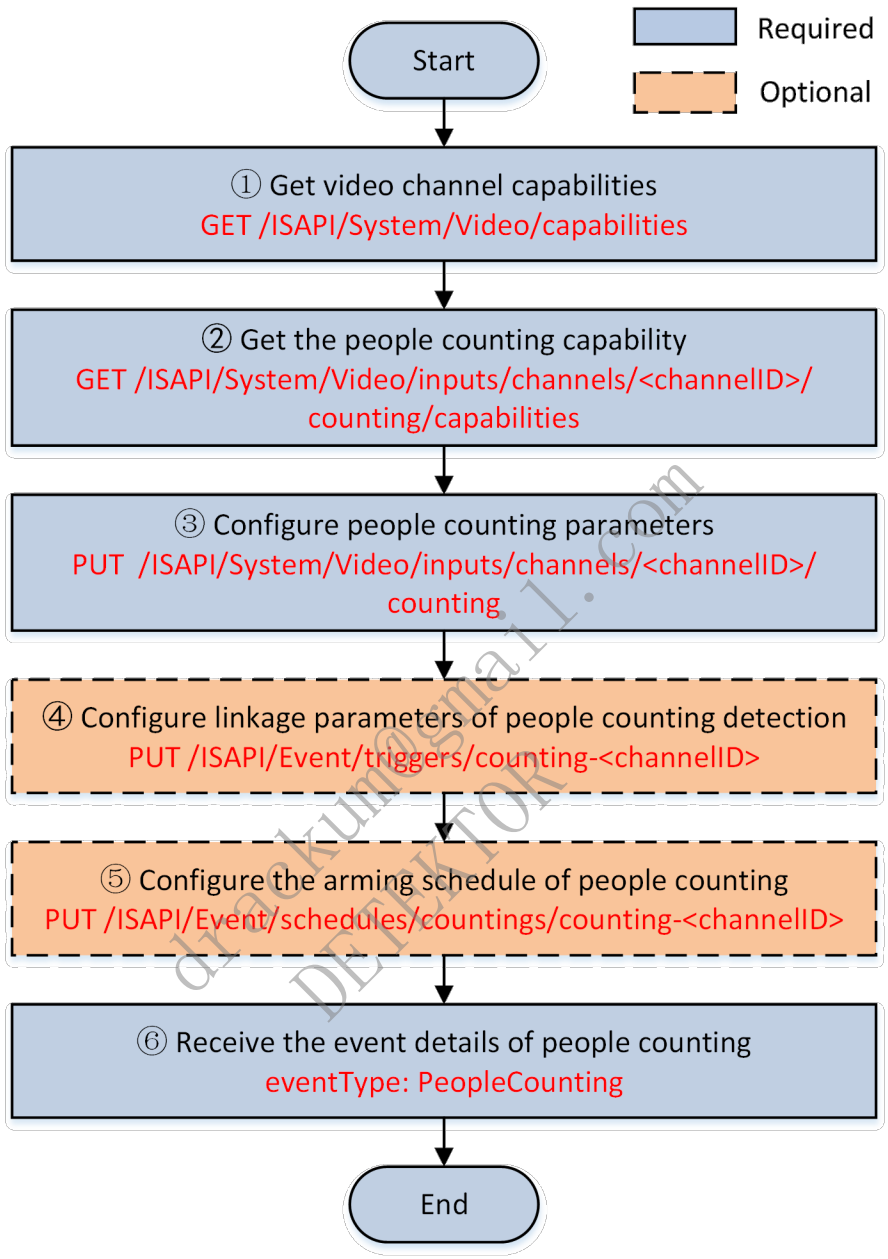
1. Get the people counting capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`. If the node `isSupportReverseAlarm` is returned and its value is `true`, it indicates that people reverse entering alarm is supported.
2. Configure parameters of the people reverse entering alarm: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/reverseAlarm?format=json`.
3. (Optional) Configure linkage parameters of the people reverse entering alarm: `PUT /ISAPI/Event/triggers/reverseEntrance-<channelID>`.
4. (Optional) Configure the arming schedule for the people reverse entering alarm: `PUT /ISAPI/Event/schedules/reverseEntrance/reverseEntrance-<channelID>`.
5. Receive events reported by device via arming or listening: `reverseEntrance`.

9.4 Real-time People Counting

9.4.1 Introduction to the Function

The function is to count line crossing persons or objects in a specified time period. It is commonly applied in the entrance and exit of the scenic spots, malls, stores, etc.

9.4.2 API Calling Flow



1. Get video channel capabilities: `GET /ISAPI/System/Video/capabilities`. If the nodes `isSupportCounting` and `isSupportMultiChannelCounting` are returned and the value is true, it indicates that the device supports real-time people counting.
2. Get the people counting capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities`.
3. Configure people counting parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting`.
4. (Optional) Configure linkage rules for people counting detection: `PUT /ISAPI/Event/triggers/counting-<channelID>`.
5. (Optional) Configure the arming schedule for people counting: `PUT /ISAPI/Event/schedules/countings/counting-<channelID>`.
6. Receive events uploaded by device in listening or arming mode: `PeopleCounting`.

Notes:

- During people counting, considering the height difference between children and adults, you can enable the children detection mode so as to improve the accuracy. a. Get the capability of people counting. If `ChildFilter` is returned, it indicates that the device supports counting the number of children. b. Enable the children counting during the configuration of people counting parameters (set `enabled` of `ChildFilter` to true). c. Other operations are same as above steps.
- The people counting detection event (`PeopleCounting`) supports the following data types: a. Real-time statistics data (`statisticalMethods` is `realTime`): after enabled, persons' entry and exit will trigger the event data uploading, and the data is the real-time people counting statistics from 12:00 A.M. to the current clock on the day. b. Statistics data by period (`statisticalMethods` is `timeRange`): the device will upload the statistics data according to the scheduled uploading interval (`dataUploadCycle`) within a specified period. c. Statistics data triggered by alarm input (`statisticalMethods` is `signalTrigger`): the device is connected to the alarm input signal, and the statistics data will be accumulated and uploaded every time the signal is triggered.
- The dimensions of statistics data are as follows: the number of people entering (`enter`), the number of people exiting (`exit`), the number of people repeating passing (`duplicatePeople`, which is only valid for people entering, and deduplication is invalid for people exiting), the number of people according to different face attributes, including face expression (`FaceExpressionList`), gender (`GenderList`), mask wearing (`MaskList`), etc. See details in the event message.
- If the client software fails to receive the people counting event uploaded by devices due to exceptions, or the connected devices have operated for a while, you can get the history people counting data via URI (POST /ISAPI/System/Video/inputs/channels/counting/search).

10 Abnormal Event Detection

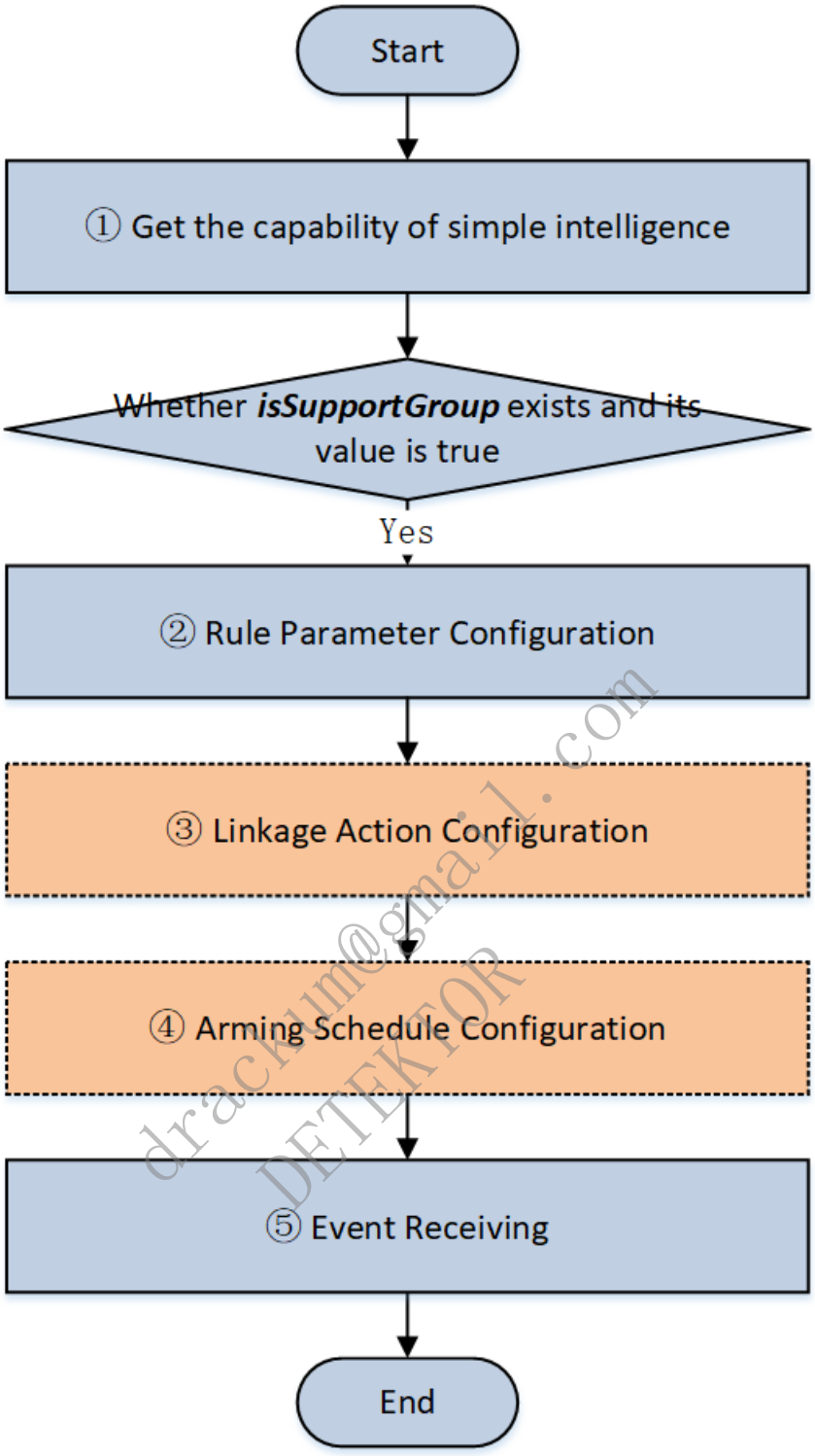
10.1 People Gathering

10.1.1 Introduction to the Function

The people gathering function is used for triggering the alarm in the fastest and best way when the people density in the set area exceeds the set threshold. The people gathering function is often used for performing the real-time monitoring of the abnormal behavior of people in the monitoring area, effectively preventing accidents caused by negligence, so as to avoid potential safety accidents and improve the supervision efficiency.

10.1.2 API Calling Flow

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1. Get the apability of AI intelligent search.

   Get the `isSupportGroup` in the capability of simple intelligence (related URI: `GET /ISAPI/Smart/capabilities`).

2. Configure rule parameters.

   Configure All Channels

- Get people gathering parameters (related URI: `GET /ISAPI/Smart/group`).
- Configure people gathering parameters (related URI: `PUT /ISAPI/Smart/group`).

  Configure Specified Channel

- Get the capability of people gathering parameters (related URI: `GET /ISAPI/Smart/group/<channelID>/capabilities`).

- Get the people gathering parameters of a specified channel (related URI: `GET /ISAPI/Smart/group/<channelID>`).
- Configure the people gathering parameters of a specified channel (related URI: `PUT /ISAPI/Smart/group/<channelID>`).

Configure All Areas of Specified Channel

- Get the capability of the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/capabilities`).
- Get the parameters of all areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions`).
- Configure the parameters of all areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions`).

Configure Specified Channel in Specified Channel

- Get the capability of the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>/capabilities`).
- Get the parameters of specified areas for people gathering (related URI: `GET /ISAPI/Smart/group/<channelID>/regions/<regionID>`).
- Configure the parameters of specified areas for people gathering (related URI: `PUT /ISAPI/Smart/group/<channelID>/regions/<regionID>`). Note: The percentage refers to the percentage of people number accounts for the alarming area. An alarm will be triggered in the system when the percentage is higher than the set percentage. Otherwise, no alarm will be triggered.

3. (Optional) Configure arming linkage parameters.

- Get the capability of event linkage configuration (related URI: `GET /ISAPI/Event/triggersCap`), whether the message `EventTriggersCap` contains the node `GroupDetectionTriggerCap`.
- Get the linkage parameters of people gathering (related URI: `GET /ISAPI/Event/triggers/group-<channelID>`).
- Configure linkage parameters of people gathering (related URI: `PUT /ISAPI/Event/triggers/group-<channelID>`).

4. (Optional) Configure arming schedule.

- Get the arming schedule of alarm outputs for all channels (related URI: `GET /ISAPI/Event/schedules/groups`).
- Configure the arming schedule of people gathering for all channels (related URI: `PUT /ISAPI/Event/schedules/groups`).
- Get the arming schedule of people gathering for a specified channel (related URI: `GET /ISAPI/Event/schedules/groups/group-<channelID>`).
- Configure the arming schedule of people gathering for a specified channel (related URI: `PUT /ISAPI/Event/schedules/groups/group-<channelID>`). Note: 1. Supports setting up to 8 time periods in one day on the timeline. 2. Overlapped time between any two time periods is not allowed when setting the arming schedule.

5. Receive event: `"eventType":"group"`, including the following. `detectionPictureTransType` refers to the transmission method of the pictures. The field value is bkgUrl for the picture URL or the picture encoding data via base64 when the value of detectionPictureTransType is "url" or "base64". The field value is the picture Content-ID when the value of detectionPictureTransType is "binary".

11 Face Capture

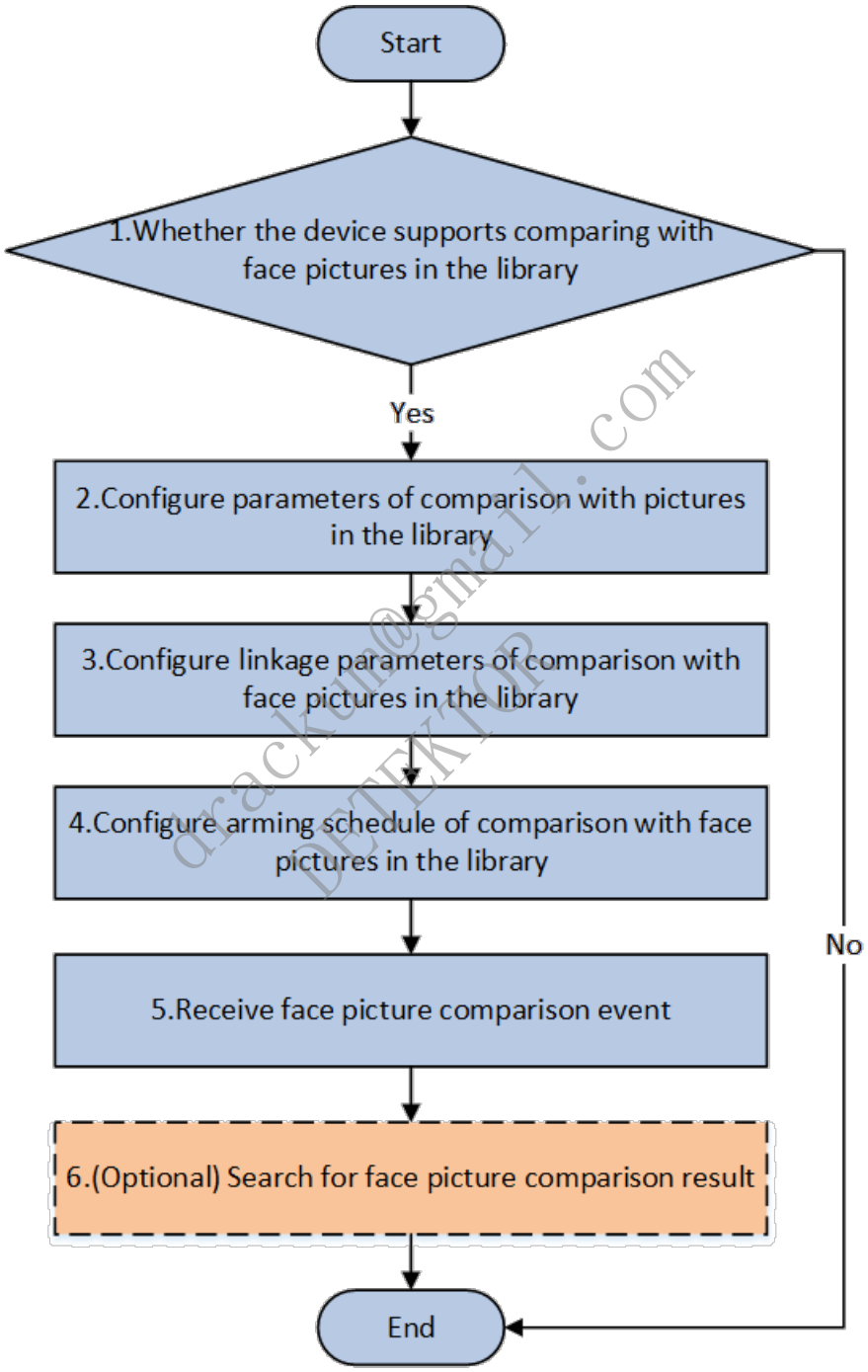
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11.1 . Comparison with Pictures in the Library

11.1.1 Introduction to the Function

Note: This feature depends on the face picture library management (refer to Face Picture Library Management Field for details). Analyze and model the face captured in the camera, then compare the face picture with pictures in the library, and lastly return the comparison similarity.

11.1.2 API Calling Flow



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1. **Whether the device supports comparing with face pictures in the library:**

   Get the capability node to evaluate whether the device supports comparing pictures with face pictures in the library, `GET /ISAPI/System/capabilities`; If the node `<isSupportFaceContrast>` is returned and its value is `true`, it indicates that the device supports comparing pictures with face pictures in the library.

2. **The parameters of comparison with pictures in the library including:**

   Get the capability of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast/capabilities`. Configure the parameters of face

picture comparison, `PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast`; Get the configuration of face picture comparison, `GET /ISAPI/Intelligent/channels/<channelID>/faceContrast`; **Note: The rule parameters support configuring the triggering threshold of face similarity event and specifying information including face picture library ID.**

3. **The linkage parameters of comparison with face pictures in the library:**

   Two sets of configurations linked to business systems are provided due to the front-end and back-end device differences:

   Linkage action of face picture library: for front-end devices (e.g., Deepinview), a business system of single channel and multiple face picture libraries. Therefore it supports configuring different arming linkage parameters via each compared face picture library. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceLibTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters, `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/triggers/faceLib-<channelID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the linkage action parameters, `GET /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow.

   Linkage action of channel: for back-end devices (e.g., DeepinMind), a business system of multiple channels and single face picture library. Therefore it supports configuring different arming linkage parameters via each video channel. Call `GET /ISAPI/Event/triggersCap` to evaluate whether it supports the event linkage configuration capability, it does if `<FaceContrastTriggerCap>` is returned, and the relevant configuration APIs are as follows: Configure the linkage action parameters of face picture comparison, `PUT /ISAPI/Event/triggers/faceContrast-<channelID>`; get the linkage action parameters of face picture comparison, `GET /ISAPI/Event/triggers/faceContrast-<channelID>`;

4. **Arming schedule of comparison with face pictures in the library**:

   Two sets of arming schedule configurations linked to business systems are provided due to the front-end and back-end device differences: Arming schedule of face picture library: Configure the single-channel arming schedule of face picture library, `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; if users have added a customized face picture library ID when creating one, they can also call `PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>/<customFaceLibID>?FDType=custom` to configure via `customFaceLibID`; get the single-channel arming schedule of face picture library, `GET /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>`; Customizing the face picture library ID follows the same flow. Arming schedule of channel: Configure the arming schedule of face picture comparison, `PUT /ISAPI/Event/schedules/faceContrast-<channelID>`; get the arming schedule of face picture comparison, `GET /ISAPI/Event/schedules/faceContrast-<channelID>`;

5. **Face picture comparison event**:

   When the face captured on the camera is compared with pictures in the library, and the similarity has reached the rule parameters including the threshold, the device will upload the face picture comparison event; the event information mainly includes comparison similarity, captured face and its attributes, and the corresponding registered face picture and its attributes, `"eventType": "alarmResult"`.

6. **Search for face picture comparison result:**

   If supported by device, the face picture comparison result will be saved to the device local storage, and you can search for comparison history via the following API: Search for face picture comparison information, `GET /ISAPI/Intelligent/FDLib/FCSearch`.

11.2 . Search Via Picture Comparison In Face Capture Library

11.2.1 Introduction to the Function

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You can search for matched face pictures in face capture library via comparison with the uploaded local face picture. The general application scenario: compare the target face picture with face pictures in the face capture library to track the target according to the comparison result.

11.2.2 Search via Picture Comparison Synchronously

11.2.1.1 Introduction to the Function

After the request of searching via picture comparison synchronously is applied, the search result will be returned directly in the response message.

11.2.1.2 API Calling Flow

1. **Check whether the device supports searching via picture comparison synchronously.**

   Get the capability of searching via picture comparison: `GET /ISAPI/SDT/Face/searchByPic/capabilities`. If the node `isSupportSyncSearch` is returned and its value is `true`, it indicates that searching via picture comparison synchronously is supported.

   Note: For some devices supporting searching via picture comparison synchronously, the API for getting the capability is invalid, you can call `POST /ISAPI/SDT/Face/searchByPic?supportSync=true` directly.

2. **Submit the task of synchronously searching via picture comparison.**

   Synchronously search for pictures via picture comparison: `POST /ISAPI/SDT/Face/searchByPic?supportSync=true`.

   Note: Searching via picture comparison synchronously supports three types of pictures (`dataType`): picture URL (`faceURL`), picture model data (`targetModelData`, which should be generated in advance via `POST /ISAPI/SDT/Face/pictureAnalysis`, and binary picture data, which is in form format and follows the JSON message.

   The Request Message Example of Binary Picture Data

```
POST /ISAPI/SDT/Face/searchByPic?supportSync=true
Host: device_ip:port
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: application/json

[JSON Message]
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture Data]
--<frontier>--
```

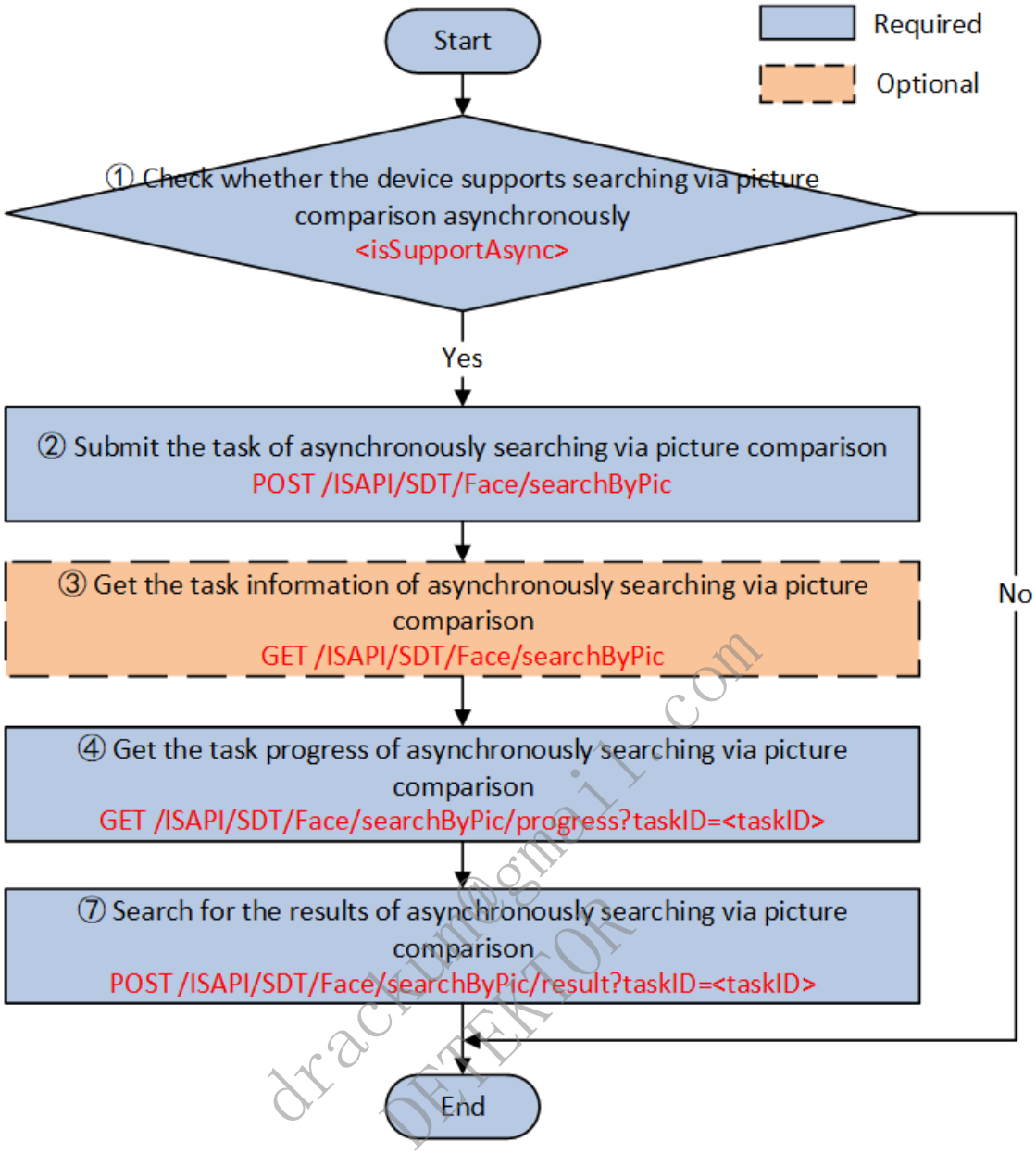
11.2.2 Search via Picture Comparison Asynchronously

11.2.2.1 Introduction to the Function

Submit the task of asynchronously searching via picture comparison and search for results by the returned task ID.

11.2.2.2 API Calling Flow

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1. **Check whether the device supports searching via picture comparison asynchronously.**

   Get capabilities of searching via picture comparison: `GET /ISAPI/SDT/Face/searchByPic/capabilities`. If the node `isSupportAsync` is returned and its value is `true`, it indicates that searching via picture comparison asynchronously is supported.

2. **Submit the task of asynchronously searching via picture comparison.**

   Asynchronously search via picture comparison: `POST /ISAPI/SDT/Face/searchByPic`. The task ID (`taskID`) is returned and will be used in the APIs below.

   Delete the task of asynchronously searching via picture comparison: `DELETE /ISAPI/SDT/Face/searchByPic?format=json&taskID=<taskID>`.

3. **(Optional) Get the task information of asynchronously searching via picture comparison.**

   `GET /ISAPI/SDT/Face/searchByPic`.

4. **Get the task progress of asynchronously searching via picture comparison.**

   Get the task progress: `GET /ISAPI/SDT/Face/searchByPic/progress?taskID=<taskID>`. When the progress (`progress`) is 100%, you can proceed to the following step.

5. **Search for the results of asynchronously searching via picture comparison.**

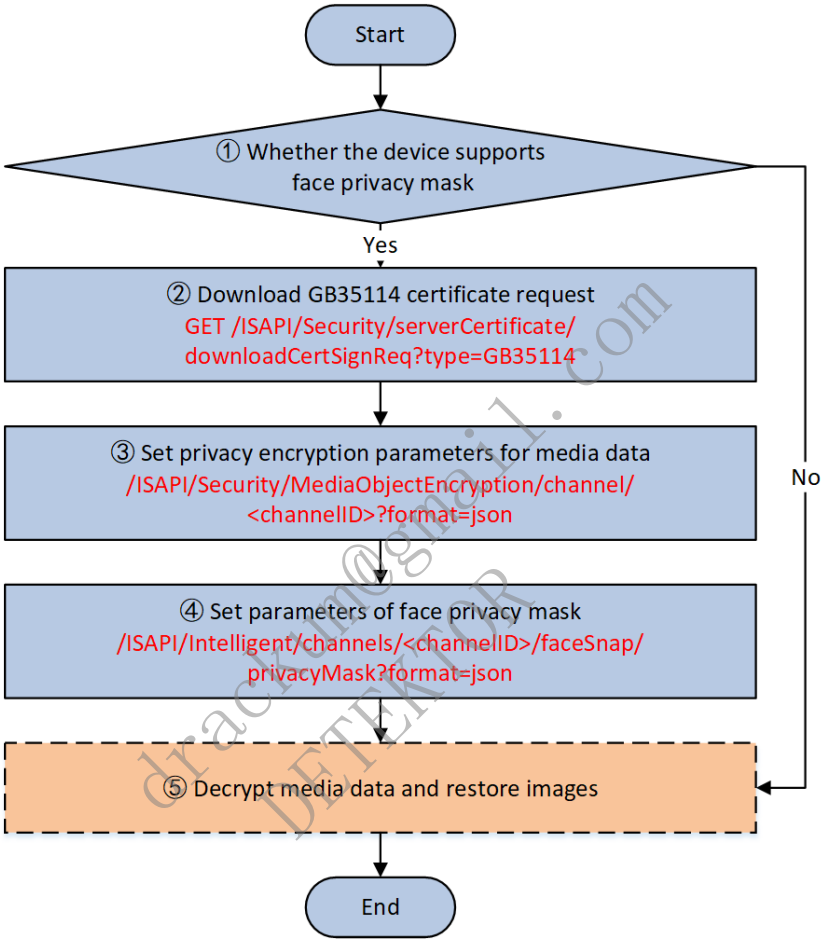
Search for results of asynchronously searching via picture comparison: `POST /ISAPI/SDT/Face/searchByPic/result?taskID=<taskID>`.

11.3 Face Capture

11.3.1 Introduction to the Function

Face capture is to detect whether face pictures show up in images. If set conditions are met and there are faces showing up, the face pictures will be captured and uploaded as alarms.

11.3.2 API Calling Flow



1. **Whether the device supports face capture:**

   Get the system capabilities: `GET /ISAPI/System/capabilities`. If the node (its parent node is) is returned as true, it indicates that the device supports face capture.

2. **Whether the device supports multi-scene face capture:**

   Get the intelligent capabilities of device: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`. If the node is returned and its value is true, it indicates that the device supports face capture.

3. **Set face picture capturing rules of multiple scenes:**

   If multi-scene capture of a channel is required, you can set the face picture capturing rules for multiple scenes.

   Set the face capture rules of a specified scene: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

   Get the face capture rules of a specified scene: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/<SID>`;

Set the face picture capturing rules of multiple scenes for a specified channel: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`;

Get the face picture capturing rules of multiple scenes for a specified channel: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene`.

4. **Set face picture capturing rules of a specific scene.**

   If specific-scene capture of a channel is required, you can set the face picture capturing rules.

   Get the capabilities of captured face picture rules `GET /ISAPI/Intelligent/channels/<channelID>/faceRule/capabilities`;

   Set the face capture rule parameters: `PUT /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the face capture rule parameters: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule`;

   Get the default parameters of face capture rule: `GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>`.

5. **Set linkage mode of face picture capture.**

   Set the linkage parameters of face capture: `PUT /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Get the linkage parameters of face capture: `GET /ISAPI/Event/triggers/faceSnap-<channelID>`;

   Delete the linkage of face capture: `DELETE /ISAPI/Event/triggers/faceSnap-<channelID>`.

6. **Configure the arming schedule of face capture.**

   Set the arming schedule of face capture for a specified channel: `PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Get the arming schedule of face capture for a specified channel: `GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>`;

   Set the arming schedule of face capture: `PUT /ISAPI/Event/schedules/faceSnap`;

   Get the arming schedule of face capture: `GET /ISAPI/Event/schedules/faceSnap`.

7. **Receive face capture and recognition events.**

   If the wanted face shows up in the set area, face recognition event will be reported. The event information includes face pictures and attribute information, etc. `"eventType": "faceCapture"`.

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11.4 Face Modeling and Face Picture Comparison

11.4.1 Face Picture Analysis

11.4.1.1 Introduction to the Function

Face picture analysis includes face detection and face modeling. After analyzing facial features (including age, gender, glasses, smiling, face angle, face frame, etc) of specified pictures, the platform performs modeling and returns model data. To perform face picture comparison in the name list, users need to submit the task of face picture comparison. For details, refer to the process of synchronous/asynchronous face picture comparison.

11.4.1.2 API Calling Flow

1. Get the capabilities of analyzing face pictures: `GET /ISAPI/SDT/Face/pictureAnalysis/capabilities`. If calling succeeded, it indicates that the device supports face picture analysis function.

2. Analyze face pictures (modeling): `POST /ISAPI/SDT/Face/pictureAnalysis`. The API will return the analysis result to the caller synchronously.

to the caller synchronously.

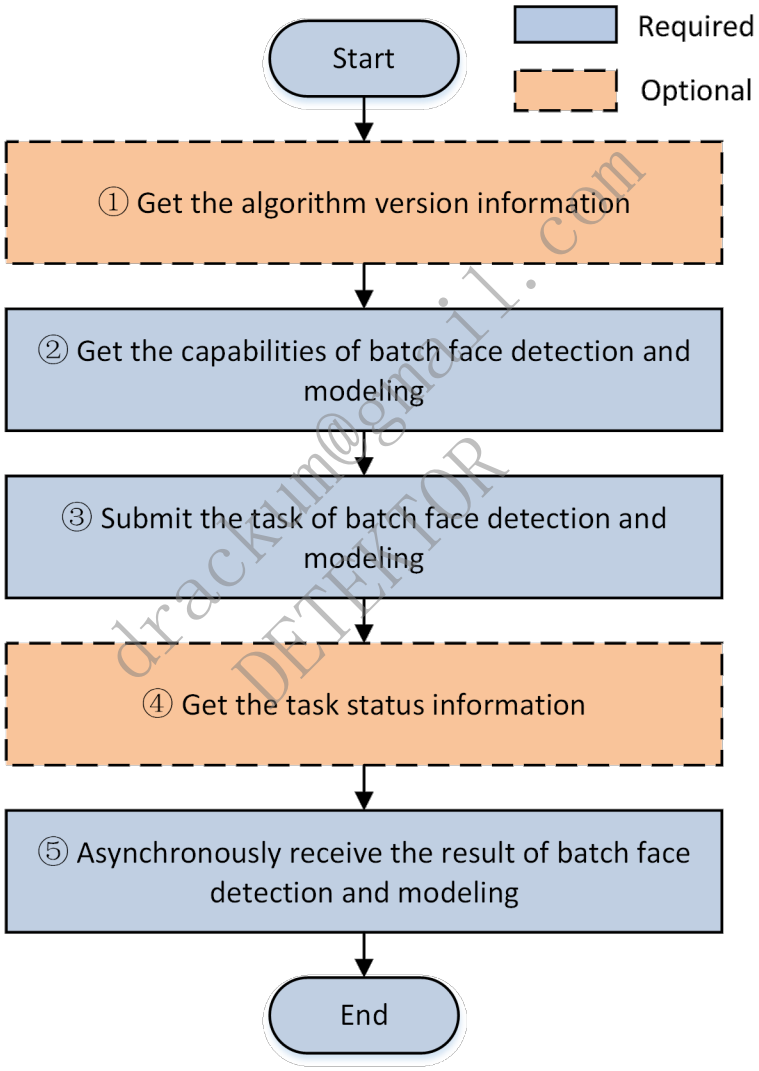
11.4.2 Batch Face Picture Analysis

11.4.2.1 Introduction to the Function

Batch face picture analysis supports detecting faces in multiple pictures, modeling and analyzing face thumbnails, uploading the analysis result via event type, and linking to corresponding tasks via task ID. Make sure the algorithm resources allocation is completed before performing batch face picture modeling detection. To perform face picture comparison in the name list, users need to submit the task of face picture comparison. For details, refer to the process of synchronous/asynchronous face picture comparison.

Batch picture analysis allows users to specify whether to import pictures into the face picture library. If importing pictures into library is supported, users can search for the status of the pictures imported to the library in a batch.

11.4.2.2 API Calling Flow



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1. Get the version information of intelligent algorithm scheduling supported by the device: `GET /ISAPI/Intelligent/algorithmInfos?format=json`.
2. Get the capabilities of batch face detection and modeling: `GET /ISAPI/SDT/Face/pictureAnalysis/batch/capabilities`. Check whether the device supports batch face detection and modeling.
3. Submit the task of batch face detection and modeling: `POST /ISAPI/SDT/Face/pictureAnalysis/batch`. It is required to upload the pictures to server or cloud storage considering only importing the face pictures by URL is allowed.
4. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.

5. Asynchronously receive the result of batch face picture analysis mode: `eventType: analysisResult`.

11.4.3 Synchronous Face Picture Comparison

11.4.3.1 Introduction to the Function

Synchronous face picture comparison analyzes the facial features in a single picture first, and then compares the picture with pictures in the face picture library, and returns matched face records. It is commonly used for the analysis of a single picture. Users can obtain the face modeling data via face picture modeling function. For details, refer to the process of (batch) face picture analysis.

11.4.3.2 API Calling Flow

1. Get the capabilities of synchronous face picture comparison task: `GET /ISAPI/SDT/Management/Task/facePicture/sync/capabilities?format=json`.
2. Submit the task of synchronous face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/sync?format=json`. The API will return the analysis result to the caller synchronously.

11.4.3.3 Exception Handling

11.4.3.3.1 Error Code

statusCode	statusString	subStatusCode	Error Code	Description	Remarks
4	Invalid Operation	noMoreTasksCanBeAdded	0x4000118A	No more tasks can be added.	No more tasks can be added. Delete the completed or unneeded tasks and try again.
4	Invalid Operation	userScheOffline	0x40001040	Task scheduler server is offline.	Please contact the technical support.
4	Invalid Operation	clusterNodeOffline	0x40001075	Cluster node is offline.	Cluster node is offline. Check the cluster status.
4	Invalid Operation	destinationaddressinvalid	0x400011c6	Invalid third-party address.	Invalid third-party address. Check again.
4	Invalid Operation	cloudAnalysisRequestInformationBuildFailed	0x4000110C	Creating request message of cloud analysis failed	Please provide related log and packet capture information and contact the

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				failed.	the technical support.
4	Invalid Operation	cloudAnalysisResponseInformationParseFailed	0x4000110D	Parsing response message of cloud analysis failed.	Please provide related log and packet capture information and contact the technical support.
4	Invalid Operation	cloudAnalysisServerReturnFailed	0x400010FC	No response is returned by cloud analytic service.	Please provide related log and packet capture information and contact the technical support.
4	Invalid Operation	numberReachedLimit	0x40001119	The number reached limit.	No more configured items can be added. Delete unneeded items and try again.
4	Invalid Operation	tsModelError	0x40001254	Forwarding service module error.	Incorrect task receiving address. Check again.
4	Invalid Operation	anModelError	0x40001255	Computing node error.	Please contact the technical support.
4	Invalid Operation	schedulingdatabaseisfull	0x400011c4	Database disk is full.	Database disk is full. Try again later or cancel related tasks and try again.
					Scheduling module

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4	Invalid Operation	ssModelError	0x4000124c	Scheduling module error.	error. Please contact the technical support.
6	Invalid Content	requestMemoryNULL	0x6000003F	Request memory is null.	Please contact the technical support.
2	Device Busy	noMemory	0x20000001	Insufficient memory.	Confirm the remaining memory by checking the value of the free field via the directory path of the linux system. Please provide related log information and contact the technical support.

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11.4.4 Asynchronous Face Picture Comparison

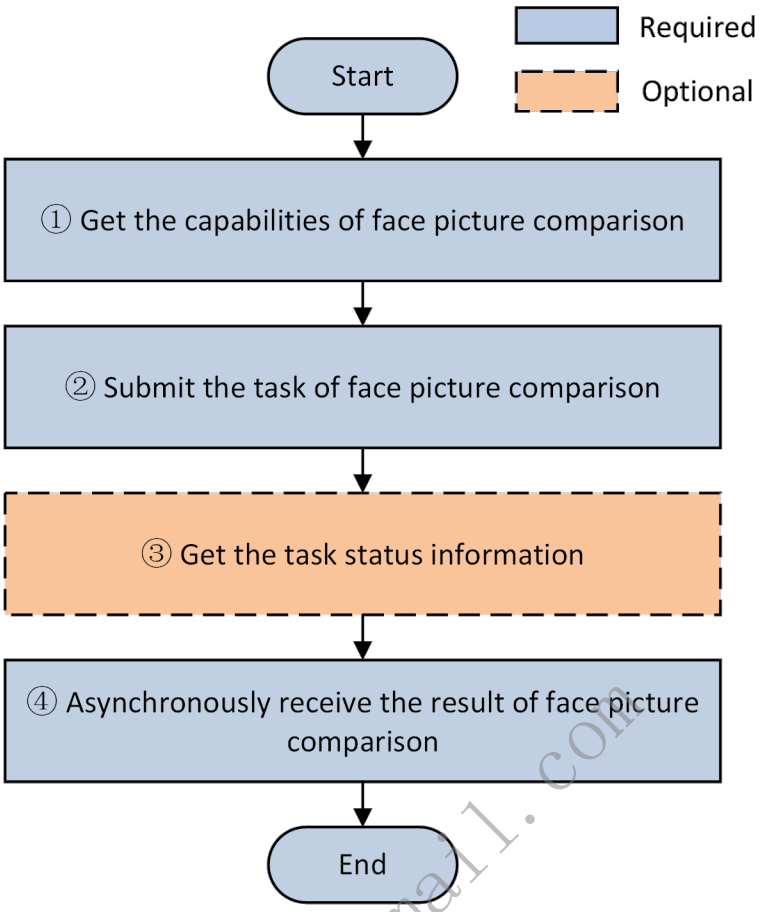
11.4.4.1 Introduction to the Function

Asynchronous face picture comparison analyzes face information in multiple pictures and compares with the faces in the face picture library, and then returns the matched face records. It is commonly used for analyzing multiple pictures. Users can obtain the face modeling data via face picture modeling function. For details, refer to the process of (batch) face picture analysis.

Asynchronous face picture comparison supports single and batch task.

11.4.4.2 API Calling Flow

11.4.4.2.1 Submit the Task of Face Picture Comparison



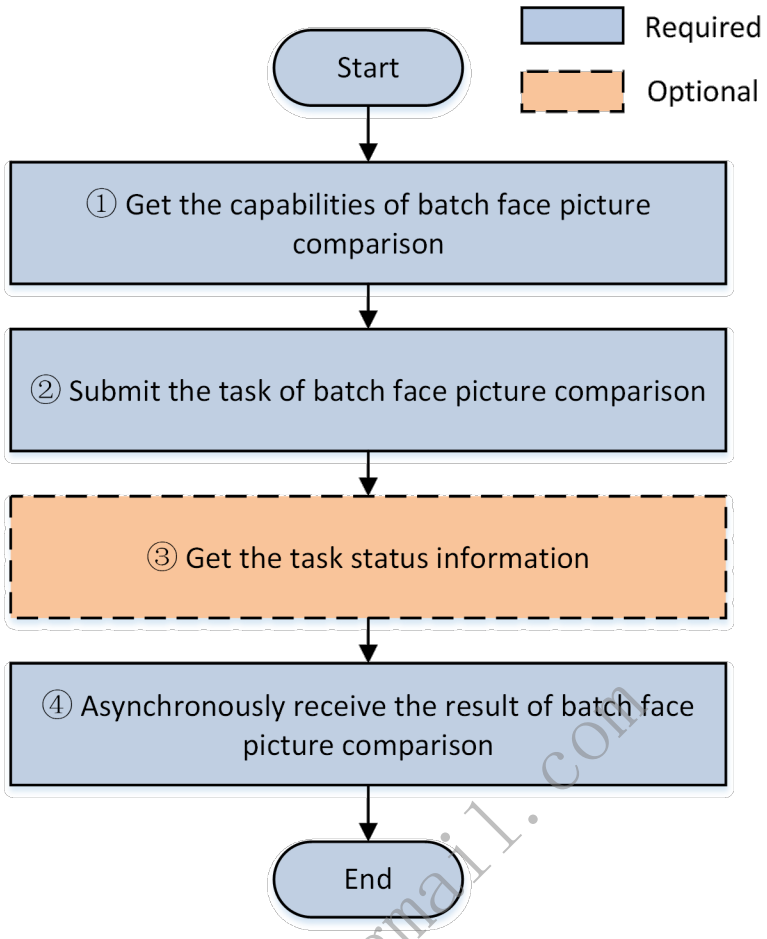
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1. Get the capability set of the face picture comparison task: `GET /ISAPI/SDT/Management/Task/facePicture/capabilities`.
2. Submit the face picture comparison task: `POST /ISAPI/SDT/Management/Task/facePicture`.

   Notes: It can be applied in scenes including blocklist alarm, stranger detection alarm, frequently appeared person alarm, and one person one archive. Users can distinguish various scenes according to the node `type` of the imported parameters `relateGroup`.
3. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.
4. Asynchronously receive the result of face picture comparison: `eventType: alarmResult`. The corresponding task is linked via the task ID.

Notes: Users can get the supported maximum number of pictures to be analyzed via the `maxPictureNum` of `GET /ISAPI/SDT/Management/Task/facePicture/capabilities`.

11.4.4.2.2 Submit the Task of Batch Face Picture Comparison



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1. Get the capabilities of the task of batch face picture comparison: `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities`.
2. Submit the task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`.

   Notes: It can be applied in scenes including blocklist alarm, stranger detection alarm, frequently appeared person alarm, and one person one archive. Users can distinguish various scenes according to the node `type` of the imported parameters `relateGroup`.
3. Get the task status information: `GET /ISAPI/SDT/Management/Task/status`.
4. Asynchronously receive the result of face picture comparison: `eventType: alarmResult`. The corresponding task is linked via the task ID.

Notes: Users can get the supported maximum number of pictures to be analyzed via the `maxPictureNum` of `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities`.

11.4.5 Transmitted Parameters

11.4.5.1 Introduction to the Function

For data center products, transmitted parameters (targetAttrs) are imported when the platform applies tasks. The transmitted parameters will not be parsed and will be returned when the analysis result is uploaded to link to corresponding alarms. Transmitted content is agreed between the platform and device.

Notes: The message length of the transmitted parameters is restricted, which depends on the corresponding capability set of the task API returned by the device.

11.4.5.2 Sample Message

11.4.5.2.1 The Task of Batch Face Picture Comparison

The task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`. The sample message of the transmitted parameters targetAttrs are as follows:

```
"targetAttrs": {
/*opt, object, the analysis device will not parse the transmitted content*/
    "deviceId": "1",
    /*opt, string, device ID, range:[,]*/
    "deviceChannel": 1,
    /*opt, int, device channel No., range:[,], step:, unit:, unitType:*/
    "deviceName": "shebei",
    /*opt, string, device name, range:[,]*/
    "faceTime": "2016-09-10T01:01:00.000Z",
    /*opt, string, capture time, range:[,]*/
    "rect": {
        /*opt, object, background picture frame information of face*/
        "height": 1.0,
        /*opt, float, height, range:[,], step:, unit:, unitType:*/
        "width": 1.0,
        /*opt, float, width, range:[,], step:, unit:, unitType:*/
        "x": 0.0,
        /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
        "y": 0.0
        /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
    },
    "bkgUrl": "http://absddsdfsdfsdk.jpg",
    /*opt, string, background picture, range:[,]*/
    "cameraIp": "10.11.41.26",
    /*req, string, camera IP address, range:[,]*/
    "cameraIndexCode": "3301023820382193",
    /*req, string, camera No., range:[,]*/
    "longitude": 116.39737,
    /*opt, float, latitude, range:[,], step:, unit:, unitType:*/
    "latitude": 116.39737,
    /*opt, float, latitude, range:[,], step:, unit:, unitType:*/
    "name": "Jack",
    /*opt, string, name, range:[,]*/
    "registerGender": "male",
    /*opt, string, gender, range:[,]*/
    "nation": 1,
    /*opt, int, nation, range:[,], step:, unit:, unitType:*/
    "bornTime ": "2016-01-30",
    /*opt, string, birthday, range:[,]*/
    "address": "New York, the United States",
    /*opt, string, address, range:[,]*/
    "certificateNumber": "111111111111111111",
    /*opt, string, ID card, range:[,]*/
    "credImgUrl": "http://XXX",
    /*opt, string, picture URL, range:[,]*/
    "matchResult": 0,
    /*opt, int, the number of matched results, range:[,], step:, unit:, unitType:*/
    "linkFaceBodyId": "XXX",
    /*opt, string, linkage ID between face and human body picture, range:[,]*/
    "linkFaceVehicleId": "XXX",
    /*opt, string, linkage ID between face and vehicle, range:[,]*/
    "linkFaceMacId": "XXX",
    /*opt, string, linkage ID between face and MAC, range:[,]*/
    "Coordinates": [
        /*req, array, coordinate, subType:object, range:[,]*/
        {
            "timeStamp": "test",
            /*req, string, timestamp, range:[,]*/
            "VT": {
                /*req, object, VT*/
                "h": 1.0,
                /*opt, float, height, range:[,], step:, unit:, unitType:*/
                "w": 1.0,
                /*opt, float, width, range:[,], step:, unit:, unitType:*/
                "x": 0.0,
                /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
                "y": 0.0
                /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
            },
            "isPTValid": true,
            /*req, bool, whether the PT is valid*/
            "PT": {
                /*req, object, PT*/
                "x": 0.0,
                /*opt, float, X coordinate, range:[,], step:, unit:, unitType:*/
                "y": 0.0,
                /*opt, float, Y coordinate, range:[,], step:, unit:, unitType:*/
                "z": 0.0
                /*opt, float, Z coordinate, range:[,], step:, unit:, unitType:*/
            },
            "humanBodyClarityType": 1,
            /*ro, opt, enum, type of body completeness, subType:int, (1#totally covered,2#upper body not covered,3#whole body not covered)*/
            "posture": "sitting",
            /*ro, opt, enum, gesture, subType:string, ("sitting", "standingUp")*/
            "motion": "yes",
            /*ro, opt, enum, whether the person is in motion, subType:string, ("yes", "no")*/
            "height": 1,
            /*ro, opt, int, height, range:[0,1000], step:1, unit:cm, unitType:length*/
            "humanRect": {
            /*ro, opt, object, coordinates of the human body in the video, desc:coordinates in the video (x,y,width,height); float (accurate to 3 decimal
places). The origin is the upper-left corner of the screen */
                "h": 1.000,
```

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```
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, width, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            },
            "faceRect": {
            /*ro, opt, object, coordinates of the human face in the video, desc: coordinates in the video (x,y,width,height); float (accurate to 3 decimal
places). The origin is the upper-left corner of the screen */
                "h": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            },
            "hsRect": {
            /*ro, opt, object, coordinates of the head and shoulder frame in the video, desc: coordinates in the video (x,y,width,height); float (accurate
to 3 decimal places). The origin is the upper-left corner of the screen */
                "h": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "w": 1.000,
                /*ro, req, float, height, range:[0.000,1.000], step:, unit:, unitType:*/
                "x": 0.000,
                /*ro, req, float, X coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
                "y": 0.000
                /*ro, req, float, Y coordinate, range:[0.000,1.000], step:, unit:, unitType:*/
            }
        }
    ],
    "targetID": "1",
    /*req, string, the tracking ID of a single camera, range:[,]*/
    "traComplete": true
    /*req, bool, boll*/
}
```

Notes: The length limit of the transmittance node `targetAttrs` depends on the `targetAttrsMaxLen` of the `GET /ISAPI/SDT/Management/Task/facePicture/batch/capabilities` (API of getting the capabilities of batch face picture comparison).

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11.4.6 Result Recipient List

11.4.6.1 Introduction to the Function

Users can specify address of the analysis result recipient by configuring the `destinationType` and `destinationUrl` of the `destination` node. Related APIs:

1. The task of face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture`.
2. The task of batch face picture comparison: `POST /ISAPI/SDT/Management/Task/facePicture/batch`.

11.4.6.2 Corresponding Relationship List

The relationship between `destinationType` and `destinationUrl` is as follows:

destinationType	destinationType Description	destinationUrl Format	Rema
ServerKafka	Kafka address.	{"topic":"test","url":"10.17.139.184:2181"}	
faceCaptureLib	Captured face picture library of big data.	{"topic":"SNAP_IMAGE_INFO_TOPIC","url":"172.10.201.156:2181"}	
faceAlarmUpload	Upload list alarms by ISAPI.	http://IP:PORT/****(****represents custom string description and empty is allowed)	
AlarmRMQ	AlarmRMQ.	{"topic":"test_face_alarm:test_face","url":"10.15.117.65:6005"}	
KafkaFaceLabel	Fusion big data tagging.	" {"topic":"LABEL_SNAP_IMAGE_INFO_TOPIC","url":"10.66.106.149:2181"}"	
faceLabelUpload	Face picture tagging.	{"topic":"SNAP_IMAGE_INFO_TOPIC","url":"172.10.201.156:2181"}	Its UR same of the faceC

11.5 Face Picture Library Management

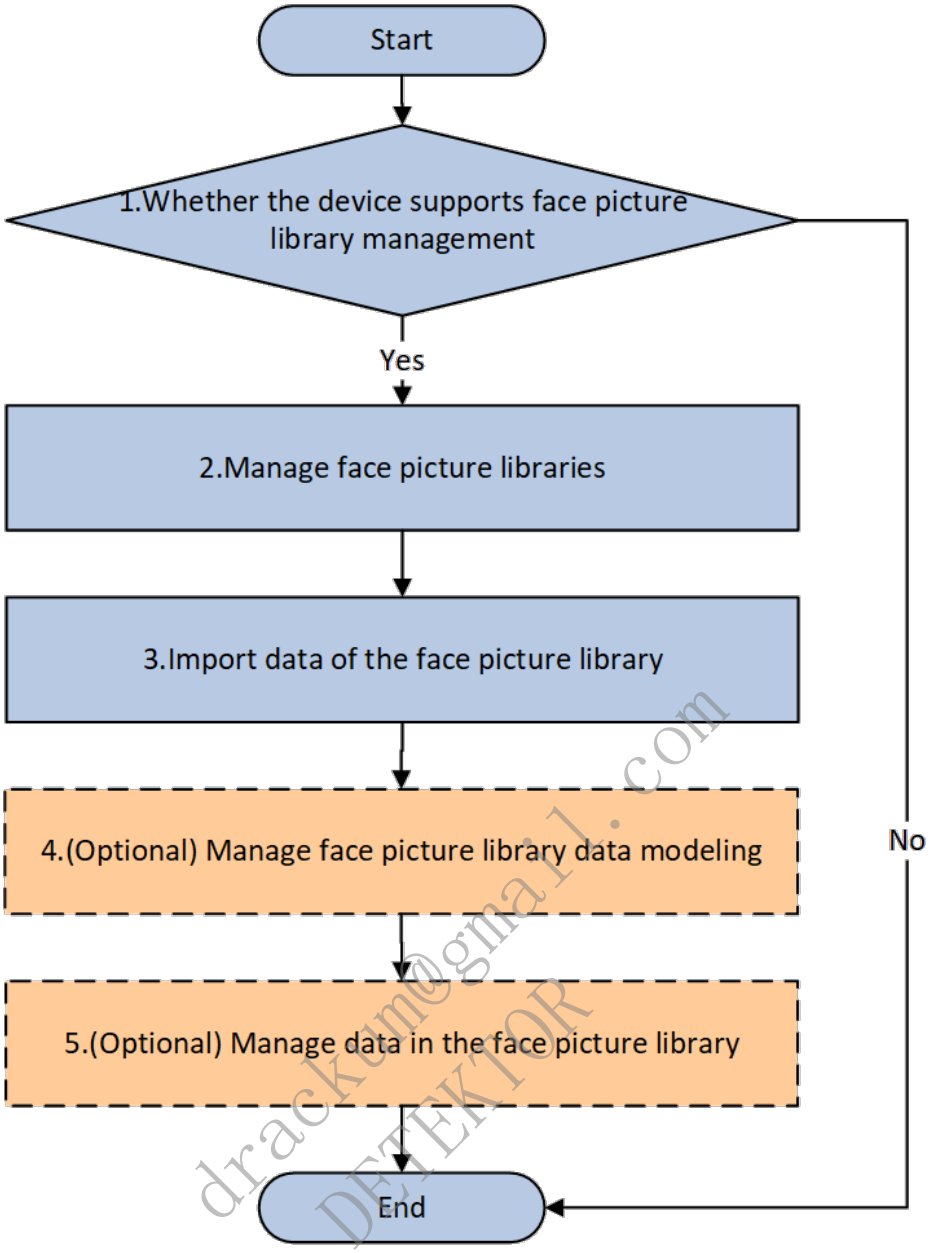
11.5.1 Face Picture Library Management

11.5.1.1 Introduction to the Function

This function is mainly used to manage operations on face picture libraries on the device, such as adding, deleting, editing, or searching for the face picture libraries, importing face pictures and additional person information to the libraries, editing, deleting, or searching for face pictures and additional person information in the libraries, and face picture modeling. This function is also the prerequisite of other face picture library functions such as face picture library comparison.

11.5.1.2 API Calling Flow

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1. Get the capability of managing face picture libraries for checking functions supported by the device: `GET /ISAPI/Intelligent/FDLib/capabilities`.
2. Manage face picture libraries.
    i. Create a face picture library: `POST /ISAPI/Intelligent/FDLib`. After the face picture library is created, the device will automatically generate the face picture library ID (that is, `FDID` is returned in the message `<FDLibInfoList>`). If you need to manage the face picture library ID by yourself, you can configure the custom face picture library ID (`customFaceLibID`) in the request message `<CreateFDLibList>` as needed. Afterwards, you can perform operations on the face picture library by the custom ID instead of `FDID`.
    ii. Edit parameters of the face picture library: `PUT /ISAPI/Intelligent/FDLib/<FDID>`. If you have added the custom face picture library ID in the above steps, you can also call `PUT /ISAPI/Intelligent/FDLib/<customFaceLibID>?FDType=custom` to configure parameters using `customFaceLibID`.
    iii. Get parameters of the face picture library: `GET /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
    iv. Delete the face picture library: `DELETE /ISAPI/Intelligent/FDLib/<FDID>`. Operations involved with the custom face picture library ID are the same as those mentioned above.
3. Import data of the face picture library.

i. Import face picture data and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`. By repeatedly applying the previous API, you can import the face picture data (including the face picture and additional face picture information `FaceAppendData`) to the corresponding face picture library; after it is successfully imported, the device will automatically model on the picture which could take some time, then the a face picture ID will be automatically generated (i.e., `PID` is returned in response parameters `MaskInfo`) by device after modeling succeeds; if you want to manually manage the face picture ID, you can add customized face picture ID (`customHumanID`) as needed when applying the request parameter `PictureUploadData`. `PID` can be replaced with customized face picture ID for subsequent face-picture-related operations.

ii. Import face picture library data in concurrent operation mode, `POST /ISAPI/Intelligent/FDLib/pictureUpload?type=concurrent`. If you want to improve the import efficiency, you can apply concurrent operation that the device will only store the picture not model on the picture. You will need to manually model on the picture (refer to Face Picture Library Data Modeling Management Flow for details). Note: Back-end DeepinMind devices will automatically perform modeling after data is imported even though under concurrent operation circumstances; if you demand to manually perform modeling, you can still apply modeling management.

4. (Optional) Manage face picture library data modeling. If you imported face picture library data in concurrent operation mode, you can select a time for device data modeling management after importing completed:
   i. Modeling on registered face pictures, `GET /ISAPI/Intelligent/FDLib/manualModeling`;
   ii. Search for modeling control results, `GET /ISAPI/Intelligent/FDLib/manualModelingRangeTask`;
   iii. Get modeling progress status, `GET /ISAPI/Intelligent/FDLib/manualModeling/progressState`.

5. (Optional) Manage data in the face picture library. You can search, edit, and delete data in the face picture library.
   i. Search for face picture data: `POST /ISAPI/Intelligent/FDLib/FDSearch`;
   ii. Edit face picture data: `PUT /ISAPI/Intelligent/FDLib/FDModify?format=json`, it is for editing data of imported pictures in binary format;
   iii. Edit face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch?format=json&FDID=<FDID>&FPID=<FPID>&faceLibType=<faceLibType>`, it is for editing data of imported pictures in URL format;
   iv. Delete face picture data in the library: `PUT /ISAPI/Intelligent/FDLib/FDSearch/Delete?format=json&FDID=<FDID>&faceLibType=<FDType>`.

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11.5.1.3 Exception Handling

11.5.1.3.1 Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Rem
1	OK	ok	0x00000001			
2	Device Busy	noMemory	0x20000001		Not enough memory.	
4	Invalid Operation	notSupport	0x40000001		Not supported.	
4	Invalid Operation	faceLibDiskFull	0x400020F4		Face picture library storage media is full.	
4	Invalid Operation	faceLibraryIDNotExis	0x400020FD		The face picture library does not exist.	
					The face	

4	Invalid Operation	faceLibInitializing	0x40002136		picture library is initializing. Please wait.	
4	Invalid Operation	notMatchFacelib	0x4000a004		Face picture library is not matched.	
4	Invalid Operation	existSameNameFacelib	0x4000a00a		Face picture library with duplicate name exists.	
4	Invalid Operation	delFacelibFailed	0x4000a00b		Deleting the face picture library failed.	
4	Invalid Operation	modifyFacelibFailed	0x4000a00c		Editing the face picture library failed.	
4	Invalid Operation	createFacelibFailed	0x4000a00d		Creating a face picture library failed.	
4	Invalid Operation	facelibAlreadyUsed	0x4000a010		The face picture library has been used.	
6	Invalid Content	faceLibraryDatabaseError	0x60000032		Database version of face picture library mismatch.	
6	Invalid Content	faceLibraryError	0x60000034		Face picture library error.	
6	Invalid Content	multipleFaceObjectError	0x60000035		Multiple faces are recognized in the pictures to be imported into the face picture library.	
6	Invalid Content	faceLibPicImportingError	0x6000003A		Importing picture into library...	
6	Invalid Content	faceLibraryNoData	0x60000069		No data found in face picture	

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					library.	
6	Invalid Content	NoDataForModelingError	0x6000006B		There is no data to be modeled in the face picture library.	
6	Invalid Content	customFaceLibIDRepeat	0x6000006C		Duplicate ID of custom face picture library.	
6	Invalid Content	customFaceDataDeleteing	0x600000A1		Deleting original customized face picture library data... The upper level cannot duplicate the customized face picture library of this ID.	
6	Invalid Content	analysisEnginesFacelibImporting	0x60001005		Importing face picture library to analysis engine...	
6	Invalid Content	facelibDataProcessing	0x6000100b		Processing face picture library data.	
6	Invalid Content	personInfoExtendValueIsTooLong	0x60001083		Customized tag for face picture library exceeded limit.	
6	Invalid Content	personInfoExtendValueContainsInvalidChar	0x60001084		Customized tag for face picture library contains invalid character(s).	

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11.5.1.4 Message Example

Import face pictures to face picture library in binary format and automatically perform modeling, `POST /ISAPI/Intelligent/FDLib/pictureUpload`:

```
POST http://10.16.96.29/ISAPI/Intelligent/FDLib/pictureUpload HTTP/1.1
If-Modified-Since: 0
Accept: */*
X-Requested-With: XMLHttpRequest
Content-Type: multipart/form-data; boundary=----------------------------7e27010d1dba
Referer: http://10.16.96.29/doc/page/config.asp
Accept-Language: zh-CN
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/5.0 (compatible; MSIE 10.0; Windows NT 6.1; WOW64; Trident/6.0)
Connection: Keep-Alive
Content-Length: 903330
DNT: 1
Host: 10.16.96.29
Pragma: no-cache
Cookie: language=zh; WebSession=48b693ca2b7301e3119d; sdMarkMenu=14%3AfaceSnapLib; sdMarkTab_1_0=0%3AsettingBasic

----------------------------7e27010d1dba
Content-Disposition: form-data; name="PictureUploadData"

<?xml version='1.0' encoding='UTF-8'?
><PictureUploadData><FDID>55869AFEA1E840958120C97054720372</FDID><FaceAppendData><name>20171222103007395_2</name></FaceAppendData></PictureUploadData>
----------------------------7e27010d1dba
Content-Disposition: form-data; name="importImage"; filename="20171222103007395_2.jpeg"
Content-Type: image/jpeg

【Picture data in binary format】
----------------------------7e27010d1dba--
```

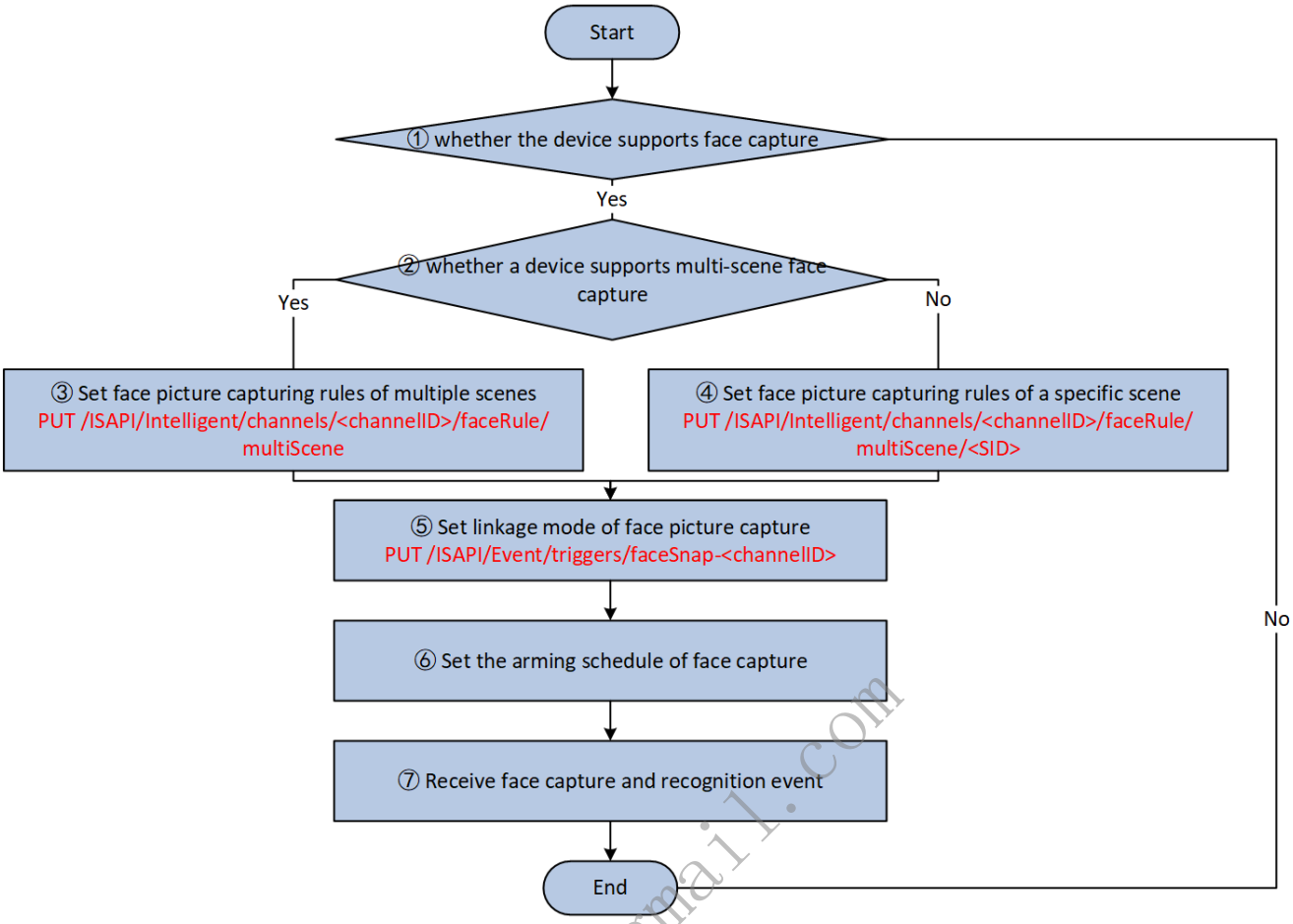
11.6 Privacy Mask on Faces

11.6.1 Introduction to the Function

Privacy mask on faces is to encrypt privacy contents for face capture and media data (see "Video Privacy Mask" API calling flow); it is the process of privacy masking for all faces of cameras and the encrypted images can be restored. First of all, all faces need to be identified, the original media data of faces in the images and masking block data will be replaced, the original media data of faces will be added to the private data of the live view stream and encrypted, then reported to the user synchronously; and the user can decrypt the private data to get the original media data and restore the face image. **Note: The current image restoration only supports private streams, which needs to be restored by integrating the Playback Library.** The following figure shows the effect of face privacy mask by mosaic processing.

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11.6.2 API Calling Flow



1. **Whether the video channel supports face privacy mask.** Get the channel intelligence capabilities: `GET /ISAPI/Intelligent/channels/<channelID>/capabilities`, if the node `<isSupportFacePrivacyMask>` is returned as true, it indicates that the device supports face privacy masking.

2. **Set privacy encryption parameters in media data.** Users apply the encryption parameters including secondary key `secretKey` and the update time of the primary key `secretKeyUpdateFrequency` to devices, etc. Get the encryption capabilities of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>/capabilities?format=json`. Set the encryption parameters of media data privacy `PUT /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.Get the encryption parameters of media data privacy: `GET /ISAPI/Security/MediaObjectEncryption/channel/<channelID>?format=json`.

3. **Set parameters of face privacy mask.** Users can enable face privacy mask for a channel `enabled`. Get the capability of face privacy masking: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json`;
Set the parameters of face privacy mask: `PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`; Get the parameters of face privacy mask: `GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json`;

4. **(Optional) Decrypt media data and restore images**. The device generates the primary key and updates the primary key periodically according to the primary key update time in the media data privacy encryption parameters, then replaces the original media data with the masking block in the specified sensitive area in the live view screen, after which the images in the sensitive area are encrypted by the primary key and added to the private data of the live view stream, and at the same time the primary key is encrypted by the secondary key and added to the private data of the preview stream. The key is also encrypted by the secondary key and added to the private data of the live view stream. The user can parse the private data in the live view stream and decrypt it with the secondary key to get the primary key, then decrypt the private data with the primary key to get the original media data and restore the image.

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Note: At present, image restoration is only supported by private streams. You need to get the private data in the live view stream by integrating Playback Library, and get the primary key by decrypting the secondary key, then decrypt the private data by the primary key to get the original media data; and finally call the Playback Library API to restore the image. You can be refer to Playback library interface integration manual for playback library related APIs.

12 Human Body Recognition

12.1 Search by Human Body Picture

12.1.1 Function Introduction

Search by human body picture is the feature-based image search. It allows users to submit a human body picture as an input, and the system retrieves the most relevant images from a large-scale image records based on the input.

12.1.2 API Calling Flow

Capability Check

- Get device system capabilities: `GET /ISAPI/System/capabilities`. If `isSupportHumanRecognition` returns true, this function is supported.
- Get human body detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`.

Human Detection and Modeling

- Human detection: `POST /ISAPI/Intelligent/humanRecognition?format=json`.
- Human modeling: `POST /ISAPI/Intelligent/humanModel?format=json`.

Synchronous Search

- Get synchronous search capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportSync` returns true, the synchronous search is supported.
- Submit a synchronous search request: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/sync?format=json`.

Asynchronous Search

- Get asynchronous search capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportAsync` returns true, the asynchronous search is supported.
- Submit an asynchronous search task: `POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json`.
- Get the asynchronous search task progress: `GET /ISAPI/Intelligent/humanRecognition/searchByPic/progress?format=json&taskID=<taskID>`.
- Get the search task result: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/result?format=json`.
- Delete the search task: `DELETE /ISAPI/Intelligent/humanRecognition/searchByPic?format=json&taskID=<taskID>`.
- Search for task information.
  - Get human detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If `isSupportTaskInfo` returns true, searching for task information is supported.
  - Get task search capabilities: `GET /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo/capabilities?format=json`.
  - Search for task information: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo?format=json`.

Export Image Search Results

- Get human detection capabilities: `GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json`. If the node `isSupportExport` returns true, exporting image search results is supported.
- Submit the exporting task: `POST /ISAPI/Intelligent/humanRecognition/searchByPic/result/export?format=json`.
- Get the progress of exporting task: `GET`

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`/ISAPI/Intelligent/humanRecognition/searchByPic/result/export/progress?format=json&taskID=<taskID>`.

- Stop the exporting task: `PUT /ISAPI/Intelligent/humanRecognition/searchByPic/result/export/stop?format=json`.

13 VCA

13.1 Face Picture Library Management

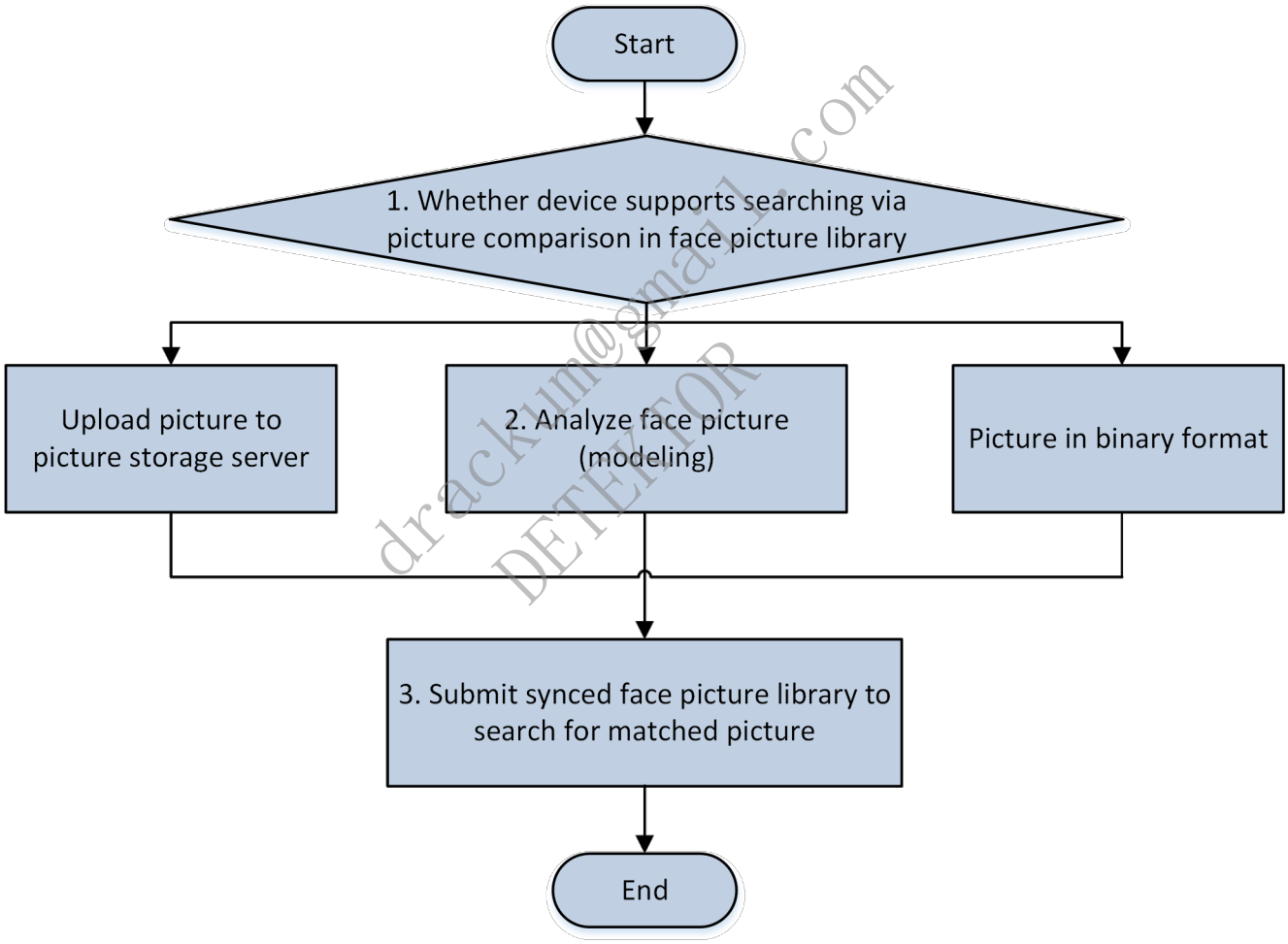
13.1.1. Search via Picture Comparison in Face Capture Library

13.1.1.1 Introduction to the Function

Picture comparison with captured picture library

Scene application: Search for similar person in face picture library, which can be used for identity confirmation.

13.1.1.2 API Calling Flow



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1. **Check whether the device supports searching by picture synchronously.**

   Get the capability of searching via picture comparison: `GET /ISAPI/Intelligent/FDLib/capabilities?format=json`. If the node `isSuportFSSearchByPic` is returned and its value is `true`, it indicates that searching via picture comparison synchronously is supported.

   Note: For some devices supporting searching via picture comparison synchronously, the API for getting the capability is invalid, you can call `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json` directly.

2. **Three methods are supported when searching picture.**

   By URL:

   Based on the picture storage server, picture can be uploaded and you can get the picture URL to download the

picture file via the URL.

By model data:

Get face picture capability: `GET /ISAPI/SDT/Face/pictureAnalysis/capabilities`;

Analyze (model) face pictures: `POSTT /ISAPI/SDT/Face/pictureAnalysis`; Return picture modeling data: `targetModelData`.

By binary picture:

Read binary picture files.

3. **Submit synced face picture library to use picture to search picture**

   Search via picture comparison in face capture library: `POST /ISAPI/Intelligent/FDLib/searchByPic?format=json`;

   Note: Input parameter node `dataType` include three modes. Picture URL corresponds to `faceURL`, model data corresponds to `targetModelData`, binary data mode is in form of form, and its JSON data is followed by binary picture data.

   The request message example of binary picture data is as follows:

```
POST /ISAPI/Intelligent/FDLib/searchByPic?format=json
Host: device_ip:port
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive

--<frontier>
Content-Disposition: form-data; name=""
Content-Type: application/json

[JSON message]
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture data]
--<frontier>--
```

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13.2 Professional Smart Abnormal Event Detection

13.2.1 Introduction to the Function

Professional Smart Abnormal Event Detection includes the following features:

- Line Crossing Detection
- Intrusion Detection
- Region Entrance Detection
- Region Exiting Detection
- Loitering Detection
- People Gathering Detection
- Rapid Movement Detection
- Parking Detection
- Unattended Baggage Detection
- Object Removal Detection
- Running Detection
- Abnormal Number of People Detection
- Person Spacing Change Detection

- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

Professional smart abnormal event detection is designed for specific scenarios: workplace activity detection and pedestrian activity detection.

Workplace activity detection includes but not limited to:

- Using a Mobile Phone
- Number of People Changes
- On/Off Duty
- Person Lingering

Pedestrian activity detection includes but not limited to:

- Running Detection
- People Gathering Detection
- Violent Motion Detection
- Person Falling Detection

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You can configure workplace activity detection and pedestrian activity detection rules and alarm linkage actions to suit your needs.

Abnormal event detection can be categorized into two types: Basic smart abnormal event detection and professional smart abnormal event detection. Their differences are as follows:

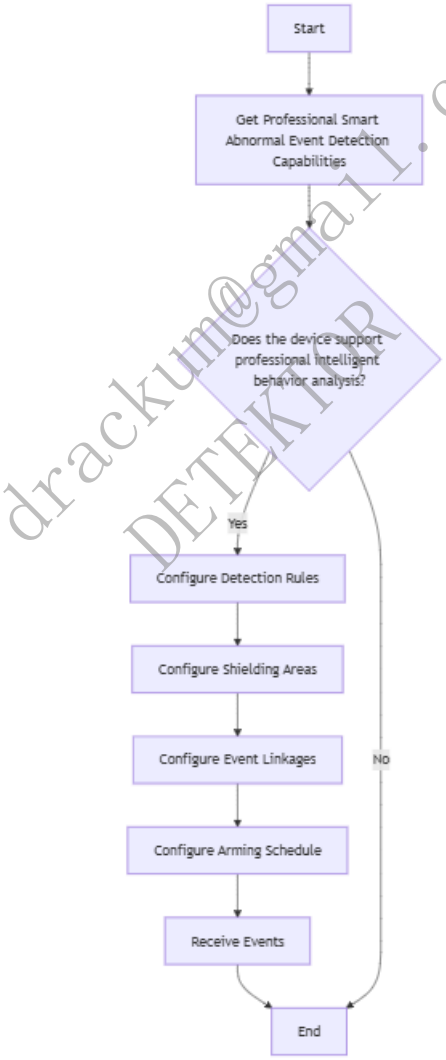
1. **Algorithms**: Traditional algorithms are used for basic smart abnormal event detection, while deep learning algorithms are used for professional smart abnormal event detection.
2. **Detection Effect**: Basic smart abnormal event detection can only analyze targets as people or vehicles. Professional smart abnormal event detection can not only distinguish between targets as people or vehicles but also identify specific behaviors, such as the number of people and human behaviors such as using a mobile phone, violent motion, and running.
3. **Supported Events**: Smart Basic Intelligent Abnormal Behavior Detection supports events such as:
    - Line Crossing Detection
    - Intrusion Detection
    - Region Entrance Detection
    - Region Exiting Detection
    - Loitering Detection
    - People Gathering Detection
    - Rapid Movement Detection
    - Parking Detection
    - Unattended Baggage Detection
    - Object Removal Detection

Professional Smart Abnormal Event Detection supports all the events mentioned above and additional events including:

- Running Detection

- Abnormal Number of People Detection
- Person Spacing Change Detection
- Violent Motion Detection
- On/Off Duty Detection
- Falling Down Detection
- Lingering Detection
- Person Density Detection
- Climbing Detection
- Trend Analysis
- In-Toilet Overtime Detection
- Sudden Change of Sound Intensity Detection
- Student Standing Up Detection
- Getting Up from Bed Detection

13.2.2 API Calling Flow



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Before taking the following steps, ensure that this smart detection has been enabled for devices.

1. **Get Professional Smart Abnormal Event Detection Capabilities**
   - Call `GET /ISAPI/Intelligent/capabilities` to get professional smart abnormal event detetection capabilities and the field `isBehaviorSupport` is returned as `true` in the response message.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/capabilities` to get professional smart abnormal event detetection capabilities for a specified channel and the field `isBehaviorSupport` is returned as `true` in the response message.

Note: `isBehaviorSupportas` `true` indicates support for professional smart abnormal event detetection. The fields of supported events are as follows:

- `isLineDetectionSupport` indicates support for **Line Crossing Detection**.
- `isFieldDetectionSupport` indicates support for **Intrusion Detection**.
- `isRegionEntranceSupport` indicates support for **Region Entrance Detection**.
- `isRegionExitingSupport` indicates support for **Region Exiting Detection**.
- `isLoiteringSupport` indicates support for **Loitering Detection**.
- `isGroupSupport` indicates support for **People Gathering Detection**.
- `isRapidMoveSupport` indicates support for **Rapid Movement Detection**.
- `isParkingSupport` indicates support for **Parking Detection**.
- `isUnattendedBaggageSupport` indicates support for **Unattended Baggage Detection**.
- `isAttendedBaggageSupport` indicates support for **Object Removal Detection**.
- `isRunningSupport` indicates support for **Running Detection**.
- `isPeopleNumChange` indicates support for **Abnormal Number of People Detection**.
- `isSpacingChangeSupport` indicates support for **Person Spacing Change Detection**.
- `isViolentMotionSupport` indicates support for **Violent Movement Detection**.
- `isLeavePositionSupport` indicates support for **On/Off Duty Detection**.
- `isFailDownSupport` indicates support for ** Falling Down Detection**.
- `isRetentionParamSupport` indicates support for **Lingering Detection**.
- `isSupportPersonDensityDetection` indicates support for **Person Density Detection**.
- `isVcaEventAdvReachHeightSupport` indicates support for **Climbing Detection**.
- `isSupportSituationAnalysis` indicates support for **Trend Analysis**.
- `isSupportToiletTarry` indicates support for **In-Toilet Overtime Detection**.
- `isSupportAudioAbnormal` indicates support for **Sudden Change of Sound Intensity Detection**.
- `isSupportStandUp` indicates support for **Student Standing Up Detection**.
- `isSupportGetUp` indicates support for **Getting Up from Bed Detection**.

2. **Configure Detection Rules**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/capabilities` to get professional smart abnormal event detection capabilities.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to get detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to configure detection rules.
   - Call `DELETE /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/rule/<ruleID>` to delete detection rules.
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to get all detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>` to configure all detection rules.

   Note:
   - The `eventType` field in the rules indicates the corresponding event type.
   - The `ruleType` field indicates the event support drawing a detection line or a polygon detection area. Some events (sudden change of sound intensity) do not support drawing a box or line, some events (line crossing, climbing) support drawing a line, and some events (except sudden change of sound intensity, line crossing, and climbing) support for polygon detection areas.

3. **(Optional) Configure Shielding Areas**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield/capabilities` to get professional smart area shielding capabilities.

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- Call `GET /ISAPI/Intelligent/channels/<channelID>/Shield` to get shielding areas for all events.
- Call `PUT /ISAPI/Intelligent/channels/<channelID>/Shield` to configure shielding areas for all events.

4. **(Optional) Configure Event Linkages**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to get event linkages.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/notifications` to configure event linkages.
5. **(Optional) Configure Arming Schedule**
   - Call `GET /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to get the schedule to disable the professional smart abnormal event detection rules.
   - Call `PUT /ISAPI/Intelligent/channels/<channelID>/behaviorRule/<SID>/schedules` to configure the schedule to disable the professional smart abnormal event detection rules.

   Note:
   - Up to eight time periods can be configured per day.
   - Each two time periods cannot overlap.
6. **Receive Events**
   - Line Crossing Detection: `"eventType":"linedetection"`
   - Intrusion Detection: `"eventType":"fielddetection"`
   - Region Entrance Detection: `"eventType":"regionEntrance"`
   - Region Exiting Detection: `"eventType":"regionExiting"`
   - Loitering Detection: `"eventType":"loitering"`
   - People Gathering Detection: `"eventType":"group"`
   - Rapid Movement Detection: `"eventType":"rapidMove"`
   - Parking Detection: `"eventType":"parking"`
   - Unattended Baggage Detection: `"eventType":"unattendedBaggage"`
   - Object Removal Detection: `"eventType":"attendedBaggage"`
   - Running Detection: `"eventType":"running"`
   - Abnormal Number of People Detection: `"eventType":"peopleNumChange"`
   - Person Spacing Change Detection: `"eventType":"spacingChange"`
   - Violent Movement Detection: `"eventType":"violentMotion"`
   - On/Off Duty Detection: `"eventType":"leavePosition"`
   - Falling Down Detection: `"eventType":"failDown"`
   - Lingering Detection: `"eventType":"retention"`
   - Person Density Event: `"eventType":"personDensityDetection"`
   - Climbing Detection: `"eventType":"advReachHeight"`
   - Trend Analysis: `"eventType":"situationAnalysis"`
   - In-Toilet Overtime Detection: `"eventType":"toiletTarry"`
   - Sudden Change of Sound Intensity Detection: `"eventType":"audioAbnormal"`
   - Student Standing Up Detection: `"eventType":"standUp"`
   - Getting Up from Bed Detection: `"eventType":"getUp"`
   - Sitting Still Detection: `"eventType":"sitQuietly"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`
   - Using Mobile Phone Detection: `"eventType":"playCellphone"`
   - Yard Lingering: `"eventType":"yardTarry"`
   - Key Target Standing Up Detection: `"eventType":"keyPersonGetUp"`

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- Sitting Still Detection: `"eventType":"sitQuietly"`
- Yard Lingering: `"eventType":"yardTarry"`
- Using Mobile Phone Detection: `"eventType":"playCellphone"`

Note:

- The content of `targetAttrs` is in JSON format. When the event format is XML, use CDATA (Character Data) to include this data in the event and transmit it to the platform.
- CDATA (Character Data) is used to include large amounts of text data in XML documents. This text may contain special characters (such as `<`, `>`, `&`), which have special meanings in XML and typically need to be escaped. CDATA blocks allow these special characters to be included directly in the document without escaping, such as including JavaScript scripts or CSS styles in XML documents.

13.3 Upgrading the Deep Learning Model for Perimeter Intrusion Detection

13.3.1 Introduction to the Function

You can upgrade the deep learning model for perimeter intrusion detection. You can also upgrade both the local training model for NVRs and the training model for network camera channels.

13.3.2 API Calling Flow

1. Determine if the device supports upgrading the deep learning model for perimeter intrusion detection: Call `GET /ISAPI/Intelligent/capabilities`. If the field `isSupportUpgradeDepthPerimeterModel` is returned as `true` in the response message, this function is supported.
2. Get the deep learning model upgrade capabilities: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/capabilities?format=json`
3. Set the model version: `PUT /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/VersionOperation?format=json`
   - You cannot specify a version to upgrade. Only rolling back to the previous version and restoring to the factory model version are currently supported.
4. Get the model upgrade progress: `GET /ISAPI/Intelligent/depthPerimeter/upgradeTrainModel/process?format=json`
5. Set the model version for a channel: `PUT /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/VersionOperation?format=json`
6. Get the model upgrade progress for a channel: `GET /ISAPI/Intelligent/depthPerimeter/channels/<channelID>/upgradeTrainModel/process?format=json`

14 Audio (General)

14.1 Audio Recording Integration Description

14.1.1 Introduction to the Function

Audio Recording includes functions such as manual audio recording, searching, downloading, and deleting audio.

14.1.2 API Calling Flow

1. **Get the capability:**

   Get device system capability: `GET /ISAPI/System/capabilities`.

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Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`.

2. **Set audio name** (placed after the default name):

   Get parameters of setting a custom audio name of a specific audio channel: `GET /ISAPI/ContentMgmt/recordAudio/<audioID>/customFileName?format=json`.

   Set parameters of setting a custom audio name of a specific audio channel: `PUT /ISAPI/ContentMgmt/recordAudio/<audioID>/customFileName?format=json`.

3. **Record Audio:**

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If the node `isSupportRecordAudio` is returned and its value is `true`, the device supports audio recording.

   Get audio recording parameter capability of a specific audio channel: `GET /ISAPI/ContentMgmt/recordAudio/<audioID>/capabilities?format=json`.

   Get audio recording parameters of a specified channel: `GET /ISAPI/ContentMgmt/recordAudio/<audioID>?format=json`.

   Set audio recording parameters of a specified channel: `PUT /ISAPI/ContentMgmt/recordAudio/<audioID>?format=json`.

   Note: In the parameters, if the value of `enabled` is `true`, the recording is on; if the value of `enabled` is `false`, the recording is off.

4. **Search the audio:**

   Search audio information by month (check whether there is any audio every day in a specific month):

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportAudioDailyDistribution` is returned and its value is `true`, the device supports searching audio information by month.

   Get monthly audio information of a specific audio channel: `POST /ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json`.

   Search audio information (search by date):

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportSearchAudio` is returned and its value is `true`, the device supports searching audio information.

   Get the capability of searching audio: `GET /ISAPI/ContentMgmt/recordAudio/search/capabilities?format=json`.

   Search audio information: `POST /ISAPI/ContentMgmt/recordAudio/search?format=json`.

5. **Download the audio:**

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportDownloadAudio` is returned and its value is `true`, the device supports downloading audio.

   Get the capability of exporting an audio file with its custom name: `GET /ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json`.

   Export an audio file with the custom name (binary): `POST /ISAPI/ContentMgmt/recordAudio/fileNameDownload/binary?format=json`.

6. **Delete the audio:**

   Get device storage capability: `GET /ISAPI/ContentMgmt/capabilities`. If `isSupportDeleteAudio` is returned and its value is `true`, the device supports deleting audio by name.

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Delete specific audio file: `PUT /ISAPI/ContentMgmt/recordAudio/deleteFile?format=json`.

Note: For the file name in the input, see the returned results in `/ISAPI/ContentMgmt/recordAudio/search?format=json`.

7. **Audio playback:**

1. The client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that an authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client resends an RTSP DESCRIBE command with the authentication information.

```
DESCRIBE rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: Accept:application/sdp
```

4. The server returns the SDP information.

```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B4802010000040005001171011040 1F000000FA0000000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends the RTSP SETUP command twice, and the server responds.

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video
RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

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```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio
RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY command, and the server starts pushing playback data stream.

```
PLAY rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

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7. The client sends an RTSP TEARDOWN command to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/audio/1/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

15 Two-Way Audio

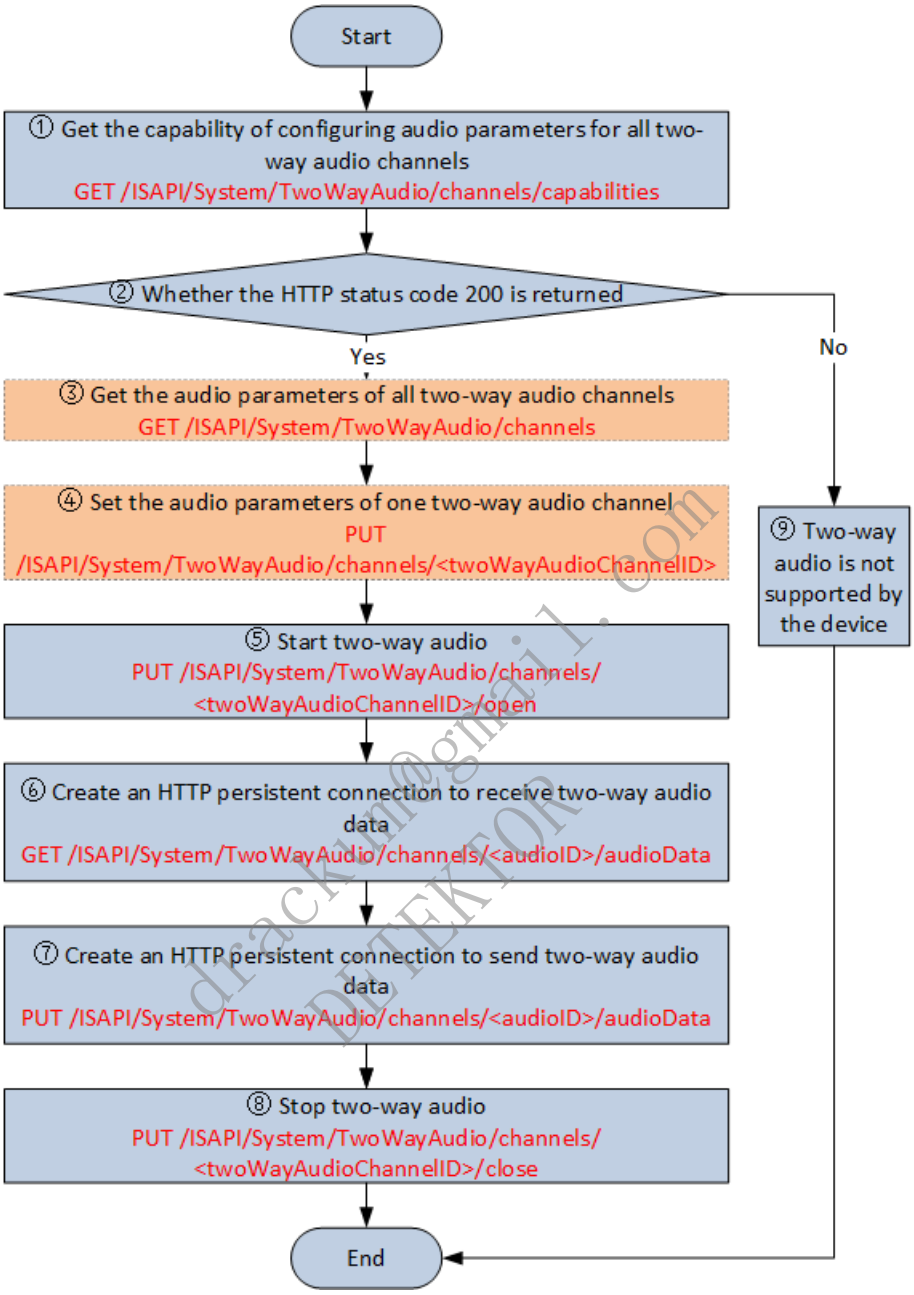
15.1 Two-Way Audio

15.1.1 Introduction to the Function

This function is for the two-way audio between the client and device. When you call the API to implement the two-way audio, it requires the client to collect and encode the local audios and to decode the device audio data.

Note: Before starting two-way audio, check whether the way of collecting device audio data, audio encoding format, volume, etc., is correct or not. The audio encoding format of the client should be consistent with the device audio encoding format.

15.1.2 API Calling Flow



1. Get the capability of configuring audio parameters for all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels/capabilities`. Check whether the device supports two-way audio. If the HTTP status code `200 OK` is returned, then the device supports two-way audio.
2. Get the audio parameters of all two-way audio channels: `GET /ISAPI/System/TwoWayAudio/channels`. Get the No. of two-way audio channel for response and parsing `<id>`, audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, sampling rate `<audioSamplingRate>` and so on, to ensure the consistent of audio format during the two-way audio process.
3. Set the audio parameters of one two-way audio channel: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>`. This step is optional, and the `<audioID>` is the channel No. `<id>` responded and parsed in step 2.
4. Start two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/open`.
5. Create an HTTP persistent connection to receive the two-way audio data: `GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.
6. Create an HTTP persistent connection to send the two-way audio data: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData`.

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7. When the two-way audio ends, the client disables the persistent connections in steps 6 and 7 and sends the message of stopping the two-way audio: `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/close`.

Note:

- Two-way audio over ISAPI supports digest authentication.
- It supports the two-way audio mode or sending only mode (the platform only call `PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData` to send the audio data to devices).
- The `<audioID>` in step 3, 4, 5, 6, and 7 is the two-way audio channel No. which starts from 1. It can be parsed from the `<id>` in the device response message of the step 2.
- The audio type `<audioCompressionType>`, frame rate `<audioBitRate>`, and sampling rate `<audioSamplingRate>` can be parsed from the device response message of the step 2 for audio playing and collection in the step 5 and 6.
- The step 6 and 7 use persistent connections. In the HTTP request headers, you don't need to set `Content-Legth`, but need to set the `Connection: keep-alive` and `Content-Type: application/octet-stream`.
- When the encoding format of audio data in the step 5 and 6 is AAC/MP2L2/MP3, the audio data length is variable, so the frame header should be lengthened by 4 bytes. When the encoding format is G.722.1/G.711alaw/G.711ulaw/G.726/G.729/G.729a/G.729b/PCM/ADPCM/Opus, the audio data length is fixed, there is no need to lengthen the frame header. For example, if the content length of a frame of MP3 audio data is 576 bytes and the header length is 4 bytes, the big end data is 0x00000240, the total length is 580 bytes.
- The damaged audio file in the AC3 format is usually incompatible when using. Currently, the audio algorithm library does not support this format.

Statistics of audio types with fixed lengths:

Audio Type	Fixed Length	Frame Interval
G.722.1	80	40ms
G.711alaw	160/320	20ms/40ms
G.711ulaw	160/320	20ms/40ms
G.726	80	40ms
G.729	10	10ms
G.729a	10	10ms
G.729b	10	10ms
PCM	1920	None
ADPCM	80	20ms
Opus	32/64	20ms/40ms

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15.1.3 Exception Handling

15.1.3.1 Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Re
4	Invalid Operation	twoWayAudioInProgressPleaseWait	0x40002068	Two-way audio in progress...Please wait.	Two-way audio in progress...Please wait.	Th op is all Tw au in pr Pl ag af tw au st

15.1.4 Message Format and Example

15.1.4.1 Receive Two-Way Audio Data

```
GET /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

HTTP/1.1 200 OK
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the device.
```

15.1.4.2 Send Two-Way Audio Data

```
PUT /ISAPI/System/TwoWayAudio/channels/1/audioData HTTP/1.1
Host: 10.17.115.128
Connection: keep-alive
Content-Type: application/octet-stream

//The following are the two-way audio data sent by the client.
```

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16 API Reference

16.1 Device (General)

16.1.1 Device Exception Detection

16.1.1.1 Get the linkage parameters of HDD error

Request URL

GET /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->diskerror
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.1.1.2 Set the linkage parameters of HDD error

Request URL

PUT /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->diskerror
  </id>
  <eventType>
    <!--req, enum, subType:string-->diskerror
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.1.3 Delete the linkage of HDD error

Request URL

DELETE /ISAPI/Event/triggers/diskerror

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.1.1.4 Get the linkage parameters of full HDD

Request URL

GET /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->diskfull
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->diskfull
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.1.1.5 Set the linkage parameters of full HDD

Request URL

PUT /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->diskfull
  </id>
  <eventType>
    <!--req, enum, subType:string-->diskfull
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.1.6 Delete the linkage of full HDD

Request URL

DELETE /ISAPI/Event/triggers/diskfull

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.1.1.7 Set the linkage parameters of illegal login

Request URL

PUT /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->illaccess
  </id>
  <eventType>
    <!--req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.1.8 Get the linkage parameters of illegal login

Request URL

GET /ISAPI/Event/triggers/illaccess

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->illaccess
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->illaccess
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.1.1.9 Get the linkage parameters of IP address conflict event

Request URL

GET /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->ipconflict
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.1.1.10 Delete the linkage of IP address conflict event

Request URL

DELETE /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.1.11 Set the linkage parameters of IP address conflict event

Request URL

PUT /ISAPI/Event/triggers/ipconflict

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->ipconflict
  </id>
  <eventType>
    <!--req, enum, subType:string-->ipconflict
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.1.12 Set the linkage parameters of log exception

Request URL

PUT /ISAPI/Event/triggers/logException

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->logException
  </id>
  <eventType>
    <!--req, enum, subType:string-->logException
  </eventType>
  <eventDescription>
    <!--opt, enum, subType:string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <targetNotificationInterval>
        <!--opt, int, range:[0,1000], unit:s-->1
      </targetNotificationInterval>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.1.13 Get the linkage parameters of log exception

Request URL

GET /ISAPI/Event/triggers/logException

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->logException
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->logException
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <targetNotificationInterval>
        <!--ro, opt, int, range:[0,1000], unit:s-->1
      </targetNotificationInterval>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.1.1.14 Get the linkage parameters of network disconnection

Request URL

GET /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->nicbroken
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--ro, opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--ro, opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.1.1.15 Set the linkage parameters of network disconnection

Request URL

PUT /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->nicbroken
  </id>
  <eventType>
    <!--req, enum, subType:string-->nicbroken
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->test
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.ptzAction.actionName,eq,preset}-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.1.16 Delete the linkage of network disconnection

Request URL

DELETE /ISAPI/Event/triggers/nicbroken

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.1.1.17 HDD error

EventType:diskerror

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"diskerror" (HDD Error)-->diskerror
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:HDD Error-->diskerror
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <diskNo>
    <!--ro, opt, int, HDD number-->1
  </diskNo>
  <HDDList>
    <!--ro, opt, array, HDD list, subType:object-->
    <HDD>
      <!--ro, opt, object, HDD-->
      <id>
        <!--ro, req, int, index, desc:it starts from 1-->1
      </id>
      <diskNumber>
        <!--ro, req, int, HDD No.-->1
      </diskNumber>
    </HDD>
  </HDDList>
</EventNotificationAlert>
```

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16.1.1.18 Invalid access

EventType:illaccess

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event parameters, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the alarm device-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the alarm device-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the alarm device-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, transfer protocol type, range:[0,64], desc:transfer protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI
protocol is transmitted via EZVIZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, No, of the channel that triggers an alarm, desc:No. of the video channel that triggers an alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"illaccess" (invalid access)-->illaccess
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for persistent events: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:illaccess-invalid access-->illaccess
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <username>
    <!--ro, opt, string, locked user name, range:[0,32], desc:the number of attempts by current user name has reached limit. The user name and IPv4 address
may be locked at the same time-->test
  </username>
  <ipV4Address>
    <!--ro, opt, string, locked IPv4 address, desc:the number of attempts by current IPv4 address has reached limit. The user name and Ipv4 address may be
locked at the same time-->test
  </ipV4Address>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:; “unlocked”, “locked”-->unlocked
  </lockStatus>
  <retryLoginTime>
    <!--ro, opt, int, remaining attempts, range:[0,65535], desc:it is valid when the value of lockStatus is “unlocked”-->0
  </retryLoginTime>
  <lockedTime>
    <!--ro, opt, int, locking duration (minutes), range:[0,65535], unit:秒, desc:it is valid when the value of lockStatus is “locked”-->0
  </lockedTime>
  <accessInfoList>
    <!--ro, opt, array, invalid access information list, subType:object-->
    <accessInfo>
      <!--ro, opt, object, invalid access information-->
      <ipV4Address>
        <!--ro, opt, string, Invalid access IP address-->test
      </ipV4Address>
      <dateTime>
        <!--ro, opt, datetime, invalid access time-->2004-05-03T17:30:08+08:00
      </dateTime>
    </accessInfo>
  </accessInfoList>
</EventNotificationAlert>
```

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16.1.1.19 Network disconnected

EventType:nicbroken

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers an alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers an alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, string, communication protocol types, range:[0,64], desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome" (the value
should be "HTTP" when ISAPI protocol is transmitted via EZ protocol, and the value should be "EHome" when ISAPI protocol is transmitted via ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, the uploading times of the same alarm, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, string, event type, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, desc:"nicbroken" (network disconnected)-->nicbroken
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DevicePositionInfo>
    <!--ro, opt, object, device location, desc:the current location of the device when triggering the event, which is mainly for on-board device-->
    <divisionEW>
      <!--ro, opt, enum, eastern/western hemisphere, subType:string, desc:"E" (eastern hemisphere), "W" (western hemisphere)-->E
    </divisionEW>
    <longitude>
      <!--ro, opt, int, longitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </longitude>
    <divisionNS>
      <!--ro, opt, enum, southern/northern hemisphere, subType:string, desc:"S" (southern hemisphere), "N" (northern hemisphere)-->N
    </divisionNS>
    <latitude>
      <!--ro, opt, int, latitude, desc:degree*3600*100 + minute*60*100 + second*100-->100
    </latitude>
    <direction>
      <!--ro, opt, int, vehicle direction, desc:actual direction (degree) (due north as 0 degree, calculated clockwise)-->100
    </direction>
    <speed>
      <!--ro, opt, int, speed, desc:cm/h-->100
    </speed>
    <height>
      <!--ro, opt, int, height, unit:cm, desc:cm-->1
    </height>
  </DevicePositionInfo>
</EventNotificationAlert>
```

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16.1.2 Event Subscription

16.1.2.1 Establish arming connection

Request URL

GET /ISAPI/Event/notification/alertStream?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	string	0 (client-side arming), 1 (real-time arming). Defaults to the client-side arming when delpoyID does not exist.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event massage, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPV4 address of the device that triggers an alarm-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPV6 address of the device that triggers an alarm-->test
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No. of the device that triggers an alarm, range:[0,65535]-->0
  </portNo>
  <protocol>
    <!--ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address, range:[0,17]-->00-16-EA-AE-3C-40
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers an alarm, range:[0,65535]-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm triggering time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, the uploading times of the same alarm, range:[0,65535]-->0
  </activePostCount>
  <eventType>
    <!--ro, req, string, heartbeat type-->test
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (triggered), "inactive" (not triggered, the heartbeat data)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description, range:[0,128]-->test
  </eventDescription>
</EventNotificationAlert>
```

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16.1.2.2 Set alarm/event subscription parameters

Request URL

PUT /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all"(upload all alarms/events), "list"(upload specified alarm/event)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, "all"-upload all alarms/events,"list"-upload specified alarm/event, subType:string, desc:"all"(upload all alarms/events), "list"(upload
specified alarm/event)-->list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
```

```
      <!--opt, object, uploading mode of specified alarm/event-->
      <type>
        <!--req, enum, alarm/event types, subType:string, desc:see details in event types: "ADAS"(advanced driving assistance system), "ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent", "CDsStatus", "DBD"(driving behavior detection) "GPSUpload", "HFPD"(frequently appeared person), "IO"(I/O Alarm), "IOTD", "LES", "LFPD"(low frequency person detection), "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "LowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, subType:object-->
        <childDevID>
          <!--opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string-->1,2,3,4
      </audioChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4...", desc:event linked channel information,and multiple channels can be linked,each channel is separated by comma,e.g.,"1,2,3,4..."-->1,2,3,4
  </channels>
  <pictureURLType>
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    <!--opt, enum, alarm picture format, subType:string, desc:"binary"(binary), "localURL"(device local URL), "cloudStorageURL"(cloud storage URL)-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:5 minute (default). If GUID is not reconnected in the internal, a new GUID will be generated since the device starts a new arm period-->300
    </interval>
    <StatusSub>
      <!--opt, object, status description-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, subType:object-->
    <childDevID>
      <!--opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
```

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  </qosLevel>
  <audioChannels>
    <!--opt, string-->1,2,3,4
  </audioChannels>
</SubscribeEvent>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

16.1.2.3 Get alarm/event subscription parameters

Request URL

GET /ISAPI/Event/notification/subscribeEvent/<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), the (subscription ID) will be returned in message SubscribeEventResponse. When an event/alarm is subscribed (related URL: POST /ISAPI/Event/notification/subscribeEvent), and if subscribeEventID is applied, it can be used to replace the in SubscribeEventResponse.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--ro, opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--ro, opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--ro, req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:alarm/event types, which are obtained from the capability, refer to Alarm/Event Types
(eventType)-->mixedTargetDetection
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
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      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--ro, opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--ro, opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--ro, opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string, desc:"success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--ro, opt, array, subType:object-->
        <childDevID>
          <!--ro, opt, string, range:[1,128]-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--ro, opt, string-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--ro, opt, string-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--ro, opt, bool-->true
      </childDeviceFormat>
    </Event>
  </EventList>
  <channels>
    <!--ro, opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--ro, opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, the <pictureURLType> of the <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, object, status description-->
      <all>
        <!--ro, opt, bool, whether to subscribe to all channels, HDDs, and capability set changes-->false
      </all>
      <channel>
        <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will
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be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm--
>true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--ro, opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--ro, opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <level>
    <!--ro, opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--ro, opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--ro, opt, array, subType:object-->
    <childDevID>
      <!--ro, opt, string, range:[1,128]-->test
    </childDevID>
  </childDevIDList>
  <radarChannels>
    <!--ro, opt, string-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object-->
    <defaultLevel>
      <!--ro, req, enum, subType:int-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, subType:object-->
      <event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, string, range:[0,64]-->test
        </eventType>
        <level>
          <!--ro, req, enum, subType:int-->1
        </level>
        <channels>
          <!--ro, opt, string, range:[0,256]-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--ro, opt, string-->1,2,3,4
  </audioChannels>
  <retransmissionID>
    <!--ro, opt, string, range:[1,64]-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--ro, opt, int, range:[1,168], unit:h-->1
  </retransmissionTimeoutDuration>
  <SubscribeISAPIMessage>
    <!--ro, opt, object-->
    <EventTypeList>
      <!--ro, req, array, subType:object-->
      <eventType>
        <!--ro, req, enum, subType:string-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--ro, opt, bool-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--ro, req, array, subType:object, range:[0,200]-->
      <Event>
        <!--ro, opt, object-->
        <eventType>
          <!--ro, req, enum, subType:string-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--ro, opt, bool-->false
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
```

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</SubscribeEvent>
```

16.1.2.4 Subscribe to events/alarms in arming mode

Request URL

POST /ISAPI/Event/notification/subscribeEvent?deployID=<deployID>

Query Parameter

Parameter Name	Parameter Type	Description
deployID	enum	0 (device local arming), 1 (real-time arming by platform). If the deployID is absent, it defaults to device local arming.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEvent xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <heartbeat>
    <!--opt, int, heartbeat interval time, range:[1,180], unit:s-->1
  </heartbeat>
  <channelMode>
    <!--opt, enum, channel subscription mode, subType:string, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel
list)-->list
  </channelMode>
  <eventMode>
    <!--req, enum, event subscription mode, subType:string, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)--
>list
  </eventMode>
  <EventList>
    <!--opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--opt, object, subscription of a specified alarm/event-->
      <type>
        <!--req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm"
(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access
controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared
person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch"
(video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection),
"TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight",
"alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind
spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus
detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm",
"faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection",
"fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector
detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event),
"highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address
conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection),
"listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event),
"mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected),
"nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange",
"peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection",
"personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone
detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection",
"radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove",
"reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention"
(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm"
(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting
detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system),
"takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature
alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection),
"unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException",
"videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "CertificateCaptureEvent"(person ID
capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event),
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status),
"certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)--
>mixedTargetDetection
      </type>
      <minorAlarm>
        <!--opt, string, minor alarm type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--opt, string, minor exception type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--opt, string, minor operation type, desc:this node is required when type is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--opt, string, minor event type, desc:this node is required when type is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
```

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```
        <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <channels>
        <!--opt, string, channel information linked to event, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <ConferenceIDList>
        <!--opt, array, video conference ID list, subType:object, desc:when this node does not exist, it indicates to subscribe to all conferences-->
        <ConferenceID>
          <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
        </ConferenceID>
      </ConferenceIDList>
      <uploadAllTarget>
        <!--opt, bool, whether to upload all detected targets, desc:the default value is false. In perimeter protection, this node is used for NVR to get all targets detected by the network camera-->false
      </uploadAllTarget>
      <uploadMEFData>
        <!--opt, bool, whether to upload the electronic fence data related to the event, desc:true by default-->true
      </uploadMEFData>
      <minorType>
        <!--opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel check-in.
3. When this node is not applied, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
      <childDevIDList>
        <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
        <childDevID>
          <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
        </childDevID>
      </childDevIDList>
      <radarChannels>
        <!--opt, string, arm by radar channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </radarChannels>
      <audioChannels>
        <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </audioChannels>
      <childDeviceFormat>
        <!--opt, bool, upload by sub device, desc:upload by sub device-->true
      </childDeviceFormat>
      <unitChannels>
        <!--opt, string, arm by optical fiber channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </unitChannels>
    </Event>
  </EventList>
  <channels>
    <!--opt, string, unified arming of all channels, desc:if this node is applied, <channels> in <Event> will be invalid. If the value of <channelMode> is all, this node should not be applied-->1,2,3,4
  </channels>
  <pictureURLType>
    <!--opt, enum, format unified configuration of alarm pictures, subType:string, desc:"binary", "localURL" (device local URL), "cloudStorageURL" (cloud storage URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->cloudStorageURL
  </pictureURLType>
  <ChangedUploadSub>
    <!--opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--opt, object, status information-->
      <all>
        <!--opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--opt, bool, status of subscribed channels, desc:it is not required if the value of <all> is true-->true
      </channel>
      <hd>
        <!--opt, bool, subscribe to HDD status, desc:it is not required if the value of <all> is true-->true
      </hd>
      <capability>
        <!--opt, bool, subscribe to changing status of capability set, desc:it is not required if the value of <all> is true-->true
      </capability>
      <ChanStatus>
        <!--ro, opt, object, channel status subscription of uploaded device status changed event, desc:it is valid when the value of <channel> is true. When this node does not exist, it indicates to upload all channel status changes. For example, the recording status is uploaded frequently, and the platform focuses more on the device online status and arming status, thus the recording status needs to be unsubscribed-->
        <online>
          <!--ro, opt, bool, whether the channel online status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </online>
        <record>
          <!--ro, opt, bool, whether the recording status (the device is recording) is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </record>
        <recordStatus>
          <!--ro, opt, bool, whether the recording status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
        </recordStatus>
        <signal>
          <!--ro, opt, bool, whether the signal status is subscribed, desc:if this node does not exist, it indicates there will be no related alarm-->true
```

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```
        </signal>
        <arming>
          <!--ro, opt, bool, whether the camera arming status (armed by NVR) is subscribed, desc:if this node does not exist, it indicates there will be no
related alarm-->true
        </arming>
      </ChanStatus>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey>
    <!--opt, string, subscription connection interaction key, range:[1,64], desc:N/A-->test
  </identityKey>
  <ConferenceIDList>
    <!--opt, array, video conference ID list, subType:object-->
    <ConferenceID>
      <!--opt, string, ID of subscribed video conference, range:[1,32]-->test
    </ConferenceID>
  </ConferenceIDList>
  <subscribeEventID>
    <!--opt, string, custom subscription ID. The platform should guarantee the uniqueness of ID when applying it. The automatic network replenishment of
events will be enabled by default after the node is applied, range:[1,128], desc:when an event/alarm is subscribed (related URL: POST
/ISAPI/Event/notification/subscribeEvent), if this node is applied, it can be used to replace <id> (subscription ID) returned in message
SubscribeEventResponse-->test
  </subscribeEventID>
  <level>
    <!--opt, enum, arming level, subType:string, desc:"high" (default value), "middle", "low"-->high
  </level>
  <middleLevelPictureEnabeld>
    <!--opt, bool, whether to enable picture uploading for middle-level arming, desc:the default value is true-->true
  </middleLevelPictureEnabeld>
  <childDevIDList>
    <!--opt, array, (sub-device local identifier) device serial No. list, subType:object, desc:arm sub device (sensor)-->
    <childDevID>
      <!--opt, string, (sub-device local identifier) device serial No. list, range:[1,128], desc:arm sub device (sensor)-->test
    </childDevID>
  </childDevIDList>
  <SubscribeISAPIMessage>
    <!--opt, object-->
    <EventTypeList>
      <!--opt, array, event list, subType:object, desc:event list-->
      <eventType>
        <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
      </eventType>
      <uploadPicEnabled>
        <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
      </uploadPicEnabled>
    </EventTypeList>
    <EventList>
      <!--req, array, event list, subType:object, range:[0,200]-->
      <Event>
        <!--opt, object, a single event-->
        <eventType>
          <!--req, enum, event type, subType:string, desc:event type-->AIOP_Video
        </eventType>
        <uploadPicEnabled>
          <!--opt, bool, whether to upload the event picture, desc:whether to upload the event picture-->false
        </uploadPicEnabled>
      </Event>
    </EventList>
  </SubscribeISAPIMessage>
  <radarChannels>
    <!--opt, string, batch arming by radar channel, desc:if this field is applied, radarChannels in Event will not take effect. To arm all radar channels,
this field should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </radarChannels>
  <qosLevel>
    <!--ro, opt, object, subscription events qos level, desc:subscription events qos level-->
    <defaultLevel>
      <!--ro, req, enum, default qos level, subType:int, desc:default qos level-->0
    </defaultLevel>
    <eventList>
      <!--ro, opt, array, custom event qos level list, subType:object-->
      <event>
        <!--ro, opt, object, custom event qos level-->
        <eventType>
          <!--ro, req, string, event type, range:[0,64], desc:its values refer to event types of different domains-->test
        </eventType>
        <level>
          <!--ro, req, enum, qos level of this event type, subType:int, desc:qos level of this event type-->1
        </level>
        <channels>
          <!--ro, opt, string, video resource channel No., range:[0,256], desc:no such node means it applies to all channels; otherwise it applies to the
channels listed in the array, and channels not listed use default qos level-->0,1,2
        </channels>
      </event>
    </eventList>
  </qosLevel>
  <audioChannels>
    <!--opt, string, arm by audio channel, desc:it supports multiple channels, which are separated by commas-->1,2,3,4
  </audioChannels>
  <unitChannels>
    <!--opt, string, batch arming by optical fiber channel, desc:if this field is applied, unitChannels in Event will not take effect. To arm all fiber
channels, this node should not be applied. To arm a part of channels, you can list the channel numbers. Multiple channels are separated by commas-->1,2,3,4
  </unitChannels>
  <retransmissionID>
```

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    <!--opt, string, ANR arming ID, range:[1,64], desc:UUID (an unique ID generated by the client). If this field is applied during establishing the event arming/subscription connection, the ANR of subscribed event will be enabled by default; when the network reconnected, the platform can apply the same UUID to tell the device to re-upload saved event information during network disconnection period-->123456
  </retransmissionID>
  <retransmissionTimeoutDuration>
    <!--opt, int, data transmission timeout for ANR, range:[1,168], unit:h, desc:data transmission timeout for ANR-->1
  </retransmissionTimeoutDuration>
  <subscribeWithoutPic>
    <!--opt, bool-->false
  </subscribeWithoutPic>
</SubscribeEvent>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SubscribeEventResponse xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, subscription result, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, subscription ID, which is returned by the device. The device should guarantee the uniqueness of ID, range:[1,64]-->1
  </id>
  <FailedEventList>
    <!--ro, opt, array, list of events, of which subscription failed, subType:object, desc:when subscription of one or multiple events failed, this node should be returned. The client software can check if all events are subscribed to by checking whether the node <FailedEventList> is returned-->
    <Event>
      <!--ro, opt, object, event, of which the subscription failed. The event/alarm type name can be obtained via the capability set-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:event type-->ADAS
      </type>
      <minorAlarm>
        <!--ro, opt, string, minor alarm type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorAlarm>
      <minorException>
        <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorException>
      <minorOperation>
        <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
      </minorOperation>
      <minorEvent>
        <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
      </minorEvent>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL).-->binary
      </pictureURLType>
      <channels>
        <!--ro, opt, string, channel No., desc:it supports multiple channels, which are separated by commas-->1,2,3,4
      </channels>
      <subStatusCode>
        <!--ro, req, enum, error code, subType:string, desc:refer to the error code dictionary-->activeNumMax
      </subStatusCode>
      <minorType>
        <!--ro, opt, enum, event sub-type, subType:string,
desc:1. It is valid when the value of <type> is "CertificateCaptureEvent" or "ncertificateCompareEvent".
2. There is the requirement to only subscribe to the failure or success of an event, for example, security inspection at the railway station and the hotel check-in.
3. When this node is not returned, it indicates to subscribe to both success and failure of the event. "success", "failed"-->success
      </minorType>
    </Event>
  </FailedEventList>
</SubscribeEventResponse>
```

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16.1.2.5 Get the alarm/event subscription capability

Request URL

GET /ISAPI/Event/notification/subscribeEventCap

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SubscribeEventCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture uploading modes of all events which contain pictures, attr:version{req, string, protocolVersion}-->
  <channelMode opt="all,list">
    <!--ro, opt, enum, channel subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all channels), "list" (subscribe to channels according to channel list)-->list
  </channelMode>
  <eventMode opt="all,list,allAndChild">
    <!--ro, req, enum, event subscription mode, subType:string, attr:opt{req, string}, desc:"all" (subscribe to all alarms/events), "list" (subscribe to specified alarms/events)-->list
  </eventMode>
  <EventList>
    <!--ro, opt, array, event type list for subscription, subType:object, desc:this node is valid when eventMode is "list"-->
    <Event>
      <!--ro, opt, object, subscription of a specified alarm/event-->
      <type>
        <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm" (advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpLoad" (GPS information upload), "HFPD"(frequently appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "LeavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention" (people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm" (angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "target(capture)", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TFS" (traffic enforcement event), "TransactionRecordEvent", "HealthInfoSyncQuery" (health information search event), "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery" (searching the consumption status), "certificateRevocation" (certificate expiry), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
      </type>
      <pictureURLType opt="binary,localURL,cloudStorageURL,multipart" def="cloudStorageURL">
        <!--ro, opt, enum, alarm picture format, subType:string, attr:opt{req, string},def{req, string}, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-->cloudStorageURL
      </pictureURLType>
      <childDeviceFormat opt="true,false">
        <!--ro, opt, bool, upload event by sub device, attr:opt{req, string}-->true
      </childDeviceFormat>
    </Event>
  </EventList>
  <ChangedUploadSub>
    <!--ro, opt, object, subscribe to the uploaded information of device status changed event-->
    <interval>
      <!--ro, opt, int, the lifecycle of arming GUID, range:[1,300], unit:s, desc:the default value is 5 (unit: minute). Within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->300
    </interval>
    <StatusSub>
      <!--ro, opt, bool, status information-->true
      <all>
        <!--ro, opt, bool, whether to subscribe to the changing status of all channels, HDDs, and capability sets-->false
      </all>
      <channel>
        <!--ro, opt, bool, status of subscribed channels, desc:reporting is not required if all is true-->true
      </channel>
      <hd>
        <!--ro, opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
      </hd>
    </StatusSub>
  </ChangedUploadSub>
  <identityKey max="64">
    <!--ro, opt, string, subscription connection interaction key, range:[1,64], attr:max{req, int}, desc:N/A-->test
  </identityKey>
</SubscribeEventCap>
```

16.1.2.6 Cancel subscribing alarm/event

Request URL

PUT /ISAPI/Event/notification/unSubscribeEvent?ID=<subscribeEventID>

Query Parameter

Parameter Name	Parameter Type	Description
subscribeEventID	string	Subscription event ID, which should match the id field in the SubscribeEventResponse of the response message from the POST /ISAPI/Event/notification/subscribeEvent?deployID=.This field is optional. When the protocol is /ISAPI/Event/notification/unSubscribeEvent, it indicates that all subscription links are to be unsubscribed. When the protocol is /ISAPI/Event/notification/unSubscribeEvent?ID=XXX, it indicates that a specific subscription link is to be unsubscribed.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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16.1.3 IoT Channel Management

16.1.3.1 Get the configuration capability of all kinds of event linkages

Request URL

GET /ISAPI/Event/triggersCap/IOT

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IOTTriggersCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
</IOTTriggersCap>
```

16.1.3.2 Get the event search capability

Request URL

GET /ISAPI/SDT/Management/EventSearch/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "startTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, start time, desc:start time,ISO8601_time*/
    "endTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, end time, desc:end time,ISO8601_time*/
    "resultMaxNum":  100,
    /*ro, req, int, supported maximum number of searching*/
    "channelIDLen": {
    /*ro, opt, object, channel ID length*/
        "min":  0,
        /*ro, req, int, minimum value, step:1*/
        "max":  32
        /*ro, req, int, maximum value, step:1*/
    },
    "eventType": {
    /*ro, opt, object, event type, desc:studentStoodUp-student stands up, accessController-access controller, videoIntercom-video intercom, Luminite-
Luminite security control panel, GJD-GJD security control panel, cameraDetector-detector, securityControlPanel-security control panel. Multiple selections
are separated by comma*/
        "@opt":  "studentStoodUp, accessController,videoIntercom, OPTEX,Luminite,GJD, cameraDetector, securityControlPanel",
        /*ro, req, string, range*/
        "#text":  "studentStoodUp"
        /*ro, opt, string*/
    },
    "type": {
    /*ro, opt, object, channel type, desc:video-video channel, IOT-IOT channel; if this node does not exist, it indicates video channel*/
        "@opt":  "video,IOT",
        /*ro, req, string*/
        "#text":  "video"
        /*ro, opt, string*/
    },
    "AccessController": {
    /*ro, opt, object, access control event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        },
        "nameLen": {
        /*ro, opt, object, name length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        },
        "cardNoLen": {
        /*ro, opt, object, card No. length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        }
    },
    "VideoIntercom": {
    /*ro, opt, object, Video Intercom Event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        },
        "nameLen": {
        /*ro, opt, object, name length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
            /*ro, req, int, maximum value, step:1*/
        },
        "cardNoLen": {
        /*ro, opt, object, card No. length*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  32
```

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            /*ro, req, int, maximum value, step:1*/
        }
    },
    "OPTEX": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "Luminite": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "GJD": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "CameraDetector": {
    /*ro, opt, object*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test",
            /*ro, req, string, range*/
            "#text":  "test"
            /*ro, opt, string*/
        }
    },
    "SecurityControlPanel": {
    /*ro, opt, object, security control panel event*/
        "eventType": {
        /*ro, opt, object, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
            "@opt":  "test"
            /*ro, req, string, range*/
        },
        "eventMainType": {
        /*ro, opt, object*/
            "@opt": ["alarm", "fault", "operation", "others"]
            /*ro, req, array, subType:string*/
        },
        "zoneNo": {
        /*ro, req, object, zone No.*/
            "@size":  1,
            /*ro, req, int, step:1*/
            "@min":  1,
            /*ro, req, int, minimum value, step:1*/
            "@max":  16,
            /*ro, req, int, maximum value, step:1*/
            "#text":  1
            /*ro, opt, int*/
        },
        "ipAddress": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,128], step:1*/
            "@max":  128
            /*ro, req, int, maximum value, range:[0,128], step:1*/
        },
        "devIndex": {
        /*ro, opt, object, maximum value*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,256], step:1*/
            "@max":  256
            /*ro, req, int, maximum value, range:[0,256], step:1*/
        },
        "code": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, minimum value, range:[0,5000], step:1*/
```

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            /*ro, req, int, minimum value, range:[0,5000], step:1*/
            "@max":  5000
            /*ro, req, int, maximum value, range:[0,5000], step:1*/
        },
        "devName": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,48], step:1*/
            "@max":  48
            /*ro, req, int, range:[1,48], step:1*/
        }
    },
    "isSupportEventExport":  true
    /*ro, opt, bool*/
}
```

16.1.3.3 Search for event information

Request URL

POST /ISAPI/SDT/Management/EventSearch?format=json

Query Parameter

None

Request Message

```
{
    "searchID":  "test",
    /*req, string, search ID, desc:this node is used to check whether the search is the same one. Due to device performance limitation, you can search
asynchronously by applying the search conditions multiple times and getting the searching progress (the searchID should be the same)*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100,
    /*req, int, the maximum number of search results this time*/
    "startTime":  "1970-01-01T08:00:00+08:00",
    /*req, datetime, start time, desc:start time,ISO8601_time*/
    "endTime":  "1970-01-01T09:00:00+08:00",
    /*req, datetime, end time, desc:end time,ISO8601_time*/
    "choiceChannel": [
    /*opt, array, camera information, subType:object*/
        {
            "channelID":  "26c92893d2a94bc1840451b78444e0f9"
            /*opt, string, camera No.*/
        }
    ],
    "eventType":  "accessController",
    /*req, enum, event type, subType:string, desc:"studentStoodUp" (student stands up), "accessController" (access controller), "videoIntercom" (video
intercom), "OPTEX" (OPTEX security control panel), "Luminite" (Luminite security control panel), "GJD" (GJD security control panel), "cameraDetector"
(detector), "securityControlPanel" (security control panel), multiple selections are separated with commas*/
    "type":  "video",
    /*opt, enum, channel type, subType:string, desc:"video" (video channel), "IOT" (IOT channel), if this node does not exist, it indicates video channel*/
    "AccessController": {
    /*opt, object*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "name":  "test",
        /*opt, string, name*/
        "cardNo":  "test"
        /*opt, string, card No.*/
    },
    "VideoIntercom": {
    /*opt, object*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "name":  "test",
        /*opt, string, name*/
        "cardNo":  "test"
        /*opt, string, card No.*/
    },
    "OPTEX": {
    /*opt, object, OPTEX security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "Luminite": {
    /*opt, object, Luminite security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    }
```

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    },
    "GJD": {
    /*opt, object, GJD security control panel*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "CameraDetector": {
    /*opt, object*/
        "eventType":  "test"
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
    },
    "SecurityControlPanel": {
    /*opt, object, security control panel*/
        "eventType":  "test",
        /*opt, string, event type, desc:event type: studentStoodUp-student stands up,accessController-access controller,videoIntercom-video
intercom,Luminite-Luminite security control panel,GJD-GJD security control panel,cameraDetector-detector,securityControlPanel-security control
panel,multiple selections are separated by comma*/
        "eventMainType":  "alarm",
        /*opt, enum, subType:string*/
        "zoneNo": [1, 2],
        /*opt, array, zone No., subType:int*/
        "ipAddress":  "10.1.200.1",
        /*opt, string, device's IPv4 address, range:[0,128]*/
        "devIndex":  "test",
        /*opt, string, device No., range:[0,256]*/
        "code":  0,
        /*opt, int, range:[0,5000]*/
        "devName":  "test"
        /*opt, string, range:[1,48]*/
    }
}
```

Response Message

```
{
    "responseStatusStrg":  "MORE",
    /*ro, req, enum, searching status, subType:string, desc:"OK" (searching ended), "MORE" (search again for more results), "NO MATCH" (no matched data
found)*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
    "eventInfo": [
    /*ro, opt, array, event information, subType:object*/
        {
            "captureTime":  "1970-01-01T08:00:00+08:00",
            /*ro, req, datetime, capture time*/
            "channelID":  "test",
            /*ro, opt, string, camera No.*/
            "channelName":  "test",
            /*ro, opt, string, camera name*/
            "picUrl":  "test",
            /*ro, opt, string, picture URL*/
            "subPicUrl":  "test",
            /*ro, opt, string, thumbnail URL*/
            "eventType":  "accessController",
            /*ro, req, enum, event type, subType:string, desc:"studentStoodUp" (student stands up), "accessController" (access controller), "videoIntercom"
(video intercom), "OPTEX" (OPTEX security control panel), "Luminite" (Luminite security control panel), "GJD" (GJD security control panel), "cameraDetector"
(detector), "securityControlPanel" (security control panel), multiple selections are separated with commas*/
            "studentStoodUp": {
            /*ro, opt, object, student stand up event information*/
                "studentsStoodUp_number":  4
                /*ro, opt, int, the number of students who are standing up*/
            },
            "AccessController": {
            /*ro, req, object, event type*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "name":  "test",
                /*ro, opt, string, name*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  "invalid",
                /*ro, opt, enum, card type, subType:string, desc:"invalid" (invalid card), "ordinary" (normal card), "disabled" (card for the disabled),
"blocklist" (card in blocklist), "patrol" (patrol card), "super" (super card), "guest" (visitor card), "remove" (removed card)*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "VideoIntercom": {
            /*ro, opt, object, video intercom*/
```

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```
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "name":  "test",
                /*ro, opt, string, name*/
                "cardNo":  "test",
                /*ro, opt, string, card No.*/
                "cardType":  "invalid",
                /*ro, opt, enum, card type, subType:string, desc:"invalid" (invalid card), "ordinary" (normal card), "disabled" (card for the disabled),
"blocklist" (card in blocklist), "patrol" (patrol card), "super" (super card), "guest" (visitor card), "remove" (removed card)*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "OPTEX": {
            /*ro, opt, object, OPTEX security control panel*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "Luminite": {
            /*ro, opt, object, Luminite security control panel*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "GJD": {
            /*ro, opt, object, GJD security control panel*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "CameraDetector": {
            /*ro, opt, object*/
                "eventType":  "test",
                /*ro, req, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ]
            },
            "SecurityControlPanel": {
            /*ro, opt, object*/
                "eventType":  "test",
                /*ro, opt, string, event type, desc:event type: studentStoodU-student stands up,videoIntercom-video intercom*/
                "eventMainType":  "alarm",
                /*ro, opt, enum, subType:string*/
                "videoChannelInfo": [
                /*ro, opt, array, video channel information, subType:object*/
                    {
                        "videoChannel":  1,
                        /*ro, req, int, video channel No.*/
                        "picUrl":  "test"
                        /*ro, opt, string, picture URL*/
                    }
                ],
                "zoneNo":  1,
                /*ro, opt, int, zone No., desc:zone No.,it is valid when eventType values "zone"*/
                "code":  1103,
                /*ro, opt, int, CID event No., desc:CID event No.,it is valid when eventType values "zone"*/
```

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                /*ro, opt, int, CID event No., desc:CID event No.,it is valid when eventType values "zone"*/
                "zoneName":  "test",
                /*ro, opt, string, zone name*/
                "ipAddress":  "test",
                /*ro, opt, string, device's IPv4 address, range:[0,128]*/
                "devIndex":  "test",
                /*ro, opt, string, device No., range:[0,256]*/
                "subSysNo":  1,
                /*ro, opt, int*/
                "recordID":  "test",
                /*ro, opt, string*/
                "deviceName":  "test"
                /*ro, opt, string, range:[1,48]*/
            }
        }
    ]
}
```

16.1.3.4 Export the template of device list

Request URL

GET /ISAPI/System/IOT/channelConfig/template?format=json

Query Parameter

None

Request Message

None

Response Message

Binary Data

16.1.3.5 Export the list of added IoT devices

Request URL

GET /ISAPI/System/IOT/channelConfig?format=json

Query Parameter

None

Request Message

None

Response Message

Binary Data

16.1.3.6 Import the list of added IoT devices

Request URL

PUT /ISAPI/System/IOT/channelConfig?format=json

Query Parameter

None

Request Message

Binary Data

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string, */
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string, */
}
```

16.1.3.7 Get information of all IoT device channels

Request URL

GET /ISAPI/System/IOT/channels?format=json&deviceInductiveType=<deviceInductiveType>

Query Parameter

Parameter Name	Parameter Type	Description
deviceInductiveType	string	IoT device types: "inductiveType1" (access control and video intercom devices), "inductiveType2" (alarm devices), "inductiveType3" (logistics code scanning devices). The specific device type value is returned in GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json.

Request Message

None

Response Message

```
{
    "IOTChannelList": [
    /*ro, opt, array, information of the added IoT device, subType:object*/
        {
            "channel":  1,
            /*ro, opt, int, channel No.*/
            "addressingFormatType":  "hostname",
            /*ro, req, enum, address type, subType:string, desc:"hostname" (host name), "ipaddress" (IP address)*/
            "hostName":  "www.baidu.com",
            /*ro, opt, string, host name, range:[1,64], desc:it is required when addressingFormatType is "hostname"*/
            "ipAddress":  "10.11.12.13",
            /*ro, opt, string, devices IPv4 address, range:[1,32], desc:it is required when addressingFormatType is "ipaddress"*/
            "ipv6Address":  "910A:2222:5498:8475:1111:3900:2020",
            /*ro, opt, string, devices IPv6 address, range:[1,128], desc:it is optional when addressingFormatType is "hostname"*/
            "protocol":  "tcp",
            /*ro, opt, enum, protocol type, subType:string, desc:"tcp" (TCP protocol), "udp" (UDP protocol), "muticast" (multicast protocol)*/
            "portNo":  80,
            /*ro, opt, int, device port No.*/
            "userName":  "admin",
            /*ro, opt, string, user name, range:[1,32]*/
            "deviceChannel":  1,
            /*ro, req, int, IoT channel No. of the added device, range:[1,16]*/
            "videoChannels": [1, 2],
            /*ro, opt, array, video channel No. of the added device, subType:int, desc:it is for device that contains both video channels and IoT channels.
If this node is configured, when adding a new channel No. to the current device, the device will automatically add it to the video channel. If this node is
not configured, the channel will be linked to the IoT channel. This function is only available for devices added via device access protocol*/
            "addAll":  true,
            /*ro, opt, bool, whether to add all channels (including video and IoT channels) of the added device*/
            "inductiveDeviceType":  "inductiveType1",
            /*ro, req, enum, type of the added IoT device, subType:string, desc:"inductiveType1" (access control and video intercom devices),
"inductiveType2" (alarm devices), "inductiveType3" (logistics code scanning devices. Inductive type 1 includes: "accessController", "videoInterCom".
Inductive type 2 includes: "OPTEX" (OPTEX security control panel protocol), "Luminite" (Luminite security control panel), "GJD" (GJD security control
panel), "securityControlPanel" (security control panel), "Adam" (Adam module protocol), "Arteco" (Arteco module protocol), "FM2466" (FM2466 protocol). If
this node is not configured, when getting channel information via an inductive protocol, the information of the added device channels will all be returned*/
            "zoneNum":  1,
            /*ro, opt, int, the number of added zones, desc:it is valid when using SIA-CID protocol as the access protocol*/
            "subnetMask":  "test",
            /*ro, opt, string, IPv4 subnet mask, range:[0,32]*/
            "DefaultGateway": {
            /*ro, opt, object, default gateway*/
                "ipAddress":  "10.11.12.13",
                /*ro, opt, string, IPv4 address, range:[0,32]*/
                "ipv6Address":  "910A:2222:5498:8475:1111:3900:2020"
                /*ro, opt, string, IPv6 address, range:[0,128]*/
```

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                /*ro, opt, string, IPv6 address, range:[0,128]*/
            },
            "regNumber":  "test",
            /*ro, opt, string, registration number, range:[0,32]*/
            "periodNumber":  1,
            /*ro, opt, int, area No., range:[0,100]*/
            "buildingNumber":  1,
            /*ro, opt, int, building No., range:[0,100]*/
            "unitNumber":  1,
            /*ro, opt, int, unit No., range:[0,100]*/
            "floorNumber":  0,
            /*ro, opt, int, floor No.*/
            "roomNumber":  0,
            /*ro, opt, int, room No.*/
            "unitType":  "indoor",
            /*ro, opt, enum, type, subType:string, desc:"indoor" (indoor station), "villa" (villa door station), "confirm" (doorphone), "outdoor" (door
station), "fence" (outer door station), "doorbell" (doorbell), "manage" (main station), "acs" (access control devices), "wardStation", "bedheadExtension",
"bedsideExtension", "terminal", "netAudio" (network sound box), "analogIndoor" (analog indoor station), "networkFireAlarmMatrix", "amplifier" (network
amplifier), "pagingMicrophone" (SIP phone)*/
            "industryType":  "builidings",
            /*ro, opt, enum, industry (scene) type, subType:string, desc:"builidings" (building), "prison", "medicalTreatment" (medical), "broadcasting",
"general"*/
            "serialNumber":  "test",
            /*ro, opt, string, device serial No., range:[0,48]*/
            "enableTiming":  true,
            /*ro, opt, bool, whether to perform time sync on the device*/
            "priority":  1,
            /*ro, opt, int, calling priority, range:[1,15], desc:this node defines the calling priority when there are multiple main stations*/
            "transportType":  "multicast",
            /*ro, opt, enum, broadcasting method, subType:string, desc:"unicast", "multicast"*/
            "audioVolume":  5,
            /*ro, opt, int, volume, range:[0,15]*/
            "callForwardingEnabled":  true,
            /*ro, opt, bool, whether to enable call transfer*/
            "callForwardingType":  "unconditional",
            /*ro, opt, enum, call transfer type, subType:string, dep:and,{$.IOTChannel.callForwardingEnabled,eq,true}, desc:"unconditional" (call transfer
under any condition), "busy" (call transfer when busy), "noReply" (call transfer when there is no reply), "noReachable" (call transfer when it is
unreachable), "calling" (call transfer when calling)*/
            "callTerminalNumber":  "test",
            /*ro, opt, string, long number of the target main station for call forwarding, range:[0,128], dep:and,
{$.IOTChannel.callForwardingEnabled,eq,true}*/
            "visitingTime":  360,
            /*ro, opt, int, visiting time, range:[0,36000], step:1, unit:s*/
            "deviceVersion":  "test",
            /*ro, opt, string, device version, range:[0,64]*/
            "customCommunicationName":  "test",
            /*ro, opt, string, custom contact name, range:[0,64], desc:this node is valid only when the value of the node "inductiveDeviceType" is
"inductiveType1". When this node is not configured, the long number of the indoor station that sends the call will be displayed on the door station and the
main station; when this node is configured, the custom contact name will be displayed*/
            "dialUpAddress":  1,
            /*ro, opt, int, dial address, range:[1,1024], desc:an analog indoor station does not have an IP address or a long number, so the dial address
becomes its ID, which can be used by the analog indoor station to send a call*/
            "alarmEnabled":  false,
            /*ro, opt, bool, whether to enable the arming function of devices*/
            "mappingRoomNumber":  "12A",
            /*ro, opt, string, mapping room No., range:[1,5], desc:the mapping room No. is unique and can be called without exposing the actual room No. for
the consideration of privacy*/
            "deviceID":  "test01",
            /*ro, opt, string, device ID, range:[0,64], desc:required for ISUP protocol access*/
            "groupID":  1,
            /*ro, opt, int, device group ID, range:[1,16], desc:device group ID that the channel belongs to. This node is returned when a device is added to
a device group. See details in /ISAPI/System/DeviceGroupMgr/AddDeviceGroup?format=json*/
            "relateTrailID":  1
            /*ro, opt, int, interrogation channel to link to the IoT device, range:[1,2], desc:No. of the interrogation channel to link to the IoT device
(such as environment monitoring screen) accessed to the interrogation device. The channel No. can be obtained from /ISAPI/Trial/channelname*/
        }
    ]
}
```


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16.1.3.8 Get the IoT devices which can be added

Request URL

GET /ISAPI/System/IOT/search?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "IOTSourceList": [
    /*ro, opt, array, source informaiton, subType:object*/
        {
            "addressingFormatType":  "hostname",
            /*ro, req, enum, address type, subType:string, desc:"hostname" (host name), "ipaddress" (ip address)*/
            "hostName":  "www.baidu.com",
            /*ro, opt, string, host name, range:[1,64], desc:host name*/
            "ipAddress":  "10.11.12.13",
            /*ro, opt, string, devices IPv4 address, range:[1,32], desc:device's IPv4 address*/
            "ipv6Address":  "910A:2222:5498:8475:1111:3900:2020",
            /*ro, opt, string, devices IPv6 address, range:[1,128], desc:device's IPv6 address*/
            "portNo":  80,
            /*ro, opt, int, device port No.*/
            "userName":  "admin",
            /*ro, opt, string, user name, range:[1,32]*/
            "password":  "12345",
            /*ro, opt, string, password, range:[1,16]*/
            "activeStatus":  true,
            /*ro, req, bool, active or not*/
            "deviceType":  "accessController",
            /*ro, opt, enum, device type, subType:string, desc:device type*/
            "macAddress":  "00:00:12:8F:9f:dd:ee:ww",
            /*ro, opt, string, device MAC address, range:[1,48]*/
            "serialNumber":  "1111111",
            /*ro, opt, string, device serial No., range:[1,48]*/
            "shortSerialNumber":  "test",
            /*ro, opt, string, , range:[1,64]*/
            "model":  "test",
            /*ro, opt, string, , range:[1,64]*/
            "firmwareVersion":  "V5.3.2build 150521",
            /*ro, opt, string, firmware version (including compiled date), range:[1,128]*/
            "networkCardName":  "test",
            /*ro, opt, string, , range:[1,128]*/
            "supportEzvizUserToken":  true
            /*ro, opt, bool, */
        }
    ]
}
```

16.1.3.9 Get access management capability of IoT device channel

Request URL

GET /ISAPI/System/IOT/sourceSupport/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "IOTSourceSupport": {
    /*ro, req, object, supported source information*/
        "sourceNum":  10,
        /*ro, req, int, the number of protocols*/
        "IOTSourceDescriptions": [
        /*ro, opt, array, protocol information, subType:object*/
            {
                "IOTProtocolType":  "CodeMaster",
                /*ro, req, enum, IOT protocol type, subType:string, desc:"CodeMaster" (CodeMaster protocol), "GJD" (GJD security control panel protocol),
"HIKVISION" (device access protocol), "Luminate" (Luminite security control panel protocol), "OPTEX"(OPTEX security control panel protocol), "SIA-CID" (SIA-
CID protocol), "logisticsScanningGun" (barcode scanner protocol), "Adam" (Adam module protocol), "Arteco" (Arteco module protocol), "ISUP" (ISUP protocol),
"Ezviz" (EZ protocol), "FM2466" (FM2466 protocol)*/
                "addressingFormatType": {
                /*ro, opt, object, address type*/
                    "@opt":  "ipaddress,hostname",
                    /*ro, req, string, range*/
                    "#text":  "ipaddress"
                    /*ro, opt, string, sample value*/
                },
                "protocol": {
                /*ro, opt, object, protocol type*/
                    "@opt":  "tcp,udp,muticast",
                    /*ro, req, string, range*/
                    "#text":  "tcp"
                    /*ro, opt, string, sample value*/
                },
                "portNo": {
                /*ro, opt, object, device port No.*/
                    "@min":  0,
                    /*ro, req, int, minimum value*/
                    "@max":  65535,
                    /*ro, req, int, maximum value*/
                    "#text":  80
                    /*ro, opt, int, sample value*/
                },
                "userName": {
                /*ro, opt, object, user name*/
                    "@min":  0,
                    /*ro, req, int, minimum value*/
                    "@max":  32,
                    /*ro, req, int, maximum value*/
                    "#text":  "admin"
                    /*ro, opt, string, sample value*/
                },
                "password": {
                /*ro, opt, object, password*/
                    "@min":  0,
                    /*ro, req, int, minimum value*/
                    "@max":  16,
                    /*ro, req, int, maximum value*/
                    "#text":  "12345"
                    /*ro, opt, string, sample value*/
                },
                "enableTiming": {
                /*ro, opt, object, whether time sync on the device is supported*/
                    "@opt": [true, false],
                    /*ro, opt, array, optional value, subType:bool*/
                },
            }
        ],
        "DeviceInductiveType": {
        /*ro, opt, object, device inductive type, desc:Multiple types of devices might need to be put into one device group for management, so inductive
type is introduced to sort out the devices.*/
            "inductiveType1": ["accessController", "videoIntercom"],
            /*ro, opt, array, inductive type 1, subType:string, desc:"accessController" (access controller), "videoIntercom" (video intercom)*/
            "inductiveType2": ["OPTEX", "Luminite", "GJD", "securityControlPanel", "Adam", "Arteco", "FM2466"],
            /*ro, opt, array, inductive type 2, subType:string, desc:"OPTEX"(OPTEX security control panel protocol), "Luminite" (Luminite security control
panel), "GJD" (GJD security control panel), "securityControlPanel" (security control panel), "Adam" (Adam module protocol), "Arteco" (Arteco module
protocol), "FM2466" (FM2466 protocol)*/
        },
        "unitType": {
        /*ro, opt, object, device type value range*/
        "IOInputNum": {
        /*ro, opt, object, number of alarm input interfaces that can be used, dep:and,{$.IOTChannel.IOTProtocolType,eq,FM2466}*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  6
            /*ro, opt, int, maximum value*/
        },
        "IOOutputNum": {
        /*ro, opt, object, number of alarm output interfaces that can be used, dep:and,{$.IOTChannel.IOTProtocolType,eq,FM2466}*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  6
            /*ro, opt, int, maximum value*/
        },
    }
}
```

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16.1.4 Online Upgrading Management

16.1.4.1 Get the device online upgrade capability

Request URL

GET /ISAPI/System/onlineUpgrade/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade, attr:version{req, string, protocolVersion}-->
  <firmwareNum max="10">
    <!--ro, req, int, number of supported online upgrade packages, attr:max{req, int}-->1
  </firmwareNum>
  <firmwareCode max="256">
    <!--ro, req, int, maximum length of firmware ID, attr:max{req, int}-->1
  </firmwareCode>
  <firmwareVersion max="64">
    <!--ro, req, int, maximum length of version, attr:max{req, int}-->1
  </firmwareVersion>
  <firmwareCodeNumOnce max="10">
    <!--ro, req, int, maximum number of obtained firmware IDs, attr:max{req, int}-->1
  </firmwareCodeNumOnce>
  <upgradePercent min="0" max="100">
    <!--ro, req, int, upgrade progress capability in percentage, attr:min{req, int},max{req, int}-->1
  </upgradePercent>
  <Version>
    <!--ro, opt, object, upgrade package version information-->
    <newVersion max="64">
      <!--ro, req, string, version information of the new upgrade package, attr:max{req, int}-->test
    </newVersion>
    <changeLog max="64">
      <!--ro, req, string, log of the new upgrade package, attr:max{req, int}-->test
    </changeLog>
  </Version>
  <rebootAfterUpgrade>
    <!--ro, opt, enum, upgrade options after device reboot, subType:string, desc:"auto" (auto upgrade), "manual" (manual upgrade)-->auto
  </rebootAfterUpgrade>
  <DeviceParameter>
    <!--ro, opt, object, online upgrade parameters-->
    <isSupportAutoDownloadPackage>
      <!--ro, opt, bool, whether it supports automatic download of upgrade package-->true
    </isSupportAutoDownloadPackage>
    <notSupportAutoUpgrade>
      <!--ro, opt, bool, whether it no longer supports automatic download of upgrade package and automatic upgrade, desc:corresponding URL: PUT
/ISAPI/System/onlineUpgrade/upgrade-->true
    </notSupportAutoUpgrade>
    <isSupportTimingUpgrade>
      <!--ro, opt, bool, whether it supports scheduled upgrade-->true
    </isSupportTimingUpgrade>
  </DeviceParameter>
  <ManualDownloadPackage>
    <!--ro, opt, object, manual download of upgrade package-->
    <supportOperation opt="start,cancel,pause,resume">
      <!--ro, opt, enum, supported operations, subType:string, attr:opt{req, string}, desc:"start", "cancel", "pause", "resume"-->start
    </supportOperation>
  </ManualDownloadPackage>
  <isSupportIgnoreCurrentVersion>
    <!--ro, opt, bool, whether it supports ignoring the current version-->true
  </isSupportIgnoreCurrentVersion>
  <UpgradePackageTask>
    <!--ro, opt, object, whether the device supports applying upgrade tasks, desc:if this node is returned, it indicates that the following URLs are also
supported: /ISAPI/System/onlineUpgrade/task?format=json, /ISAPI/System/onlineUpgrade/status/<ID>-->
    <isSupportUpgradeModules>
      <!--ro, opt, bool, whether the device supports upgrading sub modules-->true
    </isSupportUpgradeModules>
    <isSupportUpgradeChannels>
      <!--ro, opt, bool, whether the device supports upgrading sub channels-->true
    </isSupportUpgradeChannels>
    <isSupportDownloadStrategy>
      <!--ro, opt, bool, whether the device supports strategy parameters of downloading upgrade package, desc:corresponding filed is "downloadStrategy" of
URL (/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDownloadStrategy>
    <isSupportDelayedUpgradeByDateTime>
      <!--ro, opt, bool, whether the device supports delaying the upgrade by time, desc:corresponding filed is "delayedUpgradeDateTime" of URL
(/ISAPI/System/onlineUpgrade/task?format=json)-->true
    </isSupportDelayedUpgradeByDateTime>
    <upgradePackageAddressType opt="URL, ftpIPAddress, ftpDomainAddress">
      <!--ro, opt, string, attr:opt{req, string}-->URL
```

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```
      <!--ro, opt, string, attr:opt{req, string}-->URL
    </upgradePackageAddressType>
  </UpgradePackageTask>
  <isSupportPatrolInspection>
    <!--ro, opt, bool, whether the device supports patrol inspection-->true
  </isSupportPatrolInspection>
  <isSupportBatchUpgrade>
    <!--ro, opt, bool, whether the device supports batch online upgrades-->true
  </isSupportBatchUpgrade>
  <isSupportCancelUpgrade>
    <!--ro, opt, bool, whether the device supports canceling online upgrade-->true
  </isSupportCancelUpgrade>
  <isSupportPackagePause>
    <!--ro, opt, bool, whether the device supports pausing upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackagePause?format=json-->true
  </isSupportPackagePause>
  <isSupportPackageResume>
    <!--ro, opt, bool, whether the device supports resuming upgrade package download, desc:/ISAPI/System/onlineUpgrade/PackageResume?format=json-->true
  </isSupportPackageResume>
  <isSupportCancelUpgradeByTaskID>
    <!--ro, opt, bool, whether the device supports canceling online upgrade by task ID, desc:corresponding to the field assignByTaskID of operationType in
URL PUT  /ISAPI/System/onlineUpgrade/CancelUpgrade?format=json-->true
  </isSupportCancelUpgradeByTaskID>
</OnlineUpgradeCap>
```

16.1.4.2 Set the online device upgrade parameters

Request URL

PUT /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Query Parameter

None

Request Message

```
{
    "OnlineUpgradeParameter": {
    /*req, object, online upgrade parameters*/
        "autoDownloadPackage":  true,
        /*req, bool, whether to enable automatic download of upgrade package*/
        "timingUpgrade":  true,
        /*opt, bool, whether to enable scheduled upgrade*/
        "upgradeTime":  "00:00:00"
        /*opt, time, upgrade time, dep:and,{$.OnlineUpgradeParameter.timingUpgrade,eq,true}*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16.1.4.3 Get the device online upgrading parameters

Request URL

GET /ISAPI/System/onlineUpgrade/deviceParameter?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "OnlineUpgradeParameter": {
    /*ro, req, object, online upgrade parameters*/
        "autoDownloadPackage":  true,
        /*ro, req, bool, whether to enable automatic downloading of upgrade package*/
        "timingUpgrade":  true,
        /*ro, opt, bool, whether to enable scheduled upgrade*/
        "upgradeTime":  "00:00:00"
        /*ro, opt, time, upgrade time, dep:and,{$.OnlineUpgradeParameter.timingUpgrade,eq,trur}*/
    }
}
```

16.1.4.4 Get the online upgrade package downloading progress

Request URL

GET /ISAPI/System/onlineUpgrade/downloadPackage/status?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "DownloadPackageStatus": {
    /*ro, req, object, upgrade package downloading status*/
        "status":  "notDownload",
        /*ro, req, enum, download status, subType:string, desc:"notDownload", "downloading", "downloadFailed", "pause", "finish", "incorrectPackage"
(incorrect upgrade package format), "hdOperationFailed" (HD operation failed)*/
        "total":  0.0,
        /*ro, opt, float, upgrade package total size, desc:unit: MB, corrects to one decimal place*/
        "remain":  0.0,
        /*ro, opt, float, remaining space, desc:unit: MB, corrects to one decimal place*/
        "speed":  0.0,
        /*ro, opt, float, download speed, desc:unit: MB, corrects to one decimal place*/
        "remainTime":  0,
        /*ro, opt, int, estimated remaining time, unit:s*/
        "progress":  0
        /*ro, req, int, progress, range:[0,100]*/
    }
}
```

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16.1.4.5 Start to download upgrade package to device

Request URL

PUT /ISAPI/System/onlineUpgrade/downloadPackage?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

16.1.4.6 Get the online upgrade progress of device

Request URL

GET /ISAPI/System/onlineUpgrade/status

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, online upgrade status, attr:version{req, string, protocolVersion}-->
  <status>
    <!--ro, req, enum, online upgrade status, subType:string, desc:"notUpgrade" (not upgraded), "upgrading", "successful" (upgraded), "languageMismatch"
(language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade
version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error), "engineVersionMismatch" (mismatch of upgrade package engine
version)-->notUpgrade
  </status>
  <percent>
    <!--ro, req, int, current upgrade progress in percentage, range:[0,100]-->1
  </percent>
  <onlineUpdateDevInfoList>
    <!--ro, opt, object, online upgrade device information, desc:containing the online upgrade status of sub devices, sub modules, or sub channels-->
    <onlineUpdateDevInfo>
      <!--ro, opt, object, online upgrade device information-->
      <moduleType>
        <!--ro, req, enum, module type, subType:string, desc:"keypad", "detector", "cardReader" (card reader), "siren" (sounder), "relay", "transmitter"
(transmitter), "remoteCtrl" (remote control),"repeater" (repeater), "SSD", "extensionModule" (extension module)-->keypad
      </moduleType>
      <moduleID>
        <!--ro, req, int, module ID, range:[1,128], step:1-->1
      </moduleID>
      <model>
        <!--ro, opt, string, model, range:[1,32]-->test
      </model>
      <status>
        <!--ro, opt, enum, upgrade status, subType:string, desc:"allowance" (upgradeable, the upgrade command is not triggered), "ready" (waiting for
upgrade, the upgrade command has been triggered), "upgriding" (upgrading), "success" (upgraded), "failed" (upgrade failed)-->allowance
      </status>
      <percent>
        <!--ro, opt, int, current upgrade percentage, range:[0,100]-->1
      </percent>
      <updateTime>
        <!--ro, opt, datetime, upgrade completion time-->2004-05-03T17:30:08+08:00
      </updateTime>
    </onlineUpdateDevInfo>
  </onlineUpdateDevInfoList>
</OnlineUpgradeStatus>
```

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16.1.4.7 Get all of the online upgrade task progress

Request URL

GET /ISAPI/System/onlineUpgrade/status/task?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "onlineUpgradeStatusList": [
    /*ro, req, array, online upgrade status list, subType:object*/
        {
            "status":  "notUpgrade",
            /*ro, req, enum, upgrade status, subType:string, desc:“notUpgrade”, “upgrading”, “successful” (upgrading succeeded), “languageMismatch”,
“writeFlashError”, “packageTypeMismatch”, “packageVersionMismatch”, “netUnreachable” (network disconnected), “unknownError”, “waitManualTrigger”*/
            "percent":  0,
            /*ro, req, int, upgrade progress percentage, range:[0,100]*/
            "taskID":  "test",
            /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?
format=json*/
            "moudleType":  "stationHostAndScreen",
            /*ro, opt, enum, component type, subType:string, desc:"stationHostAndScreen" (dock station upgrade package, including host and screen),
"stationHost" (dock station host), "spellingControllerMainAndSubBoard" (main board and sub board of splicing controller), "spellingControllerMainBoard"
(main board of splicing controller), "spellingControllerSubBoard" (sub board of splicing controller), "bodyCamera" (body camera)*/
            "packageName":  "ZCS-H1_target_V1.0.1_build201123.zip",
            /*ro, opt, string, upgrade package name, range:[1,128]*/
            "taskTime":  "1970-01-01T00:00:00+08:00",
            /*ro, opt, datetime, applying upgrade task time*/
            "taskParam": {
            /*ro, opt, object, task parameters, desc:corresponding URL: POST /ISAPI/System/onlineUpgrade/task?format=json*/
                "URL":  "test",
                /*ro, req, string, upgrade package address, range:[1,1024]*/
                "delayedUpgrade":  true,
                /*ro, opt, bool, whether to delay the upgrade, desc:if this node is not applied, it indicates that the upgrade will be started immediately;
if this node is applied and the value is false, it indicates that the upgrade will be delayed*/
                "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00"
                /*ro, opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}*/
            }
        }
    ]
}
```

16.1.4.8 Apply the task of online upgrade

Request URL

POST /ISAPI/System/onlineUpgrade/task?format=json

Query Parameter

None

Request Message

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```
{
    "PackageDescription": {
    /*req, object, description of the upgrade package*/
        "name":  "DSJ_N3_GML_DEBUG_STD_1.2.0_190524.zip",
        /*opt, string, upgrade package name, range:[1,256]*/
        "URL":  "test",
        /*req, string, URL of the upgrade package, range:[1,1024], desc:the task ID is contained in the URL. The uploading and downloading of the upgrade package is based on the object storage protocol (version 2.0). If this node is null, the FTP address will take effect. If both the URL and FTP address are configured, the URL has priority over FTP address*/
        "MD5":  "test",
        /*opt, string, range:[0,256]*/
        "moduleType":  "cardReader",
        /*opt, enum, module type, subType:string, desc:“cardReader” (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen), "stationHostAndScreen" (dock station host and screen)*/
        "moduleID":  1,
        /*opt, int, module ID*/
        "spellingControllerSubBoardList": [
        /*opt, array, splicing controller sub board list, subType:object, range:[1,1024], dep:or, {$.PackageDescription.moduleType,eq,spellingControllerMainAndSubBoard},{$.PackageDescription.moduleType,eq,spellingControllerSubBoard}, desc:it is valid when the value of the node moduleType is "spellingControllerMainAndSubBoard" or "spellingControllerSubBoard". If it is not configured, it indicates that all the sub boards will be updated*/
            {
                "subBoardID":  1,
                /*opt, int, sub board ID, range:[1,1024]*/
                "ipV4Address":  "test"
                /*ro, opt, string, IPv4, range:[1,16], desc:when an exception occurs on the main board, the sub board can be updated through the IP address. The default sub board interface will be used*/
            }
        ],
        "channel":  1,
        /*opt, int, channel No., desc:this node is used for upgrading the IP channel of the NVR*/
        "childDevID":  "3434645611",
        /*opt, string, range:[9,32]*/
        "customTaskID":  "test",
        /*opt, string, user-defined task ID, range:[1,128], desc:if this node exists, it will be returned in the result*/
        "delayedUpgrade":  true,
        /*opt, bool, whether to delay upgrading the package, desc:if this node is not applied, it indicates that the package will be upgraded immediately; if this node is applied and the value is false, it indicates that the upgrade will be delayed. User can decide the time to upgrade the package of the dock station or body camera if the upgrade is delayed*/
        "delayedUpgradeDateTime":  "1970-01-01T00:00:00+08:00",
        /*opt, datetime, time of starting the delayed upgrade, dep:or,{$.PackageDescription.delayedUpgrade,eq,false}, desc:the related capability is "isSupportDelayedUpgradeByDateTime" in GET  /ISAPI/System/onlineUpgrade/capabilities*/
        "ftpAddress":  "test",
        /*opt, string, FTP address of the upgrade package, desc:this node will take effect when URL is null and the format is: ftp://[username]:[password]@[IP address]:[port No.]*/
        "downloadStrategy": {
        /*opt, object, device's upgrade package downloading strategy parameters*/
            "sequelEnabled":  true,
            /*opt, bool, whether to enable ANR (Automatic Network Replenishment), desc:if enabled, when downloading is interrupted, the device will try downloading the package again; if disabled, the default strategy of the device will be adopted*/
            "sequelTimeout":  3000,
            /*opt, int, timeout of a single ANR attempt, step:1, unit:ms, desc:when a single ANR attempt exceeds the timeout period but the download does not continue, the ANR attempt will be recognized as failed*/
            "maxRetryTimes":  5
            /*opt, int, ANR maximum attempt times, desc:when the device ANR exceeds the maximum attempt times but the download does not continue, the download and the upgrade will be recognized as failed*/
        },
        "upgradePackageURLType":  "full"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1-succeeded, other value-error occurred. This field is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"OK" (succeeded), other value (error occurred). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded), other value (error occurred). This field is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "test"
    /*ro, opt, string, ID of the upgrade task, desc:it can be used to get the status and progress of the task*/
}
```

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16.1.4.9 Get the new version information

Request URL

GET /ISAPI/System/onlineUpgrade/version?check=<check>

Query Parameter

Parameter Name	Parameter Type	Description
check	string	When check is false (default), the device does not need to interact with the server and only needs to return the results of periodic checks. When check is true, the device needs to interact with the server and return the latest results.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OnlineUpgradeVersion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, new version information, attr:version{req, string, protocolVersion}-->
  <newVersionAvailable>
    <!--ro, req, bool, whether the new version already is available-->true
  </newVersionAvailable>
  <newVersion>
    <!--ro, opt, string, version information of the new upgrade package-->test
  </newVersion>
  <changeLog>
    <!--ro, opt, string, updating logs of the new upgrade package version-->test
  </changeLog>
  <changeLevel>
    <!--ro, opt, enum, subType:int-->2
  </changeLevel>
</OnlineUpgradeVersion>
```

16.1.4.10 Upload online upgrade status event

EventType:onlineUpgradeStatus

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```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is an event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the face capture. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again. It can be regarded as the event triggering frequency*/
    "eventType":  "onlineUpgradeStatus",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Online Upgrade Status",
    /*ro, req, string, event description, desc:online upgrade status event*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:PUID. This field must be returned when ISAPI event information is transmitted via ISUP. It is the same as deviceID of URI (/ISAPI/System/Network/Ehome)*/
    "serialNumber":  "test",
    /*ro, opt, string, device short serial No., range:[1,9]*/
    "status":  "failure",
    /*ro, req, enum, upgrade status, subType:string, desc:"failure" (upgrade failed), "success" (upgrade succeeded), "upgrading"*/
    "failedReason":  "languageMismatch",
    /*ro, opt, enum, cause of failure, subType:string, desc:"languageMismatch" (language mismatch), "writeFlashError" (failed to write to flash), "packageTypeMismatch" (upgrade package type mismatch), "packageVersionMismatch" (upgrade version mismatch), "netUnreachable" (network disconnected), "unknownError" (unknown error)*/
    "moduleType":  "cardReader",
    /*ro, opt, enum, device component type, subType:string, desc:"cardReader" (card reader), "FPModule" (fingerprint module), "securityModule" (security module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module), "elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station host), "dockStationScreen" (dock station screen)*/
    "moduleID":  1,
    /*ro, opt, int, device component No., desc:it depends on the node moduleType*/
    "channel":  1,
    /*ro, opt, int, channel No., desc:it is used for IP channel (to upgrade digital channel managed by NVR)*/
    "taskID":  "test",
    /*ro, opt, string, task ID, range:[1,64], desc:generated when creating upgrade tasks, related URI: /ISAPI/System/onlineUpgrade/task?format=json*/
    "process":  1
    /*ro, opt, int, upgrade progress, range:[1,100], step:1, unit:%, dep:or,{$.status,eq,upgrading}, desc:progress percentage*/
}
```


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16.1.5 Port Settings

16.1.5.1 Get serial port authentication service capability

Request URL

GET /ISAPI/System/Serial/authentication/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AuthenticationCap": {
    /*ro, opt, object, the serial port authentication service*/
        "enabled": {
        /*ro, req, object, whether to enable serial port authentication service*/
            "@opt": [true, false]
            /*ro, opt, array, whether to enable serial port authentication service, subType:bool*/
        },
        "serviceClosurePeriod": {
        /*ro, opt, object, service life, desc:unit: day it is required when enabled is "false"*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  30,
            /*ro, opt, int, the maximum value*/
            "@def":  7
            /*ro, opt, int, the default value*/
        }
    }
}
```

16.1.5.2 Set parameters of serial authentication service

Request URL

PUT /ISAPI/System/Serial/authentication?format=json

Query Parameter

None

Request Message

```
{
    "Authentication": {
    /*opt, object*/
        "enabled":  false,
        /*req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*opt, int, service life, unit:d, desc:it is required when enabled is "false"*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16.1.5.3 Get parameters of serial authentication service

Request URL

GET /ISAPI/System/Serial/authentication?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "Authentication": {
    /*ro, opt, object*/
        "enabled":  false,
        /*ro, req, bool, whether to enable serial authentication service*/
        "serviceClosurePeriod":  7
        /*ro, opt, int, service life, unit:d, desc:it is required when enabled is "false"”*/
    }
}
```

16.1.5.4 Get the serial port capability of the device

Request URL

GET /ISAPI/System/Serial/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SerialCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, range of RS-485 serial port numbers supported by the device, attr:version{req, string, protocolVersion}-->
  <rs485PortNums>
    <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->0
  </rs485PortNums>
  <supportRS232Config>
    <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
  </supportRS232Config>
  <rs422PortNums>
    <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->0
  </rs422PortNums>
  <rs232PortNums>
    <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->0
  </rs232PortNums>
  <rs485WorkMode opt="Led, CaptureTrigger,transparent">
    <!--ro, opt, string, range of RS-485 serial port No. supported by the device, attr:opt{req, string}-->test
  </rs485WorkMode>
  <rs232SerialNumber opt="1,2,3">
    <!--ro, opt, int, range of RS-232 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs232SerialNumber>
  <rs485SerialNumber opt="4,5,6">
    <!--ro, opt, int, range of RS-485 serial port numbers supported by the device, attr:opt{req, string}-->1
  </rs485SerialNumber>
  <isSupportAuthenticationService>
    <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
  </isSupportAuthenticationService>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports configuring the information about the serial port device-->true
  </isSupportDeviceInfo>
  <isSupportSearchDeviceInfoRelations>
    <!--ro, opt, bool, whether the device supports searching the linkage information of the serial port device-->true
  </isSupportSearchDeviceInfoRelations>
</SerialCap>
```

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16.1.5.5 Get the capability of a specific serial port

Request URL

GET /ISAPI/System/Serial/ports/<portID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port No., attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->0
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <serialPortType opt="RS485,RS422,RS232">
    <!--ro, opt, string, serial port type, attr:opt{req, string}-->RS485
  </serialPortType>
  <duplexMode opt="half,full">
    <!--ro, opt, string, duplex mode of the serial port, attr:opt{req, string}-->half
  </duplexMode>
  <direction opt="monodirectional,bdirectional">
    <!--ro, opt, string, attr:opt{req, string}-->monodirectional
  </direction>
  <baudRate opt="1200,2400,4800,9600,19200,38400,57600,115200">
    <!--ro, opt, int, attr:opt{req, string}-->1200
  </baudRate>
  <dataBits opt="6,7,8">
    <!--ro, opt, int, attr:opt{req, string}-->6
  </dataBits>
  <parityType opt="none,even,odd,mark,space">
    <!--ro, opt, string, attr:opt{req, string}-->none
  </parityType>
  <stopBits opt="1,1.5,2">
    <!--ro, opt, string, stop bit, attr:opt{req, string}-->1
  </stopBits>
  <workMode opt="console,transparent,audiomixe,screenCtrl,ptzCtrl,keyboard,matrix,audioMixers">
    <!--ro, opt, string, working mode, attr:opt{req, string}-->console
  </workMode>
  <flowCtrl opt="none,software,hardware">
    <!--ro, opt, string, flowCtrl, attr:opt{req, string}-->none
  </flowCtrl>
</SerialPort>
```

16.1.5.6 Set the parameters of a specific serial port supported by the device

Request URL

PUT /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	--
childDevID	string	--
deviceIndex	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, serial port ID-->0
  </id>
  <enabled>
    <!--opt, bool, whether to enable the function-->true
  </enabled>
  <serialPortType>
    <!--opt, enum, serial port type: "RS485","RS422","RS232", subType:string, desc:serial port type: "RS485","RS422","RS232"-->RS485
  </serialPortType>
  <duplexMode>
    <!--opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"”-->half
  </duplexMode>
  <direction>
    <!--opt, enum, "monodirectional,bdirectional", subType:string, desc:"monodirectional, bdirectional”-->monodirectional
  </direction>
  <baudRate>
    <!--opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--opt, int-->6
  </dataBits>
  <parityType>
    <!--opt, enum, parity type, subType:string, desc:"none, even, odd, mark, space”-->none
  </parityType>
  <stopBits>
    <!--opt, string, stop bit: "1,1.5,2"-->1
  </stopBits>
  <workMode>
    <!--opt, enum, work mode, subType:string, desc:working mode: "console","transparent","audiomixer","stairsControl"-elevator control,"cardReader"-card
reader,"disabled","custom". This node is required only when <serialPortType> is set to "RS232"-->console
  </workMode>
  <flowCtrl>
    <!--opt, enum, "none,software,hardware", subType:string, desc:"none, software, hardware”-->none
  </flowCtrl>
</SerialPort>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.1.5.7 Get the parameters of a specific port supported by the device

Request URL

GET /ISAPI/System/Serial/ports/<portID>?permissionController=<indexID>&childDevID=<childDevID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
portID	string	--
indexID	string	0 (main permission controller), 1 (sub permission controller) When this parameter does not exist, it indicates getting the serial port information of the main permission controller (i.e., the device serial port information).
childDevID	string	--
deviceIndex	string	For C60S devices connected to multiple LED sending cards, the URL needs to include deviceIndex= to distinguish which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPort xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, port, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, serial port ID-->0
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the serial port-->true
  </enabled>
  <serialPortType>
    <!--ro, opt, enum, serial port type, subType:string, desc:"RS485", "RS422", "RS232"-->RS485
  </serialPortType>
  <duplexMode>
    <!--ro, opt, enum, duplex mode of the serial port, subType:string, desc:"half", "full"-->half
  </duplexMode>
  <direction>
    <!--ro, opt, enum, "monodirectional,bdirectional", subType:string, desc:"monodirectional", "bdirectional"-->monodirectional
  </direction>
  <baudRate>
    <!--ro, opt, enum, subType:int-->2400
  </baudRate>
  <dataBits>
    <!--ro, opt, int-->6
  </dataBits>
  <parityType>
    <!--ro, opt, enum, parity type, subType:string, desc:parity type: "none,even,odd,mark,space"-->none
  </parityType>
  <stopBits>
    <!--ro, opt, enum, stop bit, subType:string, desc:stop bit-->1
  </stopBits>
  <workMode>
    <!--ro, opt, enum, working mode, subType:string, desc:"console", "transparent", "audiomixer", "screenCtrl", "ptzCtrl", "keyboard", "matrix",
"audioMixers"-->console
  </workMode>
  <flowCtrl>
    <!--ro, opt, enum, "none,software,hardware", subType:string, desc:"none", "software", "hardware"-->none
  </flowCtrl>
</SerialPort>
```

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16.1.5.8 Get the list of serial ports supported by the device

Request URL

GET /ISAPI/System/Serial/ports?permissionController=<indexID>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	0 (main permission controller), 1 (sub permission controller) If this parameter does not exist, it indicates getting the serial port information of main permission controller, that is, the serial port information of the device.
deviceIndex	string	For C60S device connecting multiple LED sending cards, the URL needs to include the parameter deviceIndex= to specify which LED sending card to operate on. The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SerialPortList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SerialPort>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->0
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <serialPortType>
      <!--ro, opt, enum, subType:string-->RS485
    </serialPortType>
    <duplexMode>
      <!--ro, opt, enum, subType:string-->half
    </duplexMode>
    <direction>
      <!--ro, opt, enum, subType:string-->monodirectional
    </direction>
    <baudRate>
      <!--ro, opt, enum, subType:int-->2400
    </baudRate>
    <dataBits>
      <!--ro, opt, int-->6
    </dataBits>
    <parityType>
      <!--ro, opt, enum, subType:string-->none
    </parityType>
    <stopBits>
      <!--ro, opt, enum, subType:string-->1
    </stopBits>
    <workMode>
      <!--ro, opt, enum, subType:string-->console
    </workMode>
    <flowCtrl>
      <!--ro, opt, enum, subType:string-->none
    </flowCtrl>
  </SerialPort>
</SerialPortList>
```

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16.1.6 System Maintenance

16.1.6.1 Get the storage capability of the device

Request URL

GET /ISAPI/ContentMgmt/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

```
    <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
    <isSupportZeroChan>
      <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
    </isSupportZeroChan>
    <inputProxyNums>
      <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->0
    </inputProxyNums>
    <eSATANums>
      <!--ro, opt, int, supported number of eSATA interfaces-->0
    </eSATANums>
    <miniSASNums>
      <!--ro, opt, int, supported number of miniSAS interfaces-->0
    </miniSASNums>
    <nasNums>
      <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->0
    </nasNums>
    <ipSanNums>
      <!--ro, opt, int, supported number of IPSANs-->0
    </ipSanNums>
    <isSupportRaid>
      <!--ro, opt, bool, whether the device supports RAID-->true
    </isSupportRaid>
    <isSupportExtHdCfg>
      <!--ro, opt, bool, whether the device supports advanced HDD management-->true
    </isSupportExtHdCfg>
    <isSupportTransCode>
      <!--ro, opt, bool, whether the device supports live view transcoding-->true
    </isSupportTransCode>
    <isSupportIpcImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
    </isSupportIpcImport>
    <isSupportIpcStreamType>
      <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
    </isSupportIpcStreamType>
    <isSupportIOInputProxy>
      <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
    </isSupportIOInputProxy>
    <isSupportIOOutputProxy>
      <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
    </isSupportIOOutputProxy>
    <isSupportPTZRs485Proxy>
      <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
    </isSupportPTZRs485Proxy>
    <isSupportSMARTTest>
      <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
    </isSupportSMARTTest>
    <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,temperatureIntervalMeasurement,AIOP,tail,abnormalFace,normalFace,multiFace,callFace,sunglasses
Face,humanEnter,operateOverTime,stickUp,installScanner,objectPlacedOrTaken,fakeCard,securityChecking,pothole,crack,roadMaint,imageCapture,methaneConcentrati
onException,methaneLightIntensityException,bannerDetection,variableDataEvent,shipChannelAbnormal,radarVideoDetection,smokeAndFireDetection,smokeDetection,se
nsorAlarm,alarmResult,AIOP_Video,AIOP_Polling_Snap,swipePatrol,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMoti
onAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapture,UVPhotonDetection,DevPowerStatusReport,HFPD,audioAnalysis,IndoorPasswayRecovery,BatteryCarRec
overy,DetectorTemperatureRecovery,RadarFallenObjectsOnTracks,VegetationSpectrumDetection,personDensityDetection,ChannelClearEvent,channelOccupy,ChannelPassi
ngEvent,FallenObjectsOnTracksEvent,audioexception,CarWindow,shipsDetection,InsulatorAbnormalAlarm,PackingSpaceRecognition,smokeAlarm,SmokeAlarmRecovery,fire
SmartFireDetect,FirePointAlarmRecovery,offDuty,backDuty,sleepDuty,batteryCarDisobey,SmokingDetectionRecoverEvent,noFireExtinguisherRecog,FireExtinguisherRec
overyEvent,TelephoningBehaviorAlarmEvent,TelephoningBehaviorRecoveryEvent,GasTankMonitoringEvent,TemperatureMeasurementPreviewRecovery,TMPA,TMA,TemperatureM
easurementPreviewAlarmRecovery,TemperatureMeasurementAlarmRecovery,DetectorShelterAlarmRecovery,TemperatureUp,TemperatureDown,TemperatureEvent,TemperatureUp
Recover,TemperatureDownRecover,TemperatureEventRecover,ElectricFenceAlarmEvent,cityManagement,flowVelocity,volumeFlowRate,rainfall,partialDischarge,gasLeaka
ge,manualSnapshot,AlarmTriggeredByPanicButton,detectorSmokeAlarm,smokeAndFireSample,synthesis,FireStatusSwitch,FlameCombustionCounting,linkedTracking,smartS
hipDetection,targetModeling,SafetyWorkwearDetectionEvent,SafetyWorkwearRecoveryEvent,DeskMatOffset">
      <!--ro, opt, enum, picture search type, subType:string, attr:opt{req, string}, desc:"AllEvent", "CMR", "MOTION", "ALARM", "EDR", "ALARMANDMOTION",
"pir:PIR", "wlsensor",  "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection", "regionEntrance", "regionExiting",
"loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual", "manualSnapShot",  "playSnapShot",
"safetyHelmet", "personQueueCounting", "personQueueTime", "violentMotion", "leavePosition", "advReachHeight", "peopleNumChange", "mixedTargetDetection",
"illegalParking", "pedestrian", "construction", "roadblock", "evidence", "crosslane", "vehicleexist", "lanechange", "wrongdirection", "congestion",
"turnround", "parallelParking", "abandonedObject", "trafficAccident", "fogDetection", "smokeDetection", "personDensity", "failDown", "accessController",
"videoInterCom", "GJD", "Luminite", "OPTEX", "securityControlPanel", "getUp", "toiletTarry", "playCellphone", "vehicleMonitor", "running", "retention",
"spacingChange", "yardTarry", "keyPersonGetUp", "sitQuietly", "standup", "plateRecognition", "smoking", "onThePhone", "fatigueDriving",
"notLookingStraightAhead", "leaveThePost", "seatbelt", "sunglasses", "identityAbnormal", "ForwardCollisionWarning", "LaneDepartureWarning",
"BlindSpotDetection", "HeadwayMonitoringWarning", "PedestrianCollisionWarning", "SuddenSpeedUp", "SuddenSpeedDown", "SuddenLeftTurn", "SuddenRightTurn",
"Turnover", "NotYieldToPedestrian;allPic", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin", "smokeDetectAlarm", "temperatureIntervalMeasurement",
"AIOP", "tail", "abnormalFace", "normalFace;callFace", "sunglassesFace", "humanEnter", "operateOverTime", "stickup", "installScanner",
"objectPlacedOrTaken", "fakeCard", "securityChecking", "pothole", "crack", "roadMaint", "imageCapture", "ReID", "thermalVehicleDetection",
"methaneConcentrationException", "methaneLightIntensityException", "bannerDetection", "waterQualityDetection", "waterLevelDetection", "parallelParking",
"methaneConcentrationException", "variableDataEvent", "VOCsGasDetection", "shipChannelAbnormal", "radarVideoDetection", "AIDfogDetection:AID",
"smokeAndFireDetection", "smokeDetection", "sensorAlarm", "GasTankDetection", "alarmResult", "AIOP_Video", "AIOP_Polling_Snap", "swipePatrol",
"BatteryCarInfoEvent", "indoorPasswayBlock", "detectorTemperatureAlarm", "fireNoRegulation", "detectorMotionAlarm", "detectorShelterAlarm",
"detectorRunningStatus", "faceCapture", "UVPhotonDetection", "DevPowerStatusReport", 'HFPD"-->AllEvent
```

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```
  </pictureSearchType>
  <recordSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
,temperatureIntervalMeasurement,AIOP,securityChecking,scheduledRecordUpload,variableDataEvent,smokeAndFireDetection,smokeDetection,AIOP_Video,AIOP_Polling_S
nap,BatteryCarInfoEvent,indoorPasswayBlock,detectorTemperatureAlarm,fireNoRegulation,detectorMotionAlarm,detectorShelterAlarm,detectorRunningStatus,faceCapt
ure,UVPhotonDetection,DevPowerStatusReport,ChannelClearEvent,channelOccupy,ChannelPassingEvent,personQueueCounting,personQueueTime,leavePosition,localButton
Record,callRecord,partialDischarge,gasLeakage,smokeAndFireSample,passback,linkedTracking,CallRecordAnswered,CallRecordNotAnswered,lensMotorLocking">
    <!--ro, opt, enum, video search type, subType:string, attr:opt{req, string}, desc:"AllEvent" (all events), "CMR" (capture by schedule), "MOTION" (motion
detection), "ALARM", "EDR", "ALARMANDMOTION", "pir", "wlsensor", "callhelp", "facedetection", "FieldDetection", "scenechangedetection", "LineDetection",
"regionEntrance", "regionExiting", "loitering", "group", "rapidMove", "parking", "unattendedBaggage", "attendedBaggage", "vehicleDetection", "manual",
"manualSnapShot", "playSnapShot", "safetyHelmet",  "personDensityDetection", "smokeDetection", "ATMPanel", "ATMSurround", "ATMFace", "ATMSafetyCabin",
"temperatureIntervalMeasurement", "AIOP", "securityChecking", "scheduledRecordUpload", "thermalVehicleDetection", "waterQualityDetection",
"waterLevelDetection", "methaneConcentrationException", "variableDataEvent"-->AllEvent
  </recordSearchType>
  <isSupportActivateIpc>
    <!--ro, opt, bool, whether the device supports activating the network camera-->true
  </isSupportActivateIpc>
  <isSupportCheckIpcSecurity>
    <!--ro, opt, object, whether the device supports IPC security verification-->
    <isSupportCheckPassword>
      <!--ro, opt, bool, whether the device supports password detection-->true
    </isSupportCheckPassword>
  </isSupportCheckIpcSecurity>
  <isSupportMainAndSubRecord>
    <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
  </isSupportMainAndSubRecord>
  <isSupportSyncIPCPassword>
    <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
  </isSupportSyncIPCPassword>
  <isSupportTransferIPC>
    <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
  </isSupportTransferIPC>
  <isSupportSmartSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
  </isSupportSmartSearch>
  <isSupportPOS>
    <!--ro, opt, bool, whether the device supports POS-->true
  </isSupportPOS>
  <isSupportLinkNum>
    <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum-->true
  </isSupportLinkNum>
  <isSupportStorageExtraInfo>
    <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
  </isSupportStorageExtraInfo>
  <diskGroupNums>
    <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->0
  </diskGroupNums>
  <ExternalDevice>
    <!--ro, opt, object, external device storage-->
    <USB>
      <!--ro, opt, object, USB information-->
      <debugLogOutput>
        <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
      </debugLogOutput>
      <isSupportCapturePackage>
        <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
      </isSupportCapturePackage>
      <isSupportStreamStorage>
        <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
      </isSupportStreamStorage>
    </USB>
  </ExternalDevice>
  <customProtocolNums>
    <!--ro, opt, int, number of custom protocols for accessing network cameras, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols-->0
  </customProtocolNums>
  <isSupportIPCTiming>
    <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:relate to enableTiming on URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>-->true
  </isSupportIPCTiming>
  <isSupportGetBackupDevice>
    <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json-->true
  </isSupportGetBackupDevice>
  <isSupportAutoBackup>
    <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/auto/capabilities?format=json--
>true
  </isSupportAutoBackup>
  <certificateValidationEnabled>
    <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
  </defaultAdminPortEnabled>
  <isSupportLogDataPackage>
    <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/LogSearch/dataPackage-->true
```

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    <!--ro, opt, bool, whether the device supports exporting the log files, described URI: /ISAPI/ContentMgmt/logSearch/dataPackage-->true
  </isSupportLogDataPackage>
  <CloudStorageServerCap>
    <!--ro, opt, object, capability of the cloud storage service-->
    <isSupportCloudStorageParameter>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
    </isSupportCloudStorageParameter>
    <isSupportCloudStoragePool>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
    </isSupportCloudStoragePool>
  </CloudStorageServerCap>
  <isSupportWebPrivatePlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback (via private protocol) according to UTC-->true
  </isSupportWebPrivatePlaybackByUTC>
  <isSupportFindCommonFileByUTC>
    <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
  </isSupportFindCommonFileByUTC>
  <isSupportFindEventFileByUTC>
    <!--ro, opt, bool, whether it supports searching for event related videos (via private protocol) according to UTC-->true
  </isSupportFindEventFileByUTC>
  <isSupportSmartSearchRecordByUTC>
    <!--ro, opt, bool, whether it supports VCA search by UTC time-->true
  </isSupportSmartSearchRecordByUTC>
  <isSupportMRDSearchByTimeZone>
    <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
  </isSupportMRDSearchByTimeZone>
  <isSupportSearchRecordLabelByUTC>
    <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
  </isSupportSearchRecordLabelByUTC>
  <isSupportSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
  </isSupportSearchPictureByUTC>
  <isSupportSmartSearchPictureByUTC>
    <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
  </isSupportSmartSearchPictureByUTC>
  <isSupportFindLogByUTC>
    <!--ro, opt, bool, whether it supports searching log by UTC time-->true
  </isSupportFindLogByUTC>
  <isSupportUploadRecordByUTC>
    <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
  </isSupportUploadRecordByUTC>
  <isSupportPlaybackByUTC>
    <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
  </isSupportPlaybackByUTC>
  <isSupportFaceDetecttionAlarmByTimeZone>
    <!--ro, opt, bool, whether it supports reporting face detection by UTC time, desc:the time should be before isSupportCountingSearchByUTC and after
isSupportPlaybackByUTC-->true
  </isSupportFaceDetecttionAlarmByTimeZone>
  <isSupportCountingSearchByUTC>
    <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
  </isSupportCountingSearchByUTC>
  <eventRecordSearch>
    <!--ro, opt, object, whether the device supports searching videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
    <eventType
opt="all,batch,alarmResult,faceSnap,vehicleMatchResult,hawkResult,VMD,alarmIn,inquestAlarm,behavior,posAlarm,accessControllerEvent,IOTD,trialAlarm,rapidMove
,defocus,scenechangedetection,PIR,fireSmartFireDetect,shipsDetection,TMA,TDA,framesPeopleCounting,personQueueTime,peopleDetection,safetyHelmetDetection,mixe
dTargetDetection,TMPA,smokeDetection,smokeAndFireDetection,abnormalReboot,audioexception,ANPR,objectsThrownDetection,audioAnalysis,HFPD,shelteralarm,cityMan
agement,hideStart,hideStop,sleepOnduty,PerimeterTargetDetEvent">
      <!--ro, opt, enum, event type, subType:string, attr:opt{req, string}, desc:“vehicleMatchResult” (vehicle comparison), “hawkResult” (video analysis
task), “faceContrast” (face picture comparison), “faceSnap” (face capture)-->alarmResult
    </eventType>
    <alarmResultAuxEventType opt="all,alarmResultFailure,alarmResultSuccess,whiteList">
      <!--ro, opt, enum, sub-type of face picture comparison alarm, subType:string, dep:and,{$.RacmCap.eventRecordSearch.eventType,eq,alarmResult},
attr:opt{req, string}, desc:"alarmResultFailure" (face picture comparison failed), "alarmResultSuccess" (face picture comparison succeeded), "whiteList"
(allowlist alarm, i.e., strangers alarm)-->alarmResultFailure
    </alarmResultAuxEventType>
    <behavior>
      <!--ro, opt, object, abnormal event detection, dep:and,{$.EventSearchDescription.eventType,eq,behavior}-->
      <behaviorEventType
opt="any,linedetection,regionEntrance,regionExiting,fielddetection,loitering,parking,violentMotion,group,unattendedBaggage,attendedBaggage,audioAbnormal,lea
vePosition,peopleNumChange,retention,failDown,running,spacingChange,playCellphone,objectPlacedOrTaken,tailing,advReachHeight">
        <!--ro, req, enum, abnormal event detection event type, subType:string, attr:opt{req, string}, desc:"any", "linedetection" (line crossing
detection), "regionEntrance", "regionExiting", "fielddetection" (intrusion detection), "loitering" (loitering detection), "parking" (parking detection),
"violentMotion", "group" (people gathering), "unattendedBaggage", "attendedBaggage" (object removal), "audioAbnormal" (sudden change of sound intensity
detection)-->any
      </behaviorEventType>
    </behavior>
    <accessControllerEvent>
      <!--ro, opt, object, access control event, dep:and,{$.EventSearchDescription.eventType,eq,accessControllerEvent}-->
      <majorEventType>
        <!--ro, opt, int, alarm main type, desc:see details in macro definition-->0
      </majorEventType>
      <subEventType>
        <!--ro, opt, int, alarm sub type, desc:see details in macro definition-->0
      </subEventType>
    </accessControllerEvent>
    <IOTD>
      <!--ro, opt, object, upload events related with IoT devices, dep:and,{$.EventSearchDescription.eventType,eq,IOTD}-->
      <DeviceEventTypeList>
        <!--ro, req, array, sub event type list, subType:object-->
        <DeviceEventType>
          <!--ro, opt, object, sub event type-->
```


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            <subEventType>
              <!--ro, opt, string, sub event type, desc:see details in the node IoTEventType of Data Dictionary-->test
            </subEventType>
          </DeviceEventType>
        </DeviceEventTypeList>
      </IOTD>
    </eventRecordSearch>
    <isSupportTimeSearch>
      <!--ro, opt, bool, whether it supports searching for start and end time of recording by channel, desc:/ISAPI/ContentMgmt/time/search?format=json-->true
    </isSupportTimeSearch>
    <isSupportRecordSearchByTargetType>
      <!--ro, opt, bool, whether it supports searching for event recordings by target types (pedestrians and vehicles)-->true
    </isSupportRecordSearchByTargetType>
    <isSupportLinkToIPCWeb>
      <!--ro, opt, bool, whether it supports going to the network camera web page (the digital channel management page), desc:N/A-->true
    </isSupportLinkToIPCWeb>
    <isSupportSpare>
      <!--ro, opt, bool, whether the device supports hot spare configuration, desc:related capability set: /ISAPI/ContentMgmt/spare/capabilities?format=json--
>true
    </isSupportSpare>
    <isSupportPTEventTableTemplate>
      <!--ro, opt, bool-->true
    </isSupportPTEventTableTemplate>
    <isSupportPTEventTableFile>
      <!--ro, opt, bool-->true
    </isSupportPTEventTableFile>
    <isSupportAudioDailyDistribution>
      <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
    </isSupportAudioDailyDistribution>
    <isSupportDownloadAudio>
      <!--ro, opt, bool, whether it supports downloading audio files by file name, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
    </isSupportDownloadAudio>
    <isSupportAudioChanCap>
      <!--ro, opt, bool, whether the device supports managing audio channel resources, desc:related URL: /ISAPI/ContentMgmt/audioChannel/capabilities?
format=json-->true
    </isSupportAudioChanCap>
    <isSupportAutoRecordSynthesisParams>
      <!--ro, opt, bool-->true
    </isSupportAutoRecordSynthesisParams>
    <isSupportPicSearchByCoordinate>
      <!--ro, opt, bool, whether the device supports AcuSearch configurations, desc:GET or PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json-->true
    </isSupportPicSearchByCoordinate>
    <AcuSearchCap>
      <!--ro, opt, object, AcuSearch capability-->
      <targetType opt="face,human,vehicle">
        <!--ro, opt, enum, AcuSearch supported target type, subType:string, attr:opt{req, string}, desc:"face", "human" (human body), "vehicle"-->face
      </targetType>
      <dataType opt="url,binary,modelData">
        <!--ro, opt, enum, supported picture transmission format, subType:string, attr:opt{req, string}, desc:"URL", (picture URL), "modelData" (model data),
"binary" (picture binary form)-->url
      </dataType>
      <maxPicSize>
        <!--ro, opt, int, maximum size of image can be searched, range:[0,5], desc:unit: MB-->1
      </maxPicSize>
      <maxPicWidthSize>
        <!--ro, opt, int, maximum width of image can be searched, range:[0,4096]-->4096
      </maxPicWidthSize>
      <maxPicHeightSize>
        <!--ro, opt, int, maximum height of image can be searched, range:[0,4096]-->4096
      </maxPicHeightSize>
    </AcuSearchCap>
    <isSupportRedirectStorageUrl>
      <!--ro, opt, bool-->true
    </isSupportRedirectStorageUrl>
    <isSupportSSDTemperatureAbnormalTrigger>
      <!--ro, opt, bool-->true
    </isSupportSSDTemperatureAbnormalTrigger>
    <isSupportSSDLifeTimeAbnormalTrigger>
      <!--ro, opt, bool-->true
    </isSupportSSDLifeTimeAbnormalTrigger>
    <isSupportTestPicSearchByCoordinateChannel>
      <!--ro, opt, bool-->true
    </isSupportTestPicSearchByCoordinateChannel>
  </RacmCap>
```

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16.1.6.2 Get the system security capability

Request URL

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
userName	string	user name
deviceIndex	string	For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, system security capability, attr:version{req, string, protocolVersion}-->
  <supportUserNums>
    <!--ro, opt, int, supported maximum number of users-->0
  </supportUserNums>
  <userBondIpNums>
    <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->0
  </userBondIpNums>
  <userBondMacNums>
    <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->0
  </userBondMacNums>
  <isSupportOnlineUser>
    <!--ro, opt, bool, whether the device supports the online user configuration-->true
  </isSupportOnlineUser>
  <securityVersion opt="1,2,3,4">
    <!--ro, opt, string, encryption capability set, attr:opt{req, string}, desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is described in each protocol-->1
  </securityVersion>
  <keyIterateNum>
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends on the node securityVersion, the range is between 100 and 1000-->100
  </keyIterateNum>
  <isSupportUserCheck>
    <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, dep:or,{$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the login password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the versions that support encrypting the sensitive information-->true
  </isSupportUserCheck>
  <isSupportGUIDFileDataExport>
    <!--ro, opt, bool, whether the device supports exporting the device's GUID file, desc:related URI: /ISAPI/Security/GUIDFileData-->true
  </isSupportGUIDFileDataExport>
  <isSupportSecurityQuestionConfig>
    <!--ro, opt, bool, whether the device supports answering security questions, desc:related URI: /ISAPI/Security/questionConfiguration-->true
  </isSupportSecurityQuestionConfig>
  <isSupportGetOnlineUserListSC>
    <!--ro, opt, bool, whether the device supports searching the online user list, desc:related URI: /ISAPI/Security/onlineUser-->true
  </isSupportGetOnlineUserListSC>
  <SecurityLimits>
    <!--ro, opt, object, capability of configuring security limit parameters-->
    <LoginPasswordLenLimit min="1" max="16">
      <!--ro, opt, string, length limit of the user's login password, attr:min{req, int},max{req, int}-->1
    </LoginPasswordLenLimit>
    <SecurityAnswerLenLimit min="1" max="128">
      <!--ro, opt, string, length limit of the security question's answer, attr:min{req, int},max{req, int}-->1
    </SecurityAnswerLenLimit>
  </SecurityLimits>
  <RSAKeyLength opt="512,1024,2048" def="2048">
    <!--ro, opt, enum, HTTPS certificate length, subType:int, attr:opt{req, string},def{req, string}, desc:512, 1024, 2048-->2048
  </RSAKeyLength>
  <isSupportONVIFUserManagement>
    <!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol-->true
  </isSupportONVIFUserManagement>
  <WebCertificateCap>
    <!--ro, opt, object, HTTP authentication capability, desc:if this node is not returned, it indicates that device supports basic and digest authentication-->
    <CertificateType opt="basic,digest,digest/basic">
      <!--ro, req, enum, certificate type, subType:string, attr:opt{req, string}, desc:“basic” (basic authentication), “digest” (digest authentication), “digest/basic” (digest/basic authentication)-->basic
    </CertificateType>
    <SecurityAlgorithm>
      <!--ro, opt, object, security algorithm, dep:or,{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest},{$.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
      <algorithmType opt="MD5,SHA256,MD5/SHA256">
        <!--ro, opt, enum, algorithm type, subType:string, attr:opt{req, string}, desc:“MD5”, “SHA256”, “MD5/SHA256”-->MD5
      </algorithmType>
    </SecurityAlgorithm>
  </WebCertificateCap>
  <isSupportConfigFileImport>
```

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```
    <!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileImport>
  <isSupportConfigFileExport>
    <!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)-->true
  </isSupportConfigFileExport>
  <cfgFileSecretKeyLenLimit min="0" max="16">
    <!--ro, opt, string, length limit of the configuration file's verification key, attr:min{req, int},max{req, int}-->1
  </cfgFileSecretKeyLenLimit>
  <supportIPCActivatePassword>
    <!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
  </supportIPCActivatePassword>
  <isIrreversible>
    <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user
information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
  </isIrreversible>
  <salt>
    <!--ro, opt, string, the specific salt used by the user to log in-->test
  </salt>
  <isSupportPictureURlCertificate opt="true,false">
    <!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{req, string}, desc:if this node is not
returned, it indicates the digest authentication for picture URL by default-->true
  </isSupportPictureURlCertificate>
  <isSupportUnloggedUserPermissionConfig opt="true,false">
    <!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{req, string}-->true
  </isSupportUnloggedUserPermissionConfig>
  <isSupportUserNamePasswordCheckUpgrade>
    <!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password, desc:if this node is not returned, it indicates
that device does not support this function-->true
  </isSupportUserNamePasswordCheckUpgrade>
  <isSupportDeviceCertificatesManagement>
    <!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportDeviceCertificatesManagement>
  <isSupportDeviceSelfSignCertExport>
    <!--ro, opt, bool, whether the device supports exporting self-signed certificate-->true
  </isSupportDeviceSelfSignCertExport>
  <isSupportSecurityEmail>
    <!--ro, opt, bool, whether the device supports configuring the security email, desc:if this node is not returned, it indicates that device does not
support this function-->true
  </isSupportSecurityEmail>
  <isSupportResetChannelPassword>
    <!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
  </isSupportResetChannelPassword>
  <DoubleVerificationCap>
    <!--ro, opt, object, secondary authentication capability-->
    <isSupportUsersConfig>
      <!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
    </isSupportUsersConfig>
    <isSupportUsersPermissionConfig>
      <!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
    </isSupportUsersPermissionConfig>
  </DoubleVerificationCap>
  <isSupportEncryptCertificate>
    <!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
  </isSupportEncryptCertificate>
  <isSupportGetProtocolFirewallRulesFileImportStatus>
    <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
  </isSupportGetProtocolFirewallRulesFileImportStatus>
  <isSupportImportProtocolFirewallRulesFile>
    <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
  </isSupportImportProtocolFirewallRulesFile>
  <isSupportGetProtocolFirewallEngineUpgradeStatus>
    <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
  </isSupportGetProtocolFirewallEngineUpgradeStatus>
  <isSupportUpgradeProtocolFirewallEngine>
    <!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
  </isSupportUpgradeProtocolFirewallEngine>
  <isSupportProtocolFirewallVersionInfo>
    <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
  </isSupportProtocolFirewallVersionInfo>
  <isSupportProtocolFirewallParams>
    <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
  </isSupportProtocolFirewallParams>
  <isSupportGetIntrusionPreventionEngineUpgradeStatus>
    <!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET
/ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
  </isSupportGetIntrusionPreventionEngineUpgradeStatus>
  <isSupportIntrusionPreventionEngineVersionInfo>
    <!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET
/ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
  </isSupportIntrusionPreventionEngineVersionInfo>
  <isSupportIntrusionPreventionParams>
    <!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json--
>true
  </isSupportIntrusionPreventionParams>
  <isSupportUpgradeIntrusionPreventionEngine>
    <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT /ISAPI/Security/UpgradeIntrusionPreventionEngine?
format=json-->true
  </isSupportUpgradeIntrusionPreventionEngine>
  <isSupportPasswordProtection>
    <!--ro, opt, bool, whether it supports configuring SDK protocol authentication mode parameters, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
  </isSupportPasswordProtection>
```

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```
</SecurityCap>
```

16.1.6.3 Get device system capability

Request URL

GET /ISAPI/System/capabilities?type=<all>

Query Parameter

Parameter Name	Parameter Type	Description
all	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (Daylight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
      <isSupportWireless>
        <!--ro, req, bool, whether the device supports wireless network-->true
      </isSupportWireless>
      <isSupportWAN>
        <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
      </isSupportWAN>
      <isSupportPPPoE>
        <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
      </isSupportPPPoE>
      <isSupportBond>
        <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
      </isSupportBond>
      <isSupport802_1x>
        <!--ro, req, bool, whether the device supports 802.1x protocol-->true
      </isSupport802_1x>
      <isSupportNtp>
        <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
      </isSupportNtp>
      <isSupportFtp>
        <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
      </isSupportFtp>
      <isSupportUpnp>
        <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
      </isSupportUpnp>
      <isSupportPNP>
        <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
      </isSupportPNP>
      <isSupportDdns>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
      </isSupportDdns>
      <isSupportHttps>
        <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
      </isSupportHttps>
      <SnmpCap>
        <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
        <isSupport>
          <!--ro, req, bool, whether the device supports SNMP-->true
        </isSupport>
      </SnmpCap>
      <isSupportExtNetCfg>
        <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
      </isSupportExtNetCfg>
      <isSupportIPFilter>
        <!--ro, opt, bool, whether the device supports IP filtering-->true
      </isSupportIPFilter>
      <isSupportNetPreviewStrategy>
        <!--ro, opt, bool, whether the device supports network preview strategy-->true
      </isSupportNetPreviewStrategy>
      <isSupportEZVIZ>
        <!--ro, opt, bool, whether the device supports EZ protocol-->true
      </isSupportEZVIZ>
      <isSupportWPS>
        <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
      </isSupportWPS>
      <isWirelessMutexWithWirelessServer>
        <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
      </isWirelessMutexWithWirelessServer>
```

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```
        <isSupportMACFilter>
          <!--ro, opt, bool, whether the device supports physical address filtering-->true
        </isSupportMACFilter>
        <GB28181Cap>
          <!--ro, opt, object, N/A-->
          <isSupportGB28181Service>
            <!--ro, opt, bool, N/A-->true
          </isSupportGB28181Service>
        </GB28181Cap>
        <VerificationCodeModification>
          <!--ro, opt, object, device verification code configuration-->
          <verificationCodeType opt="normal,empty">
            <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
          </verificationCodeType>
          <isSupportDeclarationURL>
            <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
          </isSupportDeclarationURL>
          <isSupportPrivacyPolicyURL>
            <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
          </isSupportPrivacyPolicyURL>
          <verificationCodeModify opt="true,false">
            <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{opt, string}-->true
          </verificationCodeModify>
          <Hyperlinks>
            <!--ro, opt, object, hyperlink configuration-->
            <declarationURL>
              <!--ro, opt, string, hyperlink to the Terms of Service-->test
            </declarationURL>
            <privacyPolicyURL>
              <!--ro, opt, string, hyperlink to the Privacy Policy-->test
            </privacyPolicyURL>
          </Hyperlinks>
          <isSupportVerificationCodeCheck>
            <!--ro, opt, bool, whether the device supports verifying the verification code-->true
          </isSupportVerificationCodeCheck>
        </VerificationCodeModification>
        <isSupportIntegrate>
          <!--ro, opt, bool, whether the device supports access protocol configuration-->true
        </isSupportIntegrate>
        <isSupportPlatformAccess>
          <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome and GB28181 allowlist configuration))-->true
        </isSupportPlatformAccess>
        <isSupportResourceStatistics>
          <!--ro, opt, bool, whether the device supports getting the network resource information-->true
        </isSupportResourceStatistics>
        <isSupportBandwidthLimit>
          <!--ro, opt, bool, whether the device supports bandwidth limit-->true
        </isSupportBandwidthLimit>
        <isSupportPOEPortsDisableAdaptiveServer>
          <!--ro, opt, bool, whether the device supports the POE protocol-->true
        </isSupportPOEPortsDisableAdaptiveServer>
        <isSupportPOEConfiguration>
          <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
        </isSupportPOEConfiguration>
        <isSupportGetLinkSocketIP>
          <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
        </isSupportGetLinkSocketIP>
        <isSupportWebSocket>
          <!--ro, opt, bool, whether the device supports WebSocket-->true
        </isSupportWebSocket>
        <Adaption>
          <!--ro, opt, object, network self-adaption configuration-->
          <streamType opt="0,1,2,3,4,5,7,8,9,10">
            <!--ro, opt, string, stream type that supports network self-adaption during live view, attr:opt{opt, string}, desc:0 (main stream), 1 (sub-
stream), 2 (third stream), 3 (fourth stream/virtual stream), 4 (fifth stream), 5 (sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10
(tenth stream), and so forth. The index value here is the same as that in the live view structure NET_DVR_PREVIEWINFO of Device Network SDK. Note that 6 is
an invalid value, and from the seventh stream (index value: 7), the index value is the same as that for live view via RTSP-->test
          </streamType>
          <isSupportPlayback>
            <!--ro, opt, bool, whether the device supports playback-->true
          </isSupportPlayback>
        </Adaption>
        <isSupportEventDataOverWebSocket opt="true,false">
          <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
        </isSupportEventDataOverWebSocket>
        <isSupportEZVIZTiming>
          <!--ro, opt, bool, whether the device supports synchronizing time with EZ platform, desc:related URI: enabledTiming in the message of
/ISAPI/System/Network/EZVIZ-->true
        </isSupportEZVIZTiming>
        <isSupportPingTest>
          <!--ro, opt, bool, whether the device supports ping test, desc:Related URI: /ISAPI/System/Network/pingtest/capabilities-->true
        </isSupportPingTest>
        <isSupportRequestOutlookAuth>
          <!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
        </isSupportRequestOutlookAuth>
        <isSupportOutlookEmailList>
          <!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
        </isSupportOutlookEmailList>
        <isSupportDSTPServerCfg>
          <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET or PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
```

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format=json"?>true
      </isSupportDSTPServerCfg>
      <isSupportDSTPProtocolsCap>
        <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
      </isSupportDSTPProtocolsCap>
      <isSupportSearchOutlookAuthResult>
        <!--ro, opt, bool-->true
      </isSupportSearchOutlookAuthResult>
    </NetworkCap>
    <IOCap>
      <!--ro, opt, object, I/O capability, desc:related URI: /ISAPI/System/IO/capabilities-->
      <IOInputPortNums>
        <!--ro, opt, int, number of local input ports of the device-->true
      </IOInputPortNums>
      <IOOutputPortNums>
        <!--ro, opt, int, number of local output ports of the device-->true
      </IOOutputPortNums>
      <SoftIOInputPortNums>
        <!--ro, opt, int, number of soft input ports of the device, desc:this node depends on whether the URI /ISAPI/System/IO/softInputs/capabilities?
format=json is supported-->true
      </SoftIOInputPortNums>
      <isSupportIOOutputAdvanceParameter>
        <!--ro, opt, bool, whether the device supports configuring advanced parameters of the output port, desc:this node determines whether the URI
/ISAPI/System/IO/outputs/advanceParameter/capabilities?format=json is supported-->true
      </isSupportIOOutputAdvanceParameter>
      <isSupportSetAllIOOutput>
        <!--ro, opt, bool, whether the device supports configuring all output ports, desc:this node determines whether the PUT method of the URI
/ISAPI/System/IO/outputs is supported-->true
      </isSupportSetAllIOOutput>
      <isSupportCombinationAlarm>
        <!--ro, opt, bool, whether the device supports the combined alarm, desc:CombinationAlarmCap in /ISAPI/System/IO/inputs/capabilities-->true
      </isSupportCombinationAlarm>
      <enabledIOOutputPortNums>
        <!--ro, opt, int, number of enabled alarm outputs-->0
      </enabledIOOutputPortNums>
      <isSupportAlarmKeyParam>
        <!--ro, opt, bool, whether supports configuring the physical keys of one-push alarm, desc:related URI:
/ISAPI/System/IO/inputs/<IOInputID>/AlarmKeyParam/capabilities?format=json-->true
      </isSupportAlarmKeyParam>
    </IOCap>
    <SerialCap>
      <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
      <rs485PortNums>
        <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
      </rs485PortNums>
      <supportRS232Config>
        <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
      </supportRS232Config>
      <rs422PortNums>
        <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
      </rs422PortNums>
      <rs232PortNums>
        <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
      </rs232PortNums>
      <isSupportAuthenticationService>
        <!--ro, opt, bool, whether the device supports the serial port authentication service-->true
      </isSupportAuthenticationService>
    </SerialCap>
    <VideoCap>
      <!--ro, opt, object, video encoding capability, desc:related URI: /ISAPI/System/Video/capabilities-->
      <videoInputPortNums>
        <!--ro, opt, int, number of video input ports-->true
      </videoInputPortNums>
      <videoOutputPortNums>
        <!--ro, opt, int, number of video output ports-->true
      </videoOutputPortNums>
      <menuNums>
        <!--ro, opt, int, number of local menus that can be displayed-->true
      </menuNums>
      <isSupportHeatmap>
        <!--ro, opt, bool, whether the device supports heat map-->true
      </isSupportHeatmap>
      <isSupportCounting>
        <!--ro, opt, bool, whether the device supports people counting-->true
      </isSupportCounting>
      <countingType>
        <!--ro, opt, string, statistics type, desc:human (people counting); object (counting)-->test
      </countingType>
      <isSupportMultiChannelCounting>
        <!--ro, opt, bool, whether the device supports multichannel statistics-->true
      </isSupportMultiChannelCounting>
      <isSupportCountingCollection>
        <!--ro, opt, bool, whether the device supports people counting ANR-->true
      </isSupportCountingCollection>
      <isSupportHeatmapCollection>
        <!--ro, opt, bool, whether the device supports heat map data ANR-->true
      </isSupportHeatmapCollection>
      <channelFlexible opt="name,online,linknum">
        <!--ro, opt, string, this node is located after the node <isSupportHeatmapCollection>., attr:opt{opt, string}-->true
      </channelFlexible>
      <isSupportOutputsResource>
        <!--ro, opt, bool-->true
      </isSupportOutputsResource>
      <isSupportMixedChannel>
```

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        <!--ro, opt, bool, this node is located after the node <channelFlexible>.-->true
      </isSupportMixedChannel>
      <isSupportMixedChannelStatus>
        <!--ro, opt, bool, this node is located after the node <isSupportMixedChannel>.-->true
      </isSupportMixedChannelStatus>
    </VideoCap>
    <AudioCap>
      <!--ro, opt, object, audio encoding capability, desc:related URL: /ISAPI/System/Audio/capabilities-->
      <audioInputNums>
        <!--ro, req, int, number of audio inputs-->1
      </audioInputNums>
      <audioOutputNums>
        <!--ro, req, int, number of audio outputs-->1
      </audioOutputNums>
    </AudioCap>
    <isSupportHolidy>
      <!--ro, opt, bool, this node is located after the node <AudioCap>-->true
    </isSupportHolidy>
    <RebootConfigurationCap>
      <!--ro, opt, object, capability of configuring parameters for rebooting the device, desc:related URI: /ISAPI/System/reboot/capabilities. This node is
located before the node <isSupportSubscribeEvent> and after the node <isSupportHolidy>-->
      <ONVIF>
        <!--ro, opt, bool, whether to enable the ONVIF protocol-->true
      </ONVIF>
      <RTSP>
        <!--ro, opt, bool, whether to enable the RTSP protocol-->true
      </RTSP>
      <HTTP>
        <!--ro, opt, bool, whether to enable the HTTP protocol-->true
      </HTTP>
    </RebootConfigurationCap>
    <isSupportSubscribeEvent>
      <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportDiagnosedData>
      <!--ro, opt, bool, whether the device supports exporting the device diagnosis data, desc:true (support), this node is not returned (not support).
Related URI: /ISAPI/System/diagnosedData-->true
    </isSupportDiagnosedData>
    <isSupportChangedUpload>
      <!--ro, opt, bool, whether the device supports uploading changes-->true
    </isSupportChangedUpload>
    <isSupportGettingWorkingStatus>
      <!--ro, opt, bool, whether the device supports getting the device status-->true
    </isSupportGettingWorkingStatus>
    <isSupportGettingChannelInfoByCondition>
      <!--ro, opt, bool, whether the device supports getting the channel information by condition-->true
    </isSupportGettingChannelInfoByCondition>
    <isSupportDiagnosedDataParameter>
      <!--ro, opt, bool, whether the device supports enabling or disabling exporting the diagnosis information-->true
    </isSupportDiagnosedDataParameter>
    <isSupportSimpleDevStatus>
      <!--ro, opt, bool, whether the device supports getting the device's simple status, desc:related URI: /ISAPI/System/workingstatus?format=json-->true
    </isSupportSimpleDevStatus>
    <isSupportFlexible>
      <!--ro, opt, bool, whether the device supports the flexible combination configuration. Related URI: /ISAPI/System/Video/inputs/channels?format=json--
>true
    </isSupportFlexible>
    <isSupportTimeCap>
      <!--ro, opt, bool, whether the device supports the time configuration, desc:related URI: /ISAPI/System/time/capabilities-->true
    </isSupportTimeCap>
  </SysCap>
  <voicetalkNums>
    <!--ro, opt, int, number of two-way audio channels-->2
  </voicetalkNums>
  <isSupportSnapshot>
    <!--ro, opt, bool, whether the device supports capturing pictures-->true
  </isSupportSnapshot>
  <SecurityCap>
    <!--ro, opt, object, encryption capability set-->
    <supportUserNums>
      <!--ro, opt, int, supported maximum number of users-->1
    </supportUserNums>
    <userBondIpNums>
      <!--ro, opt, int, supported maximum number of IP addresses that can be bound-->1
    </userBondIpNums>
    <userBondMacNums>
      <!--ro, opt, int, supported maximum number of MAC addresses that can be bound-->1
    </userBondMacNums>
    <isSupportOnlineUser>
      <!--ro, opt, bool, whether the device supports the online user configuration-->true
    </isSupportOnlineUser>
    <securityVersion opt="1,2,3,4,7">
      <!--ro, opt, string, encryption capability, attr:opt{opt, string}, desc:the encryption capability of each version consists of two parts: encryption
algorithm and the range of encrypted nodes. Currently 1 refers to AES128 encryption and 2 refers to AES256 encryption. The range of encrypted nodes is
described in each protocol-->test
    </securityVersion>
    <keyIterateNum>
      <!--ro, opt, int, iteration times, desc:the value is usually between 100 and 1000-->100
    </keyIterateNum>
    <isSupportUserCheck>
      <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, desc:this node
depends on the node <securityVersion>, which means that it is only valid for the versions that support encrypting the sensitive information-->true
    </isSupportUserCheck>
    <isSupportGUIDFileDataExport>
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<isSupportGUIDFileDataExport>
<!--ro, opt, bool, whether the device supports exporting the device's GUID file-->true
</isSupportGUIDFileDataExport>
<isSupportSecurityQuestionConfig>
<!--ro, opt, bool, whether the device supports answering security questions-->true
</isSupportSecurityQuestionConfig>
<isSupportGetOnlineUserListSC>
<!--ro, opt, bool, whether the device supports searching the online user list-->true
</isSupportGetOnlineUserListSC>
<SecurityLimits>
<!--ro, opt, object, capability of configuring security limit parameters-->
<LoginPasswordLenLimit min="1" max="16">
<!--ro, opt, string, length limit of the user's login password, attr:min{opt, string},max{opt, string}-->test
</LoginPasswordLenLimit>
<SecurityAnswerLenLimit min="1" max="128">
<!--ro, opt, string, length limit of the security question's answer, attr:min{opt, string},max{opt, string}-->test
</SecurityAnswerLenLimit>
</SecurityLimits>
<RSAKeyLength opt="512,1024,2048,3072" def="3072">
<!--ro, opt, enum, HTTPS certificate length, subType:string, attr:opt{opt, string},def{opt, string}, desc:512, 1024, 2048-->3072
</RSAKeyLength>
<isSupportONVIFUserManagement>
<!--ro, opt, bool, whether the device supports ONVIF user management-->true
</isSupportONVIFUserManagement>
<WebCertificateCap>
<!--ro, opt, object, HTTP authentication capability-->
<CertificateType opt="basic,digest,digest/basic">
<!--ro, req, string, certificate type: basic (authentication), attr:opt{opt, string}-->test
</CertificateType>
<SecurityAlgorithm>
<!--ro, opt, object, encryption algorithm, dep:or,{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,degist},
{$.DeviceCap.SecurityCap.WebCertificateCap.CertificateType,eq,digest/basic}-->
<algorithmType opt="MD5,SHA256,MD5/SHA256">
<!--ro, opt, string, algorithm type, attr:opt{req, string}-->test
</algorithmType>
</SecurityAlgorithm>
</WebCertificateCap>
<isSupportConfigFileImport>
<!--ro, opt, bool, whether the device supports importing the configuration file-->true
</isSupportConfigFileImport>
<isSupportConfigFileExport>
<!--ro, opt, bool, whether the device supports exporting the configuration file-->true
</isSupportConfigFileExport>
<cfgFileSecretKeyLenLimit min="0" max="16">
<!--ro, opt, string, length limit of the configuration file's verification key, attr:min{opt, string},max{opt, string}-->0
</cfgFileSecretKeyLenLimit>
<supportIPCActivatePassword>
<!--ro, opt, bool, whether the device supports configuring the password for activating the network camera-->true
</supportIPCActivatePassword>
<isSupportUserNamePasswordCheckUpgrade>
<!--ro, opt, bool, whether the device supports upgrading the verification of the user name and password-->true
</isSupportUserNamePasswordCheckUpgrade>
<isSupportDeviceCertificatesManagement>
<!--ro, opt, bool, whether the device supports certificate management-->true
</isSupportDeviceCertificatesManagement>
<isSupportDeviceSelfSignCertExport>
<!--ro, opt, bool, whether the device supports exporting the self-signed certificate-->true
</isSupportDeviceSelfSignCertExport>
<isSupportSecurityEmail>
<!--ro, opt, bool, whether the device supports configuring the security email-->true
</isSupportSecurityEmail>
<isSupportPictureURlCertificate opt="true,false">
<!--ro, opt, bool, whether the device supports configuring parameters for picture URL authentication, attr:opt{opt, string}-->true
</isSupportPictureURlCertificate>
<isSupportUnloggedUserPermissionConfig opt="true,false">
<!--ro, opt, bool, whether the device supports configuring permission for users that have not logged in, attr:opt{opt, string}-->true
</isSupportUnloggedUserPermissionConfig>
<isSupportResetChannelPassword>
<!--ro, opt, bool, whether the device supports resetting the password of the channel-->true
</isSupportResetChannelPassword>
<DoubleVerificationCap>
<!--ro, opt, object, secondary authentication capability-->
<isSupportUsersConfig>
<!--ro, opt, bool, whether the device supports configuring users for secondary authentication-->true
</isSupportUsersConfig>
<isSupportUsersPermissionConfig>
<!--ro, opt, bool, whether the device supports configuring user permission for secondary authentication-->true
</isSupportUsersPermissionConfig>
</DoubleVerificationCap>
<isSupportEncryptCertificate>
<!--ro, opt, bool, whether the device supports certificate encryption, desc:/ISAPI/Security/deviceCertificate-->true
</isSupportEncryptCertificate>
<isSupportGetProtocolFirewallRulesFileImportStatus>
<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallRulesFileImportStatus?format=json-->true
</isSupportGetProtocolFirewallRulesFileImportStatus>
<isSupportImportProtocolFirewallRulesFile>
<!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/ImportProtocolFirewallRulesFile?format=json-->true
</isSupportImportProtocolFirewallRulesFile>
<isSupportGetProtocolFirewallEngineUpgradeStatus>
<!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/GetProtocolFirewallEngineUpgradeStatus?format=json-->true
</isSupportGetProtocolFirewallEngineUpgradeStatus>
<isSupportUpgradeProtocolFirewallEngine>
<!--ro, opt, bool, N/A, desc:PUT /ISAPI/Security/UpgradeProtocolFirewallEngine?format=json-->true
</isSupportUpgradeProtocolFirewallEngine>

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    <isSupportProtocolFirewallVersionInfo>
      <!--ro, opt, bool, N/A, desc:GET /ISAPI/Security/ProtocolFirewallVersionInfo?format=json-->true
    </isSupportProtocolFirewallVersionInfo>
    <isSupportProtocolFirewallParams>
      <!--ro, opt, bool, N/A, desc:GET/PUT /ISAPI/Security/ProtocolFirewallParams?format=json-->true
    </isSupportProtocolFirewallParams>
    <isSupportGetIntrusionPreventionEngineUpgradeStatus>
      <!--ro, opt, bool, whether the device supports getting the upgrade status of intrusion prevention engine library, desc:GET
/ISAPI/Security/GetIntrusionPreventionEngineUpgradeStatus?format=json-->true
    </isSupportGetIntrusionPreventionEngineUpgradeStatus>
    <isSupportIntrusionPreventionEngineVersionInfo>
      <!--ro, opt, bool, whether the device supports version information of intrusion prevention engine, desc:GET
/ISAPI/Security/IntrusionPreventionEngineVersionInfo?format=json-->true
    </isSupportIntrusionPreventionEngineVersionInfo>
    <isSupportIntrusionPreventionParams>
      <!--ro, opt, bool, whether the device supports intrusion prevention parameters, desc:GET/PUT /ISAPI/Security/IntrusionPreventionParams?format=json--
>true
    </isSupportIntrusionPreventionParams>
    <isSupportUpgradeIntrusionPreventionEngine>
      <!--ro, opt, bool, whether the device supports upgrading intrusion prevention engine library, desc:PUT
/ISAPI/Security/UpgradeIntrusionPreventionEngine?format=json-->true
    </isSupportUpgradeIntrusionPreventionEngine>
    <isSupportPasswordProtection>
      <!--ro, opt, bool, whether it supports parameter configuration of SDK protocol authentication mode, desc:PUT/GET
/ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=<security>&iv=<iv>-->true
    </isSupportPasswordProtection>
  </SecurityCap>
  <EventCap>
    <!--ro, opt, object, event capability-->
    <isSupportHDFull>
      <!--ro, opt, bool, whether the device supports HDD full alarm-->true
    </isSupportHDFull>
    <isSupportHDError>
      <!--ro, opt, bool, whether the device supports HDD error alarm-->true
    </isSupportHDError>
    <isSupportHDBadBlock>
      <!--ro, opt, bool-->true
    </isSupportHDBadBlock>
    <isSupportNicBroken>
      <!--ro, opt, bool, whether the device supports the network disconnected event-->true
    </isSupportNicBroken>
    <isSupportIpConflict>
      <!--ro, opt, bool, whether the device supports the IP address conflict event-->true
    </isSupportIpConflict>
    <isSupportIllAccess>
      <!--ro, opt, bool, whether the device supports the illegal login event-->true
    </isSupportIllAccess>
    <isSupportViException>
      <!--ro, opt, bool, whether the device supports video input exception (only for analog channels)-->true
    </isSupportViException>
    <isSupportViMismatch>
      <!--ro, opt, bool, whether the device supports the event of video standard mismatch-->true
    </isSupportViMismatch>
    <isSupportRecordException>
      <!--ro, opt, bool, whether the device supports the recording exception event-->true
    </isSupportRecordException>
    <isSupportRaidException>
      <!--ro, opt, bool, whether the device supports the array exception event-->true
    </isSupportRaidException>
    <isSupportViResMismatch>
      <!--ro, opt, bool, whether the device supports the event of video resolution mismatch-->true
    </isSupportViResMismatch>
    <isSupportPOCException>
      <!--ro, opt, bool, whether the device supports the event of powering over coaxial exception-->true
    </isSupportPOCException>
    <isSupportSmartDetection>
      <!--ro, opt, bool, whether the device supports the Smart event-->true
    </isSupportSmartDetection>
    <isSupportSpareException>
      <!--ro, opt, bool, whether the device supports hot spare exception-->true
    </isSupportSpareException>
    <isSupportHumanRecognition>
      <!--ro, opt, bool, whether the device supports target capture events-->true
    </isSupportHumanRecognition>
    <isSupportFaceContrast>
      <!--ro, opt, bool, whether the device supports face picture comparison events-->true
    </isSupportFaceContrast>
    <isSupportWhiteListFaceContrast>
      <!--ro, opt, bool, whether the device supports events of allowlist face picture comparison-->true
    </isSupportWhiteListFaceContrast>
    <isSupportStudentsStoodUp>
      <!--ro, opt, bool, whether the device supports the student standing up detection event-->true
    </isSupportStudentsStoodUp>
    <isSupportFaceSnap>
      <!--ro, opt, bool, whether the device supports face capture events-->true
    </isSupportFaceSnap>
    <isSupportPersonDensityDetection>
      <!--ro, opt, bool, whether the device supports people density detection-->true
    </isSupportPersonDensityDetection>
    <HFPD>
      <!--ro, opt, bool, whether the device supports frequently appeared person detection-->true
    </HFPD>
    <isSupportPersonQueueDetection>
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      <!--ro, opt, bool, whether the device supports queue detection events-->true
    </isSupportPersonQueueDetection>
    <isSupportSafetyHelmetDetection>
      <!--ro, opt, bool, whether the device supports hard hat detection-->true
    </isSupportSafetyHelmetDetection>
    <isSupportTeacherBehaviorDetect>
      <!--ro, opt, bool, whether the device supports teacher's behavior detection-->true
    </isSupportTeacherBehaviorDetect>
    <isSupportCityManagement>
      <!--ro, opt, bool, whether the device supports the city management function-->true
    </isSupportCityManagement>
    <isSupportMixedTargetDetection>
      <!--ro, opt, bool, whether the device supports multi-target-type detection events-->true
    </isSupportMixedTargetDetection>
    <isSupportFaceSnapModeling>
      <!--ro, opt, bool, whether the device supports face modeling events-->true
    </isSupportFaceSnapModeling>
    <isSupportTriggerCapCheck>
      <!--ro, opt, bool, whether the device supports verifying the capability of triggering linkage actions, desc:this node indicates that the capability
message returned by calling the URI GET /ISAPI/Event/triggersCap has been verified, and the event linkage actions can be displayed exactly according to the
capability-->true
    </isSupportTriggerCapCheck>
    <isSupportPTEventCfg>
      <!--ro, opt, bool, whether the device supports transparent transmission event configuration event, desc:/ISAPI/Event/PTEventCfg/capabilities?
format=json-->true
    </isSupportPTEventCfg>
  </EventCap>
  <RacmCap>
    <!--ro, opt, object, UI before picture search-->
    <isSupportZeroChan>
      <!--ro, opt, bool, whether the device supports channel-zero, desc:related URI: /ISAPI/ContentMgmt/ZeroVideo/channels-->true
    </isSupportZeroChan>
    <inputProxyNums>
      <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->1
    </inputProxyNums>
    <eSATANums>
      <!--ro, opt, int, supported number of eSATA interfaces-->1
    </eSATANums>
    <miniSASNums>
      <!--ro, opt, int, supported number of miniSAS interfaces-->1
    </miniSASNums>
    <nasNums>
      <!--ro, opt, int, supported number of NAS devices, desc:related URI: /ISAPI/ContentMgmt/Storage/nas-->1
    </nasNums>
    <ipSanNums>
      <!--ro, opt, int, supported number of IPSANs-->1
    </ipSanNums>
    <isSupportRaid>
      <!--ro, opt, bool, whether the device supports RAID-->true
    </isSupportRaid>
    <isSupportExtHdCfg>
      <!--ro, opt, bool, whether the device supports advanced HDD management-->true
    </isSupportExtHdCfg>
    <isSupportTransCode>
      <!--ro, opt, bool, whether the device supports live view transcoding-->true
    </isSupportTransCode>
    <isSupportIpcImport>
      <!--ro, opt, bool, whether the device supports importing the configuration file of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/ipcConfig-->true
    </isSupportIpcImport>
    <isSupportIpcStreamType>
      <!--ro, opt, bool, whether the device supports the stream type of the network camera-->true
    </isSupportIpcStreamType>
    <isSupportIOInputProxy>
      <!--ro, opt, bool, whether the device supports dynamic input channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/inputs-->true
    </isSupportIOInputProxy>
    <isSupportIOOutputProxy>
      <!--ro, opt, bool, whether the device supports dynamic output channels, desc:related URI: /ISAPI/ContentMgmt/IOProxy/outputs-->true
    </isSupportIOOutputProxy>
    <isSupportPTZRs485Proxy>
      <!--ro, opt, bool, whether the device supports dynamic PTZ RS-485 channels, desc:related URI: /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities-->true
    </isSupportPTZRs485Proxy>
    <isSupportSMARTTest>
      <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
    </isSupportSMARTTest>
    <pictureSearchType
opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
personQueueCounting,personQueueTime,violentMotion,leavePosition,advReachHeight,peopleNumChange,mixedTargetDetection,illegalParking,pedestrian,construction,r
oadblock,
abandonedObject,trafficAccident,fogDetection,smokeDetection,personDensity,failDown,accessController,videoInterCom,GJD,Luminite,OPTEX,securityControlPanel,ge
tUp,toiletTarry,
playCellphone,vehicleMonitor,vibrationDetection,running,retention,spacingChange,yardTarry,keyPersonGetUp,sitQuietly,standUp,plateRecognition,smoking,onThePh
one,fatigueDriving,
notLookingStraightAhead,leaveThePost,seatBelt,sunglasses,identityAbnormal,ForwardCollisionWarning,LaneDepartureWarning,BlindSpotDetection,HeadwayMonitoringW
arning,
PedestrianCollisionWarning,SuddenSpeedUp,SuddenSpeedDown,SuddenLeftTurn,SuddenRightTurn,Turnover,NotYieldToPedestrian,unregisteredStreetVendor,stallOutsideS
hop,stallOccupyingRoad,     illegalHeap,illegalParkingofNon-
MotorVehicle,illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,allPic,ATMPanel,
ATMSurround,ATMFace,ATMSafetyCabin,loitering,smokeDetectAlarm,alarmResult">
      <!--ro, opt, string, picture search type, attr:opt{opt, string}-->test
    </pictureSearchType>
    <recordSearchType
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opt="AllEvent,CMR,MOTION,ALARM,EDR,ALARMANDMOTION,Command,pir,wlsensor,callhelp,facedetection,FieldDetection,scenechangedetection,LineDetection,
regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage,vehicleDetection,manual,manualSnapShot,playSnapShot,safetyH
elmet,
failDown,personDensityDetection,smokeDetection,vibrationDetection,unregisteredStreetVendor,stallOutsideShop,stallOccupyingRoad,illegalHeap,illegalParkingofN
on-MotorVehicle,
illegalOutdoorAdvertisement,packGarbage,stallUnderUmbrella,dustbinOverflow,exposeGarbage,hangClothingAlongStreet,ATMPanel,ATMSurround,ATMFace,ATMSafetyCabin
">
      <!--ro, opt, string, video search type, attr:opt{opt, string}, desc:"llEvent" (all events), "CMR" (capture by schedule), "MOTION" (capture motion
detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures
and motion detection pictures), "Command", "pir" (pir alarm), "wlsensor" (wireless alarm), "callhelp" (panic alarm), "facedetection" (face detection),
"FieldDetection" (intrusion detection), "scenechangedetection" (scene change detection), "LineDetection" (line crossing detection)-->test
    </recordSearchType>
    <isSupportActivateIpc>
      <!--ro, opt, bool, whether the device supports activating the network camera-->true
    </isSupportActivateIpc>
    <isSupportCheckIpcSecurity>
      <!--ro, opt, object, device password detection-->
      <isSupportCheckPassword>
        <!--ro, opt, bool, whether the device supports password detection-->true
      </isSupportCheckPassword>
    </isSupportCheckIpcSecurity>
    <isSupportMainAndSubRecord>
      <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
    </isSupportMainAndSubRecord>
    <isSupportSyncIPCPassword>
      <!--ro, opt, bool, whether the device supports synchronizing the password of the network camera-->true
    </isSupportSyncIPCPassword>
    <isSupportTransferIPC>
      <!--ro, opt, bool, whether the device supports transparent transmission of the network camera-->true
    </isSupportTransferIPC>
    <isSupportSmartSearch>
      <!--ro, opt, bool, whether the device supports video dual-VCA, desc:related URI: /ISAPI/ContentMgmt/SmartSearch/capabilities-->true
    </isSupportSmartSearch>
    <isSupportPOS>
      <!--ro, opt, bool, whether the device supports POS-->true
    </isSupportPOS>
    <isSupportLinkNum>
      <!--ro, opt, bool, whether the device supports getting the number of links, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>/linkNum--
>true
    </isSupportLinkNum>
    <isSupportStorageExtraInfo>
      <!--ro, opt, bool, whether the device supports configuring additional storage information, desc:related URI:
/ISAPI/ContentMgmt/Storage/ExtraInfo/capabilities-->true
    </isSupportStorageExtraInfo>
    <diskGroupNums>
      <!--ro, opt, int, supported number of disk groups, desc:related URI: /ISAPI/ContentMgmt/Storage/diskGroup-->1
    </diskGroupNums>
    <ExternalDevice>
      <!--ro, opt, object, external device storage-->
      <USB>
        <!--ro, opt, object, USB information-->
        <debugLogOutput>
          <!--ro, opt, bool, whether to enable the debugging log, desc:related URI: /ISAPI/ContentMgmt/Storage/externalDevice/USB?format=json-->true
        </debugLogOutput>
        <isSupportCapturePackage>
          <!--ro, opt, bool, whether the device supports configuring capture and filter function of a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/capturePackage/capabilities?format=json-->true
        </isSupportCapturePackage>
        <isSupportStreamStorage>
          <!--ro, opt, bool, whether the device supports saving stream to a USB, desc:related URI:
/ISAPI/ContentMgmt/Storage/externalDevice/USB/streamStorage/capabilities?format=json-->true
        </isSupportStreamStorage>
      </USB>
    </ExternalDevice>
    <customProtocolNums>
      <!--ro, opt, int, number of custom protocols connected via IPC, desc:related URI: /ISAPI/ContentMgmt/InputProxy/customProtocols, supported number of
custom protocols-->1
    </customProtocolNums>
    <isSupportIPCTiming>
      <!--ro, opt, bool, whether the device supports synchronizing the time of the network camera, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol, whether the <enableTiming> node supports-->true
    </isSupportIPCTiming>
    <isSupportGetBackupDevice>
      <!--ro, opt, bool, whether the device supports getting the backup device, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/device?format=json--
>true
    </isSupportGetBackupDevice>
    <isSupportAutoBackup>
      <!--ro, opt, bool, whether the device supports automatic backup, desc:related URI: /ISAPI/ContentMgmt/Storage/backup/auto/capabilities?format=json--
>true
    </isSupportAutoBackup>
    <certificateValidationEnabled>
      <!--ro, opt, bool, whether to enable the certification validation, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <certificateValidationEnabled> node supports-->true
    </certificateValidationEnabled>
    <defaultAdminPortEnabled>
      <!--ro, opt, bool, whether to enable the default communication port, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>. In this protocol,
whether the <defaultAdminPortEnabled> node supports-->true
    </defaultAdminPortEnabled>
    <isSupportLogDataPackage>
      <!--ro, opt, bool, whether the device supports exporting the log files, desc:related URI: /ISAPI/ContentMgmt/logSearch/dataPackage-->true
    </isSupportLogDataPackage>
    <CloudStorageServerCap>
      <!--ro, opt, object, capability of configuring the cloud storage server-->
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      <isSupportCloudStorageParameter>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage parameters, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json-->true
      </isSupportCloudStorageParameter>
      <isSupportCloudStoragePool>
        <!--ro, opt, bool, whether the device supports configuring the cloud storage pool, desc:related URI:
/ISAPI/ContentMgmt/Storage/cloudServer/pool/capabilities?format=json-->true
      </isSupportCloudStoragePool>
    </CloudStorageServerCap>
    <isSupportWebPrivatePlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback (via private protocol) by UTC-->true
    </isSupportWebPrivatePlaybackByUTC>
    <isSupportFindCommonFileByUTC>
      <!--ro, opt, bool, whether it supports searching for ordinary files (private protocols) by UTC time-->true
    </isSupportFindCommonFileByUTC>
    <isSupportFindEventFileByUTC>
      <!--ro, opt, bool, whether it supports searching for event files (private protocols) by UTC time-->true
    </isSupportFindEventFileByUTC>
    <isSupportSmartSearchRecordByUTC>
      <!--ro, opt, bool, whether it supports smart video search by UTC time-->true
    </isSupportSmartSearchRecordByUTC>
    <isSupportMRDSearchByTimeZone>
      <!--ro, opt, bool, whether it supports searching for files (private protocols) by monthly calendar by UTC time-->true
    </isSupportMRDSearchByTimeZone>
    <isSupportSearchRecordLabelByUTC>
      <!--ro, opt, bool, whether it supports searching for video tags by UTC time-->true
    </isSupportSearchRecordLabelByUTC>
    <isSupportSearchPictureByUTC>
      <!--ro, opt, bool, whether it supports searching for pictures by UTC time-->true
    </isSupportSearchPictureByUTC>
    <isSupportSmartSearchPictureByUTC>
      <!--ro, opt, bool, whether it supports smart pictures search by UTC time-->true
    </isSupportSmartSearchPictureByUTC>
    <isSupportFindLogByUTC>
      <!--ro, opt, bool, whether it supports searching log by UTC time-->true
    </isSupportFindLogByUTC>
    <isSupportUploadRecordByUTC>
      <!--ro, opt, bool, whether it supports uploading files to cloud storage by UTC time-->true
    </isSupportUploadRecordByUTC>
    <isSupportPlaybackByUTC>
      <!--ro, opt, bool, whether it supports playback and locating by UTC time-->true
    </isSupportPlaybackByUTC>
    <isSupportFaceDetecttionAlarmByTimeZone>
      <!--ro, opt, bool, no file is found-->true
    </isSupportFaceDetecttionAlarmByTimeZone>
    <isSupportCountingSearchByUTC>
      <!--ro, opt, bool, whether it supports searching people counting results by UTC time-->true
    </isSupportCountingSearchByUTC>
    <eventRecordSearch>
      <!--ro, opt, object, whether the device supports searching for videos by event, desc:related URI: /ISAPI/ContentMgmt/eventRecordSearch?format=json-->
      <eventType opt="faceContrast,faceSnap,vehicleMatchResult,hawkResult">
        <!--ro, opt, string, event type, attr:opt{opt, string}-->test
      </eventType>
    </eventRecordSearch>
    <isSupportTimeSearch>
      <!--ro, opt, bool, whether it supports searching for recording start and end time by channel, desc:related URI: /ISAPI/ContentMgmt/time/search?
format=json-->true
    </isSupportTimeSearch>
    <isSupportCloudStorageModel>
      <!--ro, opt, bool, whether the device supports configuring the cloud storage mode,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>/capabilities?format=json-->true
    </isSupportCloudStorageModel>
    <isSupportCloudStorageModelList>
      <!--ro, opt, bool, whether the device supports configuring cloud storage modes in a batch,
desc:/ISAPI/ContentMgmt/Storage/CloudStorageModel/<channelID>?format=json-->true
    </isSupportCloudStorageModelList>
    <isSupportAudioDailyDistribution>
      <!--ro, opt, bool, whether it supports searching for audio file information by monthly calendar, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json-->true
    </isSupportAudioDailyDistribution>
    <isSupportDownloadAudio>
      <!--ro, opt, bool, whether the device supports downloading the audio by file name, desc:related URI:
/ISAPI/ContentMgmt/recordAudio/fileNameDownload/capabilities?format=json-->true
    </isSupportDownloadAudio>
  </RacmCap>
  <PTZCtrlCap>
    <!--ro, opt, object, device's PTZ capability-->
    <isSupportPatrols>
      <!--ro, opt, bool, whether the device supports patrol-->true
    </isSupportPatrols>
  </PTZCtrlCap>
  <SmartCap>
    <!--ro, opt, object, compact and smart capability, desc:/ISAPI/Smart/capabilities-->
    <isSupportROI>
      <!--ro, opt, bool, whether the device supports ROI encoding configuration-->true
    </isSupportROI>
    <isSupportFaceDetect>
      <!--ro, opt, bool, whether the device supports face detection-->true
    </isSupportFaceDetect>
    <isSupportIntelliTrace>
      <!--ro, opt, bool, whether the device supports smart tracking-->true
    </isSupportIntelliTrace>
    <isSupportFieldDetection>
      <!--ro, opt, bool, whether the device supports intrusion detection-->true
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      <!--ro, opt, bool, whether the device supports intrusion detection-->true
    </isSupportFieldDetection>
    <isSupportDefocusDetection>
      <!--ro, opt, bool, whether the device supports defocus detection-->true
    </isSupportDefocusDetection>
    <isSupportAudioDetection>
      <!--ro, opt, bool, whether the device supports audio detection-->true
    </isSupportAudioDetection>
    <isSupportSceneChangeDetection>
      <!--ro, opt, bool, whether the device supports scene change detection-->true
    </isSupportSceneChangeDetection>
    <isSupportLineDetection>
      <!--ro, opt, bool, whether the device supports line crossing detection-->true
    </isSupportLineDetection>
    <isSupportRegionEntrance>
      <!--ro, opt, bool, whether the device supports region entrance detection-->true
    </isSupportRegionEntrance>
    <isSupportRegionExiting>
      <!--ro, opt, bool, whether the device supports region exiting detection-->true
    </isSupportRegionExiting>
    <isSupportLoitering>
      <!--ro, opt, bool, whether the device supports loitering detection-->true
    </isSupportLoitering>
    <isSupportGroup>
      <!--ro, opt, bool, whether the device supports people gathering detection-->true
    </isSupportGroup>
    <isSupportRapidMove>
      <!--ro, opt, bool, whether the device supports fast moving detection-->true
    </isSupportRapidMove>
    <isSupportParking>
      <!--ro, opt, bool, whether the device supports parking detection-->true
    </isSupportParking>
    <isSupportUnattendedBaggage>
      <!--ro, opt, bool, whether the device supports unattended baggage detection-->true
    </isSupportUnattendedBaggage>
    <isSupportAttendedBaggage>
      <!--ro, opt, bool, whether the device supports object removal detection-->true
    </isSupportAttendedBaggage>
    <isSupportSmartCalibration>
      <!--ro, opt, bool, whether the device supports calibration filter size-->true
    </isSupportSmartCalibration>
    <isSupportShipsDetection>
      <!--ro, opt, bool, whether the device supports ship detection-->true
    </isSupportShipsDetection>
    <isSupportShield>
      <!--ro, opt, bool, whether the device supports configuring the compact and smart shielded regions-->true
    </isSupportShield>
    <isSupportAnalysisUnitSwitch opt="true,false">
      <!--ro, opt, bool, whether the device supports switching analysis unit, attr:opt{opt, string}-->true
    </isSupportAnalysisUnitSwitch>
    <isSupportHFPD>
      <!--ro, opt, bool, whether the device supports frequently appeared person detection, desc:related URI: /ISAPI/SDT/HFPD?format=json-->true
    </isSupportHFPD>
    <isSupportLFPD>
      <!--ro, opt, bool, whether the device supports rarely appeared person detection, desc:related URI: /ISAPI/SDT/LFPD?format=json-->true
    </isSupportLFPD>
  </SmartCap>
  <isSupportEhome>
    <!--ro, opt, bool, whether the device supports ISUP functions-->true
  </isSupportEhome>
  <isSupportStreamingEncrypt>
    <!--ro, opt, bool, whether the device supports stream encryption-->true
  </isSupportStreamingEncrypt>
  <TestCap>
    <!--ro, opt, object, test capability-->
    <isSupportEmailTest>
      <!--ro, opt, bool, whether the device supports email test-->true
    </isSupportEmailTest>
  </TestCap>
  <ThermalCap>
    <!--ro, opt, object, thermal capability, desc:related URI: /ISAPI/Thermal/capabilities-->
    <isSupportFireDetection>
      <!--ro, opt, bool, whether the device supports fire detection, desc:related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
    </isSupportFireDetection>
    <isSupportThermometry>
      <!--ro, opt, bool, whether the device supports thermography, desc:related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
    </isSupportThermometry>
    <isSupportRealtimeThermometry>
      <!--ro, opt, bool, whether the device supports uploading real-time temperature measurement data-->true
    </isSupportRealtimeThermometry>
    <isSupportNVR>
      <!--ro, opt, bool, it is used for client to check-->true
    </isSupportNVR>
    <isSupportThermometrySchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature measurement, desc:related URI: /ISAPI/Event/schedules/thermometry--
>true
    </isSupportThermometrySchedule>
    <isSupportTemperatureSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of temperature difference, desc:related URI: /ISAPI/Event/schedules/temperature--
>true
    </isSupportTemperatureSchedule>
    <isSupportFireDetectionSchedule>
      <!--ro, opt, bool, whether the device supports the arming schedule of fire detection, desc:related URI: /ISAPI/Event/schedules/fireDetections-->true
    </isSupportFireDetectionSchedule>
```

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```
    <isSupportTemperatureCollection>
      <!--ro, opt, bool, whether device supports temperature ANR, desc:related URI: ISAPI/Thermal/temperature/collection/capabilities?format=json-->true
    </isSupportTemperatureCollection>
  </ThermalCap>
  <isSupportGetmutexFuncErrMsg>
    <!--ro, opt, bool, whether the device supports the mutual exclusion information-->true
  </isSupportGetmutexFuncErrMsg>
  <isSupportTokenAuthenticate>
    <!--ro, opt, bool, whether the device supports token authentication-->true
  </isSupportTokenAuthenticate>
  <isSupportStreamDualVCA>
    <!--ro, opt, bool, whether the device supports overlaying intelligent private data on the stream, desc:related URI: in the
/ISAPI/Intelligent/channels/<ID>/intelliResource; corresponding node: <isOverlapIntelli>-->true
  </isSupportStreamDualVCA>
  <isSupportIntelligentAnalysisEngines>
    <!--ro, opt, bool, whether the device supports configuring the intelligent engine, desc:related URI: /ISAPI/Intelligent/analysisEngines/capabilities--
>true
  </isSupportIntelligentAnalysisEngines>
  <isSupportTriggerCapCheck>
    <!--ro, opt, bool, it indicates that the capability of the URL (/ISAPI/Event/triggersCap) is already verified-->true
  </isSupportTriggerCapCheck>
  <isSupportChannelEventCap>
    <!--ro, opt, bool, whether the device supports getting the event capability by channel, desc:related URI: /ISAPI/Event/channels/<ID>/capabilities-->true
  </isSupportChannelEventCap>
  <isSupportEncryption>
    <!--ro, opt, bool, stream encryption capability-->true
  </isSupportEncryption>
  <isSupportPanoramaCalibration>
    <!--ro, opt, bool, whether the device supports configuring the calibration of the panoramic camera-->true
  </isSupportPanoramaCalibration>
  <isSupportCameraAccessConfiguration>
    <!--ro, opt, bool, whether the device supports the camera access configuration, desc:related URI: /ISAPI/System/cameraAccess/configuration/capabilities?
format=json-->true
  </isSupportCameraAccessConfiguration>
  <isSupportIntelligentSearch>
    <!--ro, opt, bool, whether the device supports intelligent search, desc:related URI: /ISAPI/SDT/Management/IntelligentSearch/capabilities?format=json--
>true
  </isSupportIntelligentSearch>
  <isSupportAutoMaintenance>
    <!--ro, opt, bool, whether the device supports automatic maintenance, desc:related URI: /ISAPI/System/autoMaintenance/capabilities?format=json-->true
  </isSupportAutoMaintenance>
  <isSupportRtspOverHTTPS>
    <!--ro, opt, bool, supports tspOverHTTPS-->true
  </isSupportRtspOverHTTPS>
  <isSupportIntersectionAnalysis>
    <!--ro, opt, bool, whether the device supports the intersection analysis, desc:related URI:
/ISAPI/Intelligent/channels/<ID>/intersectionAnalysis/capabilities?format=json-->true
  </isSupportIntersectionAnalysis>
  <isSupportChannelFullEventListCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the event capability of all channels of the device-->true
  </isSupportChannelFullEventListCap>
  <isSupportAUXInfoCap>
    <!--ro, opt, bool, it indicates that whether the device supports getting the capability of getting the attributes of device's all channels-->true
  </isSupportAUXInfoCap>
  <IOTCap>
    <!--ro, opt, object, supports searching for the device status list by page, desc:not returned (not support, get the device status list via GET)-->
    <supportChannelNum opt="5000">
      <!--ro, opt, int, the number of supported IOT device channels, attr:opt{opt, string}-->1
    </supportChannelNum>
    <startChannelNo>
      <!--ro, opt, int, the channel No. of start IOT device-->1
    </startChannelNo>
    <isSupportIOTChannelCap>
      <!--ro, opt, bool, whether the device supports the channel configuration capability, desc:it returns true for support, related URI: (GET
/ISAPI/System/IOT/channels/<IoTChannelID>/capabilities?format=json)-->true
    </isSupportIOTChannelCap>
    <isSupportSearchMoreIOTSource>
      <!--ro, opt, bool, whether the device supports searching for the IoT devices which can be added (by multiple conditions), desc:related URI: POST
/ISAPI/System/IOT/channels/AddIOTChannelList?format=json-->true
    </isSupportSearchMoreIOTSource>
  </IOTCap>
  <isSupportShipsDetectionWithScene>
    <!--ro, opt, bool, whether the device supports ship detection by scene-->true
  </isSupportShipsDetectionWithScene>
  <isSupportPictureServer>
    <!--ro, opt, bool, whether the device supports configuring the picture storage server, desc:related URI: /ISAPI/System/PictureServer/capabilities?
format=json-->true
  </isSupportPictureServer>
  <isSupportSSDFileSystemUpgrade>
    <!--ro, opt, bool, whether the device supports updating the SSD file system, desc:related URI: /ISAPI/System/SSDFileSystem/upgrade?format=json-->true
  </isSupportSSDFileSystemUpgrade>
  <isSupportSSDFileSystemFormat>
    <!--ro, opt, bool, whether the device supports formatting SSD, desc:related URI: /ISAPI/System/SSDFileSystem/format?format=json-->true
  </isSupportSSDFileSystemFormat>
  <isSupportISUPHttpPassthrough>
    <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
  </isSupportISUPHttpPassthrough>
  <SHMCap>
    <!--ro, opt, object, SHM alarm-->
    <isSupportHighHDTemperature>
      <!--ro, opt, bool, whether the device supports HDD high temperature detection, desc:eventType: highHDTemperature-->true
    </isSupportHighHDTemperature>
    <isSupportLowHDTemperature>
      <!--ro, opt, bool, whether the device supports HDD low temperature detection, desc:eventType: lowHDTemperature-->true
```

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      <!--ro, opt, bool, whether the device supports HDD low temperature detection, desc:eventType: lowHDTemperature-->true
    </isSupportLowHDTemperature>
    <isSupportHDImpact>
      <!--ro, opt, bool, whether the device supports HDD impact detection, desc:eventType: hdImpact-->true
    </isSupportHDImpact>
    <isSupportHDBadBlock>
      <!--ro, opt, bool, whether the device supports HDD bad sector detection, desc:eventType: hdBadBlock-->true
    </isSupportHDBadBlock>
    <isSupportSevereHDFailure>
      <!--ro, opt, bool, whether the device supports HDD major fault detection, desc:eventType: severeHDFailure-->true
    </isSupportSevereHDFailure>
  </SHMCap>
  <isSupportDataPrealarm>
    <!--ro, opt, bool, whether the device supports network traffic pre-alarm event, desc:eventType: dataPrealarm-->true
  </isSupportDataPrealarm>
  <isSupportSnapshotAsync>
    <!--ro, opt, bool, whether the device supports capturing pictures asynchronously (the picture URL will be returned), desc:related URI:
/ISAPI/Streaming/channels/<ID>/picture/async?format=json-->true
  </isSupportSnapshotAsync>
  <isSupportInputProxyDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the information of the devices added to the proxy device channel, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<ID>/capabilities-->true
  </isSupportInputProxyDeviceInfo>
  <isSupportChannelOccupy>
    <!--ro, opt, bool, whether the device supports outdoor fire engine access occupied by vehicles, desc:eventType: channelOccupy-->true
  </isSupportChannelOccupy>
  <isSupportVibrationDetection>
    <!--ro, opt, bool, whether the device supports the vibration detection, desc:related URI: /ISAPI/System/channels/<ID>/vibrationDetection?format=json--
>true
  </isSupportVibrationDetection>
  <isSupportOnlineUpgradeTask>
    <!--ro, opt, bool, whether device online upgrading is supported-->true
  </isSupportOnlineUpgradeTask>
  <isSupportSSDFileSystemCapacity>
    <!--ro, opt, bool, whether the device supports getting the capacity distribution of the SSD system, desc:related URI:
/ISAPI/System/SSDFileSystem/capacity?format=json-->true
  </isSupportSSDFileSystemCapacity>
  <isSupportAbnormalReboot>
    <!--ro, opt, bool, whether the device supports abnormal reboot event, desc:eventType: abnormalReboot-->true
  </isSupportAbnormalReboot>
  <isSupportLogException>
    <!--ro, opt, bool, whether the device supports the log exception alarm, desc:eventType: logException-->true
  </isSupportLogException>
  <isSupportObjectsThrownDetection>
    <!--ro, opt, bool, whether the device supports objects thrown detection, desc:related URI:
/ISAPI/Intelligent/channels/<channelID>/objectsThrownDetection?format=json; corresponding event: objectsThrownDetection; this node is not returned if not
supported-->true
  </isSupportObjectsThrownDetection>
  <isSupportChannelsTimeSync>
    <!--ro, opt, bool, whether the device supports batch device time synchronization of the IPC/IOT channel, desc:related URL:
/ISAPI/System/time/channelsTimeSync/capabilities?format=json-->true
  </isSupportChannelsTimeSync>
  <isSupportSeachMutexFunction>
    <!--ro, opt, bool, whether the device supports searching for disabling the mutual exclusion function, desc:related URL: GET
/ISAPI/System/mutexFunction/capabilities?format=json; function description: whether the device supports searching for mutual exclusion function-->true
  </isSupportSeachMutexFunction>
  <isSupportCityCode>
    <!--ro, opt, bool, whether the device supports getting city code, desc:related URI: /ISAPI/System/cityCode?format=json-->true
  </isSupportCityCode>
  <isSupportPowerUpPicture>
    <!--ro, opt, bool, whether the device supports importing and exporting pictures displayed during startup,
desc:/ISAPI/System/powerUpPicture/capabilities?format=json-->true
  </isSupportPowerUpPicture>
  <isSupportStreamRecordStatus>
    <!--ro, opt, bool, whether the device supports getting the stream ID record status, desc:/ISAPI/System/Video/inputs/streams/recordStatus-->true
  </isSupportStreamRecordStatus>
  <isSupportSerialLogUpload>
    <!--ro, opt, bool, whether the device supports uploading serial port log, desc:true (support), false (not support)-->true
  </isSupportSerialLogUpload>
  <isSupportTimeTypeSTD>
    <!--ro, opt, bool, whether the device supports the time type of ISO8601, desc:if "true" is returned, it indicates that all the devices containing
ISO8601 parameters support adding the timeType=STD, such as /ISAPI/ContentMgmt/search?timeType=STD; the time will be implemented according to the standard--
>true
  </isSupportTimeTypeSTD>
  <isSupportAudioRecordPlan>
    <!--ro, opt, bool, whether it supports recording schedule, desc:GET/PUT /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json-->true
  </isSupportAudioRecordPlan>
  <isSupportResources>
    <!--ro, opt, bool, whether it supports resources, desc:GET /ISAPI/System/DynamicResource/Resources?format=json, Resources-->true
  </isSupportResources>
  <isSupportDeviceGroupMgr>
    <!--ro, opt, bool, whether it supports device application resources group management, desc:GET /ISAPI/System/DeviceGroupMgr/capabilities?format=json--
>true
  </isSupportDeviceGroupMgr>
  <isSupportEDRAccessList>
    <!--ro, opt, bool, whether the device supports accessing address parameters of EDR service, desc:GET/PUT /ISAPI/System/Network/EDRAccessList?
format=json-->true
  </isSupportEDRAccessList>
  <isSupportChildmanage>
    <!--ro, opt, bool, whether the device supports gateway sub-device topology management, desc:GET /ISAPI/IoTGateway/Childmanage/capabilities?format=json--
>true
  </isSupportChildmanage>
  <isSupportSearchResources>
    <!--ro, opt, bool, whether the device supports searching the resource list, desc:POST /ISAPI/System/DynamicResource/SearchResources?format=json-->true
```

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  </isSupportSearchResources>
  <isSupportSearchChildDevResourcesStatus>
    <!--ro, opt, bool, whether the device supports searching for sub-device resource status, desc:POST
/ISAPI/System/DynamicResource/SearchChildDevResourcesStatus?format=json-->true
  </isSupportSearchChildDevResourcesStatus>
  <isSupportDeleteResources>
    <!--ro, opt, bool, whether the device supports deleting resources, desc:PUT /ISAPI/System/DynamicResource/DeleteResources?format=json-->true
  </isSupportDeleteResources>
  <isSupportTestEDRServerConnect>
    <!--ro, opt, bool, whether the device supports testing the connection status of the EDR platform, desc:POST /ISAPI/System/Network/TestEDRServerConnect?
format=json-->true
  </isSupportTestEDRServerConnect>
  <isSupportProxyAIOP>
    <!--ro, opt, bool, whether the device supports proxy access of AIOP (AIOP devices connected with network cameras via NVR transparent transmission),
desc:when this node is supported, full AIOP functions (device AIOP capability, model package management, task rule management, detection data, etc.) can be
accessed via proxy transmission of ISAPI sub-device-->true
  </isSupportProxyAIOP>
  <isSupportOTAPService>
    <!--ro, opt, bool, whether the device supports OTAP service configuration, desc:/ISAPI/System/Network/OTAPService/capabilities?format=json-->true
  </isSupportOTAPService>
  <isSupportChannelPassingEvent>
    <!--ro, opt, bool, whether the device supports vehicles passing outdoor fire escape, desc:eventType:channelPassingEvent-->true
  </isSupportChannelPassingEvent>
  <isSupportAuthorizeMicAcquisitionParam>
    <!--ro, opt, bool, whether the device supports audio authorization information, desc:GET/PUT
/ISAPI/System/InformedConsentMgr/AuthorizeMicAcquisitionParam?format=json-->true
  </isSupportAuthorizeMicAcquisitionParam>
  <isSupportMutexFunctionClose>
    <!--ro, opt, bool, whether the device supports quickly disabling exclusive functions, desc:POST /ISAPI/System/mutexFunctionClose?format=json-->true
  </isSupportMutexFunctionClose>
  <isSupportAddChildDevResource>
    <!--ro, opt, bool, whether the device supports adding gateway resource information, desc:POST /ISAPI/System/DynamicResource/AddChildDevResource?
format=json-->true
  </isSupportAddChildDevResource>
  <isSupportModifyChildDevResource>
    <!--ro, opt, bool, whether the device supports editing gateway resource information, desc:POST /ISAPI/System/DynamicResource/ModifyChildDevResource?
format=json-->true
  </isSupportModifyChildDevResource>
  <isSupportUserLoginAbnormal>
    <!--ro, opt, bool, whether the device supports user login exception information, desc:GET/PUT /ISAPI/Security/users/UserLoginAbnormal?format=json-->true
  </isSupportUserLoginAbnormal>
  <isSupportSearchRecordStorage>
    <!--ro, opt, bool, whether it supports searching for record storage HDD information, desc:POST /ISAPI/ContentMgmt/search/SearchRecordStorage?
format=json-->true
  </isSupportSearchRecordStorage>
  <isSupportRTCTimeTrigger>
    <!--ro, opt, bool, whether the device supports RTC time exception linkage, desc:GET/PUT /ISAPI/Event/RTCTimeTrigger?format=json-->true
  </isSupportRTCTimeTrigger>
  <isSupportEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports configuring the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/EventRuleAndTriggerParams?format=json-->true
  </isSupportEventRuleAndTriggerParams>
  <isSupportSearchEventRuleAndTriggerParams>
    <!--ro, opt, bool, whether it supports searching for the rule linkage parameters of a specified event, desc:POST
/ISAPI/Event/EventBatchRulesMgr/SearchEventRuleAndTriggerParams?format=json-->true
  </isSupportSearchEventRuleAndTriggerParams>
  <isSupportDeviceRegionParams>
    <!--ro, opt, bool, whether it supports device area parameters, desc:GET/PUT /ISAPI/System/DeviceRegionParams?format=json-->true
  </isSupportDeviceRegionParams>
  <isSupportSearchRegionDefaultInfo>
    <!--ro, opt, bool, whether it supports searching for the default parameters for the region, desc:POST /ISAPI/System/SearchRegionDefaultInfo?format=json-
->true
  </isSupportSearchRegionDefaultInfo>
  <isSupportPicSearchByCoordinate>
    <!--ro, opt, bool, whether the device supports configuring AcuSearch parameters, desc:GET/PUT /ISAPI/ContentMgmt/PicSearchByCoordinate?format=json--
>true
  </isSupportPicSearchByCoordinate>
  <AcuSearchCap>
    <!--ro, opt, object, AcuSearch capability-->
    <targetType opt="face,human,vehicle">
      <!--ro, opt, enum, supported target types of AcuSearch, subType:string, attr:opt{req, string}, desc:"face", "human", "vehicle"-->face
    </targetType>
    <dataType opt="url,binary,modelData">
      <!--ro, opt, enum, supported image transmission format types, subType:string, attr:opt{req, string}, desc:"url", "binary", "modelData"-->url
    </dataType>
  </AcuSearchCap>
  <isSupportSearchRealStreamStatus>
    <!--ro, opt, bool, whether it supports searching for the real-time streaming status, desc:POST /ISAPI/Streaming/SearchRealStreamStatus?format=json--
>true
  </isSupportSearchRealStreamStatus>
  <isSupportSearchStreamBandwidthInfo>
    <!--ro, opt, bool, whether it supports searching for stream bandwidth status, desc:POST /ISAPI/Streaming/SearchStreamBandwidthInfo?format=json-->true
  </isSupportSearchStreamBandwidthInfo>
  <isSupportGetChanDetectResultQRCode>
    <!--ro, opt, bool-->true
  </isSupportGetChanDetectResultQRCode>
</DeviceCap>
```

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16.1.6.4 Export device configuration file

Request URL

GET /ISAPI/System/configurationData?secretkey=<secretkey>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
mode	enum	--

Request Message

None

Response Message

Binary Data

16.1.6.5 Get the languages supported by the device

Request URL

GET /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--ro, req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian",
"Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish",
"Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian,
"Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian", "Uzbek", "Kazak",
"Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
</DeviceLanguage>
```

16.1.6.6 Set device language parameters

Request URL

PUT /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian", "Bulgarian",
"Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese",
"Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian" (Slovenian), "Croatian",
"Lithuanian", "Serbian", "BrazilianPortuguese" (Brazilian Portuguese), "Indonesian", "Ukrainian", "EURSpanish"-->SimChinese
  </language>
</DeviceLanguage>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.6.7 Get the capability of configuring the device language

Request URL

GET /ISAPI/System/DeviceLanguage/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device language configuration, attr:version{req, string, protocolVersion}-->
  <language
opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,Dutch,Portuguese,Spanish,Romanian,Turkish,J
apanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietnamese,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indon
esian,Ukrainian,EURSpanish,Uzbek,Kazak,Kirghiz,Farsi,Azerbaidzhan,Burmese,Mongolian,Anglicism,Estonian">
    <!--ro, req, enum, language, subType:string, attr:opt{req, string}, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese),
"English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish",
"Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic",
"Sovenian"-Slovenian, "Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian",
"Uzbek", "Kazak", "Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
  <upgradeFirmWareEnabled>
    <!--ro, opt, bool, whether to enable upgrading the firmware-->true
  </upgradeFirmWareEnabled>
</DeviceLanguage>
```

16.1.6.8 Set restoring devices' default parameters

Request URL

PUT /ISAPI/System/factoryReset?mode=<mode>&childDevID=<devIndex>&loginPassword=<loginPassword>&module=<module>

Query Parameter

Parameter Name	Parameter Type	Description
mode	enum	--
devIndex	string	--
loginPassword	string	--
module	enum	--

Request Message

None

Response Message

creativun@gmail.com DIRECTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.6.9 Get the capability of exporting files from the device

Request URL

GET /ISAPI/System/fileExport/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "FileExportCap": {
    /*ro, opt, object, the capability of exporting files from the device*/
        "fileType": {
        /*ro, req, object, file type*/
            "@opt": ["serialLog", "coreDump", "debug", "selfDefine", "errorInfo", "playService", "communicationService", "attendanceService", "faceService",
"mainApp", "produceApp", "mainDsp", "daemon", "subAPP", "subDsp", "TVIDsp", "TVIApp", "SSDSerialLog"]
            /*ro, req, array, options, subType:string, desc:"serialLog"-serial port log, "coreDump"-core dump log, "debug"-debug log, "selfDefine"-custom
file type to be exported, "errorInfo"-error information*/
        },
        "selfDefineCmd": {
        /*ro, opt, object, custom file type to be exported, desc:custom file type to be exported,it is valid when fileType is set to "selfDefine"*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "devType": {
        /*ro, opt, object, device type*/
            "@opt": ["mainDev", "subDev", "D04"]
            /*ro, req, array, options, subType:string, desc:file type: "serialLog"-serial port log,"coreDump"-core dump log,"debug"-debug log,"selfDefine"-
custom file type to be exported,"errorInfo"-error information*/
        },
        "subSys": {
        /*ro, opt, object, partition No.*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "encryptionKey": {
        /*ro, opt, object, file encryption key*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "slotNo": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  0
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "channelID": {
        /*ro, opt, object, channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum length of the custom file type to be exported*/
            "@max":  1024
            /*ro, req, int, the maximum length of the custom file type to be exported*/
        },
        "fileList": {
        /*ro, opt, object*/
            "fileName": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  256
                /*ro, req, int*/
            },
            "time": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  25
                /*ro, req, int*/
            },
            "fileSize": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int*/
                "@max":  100000
                /*ro, req, int*/
            }
        }
    }
}
```


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16.1.6.10 Get the capability of mutual exclusion information

Request URL

GET /ISAPI/System/mutexFunction/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "function": {
    /*ro, req, object, mutually exclusive functions, desc:"VMDHumanVehicle" (motion detection), "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), "regionExiting" (region exiting), "faceCapture" (face capture), "faceRecognition" (face recognition), "IPEnhance" (IP enhancement), "SVCEnhance" (SVC enhancement), "4KOutput" (4K output), "ZCR" (zero channel), "misinfoFilter" (alarm filtering), "transcodeCfg" (stream transcoding), "AIOP_Video" (AI video task analysis), "mixedTargetDetection" (mixed targets detection), "RAID"*/
        "@opt": ["VMDHumanVehicle", "faceCapture", "faceRecognition", "IPEnhance", "SVCEnhance", "4KOutput", "linedetection", "fielddetection", "regionEntrance", "regionExiting", "ZCR", "misinfoFilter", "transcodeCfg", "mixedTargetDetection", "RAID", "perimeter"]
        /*ro, opt, array, options, subType:string,
        desc:"misinfoFilter" (filter false alarms. Conduct secondary recognition of human targets in perimeter alarm, e.g., line crossing detection and region entrance, so as to reduce false alarms caused by other factors such as swaying leaves, shadows, light, vehicles, animals, etc.)
            "transcodeCfg"  (transcoded stream compression configuration, including preview and playback transcoded stream. This transcoding configuration is exclusive with 4K output), "perimeter" (perimeter protection, corresponding to image_recognition_human in /ISAPI/Intelligent/analysisEngines. When the current engine mode is image_recognition_human, it indicates that perimeter protection algorithm mode is running)*/
    },
    "channelID": {
    /*ro, req, object, channel No.*/
        "@min":  1,
        /*ro, opt, int, the minimum value*/
        "@max":  16
        /*ro, opt, int, the maximum value*/
    }
}
```

16.1.6.11 Get the mutual exclusion information when device function exception occurred

Request URL

GET /ISAPI/System/mutexFunctionErrorMsg

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MutexFunctionErrorMsg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, mutual exclusion information, attr:version{req, string, protocolVersion}-->
  <MutexFunctionList size="10">
    <!--ro, opt, array, mutual exclusion information list, subType:object, attr:size{req, int}, desc:get the functions that are mutually exclusive to the enabled functions-->
    <MutexFunction>
      <!--ro, opt, object, mutual exclusion message-->
      <id>
        <!--ro, req, int, No.-->1
      </id>
      <describe>
        <!--ro, req, enum, description, subType:string, desc:"SMD", "vehicleDetection" (vehicle detection), "counting" (people counting), "heatMap" (heat map statistics), "fieldDetection" (intrusion detection), "lineDetection" (line crossing detection), "sceneChangeDetection" (scene change detection), "VMDHumanVehicle" (motion detection), "regionEntrance" (region entrance), "regionExiting" (region exiting), "faceCapture" (face capture), "faceRecognition" (face recognition), "IPEnhance" (IP enhancement), "SVCEnhance" (SVC enhancement), "4KOutput" (4K output), "ZCR" (zero channel), "misinfoFilter" (alarm filtering), "transcodeCfg" (stream transcoding), "AIOP_Video" (AI video task analysis)-->SMD
      </describe>
      <channelNo>
        <!--ro, req, string, channel No. corresponding to the mutually exclusive function, desc:multiple channel numbers are supported and should be separated by commas-->1
      </channelNo>
    </MutexFunction>
  </MutexFunctionList>
</MutexFunctionErrorMsg>
```

16.1.6.12 Get the network service capability

Request URL

GET /ISAPI/System/Network/capabilities

Query Parameter

None

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Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportWAN>
    <!--ro, opt, bool, whether the device supports WAN (Wide Area Network)-->true
  </isSupportWAN>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportPNP>
    <!--ro, opt, bool, whether the device supports plug-and-play protocol-->true
  </isSupportPNP>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <SnmpCap>
    <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
  <isSupportExtNetCfg>
    <!--ro, opt, bool, whether the device supports configuring extended network parameters-->true
  </isSupportExtNetCfg>
  <isSupportIPFilter>
    <!--ro, opt, bool, whether the device supports IP filtering-->true
  </isSupportIPFilter>
  <isSupportNetPreviewStrategy>
    <!--ro, opt, bool, whether the device supports network live view strategy-->true
  </isSupportNetPreviewStrategy>
  <isSupportEZVIZ>
    <!--ro, opt, bool, whether the device supports EZ protocol-->true
  </isSupportEZVIZ>
  <isSupportWPS>
    <!--ro, opt, bool, whether the device supports Wi-Fi Protected Setup-->true
  </isSupportWPS>
  <isWirelessMutexWithWirelessServer>
    <!--ro, opt, bool, whether the device supports mutual exclusion of wireless access and wireless service-->true
  </isWirelessMutexWithWirelessServer>
  <isSupportMACFilter>
    <!--ro, opt, bool, whether the device supports physical address filtering-->true
  </isSupportMACFilter>
  <GB28181Cap>
    <!--ro, opt, object, N/A-->
    <isSupportGB28181Service>
      <!--ro, opt, bool, N/A-->true
    </isSupportGB28181Service>
  </GB28181Cap>
  <VerificationCodeModification>
    <!--ro, opt, object, verification code editing-->
    <verificationCodeType opt="normal,empty">
      <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
    </verificationCodeType>
    <isSupportDeclarationURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Terms of Service-->true
    </isSupportDeclarationURL>
    <isSupportPrivacyPolicyURL>
      <!--ro, opt, bool, whether the device supports the hyperlink to the Privacy Policy-->true
    </isSupportPrivacyPolicyURL>
    <verificationCodeModify opt="true,false">
      <!--ro, opt, bool, whether the device verification code has been modified, attr:opt{req, string}-->true
    </verificationCodeModify>
    <Hyperlinks>
      <!--ro, opt, object, hyperlink-->
      <declarationURL>
```

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```
      <declarationURL>
        <!--ro, opt, string, hyperlink to the Terms of Service-->test
      </declarationURL>
      <privacyPolicyURL>
        <!--ro, opt, string, hyperlink to the Privacy Policy-->test
      </privacyPolicyURL>
    </Hyperlinks>
    <isSupportVerificationCodeCheck>
      <!--ro, opt, bool, whether the device supports verifying the verification code-->true
    </isSupportVerificationCodeCheck>
  </VerificationCodeModification>
  <isSupportIntegrate>
    <!--ro, opt, bool, whether the device supports access protocol configuration-->true
  </isSupportIntegrate>
  <isSupportPlatformAccess>
    <!--ro, opt, bool, whether the device supports platform access, desc:GET /ISAPI/System/Network/platformAccessFilter/capabilities (platform access IP
address filtering configuration (Ehome allowlist configuration))-->true
  </isSupportPlatformAccess>
  <isSupportResourceStatistics>
    <!--ro, opt, bool, whether the device supports getting the network resource information-->true
  </isSupportResourceStatistics>
  <isSupportBandwidthLimit>
    <!--ro, opt, bool, whether the device supports bandwidth limit-->true
  </isSupportBandwidthLimit>
  <isSupportPOEPortsDisableAdaptiveServer>
    <!--ro, opt, bool, whether the device supports the POE protocol-->true
  </isSupportPOEPortsDisableAdaptiveServer>
  <isSupportPOEConfiguration>
    <!--ro, opt, bool, whether the device supports configuring the POE NIC parameters-->true
  </isSupportPOEConfiguration>
  <isSupportGetLinkSocketIP>
    <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
  </isSupportGetLinkSocketIP>
  <isSupportWebSocket>
    <!--ro, opt, bool, whether the device supports WebSocket-->true
  </isSupportWebSocket>
  <Adaption>
    <!--ro, opt, object, network self-adaption-->
    <streamType opt="0,1,2,3,4,5,7,8,9,10">
      <!--ro, opt, enum, stream type that supports network self-adaption during live view, subType:int, attr:opt{req, string}, desc:stream types that
support network self-adaption during live view: 0 (main stream), 1 (sub-stream), 2 (third stream), 3 (fourth stream / virtual stream), 4 (fifth stream), 5
(sixth stream), 7 (seventh stream), 8 (eighth stream), 9 (ninth stream), 10 (tenth stream), and so on-->0
    </streamType>
    <isSupportPlayback>
      <!--ro, opt, bool, whether the device supports playback-->true
    </isSupportPlayback>
  </Adaption>
  <isSupportEventDataOverWebSocket opt="true,false">
    <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
  </isSupportEventDataOverWebSocket>
  <isSupportEZVIZTiming>
    <!--ro, opt, bool, whether the device supports EZ platform synchronization (enabledTiming of message returned by/ISAPI/System/Network/EZVIZ)-->true
  </isSupportEZVIZTiming>
  <isSupportPingTest>
    <!--ro, opt, bool, whether the device supports ping test, desc:related URI: /ISAPI/System/Network/pingtest/capabilities-->true
  </isSupportPingTest>
  <isSupportDSTPServerCfg>
    <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
  </isSupportDSTPServerCfg>
  <isSupportDSTPProtocolsCap>
    <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
  </isSupportDSTPProtocolsCap>
  <isSupportIPAdaptiveParams>
    <!--ro, opt, bool, whether the device supports self-adaptive IP address parameters, desc:GET/PUT /ISAPI/System/Network/IPAdaptiveParams?format=json--
>true
  </isSupportIPAdaptiveParams>
  <isSupportUSBNetworkCard>
    <!--ro, opt, bool, whether the device supports USB NIC-->true
  </isSupportUSBNetworkCard>
  <isSupportSearchOutlookAuthResult>
    <!--ro, opt, bool-->true
  </isSupportSearchOutlookAuthResult>
  <isSupportRequestOutlookAuth>
    <!--ro, opt, bool, whether the device supports applying authentication request of Outlook email, desc:POST
/ISAPI/System/Network/Email/RequestOutlookAuth?format=json-->true
  </isSupportRequestOutlookAuth>
  <isSupportOutlookEmailList>
    <!--ro, opt, bool, whether the device supports Outlook email information, desc:GET or PUT /ISAPI/System/Network/Email/OutlookEmailList?format=json--
>true
  </isSupportOutlookEmailList>
</NetworkCap>
```

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16.1.6.13 Set SSH parameters

Request URL

PUT /ISAPI/System/Network/ssh?readerID=<readerID>&type=<type>&SOCChipID=<SOCChipID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--
type	enum	--
SOCChipID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SSH xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--wo, req, bool, whether to enable the function-->true
  </enabled>
  <port>
    <!--wo, opt, int-->22
  </port>
</SSH>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional
Error, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot"
(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.1.6.14 Get the capability of importing and exporting parameters of pictures displayed during startup

Request URL

GET /ISAPI/System/powerUpPicture/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "pictureFormat": {
    /*ro, opt, object, picture format*/
        "@opt": ["GIF", "BMP", "JPG", "PNG"]
        /*ro, req, array, options, subType:string*/
    },
    "pictureSize": {
    /*ro, opt, object, picture size, desc:unit: KB*/
        "@min":  1,
        /*ro, req, int, options*/
        "@max":  2048
        /*ro, req, int, options*/
    },
    "pictureURL": {
    /*ro, opt, object, picture URL*/
        "@min":  1,
        /*ro, req, int, options*/
        "@max":  256
        /*ro, req, int, options*/
    },
    "pictureResolution": {
    /*ro, opt, object, picture resolution*/
        "@opt": ["600*1024", "800*1280", "800*480", "1024*768"]
        /*ro, req, array, options, subType:string*/
    },
    "ImportCap": {
    /*ro, opt, object, importing capability*/
        "filePathType": {
        /*ro, opt, object, file path type*/
            "@opt": ["binary", "URL"]
            /*ro, req, array, options, subType:string*/
        },
        "applyType": {
        /*ro, opt, object, application range of the imported pictures*/
            "@opt": ["all", "powerUpPicture", "noVideoPicture"],
            /*ro, req, array, options, subType:string*/
            "@def":  "powerUpPicture"
            /*ro, req, string, the default value*/
        }
    },
    "ExportCap": {
    /*ro, opt, object, exporting capability*/
        "filePathType": {
        /*ro, opt, object, file path type*/
            "@opt": ["localPath", "URL"]
            /*ro, req, array, options, subType:string*/
        }
    }
}
```

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16.1.6.15 Import startup pictures

Request URL

POST /ISAPI/System/powerUpPicture?format=json

Query Parameter

None

Request Message

```
{
    "filePathType":  "binary",
    /*req, enum, file path type, subType:string, desc:“binary”, "URL", when the value of filePathType is "URL", the message is in application/json format.
When the value of filePathType is "binary", the the message is in multipart/form-data*/
    "pictureURL":  "test",
    /*opt, string, picture URL, range:[1,256], dep:and,{$.filePathType,eq,URL}*/
    "applyType":  "powerUpPicture"
    /*opt, enum, imported picture application range, subType:string, desc:"all" (all the pictures), "powerUpPicture" (pictures displayed during startup),
"noVideoPicture" (no captured pictures of the video). If this node does not exist, it indicates that the default imported picture is used for the startup*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
powerUpPicture	[Message content]	application/json	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:this field is required when the status code is not 1 (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not "1"*/
}
```

16.1.6.16 Delete startup picture

Request URL

DELETE /ISAPI/System/powerUpPicture?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:this field is required when the status code is not 1 (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status description is not "ok" (success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not "1"*/
}
```

16.1.6.17 Get capability of startup picture parameters

Request URL

GET /ISAPI/System/powerUpPictureCfg/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "enabled": {
    /*ro, opt, object, whether to enable startup picture*/
        "@opt": [true, false]
        /*ro, req, array, range, subType:bool*/
    }
}
```

16.1.6.18 Get startup picture parameters

Request URL

GET /ISAPI/System/powerUpPictureCfg?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "enabled":  true,
    /*ro, opt, bool, whether to enable startup picture*/
    "importPowerUpPictureStatus":  "on"
    /*ro, opt, enum, string, desc:"on" (using), "off" (uploaded but not be used), "noUpload" (not uploaded)*/
}
```

16.1.6.19 Configure startup picture parameters

Request URL

PUT /ISAPI/System/powerUpPictureCfg?format=json

Query Parameter

None

Request Message

```
{
    "enabled":  true
    /*opt, bool, whether to enable startup picture*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:this field is required when the status code is not 1 (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status description is not "ok"(success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status description is not "ok"(success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not "1"*/
}
```

16.1.6.20 Reboot device

Request URL

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>&controllerID=<controllerID>

Query Parameter

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Parameter Name	Parameter Type	Description
devIndex	string	Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json.
module	enum	The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera).
loginPassword	string	--
controllerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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16.1.6.21 Get device status

Request URL

GET /ISAPI/System/status

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device status, attr:version{opt, string, protocolVersion}-->
  <currentDeviceTime>
    <!--ro, opt, datetime, device current time-->1970-01-01T00:00:00+08:00
  </currentDeviceTime>
  <deviceUpTime>
    <!--ro, opt, int, device running time, unit:s-->1200
  </deviceUpTime>
  <CPUList>
    <!--ro, opt, array, CPU data list, subType:object-->
    <CPU>
      <!--ro, opt, object, CPU data-->
      <cpuDescription>
        <!--ro, req, string, description-->test
      </cpuDescription>
      <cpuUtilization>
        <!--ro, req, int, CPU usage, range:[0,100], desc:CPU usage-->1
      </cpuUtilization>
    </CPU>
  </CPUList>
  <MemoryList>
    <!--ro, opt, array, memory list, subType:object-->
    <Memory>
      <!--ro, opt, object, memory-->
      <memoryDescription>
        <!--ro, req, string, description-->test
      </memoryDescription>
      <memoryUsage>
        <!--ro, req, float, used memory size, unit:MB-->1.0
      </memoryUsage>
      <memoryAvailable>
        <!--ro, req, float, available memory size, unit:MB-->1.0
      </memoryAvailable>
    </Memory>
  </MemoryList>
</DeviceStatus>
```

16.1.6.22 Get the video analysis task capability

Request URL

GET /ISAPI/System/Video/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->0
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->0
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of local menus that can be displayed-->0
  </menuNums>
  <externalChannelNum>
    <!--ro, opt, bool, number of extended analog channels-->true
  </externalChannelNum>
  <isSupportHeatmap>
    <!--ro, opt, bool, whether the device supports heat map-->true
  </isSupportHeatmap>
  <isSupportCounting>
    <!--ro, opt, bool, whether the device supports people counting-->true
  </isSupportCounting>
  <countingType>
    <!--ro, opt, enum, statistics type, subType:string, desc:statistics type-->human
  </countingType>
  <isSupportPicture>
    <!--ro, opt, bool, whether the device supports OSD picture overlay, desc:related URI: /ISAPI/System/Video/inputs/channels/<channelID>/image/picture--
>true
  </isSupportPicture>
  <isSupportPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view-->true
  </isSupportPreviewSwitch>
  <isSupportRecodStatus>
```

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```
    <!--ro, opt, bool, whether the device supports searching for the recording status-->true
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--ro, opt, bool, whether the device supports privacy mask-->true
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
  </isSupportBinocularPreviewSwitch>
  <isSupportCalibCheck>
    <!--ro, opt, bool, whether the device supports calibration verification-->true
  </isSupportCalibCheck>
  <isSupportPIP>
    <!--ro, opt, bool-->true
  </isSupportPIP>
  <isSupportFocusVideoMode>
    <!--ro, opt, bool-->true
  </isSupportFocusVideoMode>
  <isSupportExternalChannel>
    <!--ro, opt, bool, whether the device supports extended analog channels-->true
  </isSupportExternalChannel>
  <isSupportMultiChannelCounting>
    <!--ro, opt, bool-->true
  </isSupportMultiChannelCounting>
  <isSupportCountingCollection>
    <!--ro, opt, bool, whether the device supports people counting ANR-->true
  </isSupportCountingCollection>
  <isSupportHeatmapCollection>
    <!--ro, opt, bool, whether the device supports heat map data ANR-->true
  </isSupportHeatmapCollection>
  <channelFlexible opt="name,enable,online,linknum">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->name
  </channelFlexible>
  <isSupportOutputsResource>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/resource?format=json, desc:related URI: /ISAPI/System/Video/outputs/resource?format=json--
>true
  </isSupportOutputsResource>
  <OSDLanguage opt="GBK,EUC-KR,Hebrew" def="GBK">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->GBK
  </OSDLanguage>
  <isSupportMixedChannel>
    <!--ro, opt, bool-->true
  </isSupportMixedChannel>
  <isSupportMixedChannelStatus>
    <!--ro, opt, bool-->true
  </isSupportMixedChannelStatus>
  <isSupportOutputCourseware>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/courseware/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/courseware/capabilities?format=json-->true
  </isSupportOutputCourseware>
  <isSupportVideoInputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/inputs/mode/capabilities, desc:related URI: /ISAPI/System/Video/inputs/mode/capabilities-->true
  </isSupportVideoInputMode>
  <isSupportVideoOutputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/mode/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/mode/capabilities?format=json-->true
  </isSupportVideoOutputMode>
  <PreviewMode>
    <!--ro, opt, object-->
    <PIPType>
      <!--ro, opt, object-->
      <mainScreenChannelID>
        <!--ro, opt, int-->0
      </mainScreenChannelID>
      <subScreenChannelID>
        <!--ro, opt, int-->0
      </subScreenChannelID>
    </PIPType>
  </PreviewMode>
  <isSupportViewTag>
    <!--ro, opt, bool-->true
  </isSupportViewTag>
  <isSupportMenuStatus>
    <!--ro, opt, bool-->true
  </isSupportMenuStatus>
</VideoCap>
```

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16.1.6.23 Get the thermal capability

Request URL

GET /ISAPI/Thermal/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ThermalCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, thermal capability, attr:version{req, string, protocolVersion}-->
  <isSupportFireDetection>
    <!--ro, opt, bool, whether the device supports fire detection, desc:Related URI: /ISAPI/Thermal/channels/<ID>/fireDetection/capabilities-->true
  </isSupportFireDetection>
  <isSupportThermometry>
    <!--ro, opt, bool, whether the device supports thermography, desc:Related URI: /ISAPI/Thermal/channels/<ID>/thermometry/<SID>/capabilities-->true
  </isSupportThermometry>
  <isSupportRealtimeThermometry>
    <!--ro, opt, bool, whether supports uploading real-time temperature measurement data-->true
  </isSupportRealtimeThermometry>
  <isSupportNVR>
    <!--ro, opt, bool-->true
  </isSupportNVR>
  <isSupportThermometrySchedule>
    <!--ro, opt, bool-->true
  </isSupportThermometrySchedule>
  <isSupportTemperatureSchedule>
    <!--ro, opt, bool-->true
  </isSupportTemperatureSchedule>
  <isSupportFireDetectionSchedule>
    <!--ro, opt, bool-->true
  </isSupportFireDetectionSchedule>
  <isSupportTemperatureCollection>
    <!--ro, opt, bool, whether device supports temperature ANR, desc:whether device supports temperature ANR-->true
  </isSupportTemperatureCollection>
</ThermalCap>
```

16.1.6.24 Upload device status change

EventType:changedStatus

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded, step:1, desc:the event triggering frequency. The pulse event is an event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the face capture. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again. It can be regarded as the event triggering frequency*/
    "eventType":  "changedStatus",
    /*ro, req, string, event type*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid event or event starts), "inactive" (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "change status",
    /*ro, req, string, event description*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name), desc:it should be the same as the channelName in the message of URI (/ISAPI/Streaming/channels/<ID>)*/
    "deviceID":  "test0123",
    /*ro, opt, string, PUID, range:[1,64]*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers*/
    "ChangedStatus": {
    /*ro, req, object, status upload message structure*/
        "seq":  0,
        /*ro, req, int, alarm No., desc:a 32-character integer*/
        "devStatus":  0,
        /*ro, opt, enum, value of device status, subType:int, desc:0-normal, 1-CPU usage, higher than 85%, 2-hardware error (e.g., serial port exception)*/
        "devCapHash":  "fdsafdsfdsafdsfji",
        /*ro, opt, string, device capability full hash value, desc:generated by md5, and encoded by base 64. For NVR, the hash value of the managed network camera should be the calculation source*/
        "zeroChanEnable":  true,
        /*ro, opt, bool, whether to enable zero-channel, desc:true-enabled, false-disabled*/
        "cycleRecordEnable":  true,
        /*ro, opt, bool, whether to enable overwritten recording, desc:true-enabled, false-disabled*/
        "chanChanged": [1, 2, 3],
```

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```
        /*ro, opt, array, channel status change, subType:int, desc:if channels of the device are deleted, edited, or added, the hash value will change*/
        "IoTChanChanged": [1, 2, 3],
        /*ro, opt, array, IoT channel status change, subType:int, desc:if channels of the device are deleted, edited, or added, the hash value will change*/
        "hdChanged": [1, 2, 3],
        /*ro, opt, array, device HDD status, subType:int, desc:device HDD is deleted or added*/
        "ChanStatus": [
        /*ro, opt, array, channel change, subType:object*/
            {
                "chanNo": 1,
                /*ro, opt, int, channel No., step:1*/
                "enable": 1,
                /*ro, opt, enum, whether to enable the analog channel, subType:int, desc:0 (disable), 1 (enable). It is valid for analog channel only*/
                "online": 1,
                /*ro, opt, enum, online status, subType:int, desc:0 (offline), 1 (online), 2 (sleeping), 3 (waking up)*/
                "record": 1,
                /*ro, opt, enum, whether the device is recording, subType:int, desc:0 (not recording), 1 (recording)*/
                "recordStatus": 0,
                /*ro, opt, enum, recording status, subType:int, desc:0 (normal), 1 (abnormal, the device is not recording according to the recording
schedule), 2 (abnormal, manual recording failed), 3 (abnormal, the disk error), 4 (abnormal, the network camera is offline), 5 (abnormal, other cause)*/
                "signal": 0,
                /*ro, opt, enum, signal status, subType:int, desc:0 (normal), 1 (signal loss)*/
                "arming": 0
                /*ro, opt, enum, network camera arming status by NVR, subType:int, desc:0 (succeeded), 1 (failed)*/
            }
        ],
        "HDStatus": [
        /*ro, opt, array, HDD status list, subType:object*/
            {
                "hdNo": 1,
                /*ro, opt, int, HDD No.*/
                "status": 0
                /*ro, opt, enum, HDD status, subType:int, desc:0 (active), 1 (sleeping), 2 (abnormal), 3 (sleepy HDD error), 4 (unformatted), 5
(disconnected, for network HDD), 6 (formatting), 7 (the HDD is full), 8 (other error)*/
            }
        ],
        "CapChanged": [
        /*ro, opt, array, capability set change, subType:object*/
            {
                "name": "FireDetection",
                /*ro, req, string, root node that shows the capability set change*/
                "chan": 1,
                /*ro, opt, int, channel No., desc:it is required for getting channel capability, and it can be empty for getting device capability*/
                "url": "/ISAPI/Thermal/channels/1/fireDetection/capabilities",
                /*ro, opt, string, URL of the capability set, desc:required for ISAPI capability set*/
                "abilityType": "0x700"
                /*ro, opt, string, capability type, desc:required for capability outside ISAPI*/
            }
        ],
        "IoTChanStatus": [
        /*ro, opt, array, IoT channel status list, subType:object*/
            {
                "IoTChannel": 1,
                /*ro, req, int, IoT channel No.*/
                "online": 1,
                /*ro, req, enum, online status, subType:int, desc:0 (offline), 1 (online)*/
                "channelName": "test"
                /*ro, req, string, channel name, range:[1,32]*/
            }
        ],
        "intranetIPAddressStatus": {
        /*ro, opt, object, intranet IP address, desc:upload intranet IP address change. It is returned only in EZ link. Application scenario: the technical
support can operate devices by the IP address after remotely connecting to the LAN of the device. It can be used to reduce unnecessary communication and
improve efficiency. Currently, it is uploaded every five minutes*/
            "intranetIPAddress": [
            /*ro, opt, array, intranet IP address list, subType:object*/
                {
                    "id": 1,
                    /*ro, req, int, network interface ID, range:[1,4]*/
                    "ipVersion": "ipv4",
                    /*ro, opt, enum, IP address type, subType:string, desc:"ipv4", "ipv6"*/
                    "ipv4Address": "test",
                    /*ro, req, string, IPv4 address*/
                    "ipv6Address": "test"
                    /*ro, req, string, IPv6 address*/
                }
            ]
        }
    }
}
```

16.1.7 System Upgrade

16.1.7.1 Get the device upgrade progress

Request URL

GET /ISAPI/System/upgradeStatus?format=json

Query Parameter

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None

Request Message

None

Response Message

```
{
    "requestURL":  "/ISAPI/Streaming/channels/1",
    /*ro, opt, string, request URL*/
    "statusCode":  "test",
    /*ro, req, string, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, This field is required when the value of statusCode is not 1, and it corresponds to subStatusCode.*/
    "errorMsg":  "ok",
    /*ro, opt, string, This field is required when the value of statusCode is not 1. Detailed error description of a certain parameter can be provided*/
    "upgrading":  "TRUE",
    /*ro, opt, string, whether the device is upgrading: “TRUE" (upgrading), "FALSE" (not in upgrading)*/
    "percent":  22,
    /*ro, opt, int, upgrade progress (% complete)*/
    "idList": [
    /*ro, opt, array, ID list, subType:object*/
        {
            "id":  "test",
            /*ro, req, string, analysis unit ID*/
            "percent":  22,
            /*ro, opt, int, upgrade progress (% complete)*/
            "status":  "test"
            /*ro, opt, string, "backingUp" (backing up upgrade)*/
        }
    ]
}
```

16.1.7.2 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```

16.1.8 Time Settings

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16.1.8.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:daylight saving time configuration in time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

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16.1.8.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
    <FailedNodeInfo>
      <!--ro, opt, object, information of failed nodes-->
      <nodeID>
        <!--ro, req, string, node ID, range:[0,64]-->test
      </nodeID>
      <nodeIP>
        <!--ro, req, string, node IP, range:[0,20]-->test
      </nodeIP>
      <reason>
        <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
      </reason>
    </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

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16.1.8.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:“NTP” (NTP time synchronization), “manual” (manual time
synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or
no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:“local” (local time), “UTC” (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is
configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-
MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports
configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device’s NTP service information with the camera, desc:related URI:
/ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
  <isSupportTimeZoneListInfo>
    <!--ro, opt, bool-->true
  </isSupportTimeZoneListInfo>
  <isSupportAutoEnterSummerTime>
    <!--ro, opt, bool-->true
  </isSupportAutoEnterSummerTime>
  <ethernetPort min="0" max="255">
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
  </ethernetPort>
  <IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/D
akar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Afr
ica/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,Americ
a/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,A
merica/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Tur
k,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,
America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,
America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monter
rey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Sal
em,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-
Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,Amer
ica/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bah
rain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta
,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyan
g,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlant
ic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbou
rne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe
/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Euro
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
```

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```
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Am
erica/Lower_Princes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Qyzylorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apia">
    <!--ro, opt, enum, IANA synchronization, subType:string, attr:opt{opt, string}-->Africa/Abidjan
  </IANA>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Ekaterinburg Standard Time,Pakistan Standard Time,Qyzylorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->Dateline Standard Time
  </windowsZone>
  <isSupportTimeStateParam>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
  </isSupportTimeStateParam>
  <localTimeDST min="0" max="256">
    <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTimeDST>
  <isSupportStandard>
    <!--ro, opt, bool-->true
  </isSupportStandard>
  <isSupportPTPParams>
    <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
  </isSupportPTPParams>
  <isSupportTimeZoneLock>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
  </isSupportTimeZoneLock>
</Time>
```


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16.1.8.4 Get device local time

Request URL

GET /ISAPI/System/time/localTime

Query Parameter

None

Request Message

None

Response Message

Binary Data

16.1.8.5 Set device local time

Request URL

PUT /ISAPI/System/time/localTime

Query Parameter

None

Request Message

Binary Data

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.1.8.6 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.1.8.7 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```

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16.1.8.8 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->false
  </enabled>
</NTPServer>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.1.8.9 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```

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DETEKTOR

16.1.8.10 Get the configuration capability of a specific NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)--
>hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval,
unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>
```

16.1.8.11 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>
```

16.1.8.12 Set the device time type

Request URL

PUT /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

```
{
    "TimeType": {
    /*req, object, time type information, desc:time type information*/
        "type":  "UTC",
        /*req, enum, time type, subType:string, desc:"UTC, local "*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16.1.8.13 Get the device time type

Request URL

GET /ISAPI/System/time/timeType?format=json

Query Parameter

None

Request Message

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DETEKTOR

None

Response Message

```
{
    "TimeType": {
    /*ro, req, object*/
        "type":  "UTC",
        /*ro, req, enum, value: local,UTC, subType:string, desc:"UTC,local"*/
        "displayFormat":  "MM/dd/yyyy hh:mm"
        /*ro, opt, enum, subType:string*/
    }
}
```

16.1.8.14 Get time zone

Request URL

GET /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

None

16.1.8.15 Set time zone

Request URL

PUT /ISAPI/System/time/timeZone

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2 Video (General)

16.2.1 Backlight Compensation

16.2.1.1 Set the BLC (Backlight Compensation) parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/BLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, BLC area, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable BLC-->true
  </enabled>
  <BLCMode>
    <!--opt, enum, BLC mode, subType:string, desc:"UP", "DOWN", "LEFT", "RIGHT", "CENTER", "MULTI-AREA", "Region", "AUTO"-->UP
  </BLCMode>
  <BLCLevel>
    <!--opt, int, BLC level-->1
  </BLCLevel>
  <BLCRegionList>
    <!--opt, array, List of BLC areas, subType:object-->
    <BLCRegion>
      <!--opt, object, BLC area-->
      <id>
        <!--req, int, area No.-->1
      </id>
      <RegionCoordinatesList>
        <!--req, array, coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.1.2 Get the BLC (Backlight Compensation) parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/BLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, BLC mode: "UP, DOWN, LEFT, RIGHT, CENTER, MULTI-AREA, Region", attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <BLCMode>
    <!--ro, opt, enum, BLC mode, subType:string, desc:"UP", "DOWN", "LEFT", "RIGHT", "CENTER", "MULTI-AREA", "Region", "AUTO"-->UP
  </BLCMode>
  <BLCLevel>
    <!--ro, opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
  <switchMode>
    <!--ro, opt, enum, subType:string-->time
  </switchMode>
  <timeMode>
    <!--ro, opt, object, dep:and,{$.BLC.switchMode,eq,time}-->
    <timeRangeList>
      <!--ro, req, array, subType:object, range:[0,1]-->
      <timeRange>
        <!--ro, req, object-->
        <startTime>
          <!--ro, req, time-->08:00:00
        </startTime>
        <endTime>
          <!--ro, req, time-->18:00:00
        </endTime>
      </timeRange>
    </timeRangeList>
  </timeMode>
</BLC>
```

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DETEKTOR

16.2.1.3 Get the backlight compensation (in day mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/BLC/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <BLCMode>
    <!--ro, opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--ro, opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

16.2.1.4 Set the backlight compensation (in day mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/BLC/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <BLCMode>
    <!--opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--opt, array, subType:object-->
    <BLCRegion>
      <!--opt, object-->
      <id>
        <!--req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object-->
          <positionX>
            <!--req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.1.5 Get the backlight compensation (in night mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/BLC/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <BLCMode>
    <!--ro, opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--ro, opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--ro, opt, array, subType:object-->
    <BLCRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

16.2.1.6 Set the backlight compensation (in night mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/BLC/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<BLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <BLCMode>
    <!--opt, enum, subType:string-->UP
  </BLCMode>
  <BLCLevel>
    <!--opt, int-->1
  </BLCLevel>
  <BLCRegionList>
    <!--opt, array, subType:object-->
    <BLCRegion>
      <!--opt, object-->
      <id>
        <!--req, int-->1
      </id>
      <RegionCoordinatesList>
        <!--req, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object-->
          <positionX>
            <!--req, int, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </BLCRegion>
  </BLCRegionList>
</BLC>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.1.7 Set the WDR (Wide Dynamic Range) parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/hdr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode>
    <!--req, enum, WDR (Wide Dynamic Range) mode, subType:string, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto"-->Sdr
  </hdrMode>
  <HdrSwitch>
    <!--opt, object-->
    <hdrSwitchMode>
      <!--req, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness. <timeSwitch> nad <lightSwitch>
are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--req, object, enable WDR by time-->
      <startHour>
        <!--opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--opt, int, start time (hour)-->1
      </endHour>
      <endMinute>
        <!--opt, int, start time (minute)-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel>
    <!--opt, enum, WDR level, subType:string, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel>
    <!--opt, int-->1
  </SdrLevel>
</Hdr>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.1.8 Get the WDR (Wide Dynamic Range) parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/hdr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode>
    <!--ro, req, enum, WDR mode, subType:string, desc:"cLose", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto".-->Sdr
  </hdrMode>
  <HdrSwitch>
    <!--ro, opt, object-->
    <hdrSwitchMode>
      <!--ro, req, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness. <timeSwitch> nad
<lightSwitch> are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--ro, req, object, enable WDR by time-->
      <startHour>
        <!--ro, opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--ro, opt, int, start time (hour)-->1
      </endHour>
      <endMinute>
        <!--ro, opt, int, start time (minute)-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel>
    <!--ro, opt, enum, WDR level, subType:string, desc:"Level1", "Level2", "Level3", "Level4", "Level5", "Level6", "Level7"-->level7
  </hdrLevel>
  <SdrLevel>
    <!--ro, opt, int-->1
  </SdrLevel>
</Hdr>
```

16.2.1.9 Get the WDR (Wide Dynamic Range) capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/hdr/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode opt="close,realHdr,digitalHdr,Sdr,BLC">
    <!--ro, req, enum, WDR mode, subType:string, attr:opt{req, string}, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto".--
>Sdr
  </hdrMode>
  <HdrSwitch>
    <!--ro, opt, object-->
    <hdrSwitchMode opt="open,timeCtrl,lightCtrl">
      <!--ro, req, enum, switch mode, subType:string, attr:opt{req, string}, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness.
<timeSwitch> nad <lightSwitch> are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--ro, req, object, enable WDR by time-->
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->1
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->1
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->1
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel opt="level1,level2,level3,level4,level5,level6,level7">
    <!--ro, opt, enum, WDR level, subType:string, attr:opt{req, string}, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel min="0" max="100" def="50">
    <!--ro, opt, int, dep:and,{$.Hdr.hdrMode,eq,Sdr}, attr:min{req, int},max{req, int},def{req, int}-->1
  </SdrLevel>
</Hdr>
```

16.2.1.10 Get the highlight compensation (in auto mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/HLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <HLCMode>
    <!--ro, opt, enum, subType:string-->close
  </HLCMode>
  <HLCLevel>
    <!--ro, opt, int-->1
  </HLCLevel>
  <timePlanList>
    <!--ro, opt, array, subType:object, dep:and,{$.HLC.HLCMode,eq,timing}-->
    <startTime>
      <!--ro, req, time-->08:00:00
    </startTime>
    <endTime>
      <!--ro, req, time-->10:00:00
    </endTime>
  </timePlanList>
</HLC>
```

16.2.1.11 Set the highlight compensation (in auto mode) parameters of a specified channel

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Request URL

PUT /ISAPI/Image/channels/<channelID>/HLC

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HLC xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <HLCMode>
    <!--opt, enum, subType:string-->[close
  </HLCMode>
  <HLCLevel>
    <!--opt, int-->1
  </HLCLevel>
</HLC>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.1.12 Set the wide dynamic range (WDR) parameters in auto mode of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/WDR

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WDR xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->auto
  </mode>
  <WDRLevel>
    <!--opt, int, range:[1,100]-->50
  </WDRLevel>
  <WDRContrastLevel>
    <!--opt, int-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--opt, int-->1
  </WDRLevel1>
</WDR>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.1.13 Get the WDR (Wide Dynamic Range) parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/WDR

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WDR xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, WDR mode, subType:string, desc:WDR mode-->auto
  </mode>
  <WDRLevel>
    <!--ro, opt, int, WDR level, range:[0,100], desc:WDR level-->50
  </WDRLevel>
  <WDRContrastLevel>
    <!--ro, opt, int-->1
  </WDRContrastLevel>
  <WDRLevel1>
    <!--ro, opt, int-->1
  </WDRLevel1>
  <timeMode>
    <!--ro, opt, object, dep:and,{$.WDR.mode,eq,time}-->
    <timeRangeList>
      <!--ro, req, array, subType:object, range:[0,1]-->
      <timeRange>
        <!--ro, req, object-->
        <startTime>
          <!--ro, req, time-->08:00:00
        </startTime>
        <endTime>
          <!--ro, req, time-->18:00:00
        </endTime>
      </timeRange>
    </timeRangeList>
  </timeMode>
  <WDRStreakInhibition>
    <!--ro, opt, bool-->false
  </WDRStreakInhibition>
  <WDRCloseToOpenFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDRCloseToOpenFilterLevel>
  <WDROpenToCloseFilterLevel>
    <!--ro, opt, int, range:[1,9]-->5
  </WDROpenToCloseFilterLevel>
</WDR>
```

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DETEKTOR

16.2.2 Exposure Settings

16.2.2.1 Set the exposure mode for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/exposure

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--req, enum, exposure type, subType:string, desc:exposure type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--opt, int, automatic aperture level-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <Type>
      <!--opt, enum, "AUTO", "MANUAL", subType:string, desc:"AUTO", "MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--opt, int, focal length-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
  </OverexposeSuppress>
  <pIris>
    <!--opt, object, aperture-->
    <pIrisType>
      <!--opt, enum, aperture type, subType:string, desc:aperture type-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--opt, int, aperture level-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--opt, object, face exposure-->
    <enabled>
      <!--opt, bool, whether to enable face exposure-->true
    </enabled>
    <sensitivity>
      <!--opt, int, sensitivity of face exposure-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is available when ExposureType is set to "pIris-General"-->
    <irisLevel>
      <!--opt, int, aperture level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
</Exposure>
```

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DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.2.2 Get the exposure mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/exposure?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--ro, req, enum, lens type, subType:string, desc:lens type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--ro, opt, int-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <Type>
      <!--ro, opt, enum, this node depends on <enabled>,"AUTO,MANUAL", subType:string, desc:this node depends on <enabled>,"AUTO,MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
    <hightLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </hightLightDistanceLevel>
    <lowLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </lowLightDistanceLevel>
  </OverexposeSuppress>
  <pIris>
    <!--ro, opt, object, this node depends on <ExposureType>, "AUTO,MANUAL"-->
    <pIrisType>
      <!--ro, opt, enum, subType:string-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--ro, opt, int, this node depends on <pIrisType>-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--ro, opt, object, face exposure-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivity>
      <!--ro, opt, int, sensitivity-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--ro, opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is valid when <ExposureType> is set to "pIris-General"-->
    <irisLevel>
      <!--ro, opt, int, iris level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
  <combinationParams>
    <!--ro, opt, int, range:[0,100]-->50
  </combinationParams>
  <opticalFilter>
    <!--ro, opt, enum, subType:string-->none
  </opticalFilter>
  <lowGhostingExposure>
    <!--ro, opt, object-->
    <lowGhostingShutterLevel>
      <!--ro, opt, enum, subType:string-->1/100
    </lowGhostingShutterLevel>
  </lowGhostingExposure>
  <exposureCompensateLevel>
    <!--ro, opt, int, range:[-100,100]-->0
  </exposureCompensateLevel>
  <exposureControl>
    <!--ro, opt, enum, subType:string-->center
  </exposureControl>
</Exposure>
```

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DETEKTOR

16.2.2.3 Set the iris parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/iris

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Iris xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <IrisLevel>
    <!--req, int-->1
  </IrisLevel>
  <irisSpeed>
    <!--opt, int-->1
  </irisSpeed>
</Iris>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.2.4 Get the iris parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/iris

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Iris xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <IrisLevel>
    <!--ro, req, int-->1
  </IrisLevel>
  <irisSpeed>
    <!--ro, opt, int-->1
  </irisSpeed>
</Iris>
```

16.2.2.5 Set the manual exposure parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/manualExposure?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MExposure": {
    /*req, object*/
        "enabled":  true,
        /*req, bool*/
        "exposureLevel":  10,
        /*opt, int, range:[10,1000000]*/
        "gainLevel":  1
        /*opt, int, range:[0,100]*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

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DETEKTOR

16.2.2.6 Get the manual exposure parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/manualExposure?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.

Request Message

None

Response Message

```
{
    "MExposure": {
    /*ro, req, object*/
        "enabled":  true,
        /*ro, req, bool*/
        "exposureLevel":  10,
        /*ro, opt, int, range:[10,1000000]*/
        "gainLevel":  1
        /*ro, opt, int, range:[0,100]*/
    }
}
```

16.2.2.7 Set the multi-shutter parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/multishut

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMultishut xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <doubleShutEnable>
    <!--req, bool, whether to enable dual-shutter-->true
  </doubleShutEnable>
  <codecType>
    <!--req, enum, type, subType:string, desc:type-->all_ia
  </codecType>
  <threeShutEnable>
    <!--opt, bool, whether to enable three-shutter-->true
  </threeShutEnable>
  <doubleVideoEnable>
    <!--opt, bool, whether to enable dual-video-->true
  </doubleVideoEnable>
</ImageMultishut>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.2.8 Get the multi-shutter capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/multishut/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMultishut xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <doubleShutEnable>
    <!--ro, req, bool, whether to enable dual-shutter-->true
  </doubleShutEnable>
  <codecType>
    <!--ro, req, enum, "all_ia" (algorithm stream), "all_record" (record stream), "all_threeShut", subType:string, desc:"all_ia" (algorithm stream),
"all_record" (record stream), "all_threeShut"-->all_ia
  </codecType>
  <threeShutEnable>
    <!--ro, opt, bool-->true
  </threeShutEnable>
  <doubleVideoEnable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </doubleVideoEnable>
  <multiCapture>
    <!--ro, opt, object-->
    <iaCaptureEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </iaCaptureEnabled>
    <recordCaptureEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </recordCaptureEnabled>
    <captureCombineType opt="none,iaAndRecord">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->none
    </captureCombineType>
  </multiCapture>
</ImageMultishut>
```

16.2.2.9 Get the multi-shutter parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/multishut?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageMultishut xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <doubleShutEnable>
    <!--ro, req, bool, whether to enable dual-shutter-->true
  </doubleShutEnable>
  <codecType>
    <!--ro, req, enum, type, subType:string, desc:type-->all_ia
  </codecType>
  <threeShutEnable>
    <!--ro, opt, bool-->true
  </threeShutEnable>
  <doubleVideoEnable>
    <!--ro, opt, bool-->true
  </doubleVideoEnable>
  <multiCapture>
    <!--ro, opt, object-->
    <iaCaptureEnabled>
      <!--ro, opt, bool-->true
    </iaCaptureEnabled>
    <recordCaptureEnabled>
      <!--ro, opt, bool-->true
    </recordCaptureEnabled>
    <captureCombineType>
      <!--ro, opt, enum, subType:string-->none
    </captureCombineType>
  </multiCapture>
</ImageMultishut>
```

16.2.2.10 Set the configuration parameters of picture exposure control by video segment for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/sectionCtrl

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SectionCtrl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, exposure control, attr:version{req, string, protocolVersion}-->
  <sectionCtrlEnabled>
    <!--req, bool, whether to enable picture exposure control by video segment-->true
  </sectionCtrlEnabled>
</SectionCtrl>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.2.11 Get the configuration parameters of picture exposure control by video segment of a specific channel

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Request URL

GET /ISAPI/Image/channels/<channelID>/sectionCtrl

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SectionCtrl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <sectionCtrlEnabled>
    <!--ro, req, bool, whether to enable picture exposure control by video segment-->true
  </sectionCtrlEnabled>
</SectionCtrl>
```

16.2.2.12 Get the configuration capability of picture exposure control by video segment of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/sectionCtrl/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SectionCtrl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, exposure control, attr:version{req, string, protocolVersion}-->
  <sectionCtrlEnabled>
    <!--ro, req, bool, whether to enable picture exposure control by video segment-->true
  </sectionCtrlEnabled>
</SectionCtrl>
```

16.2.2.13 Set the exposure time parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/shutter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Shutter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <ShutterLevel>
    <!--opt, enum, exposure level, subType:string, desc:"1/1", "1/2", "1/3", "1/6", "1/12", "1/25", "1/50", "1/75", "1/100", "1/120", "1/125", "1/150",
"1/175", this node is valid when exposure type is "shutter first"-->1/1
  </ShutterLevel>
  <scheduleEnabled>
    <!--opt, bool-->true
  </scheduleEnabled>
  <scheduleShutterLevel>
    <!--opt, int-->1
  </scheduleShutterLevel>
  <ScheduleList>
    <!--opt, array, subType:object-->
    <Schedule>
      <!--opt, object-->
      <startHour>
        <!--opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--opt, int, end time (hour)-->1
      </endHour>
      <endMinute>
        <!--opt, int, end time (minute)-->1
      </endMinute>
    </Schedule>
  </ScheduleList>
  <redShutterLevel>
    <!--opt, int, range:[0,20000]-->1
  </redShutterLevel>
</Shutter>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.2.14 Get the exposure time parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/shutter?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shutter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <ShutterLevel>
    <!--ro, opt, enum, exposure level, subType:string, desc:"1/1", "1/2", "1/3", "1/6", "1/12", "1/25", "1/50", "1/75", "1/100", "1/120", "1/125", "1/150",
"1/175", this node is valid when exposure type is "shutter first"-->1/1
  </ShutterLevel>
  <scheduleEnabled>
    <!--ro, opt, bool-->true
  </scheduleEnabled>
  <scheduleShutterLevel>
    <!--ro, opt, int-->1
  </scheduleShutterLevel>
  <ScheduleList>
    <!--ro, opt, array, subType:object-->
    <Schedule>
      <!--ro, opt, object-->
      <startHour>
        <!--ro, opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--ro, opt, int, end time (hour)-->1
      </endHour>
      <endMinute>
        <!--ro, opt, int, end time (minute)-->1
      </endMinute>
    </Schedule>
  </ScheduleList>
  <redShutterLevel>
    <!--ro, opt, int, range:[0,20000]-->1
  </redShutterLevel>
  <maxShutterLevelLimit>
    <!--ro, opt, enum, subType:string-->1/1
  </maxShutterLevelLimit>
  <minShutterLevelLimit>
    <!--ro, opt, enum, subType:string-->1/1
  </minShutterLevelLimit>
</Shutter>
```

16.2.3 Focus Settings

16.2.3.1 Get the focus parameters of the specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/focusConfiguration

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<FocusConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, focus configuration, attr:version{req, string, protocolVersion}-->
  <focusStyle>
    <!--ro, req, enum, focus mode, subType:string, desc:"AUTO" (automatic focus), "MANUAL" (manual focus), "SEMIAUTOMATIC" (semi-automatic focus)-->AUTO
  </focusStyle>
  <focusLimited>
    <!--ro, opt, enum, focus limits, subType:int, desc:focus limits-->10
  </focusLimited>
  <focusPosition>
    <!--ro, opt, int, focus point position-->1
  </focusPosition>
  <focusSpeed>
    <!--ro, opt, int, focus speed-->1
  </focusSpeed>
  <focusSensitivity>
    <!--ro, opt, int, focus sensitivity-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled>
    <!--ro, opt, bool, whether to enable temperature change adaption-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos>
    <!--ro, opt, int, relative focus point position-->1
  </relativeFocusPos>
  <lensType>
    <!--ro, opt, enum, lens type, subType:string, desc:lens type-->1140
  </lensType>
  <highTemperaturePriority>
    <!--ro, opt, bool, whether to enable high temperature priority mode-->true
  </highTemperaturePriority>
  <focusRange>
    <!--ro, opt, enum, focus adjusting range, subType:string, desc:focus adjusting range-->max
  </focusRange>
  <focusDistanceMode>
    <!--ro, opt, enum, focus distance mode, subType:string, desc:focus distance mode-->compatible
  </focusDistanceMode>
  <zoomFocusFollowEnabled>
    <!--ro, opt, bool-->true
  </zoomFocusFollowEnabled>
  <focusRegion>
    <!--ro, opt, enum, subType:string, dep:or,{$.FocusConfiguration.focusStyle,eq,AUTO},{$.FocusConfiguration.focusStyle,eq,SEMIAUTOMATIC}-->center
  </focusRegion>
  <focusScene>
    <!--ro, opt, enum, subType:string-->normal
  </focusScene>
  <timeFocusEnabled>
    <!--ro, opt, bool, dep:or,{$.FocusConfiguration.focusStyle,eq,AUTO},{$.FocusConfiguration.focusStyle,eq,SEMIAUTOMATIC}-->true
  </timeFocusEnabled>
  <timeFocus>
    <!--ro, opt, int, range:[1,1440], unit:min, dep:and,{$.FocusConfiguration.timeFocusEnabled,eq,true}-->1
  </timeFocus>
  <focusDistance>
    <!--ro, opt, float, range:[0.30,99.99], step:0.01, unit:m, dep:and,{$.FocusConfiguration.focusStyle,eq,distanceCalculate}-->0.35
  </focusDistance>
  <focusStrategy>
    <!--ro, opt, enum, subType:string-->DTOF
  </focusStrategy>
</FocusConfiguration>
```

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16.2.3.2 Set the focus parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/focusConfiguration

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FocusConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, focus configuration, attr:version{req, string, protocolVersion}-->
  <focusStyle>
    <!--req, enum, focus mode, subType:string, desc:"auto" (automatic focus), "manual" (manual focus), "semiautomatic" (semiautomatic focus)-->AUTO
  </focusStyle>
  <focusLimited>
    <!--opt, enum, subType:int-->10
  </focusLimited>
  <focusPosition>
    <!--opt, int, focus point position-->1
  </focusPosition>
  <focusSpeed>
    <!--opt, int, focus speed-->1
  </focusSpeed>
  <focusSensitivity>
    <!--opt, int, focus sensitivity-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled>
    <!--opt, bool, whether to enable temperature change adaption-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos>
    <!--opt, int, relative focus point position-->1
  </relativeFocusPos>
  <lensType>
    <!--opt, enum, lens type, subType:string, desc:"1140", "1050"-->1140
  </lensType>
  <highTemperaturePriority>
    <!--opt, bool, whether to enable high temperature priority mode-->true
  </highTemperaturePriority>
  <focusRange>
    <!--opt, enum, focus adjustment range, subType:string, desc:"min" (the minimum), "medium" (the medium), "max" (the maximum)-->max
  </focusRange>
  <focusDistanceMode>
    <!--opt, enum, focus distance mode, subType:string, desc:focus distance mode-->compatible
  </focusDistanceMode>
</FocusConfiguration>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.3.3 Get the focus configuration capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/focusConfiguration/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FocusConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the focus configuration capability of a specific channel, attr:version{req, string, protocolVersion}-->
  <focusStyle opt="AUTO,MANUAL,SEMIAUTOMATIC,distanceCalculate">
    <!--ro, req, string, attr:opt{req, string}-->test
  </focusStyle>
  <focusLimited opt="10,50,100,150,300,600,1000,2000,5000,20000,50000,65535">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->10
  </focusLimited>
  <focusPosition min="1" max="10">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </focusPosition>
  <focusSpeed min="1" max="10">
    <!--ro, opt, int, range:[1,3], attr:min{req, int},max{req, int}-->1
  </focusSpeed>
  <focusSensitivity min="1" max="10">
    <!--ro, opt, int, focusing sensitivity, range:[0,2], attr:min{req, int},max{req, int}-->1
  </focusSensitivity>
  <temperatureChangeAdaptEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </temperatureChangeAdaptEnabled>
  <relativeFocusPos min="1" max="10">
    <!--ro, opt, int, range:[0,4294967295], attr:min{req, int},max{req, int}-->1
  </relativeFocusPos>
  <highTemperaturePriority opt="true,false">
    <!--ro, opt, bool, whether to enable the mode of priority of high temperature, attr:opt{req, string}-->true
  </highTemperaturePriority>
  <focusRange opt="min,medium,max">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->max
  </focusRange>
  <focusDistanceMode opt="compatible">
    <!--ro, opt, string, attr:opt{req, string}-->compatible
  </focusDistanceMode>
  <zoomFocusFollowEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </zoomFocusFollowEnabled>
  <focusRegion opt="center,pixelToPixel">
    <!--ro, opt, enum, subType:string, dep:or,{$.FocusConfiguration.focusStyle,eq,AUTO},{$.FocusConfiguration.focusStyle,eq,SEMIAUTOMATIC}, attr:opt{req,
string}-->center
  </focusRegion>
  <focusScene opt="lowIllumination,normal">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
  </focusScene>
  <timeFocusEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </timeFocusEnabled>
  <timeFocus min="1" max="1440">
    <!--ro, opt, int, range:[1,1440], attr:min{req, int},max{req, int}-->1
  </timeFocus>
  <focusDistance min="0.30" max="99.99">
    <!--ro, opt, float, range:[0.30,99.99], step:0.01, unit:m, attr:min{req, float},max{req, float}-->0.35
  </focusDistance>
  <focusStrategy opt="general,DTOF" def="general">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->DTOF
  </focusStrategy>
</FocusConfiguration>
```

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16.2.3.4 Set the regional focus parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/regionalFocus

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionalFocus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, regional focus parameters, attr:version{req, string, protocolVersion}-->
  <StartPoint>
    <!--opt, object, start point, desc:the origin is the upper-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
  </StartPoint>
  <EndPoint>
    <!--opt, object, end point, desc:the origin is the upper-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
  </EndPoint>
</RegionalFocus>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.4 Image Adjustment Settings

16.2.4.1 Get the default stitching distance of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/calibSensor/defaultParams?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No. of device.

Request Message

None

Response Message

```
{
    "factorydefaultParams": {
    /*ro, opt, object*/
        "splicingSectionDistance":  60.0
        /*ro, req, float, unit:dm*/
    }
}
```

16.2.4.2 Get image rotate mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/corridor

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<corridor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, mode, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
</corridor>
```

16.2.4.3 Set parameters of the image rotate mode for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/corridor

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<corridor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, mode, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
</corridor>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.4.4 Set the display parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/displayParamSwitch

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DisplayParamSwitch xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, display parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, mode, subType:string, desc:mode-->disable
  </mode>
  <LinkPresetList>
    <!--opt, array, subType:object-->
    <LinkPreset>
      <!--opt, object-->
      <id>
        <!--req, string, ID-->test
      </id>
      <presetID>
        <!--req, string-->test
      </presetID>
      <linkEnabled>
        <!--req, bool-->true
      </linkEnabled>
      <PeriodList size="4">
        <!--req, array, subType:object, attr:size{req, int}-->
        <Period>
          <!--opt, object-->
          <id>
            <!--req, string-->test
          </id>
          <enabled>
            <!--req, bool-->true
          </enabled>
          <TimeRange>
            <!--req, object-->
            <beginTime>
              <!--req, time, start time-->00:00:00+08:00
            </beginTime>
            <endTime>
              <!--req, time, end time-->00:00:00+08:00
            </endTime>
          </TimeRange>
          <linkScene>
            <!--req, enum, link mode, subType:string, desc:link mode-->indoor
          </linkScene>
        </Period>
      </PeriodList>
    </LinkPreset>
  </LinkPresetList>
  <TimingSwitch>
    <!--opt, object, dep:and,{$.DisplayParamSwitch.mode,eq,timingSwitch}-->
    <PeriodList size="4">
      <!--req, array, subType:object, attr:size{req, int}-->
      <Period>
        <!--opt, object-->
        <id>
          <!--req, string-->test
        </id>
        <enabled>
          <!--req, bool-->true
        </enabled>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
        <linkScene>
          <!--req, enum, link mode, subType:string, desc:link mode-->indoor
        </linkScene>
      </Period>
    </PeriodList>
  </TimingSwitch>
  <Month>
    <!--opt, object, dep:and,{$.DisplayParamSwitch.mode,eq,month}-->
    <PeriodList size="4">
      <!--req, array, subType:object, attr:size{req, int}-->
      <Period>
        <!--opt, object-->
        <id>
          <!--req, string-->test
        </id>
        <enabled>
          <!--req, bool-->true
```

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```
        </enabled>
        <TimeRangeList size="4">
          <!--req, array, subType:object, attr:size{req, int}-->
          <TimeRange>
            <!--req, object-->
            <beginTime>
              <!--req, time, start time-->00:00:00+08:00
            </beginTime>
            <endTime>
              <!--req, time, end time-->00:00:00+08:00
            </endTime>
            <linkScene>
              <!--req, enum, link mode, subType:string, desc:link mode-->indoor
            </linkScene>
          </TimeRange>
        </TimeRangeList>
      </Period>
    </PeriodList>
  </Month>
</DisplayParamSwitch>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.4.5 Get the gamma correction parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/gammaCorrection?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<gammaCorrection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <gammaCorrectionEnabled>
    <!--ro, opt, bool-->true
  </gammaCorrectionEnabled>
  <gammaCorrectionLevel>
    <!--ro, opt, int-->1
  </gammaCorrectionLevel>
</gammaCorrection>
```

16.2.4.6 Get the lens calibration capability

Request URL

GET /ISAPI/Image/channels/<channelID>/lensCalibration/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
}
```

16.2.4.7 Set the lens calibration parameters

Request URL

PUT /ISAPI/Image/channels/<channelID>/lensCalibration?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "LensCalibration": {
    /*opt, object*/
        "Motor": {
        /*opt, object*/
            "rotation": "turning",
            /*opt, enum, subType:string*/
            "step": 1
            /*opt, int, range:[0,50000]*/
        },
        "focus": 1,
        /*opt, int, range:[0,65535]*/
        "zoom": 1,
        /*opt, int, range:[0,65535]*/
        "returnDifference": 1,
        /*opt, int, range:[0,50000]*/
        "distance": 1,
        /*opt, int, range:[0,50000], unit:mm*/
        "focusCalibrationMode": "DCoefficient",
        /*opt, enum, subType:string*/
        "strokeValue": 10000,
        /*opt, int, range:[0,65535]*/
        "lensStrokeDetectionEnabled": true,
        /*opt, bool*/
        "transmittance": 0.000
        /*opt, float, range:[0.000,1.000]*/
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int*/
    "statusString": "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode": "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode": 1,
    /*ro, opt, int*/
    "errorMsg": "ok"
    /*ro, opt, string*/
}
```

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DETEKTOR

16.2.4.8 Get the lens calibration parameters

Request URL

GET /ISAPI/Image/channels/<channelID>/lensCalibration?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
{
    "LensCalibration": {
    /*ro, opt, object*/
        "Motor": {
        /*ro, opt, object*/
            "rotation": "turning",
            /*ro, opt, enum, subType:string*/
            "step": 1
            /*ro, opt, int, range:[0,50000]*/
        },
        "focus": 1,
        /*ro, opt, int, range:[0,65535]*/
        "zoom": 1,
        /*ro, opt, int, range:[0,65535]*/
        "returnDifference": 1,
        /*ro, opt, int, range:[0,50000]*/
        "distance": 1,
        /*ro, opt, int, range:[0,50000], unit:mm*/
        "focusCalibrationMode": "DCoefficient",
        /*ro, opt, enum, subType:string*/
        "focalLengthStatus": "farFocus",
        /*ro, opt, enum, subType:string*/
        "lensStrokeDetection": true,
        /*ro, opt, bool*/
        "piStatus": true,
        /*ro, opt, bool*/
        "strokeValue": 10000,
        /*ro, opt, int, range:[0,65535]*/
        "lensID": 1,
        /*ro, opt, int, range:[0,65535]*/
        "teleconverterID": 1,
        /*ro, opt, int, range:[0,65535]*/
        "TECGear": ["TEC1", "TEC2"],
        /*ro, opt, array, subType:string*/
        "pixelSize": 1,
        /*ro, opt, int, range:[0,1000], unit:μm*/
        "calibrateTimes": 1,
        /*ro, opt, int, range:[0,100]*/
        "lensTemperature": 20,
        /*ro, opt, int, range:[-50,100], unit:℃*/
        "fieldFocalLengthParam": [
        /*ro, opt, array, subType:object*/
            {
                "focalLengthID": 1,
                /*ro, req, int, range:[1,2]*/
                "opticsScale": 1.71
                /*ro, opt, float, range:[1.00,10.00]*/
            }
        ],
        "teleconverterMultiplicationFactor": [
        /*ro, opt, array, subType:object*/
            {
                "factorID": 1,
                /*ro, req, int, range:[1,4]*/
                "value": 50000
                /*ro, req, int, range:[-1000000,1000000]*/
            }
        ],
        "teleconverterAdditiveFactor": [
        /*ro, opt, array, subType:object*/
            {
                "factorID": 1,
                /*ro, req, int, range:[1,4]*/
                "value": 50000
                /*ro, req, int, range:[-1000000,1000000]*/
            }
        ],
        "distanceCalibrationCoefficient": 0.00,
        /*ro, opt, float, range:[0.00,9999.99]*/
        "zoomFocusFollowFactor": [
        /*ro, opt, array, subType:object, range:[1,10]*/
            {
                "factorID": 1,
                /*ro, req, int, range:[1,10]*/
                "value": 50000
                /*ro, req, int, range:[-1000000,1000000]*/
            }
        ]
    }
}
```

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16.2.4.9 Get the teleconverter parameters capability of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/teleconverter/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "TeleconverterParamCap": {
    /*ro, opt, object*/
        "serialNo": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  20
            /*ro, opt, int*/
        },
        "zoom": {
        /*ro, opt, object*/
            "@opt": ["0,1,2,3"],
            /*ro, req, array, subType:string*/
            "@def":  "0"
            /*ro, req, string*/
        },
        "manufacturer": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  256
            /*ro, opt, int*/
        },
        "TeleconverterCalibration": {
        /*ro, opt, object*/
            "@size":  4,
            /*ro, req, int*/
            "teleconverter": {
            /*ro, req, object*/
                "@opt": ["X0.5", "X1", "X2", "X3"]
                /*ro, req, array, subType:string*/
            },
            "calibrateStatus": {
            /*ro, req, object*/
                "@opt": ["calibrated", "noCalibrate"]
                /*ro, req, array, subType:string*/
            }
        },
        "interchangeableLensMount":  "lensGroup",
        /*ro, opt, enum, subType:string*/
        "interchangeableLensTemplate":  "singleTemplate",
        /*ro, opt, enum, subType:string*/
        "interchangeableLensRecognition":  "interchangeableLensNotSupported",
        /*ro, opt, enum, subType:string*/
        "thermometryTemplateParamSetID":  1
        /*ro, opt, int, range:[1,5]*/
    }
}
```

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16.2.4.10 Set the teleconverter parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/teleconverter?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "TeleconverterParam": {
    /*opt, object*/
        "serialNo":  "test",
        /*opt, string, range:[1,20]*/
        "zoom":  0,
        /*opt, enum, subType:int*/
        "manufacturer":  "test"
        /*opt, string, range:[0,256]*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.2.4.11 Get the teleconverter parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/teleconverter?format=json&lenType=<lenType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
lenType	string	--

Request Message

None

Response Message

```
{
    "TeleconverterParam": {
    /*ro, opt, object*/
        "targetType":  "outside",
        /*ro, opt, enum, subType:string*/
        "serialNo":  "test",
        /*ro, opt, string, range:[1,20]*/
        "zoom":  0,
        /*ro, opt, enum, subType:int*/
        "manufacturer":  "test",
        /*ro, opt, string, range:[0,256]*/
        "TeleconverterCalibration": [
        /*ro, opt, array, subType:object*/
            {
                "teleconverter":  "X1",
                /*ro, req, enum, subType:string*/
                "calibrateStatus":  "calibrated"
                /*ro, req, enum, subType:string*/
            }
        ]
    }
}
```

16.2.5 Image Adjustment

16.2.5.1 Set image parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <WDR>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->open
    </mode>
    <WDRLevel>
      <!--opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
  </BLC>
  <Color>
    <!--opt, object-->
    <brightnessLevel>
      <!--opt, int, brightness, range:[0,100]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--opt, int, contrast, range:[0,100]-->0
    </contrastLevel>
    <saturationLevel>
      <!--opt, int, saturation, range:[0,100]-->0
    </saturationLevel>
  </Color>
  <HLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <HLCLevel>
      <!--opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <enableImageLossDetection>
    <!--opt, bool-->true
  </enableImageLossDetection>
</ImageChannel>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.5.2 Get the automatic contrast parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/autoContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AutoContrast xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, automatic contrast parameters, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool, whether to enable automatic contrast-->true
  </enable>
  <level>
    <!--ro, req, int, automatic contrast level, range:[0,100], desc:the default value is 50-->50
  </level>
</AutoContrast>
```

16.2.5.3 Set the automatic contrast parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/autoContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AutoContrast xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--req, bool, whether to enable automatic contrast-->true
  </enable>
  <level>
    <!--req, int, automatic contrast level, range:[0,100], desc:the default value is 50-->50
  </level>
</AutoContrast>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.5.4 Get the automatic contrast capability of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/autoContrast/capabilities

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DETEKTOR

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AutoContrast xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, automatic contrast capability, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool, whether to enable automatic contrast-->true
  </enable>
  <level min="0" max="100" def="50">
    <!--ro, req, int, automatic contrast level, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->50
  </level>
</AutoContrast>
```

16.2.5.5 Set the brightness enhancement parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/brightEnhance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, brightness enhancement parameters, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--req, bool, whether to enable-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.5.6 Get the brightness enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/brightEnhance?parameterType=<parameterType>

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--ro, req, bool, whether to enable brightness enhancement-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--ro, req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

16.2.5.7 Get image configuration capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, temperature range capability, see XML_Cap_tempRange for details, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <ImageFlipStyle opt="LEFTRIGHT,UPDOWN,CENTER,AUTO">
      <!--ro, opt, string, type, attr:opt{req, string}-->LEFTRIGHT
    </ImageFlipStyle>
  </ImageFlip>
  <WDR>
    <!--ro, opt, object-->
    <mode opt="open,close,auto,SDR,time">
      <!--ro, req, enum, WDR mode: "open,close,auto", subType:string, attr:opt{req, string}, desc:WDR mode: "open,close,auto"-->open
    </mode>
    <WDRLevel min="1" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <BLCMode opt="UP,DOWN,LEFT,RIGHT,CENTER,MULTI-AREA,Region,AUTO">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->UP
    </BLCMode>
  </BLC>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel
opt="1/1,1/2,1/3,1/6,1/12,1/25,1/50,1/60,1/75,1/100,1/120,1/125,1/150,1/175,1/15,1/20,1/30,1/60,1/180,1/200,1/250,1/500,1/1000,1/2000,1/4000,1/8000,1/5000,1
/10000,1/225,1/300,1/400,,1/425,1/600,1/750,1/1250,1/1750,1/2500,1/3500,1/6000,1/30000,1/100000">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </ShutterLevel>
  </Shutter>
  <Color>
    <!--ro, opt, object, image adjustment capability-->
    <brightnessLevel min="0" max="255">
      <!--ro, opt, int, brightness, attr:min{req, int},max{req, int}-->0
    </brightnessLevel>
    <contrastLevel min="0" max="255">
      <!--ro, opt, int, contrast, attr:min{req, int},max{req, int}-->0
    </contrastLevel>
    <saturationLevel min="0" max="255">
      <!--ro, opt, int, saturation, attr:min{req, int},max{req, int}-->0
    </saturationLevel>
  </Color>
  <HLC>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <HLCLevel min="0" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </HLCLevel>
  </HLC>
  <enableImageLossDetection opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enableImageLossDetection>
  <SupplementLight>
    <!--ro, opt, object-->
    <supplementLightMode
opt="mixed,whiteLight,colorVuWhiteLight,irLight,dualLight,close,eventIntelligence,multiAzimuthMixed,multiAzimuthWhiteLight,multiAzimuthIrLight,panoramicWhit
eLight,panoramicIrLight">
      <!--ro, opt, enum, mode of supplement light, subType:string, attr:opt{req, string}, desc:mode of supplement light-->mixed
    </supplementLightMode>
    <irLightBrightness min="0" max="100">
      <!--ro, opt, int, brightness of IR supplement light, range:[0,100], attr:min{req, int},max{req, int}, desc:this node is valid when supplementLightMode
is set to "mixed" and mixedLightBrightnessRegulatMode is set to "manual"-->80
    </irLightBrightness>
    <mixedLightBrightnessRegulatMode opt="manual,auto">
      <!--ro, opt, enum, brightness adjustment mode of mixed supplement light, subType:string, attr:opt{req, string}, desc:"manual" (in this mode, you need
to set the parameters irLightBrightness and brightnessLimit), "auto" (brightness-related parameters are not required). This node is valid when
supplementLightMode is set to "mixed"-->auto
    </mixedLightBrightnessRegulatMode>
    <isAutoModeBrightnessCfg>
      <!--ro, opt, bool-->true
    </isAutoModeBrightnessCfg>
  </SupplementLight>
</ImageChannel>
```

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16.2.5.8 Set the image adjustment parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--opt, object, gray scale-->
    <grayScaleMode>
      <!--opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, bool, whether to enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted--
>true
  </nightMode>
  <redBrightnessLevel>
    <!--opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--opt, int, range:[0,100]-->0
  </sharpnessLevel>
</Color>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.5.9 Get the image adjustment parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/color?parameterType=<parameterType>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool, whether to enable night mode-->true
  </nightMode>
  <redBrightnessLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
</Color>
```

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DETEKTOR

16.2.5.10 Set the device series parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/deviceSeries

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceSeries xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <devicType>
    <!--req, string-->test
  </devicType>
</DeviceSeries>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.5.11 Get the device series parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/deviceSeries

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceSeries xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <devicType>
    <!--ro, req, string-->test
  </devicType>
</DeviceSeries>
```

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DETEKTOR

16.2.5.12 Get the capability of device series parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/deviceSeries/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceSeriesCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <devicType opt="superDarkFighterSmartColorVu,colorVu,superDarkFighter">
    <!--ro, req, string, attr:opt{req, enumarray, subType:string}-->test
  </devicType>
</DeviceSeriesCap>
```

16.2.5.13 Set the gray scale range parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/grayRange

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GrayRange xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, gray scale range, attr:version{req, string, protocolVersion}-->
  <grayValueType>
    <!--req, enum, gray scale value, subType:string, desc:"hightContrast"-high contrast with less details in shadows, the value ranges from 0 to 255;
"moreDetail"-more detail in shadows but low contrast, the value ranges from 16 to 235. "hightContrast" (high contrast with less details in shadows),
"moreDetail" (more detail in shadows but low contrast)-->hightContrast
  </grayValueType>
</GrayRange>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.5.14 Get the grayscale range capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/grayRange/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GrayRange xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <grayValueType opt="hightContrast,moreDetail">
    <!--ro, req, string, grayscale value, attr:opt{req, string}, desc:"hightContrast"-high contrast with less details in shadows, the value ranges from 0 to
255; "moreDetail"-more detail in shadows but low contrast,-->test
  </grayValueType>
</GrayRange>
```

16.2.5.15 Get the grayscale range parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/grayRange?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GrayRange xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <grayValueType>
    <!--ro, req, enum, grayscale value, subType:string, desc:"hightContrast"-high contrast with less details in shadows, the value ranges from 0 to 255;
"moreDetail"-more detail in shadows but low contrast, the value ranges from 16 to 235. "hightContrast" (high contrast with less details in shadows),
"moreDetail" (more detail in shadows but low contrast)-->hightContrast
  </grayValueType>
</GrayRange>
```

16.2.5.16 Get the image mode parameters (capture camera) of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/imageModes

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageModeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, image mode list, subType:object, attr:version{req, string, protocolVersion}-->
  <ImageMode>
    <!--ro, opt, object, image mode-->
    <type>
      <!--ro, req, enum, type, subType:string, desc:type-->standard
    </type>
    <recommendation>
      <!--ro, req, object-->
      <brightnessLevel>
        <!--ro, opt, int, brightness, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, contrast, range:[0,100]-->1
      </contrastLevel>
      <sharpnessLevel>
        <!--ro, opt, int, sharpness, range:[0,100]-->1
      </sharpnessLevel>
      <saturationLevel>
        <!--ro, opt, int, saturation, range:[0,100]-->1
      </saturationLevel>
      <hueLevel>
        <!--ro, opt, int, hue, range:[0,100]-->1
      </hueLevel>
      <deNoiseLevel>
        <!--ro, opt, int, noise reduction, range:[0,100]-->1
      </deNoiseLevel>
    </recommendation>
  </ImageMode>
</ImageModeList>
```

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DETEKTOR

16.2.5.17 Get the image brightness limit parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/irLight

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrLight xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->auto
  </mode>
  <brightnessLevel>
    <!--ro, opt, int-->1
  </brightnessLevel>
  <sensitivityLevel>
    <!--ro, opt, int-->1
  </sensitivityLevel>
  <brightnessLimit>
    <!--ro, opt, int-->1
  </brightnessLimit>
  <triggerMode>
    <!--ro, opt, enum, subType:string-->camera
  </triggerMode>
</IrLight>
```

16.2.5.18 Set the image brightness limit parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/irLight

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrLight xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->auto
  </mode>
  <brightnessLevel>
    <!--opt, int-->1
  </brightnessLevel>
  <sensitivityLevel>
    <!--opt, int-->1
  </sensitivityLevel>
  <brightnessLimit>
    <!--opt, int-->1
  </brightnessLimit>
  <triggerMode>
    <!--opt, enum, subType:string-->camera
  </triggerMode>
</IrLight>
```

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.5.19 Get the contrast enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lse

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--ro, opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

16.2.5.20 Set the contrast enhancement parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/lse

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--opt, enum, LSE switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.5.21 Get the contrast enhancement capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lse/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, req, int, contrast enhancement level, range:[0,100]-->50
  </lseLevel>
  <lseHaloLevel>
    <!--ro, req, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, opt, int, brightness level, range:[0,100], dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

16.2.5.22 Set the cursor parameters for manually measuring the distance in the image of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/manualRanging/cursor?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "enable": true,
    /*req, bool*/
    "mode": "continuous"
    /*opt, enum, subType:string*/
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int*/
    "statusString": "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode": "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode": 1,
    /*ro, opt, int*/
    "errorMsg": "ok"
    /*ro, opt, string*/
}
```

16.2.5.23 Set the mounting scenario mode of a camera

Request URL

PUT /ISAPI/Image/channels/<channelID>/mountingScenario

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor“, “outdoor”, “day”, “night”, “morning”, “nightfall”, “mode1”, “mode2”, “mode3”, “mode4”, “street”, “lowIllumination”-low illumination, “custom1”, “custom2”, “normal”, “road”, “faceSnap”-face picture capture, “highway”, “road2”, “backlight”-back light, “frontlight”-front light, “facenight”, “fastRoadAndLight”, “fastRoadNoLihgt”, “normalRoadAndLight”, “normalRoadNoLight”, “cityRoadAndBright”-->indoor
  </mode>
</MountingScenario>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.5.24 Get the capability of configurations of mounting scenario mode

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR,sportSnap,dustMoisture,rampWay">
    <!--ro, req, enum, Mode, subType:string, attr:opt{req, string}, desc:"lowIllumination"-low illumination, "backlight"-back light, "frontlight"-front
light, "faceSnap"-face picture capture-->indoor
  </mode>
  <isSupportRecommendation>
    <!--ro, opt, bool-->true
  </isSupportRecommendation>
  <modeNameList size="2">
    <!--ro, opt, object, attr:size{req, int}-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

16.2.5.25 Get the mounting scenario mode of a camera

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	Parameter type. When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured, it will be the value of the parameter configured on the device.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-
low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor”, “outdoor”, “day”, “night”,
“morning”, “nightfall”, “mode1”, “mode2”,“ mode3”, “mode4”, “highway”, “road2”, “faceSnap”, “backlight”, “frontlight”, “lowIllumination”-low illumination,
“backlight”-back light, “frontlight”-front light, “faceSnap”-face picture capture-->indoor
  </mode>
  <modeNameList>
    <!--ro, opt, array, subType:object-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode>
        <!--ro, req, enum, subType:string-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

16.2.5.26 Set the license plate brightness compensation parameters of a specific channel

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DETEKTOR

Request URL

PUT /ISAPI/Image/channels/<channelID>/plateBright

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateBright xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, license plate brightness compensation, attr:version{req, string, protocolVersion}-->
  <plateBrightEnabled>
    <!--req, bool, whether to enable license plate brightness compensation-->true
  </plateBrightEnabled>
  <plateBrightSensitivity>
    <!--opt, int, license plate brightness compensation sensitivity, range:[1,100]-->1
  </plateBrightSensitivity>
  <correctFactorEnabled>
    <!--req, bool, whether to enable supplement light correction factor-->true
  </correctFactorEnabled>
  <correctFactor>
    <!--opt, int, supplement light correction factor, range:[0,100]-->1
  </correctFactor>
</PlateBright>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.5.27 Get the license plate brightness compensation capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/plateBright/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateBright xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, license plate brightness compensation parameters, attr:version{req, string, protocolVersion}-->
  <plateBrightEnabled>
    <!--ro, req, bool, whether to enable license plate brightness compensation-->true
  </plateBrightEnabled>
  <plateBrightSensitivity min="0" max="100">
    <!--ro, opt, int, license plate brightness compensation sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
  </plateBrightSensitivity>
  <correctFactorEnabled>
    <!--ro, opt, bool, whether to enable supplement light correction factor-->true
  </correctFactorEnabled>
  <correctFactor min="0" max="100">
    <!--ro, opt, int, supplement light correction factor, range:[0,100], attr:min{req, int},max{req, int}-->1
  </correctFactor>
</PlateBright>
```

16.2.5.28 Get the license plate brightness compensation parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/plateBright?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateBright xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, license plate brightness compensation parameters, attr:version{req, string, protocolVersion}-->
  <plateBrightEnabled>
    <!--ro, req, bool, whether to enable license plate brightness compensation-->true
  </plateBrightEnabled>
  <plateBrightSensitivity>
    <!--ro, opt, int, license plate brightness compensation sensitivity, range:[0,100]-->1
  </plateBrightSensitivity>
  <correctFactorEnabled>
    <!--ro, opt, bool, whether to enable supplement light correction factor-->true
  </correctFactorEnabled>
  <correctFactor>
    <!--ro, opt, int, supplement light correction factor, range:[0,100]-->1
  </correctFactor>
</PlateBright>
```

16.2.5.29 Reset the image parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/reset

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.5.30 Restore the image parameters of a specific channel to default settings

Request URL

PUT /ISAPI/Image/channels/<channelID>/restore

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.5.31 Get the scene parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/scene

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DETEKTOR

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Scene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->indoor
  </mode>
</Scene>
```

16.2.5.32 Set the scene parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/scene

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Scene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->indoor
  </mode>
</Scene>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.5.33 Set the sharpness control parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/sharpness

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

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Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.5.34 Get the sharpness control parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/sharpness?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--ro, opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--ro, req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

16.2.5.35 Set the part contrast parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/svce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Svce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, part contrast parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable part contrast-->true
  </enabled>
  <svceLevel>
    <!--opt, int, part contrast level, range:[0,100], desc:the default value is 50-->1
  </svceLevel>
</Svce>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.5.36 Get the part contrast parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/svce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Svce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable part contrast-->true
  </enabled>
  <svceLevel>
    <!--ro, opt, int, part contrast level, range:[0,100], desc:part contrast level-->1
  </svceLevel>
</Svce>
```

16.2.5.37 Get the part contrast capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/svce/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Svce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable part contrast-->true
  </enabled>
  <svceLevel min="0" max="100">
    <!--ro, opt, int, part contrast level, range:[0,100], attr:min{req, int},max{req, int}, desc:part contrast level-->1
  </svceLevel>
</Svce>
```

16.2.5.38 Get the image processing parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Video channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <videoInputID>
    <!--ro, req, int-->1
  </videoInputID>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </ImageFlip>
  <WDR>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->open
    </mode>
    <WDRLevel>
      <!--ro, opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <BLCMode>
```

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DEFEKTOR

```
      <!--ro, opt, enum, subType:string-->UP
    </BLCMode>
    <BLCRegionList>
      <!--ro, opt, array, subType:object-->
      <BLCRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int, range:[1,10]-->1
        </id>
        <RegionCoordinatesList>
          <!--ro, req, array, subType:object, range:[0,4]-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,704]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,576]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </BLCRegion>
    </BLCRegionList>
  </BLC>
  <NoiseReduce>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel>
        <!--ro, req, int, range:[0,100]-->1
      </generalLevel>
    </GeneralMode>
  </NoiseReduce>
  <WhiteBalance>
    <!--ro, opt, object-->
    <WhiteBalanceStyle>
      <!--ro, opt, enum, subType:string-->auto
    </WhiteBalanceStyle>
    <WhiteBalanceRed>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceRed>
    <WhiteBalanceBlue>
      <!--ro, opt, int, range:[-127,127]-->1
    </WhiteBalanceBlue>
    <WhiteBalanceRedFineTuning>
      <!--ro, opt, int, range:[0,100]-->1
    </WhiteBalanceRedFineTuning>
    <WhiteBalanceBlueFineTuning>
      <!--ro, opt, int, range:[0,100]-->1
    </WhiteBalanceBlueFineTuning>
  </WhiteBalance>
  <Exposure>
    <!--ro, opt, object-->
    <ExposureType>
      <!--ro, req, enum, subType:string-->auto
    </ExposureType>
    <OverexposeSuppress>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
    </OverexposeSuppress>
  </Exposure>
  <Sharpness>
    <!--ro, opt, object-->
    <SharpnessLevel>
      <!--ro, req, int, range:[0,100]-->1
    </SharpnessLevel>
  </Sharpness>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel>
      <!--ro, opt, enum, subType:string-->1/1
    </ShutterLevel>
    <maxShutterLevelLimit>
      <!--ro, opt, enum, subType:string-->1/1
    </maxShutterLevelLimit>
    <minShutterLevelLimit>
      <!--ro, opt, enum, subType:string-->1/1
    </minShutterLevelLimit>
  </Shutter>
  <powerLineFrequency>
    <!--ro, opt, object-->
    <powerLineFrequencyMode>
      <!--ro, opt, enum, subType:string-->50hz
    </powerLineFrequencyMode>
  </powerLineFrequency>
  <Gain>
    <!--ro, opt, object-->
    <GainLevel>
```

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DETEKTOR

```
      <!--ro, opt, string, range:[0,10]-->test
    </GainLevel>
    <GainLimit>
      <!--ro, opt, enum, subType:string-->0
    </GainLimit>
  </Gain>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </contrastLevel>
    <saturationLevel>
      <!--ro, opt, int, range:[0,255]-->0
    </saturationLevel>
  </Color>
  <IrcutFilter>
    <!--ro, opt, object-->
    <IrcutFilterType>
      <!--ro, opt, enum, subType:string-->auto
    </IrcutFilterType>
    <nightToDayFilterLevel>
      <!--ro, opt, int, range:[0,7]-->4
    </nightToDayFilterLevel>
    <nightToDayFilterTime>
      <!--ro, opt, int, range:[0,10]-->1
    </nightToDayFilterTime>
  </IrcutFilter>
  <PTZ>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
  </PTZ>
  <HLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <HLCLevel>
      <!--ro, opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <Dehaze>
    <!--ro, opt, object-->
    <DehazeMode>
      <!--ro, opt, enum, subType:string-->open
    </DehazeMode>
    <DehazeLevel>
      <!--ro, opt, int, range:[0,10]-->0
    </DehazeLevel>
  </Dehaze>
  <enableImageLossDetection>
    <!--ro, opt, bool-->true
  </enableImageLossDetection>
  <CaptureMode>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
  </CaptureMode>
  <SupplementLight>
    <!--ro, opt, object-->
    <supplementLightMode>
      <!--ro, opt, enum, subType:string-->mixed
    </supplementLightMode>
    <irLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
    </irLightBrightness>
    <mixedLightBrightnessRegulatMode>
      <!--ro, opt, enum, subType:string-->auto
    </mixedLightBrightnessRegulatMode>
    <whiteLightBrightness>
      <!--ro, opt, int, range:[0,100]-->80
    </whiteLightBrightness>
    <EventIntelligenceModeCfg>
      <!--ro, opt, object-->
      <brightnessRegulatMode>
        <!--ro, opt, enum, subType:string-->auto
      </brightnessRegulatMode>
      <whiteLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </whiteLightBrightness>
      <irLightBrightness>
        <!--ro, opt, int, range:[0,100]-->80
      </irLightBrightness>
      <irLightbrightLimit>
        <!--ro, opt, int, range:[0,100]-->80
      </irLightbrightLimit>
      <whiteLightbrightLimit>
        <!--ro, opt, int, range:[0,100]-->80
      </whiteLightbrightLimit>
```

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DETEKTOR

```
    </EventIntelligenceModeCfg>
    <IrLightModeCfg>
      <!--ro, opt, object-->
      <irLightbrightLimit>
        <!--ro, opt, int, range:[0,100]-->80
      </irLightbrightLimit>
    </IrLightModeCfg>
    <colorVuWhiteLightModeCfg>
      <!--ro, opt, object-->
      <whiteLightbrightLimit>
        <!--ro, opt, int, range:[0,100], dep:and,{$.ImageChannel.SupplementLight.mixedLightBrightnessRegulatMode,eq,auto}-->100
      </whiteLightbrightLimit>
    </colorVuWhiteLightModeCfg>
  </SupplementLight>
  <DigitalZoom>
    <!--ro, opt, object-->
    <ZoomRatio>
      <!--ro, opt, enum, subType:string-->1x
    </ZoomRatio>
  </DigitalZoom>
  <Palettes>
    <!--ro, opt, object-->
    <mode>
      <!--ro, opt, enum, subType:string-->WhiteHot
    </mode>
  </Palettes>
  <LaserLight>
    <!--ro, opt, object-->
    <mode>
      <!--ro, opt, enum, subType:string-->auto
    </mode>
  </LaserLight>
</ImageChannel>
```

16.2.5.39 Get the image mode parameters of all channels

Request URL

GET /ISAPI/Image/channels/imageModes

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageModeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, image mode configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <ImageMode>
    <!--ro, opt, object, image mode-->
    <type>
      <!--ro, req, enum, type, subType:string, desc:"standard", "indoor", "outdoor", "dimLight", "bright", "soft", "vivid"-->standard
    </type>
    <recommendation>
      <!--ro, req, object, recommended configurations-->
      <brightnessLevel>
        <!--ro, opt, int, brightness, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, contrast, range:[0,100]-->1
      </contrastLevel>
      <sharpnessLevel>
        <!--ro, opt, int, sharpness, range:[0,100]-->1
      </sharpnessLevel>
      <saturationLevel>
        <!--ro, opt, int, saturation, range:[0,100]-->1
      </saturationLevel>
      <hueLevel>
        <!--ro, opt, int, hue, range:[0,100]-->1
      </hueLevel>
      <deNoiseLevel>
        <!--ro, opt, int, noise reduction, range:[0,100]-->1
      </deNoiseLevel>
    </recommendation>
  </ImageMode>
</ImageModeList>
```

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16.2.6 Image Enhancement

16.2.6.1 Set the dark space enhancement parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/DarkEnhance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DarkEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, dark space enhancement, attr:version{req, string, protocolVersion}-->
  <darkEnhanceEnabled>
    <!--req, bool, whether to enable dark space enhancement-->true
  </darkEnhanceEnabled>
  <darkEnhanceLevel>
    <!--opt, int, dark space enhancement level, range:[1,100]-->1
  </darkEnhanceLevel>
</DarkEnhance>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.6.2 Get the dark space enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/DarkEnhance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DarkEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, dark space enhancement, attr:version{req, string, protocolVersion}-->
  <darkEnhanceEnabled>
    <!--ro, req, bool, whether to enable dark space enhancement-->true
  </darkEnhanceEnabled>
  <darkEnhanceLevel>
    <!--ro, opt, int, dark space enhancement level, range:[1,100]-->1
  </darkEnhanceLevel>
</DarkEnhance>
```

16.2.6.3 Get the dark space enhancement capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/DarkEnhance/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DarkEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <darkEnhanceEnabled>
    <!--ro, req, bool, whether to enable dark space enhancement-->true
  </darkEnhanceEnabled>
  <darkEnhanceLevel min="0" max="100">
    <!--ro, opt, int, dark space enhancement level, attr:min{req, int, range:[1,100]},max{req, int, range:[1,100]}-->1
  </darkEnhanceLevel>
</DarkEnhance>
```

16.2.6.4 Get the electronic defogging parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/defog

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Defog xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</Defog>
```

16.2.6.5 Set the electronic defogging parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/defog

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<Defog xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, defog configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
</Defog>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.6.6 Get defog mode parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/dehaze

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Dehaze xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, "open,close,auto", attr:version{req, string, protocolVersion}-->
  <DehazeMode>
    <!--ro, opt, enum, defog mode, subType:string, desc:"open", "close", "auto"-->open
  </DehazeMode>
  <DehazeLevel>
    <!--ro, opt, int, defog level-->1
  </DehazeLevel>
  <dehazeIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.Dehaze.DehazeMode,eq,advancedMode}-->0
  </dehazeIntensity>
  <smoothIntensity>
    <!--ro, opt, int, range:[0,100], dep:and,{$.Dehaze.DehazeMode,eq,advancedMode}-->0
  </smoothIntensity>
</Dehaze>
```

16.2.6.7 Set defog mode parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/dehaze

Query Parameter

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DETEKTOR

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Dehaze xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, "open,close,auto", attr:version{req, string, protocolVersion}-->
  <DehazeMode>
    <!--opt, enum, defog mode, subType:string, desc:"open", "close", "auto"-->open
  </DehazeMode>
  <DehazeLevel>
    <!--opt, int, defog level-->1
  </DehazeLevel>
</Dehaze>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.6.8 Set the picture gain parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/gain

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Gain xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, gain, attr:version{req, string, protocolVersion}-->
  <GainLevel>
    <!--opt, int, gain level-->1
  </GainLevel>
  <GainWindow>
    <!--opt, object-->
    <RegionCoordinatesList>
      <!--opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
  <scheduleEnabled>
    <!--opt, bool-->true
  </scheduleEnabled>
  <scheduleGainLevel>
    <!--opt, int, dep:and,{$.Gain.scheduleEnabled,eq,true}-->1
  </scheduleGainLevel>
  <Schedule>
    <!--opt, object, dep:and,{$.Gain.scheduleEnabled,eq,true}-->
    <startHour min="0" max="23">
      <!--opt, int, start time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </startHour>
    <startMinute min="0" max="59">
      <!--opt, int, start time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </startMinute>
    <endHour min="0" max="23">
      <!--opt, int, end time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </endHour>
    <endMinute min="0" max="59">
      <!--opt, int, end time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </endMinute>
  </Schedule>
  <redGainLevel>
    <!--opt, int, range:[0,100]-->1
  </redGainLevel>
</Gain>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.6.9 Get the picture gain parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/gain?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Gain xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, gain, attr:version{req, string, protocolVersion}-->
  <GainLevel>
    <!--ro, opt, int, gain level-->1
  </GainLevel>
  <GainWindow>
    <!--ro, opt, object-->
    <RegionCoordinatesList>
      <!--ro, opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object, the origin is the upper-left corner of the screen, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GainWindow>
  <scheduleEnabled>
    <!--ro, opt, bool-->true
  </scheduleEnabled>
  <scheduleGainLevel>
    <!--ro, opt, int, dep:and,{$.Gain.scheduleEnabled,eq,true}-->1
  </scheduleGainLevel>
  <Schedule>
    <!--ro, opt, object, time range, dep:and,{$.Gain.scheduleEnabled,eq,true}-->
    <startHour min="0" max="23">
      <!--ro, opt, int, start time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </startHour>
    <startMinute min="0" max="59">
      <!--ro, opt, int, start time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </startMinute>
    <endHour min="0" max="23">
      <!--ro, opt, int, end time, range:[0,23], attr:min{req, int},max{req, int}-->1
    </endHour>
    <endMinute min="0" max="59">
      <!--ro, opt, int, end time (minute), range:[0,59], attr:min{req, int},max{req, int}-->1
    </endMinute>
  </Schedule>
  <redGainLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </redGainLevel>
  <minExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </minExposureGainLevel>
  <maxExposureGainLevel>
    <!--ro, opt, int, range:[0,10]-->0
  </maxExposureGainLevel>
</Gain>
```

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DETEKTOR

16.2.6.10 Set the DDE (Digital Detail Enhancement) parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/LSEDetail

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSEDetail xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable DDE-->true
  </enabled>
  <LSEDetailLevel>
    <!--opt, int, DDE level, range:[0,100]-->1
  </LSEDetailLevel>
</LSEDetail>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.6.11 Get the DDE (Digital Detail Enhancement) parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/LSEDetail

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSEDetail xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable DDE-->true
  </enabled>
  <LSEDetailLevel>
    <!--ro, opt, int, DDE level, range:[0,100]-->1
  </LSEDetailLevel>
</LSEDetail>
```

16.2.6.12 Get the DDE (Digital Detail Enhancement) capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/LSEDetail/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

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Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSEDetail xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable DDE-->true
  </enabled>
  <LSEDetailLevel>
    <!--ro, opt, int, DDE level, range:[0,100]-->1
  </LSEDetailLevel>
</LSEDetail>
```

16.2.6.13 Get the memory color enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/mce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MCE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable memory color enhancement-->true
  </enabled>
  <mceLevel>
    <!--ro, opt, int, memory color enhancement level, range:[0,100]-->1
  </mceLevel>
</MCE>
```

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DETEKTOR

16.2.6.14 Set the memory color enhancement parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/mce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MCE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, the memory color enhancement parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable memory color enhancement-->true
  </enabled>
  <mceLevel>
    <!--opt, int, memory color enhancement level, range:[0,100]-->1
  </mceLevel>
</MCE>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.6.15 Get the memory color enhancement capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/mce/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MCE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable memory color enhancement-->true
  </enabled>
  <mceLevel min="0" max="100" def="50">
    <!--ro, opt, int, memory color enhancement level, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->1
  </mceLevel>
</MCE>
```

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DETEKTOR

16.2.6.16 Set the DNR (Digital Noise Reduction) parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/noiseReduce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--opt, object, mode-->
    <generalLevel>
      <!--req, int, level-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, advanced mode-->
    <FrameNoiseReduceLevel>
      <!--req, int-->1
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--req, int-->1
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.6.17 Get the DNR (Digital Noise Reduction) parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduce

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, this node is valid only when <mode> is "general"-->
    <generalLevel>
      <!--ro, req, int-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, advanced mode-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int-->1
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--ro, req, int-->1
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

16.2.6.18 Set the digital noise reduction (DNR) parameters in day mode of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/noiseReduce/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.6.19 Get the digital noise reduction (DNR) parameters in day mode of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduce/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--ro, req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

16.2.6.20 Set the digital noise reduction (DNR) parameters in night mode a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/noiseReduce/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.6.21 Get the digital noise reduction (DNR) parameters in day mode of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduce/night

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, subType:string-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,general}-->
    <generalLevel>
      <!--ro, req, int, range:[0,100]-->0
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, dep:and,{$.NoiseReduce.mode,eq,advanced}-->
    <FrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </FrameNoiseReduceLevel>
    <InterFrameNoiseReduceLevel>
      <!--ro, req, int, range:[0,100]-->0
    </InterFrameNoiseReduceLevel>
    <secondarySpatialNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondarySpatialNoiseReduceLevel>
    <secondaryTemporalNoiseReduceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </secondaryTemporalNoiseReduceLevel>
  </AdvancedMode>
</NoiseReduce>
```

16.2.6.22 Set the 2D noise reduction parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/NoiseReduce2D

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce2D xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, 2D noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <noiseReduce2DEnable>
    <!--req, bool, whether to enable 2D noise reduction-->true
  </noiseReduce2DEnable>
  <noiseReduce2DLevel min="0" max="100">
    <!--opt, int, 2D noise reduction level, attr:min{req, int},max{req, int}-->1
  </noiseReduce2DLevel>
</NoiseReduce2D>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.6.23 Get 2D noise reduction parameters of a specific channel

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Request URL

GET /ISAPI/Image/channels/<channelID>/NoiseReduce2D?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduce2D xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <noiseReduce2DEnable>
    <!--ro, req, bool-->true
  </noiseReduce2DEnable>
  <noiseReduce2DLevel min="0" max="255">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </noiseReduce2DLevel>
</NoiseReduce2D>
```

16.2.6.24 Set the 3D noise reduction parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/noiseReduceExt

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduceExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--opt, object, general mode-->
    <generalLevel>
      <!--opt, int, general level, range:[0,100]-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--opt, object, advanced mode-->
    <spatialLevel>
      <!--opt, int, spatial level, range:[0,100]-->1
    </spatialLevel>
    <temporalLevel>
      <!--opt, int, level, range:[0,100]-->1
    </temporalLevel>
  </AdvancedMode>
</NoiseReduceExt>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.6.25 Get 3D noise reduction parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/noiseReduceExt?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NoiseReduceExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, noise reduction parameters, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, mode, subType:string, desc:"close", "general", "advanced"-->close
  </mode>
  <GeneralMode>
    <!--ro, opt, object, general mode-->
    <generalLevel>
      <!--ro, opt, int, general level, range:[0,100]-->1
    </generalLevel>
  </GeneralMode>
  <AdvancedMode>
    <!--ro, opt, object, advanced mode-->
    <spatialLevel>
      <!--ro, opt, int, spatial level, range:[0,100]-->1
    </spatialLevel>
    <temporalLevel>
      <!--ro, opt, int, temporal level, range:[0,100]-->1
    </temporalLevel>
  </AdvancedMode>
  <trailingNoiseBalance>
    <!--ro, opt, object-->
    <level>
      <!--ro, opt, int, range:[0,100]-->1
    </level>
  </trailingNoiseBalance>
</NoiseReduceExt>
```

16.2.7 Image Light Supplement

16.2.7.1 Get the parameters of supplement light alarm output of a specified channel

Request URL

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GET /ISAPI/Event/triggers/notifications/channels/<channelID>/supplementLightAlarm/<eventType>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
eventType	string	Supported event types: "lineDetection" (line crossing), "fieldDetection" (intrusion detection), "regionEntrance", "regionExiting", "VMD", "IO" (alarm input)

Request Message

None

Response Message

```
{
    "SupplementLightAlarm": {
    /*ro, req, object*/
        "durationTime":  15
        /*ro, opt, int, unit:s*/
    }
}
```

16.2.7.2 Set the parameters of supplement light alarm output of a specified channel

Request URL

PUT /ISAPI/Event/triggers/notifications/channels/<channelID>/supplementLightAlarm/<eventType>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
eventType	string	--

Request Message

```
{
    "SupplementLightAlarm": {
    /*req, object*/
        "durationTime":  15
        /*opt, int*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.2.7.3 Set the day/night mode switching parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/dayNightGate

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DayNightGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, day/night mode switching parameters, attr:version{req, string, protocolVersion}-->
  <brightnessThreld>
    <!--opt, int, brightness-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--opt, object, day/night mode control-->
    <nightModeCtrlMode>
      <!--opt, enum, day/night switching mode, subType:string, desc:"manual" (manual switch), "time" (timed switch), "auto" (auto switch)-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--opt, object, manual switch-->
      <manualModeVal>
        <!--opt, enum, manual switching mode, subType:string, desc:"day", "night"-->day
      </manualModeVal>
    </manualMode>
    <timeMode>
      <!--opt, object, timed switch-->
      <switchList>
        <!--opt, array, subType:object-->
        <switch>
          <!--opt, object-->
          <nightModeTmeId>
            <!--opt, int-->1
          </nightModeTmeId>
          <nightModeStartHour>
            <!--opt, int, range:[0,23]-->1
          </nightModeStartHour>
          <nightModeStartMinute>
            <!--opt, int, range:[0,59]-->1
          </nightModeStartMinute>
          <nightModeEndHour>
            <!--opt, int, range:[0,23]-->1
          </nightModeEndHour>
          <nightModeEndMinute>
            <!--opt, int, range:[0,59]-->1
          </nightModeEndMinute>
        </switch>
      </switchList>
    </timeMode>
    <autoMode>
      <!--opt, object, auto switch-->
      <autoSwtichThresholdVal>
        <!--opt, int, threshold for switching, range:[1,100]-->1
      </autoSwtichThresholdVal>
    </autoMode>
  </nightModeCtrl>
</DayNightGate>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.7.4 Get the day/night switch parameters of the specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/dayNightGate

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DayNightGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, day/night switch, attr:version{req, string, protocolVersion}-->
  <brightnessThreld>
    <!--ro, opt, int, brightness-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--ro, opt, object, day/night mode switching control-->
    <nightModeCtrlMode>
      <!--ro, opt, enum, day/night switch mode, subType:string, desc:"manual" (manual switch mode), "time" (timed switch mode), "auto" (automatic switch
mode)-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--ro, opt, object, manual switch-->
      <manualModeVal>
        <!--ro, opt, enum, "day", "night", subType:string, desc:"day", "night"-->day
      </manualModeVal>
    </manualMode>
    <timeMode>
      <!--ro, opt, object, timed switch-->
      <scheduleType>
        <!--ro, opt, enum, subType:string-->day
      </scheduleType>
      <switchList>
        <!--ro, opt, array, subType:object-->
        <switch>
          <!--ro, opt, object, timed switch settings-->
          <nightModeTmeId>
            <!--ro, opt, int, night mode ID-->1
          </nightModeTmeId>
          <nightModeStartHour>
            <!--ro, opt, int, night mode starting hour, range:[0,23]-->1
          </nightModeStartHour>
          <nightModeStartMinute>
            <!--ro, opt, int, night mode starting minute, range:[0,59]-->1
          </nightModeStartMinute>
          <nightModeEndHour>
            <!--ro, opt, int, night mode ending hour, range:[0,23]-->1
          </nightModeEndHour>
          <nightModeEndMinute>
            <!--ro, opt, int, night mode ending minute, range:[0,59]-->1
          </nightModeEndMinute>
        </switch>
      </switchList>
      <timeScheduleList>
        <!--ro, opt, array, subType:object, range:[0,2]-->
        <timeSchedule>
          <!--ro, opt, object-->
          <triggerData>
            <!--ro, opt, object-->
            <startTime>
              <!--ro, opt, object-->
              <monthOfYear>
                <!--ro, opt, int, range:[1,12]-->0
              </monthOfYear>
            </startTime>
            <endTime>
              <!--ro, opt, object-->
              <monthOfYear>
                <!--ro, opt, int, range:[1,12]-->0
              </monthOfYear>
            </endTime>
          </triggerData>
          <timeSwitchList>
            <!--ro, opt, array, subType:object, range:[0,1]-->
            <timeSwitch>
              <!--ro, opt, object-->
              <startHour>
                <!--ro, opt, int, range:[0,23]-->1
              </startHour>
              <startMinute>
                <!--ro, opt, int, range:[0,59]-->1
              </startMinute>
              <endHour>
```

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DETEKTOR

```
            <!--ro, opt, int, range:[0,23]-->1
          </endHour>
          <endMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </endMinute>
        </timeSwitch>
      </timeSwitchList>
    </timeSchedule>
  </timeScheduleList>
  <dayNightStatus>
    <!--ro, opt, enum, subType:string-->day
  </dayNightStatus>
</timeMode>
<autoMode>
  <!--ro, opt, object, automatic switch-->
  <autoSwtichThresholdVal>
    <!--ro, opt, int, threshold for automatic switch, range:[0,100]-->1
  </autoSwtichThresholdVal>
  <autoNightSwtichThresholdVal>
    <!--ro, opt, int, range:[0,100]-->1
  </autoNightSwtichThresholdVal>
  <autoDayToNightDelayTime>
    <!--ro, opt, int, range:[0,120]-->1
  </autoDayToNightDelayTime>
  <autoNightToDayDelayTime>
    <!--ro, opt, int, range:[0,120]-->1
  </autoNightToDayDelayTime>
  <ICRChange>
    <!--ro, opt, bool-->false
  </ICRChange>
  <lightChange>
    <!--ro, opt, bool-->false
  </lightChange>
  <WDRChange>
    <!--ro, opt, bool-->false
  </WDRChange>
  <dayNightStatus>
    <!--ro, opt, enum, subType:string-->day
  </dayNightStatus>
</autoMode>
</nightModeCtrl>
</DayNightGate>
```

16.2.7.5 Set the IR-cut filter parameters for cameras of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/icr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ICR configuration parameters, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--opt, object, IR-cut filter-->
    <ICRCtrlMode>
      <!--req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-manually
switch, "time"-switch by schedule, "auto"-automatically switch-->close
    </ICRCtrlMode>
    <ManualMode>
      <!--opt, object, manually switch mode-->
      <ManualPresetVal>
        <!--req, enum, manually switch, subType:string, desc:"day", "night”-->day
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--opt, object, switch by schedule-->
      <SwitchList>
        <!--opt, array, schedule list for switching, subType:object, desc:up to two time period control can be supported. For example: time period is from
6:00  to 18:00, the preset ID is 1-->
        <Preset>
          <!--opt, object, preset-->
          <PresetId>
            <!--req, int, preset ID-->1
          </PresetId>
          <PresetVal>
            <!--opt, enum, "day","night", subType:string, desc:"day", "night”-->day
          </PresetVal>
        </Preset>
```

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DETEKTOR

```
        <TimeSwitch>
          <!--opt, array, time period, subType:object-->
          <timeId>
            <!--req, int, time period ID, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--opt, int, start hour, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--opt, int, start minute, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--opt, int, end hour, range:[0,23]-->1
          </endHour>
          <endMinute>
            <!--opt, int, end minute, range:[0,59]-->1
          </endMinute>
        </TimeSwitch>
      </SwitchList>
    </TimeMode>
    <AutoMode>
      <!--opt, object, automatically switch according to the light change-->
      <DayNightFilterTh>
        <!--req, int, day/night auto switch sensitivity, range:[0,100]-->1
      </DayNightFilterTh>
      <ICRAutoSwitch min="0" max="100">
        <!--opt, int, ICR auto-switch and threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
      </ICRAutoSwitch>
    </AutoMode>
  </ICRCtrl>
  <ECtrl>
    <!--opt, object, E device-->
    <ECtrlMode>
      <!--req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"
(manually switch), "time" (switch by schedule), "auto" (automatically switch)-->close
    </ECtrlMode>
    <ManualMode>
      <!--opt, object, manually switch-->
      <ManualPresetVal>
        <!--req, int, manually switch: "day","night", range:[0,100]-->1
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--opt, object, switch by schedule-->
      <SwitchList>
        <!--opt, array, schedule list, subType:object, desc:up to two time period control can be supported. For example: time period is from 6:00  to
18:00,the preset ID is 1-->
        <Preset>
          <!--opt, object, preset-->
          <PresetId>
            <!--req, int, preset ID,which ranges from 0 to 1-->1
          </PresetId>
          <PresetVal>
            <!--opt, enum, "day", "night", subType:string, desc:"day", "night"-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--opt, array, time period, subType:object-->
          <timeId>
            <!--req, int, time period ID, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--opt, int, start hour, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--opt, int, start minute, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--opt, int, end hour, range:[0,23]-->1
          </endHour>
          <endMinute>
            <!--opt, int, end minute, range:[0,59]-->1
          </endMinute>
        </TimeSwitch>
      </SwitchList>
    </TimeMode>
    <AutoMode>
      <!--opt, object, automatically switch according to the light change-->
      <DayNightFilterTh>
        <!--req, int, day/night auto switch sensitivity, range:[0,100]-->1
      </DayNightFilterTh>
      <ExceptionCatchTh>
        <!--req, int, exception detection sensitivity, range:[0,100]-->1
      </ExceptionCatchTh>
    </AutoMode>
  </ECtrl>
</ImageIcrE>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.7.6 Get the IR-cut filter parameters for cameras of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/icr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ICR parameters configuration, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--ro, opt, object, IR-cut filter-->
    <ICRCtrlMode>
      <!--ro, req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-
manually switch, "time"-switch by schedule, "auto"-automatically switch-->close
    </ICRCtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,manual}-->
      <ManualPresetVal>
        <!--ro, req, enum, manually switch, subType:string, desc:"day", "night"-->day
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,time}-->
      <scheduleType>
        <!--ro, opt, enum, subType:string-->day
      </scheduleType>
      <SwitchList>
        <!--ro, opt, array, schedule list for switching, subType:object, range:[0,2], dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,day}, desc:up to
two time period control can be supported. For example: time period is from 6:00  to 18:00,the preset ID is 1-->
        <Preset>
          <!--ro, opt, object, preset-->
          <PresetId>
            <!--ro, req, int, preset ID-->1
          </PresetId>
          <PresetVal>
            <!--ro, opt, enum, "day","night", subType:string, desc:"day", "night"-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId>
            <!--ro, req, int, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--ro, opt, int, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--ro, opt, int, range:[0,23]-->1
          </endHour>
```

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```
            <endMinute>
              <!--ro, opt, int, range:[0,59]-->1
            </endMinute>
            <PresetVal>
              <!--ro, opt, enum, subType:string-->day
            </PresetVal>
          </TimeSwitch>
        </SwitchList>
        <dayNightStatus>
          <!--ro, opt, enum, subType:string-->day
        </dayNightStatus>
        <timeScheduleList>
          <!--ro, opt, array, subType:object, range:[0,2], dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,range}-->
          <timeSchedule>
            <!--ro, opt, object-->
            <triggerData>
              <!--ro, opt, object-->
              <startTime>
                <!--ro, opt, object-->
                <monthOfYear>
                  <!--ro, opt, int, range:[1,12]-->1
                </monthOfYear>
              </startTime>
              <endTime>
                <!--ro, opt, object-->
                <monthOfYear>
                  <!--ro, opt, int, range:[1,12]-->1
                </monthOfYear>
              </endTime>
            </triggerData>
            <timeSwitchList>
              <!--ro, opt, array, subType:object, range:[0,2]-->
              <timeSwitch>
                <!--ro, opt, object-->
                <timeID>
                  <!--ro, req, int, range:[0,1]-->1
                </timeID>
                <startHour>
                  <!--ro, opt, int, range:[0,23]-->1
                </startHour>
                <startMinute>
                  <!--ro, opt, int, range:[0,59]-->1
                </startMinute>
                <endHour>
                  <!--ro, opt, int, range:[0,23]-->1
                </endHour>
                <endMinute>
                  <!--ro, opt, int, range:[0,59]-->1
                </endMinute>
                <dayNightVal>
                  <!--ro, opt, enum, subType:string-->day
                </dayNightVal>
              </timeSwitch>
            </timeSwitchList>
          </timeSchedule>
        </timeScheduleList>
      </TimeMode>
      <AutoMode>
        <!--ro, opt, object, automatically switch according to the light change, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,auto}-->
        <DayNightFilterTh>
          <!--ro, req, int, day/night auto switch sensitivity, range:[0,100]-->1
        </DayNightFilterTh>
        <ICRAutoSwitch min="0" max="100">
          <!--ro, opt, int, ICR auto-switch and threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
        </ICRAutoSwitch>
        <dayNightStatus>
          <!--ro, opt, enum, subType:string-->day
        </dayNightStatus>
      </AutoMode>
    </ICRCtrl>
    <ECtrl>
      <!--ro, opt, object, -E device,halo inhibition filter,which can be used to inhibit the halo-->
      <ECtrlMode>
        <!--ro, req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-
manually switch, "time"-switch by schedule, "auto"-automatically switch-->close
      </ECtrlMode>
      <ManualMode>
        <!--ro, opt, object, manually switch, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,manual}-->
        <ManualPresetVal>
          <!--ro, req, int, manually switch, range:[0,100]-->1
        </ManualPresetVal>
      </ManualMode>
      <TimeMode>
        <!--ro, opt, object, switch by schedule, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,time}-->
        <SwitchList>
          <!--ro, opt, array, schedule list for switching, subType:object, range:[0,2], desc:up to two time period control can be supported. For example: time
period is from 6:00  to 18:00,the preset ID is 1-->
          <Preset>
            <!--ro, opt, object, preset-->
            <PresetId>
              <!--ro, req, int, preset ID-->1
            </PresetId>
            <PresetVal>
              <!--ro, opt, enum, "day", "night", subType:string, desc:"day", "night"-->day
```

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```
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId>
            <!--ro, req, int, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--ro, opt, int, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--ro, opt, int, range:[0,23]-->1
          </endHour>
          <endMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </endMinute>
          <PresetVal>
            <!--ro, opt, enum, subType:string-->day
          </PresetVal>
        </TimeSwitch>
      </SwitchList>
    </TimeMode>
    <AutoMode>
      <!--ro, opt, object, automatically switch according to the light change, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,auto}-->
      <DayNightFilterTh>
        <!--ro, req, int, day/night auto switch sensitivity, range:[0,100]-->1
      </DayNightFilterTh>
      <ExceptionCatchTh>
        <!--ro, req, int, exception detection sensitivity, range:[0,100]-->1
      </ExceptionCatchTh>
    </AutoMode>
  </ECtrl>
</ImageIcrE>
```

16.2.7.7 Get the IR-cut filter capability for cameras of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/icr/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ICR parameters configuration, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--ro, opt, object, IR-cut filter-->
    <ICRCtrlMode opt="close,manual,time,auto">
      <!--ro, req, string, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, attr:opt{req, string}-->test
    </ICRCtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch-->
      <ManualPresetVal opt="day,night">
        <!--ro, req, string, manually switch, attr:opt{req, string}-->test
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule-->
      <scheduleType opt="day,range">
        <!--ro, opt, string, attr:opt{req, string}-->day
      </scheduleType>
      <SwitchList size="2">
        <!--ro, opt, array, subType:object, dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,day}, attr:size{req, int}-->
        <Preset>
          <!--ro, req, object-->
          <PresetId min="0" max="1">
            <!--ro, req, int, attr:min{req, int},max{req, int}-->1
          </PresetId>
          <PresetVal opt="day,night">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
```

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```
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId min="0" max="1">
            <!--ro, req, int, attr:min{req, int},max{req, int}-->1
          </timeId>
          <startHour min="0" max="23">
            <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
          </startHour>
          <startMinute min="0" max="59">
            <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
          </startMinute>
          <endHour min="0" max="23">
            <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
          </endHour>
          <endMinute min="0" max="59">
            <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
          </endMinute>
          <PresetVal opt="day,night">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
          </PresetVal>
        </TimeSwitch>
      </SwitchList>
      <timeScheduleList size="2">
        <!--ro, opt, object, dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,range}, attr:size{req, int}-->
        <timeSchedule>
          <!--ro, opt, object-->
          <triggerData>
            <!--ro, opt, object-->
            <startTime>
              <!--ro, opt, object-->
              <monthOfYear min="1" max="12">
                <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->0
              </monthOfYear>
            </startTime>
            <endTime>
              <!--ro, opt, object-->
              <monthOfYear min="1" max="12">
                <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->0
              </monthOfYear>
            </endTime>
          </triggerData>
          <timeSwitchList size="2">
            <!--ro, opt, object, attr:size{req, int}-->
            <timeSwitch>
              <!--ro, opt, object-->
              <timeID min="0" max="1">
                <!--ro, req, int, range:[0,1], attr:min{req, int},max{req, int}-->1
              </timeID>
              <startHour min="0" max="23">
                <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
              </startHour>
              <startMinute min="0" max="59">
                <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
              </startMinute>
              <endHour min="0" max="23">
                <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
              </endHour>
              <endMinute min="0" max="59">
                <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
              </endMinute>
              <dayNightVal opt="day,night">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
              </dayNightVal>
            </timeSwitch>
          </timeSwitchList>
        </timeSchedule>
      </timeScheduleList>
    </TimeMode>
    <AutoMode>
      <!--ro, opt, object, automatically switch according to the light change-->
      <DayNightFilterTh min="0" max="100">
        <!--ro, req, int, day/night auto switch sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
      </DayNightFilterTh>
      <ICRAutoSwitch min="0" max="100">
        <!--ro, opt, int, ICR auto-switch and threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
      </ICRAutoSwitch>
    </AutoMode>
  </ICRCtrl>
  <ECtrl>
    <!--ro, opt, object, -E device,halo inhibition filter,which can be used to inhibit the halo-->
    <ECtrlMode opt="close,manual,time,auto">
      <!--ro, req, string, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, attr:opt{req, string}-->test
    </ECtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch-->
      <ManualPresetVal min="0" max="100">
        <!--ro, req, int, manually switch, range:[0,100], attr:min{req, int},max{req, int}-->1
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule-->
      <SwitchList size="2">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <Preset>
```

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```
            <!--ro, req, object-->
            <PresetId min="0" max="1">
              <!--ro, req, int, attr:min{req, int},max{req, int}-->1
            </PresetId>
            <PresetVal opt="day,night">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
            </PresetVal>
          </Preset>
          <TimeSwitch>
            <!--ro, opt, object-->
            <timeId min="0" max="1">
              <!--ro, req, int, attr:min{req, int},max{req, int}-->1
            </timeId>
            <startHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </startHour>
            <startMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </startMinute>
            <endHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </endHour>
            <endMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </endMinute>
            <PresetVal opt="day,night">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
            </PresetVal>
          </TimeSwitch>
        </SwitchList>
      </TimeMode>
      <AutoMode>
        <!--ro, opt, object, automatically switch according to the light change-->
        <DayNightFilterTh min="0" max="100">
          <!--ro, req, int, day/night auto switch sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
        </DayNightFilterTh>
        <ExceptionCatchTh min="0" max="100">
          <!--ro, req, int, "0-100", range:[0,100], attr:min{req, int},max{req, int}-->1
        </ExceptionCatchTh>
      </AutoMode>
    </ECtrl>
  </ImageIcrE>
```

16.2.7.8 Get day/night auto switch parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/ircutFilter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--ro, opt, enum, day/night auto switch mode, subType:string, desc:"auto, day, night, schedule, eventTrigger, darkFighterX, darkFighterXAuto,
darkFighterXSchedule"-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--ro, opt, int, level of switching day to night, range:[0,7]-->4
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--ro, opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--ro, opt, int, level of switching night to day, range:[0,7]-->4
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--ro, opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--ro, opt, object-->
    <scheduleType>
      <!--ro, req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--ro, req, object, time period on schedule-->
      <beginTime>
        <!--ro, req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--ro, opt, object-->
    <eventType>
      <!--ro, req, enum, event type, subType:string, desc:event type-->IO
    </eventType>
    <IrcutFilterAction>
      <!--ro, req, enum, "day, night", subType:string, desc:"day", "night"-->day
    </IrcutFilterAction>
  </EventTrigger>
  <dayNightStatus>
    <!--ro, opt, enum, subType:string-->day
  </dayNightStatus>
</IrcutFilter>
```

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16.2.7.9 Set day/night auto switch parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/ircutFilter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrcutFilter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, IrcutFilter, attr:version{req, string, protocolVersion}-->
  <IrcutFilterType>
    <!--opt, enum, IrcutFilter type, subType:string, desc:"auto", "day", "night", "schedule" , "eventTrigger", "videoAuto", "darkFighterX",
"darkFighterXAuto", "darkFighterXSchedule", "colorVuAuto"-->auto
  </IrcutFilterType>
  <dayToNightFilterLevel>
    <!--opt, enum, level of switching day to night, subType:string, desc:"low", "normal", "high"-->low
  </dayToNightFilterLevel>
  <dayToNightFilterTime>
    <!--opt, int, time interval of switching day to night-->1
  </dayToNightFilterTime>
  <nightToDayFilterLevel>
    <!--opt, enum, level of switching night to day, subType:string, desc:"low", "normal", "high"-->low
  </nightToDayFilterLevel>
  <nightToDayFilterTime>
    <!--opt, int, time interval of switching night to day-->1
  </nightToDayFilterTime>
  <Schedule>
    <!--opt, object, schedule-->
    <scheduleType>
      <!--req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--req, object, time period on schedule-->
      <beginTime>
        <!--req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
  <EventTrigger>
    <!--opt, object, event trigger-->
    <eventType>
      <!--req, enum, event type, subType:string, desc:"IO", "VMD"-->IO
    </eventType>
    <IrcutFilterAction>
      <!--req, enum, "day,night", subType:string, desc:"day", "night"-->day
    </IrcutFilterAction>
  </EventTrigger>
</IrcutFilter>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.7.10 Set the day/night mode parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/ISPMode

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ISPMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ISP mode, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--opt, enum, "auto,schedule", subType:string, desc:“auto”, “schedule”-->schedule
  </mode>
  <Schedule>
    <!--opt, object-->
    <scheduleType>
      <!--req, enum, "day,night", subType:string, desc:“day”, “night”-->day
    </scheduleType>
    <TimeRange>
      <!--req, object, time range-->
      <beginTime>
        <!--req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
</ISPMode>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.7.11 Get the ISP mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/ISPMode

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ISPMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ISP mode, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, opt, enum, mode, subType:string, desc:"auto", "schedule"-->schedule
  </mode>
  <Schedule>
    <!--ro, opt, object, schedule information-->
    <scheduleType>
      <!--ro, req, enum, schedule type, subType:string, desc:"day", "night"-->day
    </scheduleType>
    <TimeRange>
      <!--ro, req, object, time range-->
      <beginTime>
        <!--ro, req, time, start time-->00:00:00+08:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->00:00:00+08:00
      </endTime>
    </TimeRange>
  </Schedule>
</ISPMode>
```

16.2.7.12 Set the laser spot adjustment parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/laserLight/laserSpotAdjustment

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<laserSpotAdjustment xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <laserSpotSize>
    <!--req, enum, subType:string-->increase
  </laserSpotSize>
</laserSpotAdjustment>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.7.13 Get the capability of laser spot adjustment of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/laserLight/laserSpotAdjustment/capabilities

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<laserSpotAdjustment xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <laserSpotSize opt="increase,decrease">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->increase
  </laserSpotSize>
</laserSpotAdjustment>
```

16.2.8 Manual Capture

16.2.8.1 Capture pictures via digital channel

Request URL

GET /ISAPI/ContentMgmt/StreamingProxy/channels/<trackStreamID>/Picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--
snapShotImageType	enum	--
videoResolutionWidth	string	--
videoResolutionHeight	string	--

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Request Message

None

Response Message

Binary Data

16.2.8.2 Asynchronously get the picture captured manually

Request URL

GET /ISAPI/Streaming/channels/<channelID>/picture/async?format=json&imageType=<imageType>&URLType=<URLType>&channelFormat=<channelFormat>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
imageType	enum	--
URLType	enum	--
channelFormat	string	--

Request Message

None

Response Message

```
{
    "PictureData": {
    /*ro, opt, object, data of the captured picture*/
        "url":  "test"
        /*ro, req, string, URL of the pictured stored in the picture storage server, range:[1,256]*/
    }
}
```

16.2.9 Motion Detection

16.2.9.1 Get the linkage parameters of motion detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupporttriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, string-->VMD
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <triggerEnabled>
        <!--ro, opt, bool-->true
      </triggerEnabled>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->[email
      </notificationMethod>
      <notificationRecurrence>
```

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```
    <!--ro, opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--ro, opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--ro, opt, string--> 1
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--ro, opt, string--> 1
  </dynOutputIOPortID>
  <videoInputID>
    <!--ro, opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--ro, opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--ro, opt, object-->
    <ptzChannelID>
      <!--ro, req, string-->1
    </ptzChannelID>
    <actionName>
      <!--ro, req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--ro, opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--ro, opt, int, unit:s-->1
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--ro, opt, string, range:[1,11]-->test
  </cellphoneNumber>
  <LightAudioByIPAction>
    <!--ro, opt, object-->
    <notificationType>
      <!--ro, req, enum, subType:string-->audio
    </notificationType>
    <notificationName>
      <!--ro, opt, string-->test
    </notificationName>
  </LightAudioByIPAction>
  <SupplementLightAlarm>
    <!--ro, opt, object-->
    <durationTime>
      <!--ro, opt, int, range:[0,90], unit:s-->5
    </durationTime>
  </SupplementLightAlarm>
  <preRecordTime>
    <!--ro, opt, int, range:[0,10], unit:s-->1
  </preRecordTime>
  <recordTime>
    <!--ro, opt, int, range:[0,180], unit:s-->5
  </recordTime>
  <WhiteLightAction>
    <!--ro, opt, object-->
    <whiteLightDurationTime>
      <!--ro, opt, int, range:[1,60], unit:s-->1
    </whiteLightDurationTime>
    <lightFlashFrequency>
      <!--ro, opt, enum, subType:string-->high
    </lightFlashFrequency>
    <brightness>
      <!--ro, opt, int, range:[0,100]-->1
    </brightness>
    <timeRangeList>
      <!--ro, opt, array, subType:object-->
      <timeRangeItem>
        <!--ro, opt, object-->
        <week>
          <!--ro, req, enum, subType:int-->1
        </week>
        <timeRange>
          <!--ro, req, array, subType:object-->
          <timeItem>
            <!--ro, req, object-->
            <timeRangeID>
              <!--ro, req, int-->1
            </timeRangeID>
            <beginTime>
              <!--ro, req, string-->08:00
            </beginTime>
            <endTime>
              <!--ro, req, string-->24:00
            </endTime>
          </timeItem>
        </timeRange>
      </timeRangeItem>
    </timeRangeList>
  </WhiteLightAction>
  <AudioAction>
```

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```
        <!--ro, opt, object-->
        <audioID>
          <!--ro, req, int-->1
        </audioID>
        <alarmTimes>
          <!--ro, opt, int, range:[0,9]-->1
        </alarmTimes>
        <childDevID>
          <!--ro, opt, string, range:[9,32]-->3434645611
        </childDevID>
      </AudioAction>
      <audioAlarmTriggerParam>
        <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
        <audioIndexList>
          <!--ro, opt, array, subType:object, range:[1,3]-->
          <audioIndexItem>
            <!--ro, req, object-->
            <indexID>
              <!--ro, req, int, range:[1,3]-->1
            </indexID>
            <indexName>
              <!--ro, opt, string, range:[1,128]-->test
            </indexName>
            <alarmTimes>
              <!--ro, opt, int, range:[1,50]-->1
            </alarmTimes>
            <audioID>
              <!--ro, opt, enum, subType:int-->1
            </audioID>
          </audioIndexItem>
        </audioIndexList>
        <timeRangeList>
          <!--ro, opt, array, subType:object-->
          <timeRangeItem>
            <!--ro, opt, object-->
            <week>
              <!--ro, req, enum, subType:int-->1
            </week>
            <timeRange>
              <!--ro, req, array, subType:object-->
              <timeItem>
                <!--ro, req, object-->
                <timeRangeID>
                  <!--ro, req, int-->1
                </timeRangeID>
                <audioID>
                  <!--ro, opt, enum, subType:int-->1
                </audioID>
                <beginTime>
                  <!--ro, req, string-->08:00
                </beginTime>
                <endTime>
                  <!--ro, req, string-->24:00
                </endTime>
              </timeItem>
            </timeRange>
          </timeRangeItem>
        </timeRangeList>
      </audioAlarmTriggerParam>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.2.9.2 Set the linkage parameters of motion detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
```

```
<eventType>
  <!--req, enum, subType:string-->VMD
</eventType>
<eventDescription>
  <!--opt, string-->test
</eventDescription>
<inputIOPortID>
  <!--opt, string-->1
</inputIOPortID>
<dynInputIOPortID>
  <!--opt, string-->1
</dynInputIOPortID>
<videoInputChannelID>
  <!--opt, string-->1
</videoInputChannelID>
<dynVideoInputChannelID>
  <!--opt, string-->1
</dynVideoInputChannelID>
<intervalBetweenEvents>
  <!--opt, int, unit:s-->1
</intervalBetweenEvents>
<WLSensorID>
  <!--opt, string-->1
</WLSensorID>
<EventTriggerNotificationList>
  <!--opt, array, subType:object-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <notificationMethod>
      <!--req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
      <presetDurationTime>
        <!--opt, int, unit:s-->1
      </presetDurationTime>
    </ptzAction>
    <cellphoneNumber>
      <!--opt, string, range:[1,11]-->test
    </cellphoneNumber>
    <LightAudioByIPAction>
      <!--opt, object-->
      <notificationType>
        <!--req, enum, subType:string-->audio
      </notificationType>
      <notificationName>
        <!--opt, string-->test
      </notificationName>
    </LightAudioByIPAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.9.3 Motion detection

EventType:VMD

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, EventNotificationAlert, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 address-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->1
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->2017-04-22T15:39:01+08:0
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"VMD" (motion detection)-->VMD
  </eventType>
  <eventState>
    <!--ro, req, enum, event status (for persistent event), subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <DetectionRegionList>
    <!--ro, opt, array, detection area parameter list (for expert mode only), subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, parameters of a detection area (for expert mode only), desc:parameters of a detection area (for expert mode only)-->
      <regionID>
        <!--ro, req, string, detection area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, req, array, detection area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, detection area origin coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
```

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```
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID-->1
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, req, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, req, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <targetType>
    <!--ro, opt, string, target classification, desc:"human", "vehicle", "human,vehicle" (human and vehicle). If detection target classification is not
supported, this field is not returned-->human
  </targetType>
  <bkgUrl>
    <!--ro, opt, string, background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (it is for the cloud storage protocol), "digest" (digest authentication,
it is for the device local storage)-->no
  </URLCertificationType>
</EventNotificationAlert>
```

16.2.10 OSD Settings

16.2.10.1 Get text overlay parameters of a specific video digital input channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[0,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[0,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent>
      <!--ro, req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--ro, req, bool, Flashing Status-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--ro, opt, object, list of overlay information-->
    <TextOverlay>
      <!--ro, opt, object, list-->
      <id>
        <!--ro, req, string, text No., range:[1,8]-->test
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
```

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```
      </positionY>
      <displayText>
        <!--ro, req, string, overlaid text, range:[1,32]-->test
      </displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <dateStyle>
      <!--ro, opt, enum, date format, subType:string, desc:date format-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--ro, opt, enum, time format, subType:string, desc:time format-->12hour
    </timeStyle>
    <displayWeek>
      <!--ro, opt, bool, whether to display the day of the week-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--ro, opt, enum, font size, subType:string, desc:font size-->adaptive
  </fontSize>
  <frontColorMode>
    <!--ro, opt, enum, font color mode, subType:string, desc:font color mode-->auto
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, dep:and,{$.VideoOverlay.frontColorMode,eq,customize}, desc:font color-->test
  </frontColor>
  <BatteryPowerOverlay>
    <!--ro, opt, object, battery power overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </BatteryPowerOverlay>
  <alignment>
    <!--ro, opt, enum, alignment mode, subType:string, desc:alignment mode-->customize
  </alignment>
  <publicSecurity>
    <!--ro, req, bool, public security-->true
  </publicSecurity>
</VideoOverlay>
```

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16.2.10.2 Set text overlay parameters of a specific video digital input channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--req, object, normalized coordinates-->
```

```
      <normalizedScreenWidth>
        <!--req, int, normalized width, range:[0,1000]-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height, range:[0,1000]-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <attribute>
      <!--opt, object-->
      <transparent>
        <!--req, bool, whether it is transparent-->true
      </transparent>
      <flashing>
        <!--req, bool, whether it is flickering-->true
      </flashing>
    </attribute>
    <TextOverlayList>
      <!--opt, object-->
      <TextOverlay>
        <!--opt, object, list-->
        <id>
          <!--req, string, text No., range:[1,8]-->test
        </id>
        <enabled>
          <!--req, bool, whether to enable-->true
        </enabled>
        <positionX>
          <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
        </positionX>
        <positionY>
          <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
        </positionY>
        <displayText>
          <!--req, string, overlaid text, range:[1,32]-->test
        </displayText>
      </TextOverlay>
    </TextOverlayList>
    <DateTimeOverlay>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to display the date-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
      </positionY>
      <dateStyle>
        <!--opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
      </dateStyle>
      <timeStyle>
        <!--opt, enum, time format, subType:string, desc:time format: "12hour","24hour"-->12hour
      </timeStyle>
      <displayWeek>
        <!--opt, bool, whether to display the day of the week-->true
      </displayWeek>
    </DateTimeOverlay>
    <channelNameOverlay>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
      </positionY>
    </channelNameOverlay>
    <fontSize>
      <!--opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64"-->adaptive
    </fontSize>
    <frontColorMode>
      <!--opt, enum, font color mode, subType:string, desc:font color mode: "auto","customize"-->auto
    </frontColorMode>
    <frontColor>
      <!--opt, string, font color, dep:and,{$.VideoOverlay.frontColorMode,eq,customize}, desc:font color-->test
    </frontColor>
    <BatteryPowerOverlay>
      <!--opt, object, battery power overlay-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:Y-coordinate-->1
      </positionY>
    </BatteryPowerOverlay>
    <alignment>
```

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```
    <!--opt, enum, alignment mode, subType:string, desc:alignment mode: "customize"-custom,"alignRight"-align to right,"alignLeft"-align to left-->customize
  </alignment>
  <publicSecurity>
    <!--req, bool, public security-->true
  </publicSecurity>
</VideoOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

16.2.10.3 Get the text overlay configuration capability of a specific video digital input channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, number of characters in the lower-right corner, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, normalized coordinates-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object-->
    <transparent opt="true,false">
      <!--ro, req, bool, whether it is transparent, attr:opt{req, string}-->true
    </transparent>
    <flashing opt="true,false">
      <!--ro, req, bool, Flashing Status, attr:opt{req, string}-->true
    </flashing>
  </attribute>
  <TextOverlayList size="8">
    <!--ro, opt, array, subType:object, range:[0,8], attr:size{req, int}-->
    <TextOverlay>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <positionX>
        <!--ro, req, int, range:[0,704]-->1
      </positionX>
      <positionY>
```

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DETEKTOR

```
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayText>
        <!--ro, req, string-->test
      </displayText>
      <isPersistentText>
        <!--ro, opt, bool-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
      <!--ro, opt, enum, date format, subType:string, attr:opt{req, string}, desc:date format-->YYYY-MM-DD
    </dateStyle>
    <timeStyle opt="12hour,24hour">
      <!--ro, opt, enum, time format, subType:string, attr:opt{req, string}, desc:time format-->12hour
    </timeStyle>
    <displayWeek opt="true,false">
      <!--ro, opt, bool, Display Day, attr:opt{req, string}-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, Show Channel Name-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </channelNameOverlay>
  <fontSize opt="16*16,32*32,48*48,64*64,adaptive">
    <!--ro, opt, enum, font size, subType:string, attr:opt{req, string}, desc:font size,unit: pixel-->adaptive
  </fontSize>
  <frontColorMode opt="auto,customize">
    <!--ro, opt, enum, font color mode, subType:string, attr:opt{req, string}, desc:font color mode-->auto
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, dep:and,{$.VideoOverlay.frontColorMode,eq,customize}-->test
  </frontColor>
  <BatteryPowerOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </BatteryPowerOverlay>
  <DeviceStatusDisplay>
    <!--ro, opt, object, device status display-->
    <batteryCapacityDisplay>
      <!--ro, opt, bool, whether to display battery capacity-->true
    </batteryCapacityDisplay>
    <chargingStatusDisplay>
      <!--ro, opt, bool, whether to display charging status-->true
    </chargingStatusDisplay>
    <bluetoothStatusDisplay>
      <!--ro, opt, bool, whether to display bluetooth status-->true
    </bluetoothStatusDisplay>
    <dialStatusDisplay>
      <!--ro, opt, bool, whether to display dial-up status-->true
    </dialStatusDisplay>
  </DeviceStatusDisplay>
  <alignment opt="customize,alignRight,alignLeft,GB,allRight,allLeft">
    <!--ro, opt, enum, alignment mode, subType:string, attr:opt{req, string}, desc:alignment mode: "customize"-custom,"alignRight"-align to
right,"alignLeft"-align to left,"allRight"-align all to right (align all OSD texts to the right,including characters,time,title,and so on),"allLeft"-align
all to left (align all OSD texts to the left,including characters,time,title,and so on)-->customize
  </alignment>
  <lowRightTextNumber min="1" max="6">
    <!--ro, opt, int, number of characters in the lower-left corner, attr:min{req, int},max{req, int}-->1
  </lowRightTextNumber>
  <lowLeftTextNumber min="7" max="8">
    <!--ro, opt, int, number of characters in the lower-right corner, attr:min{req, int},max{req, int}-->1
  </lowLeftTextNumber>
  <boundary min="0" max="2">
    <!--ro, opt, int, boundary, attr:min{req, int},max{req, int}-->1
  </boundary>
  <TextOverlaySize min="0" max="0">
```

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DETEKTOR

```
    <!--ro, opt, int, number of overlaid texts, attr:min{req, int},max{req, int}-->1
  </TextOverlaySize>
  <FloorNumOverlay>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
  </FloorNumOverlay>
  <upDownboudary>
    <!--ro, opt, int, minimum size of the upper or lower boundary, range:[0,2], desc:unit: number of characters, range: [0,2] (the value can be 0,1,or 2)--
>1
  </upDownboudary>
  <leftRightboudary>
    <!--ro, opt, int, minimum size of the left or right boundary, range:[0,2], desc:unit: number of characters, range: [0,2] (the value can be 0,1,or 2)-->1
  </leftRightboudary>
  <Angle>
    <!--ro, opt, object, device angle-->
    <TiltAngle>
      <!--ro, opt, object, pitch angle-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
      </enabled>
      <positionX min="0" max="1000">
        <!--ro, req, int, X-coordinate, range:[0,1000], attr:min{req, int},max{req, int}, desc:the origin is the lower-left corner of the screen-->1
      </positionX>
      <positionY min="0" max="1000">
        <!--ro, req, int, Y-coordinate, range:[0,1000], attr:min{req, int},max{req, int}, desc:the origin is the lower-left corner of the screen-->1
      </positionY>
    </TiltAngle>
  </Angle>
</VideoOverlay>
```

16.2.10.4 Set the channel name overlay parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, range:[0,576]-->1
  </positionY>
</channelNameOverlay>
```

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.10.5 Get the channel name overlay parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <positionX>
    <!--ro, req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--ro, req, int, range:[0,576]-->1
  </positionY>
</channelNameOverlay>
```

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DETEKTOR

16.2.10.6 Get the time information overlay parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <positionX>
    <!--ro, req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--ro, req, int, range:[0,576]-->1
  </positionY>
  <dateStyle>
    <!--ro, opt, enum, subType:string-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, subType:string-->12hour
  </timeStyle>
  <displayWeek>
    <!--ro, opt, bool-->true
  </displayWeek>
</DateTimeOverlay>
```

16.2.10.7 Set the time information overlay parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, range:[0,576]-->1
  </positionY>
  <dateStyle>
    <!--opt, enum, subType:string-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--opt, enum, subType:string-->12hour
  </timeStyle>
  <displayWeek>
    <!--opt, bool-->true
  </displayWeek>
</DateTimeOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.10.8 Get all text overlay parameters of a specified digital channel

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DETEKTOR

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, range:[1,8]-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,576]-->1
    </positionY>
    <displayText>
      <!--ro, req, string, range:[1,32]-->test
    </displayText>
  </TextOverlay>
</TextOverlayList>
```

16.2.10.9 Set all text overlay parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--opt, object-->
    <id>
      <!--req, string, range:[1,8]-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--req, int, range:[0,576]-->1
    </positionY>
    <displayText>
      <!--req, string, range:[1,32]-->test
    </displayText>
  </TextOverlay>
</TextOverlayList>
```

Response Message

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DEFEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.10.10 Get the text overlay parameters of the specific digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, text ID, range:[1,8]-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable text overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-
->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-
->1
  </positionY>
  <displayText>
    <!--ro, req, string, overlaid text, range:[1,32]-->test
  </displayText>
</TextOverlay>
```

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DETEKTOR

16.2.10.11 Set parameters of a specific digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, text No., range:[1,8]-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, overlaid text, range:[1,32]-->test
  </displayText>
</TextOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.10.12 Get the OSD parameters of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, screen size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent>
      <!--ro, req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--ro, req, bool, whether it is flickering-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--ro, opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--ro, opt, object, text content-->
      <id>
        <!--ro, req, string, text No.-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--ro, req, string, overlay text-->test
      </displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object, channel time information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <dateStyle>
      <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY",
"CHR-YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--ro, opt, enum, time format, subType:string, desc:"12hour", "24hour"-->12hour
    </timeStyle>
    <displayWeek>
      <!--ro, opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object, channel name overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
</VideoOverlay>
```

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16.2.10.13 Set OSD parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--req, object, screen size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--opt, object, attributes-->
    <transparent>
      <!--req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--req, bool, whether it is flickering-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--opt, array, text content list, subType:object-->
  </TextOverlayList>
  <DateTimeOverlay>
    <!--opt, object, channel time information overlay-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <dateStyle>
      <!--opt, enum, date, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
    </dateStyle>
    <timeStyle>
      <!--opt, enum, time format, subType:string, desc:time format-->12hour
    </timeStyle>
    <displayWeek>
      <!--opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--opt, object, channel name overlay-->
    <enabled>
      <!--req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
</VideoOverlay>
```

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DEFEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, it is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

16.2.10.14 Get the OSD capability of a specific video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_BatteryPowerOverlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent opt="true,false">
      <!--ro, req, bool, whether it is transparent, attr:opt{req, string}-->true
    </transparent>
    <flashing opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </flashing>
  </attribute>
  <TextOverlayList size="8">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <TextOverlay>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, listening host ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayText min="0" max="64">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->test
      </displayText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,576]-->1
    </positionY>
    <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
      <!--ro, opt, string, attr:opt{req, string}-->YYYY-MM-DD
    </dateStyle>
    <timeStyle opt="12hour,24hour">
      <!--ro, opt, string, time format, attr:opt{req, string}-->12hour
    </timeStyle>
    <displayWeek opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, opt, int, range:[0,1920]-->1
    </positionX>
    <positionY>
      <!--ro, opt, int, range:[0,1920]-->1
    </positionY>
  </channelNameOverlay>
</VideoOverlay>
```

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DETEKTOR

16.2.10.15 Get the channel name overlay parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, channel name overlay parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--ro, opt, string, channel name, range:[0,128]-->Camera 01
  </name>
  <OverlayRegion>
    <!--ro, opt, object, coordinates of overlaid string frame (the origin is the is the upper-left corner of the screen)-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

16.2.10.16 Set the channel name overlay parameters for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--opt, string, channel name, range:[0,32]-->Camera 01
  </name>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.10.17 Set the parameters of date and time information overlay for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, configurations of date and time information overlay of video input channels, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable date and time information overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--opt, enum, time format, subType:string, desc:"12hour" (12-hour system), "24hour" (24-hour system)-->12hour
  </timeStyle>
  <displayWeek>
    <!--opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--opt, bool, whether to display date-->true
  </displayDate>
  <displayTime>
    <!--opt, bool, whether to display time-->true
  </displayTime>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.10.18 Get the parameters of date and time information overlay of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable date and time overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlay area, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlay area, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY"--
>YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, time format, subType:string, desc:time format-->12hour
  </timeStyle>
  <displayWeek>
    <!--ro, opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--ro, opt, bool-->true
  </displayDate>
  <displayTime>
    <!--ro, opt, bool-->true
  </displayTime>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

16.2.10.19 Delete the text overlay parameters of a specific video input channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.10.20 Set the text overlay parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_TextOverlay, subType:object, attr:version{opt, string, protocolVersion}-->
  <TextOverlay>
    <!--opt, object-->
    <id>
      <!--req, string, text No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate-->1
    </positionY>
    <displayText>
      <!--req, string, overlay character-->test
    </displayText>
    <OverlayRegion>
      <!--opt, object-->
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.10.21 Get parameters of text information overlay of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, text information overlay, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--ro, opt, object, text overlay configuration-->
    <id>
      <!--ro, req, string, text No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the
screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the
screen-->1
    </positionY>
    <displayText>
      <!--ro, req, string, overlaid text-->test
    </displayText>
    <OverlayRegion>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

16.2.10.22 Add a text to overlay on the video for a specific video input channel

Request URL

POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

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DETEKTOR

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay settings, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, text overlay-->test
  </displayText>
</TextOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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DETEKTOR

16.2.10.23 Set the parameters of a specific piece of text overlay information of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, text No.-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, overlay character-->test
  </displayText>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.10.24 Get the text overlay parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	Text overlay ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay configuration, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, text No.-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-
->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-
->1
  </positionY>
  <displayText>
    <!--ro, req, string, overlaid text-->test
  </displayText>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

16.2.10.25 Delete a specified piece of text overlay of a video input channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	Text overlay ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.11 Privacy Mask

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DETEKTOR

16.2.11.1 Get the privacy mask parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, privacy mask, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--ro, opt, array, privacy mask area configurations, subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--ro, opt, object, configurations of a privacy mask area-->
      <id>
        <!--ro, req, int, area ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--ro, opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mosaicThick>
        <!--ro, opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMask.PrivacyMaskRegionList[*].PrivacyMaskRegion.maskType,eq,mosaic}, desc:it
is valid when the value of maskType is "mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
      </mosaicThick>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--ro, opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

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16.2.11.2 Set the privacy mask parameters of the specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, privacy mask, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--opt, array, privacy mask area configurations, subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--opt, object, configurations of a privacy mask area-->
      <id>
        <!--req, int, area ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:The origin is the lower-left corner of the screen.-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mosaicThick>
        <!--opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMask.PrivacyMaskRegionList[*].PrivacyMaskRegion.maskType,eq,mosaic}, desc:It is
valid when the value of maskType is "mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
      </mosaicThick>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.11.3 Get the privacy mask area parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--ro, opt, object, configurations of a privacy mask area-->
    <id>
      <!--ro, req, int, area ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--ro, req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--ro, opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mosaicThick>
      <!--ro, opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMaskRegion.maskType,eq,mosaic}, desc:it is valid when the value of maskType is
"mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
    </mosaicThick>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

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16.2.11.4 Set parameters of all privacy mask areas of the specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--opt, object, configurations of a privacy mask area-->
    <id>
      <!--req, int, area ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mosaicThick>
      <!--opt, enum, mosaic blur strength, subType:int, dep:and,{$.PrivacyMaskRegion.maskType,eq,mosaic}, desc:it is valid when the value of maskType is
"mosaic". 1 (low, the block size is 8*8), 2 (medium, the block size is 16*16), 3 (high, the block size is 32*32)-->1
    </mosaicThick>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0:OK;1:OK;2:Device Busy;3:Device Error;4:Invalid Operation;5:Invalid XML Format;6:Invalid XML
Content;7:Reboot Required;-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:OK, Device Busy, Device Error, Invalid Operation, Invalid XML Format, Invalid XML Content,
Reboot-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.11.5 Get the parameters of all privacy mask areas of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--ro, opt, object, configurations of a privacy mask area-->
    <id>
      <!--ro, req, int, area ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,576]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--ro, opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mediaObjectEncryption>
      <!--ro, opt, object, encryption of the private target in media data, desc:it is disabled when this field is not returned. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
      </maskType>
    </mediaObjectEncryption>
    <regionRangeType>
      <!--ro, opt, enum, subType:string-->region
    </regionRangeType>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

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DETEKTOR

16.2.11.6 Set the parameters of all privacy mask areas of a specified channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMaskRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, privacy mask area configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <PrivacyMaskRegion>
    <!--opt, object, configurations of a privacy mask area-->
    <id>
      <!--req, int, area ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the area-->true
    </enabled>
    <RegionCoordinatesList>
      <!--req, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,704]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <privacymaskName>
      <!--opt, string, privacy mask name-->test
    </privacymaskName>
    <maskType>
      <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
    </maskType>
    <zoomdoorlimit>
      <!--opt, int, zooming in limit, when the zooming in value is larger than the limit, the private area will be displayed, range:[10,1000]-->1
    </zoomdoorlimit>
    <mediaObjectEncryption>
      <!--opt, object, encryption of the private target in media data, desc:it is disabled when this field is not configured. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
      </maskType>
    </mediaObjectEncryption>
  </PrivacyMaskRegion>
</PrivacyMaskRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.11.7 Get the privacy mask parameters of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
switchType	string	Analysis unit switch: "local" (analysis by local device, e.g., executed locally by NVR), "remote" (the device transmits data to sub-device for analysis, e.g., from the NVR to the network camera).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--ro, opt, array, list of video tampering region(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--ro, opt, object, configurations of a privacy mask region-->
      <id>
        <!--ro, req, int, region ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the region-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--ro, opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--ro, opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--ro, opt, int, zooming in limit, when the zooming in value is larger than the limit, the private region will be displayed, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--ro, opt, object, encryption of the private target in media data, desc:If it is enabled, the main steam of corresponding region will be covered,
but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?format=json). You can use the key to decrypt the
encrypted region.-->
        <enabled>
          <!--ro, req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--ro, opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
      <regionRangeType>
        <!--ro, opt, enum, subType:string-->region
      </regionRangeType>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--ro, opt, enum, region type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

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16.2.11.8 Set the privacy mask parameters of a specified channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/privacyMask?switch=<switchType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
switchType	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PrivacyMask xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, privacy mask configurations, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the privacy mask-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <PrivacyMaskRegionList size="8">
    <!--opt, array, list of video privacy mask area(s), subType:object, attr:size{req, int}-->
    <PrivacyMaskRegion>
      <!--opt, object, configurations of a privacy mask area-->
      <id>
        <!--req, int, area ID-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable the area-->true
      </enabled>
      <RegionCoordinatesList>
        <!--req, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,704]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,576]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <privacymaskName>
        <!--opt, string, privacy mask name-->test
      </privacymaskName>
      <maskType>
        <!--opt, enum, mask type, subType:string, desc:"gray", "red", "yellow", "blue", "orange", "green", "transparent", "half-transparent", "mosaic",
"black"-->gray
      </maskType>
      <zoomdoorlimit>
        <!--opt, int, zooming value limit range, range:[10,1000]-->1
      </zoomdoorlimit>
      <mediaObjectEncryption>
        <!--opt, object, encryption of the private target in media data, desc:it is disabled when this field is not configured. If it is enabled, the main
steam of corresponding area will be covered, but the mask can be removed via keys (related URI: /ISAPI/Security/MediaObjectEncryption/channel/<videoID>?
format=json). You can use the key to decrypt the encrypted area-->
        <enabled>
          <!--req, bool, whether to enable-->true
        </enabled>
        <maskType>
          <!--opt, enum, mask type, subType:string, desc:"mosaic"-->mosaic
        </maskType>
      </mediaObjectEncryption>
    </PrivacyMaskRegion>
  </PrivacyMaskRegionList>
  <regionType>
    <!--opt, enum, area type, subType:string, desc:"quadrilateral"-->quadrilateral
  </regionType>
</PrivacyMask>
```

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DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.12 Real-Time Streaming

16.2.12.1 Get multicast parameters

Request URL

GET /ISAPI/System/Network/extension

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<networkExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <multicastAddress>
    <!--ro, opt, object-->
    <ipVersion>
      <!--ro, req, enum, IP address version, subType:string, desc:IP address version-->v4
    </ipVersion>
    <ipAddress>
      <!--ro, opt, string, IPv4 address-->172.6.64.7
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address-->1080:0:0:0:8:800:200C:417A
    </ipv6Address>
  </multicastAddress>
  <enVirtualHost>
    <!--ro, opt, bool-->true
  </enVirtualHost>
</networkExtension>
```

16.2.12.2 Set the multicast address

Request URL

PUT /ISAPI/System/Network/extension

Query Parameter

None

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<networkExtension xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, object, advanced network parameters, attr:version{req, string, protocolVersion}-->
  <multicastAddress>
    <!--wo, opt, object, multicast address-->
    <ipVersion>
      <!--wo, req, enum, IP address version, subType:string, desc:"v4" (IPv4), "v6" (IPv6), "dual" (IPv4 and IPv6)-->v4
    </ipVersion>
    <ipAddress>
      <!--wo, opt, string, IPv4 address-->172.6.64.7
    </ipAddress>
    <ipv6Address>
      <!--wo, opt, string, IPv6 address-->1080:0:0:0:8:800:200C:417A
    </ipv6Address>
  </multicastAddress>
  <enVirtualHost>
    <!--wo, opt, bool, whether to enable the virtual host-->true
  </enVirtualHost>
</networkExtension>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.13 Scene Change Detection

16.2.13.1 Set the arming schedule of scene change detection for all channels

Request URL

PUT /ISAPI/Event/schedules/sceneChangeDetections

Query Parameter

None

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<SceneChangeDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->sceneChangeDetection_video1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->sceneChangeDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SceneChangeDetectionScheduleList>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.13.2 Get the arming schedule of scene change detection for all channels

Request URL

GET /ISAPI/Event/schedules/sceneChangeDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SceneChangeDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->sceneChangeDetection_video1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->sceneChangeDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SceneChangeDetectionScheduleList>
```

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DETEKTOR

16.2.13.3 Get the arming schedule of scene change detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->sceneChangeDetection_video1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->sceneChangeDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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DETEKTOR

16.2.13.4 Set the arming schedule of scene change detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/sceneChangeDetections/sceneChangeDetection_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->sceneChangeDetection_video1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->sceneChangeDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com
DETEKTOR

16.2.13.5 Sudden scene change detection

EventType:scenechangedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"scenechangedetection" (sudden scene change detection)-->scenechangedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
</EventNotificationAlert>
```

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DETEKTOR

16.2.14 Video Channel Resource Management

16.2.14.1 Set the image parameters of all channels

Request URL

PUT /ISAPI/ContentMgmt/ImageProxy/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageChannellist xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <ImageChannel>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <inputPort>
      <!--req, int-->1
    </inputPort>
    <Color>
      <!--opt, object-->
      <brightnessLevel>
        <!--opt, int, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--opt, int, range:[0,100]-->1
      </contrastLevel>
      <saturationLevel>
        <!--opt, int, range:[0,100]-->1
      </saturationLevel>
    </Color>
    <enableImageLossDetection>
      <!--opt, bool-->true
    </enableImageLossDetection>
  </ImageChannel>
</ImageChannellist>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.14.2 Get the image parameters of all channels

Request URL

GET /ISAPI/ContentMgmt/ImageProxy/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageChannellist xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <ImageChannel>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->1
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <inputPort>
      <!--ro, req, int-->1
    </inputPort>
    <Color>
      <!--ro, opt, object-->
      <brightnessLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </contrastLevel>
      <saturationLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </saturationLevel>
    </Color>
    <enableImageLossDetection>
      <!--ro, opt, bool-->true
    </enableImageLossDetection>
  </ImageChannel>
</ImageChannellist>
```

16.2.14.3 Get the image parameters of a specified channel

Request URL

GET /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <inputPort>
    <!--ro, req, int-->1
  </inputPort>
  <WDR>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->[open
    </mode>
    <WDRLevel>
      <!--ro, opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
  </BLC>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </contrastLevel>
    <saturationLevel>
      <!--ro, opt, int, range:[0,100]-->0
    </saturationLevel>
  </Color>
  <HLC>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <HLCLevel>
      <!--ro, opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <enableImageLossDetection>
    <!--ro, opt, bool-->true
  </enableImageLossDetection>
</ImageChannel>
```

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DETEKTOR

16.2.14.4 Set the image parameters of a specified channel

Request URL

PUT /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <inputPort>
    <!--req, int-->1
  </inputPort>
  <WDR>
    <!--opt, object-->
    <mode>
      <!--req, enum, subType:string-->[open
    </mode>
    <WDRLevel>
      <!--opt, int, range:[1,100]-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
  </BLC>
  <Color>
    <!--opt, object-->
    <brightnessLevel>
      <!--opt, int, range:[0,100]-->0
    </brightnessLevel>
    <contrastLevel>
      <!--opt, int, range:[0,100]-->0
    </contrastLevel>
    <saturationLevel>
      <!--opt, int, range:[0,100]-->0
    </saturationLevel>
  </Color>
  <HLC>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <HLCLevel>
      <!--opt, int, range:[0,10]-->0
    </HLCLevel>
  </HLC>
  <enableImageLossDetection>
    <!--opt, bool-->true
  </enableImageLossDetection>
</ImageChannel>
```

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DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.5 Get the image parameters capability of a specified channel

Request URL

GET /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <inputPort>
    <!--ro, req, int-->1
  </inputPort>
  <ImageFlip>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <ImageFlipStyle opt="LEFTRIGHT,UPDOWN,CENTER,AUTO">
      <!--ro, opt, string, attr:opt{req, string}-->LEFTRIGHT
    </ImageFlipStyle>
  </ImageFlip>
  <WDR>
    <!--ro, opt, object-->
    <mode opt="open,close,auto,SDR">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->[open
    </mode>
    <WDRLevel min="1" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </WDRLevel>
  </WDR>
  <BLC>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <BLCMode opt="UP,DOWN,LEFT,RIGHT,CENTER,MULTI-AREA,Region,AUTO">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->UP
    </BLCMode>
  </BLC>
  <NoiseReduce>
    <!--ro, opt, object-->
    <mode opt="close,general,advanced">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->close
    </mode>
    <GeneralMode>
      <!--ro, opt, object-->
      <generalLevel min="0" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
      </generalLevel>
    </GeneralMode>
    <AdvancedMode>
      <!--ro, opt, object-->
      <FrameNoiseReduceLevel min="0" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
      </FrameNoiseReduceLevel>
      <InterFrameNoiseReduceLevel min="0" max="10">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
      </InterFrameNoiseReduceLevel>
    </AdvancedMode>
  </NoiseReduce>
  <Shutter>
    <!--ro, opt, object-->
    <ShutterLevel opt="1/25,1/50,1/75,1/100,1/125,1/150,1/175,1/200,1/225,1/250,1/300,1/400,1/500,1/750,1/1000,1/2000,1/4000,1/10000,1/100000">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </ShutterLevel>
    <maxShutterLevelLimit
opt="1/25,1/50,1/75,1/100,1/120,1/150,1/175,1/200,1/225,1/250,1/300,1/425,1/600,1/1000,1/1250,1/1750,1/2500,1/3500,1/6000,1/10000,1/30000">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </maxShutterLevelLimit>
    <minShutterLevelLimit
opt="1/25,1/50,1/75,1/100,1/120,1/150,1/175,1/200,1/225,1/250,1/300,1/425,1/600,1/1000,1/1250,1/1750,1/2500,1/3500,1/6000,1/10000,1/30000">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </minShutterLevelLimit>
  </Shutter>
  <Color>
    <!--ro, opt, object-->
    <brightnessLevel def="50" min="0" max="100">
      <!--ro, opt, int, attr:def{req, int},min{req, int},max{req, int}-->1
    </brightnessLevel>
    <contrastLevel def="50" min="0" max="100">
      <!--ro, opt, int, attr:def{req, int},min{req, int},max{req, int}-->1
    </contrastLevel>
    <saturationLevel def="50" min="0" max="100">
      <!--ro, opt, int, attr:def{req, int},min{req, int},max{req, int}-->1
    </saturationLevel>
  </Color>
  <IrcutFilter>
    <!--ro, opt, object-->
```

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```
      <IrcutFilterType opt="auto,day,night,schedule,eventTrigger,videoAuto,IrcutFilterType,darkFighterX,darkFighterXAuto,darkFighterXSchedule">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
      </IrcutFilterType>
      <nightToDayFilterLevel opt="low,normal,high">
        <!--ro, opt, string, attr:opt{req, string}-->test
      </nightToDayFilterLevel>
      <nightToDayFilterTime min="0" max="10">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
      </nightToDayFilterTime>
      <Schedule>
        <!--ro, opt, object-->
        <scheduleType opt="day,night">
          <!--ro, req, enum, subType:string, attr:opt{req, string}-->day
        </scheduleType>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </Schedule>
    </IrcutFilter>
    <HLC>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <HLCLevel min="0" max="10">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
      </HLCLevel>
    </HLC>
    <enableImageLossDetection opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </enableImageLossDetection>
    <SupplementLight>
      <!--ro, opt, object-->
      <supplementLightMode
  opt="mixed,whiteLight,colorVuWhiteLight,irLight,dualLight,close,eventIntelligence,multiAzimuthMixed,multiAzimuthWhiteLight,multiAzimuthIrLight,panoramicWhit
  eLight,panoramicIrLight">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->mixed
      </supplementLightMode>
      <irLightBrightness min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
      </irLightBrightness>
      <mixedLightBrightnessRegulatMode opt="manual,auto">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
      </mixedLightBrightnessRegulatMode>
      <whiteLightBrightness min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
      </whiteLightBrightness>
      <isAutoModeBrightnessCfg>
        <!--ro, opt, bool-->true
      </isAutoModeBrightnessCfg>
    </SupplementLight>
  </ImageChannel>
```

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16.2.14.6 Get the rotate mode parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>/corridor

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<corridor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
</corridor>
```

16.2.14.7 Set the rotate mode parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/ImageProxy/channels/<channelID>/corridor

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<corridor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
</corridor>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.14.8 Delete a specific digital channel

Request URL

DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.14.9 Get the configuration capability of a specific digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, network parameters of the digital channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <name>
    <!--ro, opt, string, name-->test
  </name>
  <sourceInputPortDescriptor>
    <!--ro, req, object, input channel description-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress", it is invalid when the protocol type is GB28181-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, domain name, desc:it is invalid when the protocol type is GB28181-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, desc:it is invalid when the protocol type is GB28181-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, desc:it is invalid when the protocol type is GB28181-->test
    </ipv6Address>
    <managePortNo min="2000" max="65535">
      <!--ro, req, int, port No. under management, attr:min{req, int},max{req, int}-->1
    </managePortNo>
    <srcInputPort>
      <!--ro, req, string, device channel No.-->test
    </srcInputPort>
    <userName>
      <!--ro, req, string, user name-->test
    </userName>
    <password>
      <!--ro, req, string, password-->test
    </password>
    <connMode>
      <!--ro, opt, enum, connect mode, subType:string, desc:“plugplay” (plug and play), “manual”-->plugplay
    </connMode>
    <streamType>
```

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```
      <!--ro, opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
    </streamType>
    <deviceID>
      <!--ro, opt, string, device ID-->test
    </deviceID>
    <getStreamMode>
      <!--ro, opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
    </getStreamMode>
    <cameraType>
      <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating), "teacherFull"
(teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "IPCamera", "IPDome", "none"-->none
    </cameraType>
    <isDoubleAdd>
      <!--ro, opt, bool, add two streams of the same camera, desc:it is used when DeepinView is added to education sharing server-->true
    </isDoubleAdd>
    <streamUrl>
      <!--ro, opt, string, streaming URL, desc:if this node exists, use this URL for streaming; if not, follow the original rule-->test
    </streamUrl>
    <activateEnabled opt="true,false">
      <!--ro, opt, bool, whether to activate the device, attr:opt{req, string}, desc:if this node does not exist, it indicates that the device has already
been activated. Application scenario: supports adding an unactivated device to the NVR channel. If the device is unactivated (this node is true), then it
needs to be activated by the NVR. The NVR will auto-activate the device (MAC address is needed because IP access is unavailable for unactivated devices),
and will add the device to NVR.-->true
    </activateEnabled>
    <macAddress min="0" max="48">
      <!--ro, opt, string, MAC address, range:[0,48], attr:min{req, int},max{req, int}, desc:it is required when the value of activateEnabled is true-->test
    </macAddress>
    <streamMode>
      <!--ro, opt, enum, stream type, subType:string, desc:"main", "sub"-->main
    </streamMode>
    <getSubStream opt="true,false">
      <!--ro, opt, bool, whether to get sub-stream, attr:opt{req, string}, desc:when this node is not configured, its value is true by default, which means
the sub-streaming will start after the device is connected-->true
    </getSubStream>
  </sourceInputPortDescriptor>
  <certificateValidationEnabled>
    <!--ro, opt, bool, whether to enable the certification validation-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--ro, opt, bool, whether to enable the default communication port-->true
  </defaultAdminPortEnabled>
  <enableAnr opt="true,false">
    <!--ro, opt, bool, whether to enable ANR (Automatic Network Replenishment), attr:opt{req, string}-->true
  </enableAnr>
  <enableTiming>
    <!--ro, opt, bool, whether to perform time sync on the device-->true
  </enableTiming>
  <isSupportChannelDeviceInfo>
    <!--ro, opt, bool, whether it supports getting device information in the digital channel, desc:related URI:
/ISAPI/ContentMgmt/InputProxy/channels/<channelID>/deviceInfo-->true
  </isSupportChannelDeviceInfo>
  <NVRInfo>
    <!--ro, opt, object, NVR information-->
    <ipAddressNVR>
      <!--ro, opt, string, IP address of NVR-->test
    </ipAddressNVR>
    <portNVR min="2000" max="65535">
      <!--ro, opt, int, port No. of NVR, attr:min{req, int},max{req, int}-->1
    </portNVR>
    <ipcChannelNo min="1" max="10">
      <!--ro, opt, int, the channel No. of the network camera in NVR, attr:min{req, int},max{req, int}-->1
    </ipcChannelNo>
  </NVRInfo>
  <anrInterval min="1" max="24" def="1">
    <!--ro, opt, int, time interval for automatic network replenishment, attr:min{req, int},max{req, int},def{req, int}-->1
  </anrInterval>
</InputProxyChannel>
```

16.2.14.10 Set the device parameters of a digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/deviceInfo

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--req, string, device name-->test
```

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```
</deviceName>
<deviceID>
  <!--req, string, UUID device No.-->fb8d4000-8158-11b5-8450-b4a38245654c
</deviceID>
<deviceDescription>
  <!--opt, string, device description-->test
</deviceDescription>
<deviceLocation>
  <!--opt, string, device location-->hangzhou
</deviceLocation>
<deviceStatus>
  <!--opt, enum, device status, subType:string, desc:"normal", "abnormal"-->normal
</deviceStatus>
<DetailAbnormalStatus>
  <!--opt, object, error status details-->
  <hardDiskFull>
    <!--opt, bool, the disk is full-->true
  </hardDiskFull>
  <hardDiskError>
    <!--opt, bool, the disk error-->true
  </hardDiskError>
  <ethernetBroken>
    <!--opt, bool, network disconnected-->true
  </ethernetBroken>
  <ipaddrConflict>
    <!--opt, bool, IP address is conflicted-->true
  </ipaddrConflict>
  <illegalAccess>
    <!--opt, bool, illegal access-->true
  </illegalAccess>
  <recordError>
    <!--opt, bool, recording exception-->true
  </recordError>
  <raidLogicDiskError>
    <!--opt, bool, virtual disk exception in the array-->true
  </raidLogicDiskError>
  <spareWorkDeviceError>
    <!--opt, bool, backup device exception-->true
  </spareWorkDeviceError>
</DetailAbnormalStatus>
<systemContact>
  <!--opt, string, manufacturer-->STD-CGI
</systemContact>
<model>
  <!--req, string, device model-->iDS-9632NX-I8/X
</model>
<serialNumber>
  <!--req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
</serialNumber>
<macAddress>
  <!--req, string, MAC address-->44:47:cc:c8:d9:e4
</macAddress>
<firmwareVersion>
  <!--req, string, firmware version No.-->V4.1.40
</firmwareVersion>
<firmwareReleasedDate>
  <!--opt, string, firmware release date-->build 181129
</firmwareReleasedDate>
<encoderVersion>
  <!--opt, string, device encoder version No.-->V7.3
</encoderVersion>
<encoderReleasedDate>
  <!--opt, string, compile date of the device encoder version-->build 20161223
</encoderReleasedDate>
<bootVersion>
  <!--opt, string, boot version-->V1.3.4
</bootVersion>
<bootReleasedDate>
  <!--opt, string, release date of boot-->100316
</bootReleasedDate>
<panelVersion>
  <!--opt, string, panel version-->V1.0
</panelVersion>
<hardwareVersion>
  <!--opt, string, hardware version-->0x0
</hardwareVersion>
<decoderVersion>
  <!--opt, string, device decoder version-->V1.0
</decoderVersion>
<decoderReleasedDate>
  <!--opt, string, compile date of the device decoder version-->build 20161223
</decoderReleasedDate>
<softwareVersion>
  <!--opt, string, software version-->build 181129
</softwareVersion>
<capacity>
  <!--opt, int, device capacity, unit: MB-->10240
</capacity>
<usedCapacity>
  <!--opt, int, capacity usage, unit: MB-->5120
</usedCapacity>
<telecontrolID>
  <!--opt, int, remote control control ID, range:[1,255]-->1
</telecontrolID>
```

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```
<supportBeep>
  <!--opt, bool, whether to support buzzer-->true
</supportBeep>
<supportVideoLoss>
  <!--opt, bool, whether to support video loss-->true
</supportVideoLoss>
<firmwareVersionInfo>
  <!--opt, string, firmware version information-->1
</firmwareVersionInfo>
<actualFloorNum>
  <!--opt, int, actual number of floors, range:[1,255]-->1
</actualFloorNum>
<localZoneNum>
  <!--opt, int, number of local zones-->1
</localZoneNum>
<alarmOutNum>
  <!--opt, int, number of alarm outputs-->1
</alarmOutNum>
<distanceResolution>
  <!--opt, float, distance resolution-->0.000
</distanceResolution>
<angleResolution>
  <!--opt, float, angle resolution-->0.000
</angleResolution>
<speedResolution>
  <!--opt, float, speed resolution-->0.000
</speedResolution>
<detectDistance>
  <!--opt, float, detection distance-->0.000
</detectDistance>
<relayNum>
  <!--opt, int, number of local relays-->1
</relayNum>
<electroLockNum>
  <!--opt, int, number of local electronic locks-->1
</electroLockNum>
<RS485Num>
  <!--opt, int, number of local RS-485-->1
</RS485Num>
<radarVersion>
  <!--opt, string, radar version-->test
</radarVersion>
<cameraModuleVersion>
  <!--opt, string, camera module version-->test
</cameraModuleVersion>
<mainversion>
  <!--opt, int, main version No., range:[1,255]-->1
</mainversion>
<subversion>
  <!--opt, int, sub version No., range:[1,255]-->1
</subversion>
<upgradeversion>
  <!--opt, int, updated version No., range:[1,255]-->1
</upgradeversion>
<customizeversion>
  <!--opt, int, customized version No., range:[1,255]-->1
</customizeversion>
<companyName>
  <!--opt, string, manufacturer name-->test
</companyName>
<copyright>
  <!--opt, string, copyright Information-->test
</copyright>
<systemName>
  <!--opt, enum, storage system name, subType:string, desc:"storageManagement" (storage management system), "distributedStorageManagement" (distributed
storage management system)-->storageManagement
</systemName>
<systemStatus>
  <!--opt, enum, system status, subType:string, desc:"configured", "unConfigured" (not configured)-->configured
</systemStatus>
<isLeaderDevice>
  <!--opt, bool, whether it is the corresponding device of the resource IP address-->true
</isLeaderDevice>
<clusterVersion>
  <!--opt, string, system cluster version, desc:this node is valid when the value of isLeaderDevice is true-->test
</clusterVersion>
<centralStorageVersion>
  <!--opt, string, center storage version, range:[1,16]-->test
</centralStorageVersion>
<powerOnMode>
  <!--opt, enum, startup mode, subType:string, desc:"button" (press button to power on), "adapter" (connect adapter to power on)-->button
</powerOnMode>
<customizedInfo>
  <!--opt, string, customized project No., desc:order No. of the customization project. For baseline devices,this node is empty; for custom devices,the
order No. of the customization project will be returned by this node-->test
</customizedInfo>
<charEncodeFormat>
  <!--opt, enum, character encoding, subType:string, desc:GB2312:GB2312;GBK:GBK-->GB2312
</charEncodeFormat>
<languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech">
  <!--opt, enum, language type, subType:string, attr:opt{req, string}, desc:"Chinese", "English", "Spanish", "Portuguese", "Italian", "French", "Russian",
"Turkish", "Greek", "Czech"-->chinese
</languageType>
```

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```
  <DockStation>
    <!--opt, object, dock station information configuration-->
    <Platform>
      <!--opt, object, platform dock station information configuration-->
      <ip>
        <!--opt, string, IP address-->test
      </ip>
      <port>
        <!--opt, int, communication port-->1
      </port>
      <userName>
        <!--req, string, user name-->test
      </userName>
      <password>
        <!--req, string, password-->test
      </password>
    </Platform>
    <centralStorageBackupEnabled>
      <!--opt, bool, whether to enable center storage backup-->true
    </centralStorageBackupEnabled>
  </DockStation>
  <webVersion>
    <!--opt, string, Web version No.-->test
  </webVersion>
  <deviceRFProgramVersion>
    <!--opt, string, version No. of the device's RF (Radio Frequency) program-->test
  </deviceRFProgramVersion>
  <securityModuleSerialNo>
    <!--opt, string, security module serial No.-->test
  </securityModuleSerialNo>
  <securityModuleVersion>
    <!--opt, string, security module version No.-->test
  </securityModuleVersion>
  <securityChipVersion>
    <!--opt, string, security chip version No.-->test
  </securityChipVersion>
  <securityModuleKeyVersion>
    <!--opt, string, security module key version No.-->test
  </securityModuleKeyVersion>
  <UIDLampRecognition>
    <!--opt, object, information of the UID lamp recognition device-->
    <enabled>
      <!--opt, bool, whether to enable the function-->true
    </enabled>
  </UIDLampRecognition>
  <confDeviceIdPrefix>
    <!--opt, bool, whether the meeting uses the device name as its prefix-->true
  </confDeviceIdPrefix>
  <OEMCode>
    <!--opt, enum, manufacturer OME code, subType:int, desc:1 (Hikvision device), 0 (neutral device)-->1
  </OEMCode>
  <bootTime>
    <!--opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
  </bootTime>
  <intelligentAnalysisEngineModel>
    <!--opt, string, engine model-->test
  </intelligentAnalysisEngineModel>
</DeviceInfo>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

16.2.14.11 Get the device parameters of a digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/deviceInfo

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--ro, req, string, device name-->test
  </deviceName>
  <deviceID>
    <!--ro, req, string, UUID device No.-->fb8d4000-8158-11b5-8450-b4a38245654c
  </deviceID>
  <model>
    <!--ro, req, string, device model-->iDS-9632NX-I8/X
  </model>
  <serialNumber>
    <!--ro, req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress>
    <!--ro, req, string, MAC address-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion>
    <!--ro, req, string, firmware version No.-->V4.1.40
  </firmwareVersion>
</DeviceInfo>
```

16.2.14.12 Get the network parameters of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/netParam

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelNetParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string-->test
  </ipv6Address>
  <managePortNo>
    <!--ro, opt, int-->1
  </managePortNo>
</InputProxyChannelNetParam>
```

16.2.14.13 Set parameters of a specific digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/password

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelPassword xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <oldPassword>
    <!--req, string, old password-->test
  </oldPassword>
  <newPassword>
    <!--req, string, new password-->test
  </newPassword>
</InputProxyChannelPassword>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.14.14 Get status of a specific digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/status

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, status of a single digital channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <sourceInputPortDescriptor>
    <!--ro, req, object, input channel source description-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress". It is invalid when the protocol type is GB28181-->hostname
    </addressingFormatType>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
    </ipAddress>
    <managePortNo>
      <!--ro, req, int, port No. under management, range:[1,65535]-->1
    </managePortNo>
    <srcInputPort>
      <!--ro, req, string, device channel No., range:[1,16]-->test
    </srcInputPort>
    <userName>
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
    <password>
```

```
        <!--ro, req, string, password, range:[1,16], desc:this node is empty in the response message-->test
      </password>
      <connMode>
        <!--ro, opt, enum, connection mode, subType:string, desc:"plugplay" (plug and play), "manual"-->plugplay
      </connMode>
      <streamType>
        <!--ro, opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
      </streamType>
      <deviceID>
        <!--ro, opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
      </deviceID>
    </sourceInputPortDescriptor>
    <online>
      <!--ro, req, bool, whether the camera is online-->true
    </online>
    <streamingProxyChannelIdList>
      <!--ro, req, array, streaming channel list, subType:object-->
      <streamingProxyChannelId>
        <!--ro, opt, string, streaming channel No.-->test
      </streamingProxyChannelId>
    </streamingProxyChannelIdList>
    <relatedIOProxy>
      <!--ro, opt, object, linked IO port-->
      <inputProxyPortIdList>
        <!--ro, opt, array, input port No. list, subType:object-->
        <inputProxyPortId>
          <!--ro, opt, string, input port No.-->test
        </inputProxyPortId>
      </inputProxyPortIdList>
      <outputProxyPortIdList>
        <!--ro, opt, array, output port No. list, subType:object-->
        <outputProxyPortId>
          <!--ro, opt, string, output port No.-->test
        </outputProxyPortId>
      </outputProxyPortIdList>
    </relatedIOProxy>
    <chanDetectResult>
      <!--ro, opt, enum, channel detection status, subType:string, desc:"connect" (connected), "overSysBandwidth" (exceeded the system bandwidth),
"domainError" (domain error), "ipcStreamFail" (network camera streaming failed), "connecting" (connecting), "chanNoError" (incorrect channel number),
"ipAddrConflictWithDev" (IP address is conflicted with current device's), "ipAddrConflicWithIpc" (IP address is conflicted with other device's.),
"errorUserNameOrPasswd" (incorrect user name or password), "netUnreachable" (network is unreachable), "unknownError" (unknown error), "notExist" (not
exist), "ipcStreamTypeNotSupport" (the network camera stream type is not supported), "ipcResolutionNotSupport" (the network camera resolution is not
supported), "userLocked" (the user is locked), "userNotExist" (the user does not exist), "ipcUnregistered" (network camera is not registered),
"ipcNotActivated" (network camera is inactivated), "poePortDetecting" (detecting POE port), "uploadingCloudFile" (uploading files to cloud storage),
"certificateValidationError" (certification validation failed), "ONVIFRegistrationError" (ONVIF registration failed), "streamingParameters" (streaming
parameter error), "sleep", "wakeUping" (waking up)-->connect
    </chanDetectResult>
    <SecurityStatus>
      <!--ro, opt, object, security status-->
      <PasswordStatus>
        <!--ro, opt, enum, password status, subType:string, desc:"ipcNotActivated" (IPC is not activated), "risk", "weak", "medium", "strong", "invalid"--
>ipcNotActivated
      </PasswordStatus>
    </SecurityStatus>
    <channelName>
      <!--ro, opt, string, channel name, range:[1,32]-->--
    </channelName>
    <bitStreamType>
      <!--ro, req, enum, stream type, subType:string, desc:"mainStream", "subStream"-->mainStream
    </bitStreamType>
    <rateType>
      <!--ro, req, enum, bit rate type, subType:string, desc:"CBR" (constant bit rate), "VBR" (variable bit rate)-->CBR
    </rateType>
    <codeParam>
      <!--ro, req, enum, encoding parameters, subType:string, desc:"H.264", "H.265", "unknown"-->H.264
    </codeParam>
  </InputProxyChannelStatus>
```

16.2.14.15 Get parameters of a specific digital channel

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

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```
<InputProxyChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, channel information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <name>
    <!--ro, opt, string, name-->test
  </name>
  <sourceInputPortDescriptor>
    <!--ro, req, object, input channel description-->
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:it is invalid when the protocol type is GB28181-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
    </ipv6Address>
    <managePortNo>
      <!--ro, req, int, port No. under management, range:[1,65535]-->1
    </managePortNo>
    <srcInputPort>
      <!--ro, req, string, device channel No., range:[1,16]-->test
    </srcInputPort>
    <userName>
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--ro, req, string, password, range:[0,16], desc:this node is empty in the response message-->test
    </password>
    <connMode>
      <!--ro, opt, enum, connect mode, subType:string, desc:"plugplay" (plug and play), "manual"-->plugplay
    </connMode>
    <streamType>
      <!--ro, opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
    </streamType>
    <deviceID>
      <!--ro, opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
    </deviceID>
    <getStreamMode>
      <!--ro, opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
    </getStreamMode>
    <cameraType>
      <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating), "teacherFull"
(teacher full view), "studentFull" (student full view), "media", "blackboard", "blackboardLocation", "IPCamera", "IPDome", "none"-->none
    </cameraType>
    <isDoubleAdd>
      <!--ro, opt, bool, add two streams of the same camera, desc:it is used when DeepinView is added to education sharing server-->true
    </isDoubleAdd>
    <streamUrl>
      <!--ro, opt, string, RTSP streaming URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule-->test
    </streamUrl>
    <transMode>
      <!--ro, opt, enum, subType:int-->0
    </transMode>
    <getStream>
      <!--ro, opt, bool, whether to stream, desc:its value is true by default, which means getting device information after the device is connected and not
recording video-->true
    </getStream>
    <RTSP>
      <!--ro, opt, object, RTSP related nodes-->
      <userName>
        <!--ro, req, string, user name, range:[1,64], desc:"admin" by default-->test
      </userName>
      <password>
        <!--ro, req, string, password, range:[0,64], desc:empty by default-->test
      </password>
      <portNo>
        <!--ro, opt, int, RTSP streaming port, range:[1,65535], step:1, desc:the default port is 554-->554
      </portNo>
    </RTSP>
    <customChanID>
      <!--ro, opt, string, custom camera channel No., range:[1,64], desc:support configuring remotely. When adding sub devices, the platform applies a
custom ID to the accessed digital channel. The ID is the unique identifier with 63 bytes. After the platform applies, the digital channel will upload the ID
when an alarm is triggered, and the platform can find the corresponding device through the ID-->test
    </customChanID>
    <streamMode>
      <!--ro, opt, enum, stream type, subType:string, desc:"main", "sub"-->main
    </streamMode>
    <getSubStream>
      <!--ro, opt, bool, whether to get sub-stream, desc:when this node is not configured, its value is true by default, which means the sub-streaming will
start after the device is connected-->true
    </getSubStream>
  </sourceInputPortDescriptor>
  <certificateValidationEnabled>
    <!--ro, opt, bool, whether to enable the certification validation-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--ro, opt, bool, whether to enable the default communication port-->true
  </defaultAdminPortEnabled>
```

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```
  </defaultAdminPortEnabled>
  <enableAnr>
    <!--ro, opt, bool, whether to enable ANR (Automatic Network Replenishment), desc:after the camera goes offline, it will reload all the video files during the offline period once it goes online again-->true
  </enableAnr>
  <enableTiming>
    <!--ro, opt, bool, whether to perform time sync on the device-->true
  </enableTiming>
  <streamID>
    <!--ro, opt, string, stream ID, range:[1,32], desc:if this node is configured, the stream ID of the current channel can be specified-->test
  </streamID>
  <enabled>
    <!--ro, opt, bool, whether to enable the channel, desc:the default value is true-->true
  </enabled>
  <devIndex>
    <!--ro, opt, string, sub-device ID, range:[0,64], desc:used to specify a sub-device to which ISAPI is transmitted-->test
  </devIndex>
  <boundAudioChannelID>
    <!--ro, opt, int, linked audio channel No., desc:if this node does not exist, no audio channel is linked to-->1
  </boundAudioChannelID>
  <anrInterval>
    <!--ro, opt, int, time interval for automatic network replenishment, range:[1,24], unit:h, dep:and,{$.InputProxyChannel.enableAnr,eq,true}, desc:NVR will auto-reload the video according to the time interval-->1
  </anrInterval>
  <twoWayAudioChannelIDList>
    <!--ro, opt, array, two-way audio channel No. list of the sub devices managed by NVR, subType:object, desc:this node is valid when devIndex exists-->
    <twoWayAudioChannelID>
      <!--ro, opt, int, two-way audio channel No. of the sub devices managed by NVR-->3
    </twoWayAudioChannelID>
  </twoWayAudioChannelIDList>
  <streamTypeReflectList>
    <!--ro, opt, array, subType:object, range:[0,1]-->
    <streamTypeReflect>
      <!--ro, opt, object-->
      <streamType>
        <!--ro, req, enum, subType:string-->sub
      </streamType>
      <streamReflectType>
        <!--ro, req, enum, subType:string-->sub
      </streamReflectType>
    </streamTypeReflect>
  </streamTypeReflectList>
  <recordTranscodeEnabled>
    <!--ro, opt, bool-->true
  </recordTranscodeEnabled>
  <recordTranscodeConfig>
    <!--ro, opt, bool-->true
  </recordTranscodeConfig>
</InputProxyChannel>
```

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16.2.14.16 Set parameters of a specific digital channel

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, channel information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index-->test
  </id>
  <name>
    <!--opt, string, name-->test
  </name>
  <sourceInputPortDescriptor>
    <!--req, object, input channel description-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress", it is invalid when the protocol type is GB28181-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
    </ipv6Address>
    <managePortNo>
```

```
    <managePortNo>
      <!--req, int, port No. under management, range:[1,65535]-->1
    </managePortNo>
    <srcInputPort>
      <!--req, string, channel No., range:[1,16]-->test
    </srcInputPort>
    <userName>
      <!--req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--req, string, password, range:[1,16]-->test
    </password>
    <connMode>
      <!--opt, enum, connect mode, subType:string, desc:"plugplay" (plug and play), "manual"-->plugplay
    </connMode>
    <streamType>
      <!--opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
    </streamType>
    <deviceID>
      <!--opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
    </deviceID>
    <getStreamMode>
      <!--opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
    </getStreamMode>
    <cameraType>
      <!--opt, enum, camera type, subType:string, desc:"", "", "teacherFull", "studentFull", "media", "blackboard", "teacherLocation", "studentLocation",
"blackboardLocation", "none"-->none
    </cameraType>
    <isDoubleAdd>
      <!--opt, bool, add two streams of the same camera, desc:it is used when DeepinView is added to education sharing server-->true
    </isDoubleAdd>
    <streamUrl>
      <!--opt, string, RTSP streaming URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule-->test
    </streamUrl>
    <transMode>
      <!--opt, enum, subType:int-->0
    </transMode>
    <getStream>
      <!--opt, bool, whether to stream, desc:its value is true by default, which means getting device information after the device is connected and not
recording video-->true
    </getStream>
    <RTSP>
      <!--opt, object, RTSP related nodes-->
      <userName>
        <!--req, string, user name, range:[1,64], desc:"admin" by default-->test
      </userName>
      <password>
        <!--req, string, password, range:[0,64], desc:empty by default-->test
      </password>
      <portNo>
        <!--ro, opt, int, RTSP streaming port, range:[1,65535], step:1, desc:the default port is 554-->554
      </portNo>
    </RTSP>
    <customChanID>
      <!--opt, string, custom camera channel No., range:[1,64], desc:support configuring remotely. When adding sub devices, the platform applies a custom ID
to the accessed digital channel. The ID is the unique identifier with 63 bytes. After the platform applies, the digital channel will upload the ID when an
alarm is triggered, and the platform can find the corresponding device through the ID-->test
    </customChanID>
    <streamMode>
      <!--ro, opt, enum, stream type, subType:string, desc:"main", "sub"-->main
    </streamMode>
    <getSubStream>
      <!--opt, bool, whether to get sub-stream, desc:when this node is not configured, its value is true by default, which means the sub-streaming will
start after the device is connected-->true
    </getSubStream>
    <cloudStorageCfg>
      <!--opt, object, access parameters of cloud storage, dep:or,{$.InputProxyChannel.sourceInputPortDescriptor.proxyProtocol,eq,cloudStrage}-->
      <accessKey>
        <!--req, string, password, range:[1,64], desc:accesskey is the key that can access the cloud storage. Key is generated by cloud storage management
system and it is the secret key for users in the resource pool. Cloud storage introduction (version 2.0)-->test
      </accessKey>
      <secretKey>
        <!--req, string, secret key, range:[1,64], desc:secretkey is used for the cloud protocol;key is generated by cloud storage management system and it
is the secret key for users in the resource pool;cloud storage introduction (version 2.0)-->test
      </secretKey>
      <encoderID>
        <!--opt, string, encoder ID, range:[1,64], desc:encoder ID is the camera ID that has accessed the cloud storage;you can obtain the encoder ID on the
cloud storage platform which has been assigned by the cloud storage;add the encoder ID when you access the cloud storage server-->test
      </encoderID>
    </cloudStorageCfg>
  </sourceInputPortDescriptor>
  <certificateValidationEnabled>
    <!--opt, bool, whether to enable the certification validation-->true
  </certificateValidationEnabled>
  <defaultAdminPortEnabled>
    <!--opt, bool, whether to enable the default communication port-->true
  </defaultAdminPortEnabled>
  <enableAnr>
    <!--opt, bool, whether to enable ANR function, desc:after the camera goes offline, it will reload all the video files during the offline period once it
goes online again-->true
  </enableAnr>
  <enableTiming>
    <!--opt, bool, whether to perform time sync on the device-->true
  </enableTiming>
```

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```
  <streamID>
    <!--opt, string, stream ID, range:[1,32], desc:if this node is transmitted, the stream ID of the current channel can be specified-->test
  </streamID>
  <enabled>
    <!--opt, bool, whether to enable the channel, desc:the default value is true-->true
  </enabled>
  <anrInterval>
    <!--ro, opt, int, time interval for automatic network replenishment, range:[1,24], unit:h, dep:and,{$.InputProxyChannel.enableAnr,eq,true}, desc:NVR
will auto-reload the video according to the time interval-->1
  </anrInterval>
  <streamTypeReflectList>
    <!--opt, array, subType:object, range:[0,1]-->
    <streamTypeReflect>
      <!--opt, object-->
      <streamType>
        <!--req, enum, subType:string-->sub
      </streamType>
      <streamReflectType>
        <!--req, enum, subType:string-->sub
      </streamReflectType>
    </streamTypeReflect>
  </streamTypeReflectList>
</InputProxyChannel>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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16.2.14.17 Get the capability of activating digital channels

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/activate/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceActivationCapability xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <ActivateByChannel>
    <!--ro, opt, object-->
  </ActivateByChannel>
  <ActivateByAddress>
    <!--ro, opt, object-->
  </ActivateByAddress>
</VideoSourceActivationCapability>
```

16.2.14.18 Activate the network devices

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels/activate

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceActivation xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <userName>
    <!--req, string, user name, range:[1,32]-->test
  </userName>
  <passwordType>
    <!--opt, enum, password types, subType:string, desc:"default" (NVR password), "selfDefine", "IPCActivatePasswd" (activation password)-->default
  </passwordType>
  <password>
    <!--req, string, password, range:[1,16]-->test
  </password>
  <ActivationDevice>
    <!--opt, object-->
    <mode>
      <!--req, enum, "channel,descriptor", subType:string, desc:"channel, descriptor"-->channel
    </mode>
    <channel>
      <!--opt, object, channel information, desc:channel information-->
      <dynVideoInputChannelID>
        <!--req, int-->1
      </dynVideoInputChannelID>
    </channel>
    <descriptor>
      <!--opt, object-->
      <macAddress>
        <!--req, string, MAC address, range:[1,48]-->test
      </macAddress>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"hostname" (host name), "ipaddress" (ip address)-->hostname
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,64]-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128]-->test
      </ipv6Address>
      <portNo>
        <!--req, int, port No., range:[1,65535]-->1
      </portNo>
    </descriptor>
  </ActivationDevice>
</VideoSourceActivation>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.19 Get the digital channel access capability

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelListCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, digital channel access management capability, attr:version{req, string, protocolVersion}-->
  <InputProxyChannel>
    <!--ro, opt, object, input channel-->
    <id>
      <!--ro, req, string, index-->test
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
    <sourceInputPortDescriptor>
      <!--ro, req, object, input channel description-->
      <addressingFormatType opt="ipaddress,hostname">
        <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"hostname", "ipaddress". It is invalid when the protocol type is
GB28181-->hostname
      </addressingFormatType>
      <hostName min="1" max="64">
        <!--ro, opt, string, domain name, range:[1,64], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </hostName>
      <ipAddress min="1" max="32">
        <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <ipv6Address min="1" max="128">
        <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </ipv6Address>
      <managePortNo min="1" max="65535">
        <!--ro, req, int, port No. under management, attr:min{req, int},max{req, int}-->1
      </managePortNo>
      <srcInputPort min="1" max="256">
        <!--ro, req, string, device channel No., range:[1,256], attr:min{req, int},max{req, int}-->test
      </srcInputPort>
      <userName min="1" max="32">
        <!--ro, req, string, user name, range:[1,32], attr:min{req, int},max{req, int}-->test
      </userName>
      <password min="1" max="16">
        <!--ro, req, string, password, range:[1,16], attr:min{req, int},max{req, int}-->test
      </password>
      <streamType opt="auto,tcp,udp">
        <!--ro, opt, enum, streaming protocol type, subType:string, attr:opt{req, string}, desc:"auto", "TCP", "UDP"-->auto
      </streamType>
      <deviceID min="1" max="32">
        <!--ro, opt, string, device ID, range:[1,32], attr:min{req, int},max{req, int}-->test
      </deviceID>
      <activateEnabled opt="true,false">
        <!--ro, opt, bool, whether the device needs to be activated, attr:opt{req, string}, desc:if this node does not exist, it indicates that the device
has already been activated. Application scenario: supports adding an unactivated device to the NVR channel. If this node is true, the device is unactivated
and needs to be activated by the NVR. The NVR will auto-activate the device (MAC address is needed because IP access is unavailable for unactivated
devices), and will add the device to NVR-->true
      </activateEnabled>
      <macAddress min="0" max="48">
        <!--ro, opt, string, MAC address, range:[0,48], attr:min{req, int},max{req, int}, desc:it is required when the value of activateEnabled is true--
>test
      </macAddress>
      <cameraType>
        <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating),
"teacherFull" (teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "IPCamera", "IPDome", "none"-->none
      </cameraType>
      <streamMode opt="main,sub">
        <!--ro, opt, enum, stream type, subType:string, attr:opt{req, string}, desc:"main", "sub"-->main
      </streamMode>
      <getSubStream opt="true,false">
        <!--ro, opt, bool, whether to get sub-stream, attr:opt{req, string}, desc:when this node is not configured, its value is true by default, which
means the sub-streaming will start after the device is connected-->true
      </getSubStream>
    </sourceInputPortDescriptor>
    <online opt="true,false">
      <!--ro, opt, bool, whether the channel is online, attr:opt{req, string}-->true
    </online>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the channel, attr:opt{req, string}, desc:the default value is true-->true
    </enabled>
    <anrInterval min="1" max="24">
      <!--ro, opt, int, time interval for automatic network replenishment, attr:min{req, int},max{req, int}-->1
    </anrInterval>
  </InputProxyChannel>
  <FilterchannelsByIp>
    <!--ro, opt, bool, whether filtering channel information by IP address is supported-->true
  </FilterchannelsByIp>
  <FilterchannelsByStreamID>
    <!--ro, opt, bool, whether filtering channel information by stream ID is supported-->true
  </FilterchannelsByStreamID>
  <isSupportSmartConfig>
```

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```
    <!--ro, opt, bool, -ro,
desc:search for connected wireless network camera automatically and then activate and add the network camera. The automatic match period is 120 seconds. The
match process will end as soon as the number of channels reaches the upper limit.
    If its value is true, it indicates that the following URLs are supported:
PUT /ISAPI/ContentMgmt/InputProxy/channels/smartConfig?format=json;
GET ISAPI/ContentMgmt/InputProxy/channels/smartConfig/status?format=json;
PUT /ISAPI/ContentMgmt/InputProxy/channels/smartConfig/stop?format=json-->true
  </isSupportSmartConfig>
  <isSupportDevIndex>
    <!--ro, opt, bool, whether transparent transmission of sub-devices ID is supported, desc:related URLs: /ISAPI/ContentMgmt/InputProxy/channels. Each
channel corresponds to a devIndex node. The sub-devices do not include the gateway sub-devices in /ISAPI/ContentMgmt/DeviceMgmt/addDevice?format=json-->true
  </isSupportDevIndex>
  <displayGetStream>
    <!--ro, opt, bool, whether the node configuration is displayed on the web channel configuration page, desc:currently, the configuration is not displayed
on the NVR channel page, and is displayed on the AI DeepinMind channel page-->true
  </displayGetStream>
</InputProxyChannelListCap>
```

16.2.14.20 Get status of all digital channels

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/status

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelStatusList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, channel status list, subType:object, attr:version{req, string, protocolVersion}-->
  <InputProxyChannelStatus>
    <!--ro, opt, object, channel status-->
    <id>
      <!--ro, req, string, index-->test
    </id>
    <sourceInputPortDescriptor>
      <!--ro, req, object, input channel description-->
      <addressingFormatType>
        <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress". It is invalid when the protocol type is GB28181-->hostname
      </addressingFormatType>
      <hostName>
        <!--ro, opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
      </hostName>
      <ipAddress>
        <!--ro, opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
      </ipv6Address>
      <managePortNo>
        <!--ro, req, int, port No. under management, range:[0,65535]-->0
      </managePortNo>
      <srcInputPort>
        <!--ro, req, string, device channel No., range:[1,16]-->test
      </srcInputPort>
      <userName>
        <!--ro, req, string, user name, range:[0,32]-->test
      </userName>
      <password>
        <!--ro, req, string, password, range:[0,16], desc:this node is empty in the response message-->test
      </password>
      <connMode>
        <!--ro, opt, enum, connect mode, subType:string, desc:"plugplay" (plug and play), "manual"-->plugplay
      </connMode>
      <streamType>
        <!--ro, opt, enum, streaming protocol type, subType:string, desc:“auto”, “tcp”, “udp”-->auto
      </streamType>
      <deviceID>
        <!--ro, opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
      </deviceID>
      <getStreamMode>
        <!--ro, opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
      </getStreamMode>
      <cameraType>
        <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating),
"teacherFull" (teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "none"-->none
      </cameraType>
      <isDoubleAdd>
        <!--ro, opt, bool, add two streams of the same camera, desc:it is used when DeepinView is added to education sharing server-->true
      </isDoubleAdd>
      <streamUrl>
        <!--ro, opt, string, RTSP streaming URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule--
```

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```
>test
      </streamUrl>
      <macAddress>
        <!--ro, opt, string, MAC address, range:[1,48]-->test
      </macAddress>
      <deviceName>
        <!--ro, opt, string, device name-->DS-2XM6736FWD-IM/ND
      </deviceName>
      <firmwareVersion>
        <!--ro, opt, string, software version-->V5.5.62 build 180822
      </firmwareVersion>
      <SDKMapPort>
        <!--ro, opt, int-->8000
      </SDKMapPort>
      <SSHMapPort>
        <!--ro, opt, int-->22
      </SSHMapPort>
      <cloudStrageCfg>
        <!--ro, opt, object, dep:or,{$.InputProxyChannel.sourceInputPortDescriptor.proxyProtocol,eq,cloudStrage}-->
        <accessKey>
          <!--ro, opt, string, range:[1,64]-->test
        </accessKey>
        <secretKey>
          <!--ro, opt, string, range:[1,64]-->test
        </secretKey>
        <encoderID>
          <!--ro, opt, string, range:[1,64]-->test
        </encoderID>
        <protocolType>
          <!--ro, opt, string, range:[1,64]-->V3.1
        </protocolType>
      </cloudStrageCfg>
    </sourceInputPortDescriptor>
    <online>
      <!--ro, req, bool, whether the channel is online-->true
    </online>
    <supportCreateStream>
      <!--ro, opt, bool, whether creating stream is supported-->true
    </supportCreateStream>
    <streamingProxyChannelIdList>
      <!--ro, req, array, streaming channel list, subType:object-->
      <streamingProxyChannelId>
        <!--ro, opt, string, streaming channel-->test
      </streamingProxyChannelId>
    </streamingProxyChannelIdList>
    <relatedIOProxy>
      <!--ro, opt, object, linked IO port-->
      <inputProxyPortIdList>
        <!--ro, opt, array, input port No. list, subType:object-->
        <inputProxyPortId>
          <!--ro, opt, string, input port No.-->test
        </inputProxyPortId>
      </inputProxyPortIdList>
      <outputProxyPortIdList>
        <!--ro, opt, array, output port No. list, subType:object-->
        <outputProxyPortId>
          <!--ro, opt, string, output port No.-->test
        </outputProxyPortId>
      </outputProxyPortIdList>
    </relatedIOProxy>
    <chanDetectResult>
      <!--ro, opt, enum, channel detection status, subType:string, desc:"connect" (connected), "overSysBandwidth" (exceeded the system bandwidth),
"domainError" (domain error), "ipcStreamFail" (network camera streaming failed), "connecting" (connecting), "chanNoError" (incorrect channel number),
"ipAddrConflictWithDev" (IP address is conflicted with current device's), "ipAddrConflicWithIpc" (IP address is conflicted with other device's.),
"errorUserNameOrPasswd" (incorrect user name or password), "netUnreachable" (network is unreachable), "unknownError" (unknown error), "notExist" (not
exist), "ipcStreamTypeNotSupport" (the network camera stream type is not supported), "ipcResolutionNotSupport" (the network camera resolution is not
supported), "userLocked" (the user is locked), "userNotExist" (the user does not exist), "ipcUnregistered" (network camera is not registered),
"ipcNotActivated" (network camera is inactivated), "poePortDetecting" (detecting POE port), "uploadingCloudFile" (uploading files to cloud storage),
"certificateValidationError" (certification validation failed), "ONVIFRegistrationError" (ONVIF registration failed), "streamingParameters" (streaming
parameter error), "sleep", "wakeUping" (waking up)-->connect
    </chanDetectResult>
    <SecurityStatus>
      <!--ro, opt, object, security status-->
      <PasswordStatus>
        <!--ro, opt, enum, password status, subType:string, desc:"ipcNotActivated" (IPC is not activated), "risk", "medium", "strong:", "invalid"--
>ipcNotActivated
      </PasswordStatus>
    </SecurityStatus>
    <chanStatus>
      <!--ro, opt, enum, channel status, subType:string, desc:"locate" (locating), "getStream" (getting stream)-->locate
    </chanStatus>
    <channelName>
      <!--ro, opt, string, channel name, range:[1,32]-->--
    </channelName>
    <IP>
      <!--ro, opt, string, IP address, range:[1,128]-->test
    </IP>
    <deviceName>
      <!--ro, opt, string, device name-->test
    </deviceName>
    <recordInformation>
      <!--ro, opt, object, recording parameters-->
      <status>
        <!--ro, req, enum, recording status, subType:string, desc:"recording", "noVideo" (no video recording), "recordingFailed"-->neconding
```

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```
        <!--ro, req, enum, recording status, subType:string, desc:"recording", "novideo" (no video recording), "recordingfailed"-->recording
      </status>
      <recordType>
        <!--ro, opt, enum, recording type, subType:string, desc:"CMR" (scheduled recording), "MOTION" (motion detection recording), "AllEvent" (event
recording), "MANUAL" (manual recording)-->CMR
      </recordType>
      <failedReason>
        <!--ro, opt, enum, recording failed reason, subType:string, desc:"diskFull", "diskError", "noAssociatedDiskGroup", "other"-->diskFull
      </failedReason>
    </recordInformation>
    <isSettingRecordPlan>
      <!--ro, opt, bool, whether to configure recording schedule-->true
    </isSettingRecordPlan>
    <videoPlanStatus>
      <!--ro, opt, enum, recording schedule status, subType:string, desc:"normal", "busy", "abnormal", "planless"-->normal
    </videoPlanStatus>
    <videoPassBackStatus>
      <!--ro, opt, enum, recording copy-back status, subType:string, desc:"normal", "busy", "abnormal", "planless"-->normal
    </videoPassBackStatus>
    <videoPassBackDetail>
      <!--ro, opt, string, recording copy-back details-->test
    </videoPassBackDetail>
    <videoFirstStartTime>
      <!--ro, opt, datetime, earliest recording time-->1970-01-01T00:00:00+08:00
    </videoFirstStartTime>
    <nodeId>
      <!--ro, opt, string, node ID-->test
    </nodeId>
    <bitStreamType>
      <!--ro, req, enum, stream type, subType:string, desc:"mainStream", "subStream"-->mainStream
    </bitStreamType>
    <rateType>
      <!--ro, req, enum, bit rate type, subType:string, desc:"CBR" (constant bit rate), "VBR" (variable bit rate)-->CBR
    </rateType>
    <codeParam>
      <!--ro, req, enum, encoding parameters, subType:string, desc:"H.264", "H.265", "HK.264", "MPEG4", "MJPEG", "SVAC", "UNKNOWN"-->H.264
    </codeParam>
    <enabled>
      <!--ro, opt, bool, whether to enable the channel, desc:the default value is true-->true
    </enabled>
    <chanExceptionDetails>
      <!--ro, opt, string, channel detection status detailed information, dep:and,
{$.InputProxyChannelStatusList[*].InputProxyChannelStatus.chanDetectResult,de,connect}, desc:it can be used to locate channel access exceptions-->incorrect
username or password
    </chanExceptionDetails>
    <batteryPercentage>
      <!--ro, opt, int, channel camera's remaining battery percentage, range:[0,100], desc:it is valid when a solar camera is accessed-->0
    </batteryPercentage>
    <localVerificationAnalysis>
      <!--ro, opt, object-->
      <mode>
        <!--ro, opt, enum, subType:string-->perimeter
      </mode>
      <status>
        <!--ro, req, enum, subType:string-->analysising
      </status>
    </localVerificationAnalysis>
  </InputProxyChannelStatus>
</InputProxyChannelStatusList>
```

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16.2.14.21 Set parameters of all digital channels

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, digital channel input information list, subType:object, attr:version{req, string, protocolVersion}-->
  <InputProxyChannel>
    <!--opt, object, digital channel input information-->
    <id>
      <!--req, string, index-->test
    </id>
    <name>
      <!--opt, string, name-->test
    </name>
    <sourceInputPortDescriptor>
      <!--req, object, input channel source description-->
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress", it is invalid when the protocol type is GB28181-->hostname
      </addressingFormatType>
      <hostName>
        <!--opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
```

```
      <!--opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
    </ipv6Address>
    <managePortNo>
      <!--req, int, port No., range:[1,65535]-->1
    </managePortNo>
    <srcInputPort>
      <!--req, string, device channel No., range:[1,16]-->test
    </srcInputPort>
    <userName>
      <!--req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--req, string, password, range:[1,16], desc:this node is empty in the response message-->test
    </password>
    <connMode>
      <!--opt, enum, connect mode, subType:string, desc:“plugplay” (plug and play), “manual”-->plugplay
    </connMode>
    <streamType>
      <!--opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
    </streamType>
    <deviceID>
      <!--opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
    </deviceID>
    <getStreamMode>
      <!--opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
    </getStreamMode>
    <cameraType>
      <!--opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating), "teacherFull"
(teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "none"-->none
    </cameraType>
    <isDoubleAdd>
      <!--opt, bool, add two streams of a same camera, desc:it is used when DeepinView is added to education sharing server-->true
    </isDoubleAdd>
    <streamUrl>
      <!--opt, string, RTSP streaming URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule-->test
    </streamUrl>
    <getStream>
      <!--opt, bool, whether to stream, desc:its value is true by default, which means getting device information after the device is connected but not
recording video-->true
    </getStream>
    <RTSP>
      <!--opt, object, RTSP related nodes-->
      <userName>
        <!--req, string, user name, range:[1,64], desc:"admin" by default-->test
      </userName>
      <password>
        <!--req, string, password, range:[0,64], desc:empty by default-->test
      </password>
      <portNo>
        <!--ro, opt, int, RTSP streaming port, range:[1,65535], step:1, desc:default port is 554-->554
      </portNo>
    </RTSP>
    <customChanID>
      <!--opt, string, custom camera channel No., range:[1,64], desc:support configuring remotely. When adding sub devices, the platform applies a custom
ID to the accessed digital channel. The ID is the unique identifier with 63 bytes. After the platform applies, the digital channel will upload the ID when
an alarm is triggered, and the platform can find the corresponding device through the ID-->test
    </customChanID>
    <model>
      <!--ro, opt, string, device model, range:[1,64]-->iDS-9632NX-I8/X
    </model>
    <serialNumber>
      <!--ro, opt, string, serial No., range:[0,64]-->test
    </serialNumber>
    <firmwareVersion>
      <!--ro, opt, string, firmware version, range:[0,64]-->test
    </firmwareVersion>
    <getSubStream>
      <!--opt, bool, whether to get sub-stream, desc:when this node is not configured, its value is true by default, which means the sub-streaming will
start after the device is connected-->true
    </getSubStream>
    <cloudStorageCfg>
      <!--opt, object, dep:or,{$.InputProxyChannel.sourceInputPortDescriptor.proxyProtocol,eq,cloudStrage}-->
      <accessKey>
        <!--req, string, range:[1,64]-->test
      </accessKey>
      <secretKey>
        <!--req, string, range:[1,64]-->test
      </secretKey>
      <encoderID>
        <!--opt, string, range:[1,64]-->test
      </encoderID>
    </cloudStorageCfg>
  </sourceInputPortDescriptor>
  <enableAnr>
    <!--opt, bool, GB23181 push-flow port, desc:return to the push-flow port after adding the GB23181-->true
  </enableAnr>
  <enableTiming>
    <!--opt, bool, whether to enable synchronizing the time of the device-->true
```

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```
    </enableTiming>
    <streamID>
      <!--opt, string, stream ID, range:[1,32], desc:if this node is transmitted, the stream ID of the current channel can be specified-->test
    </streamID>
    <enabled>
      <!--opt, bool, whether to enable channel ID, desc:the default value is true-->true
    </enabled>
    <anrInterval>
      <!--ro, opt, int, time interval for automatic network replenishment, range:[1,24], unit:h, dep:and,{$.InputProxyChannel.enableAnr,eq,true}, desc:NVR
will auto-reload the video according to the time interval-->1
    </anrInterval>
  </InputProxyChannel>
</InputProxyChannelList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

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DETEKTOR

16.2.14.22 Get access parameters of all digital channels

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, channel list, subType:object, attr:version{req, string, protocolVersion}-->
  <InputProxyChannel>
    <!--ro, opt, object, channel information-->
    <id>
      <!--ro, req, string, index-->test
    </id>
    <name>
      <!--ro, opt, string, name-->test
    </name>
    <sourceInputPortDescriptor>
      <!--ro, req, object, input channel description-->
      <addressingFormatType>
        <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress". It is invalid when the protocol type is GB28181-->hostname
      </addressingFormatType>
      <hostName>
        <!--ro, opt, string, domain name, range:[1,64], desc:it is invalid when the protocol type is GB28181-->test
      </hostName>
      <ipAddress>
        <!--ro, opt, string, IPv4 address, range:[1,32], desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address, range:[1,128], desc:it is invalid when the protocol type is GB28181-->test
      </ipv6Address>
      <managePortNo>
        <!--ro, req, int, port No. under management, range:[1,65535]-->1
      </managePortNo>
      <srcInputPort>
        <!--ro, req, string, device channel No., range:[1,16]-->test
      </srcInputPort>
```

```
      <userName>
        <!--ro, req, string, user name, range:[1,32]-->test
      </userName>
      <password>
        <!--ro, req, string, password, range:[0,16], desc:the node is empty in the returned results-->test
      </password>
      <connMode>
        <!--ro, opt, enum, connection mode, subType:string, desc:"plugplay" (plug and play), "manual"-->plugplay
      </connMode>
      <streamType>
        <!--ro, opt, enum, streaming protocol type, subType:string, desc:"auto", "tcp", "udp"-->auto
      </streamType>
      <deviceID>
        <!--ro, opt, string, device ID, range:[1,20], desc:the node exists only when the protocol type is GB28181-->test
      </deviceID>
      <getStreamMode>
        <!--ro, opt, enum, streaming mode, subType:string, desc:"active", "passive". The node exists only when the protocol type is GB28181-->active
      </getStreamMode>
      <cameraType>
        <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating),
"teacherFull" (teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "IPCamera", "IPDome", "none"-->none
      </cameraType>
      <isDoubleAdd>
        <!--ro, opt, bool, add two streams of the same camera, desc:it is used when DeepinView is added to education sharing server-->true
      </isDoubleAdd>
      <streamUrl>
        <!--ro, opt, string, RTSP stream URL, range:[1,256], desc:if this node exists, use this URL for streaming; if not, follow the original rule-->test
      </streamUrl>
      <transMode>
        <!--ro, opt, enum, subType:int-->0
      </transMode>
      <getStream>
        <!--ro, opt, bool, whether to stream, desc:its value is true by default, which means getting device information after the device is connected and
not recording video-->true
      </getStream>
      <RTSP>
        <!--ro, opt, object, RTSP related nodes-->
        <userName>
          <!--ro, req, string, user name, range:[1,64], desc:"admin" by default-->test
        </userName>
        <password>
          <!--ro, req, string, password, range:[0,64], desc:empty by default-->test
        </password>
        <portNo>
          <!--ro, opt, int, RTSP streaming port, range:[1,65535], step:1, desc:default port is 554-->554
        </portNo>
      </RTSP>
      <customChanID>
        <!--ro, opt, string, custom camera channel ID, range:[1,64], desc:support configuring remotely. When adding sub devices, the platform applies a
custom ID to the accessed digital channel. The ID is the unique identifier with 63 bytes. After the platform applies, the digital channel will upload the ID
when an alarm is triggered, and the platform can find the corresponding device through the ID-->test
      </customChanID>
      <model>
        <!--ro, opt, string, device model, range:[1,64]-->iDS-9632NX-I8/X
      </model>
      <serialNumber>
        <!--ro, opt, string, serial No., range:[1,64]-->test
      </serialNumber>
      <firmwareVersion>
        <!--ro, opt, string, firmware version, range:[1,64]-->test
      </firmwareVersion>
      <streamMode>
        <!--ro, opt, enum, stream type, subType:string, desc:"main", "sub"-->main
      </streamMode>
      <getSubStream>
        <!--ro, opt, bool, whether to get sub-stream, desc:when this node is not configured, its value is true by default, which means the sub-streaming
will start after the device is connected-->true
      </getSubStream>
    </sourceInputPortDescriptor>
    <certificateValidationEnabled>
      <!--ro, opt, bool, whether to enable the certification validation-->true
    </certificateValidationEnabled>
    <defaultAdminPortEnabled>
      <!--ro, opt, bool, whether to enable the default communication port-->true
    </defaultAdminPortEnabled>
    <enableAnr>
      <!--ro, opt, bool, whether to enable ANR (Automatic Network Replenishment), desc:after the camera goes offline, it will reload all the video files
during the offline period once it goes online again-->true
    </enableAnr>
    <enableTiming>
      <!--ro, opt, bool, whether to perform time sync on the device-->true
    </enableTiming>
    <streamID>
      <!--ro, opt, string, stream ID, range:[1,32], desc:if this node is transmitted, the stream ID of the current channel can be specified-->test
    </streamID>
    <online>
      <!--ro, opt, bool, whether the channel is online-->true
    </online>
    <NVRInfo>
      <!--ro, opt, object-->
      <ipAddressNVR>
        <!--ro, opt, string, range:[1,32]-->192.168.1.100
      </ipAddressNVR>
```

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DEIKTOR

```
        <portNVR>
          <!--ro, opt, int, range:[0,65535]-->1
        </portNVR>
        <ipcChannelNo>
          <!--ro, opt, int, range:[0,64]-->1
        </ipcChannelNo>
      </NVRInfo>
      <enabled>
        <!--ro, opt, bool, whether to enable the channel, desc:the default value is true-->true
      </enabled>
      <devIndex>
        <!--ro, opt, string, sub-device ID, range:[0,64], desc:used to specify a sub-device to which ISAPI is transmitted-->test
      </devIndex>
      <boundAudioChannelID>
        <!--ro, opt, int, linked audio channel No., desc:if this node does not exist, no audio channel is linked to-->1
      </boundAudioChannelID>
      <anrInterval>
        <!--ro, opt, int, time interval for automatic network replenishment, range:[1,24], unit:h, dep:and,{$.InputProxyChannel.enableAnr,eq,true}, desc:NVR
will auto-reload the video according to the time interval-->1
      </anrInterval>
      <offlineReason>
        <!--ro, opt, enum, channel offline reason, subType:string, dep:and,{$.InputProxyChannelList[*].InputProxyChannel.online,eq,false}, desc:passwordError:
incorrect password-->passwordError
      </offlineReason>
      <streamTypeReflectList>
        <!--ro, opt, array, subType:object, range:[0,1]-->
        <streamTypeReflect>
          <!--ro, opt, object-->
          <streamType>
            <!--ro, req, enum, subType:string-->sub
          </streamType>
          <streamReflectType>
            <!--ro, req, enum, subType:string-->sub
          </streamReflectType>
        </streamTypeReflect>
      </streamTypeReflectList>
      <recordTranscodeEnabled>
        <!--ro, opt, bool-->true
      </recordTranscodeEnabled>
      <recordTranscodeConfig>
        <!--ro, opt, bool-->true
      </recordTranscodeConfig>
    </InputProxyChannel>
  </InputProxyChannelList>
```

16.2.14.23 Set parameters of all custom protocols

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/customProtocols

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<customProtocolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <customProtocol>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <protocalName>
      <!--req, string-->test
    </protocalName>
    <streamingList>
      <!--req, array, subType:object-->
      <streaming>
        <!--opt, object-->
        <id>
          <!--req, string-->test
        </id>
        <enableStream>
          <!--req, bool-->true
        </enableStream>
        <streamProType>
          <!--req, enum, subType:string-->RTSP
        </streamProType>
        <streamTransMode>
          <!--req, enum, subType:string-->UDP
        </streamTransMode>
        <streamPort>
          <!--req, int-->1
        </streamPort>
        <streamPath>
          <!--req, string-->test
        </streamPath>
      </streaming>
    </streamingList>
  </customProtocol>
</customProtocolList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.14.24 Get configurations of all custom protocols

Request URL

GET /ISAPI/ContentMgmt/InputProxy/customProtocols

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<customProtocolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <customProtocol>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <protocalName>
      <!--ro, req, string-->test
    </protocalName>
    <streamingList>
      <!--ro, req, array, subType:object-->
      <streaming>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->test
        </id>
        <enableStream>
          <!--ro, req, bool-->true
        </enableStream>
        <streamProType>
          <!--ro, req, enum, subType:string-->RTSP
        </streamProType>
        <streamTransMode>
          <!--ro, req, enum, subType:string-->UDP
        </streamTransMode>
        <streamPort>
          <!--ro, req, int-->1
        </streamPort>
        <streamPath>
          <!--ro, req, string-->test
        </streamPath>
      </streaming>
    </streamingList>
  </customProtocol>
</customProtocolList>
```

16.2.14.25 Get the custom protocol configuration capability

Request URL

GET /ISAPI/ContentMgmt/InputProxy/customProtocols/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<customProtocol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <protocalName>
    <!--ro, req, string-->test
  </protocalName>
  <streamingList>
    <!--ro, req, array, subType:object-->
    <streaming>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <enableStream opt="true">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enableStream>
      <streamProType opt="RTSP">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->RTSP
      </streamProType>
      <streamTransMode opt="auto,UDP,RTP Over RTSP,RTSP Over HTTP,RTSP Over HTTPS">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->UDP
      </streamTransMode>
      <streamPort min="1" max="65535">
        <!--ro, req, int, attr:min{req, int},max{req, int}-->1
      </streamPort>
      <streamPath>
        <!--ro, req, string-->test
      </streamPath>
    </streaming>
  </streamingList>
  <RTSPBetterStorageEffectMaxBitRate>
    <!--ro, opt, int-->6
  </RTSPBetterStorageEffectMaxBitRate>
</customProtocol>
```

16.2.14.26 Get the parameters of a specified custom protocol

Request URL

GET /ISAPI/ContentMgmt/InputProxy/customProtocols/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	Custom protocol index

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<customProtocol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <protocalName>
    <!--ro, req, string-->test
  </protocalName>
  <streamingList>
    <!--ro, req, array, subType:object-->
    <streaming>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <enableStream>
        <!--ro, req, bool-->true
      </enableStream>
      <streamProType>
        <!--ro, req, enum, subType:string-->RTSP
      </streamProType>
      <streamTransMode>
        <!--ro, req, enum, subType:string-->UDP
      </streamTransMode>
      <streamPort>
        <!--ro, req, int-->1
      </streamPort>
      <streamPath>
        <!--ro, req, string-->test
      </streamPath>
    </streaming>
  </streamingList>
</customProtocol>
```

16.2.14.27 Set the parameters of a specified custom protocol

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/customProtocols/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<customProtocol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <protocalName>
    <!--req, string-->test
  </protocalName>
  <streamingList>
    <!--req, array, subType:object-->
    <streaming>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <enableStream>
        <!--req, bool-->true
      </enableStream>
      <streamProType>
        <!--req, enum, subType:string-->RTSP
      </streamProType>
      <streamTransMode>
        <!--req, enum, subType:string-->UDP
      </streamTransMode>
      <streamPort>
        <!--req, int-->1
      </streamPort>
      <streamPath>
        <!--req, string-->test
      </streamPath>
    </streaming>
  </streamingList>
</customProtocol>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.14.28 Search for device information of all digital channels

Request URL

POST /ISAPI/ContentMgmt/InputProxy/deviceInfo

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ProxyDevicesInfoCondition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time-->1
  </maxResults>
</ProxyDevicesInfoCondition>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ProxyDevicesInfoResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <responseStatus>
    <!--ro, req, bool, search status-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, search status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching
completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <DeviceInfoList>
    <!--ro, opt, array, refer to XML_DeviceInfo, subType:object-->
    <DeviceInfo>
      <!--ro, opt, object-->
      <deviceName>
        <!--ro, req, string, device name-->test
      </deviceName>
      <deviceID>
        <!--ro, req, string, uuid-->fb8d4000-8158-11b5-8450-b4a38245654c
      </deviceID>
      <deviceDescription>
        <!--ro, opt, string, device description-->test
      </deviceDescription>
      <deviceLocation>
        <!--ro, opt, string, device location-->hangzhou
      </deviceLocation>
      <deviceStatus>
        <!--ro, opt, enum, device status, desc:"normal", "abnormal"-->normal
      </deviceStatus>
      <DetailAbnormalStatus>
        <!--ro, opt, object, abnormal status-->
        <hardDiskFull>
          <!--ro, opt, bool, whether disk is full-->true
        </hardDiskFull>
        <hardDiskError>
          <!--ro, opt, bool, whether disk has error-->true
        </hardDiskError>
        <ethernetBroken>
          <!--ro, opt, bool, whether it is disconnected-->true
        </ethernetBroken>
        <ipaddrConflict>
          <!--ro, opt, bool, whether IP address conflicts-->true
        </ipaddrConflict>
        <illegalAccess>
          <!--ro, opt, bool, whether it is illegal access-->true
        </illegalAccess>
        <recordError>
          <!--ro, opt, bool, whether video exception occurred-->true
        </recordError>
        <raidLogicDiskError>
          <!--ro, opt, bool, whether raid logic disk exception occurred-->true
        </raidLogicDiskError>
        <spareWorkDeviceError>
          <!--ro, opt, bool, whether spare device exception occurred-->true
        </spareWorkDeviceError>
      </DetailAbnormalStatus>
      <systemContact>
        <!--ro, opt, string, manufacturer-->STD-CGI
      </systemContact>
      <model>
        <!--ro, req, string, device model-->iDS-9632NX-I8/X
      </model>
      <serialNumber>
        <!--ro, req, string, serial No.-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
      </serialNumber>
      <macAddress>
        <!--ro, req, string, physical address-->44:47:cc:c8:d9:e4
      </macAddress>
      <firmwareVersion>
        <!--ro, req, string, firmware version-->V4.1.40
      </firmwareVersion>
      <firmwareReleasedDate>
        <!--ro, opt, string-->build 181129
      </firmwareReleasedDate>
      <encoderVersion>
        <!--ro, opt, string, device encoder version No.-->V7.3
      </encoderVersion>
      <encoderReleasedDate>
        <!--ro, opt, string-->build 20161223
      </encoderReleasedDate>
      <bootVersion>
        <!--ro, opt, string, boot version-->V1.3.4
      </bootVersion>
      <bootReleasedDate>
        <!--ro, opt, string, release date of boot-->100316
```

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```
        </bootReleasedDate>
        <panelVersion>
          <!--ro, opt, string, panel version-->V1.0
        </panelVersion>
        <hardwareVersion>
          <!--ro, opt, string, hardware version-->0x0
        </hardwareVersion>
        <decoderVersion>
          <!--ro, opt, string, device decoder version-->V1.0
        </decoderVersion>
        <decoderReleasedDate>
          <!--ro, opt, string, compile date of the device decoder version-->build 20161223
        </decoderReleasedDate>
        <softwareVersion>
          <!--ro, opt, string, software version-->build 181129
        </softwareVersion>
        <capacity>
          <!--ro, opt, int, device capacity (MB)-->10240
        </capacity>
        <usedCapacity>
          <!--ro, opt, int, used capacity (MB)-->5120
        </usedCapacity>
        <deviceType>
          <!--ro, req, enum, device type, subType:string, desc:read-only,"IPCamera,IPDome,DVR,HybirdNVR,NVR,DVS,IPZoom,CVR"-->Blade
        </deviceType>
        <telecontrolID>
          <!--ro, opt, int, "1-255", range:[1,255]-->1
        </telecontrolID>
        <supportBeep>
          <!--ro, opt, bool, whether to support buzzer-->true
        </supportBeep>
        <supportVideoLoss>
          <!--ro, opt, bool, whether it supports video loss-->true
        </supportVideoLoss>
        <firmwareVersionInfo>
          <!--ro, opt, string, firmware version information-->1
        </firmwareVersionInfo>
        <actualFloorNum>
          <!--ro, opt, int, "1-128", range:[1,255]-->1
        </actualFloorNum>
        <localZoneNum>
          <!--ro, opt, int, number of local zones-->1
        </localZoneNum>
        <alarmOutNum>
          <!--ro, opt, int, number of alarm outputs-->1
        </alarmOutNum>
        <distanceResolution>
          <!--ro, opt, float, distance resolution-->0.000
        </distanceResolution>
        <angleResolution>
          <!--ro, opt, float, angle resolution-->0.000
        </angleResolution>
        <speedResolution>
          <!--ro, opt, float, speed resolution-->0.000
        </speedResolution>
        <detectDistance>
          <!--ro, opt, float, detection distance-->0.000
        </detectDistance>
        <relayNum>
          <!--ro, opt, int, number of local relays-->1
        </relayNum>
        <electroLockNum>
          <!--ro, opt, int, number of local electronic locks-->1
        </electroLockNum>
        <RS485Num>
          <!--ro, opt, int, number of local RS-485-->1
        </RS485Num>
        <radarVersion>
          <!--ro, opt, string, radar version-->test
        </radarVersion>
        <cameraModuleVersion>
          <!--ro, opt, string, camera module version-->test
        </cameraModuleVersion>
        <mainversion>
          <!--ro, opt, int, main version No., range:[1,255]-->1
        </mainversion>
        <subversion>
          <!--ro, opt, int, sub version No., range:[1,255]-->1
        </subversion>
        <upgradeversion>
          <!--ro, opt, int, update version No., range:[1,255]-->1
        </upgradeversion>
        <customizeversion>
          <!--ro, opt, int, custom version No., range:[1,255]-->1
        </customizeversion>
        <companyName>
          <!--ro, opt, string, manufacturer name-->test
        </companyName>
        <copyright>
          <!--ro, opt, string, copyright Information-->test
        </copyright>
        <systemName>
          <!--ro, opt, enum, storage system name, subType:string, desc:"storageManagement (storage management system), "distributedStorageManagement"
(distributed storage management system)-->storageManagement
```

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```
(distributed storage management system) -->storageManagement
    </systemName>
    <systemStatus>
      <!--ro, opt, enum, system status, subType:string, desc:"configured" (system has been configured), "unConfigured" (system has not been configured)--
>configured
    </systemStatus>
    <isLeaderDevice>
      <!--ro, opt, bool, whether it is the corresponding device of the resource IP-->true
    </isLeaderDevice>
    <clusterVersion>
      <!--ro, opt, string, system cluster version, desc:this field is valid only when isLeaderDevice is "true"-->test
    </clusterVersion>
    <manufacturer>
      <!--ro, opt, enum, manufacturer information, subType:string, desc:it is used to differentiate OEM devices-->test
    </manufacturer>
    <centralStorageVersion>
      <!--ro, opt, string, center storage version, range:[1,16]-->test
    </centralStorageVersion>
    <powerOnMode>
      <!--ro, opt, enum, startup mode, subType:string, desc:“button” (press button to power on), “adapter” (connect adapter to power on)-->button
    </powerOnMode>
    <customizedInfo>
      <!--ro, opt, string, custom project No., desc:custom project information-->test
    </customizedInfo>
    <charEncodeFormat>
      <!--ro, opt, enum, subType:string-->GB2312
    </charEncodeFormat>
    <languageType opt="chinese,english,spanish,portuguese,italian,french,russian,turkish,greek,czech">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->chinese
    </languageType>
    <DockStation>
      <!--ro, opt, object, dock station information configuration-->
      <Platform>
        <!--ro, opt, object, platform dock station information configuration-->
        <type>
          <!--ro, req, enum, platform which is connected to the dock station, subType:string, desc:“none”, “9533”, “8618”, “ISAPI”-->none
        </type>
        <ip>
          <!--ro, opt, string, IP address-->test
        </ip>
        <port>
          <!--ro, opt, int, communication port-->1
        </port>
        <userName>
          <!--ro, req, string, user name-->test
        </userName>
        <password>
          <!--ro, req, string, password-->test
        </password>
      </Platform>
      <centralStorageBackupEnabled>
        <!--ro, opt, bool-->true
      </centralStorageBackupEnabled>
    </DockStation>
    <webVersion>
      <!--ro, opt, string, Web version No.-->test
    </webVersion>
    <deviceRFProgramVersion>
      <!--ro, opt, string-->test
    </deviceRFProgramVersion>
    <securityModuleSerialNo>
      <!--ro, opt, string, security module serial No.-->test
    </securityModuleSerialNo>
    <securityModuleVersion>
      <!--ro, opt, string-->test
    </securityModuleVersion>
    <securityChipVersion>
      <!--ro, opt, string-->test
    </securityChipVersion>
    <securityModuleKeyVersion>
      <!--ro, opt, string-->test
    </securityModuleKeyVersion>
    <UIDLampRecognition>
      <!--ro, opt, object, recognize the device information by UID-->
      <enabled>
        <!--ro, opt, bool, whether to enable the function-->true
      </enabled>
    </UIDLampRecognition>
    <confDeviceIdPrefix>
      <!--ro, opt, bool, whether the meeting uses the device name as its prefix-->true
    </confDeviceIdPrefix>
    <OEMCode>
      <!--ro, opt, enum, manufacturer OME code, subType:int, desc:1 (standard device), 0 (neutral device)-->1
    </OEMCode>
    <bootTime>
      <!--ro, opt, datetime, system boot time-->1970-01-01T00:00:00+08:00
    </bootTime>
    <intelligentAnalysisEngineModel>
      <!--ro, opt, string, engine model-->test
    </intelligentAnalysisEngineModel>
    <channel>
      <!--ro, opt, int, channel No. of the NVR with the network camera-->1
    </channel>
  </DeviceInfo>
```

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```
  </DeviceInfoList>
</ProxyDevicesInfoResult>
```

16.2.14.29 Set the digital channel configuration file

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/ipcConfig

Query Parameter

None

Request Message

Binary Data

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImportIpcCfgError xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <existError>
    <!--ro, opt, bool-->true
  </existError>
  <errorCode>
    <!--ro, opt, enum, subType:string-->chanNumReachLimit
  </errorCode>
  <errorRowNo>
    <!--ro, opt, string, range:[1,2]-->4
  </errorRowNo>
  <errorColNo>
    <!--ro, opt, string, range:[1,12]-->4
  </errorColNo>
  <IpcErrorList>
    <!--ro, opt, object-->
    <IpcError>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, int-->1
      </id>
      <errorRowNo>
        <!--ro, opt, string-->4
      </errorRowNo>
      <errorType>
        <!--ro, opt, enum, subType:string-->adminPortError
      </errorType>
    </IpcError>
  </IpcErrorList>
</ImportIpcCfgError>
```

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16.2.14.30 Get the digital channel configuration file

Request URL

GET /ISAPI/ContentMgmt/InputProxy/ipcConfig

Query Parameter

None

Request Message

None

Response Message

Binary Data

16.2.14.31 Search for accessible network cameras on LAN (Local Area Network)

Request URL

GET /ISAPI/ContentMgmt/InputProxy/search?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	Device short serial No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoSourceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, accessible network cameras on LAN, subType:object, attr:version{req, string, protocolVersion}-->
  <VideoSourceDescriptor>
    <!--ro, opt, object, video source description-->
    <id>
      <!--ro, req, string, index No.-->test
    </id>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:; “hostname” (domain name), “ipaddress” (IP address)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host server name, range:[0,128]-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[0,32]-->test
    </ipAddress>
    <subnetMask>
      <!--ro, opt, string, subnet mask for IPv4 address, range:[0,32]-->test
    </subnetMask>
    <gateway>
      <!--ro, opt, string, IPv4 address of gateway, range:[0,32]-->test
    </gateway>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[0,128]-->test
    </ipv6Address>
    <bitMask>
      <!--ro, opt, string, subnet mask for IPv6 address, range:[0,128]-->test
    </bitMask>
    <serialNumber>
      <!--ro, opt, string, serial No., range:[0,64]-->test
    </serialNumber>
    <macAddress>
      <!--ro, opt, string, MAC address, range:[0,48]-->test
    </macAddress>
    <firmwareVersion>
      <!--ro, req, string, firmware version, range:[0,64]-->test
    </firmwareVersion>
    <managePortNo>
      <!--ro, opt, int, management port No., range:[0,65535]-->1
    </managePortNo>
    <userName>
      <!--ro, opt, string, user name, range:[0,32]-->test
    </userName>
    <password>
      <!--ro, opt, string, password, range:[0,16], desc:this is a default password from NVR, it may not be the actual password scenario: the device will use
the default password to batch add network cameras; if the default password is not the same as the camera, you can manual edit the camera password
thereafter-->test
    </password>
    <srcInputPortNums>
      <!--ro, req, int, the number of input ports-->1
    </srcInputPortNums>
    <deviceID>
      <!--ro, opt, string, device No., range:[1,64], desc:device No.-->test
    </deviceID>
    <deviceModel>
      <!--ro, opt, string, device model, range:[0,64]-->test
    </deviceModel>
    <softwareVersion>
      <!--ro, opt, string, software version, range:[0,64]-->test
    </softwareVersion>
    <activated>
      <!--ro, opt, bool, device activation status-->true
    </activated>
    <registerStatus>
      <!--ro, opt, enum, ISUP registration status, subType:string, desc:; “unknown” (unknown”; “unregister” (not registered); “registered” (registered)--
>unknown
    </registerStatus>
    <isSupportGB35114>
      <!--ro, opt, bool,   -->true
    </isSupportGB35114>
    <readerID>
      <!--ro, opt, int, card reader No., range:[1,8], desc:supports being searched over SADP-->1
    </readerID>
    <POEID>
      <!--ro, opt, int, PoE port No., range:[1,8], step:1, desc:the No. of PoE port connected with the device-->1
    </POEID>
    <isIPConflict>
      <!--ro, opt, bool-->true
    </isIPConflict>
    <personnelChannelInfo>
      <!--ro, opt, object, dep:or,{$.VideoSourceList[*].VideoSourceDescriptor.deviceType,eq,PersonnelChannel}-->
      <isMainPermissionController>
        <!--ro, req, bool-->true
      </isMainPermissionController>
      <subPermissionControllerSerialNumber>
```

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```
      <subPermissionControllerSerialNumber>
        <!--ro, opt, string, range:[9,9], dep:or,{$.VideoSourceList[*].VideoSourceDescriptor.personnelChannelInfo.isMainPermissionController,eq,true}-->test
      </subPermissionControllerSerialNumber>
    </personnelChannelInfo>
    <deviceName>
      <!--ro, opt, string, range:[0,64]-->test
    </deviceName>
    <addStatus>
      <!--ro, opt, enum, subType:string-->added
    </addStatus>
  </VideoSourceDescriptor>
</VideoSourceList>
```

16.2.14.32 Get the capability of accessing protocol

Request URL

POST /ISAPI/ContentMgmt/InputProxy/sourceCapability

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<sourceDescriptor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address-->test
  </ipv6Address>
  <adminPortNo>
    <!--req, int-->1
  </adminPortNo>
  <managePortNo>
    <!--req, int, port number-->1
  </managePortNo>
  <userName>
    <!--req, string, user name-->test
  </userName>
  <password>
    <!--req, string, password-->test
  </password>
  <isDoubleAdd>
    <!--opt, bool-->true
  </isDoubleAdd>
</sourceDescriptor>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<sourceCapability xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <videoInputNums>
    <!--ro, req, int-->1
  </videoInputNums>
  <audioInputNums>
    <!--ro, opt, int-->1
  </audioInputNums>
  <channelInfoList>
    <!--ro, opt, array, subType:object-->
    <channelInfo>
      <!--ro, opt, object-->
      <channelID>
        <!--ro, opt, int, range:[1,256]-->1
      </channelID>
      <channelName>
        <!--ro, opt, string, range:[1,128]-->test
      </channelName>
      <status>
        <!--ro, opt, enum, subType:string-->online
      </status>
    </channelInfo>
  </channelInfoList>
</sourceCapability>
```

16.2.14.33 Set the network camera's password to its activation password

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/syncIPCActivatePasswd?loginPassword=<loginPassword>

Query Parameter

Parameter Name	Parameter Type	Description
loginPassword	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
	[Message content]	application/json	--	--	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.14.34 Synchronize the device password to the digital channel password

Request URL

PUT /ISAPI/ContentMgmt/InputProxy/syncIPCPasswd

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.2.14.35 Get PTZ control capability of digital channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZCtrlCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, PTZ control capability of digital channel, attr:version{req, string, protocolVersion}-->
  <maxPTZChannleNum>
    <!--ro, opt, int, maximum number of PTZ channels-->0
  </maxPTZChannleNum>
  <isSupportPTZWithPreview>
    <!--ro, opt, bool, whether the device supports live view, desc:;;;-->true
  </isSupportPTZWithPreview>
</PTZCtrlCap>
```

16.2.14.36 Get parameters of all digital PTZ channels

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZChannel>
    <!--ro, req, object-->
    <id>
      <!--ro, req, int, ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <serialNumber>
      <!--ro, req, int, serial port No. of channel, desc:if the value is 0, it indicates that no serial port is linked-->1
    </serialNumber>
    <videoInputID>
      <!--ro, req, int-->1
    </videoInputID>
    <panMaxSpeed>
      <!--ro, opt, int, panning speed-->1
    </panMaxSpeed>
    <tiltMaxSpeed>
      <!--ro, opt, int, tilting speed-->1
    </tiltMaxSpeed>
    <presetSpeed>
      <!--ro, opt, int, preset speed, range:[1,8]-->1
    </presetSpeed>
    <autoPatrolSpeed>
      <!--ro, opt, int, patrol speed, range:[0,100]-->1
    </autoPatrolSpeed>
    <keyBoardControlSpeed>
      <!--ro, opt, int, PTZ control speed, range:[0,100]-->1
    </keyBoardControlSpeed>
    <controlAddress>
      <!--ro, req, object-->
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <Address>
        <!--ro, opt, string, address, range:[1,255]-->1
      </Address>
    </controlAddress>
    <defaultPresetID>
      <!--ro, opt, string, default preset No.-->test
    </defaultPresetID>
    <PTZRs485Para>
      <!--ro, req, object-->
      <baudRate>
        <!--ro, req, int, baud rate-->9600
      </baudRate>
      <dataBits>
        <!--ro, req, int, data bit-->1
      </dataBits>
      <parityType>
        <!--ro, req, enum, parity type, subType:string, desc:“none”, “even”, “odd”, “mark”, “space”-->space
      </parityType>
      <stopBits>
        <!--ro, req, enum, stop bit, subType:string, desc:“1”, “1.5”, “2”-->1
      </stopBits>
      <flowCtrl>
        <!--ro, req, enum, encoding or decoding type, subType:string, desc:“none”, “software”, “hardware”-->hardware
      </flowCtrl>
    </PTZRs485Para>
    <manualControlSpeed>
      <!--ro, opt, enum, manual control speed type, subType:string, desc:“pedestrian”, “nonMotorVehicle”, “motorVehicle”, “selfadaptive”, “compatible”--
>pedestrian
    </manualControlSpeed>
  </PTZChannel>
</PTZChannelList>
```

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16.2.14.37 Set parameters of all digital PTZ channels

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, PTZ channels' configuration list, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZChannel>
    <!--req, object, PTZ channel configuration-->
    <id>
      <!--req, int, ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <serialNumber>
      <!--req, int, serial port No., desc:this field should be supported by both GET and PUT methods. If the value is 0, it indicates that no serial port is
linked-->1
    </serialNumber>
    <videoInputID>
      <!--req, int, video input-->1
    </videoInputID>
    <panMaxSpeed>
      <!--opt, int, panning speed, unit: degree/second-->1
    </panMaxSpeed>
    <tiltMaxSpeed>
      <!--opt, int, tilting speed, unit: degree/second-->1
    </tiltMaxSpeed>
    <presetSpeed>
      <!--opt, int, preset speed, range:[1,8]-->1
    </presetSpeed>
    <autoPatrolSpeed>
      <!--opt, int, auto patrol speed, range:[0,100]-->1
    </autoPatrolSpeed>
    <keyBoardControlSpeed>
      <!--opt, int, keypad control speed, range:[0,100]-->1
    </keyBoardControlSpeed>
    <controlAddress>
      <!--req, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <Address>
        <!--opt, string, range: [1,255], range:[1,255]-->1
      </Address>
    </controlAddress>
    <defaultPresetID>
      <!--opt, string, default preset No.-->test
    </defaultPresetID>
    <PTZRs485Para>
      <!--req, object-->
      <baudRate>
        <!--req, int-->9600
      </baudRate>
      <dataBits>
        <!--req, int, data bit-->1
      </dataBits>
      <parityType>
        <!--req, enum, parity type, subType:string, desc:"none", "even", "odd", "mark", "space"-->space
      </parityType>
      <stopBits>
        <!--req, enum, stop bit, subType:string, desc:1, 1.5, 2-->1
      </stopBits>
      <flowCtrl>
        <!--req, enum, encoding or decoding type, subType:string, desc:"none", "software", "hardware"-->hardware
      </flowCtrl>
    </PTZRs485Para>
    <manualControlSpeed>
      <!--opt, enum, manual control speed type, subType:string, desc:"pedestrian", "nonMotorVehicle" (non-motor vehicles), "motorVehicle" (vehicle),
"selfadaptive" (self-adaptive), "compatible"-->pedestrian
    </manualControlSpeed>
  </PTZChannel>
</PTZChannelList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.38 Get parameters of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, PTZ channel configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <serialNumber>
    <!--ro, req, int, serial No., desc:serial No.-->1
  </serialNumber>
  <videoInputID>
    <!--ro, req, int, video input-->1
  </videoInputID>
  <panMaxSpeed>
    <!--ro, opt, int, the maximum panning speed-->1
  </panMaxSpeed>
  <tiltMaxSpeed>
    <!--ro, opt, int, the maximum tilting speed-->1
  </tiltMaxSpeed>
  <presetSpeed>
    <!--ro, opt, int, predefined speed, range:[1,8]-->1
  </presetSpeed>
  <autoPatrolSpeed>
    <!--ro, opt, int, patrol speed, range:[0,100]-->1
  </autoPatrolSpeed>
  <keyBoardControlSpeed>
    <!--ro, opt, int, range:[0,100]-->1
  </keyBoardControlSpeed>
  <controlAddress>
    <!--ro, req, object, control address-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <Address>
      <!--ro, opt, string, address, range:[1,255]-->1
    </Address>
  </controlAddress>
  <defaultPresetID>
    <!--ro, opt, string, the default ID-->test
  </defaultPresetID>
  <PTZRs485Para>
    <!--ro, req, object-->
    <baudRate>
      <!--ro, req, int, baud rate-->9600
    </baudRate>
    <dataBits>
      <!--ro, req, int, data bit-->1
    </dataBits>
    <parityType>
      <!--ro, req, enum, verification type, subType:string, desc:verification type-->space
    </parityType>
    <stopBits>
      <!--ro, req, enum, stop bit, subType:string, desc:stop bit-->1
    </stopBits>
    <flowCtrl>
      <!--ro, req, enum, "none, software, hardware", subType:string, desc:"none, software, hardware"-->hardware
    </flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--ro, opt, enum, "pedestrian, nonMotorVehicle, motorVehicle, selfadaptive, compatible", subType:string, desc:"pedestrian, nonMotorVehicle,
motorVehicle, selfadaptive, compatible"-->pedestrian
  </manualControlSpeed>
  <panSupport>
    <!--ro, opt, bool, whether it supports panning-->true
  </panSupport>
  <tiltSupport>
    <!--ro, opt, bool, whether it supports tilting-->true
  </tiltSupport>
  <zoomSupport>
    <!--ro, opt, bool, whether it supports zooming-->true
  </zoomSupport>
</PTZChannel>
```

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16.2.14.39 Set the PTZ parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int-->1
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <serialNumber>
    <!--req, int-->1
  </serialNumber>
  <videoInputID>
    <!--req, int-->1
  </videoInputID>
  <panMaxSpeed>
    <!--opt, int-->1
  </panMaxSpeed>
  <tiltMaxSpeed>
    <!--opt, int-->1
  </tiltMaxSpeed>
  <presetSpeed>
    <!--opt, int, range:[1,8]-->1
  </presetSpeed>
  <autoPatrolSpeed>
    <!--opt, int, range:[0,100]-->1
  </autoPatrolSpeed>
  <keyBoardControlSpeed>
    <!--opt, int, range:[0,100]-->1
  </keyBoardControlSpeed>
  <controlAddress>
    <!--req, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <Address>
      <!--opt, string, range:[1,255]-->1
    </Address>
  </controlAddress>
  <defaultPresetID>
    <!--opt, string-->test
  </defaultPresetID>
  <PTZRs485Para>
    <!--req, object-->
    <baudRate>
      <!--req, int-->9600
    </baudRate>
    <dataBits>
      <!--req, int-->1
    </dataBits>
    <parityType>
      <!--req, enum, subType:string-->space
    </parityType>
    <stopBits>
      <!--req, enum, subType:string-->1
    </stopBits>
    <flowCtrl>
      <!--req, enum, subType:string-->hardware
    </flowCtrl>
  </PTZRs485Para>
  <manualControlSpeed>
    <!--opt, enum, subType:string-->pedestrian
  </manualControlSpeed>
  <panSupport>
    <!--opt, bool-->true
  </panSupport>
  <tiltSupport>
    <!--opt, bool-->true
  </tiltSupport>
  <zoomSupport>
    <!--opt, bool-->true
  </zoomSupport>
</PTZChannel>
```

drackum@gmail.com
DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.40 Get the PTZ capability of a specified digital channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZChannelCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id min="1" max="16">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </id>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <controlAddress>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <Address>
      <!--ro, opt, object-->
    </Address>
  </controlAddress>
  <isSupportPosition3D>
    <!--ro, opt, bool-->true
  </isSupportPosition3D>
</PTZChannelCap>
```

16.2.14.41 Manually control PTZ

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/continuous

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <pan>
    <!--opt, int, pan range, range:[-100,100]-->60
  </pan>
  <tilt>
    <!--opt, int, tilt range, range:[-100,100]-->60
  </tilt>
  <zoom>
    <!--opt, int, zoom range, range:[-100,100]-->60
  </zoom>
</PTZData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

drackum@gmail.com DETEKTOR

16.2.14.42 Perform scheduled PTZ control via digital channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/momentary

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <pan>
    <!--opt, int, pan range, range:[-100,100]-->1
  </pan>
  <tilt>
    <!--opt, int, tilt range, range:[-100,100]-->1
  </tilt>
  <zoom>
    <!--opt, int, zoom range, range:[-100,100]-->1
  </zoom>
  <Momentary>
    <!--opt, object-->
    <duration>
      <!--opt, int, unit: millisecond, unit:ms-->1
    </duration>
  </Momentary>
</PTZData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.43 Perform one-touch focus via a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/onepushfoucs/start

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.14.44 Set all patrols of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--ro, opt, object, patrol parameters-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <patrolName>
      <!--ro, req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--ro, req, array, subType:object-->
      <PatrolSequence>
        <!--ro, req, object, list-->
        <presetID>
          <!--ro, req, int, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--ro, req, int, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--ro, req, int, delay time, unit:ms-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

16.2.14.45 Add a patrol of a digital channel

Request URL

POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type, subType:string, desc:“relative”, “absolute”-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, subType:object-->
    <PatrolSequence>
      <!--req, object-->
      <presetID>
        <!--req, string, preset ID-->test
      </presetID>
      <seqSpeed>
        <!--req, string, patrol speed-->test
      </seqSpeed>
      <delay>
        <!--req, int, delay time, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

Response Message

drackum@gmail.com DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.46 Delete all patrols of a digital PTZ channel

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.14.47 Set all patrols' parameters of a specified digital channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <id>
      <!--req, string-->test
    </id>
    <patrolName>
      <!--req, string-->test
    </patrolName>
    <resumeType>
      <!--opt, enum, subType:string-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--req, array, subType:object-->
      <PatrolSequence>
        <!--req, object-->
        <presetID>
          <!--req, int-->1
        </presetID>
        <seqSpeed>
          <!--req, int-->1
        </seqSpeed>
        <delay>
          <!--req, int, unit:ms-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.2.14.48 Get the patrol parameters of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <patrolName>
    <!--ro, req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--ro, req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--ro, req, object, patrol result-->
      <presetID>
        <!--ro, req, int, preset ID in the patrol-->1
      </presetID>
      <seqSpeed>
        <!--ro, req, int, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--ro, req, int, delay time, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

16.2.14.49 Set a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <id>
    <!--req, string, ID-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type: relative,absolute, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, subType:object-->
    <PatrolSequence>
      <!--req, object, list-->
      <presetID>
        <!--req, int, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--req, int, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--req, int, delay time, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.50 Delete a patrol of a digital PTZ channel

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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DETEKTOR

16.2.14.51 Pause calling a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/pause

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.52 Start calling a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/start

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.53 Search a patrol status of a digital channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/status

Query Parameter

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DETEKTOR

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPatrolStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, patrol status, attr:version{req, string, protocolVersion}-->
  <patrolID>
    <!--ro, req, string, patrol ID-->test
  </patrolID>
  <patrolStatus>
    <!--ro, req, enum, patrol status, subType:string, desc:"running", "stopped", "paused"-->running
  </patrolStatus>
</PTZPatrolStatus>
```

16.2.14.54 Stop calling a patrol of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/patrols/<patrolID>/stop

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.55 Perform 3D positioning

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/position3D

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<Position3D xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <StartPoint>
    <!--opt, object, start point, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, x-coordinate, range:[0,255]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,255]-->1
    </positionY>
  </StartPoint>
  <EndPoint>
    <!--opt, object, end point, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, x-coordinate, range:[0,255]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,255]-->1
    </positionY>
  </EndPoint>
</Position3D>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

drackun@gmail.com
DETEKTOR

16.2.14.56 Set presets of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPresetList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in XML_PTZPreset, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPreset>
    <!--opt, object, PTZ preset-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <id>
      <!--req, string-->test
    </id>
    <presetName>
      <!--req, string, preset name-->test
    </presetName>
  </PTZPreset>
</PTZPresetList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.57 Get parameters of all presets of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPresetList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, presets list, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPreset>
    <!--ro, opt, object, PTZ preset-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <presetName>
      <!--ro, req, string, preset name-->test
    </presetName>
  </PTZPreset>
</PTZPresetList>
```

dracknum@gmail.com
DETEKTOR

16.2.14.58 Delete presets of a digital PTZ channel.

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.14.59 Add a preset to a digital PTZ channel

Request URL

POST /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, PTZ preset, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, listening host ID-->test
  </id>
  <presetName>
    <!--req, string, preset name-->test
  </presetName>
</PTZPreset>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.60 Delete a preset of a digital PTZ channel

drackum@gmail.com DETEKTOR

Request URL

DELETE /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.61 Set a preset of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, PTZ preset, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable preset configuration-->true
  </enabled>
  <id>
    <!--req, string, preset No.-->test
  </id>
  <presetName>
    <!--req, string, preset name-->test
  </presetName>
</PTZPreset>
```

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.62 Get parameters of a preset of a digital PTZ channel

Request URL

GET /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PTZPreset xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, PTZ preset, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <presetName>
    <!--ro, req, string, preset name-->test
  </presetName>
</PTZPreset>
```

drackum@gmail.com
DETEKTOR

16.2.14.63 Call a preset of a digital PTZ channel

Request URL

PUT /ISAPI/ContentMgmt/PTZCtrlProxy/channels/<channelID>/presets/<presetID>/goto

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
presetID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.14.64 Get access protocols supported by the device

Request URL

GET /ISAPI/ContentMgmt/sourceSupport?send=<send>&mfgr=<mfgr>

Query Parameter

Parameter Name	Parameter Type	Description
send	string	Value range: "head", "top", "middle", "bottom", when set to "head". The device only returns the total number of supported protocols, corresponding to the mediaDeviceListSize field; when set to "top", "middle" or "bottom", the device returns the first 1/3, middle 1/3, and last 1/3 of the supported protocol information respectively.
mfgr	string	Search for specified types of protocols and it supports searching for multiple types simultaneously (GET /ISAPI/ContentMgmt/souceSupport?mfgr=HIKVISION&mfgr=ONVIF). Value range: "HIKVISION", "ONVIF", "GB28181", "ClearView", "ISAPI".

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSourceSupport xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <mediaDeviceListSize>
    <!--ro, req, int-->1
  </mediaDeviceListSize>
  <supportedMediaDeviceSourceList>
    <!--ro, opt, array, subType:object-->
    <MediaDeviceSource>
      <!--ro, opt, object-->
      <MediaDeviceMfgr>
        <!--ro, req, enum, subType:string-->
      </MediaDeviceMfgr>
      <MediaDeviceMgmtMode>
        <!--ro, req, string-->test
      </MediaDeviceMgmtMode>
      <MediaDeviceModel>
        <!--ro, req, string-->test
      </MediaDeviceModel>
    </MediaDeviceSource>
  </supportedMediaDeviceSourceList>
</CMSourceSupport>
```

16.2.15 Video Tampering

16.2.15.1 Delete all video tampering detection areas of a specified digital channel

Request URL

DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection/regions

dracl_un@gmail.com DETEKTOR

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.15.2 Delete a specified area of tampering detection of a digital channel

Request URL

DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/video/tamperDetection/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

drackun@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.2.15.3 Get the linkage parameters of video tampering detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->tamper-1
  </id>
  <eventType>
    <!--ro, req, string-->tamper
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--ro, opt, int, unit:s-->0
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--ro, opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <modbus>
        <!--ro, opt, object-->
        <outputPortID>
          <!--ro, opt, int, range:[0,15]-->1
        </outputPortID>
        <delayTime>
          <!--ro, opt, int, range:[1,10], unit:s-->1
        </delayTime>
        <alarmStatus>
          <!--ro, opt, enum, subType:string-->open
        </alarmStatus>
      </modbus>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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DETEKTOR

16.2.15.4 Delete the linkage of video tampering detection for a specified channel

Request URL

DELETE /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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DETEKTOR

16.2.15.5 Set the linkage parameters of video tampering detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/tamper-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->tamper-1
  </id>
  <eventType>
    <!--req, string-->tamper
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->test
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->0
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <modbus>
        <!--opt, object-->
        <outputPortID>
          <!--opt, int, range:[0,15]-->1
        </outputPortID>
        <delayTime>
          <!--opt, int, range:[1,10], unit:s-->1
        </delayTime>
        <alarmStatus>
          <!--opt, enum, subType:string-->open
        </alarmStatus>
      </modbus>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.15.6 Delete the linkage triggering parameters of video tampering detection for a specified channel

Request URL

DELETE /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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16.2.15.7 Set the linkage triggering parameters of video tampering detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <notificationMethod>
    <!--req, enum, subType:string-->FTP
  </notificationMethod>
  <notificationRecurrence>
    <!--opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--opt, string-->test
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--opt, string-->test
  </dynOutputIOPortID>
  <videoInputID>
    <!--opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--opt, object-->
    <ptzChannelID>
      <!--req, string-->1
    </ptzChannelID>
    <actionName>
      <!--req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--opt, int, unit:s-->0
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--opt, string, range:[0,11]-->test
  </cellphoneNumber>
  <modbus>
    <!--opt, object-->
    <outputPortID>
      <!--opt, int, range:[0,15]-->1
    </outputPortID>
    <delayTime>
      <!--opt, int, range:[1,10], unit:s-->1
    </delayTime>
    <alarmStatus>
      <!--opt, enum, subType:string-->open
    </alarmStatus>
  </modbus>
</EventTriggerNotification>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.15.8 Get the linkage triggering parameters of video tampering detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/tamper-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <notificationMethod>
    <!--ro, req, enum, subType:string-->FTP
  </notificationMethod>
  <notificationRecurrence>
    <!--ro, opt, enum, subType:string-->beginning
  </notificationRecurrence>
  <notificationInterval>
    <!--ro, opt, int, unit:ms-->1
  </notificationInterval>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <dynOutputIOPortID>
    <!--ro, opt, string-->test
  </dynOutputIOPortID>
  <videoInputID>
    <!--ro, opt, string-->1
  </videoInputID>
  <dynVideoInputID>
    <!--ro, opt, string-->1
  </dynVideoInputID>
  <ptzAction>
    <!--ro, opt, object-->
    <ptzChannelID>
      <!--ro, req, string-->1
    </ptzChannelID>
    <actionName>
      <!--ro, req, enum, subType:string-->preset
    </actionName>
    <actionNum>
      <!--ro, opt, int-->1
    </actionNum>
    <presetDurationTime>
      <!--ro, opt, int, unit:s-->0
    </presetDurationTime>
  </ptzAction>
  <cellphoneNumber>
    <!--ro, opt, string, range:[0,11]-->test
  </cellphoneNumber>
  <modbus>
    <!--ro, opt, object-->
    <outputPortID>
      <!--ro, opt, int, range:[0,15]-->1
    </outputPortID>
    <delayTime>
      <!--ro, opt, int, range:[1,10], unit:s-->1
    </delayTime>
    <alarmStatus>
      <!--ro, opt, enum, subType:string-->open
    </alarmStatus>
  </modbus>
</EventTriggerNotification>
```

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16.2.15.9 Video tampering alarm

EventType:shelteralarm

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event details, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 Address-->10.10.10.10
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->ABCD:EF01:2345:6789:ABCD:EF01:2345:6789
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:"HTTP, HTTPS, EHome". The value should be "HTTP" when ISAPI protocol
is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP.-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type: shelteralarm, subType:string, desc:event type: shelteralarm-->shelteralarm
  </eventType>
  <eventState>
    <!--ro, req, enum, event status (for persistent event), subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID-->test
  </deviceID>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, req, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, req, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <Image>
    <!--ro, opt, object, event related pictures-->
    <resourcesContentType>
      <!--ro, opt, enum, resource transmission type, subType:string, desc:"url" (transmitted via URL), "binary" (transmitted in binary format). When the
returned alarm message contains additional resource (e.g., pictures), this field should be returned.-->binary
    </resourcesContentType>
    <resourcesContent>
      <!--ro, opt, string, resource ID, range:[1,128], desc:when the returned alarm message contains additional resources (e.g., pictures), this field
should be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of
resourcesContentType is "url", its value is the resource URL-->image
    </resourcesContent>
  </Image>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
EventNotificationAlert	[Message content]	application/xml	--	--	--
image	[Binary picture data]	image/jpeg	image	image.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

16.2.16 White Balance

16.2.16.1 Set the white balance parameters of s specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/whiteBalance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--opt, enum, white balance type, subType:string, desc:"auto", "atuo1", "auto2" "manual", "indoor", "outdoor", "autotrace", "onece", "sodiumlight",
"mercurylight"-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceBlue>
</WhiteBalance>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.16.2 Get the white balance (in day mode) parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/whiteBalance/day

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--ro, opt, enum, subType:string-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--ro, req, int-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--ro, req, int-->1
  </WhiteBalanceBlue>
  <whiteBalanceLevel>
    <!--ro, opt, int-->1
  </whiteBalanceLevel>
  <customColorTemperatureLevel>
    <!--ro, opt, int, range:[2700,7500], dep:and,{$.WhiteBalance,eq,customColorTemperature}-->1
  </customColorTemperatureLevel>
</WhiteBalance>
```

16.2.16.3 Set the white balance (in day mode) parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/whiteBalance/day

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--opt, enum, subType:string-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--req, int-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--req, int-->1
  </WhiteBalanceBlue>
  <customColorTemperatureLevel>
    <!--opt, int, range:[2700,7500], dep:and,{$.WhiteBalance,eq,customColorTemperature}-->1
  </customColorTemperatureLevel>
</WhiteBalance>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.2.16.4 Get the white balance parameters of s specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/whiteBalance?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--ro, opt, enum, "auto,manual,indoor,outdoor,autotrace,onece,sodiumlight,mercurylight,auto0,atuo1,fluorescent,natural,warm,incandescent",
subType:string, desc:"auto", "auto1", "atuo2", "manual", "indoor", "outdoor", "autotrace", "onece", "sodiumlight", "mercurylight"-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceBlue>
  <whiteBalanceLevel>
    <!--ro, opt, int, range:[0,100], dep:or,{$.WhiteBalance.WhiteBalanceStyle,eq,auto},{$.WhiteBalance.WhiteBalanceStyle,eq,auto1},
{$.WhiteBalance.WhiteBalanceStyle,eq,auto2}-->1
  </whiteBalanceLevel>
  <customColorTemperatureLevel>
    <!--ro, opt, int, range:[1800,10000]-->1
  </customColorTemperatureLevel>
  <WhiteBalanceRedFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceRedFineTuning>
  <WhiteBalanceBlueFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceBlueFineTuning>
  <autoMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,auto}-->video
  </autoMode>
  <manualMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,manual}-->colorFineTuning
  </manualMode>
</WhiteBalance>
```

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16.3 Video Management

16.3.1 Cloud Storage Management

16.3.1.1 Get the cloud storage access capability

Request URL

GET /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/<indexID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.
indexID	string	Storage server index, which starts from 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, cloud storage service parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index-->test
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, address types, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port min="1" max="65535">
    <!--ro, req, int, port No., range:[1,65535], attr:min{req, int},max{req, int}-->1
  </port>
  <userName min="1" max="128">
    <!--ro, opt, string, range:[1,128], attr:min{req, int},max{req, int}-->test
  </userName>
  <password min="1" max="32">
    <!--ro, opt, string, password, attr:min{req, int},max{req, int}, desc:password-->null
  </password>
  <postPoolID min="0" max="4294967295">
    <!--ro, opt, int, checkpoint picture pool ID, attr:min{req, int},max{req, int}-->1
  </postPoolID>
  <illegalPoolID min="0" max="4294967295">
    <!--ro, opt, int, violation picture pool ID, attr:min{req, int},max{req, int}-->1
  </illegalPoolID>
  <vehicleDetectionID min="0" max="4294967295">
    <!--ro, opt, int, ID of vehicle detection picture pool, attr:min{req, int},max{req, int}-->1
  </vehicleDetectionID>
  <cloudStorageType opt="0,1,2">
    <!--ro, opt, enum, storage type, subType:int, attr:opt{req, string}, desc:0-cloud storage (default), 1-CVR, 2-pcnvr2.0; it is the cloud storage by
default if this node is empty-->1
  </cloudStorageType>
  <V2.0>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address types, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port min="0" max="65535">
      <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <isSupportAccessKey>
```

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```
      <!--ro, opt, bool, whether it supports access key configuration-->true
    </isSupportAccessKey>
    <isSupportSecretKey>
      <!--ro, opt, bool, whether it supports secret key configuration-->true
    </isSupportSecretKey>
    <postPoolID min="0" max="4294967295">
      <!--ro, opt, int, checkpoint picture pool ID, attr:min{req, int},max{req, int}, desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID min="0" max="4294967295">
      <!--ro, opt, int, violation picture pool ID, attr:min{req, int},max{req, int}-->1
    </illegalPoolID>
    <vehicleDetectionID min="0" max="4294967295">
      <!--ro, opt, int, ID of vehicle detection picture pool, attr:min{req, int},max{req, int}-->1
    </vehicleDetectionID>
    <accessKeyLen min="1" max="64">
      <!--ro, opt, int, access key length, dep:and,{$.CloudStorage.V2.0.isSupportAccessKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </accessKeyLen>
    <secretKeyLen min="1" max="64">
      <!--ro, opt, int, secret key length, dep:and,{$.CloudStorage.V2.0.isSupportSecretKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </secretKeyLen>
    <resourceID min="0" max="32">
      <!--ro, opt, string, resource pool ID, attr:min{req, int},max{req, int}, desc:resource pool ID-->test
    </resourceID>
    <transportProtocolType opt="HTTP,HTTPS" def="HTTPS">
      <!--ro, opt, enum, "HTTP", "HTTPS", subType:string, attr:opt{req, string},def{req, string}, desc:"HTTP", "HTTPS"-->HTTP
    </transportProtocolType>
  </V2.0>
  <ImageAndVideoCloud>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, address types, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port min="0" max="65535">
      <!--ro, req, int, port No., attr:min{req, int},max{req, int}-->1
    </port>
    <isSupportAccessKey>
      <!--ro, opt, bool, whether it supports access key configuration-->true
    </isSupportAccessKey>
    <isSupportSecretKey>
      <!--ro, opt, bool, whether it supports secret key configuration-->true
    </isSupportSecretKey>
    <postPoolID min="0" max="4294967295">
      <!--ro, opt, int, checkpoint picture pool ID, attr:min{req, int},max{req, int}, desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID min="0" max="4294967295">
      <!--ro, opt, int, violation picture pool ID, attr:min{req, int},max{req, int}-->1
    </illegalPoolID>
    <vehicleDetectionID min="0" max="4294967295">
      <!--ro, opt, int, ID of vehicle detection picture pool, attr:min{req, int},max{req, int}-->1
    </vehicleDetectionID>
    <accessKeyLen min="1" max="64">
      <!--ro, opt, int, access key length, dep:and,{$.CloudStorage.ImageAndVideoCloud.isSupportAccessKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </accessKeyLen>
    <secretKeyLen min="1" max="64">
      <!--ro, opt, int, secret key length, dep:and,{$.CloudStorage.ImageAndVideoCloud.isSupportSecretKey,eq,true}, attr:min{req, int},max{req, int}-->1
    </secretKeyLen>
  </ImageAndVideoCloud>
  <protocolType opt="V1.0,V2.0,Image and Video Cloud,objectStorage,videoStreamingOS">
    <!--ro, opt, enum, cloud storage protocol type, subType:string, attr:opt{req, string}, desc:"V1.0", "V2.0", "Image and Video Cloud"-->V1.0
  </protocolType>
  <isSupportCloudStorageServerParam>
    <!--ro, opt, bool, whether it supports getting the parameters of cloud storage server, desc:whether it supports getting the parameters of cloud storage
server-->true
  </isSupportCloudStorageServerParam>
  <bucketName>
    <!--ro, opt, string, bucket name of cloud storage-->test
  </bucketName>
  <associateToCenter>
    <!--ro, opt, object-->
    <centerType opt="all,ISUP,OTAP">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->ISUP
    </centerType>
    <centerID min="1" max="4">
      <!--ro, req, int, range:[1,4], attr:min{req, int},max{req, int}-->0
    </centerID>
  </associateToCenter>
</CloudStorage>
```

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16.3.1.2 Get the access parameters of specified cloud storage

Request URL

GET /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.
indexID	string	Storage server index, which starts from 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, cloud storage service, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index-->test
  </id>
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port>
    <!--ro, req, int, port No., range:[1,65535]-->1
  </port>
  <userName>
    <!--ro, req, string, range:[1,64]-->test
  </userName>
  <postPoolID>
    <!--ro, opt, int, checkpoint picture pool ID, range:[0,4294967295]-->1
  </postPoolID>
  <illegalPoolID>
    <!--ro, opt, int, violation picture pool ID, range:[0,4294967295]-->1
  </illegalPoolID>
  <vehicleDetectionID>
    <!--ro, opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
  </vehicleDetectionID>
  <protocolType>
    <!--ro, opt, enum, cloud storage protocol type, subType:string, desc:"V1.0", "V2.0", "Image and Video Cloud"-->V1.0
  </protocolType>
  <cloudStorageType>
    <!--ro, opt, enum, 0-cloud storage (default),1-CVR,2-pcnvr2.0; if this node is empty,it is the cloud storage by default, subType:int, desc:0-cloud
storage (default), 1-CVR, -pcnvr2.0; if this node is empty, it is the cloud storage by default-->0
  </cloudStorageType>
  <V2.0>
    <!--ro, opt, object, this node in valid only when the vaule of protocolType is V2.0, dep:and,{$.CloudStorage.protocolType,eq,V2.0}, desc:this node in
valid only when the vaule of protocolType is V2.0-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType>
      <!--ro, req, enum, address type: "ipaddress","hostname", subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--ro, req, int, port No., range:[1,65535]-->1
    </port>
    <postPoolID>
      <!--ro, opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
```

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```
    <illegalPoolID>
      <!--ro, opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--ro, opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <resourceID>
      <!--ro, opt, string, resource pool ID, range:[1,32], desc:resource pool ID-->test
    </resourceID>
    <transportProtocolType>
      <!--ro, opt, enum, "HTTP", "HTTPS", subType:string, desc:"HTTP", "HTTPS"-->HTTP
    </transportProtocolType>
  </V2.0>
  <ImageAndVideoCloud>
    <!--ro, opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--ro, req, int, port No., range:[1,65535]-->1
    </port>
    <postPoolID>
      <!--ro, opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--ro, opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--ro, opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
    </vehicleDetectionID>
  </ImageAndVideoCloud>
  <bucketName>
    <!--ro, opt, string, bucket name of cloud storage, desc:bucket name of cloud storage-->test
  </bucketName>
  <status>
    <!--ro, opt, enum, cloud storage status, subType:string, desc:only supports getting this node. "ok" (normal), "connectServerFail" (connecting to server
failed), "passwordError" (incorrect password), "exchangeServerFail", "createDirectoryFailed", "noWritePermission" (no writing permission), "noBucket"
(bucket does not exist)-->ok
  </status>
  <associateToCenter>
    <!--ro, opt, object-->
    <centerType>
      <!--ro, req, enum, subType:string-->ISUP
    </centerType>
    <centerID>
      <!--ro, req, int, range:[1,4]-->0
    </centerID>
  </associateToCenter>
</CloudStorage>
```

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16.3.1.3 Set the access parameters of specified cloud storage

Request URL

PUT /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
indexID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, cloud storage service, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index-->test
  </id>
  <enabled>
    <!--req, bool-->true
```

```
    <!--req, bool-->true
  </enabled>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:ipaddress: IP address; hostname: host server name;-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host server name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <port>
    <!--req, int, port No., range:[1,65535]-->1
  </port>
  <userName>
    <!--req, string, range:[1,64]-->test
  </userName>
  <password>
    <!--opt, string, password, range:[1,32]-->test
  </password>
  <postPoolID>
    <!--opt, int, checkpoint picture pool ID, range:[0,4294967295]-->1
  </postPoolID>
  <illegalPoolID>
    <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
  </illegalPoolID>
  <vehicleDetectionID>
    <!--opt, int, vehicle detection picture pool ID, range:[0,4294967295]-->1
  </vehicleDetectionID>
  <protocolType>
    <!--opt, enum, cloud storage protocol type, subType:string, desc:"V1.0"; "V2.0"; "Image and Video Cloud"-->V1.0
  </protocolType>
  <cloudStorageType>
    <!--opt, enum, storage type, subType:int, desc:0-cloud storage (default),1-CVR,2-pcnvr2.0; if this node is empty,it is the cloud storage by default-->1
  </cloudStorageType>
  <V2.0>
    <!--opt, object, this node in valid only when the vaule of protocolType is V2.0, dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:ipaddress: IP address; hostname: host server name;-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, port No., range:[1,65535]-->1
    </port>
    <accessKey>
      <!--opt, string, access key, range:[1,64]-->test
    </accessKey>
    <secretKey>
      <!--opt, string, secret key, range:[1,64]-->test
    </secretKey>
    <postPoolID>
      <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, vehicle detection picture pool ID, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <resourceID>
      <!--opt, string, capability set of resource pool ID, range:[1,32], desc:resource pool ID-->test
    </resourceID>
    <transportProtocolType>
      <!--opt, enum, cloud storage transfer protocol type, subType:string, desc:HTTP; HTTPS;-->HTTP
    </transportProtocolType>
  </V2.0>
  <ImageAndVideoCloud>
    <!--opt, object, cloud storage protocol of picture and video library, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:ipaddress: IP address; hostname: host server name;-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
```

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```
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, port No., range:[1,65535]-->1
    </port>
    <accessKey>
      <!--opt, string, access key, range:[1,64]-->test
    </accessKey>
    <secretKey>
      <!--opt, string, secret key, range:[1,64]-->test
    </secretKey>
    <postPoolID>
      <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, vehicle detection picture pool ID, range:[0,4294967295]-->1
    </vehicleDetectionID>
  </ImageAndVideoCloud>
  <bucketName>
    <!--opt, string, name of cloud storage bucket, desc:bucket name of cloud storage-->test
  </bucketName>
</CloudStorage>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.3.1.4 Search for test accessing cloud storage for test information

Request URL

POST /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage/test

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageTestDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, test conditions, attr:version{req, string, protocolVersion}-->
  <addressingFormatType>
    <!--req, enum, address types, subType:string, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <portNo>
    <!--req, int, range:[1,65535]-->1
  </portNo>
  <username>
    <!--opt, string, user name, range:[1,64]-->test
```

```
    </username>
    <password>
      <!--opt, string, password, range:[1,32]-->test
    </password>
    <postPoolID>
      <!--opt, int, checkpoint picture pool ID, range:[0,4294967295]-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <protocolType>
      <!--req, enum, cloud storage protocol type, subType:string, desc:cloud storage protocol type: "V1.0","V2.0","Image and Video Cloud"-->V1.0
    </protocolType>
    <cloudStorageType>
      <!--opt, enum, storage server type, subType:int, desc:0-cloud storage (default),1-CVR,2-pcnvr2.0; if this node is empty,it is the cloud storage by
default-->1
    </cloudStorageType>
    <V2.0>
      <!--opt, object, it is valid when <protocolType> is "V2.0", dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--req, int, range:[1,65535]-->1
      </port>
      <accessKey>
        <!--opt, string, password, range:[1,64], desc:password-->test
      </accessKey>
      <secretKey>
        <!--opt, string, secret key, range:[1,64], desc:secret key-->test
      </secretKey>
      <postPoolID>
        <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
      </postPoolID>
      <illegalPoolID>
        <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
      </vehicleDetectionID>
      <resourceID>
        <!--opt, string, both resourceID and cloudPoolID can be used to configure the cloud storage poolID. But different from cloudPoolId, the value of
resourceID can be in string format. When both resourceID and cloudPoolId are configured, cloudPoolId has priority over resourceID. If this node is not
supported, you can configure the pool ID via cloudPoolId, range:[1,32], desc:both resourceID and cloudPoolID can be used to configure the cloud storage
poolID. But different from cloudPoolId, the value of resourceID can be in string format. When both resourceID and cloudPoolId are configured, cloudPoolId
has priority over resourceID. If this node is not supported, you can configure the pool ID via cloudPoolId-->test
      </resourceID>
      <transportProtocolType>
        <!--opt, enum, cloud storage server type, subType:string, desc:cloud storage server type-->HTTP
      </transportProtocolType>
    </V2.0>
    <ImageAndVideoCloud>
      <!--opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--req, int, port No., range:[1,65535]-->1
      </port>
      <accessKey>
        <!--opt, string, password, range:[1,64], desc:password-->test
      </accessKey>
      <secretKey>
        <!--opt, string, secret key, range:[1,64], desc:secret key-->test
      </secretKey>
      <postPoolID>
        <!--opt, int, checkpoint picture pool ID, range:[0,4294967295], desc:checkpoint picture pool ID-->1
      </postPoolID>
      <illegalPoolID>
        <!--opt, int, violation picture pool ID, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--opt, int, ID of vehicle detection picture pool, range:[0,4294967295]-->1
```

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```
    </vehicleDetectionID>
  </ImageAndVideoCloud>
  <bucketName>
    <!--opt, string, bucket name of cloud storage, desc:bucket name of cloud storage-->test
  </bucketName>
</CloudStorageTestDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageTestResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, test result, attr:version{req, string, protocolVersion}-->
  <errorDescription>
    <!--ro, req, enum, error description, subType:string, desc:error description-->ok
  </errorDescription>
</CloudStorageTestResult>
```

16.3.1.5 Get the access parameters of cloud storage in a batch

Request URL

GET /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No. The default value is 1.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, cloud storage service list, subType:object, attr:version{req, string, protocolVersion}-->
  <CloudStorage>
    <!--ro, opt, object, cloud storage service-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--ro, req, int, port No., range:[1,65535]-->1
    </port>
    <userName>
      <!--ro, opt, string, range:[1,64]-->test
    </userName>
    <postPoolID>
      <!--ro, opt, int, range:[0,4294967295]-->1
    </postPoolID>
    <illegalPoolID>
      <!--ro, opt, int, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--ro, opt, int, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <protocolType>
      <!--ro, opt, enum, subType:string-->V1.0
    </protocolType>
    <cloudStorageType>
      <!--ro, opt, enum, storage server type, subType:int, desc:storage server type-->1
    </cloudStorageType>
    <V2.0>
      <!--ro, opt, object, dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
      <enabled>
```

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```
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <addressingFormatType>
        <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--ro, req, int, port No., range:[1,65535]-->1
      </port>
      <postPoolID>
        <!--ro, opt, int, range:[0,4294967295]-->1
      </postPoolID>
      <illegalPoolID>
        <!--ro, opt, int, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--ro, opt, int, range:[0,4294967295]-->1
      </vehicleDetectionID>
      <resourceID>
        <!--ro, opt, string, range:[1,32]-->test
      </resourceID>
      <transportProtocolType>
        <!--ro, opt, enum, subType:string-->HTTP
      </transportProtocolType>
    </V2.0>
    <ImageAndVideoCloud>
      <!--ro, opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <addressingFormatType>
        <!--ro, req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--ro, opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--ro, opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
        <!--ro, opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--ro, req, int, port No., range:[1,65535]-->1
      </port>
      <postPoolID>
        <!--ro, opt, int, range:[0,4294967295]-->1
      </postPoolID>
      <illegalPoolID>
        <!--ro, opt, int, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--ro, opt, int, range:[0,4294967295]-->1
      </vehicleDetectionID>
    </ImageAndVideoCloud>
    <bucketName>
      <!--ro, opt, string-->test
    </bucketName>
    <associateToCenter>
      <!--ro, opt, object-->
      <centerType>
        <!--ro, req, enum, subType:string-->ISUP
      </centerType>
      <centerID>
        <!--ro, req, int, range:[1,4]-->0
      </centerID>
    </associateToCenter>
  </CloudStorage>
</CloudStorageList>
```

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16.3.1.6 Set the access parameters of cloud storage in a batch

Request URL

PUT /ISAPI/ContentMgmt/channels/<channelID>/cloudStorage

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CloudStorageList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, cloud storage service list, subType:object, attr:version{req, string, protocolVersion}-->
  <CloudStorage>
    <!--opt, object, cloud storage service-->
    <id>
      <!--req, string, ID-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
    <port>
      <!--req, int, port No., range:[1,65535]-->1
    </port>
    <userName>
      <!--opt, string, range:[1,64]-->test
    </userName>
    <password>
      <!--opt, string, password, range:[1,32]-->test
    </password>
    <postPoolID>
      <!--opt, int, range:[0,4294967295]-->1
    </postPoolID>
    <illegalPoolID>
      <!--opt, int, range:[0,4294967295]-->1
    </illegalPoolID>
    <vehicleDetectionID>
      <!--opt, int, range:[0,4294967295]-->1
    </vehicleDetectionID>
    <protocolType>
      <!--opt, enum, subType:string-->V1.0
    </protocolType>
    <cloudStorageType>
      <!--opt, enum, storage server type, subType:int-->1
    </cloudStorageType>
    <V2.0>
      <!--opt, object, dep:and,{$.CloudStorage.protocolType,eq,V2.0}-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.V2.0.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--req, int, port No., range:[1,65535]-->1
      </port>
      <accessKey>
        <!--opt, string, password, range:[1,64]-->test
      </accessKey>
      <secretKey>
        <!--opt, string, secret key, range:[1,64]-->test
      </secretKey>
      <postPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </postPoolID>
      <illegalPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--opt, int, range:[0,4294967295]-->1
      </vehicleDetectionID>
      <resourceID>
```

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```
        <!--opt, string, range:[1,32]-->test
      </resourceID>
      <transportProtocolType>
        <!--opt, enum, subType:string-->HTTP
      </transportProtocolType>
    </V2.0>
    <ImageAndVideoCloud>
      <!--opt, object, dep:and,{$.CloudStorage.protocolType,eq,Image and Video Cloud}-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <addressingFormatType>
        <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
      </addressingFormatType>
      <hostName>
        <!--opt, string, host name, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,hostname}-->test
      </hostName>
      <ipAddress>
        <!--opt, string, IPv4 address, range:[1,16], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, IPv6 address, range:[1,128], dep:and,{$.CloudStorage.ImageAndVideoCloud.addressingFormatType,eq,ipaddress}-->test
      </ipv6Address>
      <port>
        <!--req, int, port No., range:[1,65535]-->1
      </port>
      <accessKey>
        <!--opt, string, password, range:[1,64]-->test
      </accessKey>
      <secretKey>
        <!--opt, string, secret key, range:[1,64]-->test
      </secretKey>
      <postPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </postPoolID>
      <illegalPoolID>
        <!--opt, int, range:[0,4294967295]-->1
      </illegalPoolID>
      <vehicleDetectionID>
        <!--opt, int, range:[0,4294967295]-->1
      </vehicleDetectionID>
    </ImageAndVideoCloud>
    <bucketName>
      <!--opt, string-->test
    </bucketName>
  </CloudStorage>
</CloudStorageList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.3.1.7 Get the information about a specified private storage pool

Request URL

GET /ISAPI/ContentMgmt/CloudStorageServer/pool/<poolID>

Query Parameter

Parameter Name	Parameter Type	Description
poolID	string	Storage pool ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<StoragePool xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int-->1
  </id>
  <poolName>
    <!--ro, req, string-->test
  </poolName>
  <poolType>
    <!--ro, req, enum, subType:string-->general
  </poolType>
  <coverType>
    <!--ro, req, enum, subType:string-->noCover
  </coverType>
  <poolCycle>
    <!--ro, opt, int, unit:d, dep:and,{$.StoragePoolList.StoragePool.coverType,eq,periodic}-->1
  </poolCycle>
  <poolSize>
    <!--ro, req, int, unit:MB-->1
  </poolSize>
  <disperseType>
    <!--ro, opt, enum, subType:string-->collect
  </disperseType>
  <lockLimit>
    <!--ro, opt, int-->1
  </lockLimit>
  <drawFrame>
    <!--ro, opt, bool-->true
  </drawFrame>
  <originalDataCycle>
    <!--ro, opt, int, unit:d-->1
  </originalDataCycle>
  <domainID>
    <!--ro, opt, int-->1
  </domainID>
  <freeSize>
    <!--ro, req, int, unit:MB-->1
  </freeSize>
  <lockedSize>
    <!--ro, opt, int, unit:MB-->1
  </lockedSize>
  <firstDataTime>
    <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
  </firstDataTime>
  <lastDataTime>
    <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
  </lastDataTime>
  <firstCoverTime>
    <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
  </firstCoverTime>
  <lastCoverTime>
    <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
  </lastCoverTime>
  <lastDrawFrameTime>
    <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
  </lastDrawFrameTime>
  <UserList>
    <!--ro, opt, object-->
    <userEntry>
      <!--ro, opt, object-->
      <userName>
        <!--ro, opt, string-->test
      </userName>
    </userEntry>
  </UserList>
</StoragePool>
```

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16.3.1.8 Set the third-party cloud access parameters

Request URL

PUT /ISAPI/ContentMgmt/Storage/cloud

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Cloud xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--req, bool-->true
  </enable>
  <type opt="OneDrive,GoogleDrive ,DropBox">
    <!--req, enum, subType:string, attr:opt{req, string}-->OneDrive
  </type>
  <status opt="Online,Offline">
    <!--req, enum, subType:string, attr:opt{req, string}-->Online
  </status>
  <authCode min="0" max="64">
    <!--req, string, attr:min{req, int},max{req, int}-->test
  </authCode>
  <alias min="0" max="32">
    <!--opt, string, attr:min{req, int},max{req, int}-->test
  </alias>
  <tolalCapacity>
    <!--opt, int, unit:MB-->1
  </tolalCapacity>
  <usedSpace>
    <!--opt, int, unit:MB-->1
  </usedSpace>
</Cloud>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.3.1.9 Get the third-party cloud access parameters

Request URL

GET /ISAPI/ContentMgmt/Storage/cloud

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Cloud xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool-->true
  </enable>
  <type opt="OneDrive,GoogleDrive ,DropBox">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->OneDrive
  </type>
  <status opt="Online,Offline">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->Online
  </status>
  <authCode min="0" max="64">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test
  </authCode>
  <alias min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </alias>
  <tolalCapacity>
    <!--ro, opt, int, unit:MB-->1
  </tolalCapacity>
  <usedSpace>
    <!--ro, opt, int, unit:MB-->1
  </usedSpace>
</Cloud>
```

16.3.1.10 Get the third-party cloud access capability

Request URL

GET /ISAPI/ContentMgmt/Storage/cloud/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Cloud xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool-->true
  </enable>
  <type opt="OneDrive,GoogleDrive ,DropBox">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->OneDrive
  </type>
  <status opt="Online,Offline">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->Online
  </status>
  <authCode min="0" max="64">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test
  </authCode>
  <alias min="0" max="32">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </alias>
  <tolalCapacity>
    <!--ro, opt, int, unit:MB-->1
  </tolalCapacity>
  <usedSpace>
    <!--ro, opt, int, unit:MB-->1
  </usedSpace>
</Cloud>
```

16.3.1.11 Get the capability of configuring cloud storage service

Request URL

GET /ISAPI/ContentMgmt/Storage/cloudServer/capabilities?format=json

Query Parameter

None

Request Message

None

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Response Message

```
{
    "CloudServer": {
    /*ro, opt, object*/
        "enabled":  "true,false",
        /*ro, req, string*/
        "version": {
        /*ro, req, object*/
            "@min":  0,
            /*ro, req, int*/
            "@max":  32,
            /*ro, req, int*/
            "#text":  "V2.0"
            /*ro, opt, string*/
        },
        "accessKey": {
        /*ro, req, object*/
            "@min":  32,
            /*ro, req, int*/
            "@max":  63
            /*ro, req, int*/
        },
        "secretKey": {
        /*ro, req, object*/
            "@min":  32,
            /*ro, req, int*/
            "@max":  63
            /*ro, req, int*/
        }
    }
}
```

16.3.1.12 Set the cloud storage service parameters

Request URL

PUT /ISAPI/ContentMgmt/Storage/cloudServer?format=json

Query Parameter

None

Request Message

```
{
    "CloudServer": {
    /*opt, object*/
        "enabled":  true,
        /*req, bool*/
        "version":  "V2.0",
        /*req, string, range:[1,32]*/
        "accessKey":  "12345678901234567890123456789012",
        /*req, string, range:[32,63]*/
        "secretKey":  "12345678901234567890123456789012"
        /*req, string, range:[32,63]*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.3.1.13 Get the cloud storage service parameters

Request URL

GET /ISAPI/ContentMgmt/Storage/cloudServer?format=json

Query Parameter

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None

Request Message

None

Response Message

```
{
    "CloudServer": {
    /*ro, opt, object*/
        "enabled":  true,
        /*ro, req, bool*/
        "version":  "V2.0"
        /*ro, req, string, range:[1,32]*/
    }
}
```

16.3.2 Manual Recording

16.3.2.1 Start the manual recording

Request URL

PUT /ISAPI/ContentMgmt/record/control/manual/start/tracks/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.3.2.2 Stop the manual recording

Request URL

PUT /ISAPI/ContentMgmt/record/control/manual/stop/tracks/<trackStreamID>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.3.3 Scheduled Recording

16.3.3.1 Get all parameters of a recording schedule

Request URL

GET /ISAPI/ContentMgmt/record/tracks

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TrackList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, get all parameters of a recording schedule, subType:object, attr:version{req, string, protocolVersion}-->
  <Track>
    <!--ro, opt, object, recording schedule configuration-->
    <id>
      <!--ro, opt, int, channel number corresponding to stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should be subject to that of SrcUrl-->201
    </id>
    <Channel>
      <!--ro, req, int, channel No.-->1
    </Channel>
    <Enable>
      <!--ro, req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
    </Enable>
    <Description>
      <!--ro, req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
    </Description>
    <TrackGUID>
      <!--ro, req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
    </TrackGUID>
    <Size>
      <!--ro, opt, int, recording size-->1
    </Size>
    <Duration min="0" max="750">
      <!--ro, opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours; 0 indicates the video is valid until it is recurrently overwritten-->P10DT15H
    </Duration>
    <DefaultRecordingMode>
      <!--ro, req, enum, default record type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
    </DefaultRecordingMode>
    <LoopEnable>
      <!--ro, opt, bool, whether to support recurrently overwriting-->true
    </LoopEnable>
    <SrcDescriptor>
      <!--ro, opt, object, video source description-->
      <SrcGUID>
        <!--ro, req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
      </SrcGUID>
      <SrcChannel>
```

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```
        <!--ro, req, int, source channel-->1
      </SrcChannel>
      <StreamHint>
        <!--ro, req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
      </StreamHint>
      <SrcDriver>
        <!--ro, req, string, stream executable driver name-->RTP/RTSP
      </SrcDriver>
      <SrcType>
        <!--ro, opt, string, source type-->mp4 video
      </SrcType>
      <SrcUrl>
        <!--ro, req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID>
refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main
stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
      </SrcUrl>
      <SrcUrlMethods>
        <!--ro, req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
      </SrcUrlMethods>
      <SrcLogin>
        <!--ro, req, string, login source password-->admin:admin
      </SrcLogin>
    </SrcDescriptor>
    <TrackSchedule>
      <!--ro, opt, object, recording schedule-->
      <ScheduleBlockList>
        <!--ro, opt, array, recording schedule list, subType:object-->
        <ScheduleBlock>
          <!--ro, opt, object, recording schedule module-->
          <ScheduleBlockGUID>
            <!--ro, opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--ro, req, string, schedule type-->/ISAPI/recording/schedule/default
          </ScheduleBlockType>
          <ScheduleAction>
            <!--ro, opt, array, schedule, subType:object, desc:N/A-->
            <id>
              <!--ro, req, int, ID-->1
            </id>
            <ScheduleActionStartTime>
              <!--ro, opt, object, scheduled start time-->
              <DayOfWeek>
                <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
              </DayOfWeek>
              <TimeOfDay>
                <!--ro, opt, time, time-->08:00:00
              </TimeOfDay>
            </ScheduleActionStartTime>
            <ScheduleActionEndTime>
              <!--ro, opt, object, scheduled end time-->
              <DayOfWeek>
                <!--ro, req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
              </DayOfWeek>
              <TimeOfDay>
                <!--ro, opt, time, time-->08:00:00
              </TimeOfDay>
            </ScheduleActionEndTime>
            <ScheduleDSTEnable>
              <!--ro, req, bool, whether to enable Daylight Saving Time (DST)-->true
            </ScheduleDSTEnable>
            <Description>
              <!--ro, req, string, description-->PreMorning (Midnight to 8am,local time)
            </Description>
            <Actions>
              <!--ro, opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
              <Record>
                <!--ro, opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </Record>
              <Log>
                <!--ro, opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </Log>
              <SaveImg>
                <!--ro, opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </SaveImg>
              <ActionRecordingMode>
                <!--ro, req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
              </ActionRecordingMode>
              <PreRecordTimeSeconds>
                <!--ro, opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
              </PreRecordTimeSeconds>
              <PostRecordTimeSeconds>
                <!--ro, opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
              </PostRecordTimeSeconds>
            </Actions>
          </ScheduleAction>
        </ScheduleBlock>
      </ScheduleBlockList>
```

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```
    </TrackSchedule>
    <CustomExtensionList>
      <!--ro, opt, array, custom extension list, subType:object-->
      <CustomExtension>
        <!--ro, opt, object, custom extension-->
        <CustomExtensionName>
          <!--ro, opt, string, custom extension name-->www.xxxx.com/RaCM/trackExt/ver10
        </CustomExtensionName>
        <enableSchedule>
          <!--ro, opt, bool, whether to enable recording schedule configuration-->true
        </enableSchedule>
        <SaveAudio>
          <!--ro, opt, bool, whether to enable audio recording-->true
        </SaveAudio>
        <PreRecordTimeSeconds>
          <!--ro, opt, int, pre-record, unit:s-->1
        </PreRecordTimeSeconds>
        <PostRecordTimeSeconds>
          <!--ro, opt, int, post-record, unit:s-->1
        </PostRecordTimeSeconds>
        <HolidaySchedule>
          <!--ro, opt, object, holiday schedule-->
          <ScheduleBlock>
            <!--ro, opt, object, schedule module-->
            <ScheduleBlockGUID>
              <!--ro, req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
            </ScheduleBlockGUID>
            <ScheduleBlockType>
              <!--ro, opt, string, schedule module type-->www.xxxx.com/racm/schedule/ver10
            </ScheduleBlockType>
          </ScheduleBlock>
        </HolidaySchedule>
      </CustomExtension>
    </CustomExtensionList>
    <IntelligentRecord>
      <!--ro, opt, bool, whether to enable VCA recording function-->false
    </IntelligentRecord>
    <delayTime>
      <!--ro, opt, int, capture delay time, unit:s-->1
    </delayTime>
    <durationEnabled>
      <!--ro, opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time
via the node "Duration" when it is supported-->true
    </durationEnabled>
    <redundancyRec>
      <!--ro, opt, bool, whether it is redundant recording, desc:it is false when this node does not exist-->false
    </redundancyRec>
    <passbackRecord>
      <!--ro, opt, bool, whether to copy back recording, desc:it is false when this node does not exist-->false
    </passbackRecord>
    <lockDuration>
      <!--ro, opt, int, recording locking duration, unit:h, desc:0-unlocked or this node does not exist, 0xffff-locked permanently; if the total recording
is longer than the locked part, it will not be locked-->0
    </lockDuration>
    <recordBackup>
      <!--ro, opt, bool, whether to enable video archiving, desc:it is false when this node does not exist-->false
    </recordBackup>
    <SVCLevel>
      <!--ro, opt, enum, SVC frame extracting type, subType:int, desc:0-not extracted or this node does not exist, 1-half extracted, 2-three quarters
extracted-->0
    </SVCLevel>
    <recordManage>
      <!--ro, opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording
schedule is still in use, e.g., motion detection and copy-back); it is false when this node does not exist-->false
    </recordManage>
    <extraSaveAudio>
      <!--ro, opt, bool, whether to enable separate audio storage, desc:it is false when this node does not exist-->false
    </extraSaveAudio>
  </Track>
</TrackList>
```

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16.3.3.2 Set all recording schedules

Request URL

PUT /ISAPI/ContentMgmt/record/tracks

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TrackList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, all recording schedules parameters, subType:object, attr:version{req, string, protocolVersion}-->
  <Track>
    <!--opt, object, recording schedule configuration-->
```

```
    <id>
      <!--opt, int, channel corresponding to the stream type, desc:the values of <id> and <Channel> are the same. For example, 201 refers to the main stream of channel 2, 202 refers to the sub-stream of channel 2, and so on. If the field SrcUrl of video source description is valid, the channel stream type should be subject to that of SrcUrl-->201
    </id>
    <Channel>
      <!--req, int, channel No.-->1
    </Channel>
    <Enable>
      <!--req, bool, whether to enable, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
    </Enable>
    <Description>
      <!--req, string, recording description-->trackType=standard,sourceTag=AXIS210a,contentType=video,codecType=MPEG4-SP,resolution=640x480,frameRate=20 fps,bitrate=6000 kbps
    </Description>
    <TrackGUID>
      <!--req, string, GUID, desc:generated by the client-->{A01AAAAA-BBBB-CCCC-DDDD-033595353625}
    </TrackGUID>
    <Size>
      <!--opt, int, video size-->1
    </Size>
    <Duration min="0" max="750">
      <!--opt, string, video expiry time, attr:min{req, int},max{req, int}, desc:e.g., P10DT15H indicates the video will be expired in 10 days and 15 hours; 0 indicates the video is valid until it is recurrently overwritten-->P10DT15H
    </Duration>
    <DefaultRecordingMode>
      <!--req, enum, default recording type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures), "ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection pictures), "AllEvent" (all events), "POS" (POS recording), "PECCANCY" (forensic marking recording), "SMART" (smart recording), "temperatureIntervalMeasurement" (temperature range detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default), "thermalVehicleDetection" (thermal imaging vehicle detection), "fishingShipDetection" (fishing ship detection), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "parallelParking" (vehicle entering or exiting during parallel parking, "parallelParkingContinuousRecord" (from vehicl entering to vehicle exiting for parallel parking), "linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay detection), "failDown" (getting up or falling down detection)-->CMR
    </DefaultRecordingMode>
    <LoopEnable>
      <!--opt, bool, whether to support recurrently overwriting-->true
    </LoopEnable>
    <SrcDescriptor>
      <!--opt, object, video source description-->
      <SrcGUID>
        <!--req, string, GUID generated by the client-->{E800A543-9D53-4520-8BB8-9509062C692D}
      </SrcGUID>
      <SrcChannel>
        <!--req, int, source channel-->1
      </SrcChannel>
      <StreamHint>
        <!--req, string, stream description-->video,mp4,640x480,20 fps,6000 kbps
      </StreamHint>
      <SrcDriver>
        <!--req, string, stream executable driver name-->RTP/RTSP
      </SrcDriver>
      <SrcType>
        <!--opt, string, source type-->mp4 video
      </SrcType>
      <SrcUrl>
        <!--req, string, source URL, desc:the format of source URL is: rtsp://localhost/ISAPI/Streaming/channels/<trackStreamID>. The <trackStreamID> refers to the channel stream ID = channel number * 100 + stream type (1 indicates main stream, 2 indicates sub stream). For example, 201 indicates the main stream of channel 2-->rtsp://localhost/ISAPI/Streaming/channels/101
      </SrcUrl>
      <SrcUrlMethods>
        <!--req, enum, methods supported by the source, subType:string, desc:"DESCRIBE" (subscribe), "SETUP", "PLAY", "TEARDOWN" (stop)-->DESCRIBE
      </SrcUrlMethods>
      <SrcLogin>
        <!--req, string, login source password-->admin:admin
      </SrcLogin>
    </SrcDescriptor>
    <TrackSchedule>
      <!--opt, object, recording schedule-->
      <ScheduleBlockList>
        <!--opt, array, recording schedule list, subType:object-->
        <ScheduleBlock>
          <!--opt, object, recording schedule module-->
          <ScheduleBlockGUID>
            <!--opt, string, schedule GUID-->{ABC12345-CDEF-4520-8BB8-7135789C8790}
          </ScheduleBlockGUID>
          <ScheduleBlockType>
            <!--req, string, schedule type-->/ISAPI/recording/schedule/default
          </ScheduleBlockType>
          <ScheduleAction>
            <!--opt, array, schedule, subType:object, desc:N/A-->
            <id>
              <!--req, int, ID-->1
            </id>
            <ScheduleActionStartTime>
              <!--opt, object, scheduled start time-->
              <DayOfWeek>
                <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
              </DayOfWeek>
              <TimeOfDay>
                <!--opt, time, time-->08:00:00
              </TimeOfDay>
            </ScheduleActionStartTime>
```

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```
            <ScheduleActionEndTime>
              <!--opt, object, scheduled end time-->
              <DayOfWeek>
                <!--req, enum, day of a week, subType:string, desc:"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"-->Monday
              </DayOfWeek>
              <TimeOfDay>
                <!--opt, time, time-->08:00:00
              </TimeOfDay>
            </ScheduleActionEndTime>
            <ScheduleDSTEnable>
              <!--req, bool, whether to enable Daylight Saving Time (DST)-->true
            </ScheduleDSTEnable>
            <Description>
              <!--req, string, description-->PreMorning (Midnight to 8am,local time)
            </Description>
            <Actions>
              <!--opt, object, trigger action, desc:alarm or motion detection that triggers a recording-->
              <Record>
                <!--opt, bool, record, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </Record>
              <Log>
                <!--opt, bool, log, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </Log>
              <SaveImg>
                <!--opt, bool, save image, desc:this node is retained, and returned by some devices, but has no practical meaning-->true
              </SaveImg>
              <ActionRecordingMode>
                <!--req, enum, recording schedule type, subType:string, desc:"CMR" (capture by schedule), "MOTION" (capture motion detection pictures),
"ALARM" (capture alarm pictures), "EDR" (capture alarm pictures or motion detection pictures), "ALARMANDMOTION" (capture alarm pictures and motion detection
pictures), "COMMAND" (recording by commands), "SMART" (smart recording), "thermalVehicleDetection" (thermal imaging vehicle detection),
"fishingShipDetection" (fishing ship detection), "methaneConcentrationException" (dangerous gas detection: methane concentration exception by default),
"AllEvent" (all events), "consultation", "panicAlarm" (panic alarm), "zoneAlarm" (zone alarm), "temperatureIntervalMeasurement" (temperature range
detection), "Linedetection" (line crossing detection), "toiletTarry" (in-toilet overtime detection), "getUp" (get-off-bed detection), "retention" (overstay
detection), "failDown" (getting up or falling down detection)-->CMR
              </ActionRecordingMode>
              <PreRecordTimeSeconds>
                <!--opt, int, pre-record duration, unit:s, desc:from 0 to 10 seconds; default: 5 seconds-->5
              </PreRecordTimeSeconds>
              <PostRecordTimeSeconds>
                <!--opt, int, post-record duration, unit:s, desc:from 0 to 20 seconds; default: 5 seconds-->5
              </PostRecordTimeSeconds>
            </Actions>
          </ScheduleAction>
        </ScheduleBlock>
      </ScheduleBlockList>
    </TrackSchedule>
    <CustomExtensionList>
      <!--opt, array, custom extension list, subType:object-->
      <CustomExtension>
        <!--opt, object, custom extension-->
        <CustomExtensionName>
          <!--opt, string, custom extension name-->www.xxx.com/RaCM/trackExt/ver10
        </CustomExtensionName>
        <enableSchedule>
          <!--opt, bool, whether to enable recording schedule configuration-->true
        </enableSchedule>
        <SaveAudio>
          <!--opt, bool, whether to enable audio recording-->true
        </SaveAudio>
        <PreRecordTimeSeconds>
          <!--opt, int, pre-record, unit:s-->1
        </PreRecordTimeSeconds>
        <PostRecordTimeSeconds>
          <!--opt, int, post-record, unit:s-->1
        </PostRecordTimeSeconds>
        <HolidaySchedule>
          <!--opt, object, holiday schedule-->
          <ScheduleBlock>
            <!--opt, object, schedule module-->
            <ScheduleBlockGUID>
              <!--req, string, schedule module GUID-->{00000000-0000-0000-0000-000000000000}
            </ScheduleBlockGUID>
            <ScheduleBlockType>
              <!--opt, string, schedule module type-->www.xxx.com/racm/schedule/ver10
            </ScheduleBlockType>
          </ScheduleBlock>
        </HolidaySchedule>
      </CustomExtension>
    </CustomExtensionList>
    <IntelligentRecord>
      <!--opt, bool, whether to enable VCA recording function-->false
    </IntelligentRecord>
    <delayTime>
      <!--opt, int, capture delay time, unit:s-->1
    </delayTime>
    <durationEnabled>
      <!--opt, bool, whether to enable video expiry time, desc:this node is not returned when it is not supported; you can configure video expiry time via
the node Duration when it is supported-->true
    </durationEnabled>
    <redundancyRec>
      <!--opt, bool, whether it is redundant recording, desc:it is false when this node does not exist-->false
    </redundancyRec>
    <passbackRecord>
      <!--opt, bool, whether to copy back recording, desc:it is false when this node does not exist-->false
```

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```
      <!--opt, bool, whether to copy back recording, desc:it is false when this node does not exist-->false
    </passbackRecord>
    <lockDuration>
      <!--opt, int, recording locking duration, unit:h, desc:0-unlocked or this Field does not exist, 0xffff-locked permanently; if the total recording is longer than the locked part, it will not be locked-->0
    </lockDuration>
    <recordBackup>
      <!--opt, bool, whether to enable video archiving, desc:it is false when this node does not exist-->false
    </recordBackup>
    <SVCLevel>
      <!--opt, enum, SVC frame extracting type, subType:int, desc:0-not extracted or this node does not exist, 1-half extracted, 2-three quarters extracted-->0
    </SVCLevel>
    <recordManage>
      <!--opt, bool, whether to enable video scheduling, desc:true (enabled scheduled recording), false (no scheduled recording but the recording schedule is still in use, e.g., motion detection and copy-back); it is false when this node does not exist-->false
    </recordManage>
    <extraSaveAudio>
      <!--opt, bool, whether to enable separate audio storage, desc:it is false when this node does not exist-->false
    </extraSaveAudio>
  </Track>
</TrackList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

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16.3.4 Video Download

16.3.4.1 Get the capability set of downloading videos

Request URL

GET /ISAPI/ContentMgmt/download/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DownloadAbility xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the capability of downloading, attr:version{req, string, protocolVersion}-->
  <isSupportDownloadbyTime>
    <!--ro, opt, bool, whether it supports downloading by time-->true
  </isSupportDownloadbyTime>
  <isSupportDownloadbyFileName>
    <!--ro, opt, bool, whether it supports downloading by file name-->true
  </isSupportDownloadbyFileName>
  <isSupportDownloadCourseFile>
    <!--ro, opt, bool, whether it supports downloading class files-->true
  </isSupportDownloadCourseFile>
  <isSupportDownloadToUSB>
    <!--ro, opt, bool, whether it supports exporting files to devices via USB-->true
  </isSupportDownloadToUSB>
</DownloadAbility>
```

16.3.5 Video Search

16.3.5.1 Refresh the video index manually before playback

Request URL

PUT /ISAPI/ContentMgmt/record/control/manualRefresh/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to
quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

16.3.5.2 Get the capability of manually refreshing video index

Request URL

GET /ISAPI/ContentMgmt/record/control/manualRefresh/channels/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "isSupportManualRefresh":  true
    /*ro, opt, bool, whether the device supports limiting the frequency of refreshing video index, desc:device of new version will limit the frequency of refreshing video index to less than once every 10 seconds by default; device of old version will not limit the frequency of refreshing video index, but frequent refresh will cause device exception*/
}
```

16.3.5.3 Get video track type

Request URL

GET /ISAPI/ContentMgmt/record/profile

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CMRecordProfile xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <trackType>
    <!--ro, opt, enum, video track type, subType:string, desc:video track type-->standard
  </trackType>
</CMRecordProfile>
```

16.3.5.4 Search recorded video files by date

Request URL

POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel No. * 100 + stream type (-1: main stream, -2: sub-stream).
type	enum	"channelID" (search for recordings by channel ID), "trackStreamID" (channel No.), "streamID" (search for recordings by stream ID), "trackStreamID" (stream ID). When there is no 'type" node or "type" is "channelID", it indicates searching by channel ID, and "trackStreamID" is the channel No.; when "type" is "streamID", it indicates searching by stream ID, and "trackStreamID" is the stream ID.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<trackDailyParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <year>
    <!--req, int, year-->1
  </year>
  <monthOfYear>
    <!--req, int, month of the year-->1
  </monthOfYear>
  <drawFrame>
    <!--opt, bool-->false
  </drawFrame>
</trackDailyParam>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<trackDailyDistribution xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <dayList>
    <!--ro, opt, array, subType:object-->
    <day>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, ID-->1
      </id>
      <dayOfMonth>
        <!--ro, req, int, day of the month,starts from 1st-->1
      </dayOfMonth>
      <record>
        <!--ro, req, bool, true-with video,false-no video-->true
      </record>
      <recordType>
        <!--ro, opt, enum, subType:string-->event
      </recordType>
    </day>
  </dayList>
</trackDailyDistribution>
```

16.3.5.5 Get the capability of searching for video and picture files

Request URL

GET /ISAPI/ContentMgmt/search/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search conditions, attr:version{opt, string, protocolVersion}-->
  <objectsThrownDetectionSearchCap>
    <!--ro, opt, object, capability of object thrown from building search condition-->
    <alarmLevel min="1" max="3">
      <!--ro, opt, int, alarm level, range:[1,3], attr:min{req, int},max{req, int}-->3
    </alarmLevel>
    <lowFloorID min="-100" max="100">
      <!--ro, opt, int, low floor, range:[-100,100], attr:min{req, int},max{req, int}-->0
    </lowFloorID>
    <highFloorID min="-100" max="100">
      <!--ro, opt, int, high floor, range:[-100,100], attr:min{req, int},max{req, int}-->0
    </highFloorID>
  </objectsThrownDetectionSearchCap>
  <alarmResultSearchCap>
    <!--ro, opt, object, capability of face picture comparison search condition-->
    <faceContrastType opt="all, faceContrast, whiteFaceContrast,faceContrastFailure">
      <!--ro, req, enum, face picture comparison type, subType:string, attr:opt{req, string}, desc:"all", "faceContrast" (face picture comparison
succeeded), "whiteFaceContrast" (allowlist face picture comparison), "faceContrastFailure" (face picture comparison failed)-->faceContrast
    </faceContrastType>
  </alarmResultSearchCap>
</CMSearchDescription>
```

16.3.5.6 Get video search conditions

Request URL

GET /ISAPI/ContentMgmt/search/profile

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<CMSearchProfile xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <searchProfile>
    <!--ro, opt, enum, video search conditions, subType:string, desc:video search conditions-->basic
  </searchProfile>
  <textSearch>
    <!--ro, opt, enum, text type to be searched, such as recording information text and metadata text, subType:string, desc:text type to be searched,such as
recording information text and metadata text-->true
  </textSearch>
  <maxSearchTimespans>
    <!--ro, req, int, maximum time period of results returned in each search-->1
  </maxSearchTimespans>
  <maxSearchTracks>
    <!--ro, opt, int, maximum tracks can be searched-->1
  </maxSearchTracks>
  <maxSearchSources>
    <!--ro, opt, int, maximum recording channels can be searched-->1
  </maxSearchSources>
  <maxSearchMetadatas>
    <!--ro, opt, int, maximum number of metadata can be searched-->1
  </maxSearchMetadatas>
  <maxSearchMatchResults>
    <!--ro, req, int, maximum search results can be returned-->1
  </maxSearchMatchResults>
  <maxSearchTimeout>
    <!--ro, req, int, maximum search timeout-->1
  </maxSearchTimeout>
  <maxConcurrentSearches>
    <!--ro, req, int, maximum concurrent search operations-->1
  </maxConcurrentSearches>
</CMSearchProfile>
```

16.3.5.7 Search for video and picture files

Request URL

POST /ISAPI/ContentMgmt/search?timeType=<timeType>

Query Parameter

Parameter Name	Parameter Type	Description
timeType	enum	1. If "isSupportTimeTypeSTD" is true in GET /ISAPI/System/capabilities, it indicates that the device supports parsing the optional parameter timeType=STD in the URL.2. If the parameter timeType=STD is included in the URL, time-related parameters (e.g., "datetime") is processed in ISO8601 time format.3. If the parameter timeType=STD is not included in the URL, time-related parameters is processed according to the original time format set by the device.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search conditions, attr:version{opt, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searches, the search history will be saved in the memory to speed up next search-->test
  </searchID>
  <trackIDList>
    <!--opt, array, channel list, subType:object-->
    <trackID>
      <!--opt, int, channel stream ID, desc:"XX01" (main stream videos of XX channel), "XX03" (search for XX channel pictures), 1 by default-->101
    </trackID>
  </trackIDList>
  <timeSpanList>
    <!--opt, array, list of time periods, subType:object-->
    <timeSpan>
      <!--opt, object, time period-->
      <startTime>
        <!--req, datetime, start time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:00:00Z
      </startTime>
      <endTime>
        <!--req, datetime, end time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T01:00:00Z
      </endTime>
      <laneNumber>
        <!--opt, int, lane No.-->1
      </laneNumber>
      <carType>
```

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```
        <!--opt, enum, vehicle type, subType:string, desc:"coach", "truck", "car", "minibus", "smalltruck" (small-sized truck), "bigcar" (large-sized vehicle), "smallcar" (small-sized vehicle), "human", "tumbrel" (two wheeler), "trike" (tricycle), "suv", "midcar" (middle-sized bus), "all", "unknown"-->coach
      </carType>
      <illegalType>
        <!--opt, enum, violation type, subType:string, desc:"all", "normal", "lowSpeed" (driving in low speed), "highSpeed" (driving in high speed), "reverse" (wrong-way driving), "redLight" (red light running), "driveLine" (driving on the lane line), "direction" (driving against direction guidance), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "changeLane" (illegal lane change), "wrongLine" (occupy dedicated lane), "badSign" (prohibition violation), "intersectionStop" (stop vehicle at intersection), "greenLightStop (stop vehicle when green light), "overYello" (over yellow line), "illegalParking" (illegal parking), "unTurn" (illegal U-turn), "emergencyLane" (occupy emergency lane), "banRight" (no right turn), "banLeft" (no left turn), "failSafeBelt" (seatbelt unbuckled), "pedRedLight" (pedestrian red light running), "failYieldPed" (not yield to pedestrian), "overStop" (over stop line), "bigCarOccupy" (lane occupation by large-sized vehicle), "leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "gasser", "threeWheel", "Passenger", "uphone", "congestion", "accident", "runGreeLight", "nonMotorExist", "failHelmet", "smoke", "playPhone", "failSafeBeltVice", "roadBlock", "construction", "notKeepDistance", "overtakeRightSide", "dragRacing", "changeLaneContinuously", "SSharpDriving", "abandonedObject", "pedestrian", "notSlowZebraCrossing", "overThermal", "turnRightStop", "occupyDedicatedLane", "notDriveInDedicatedLane", "smog", "conflagration", "animalsOccupy", "irregularSafeBeltVic", "notYieldToRightRoad", "rampNotYieldToMainRoad", "enterNotYieldToInRoundabout", "TruckOccupyOvertakingLane", "noAlterTraffic", "banStraight", "followClose"-->all
      </illegalType>
      <vehicleTypeByWeight>
        <!--opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1 (7 and fewer seats passenger vehicle / 2 tons or fewer truck), 2 (8 to 19 seats passenger vehicle / 2 to 5 tons truck), 3 (20 to 39 seats passenger vehicle/5 to 10 tons truck), 4 (40 and more seats passenger vehicle/10 to 15 tons truck), 5 (15 and more tons truck)-->4
      </vehicleTypeByWeight>
      <mediaID>
        <!--opt, int, file ID, desc:file ID (media ID), which is used to return video file information after segmented by startTime and endTime-->0
      </mediaID>
      <listType>
        <!--ro, opt, enum, name list type, subType:string, desc:"black" (blocklist). If this node is empty, it indicates that the name list type is not limited, corresponding to the listType in ANPR-->black
      </listType>
    </timeSpan>
  </timeSpanList>
  <sourceIDList>
    <!--opt, array, stream ID list, subType:object-->
    <sourceID>
      <!--req, string, GUID, range:[0,64]-->test
    </sourceID>
  </sourceIDList>
  <searchTypeList>
    <!--opt, array, search type list, subType:object, range:[0,20]-->
    <searchKeyword>
      <!--opt, string, keyword-->test
    </searchKeyword>
    <keywordType>
      <!--opt, enum, keyword type, subType:string, desc:“policeCode” (police No.), “shortSerialNumber” (device serial No.); when searching for videos stored in dock stations via the platform, videos can be filtered according to the serial No. of body camera; if the dock station does not send this parameter, the policeCode will be used by default-->shortSerialNumber
    </keywordType>
  </searchTypeList>
  <contentTypeList>
    <!--opt, array, file type list for search, subType:object-->
    <contentType>
      <!--opt, enum, file type for search, subType:string, desc:"video", "audio", "metadata", "text", "mixed", "other". Note: "metadata" is used when searching for pictures-->video
    </contentType>
  </contentTypeList>
  <uploadState>
    <!--opt, enum, uploading status, subType:string, desc:"all", "uploaded", "nonUploaded” (not uploaded), "uploadFailed” (uploading failed)-->all
  </uploadState>
  <lockState>
    <!--opt, enum, locking status, subType:string, desc:"lock” (locked), "unlock” (unlocked), "all"-->lock
  </lockState>
  <maxResults>
    <!--opt, int, the maximum number of results returned for one time, range:[1,500]-->500
  </maxResults>
  <searchResultPostion>
    <!--opt, int, start position for search, desc:the node of body camera is in front of <maxResults>-->1
  </searchResultPostion>
  <metadataList>
    <!--req, array, search conditions, subType:object, desc:only supports a single description list-->
    <metadataDescriptor>
      <!--req, enum, video or file type for search, subType:string, desc:1. the value of CMSearchDescription->contentTypeList->contentType indicates the type (video or picture) for search; 2. the video and picture file type supported by the device will be returned by the <recordSearchType> node and the <pictureSearchType> node in the RacmCap message returned by calling /ISAPI/ContentMgmt/capabilities; 3. transmit the data in the format of “recordType.meta..com/[type]” to search for videos or pictures; if /[type] is not contained, it indicates that all types of video files will be searched; if the [type] is set to “allPic”, it indicates that all types of picture files will be searched; the [type] refers to the followings: “imageCapture” (scheduled capture by channel), “pothole”, “crack”, “roadMaint” (road maintenance), “scheduledRecordUpload” (scheduled recording by channel), "ReID" (target pattern detection), “thermalVehicleDetection” (thermal-based vehicle detection), “objectsThrownDetection”, “methaneConcentrationException”, “methaneLightIntensityException” (hazardous gas detection), “bannerDetection”, “waterQualityDetection”, “waterLevelDetection”, “parallelParking”, “waterLevelAndRainfall” (water level and rainfall), “AIOpenPlatform” (events on AI open platform), “fishingShipDetection”, “variableDataEvent”, “VOCsGasDetection”, “angleDeviationDetection” (tilt detection), “pir”, “shipChannelAbnormal” (ship deviation), “radarVideoDetection” (radar-assisted target detection), “AIDfogDetection”, “consultation”, “panicAlarm”, “zoneAlarm”, “smokeAndFireDetection”, “smokeDetection”, “facedetection”, “FieldDetection” (intrusion detection), “scenechangedetection”, “LineDetection” (line crossing detection), “regionEntrance”, “regionExiting”, “Loitering”, “group” (people gathering), “rapidMove” (fast moving), “parking”, “unattendedBaggage”, “attendedBaggage”, “vehicleDetection”, “manual”, “manualSnapShot” (manual capture), “playSnapShot” (capture in playback), “safetyHelmet” (hard hat detection), “personQueueCounting” (people queuing-up detection), “personQueueTime” (waiting time in queue detection), “violentMotion”, “LeavePosition” (absent detection), “advReachHeight” (climing detection), “peopleNumChange” (people count change), “mixedTargetDetection” (multi-target-type detection), “illegalParking” (parking violation), “pedestrian” (pedestrian detection), “construction”, “roadblock”, “evidence” (manual evidence capture), “crosslane” (driving on the lane line), “vehicleexist” (motor vehicle on non-motor vehicle lane), “Lanechange”, “wrongdirection” (wron-way driving), “congestion”, “turnround”, “abandonedObject”, “trafficAccident”, “fogDetection”, “sensorAlarm”, “parallelParkingContinuousRecord” (continuous recording for parallel parking), “GasTankDetection”, “swipePatrol” (patrolling by card), “alarmResult” (face picture comparison), “UVPhotonDetection”, “DevPowerStatusReport” (device power status uploading), “playCellphone” (using mobile phone detection), “HFPD”, “audioAnalysis”, “PeopleFallingDownDetection”, “PeopleSedentary”, “RadarFallenObjectsOnTracks”, “VegetationSpectrumDetection”, “personDensityDetection”, “ChannelClearEvent”, “channelOccupy”, ChannelPassingEvent”, and “FallenObjectsOnTracksEvent”-->recordType.meta..com/imageCapture
```

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```
    </metadataDescriptor>
    <SearchProperity>
      <!--opt, object, search by features-->
      <plateSearchMask>
        <!--opt, enum, license plate-->test
      </plateSearchMask>
      <stateOrProvince>
        <!--opt, int, province, desc:"0xffff" (undefined)-->0xffff
      </stateOrProvince>
      <country>
        <!--opt, enum, country/region, subType:string, desc:0: Algorithm library does not support the national identification card; 1: CZ - Czech Republic;
2: FRA - France; 3: DE - Germany; 4: E - Spain; 5: IT - Italy; 6: NL - Netherlands; 7: PL - Poland; 8: SVK - Slovakia; 9: BY - Belorussia; 10: MDA -
Moldova; 11: RU - Russia; 12: UA - Ukraine; 169: Canada; 172: US; 227: Australia; 0xff: All-->0
      </country>
      <subType>
        <!--opt, enum, sub types, subType:string, desc:only supported by vehicleDetection and HVTVehicleDetection. "all", "motorVehicle", "nonMotorVehicle",
"pedestrian"-->all
      </subType>
      <capTypeDescriptor>
        <!--opt, enum, types of captured pictures, subType:string, desc:"event", "evidence". Events include parkEvent, crosslane, vehicleexist, lanechange,
wrongdirection, congestion, turnaround, edfManual. Evidences include parkEvent, crosslane, vehicleexist, lanechange, wrongdirection, and anyType-->anyType
      </capTypeDescriptor>
      <nation>
        <!--opt, enum, area, subType:string, desc:EU: Europe Region; ER: Middle East; AP: Asia Pacific; AfricaAndAmerica: Africa and America-->EU
      </nation>
      <vehicleType>
        <!--opt, enum, vehicle type, subType:string, desc:"unknown", "bus", "truck", "vehicle", "van", "buggy", "SUVMPV": SUV/MPV, "mediumBus"-->bus
      </vehicleType>
      <absoulteParkingNum>
        <!--opt, string, parking space No., desc:the parking space No. is the same as the number on the actual ground surface, which can be captured by the
camera-->test
      </absoulteParkingNum>
    </SearchProperity>
  </metadataList>
  <cardNumber>
    <!--opt, string, card No., range:[0,32], desc:if this node is empty or does not exist, it indicates that the card No. is not restricted-->test
  </cardNumber>
  <drawFrame>
    <!--opt, bool, whether to enable frame extraction, desc:the value of this node is false by default-->false
  </drawFrame>
  <AlarmInNoList>
    <!--opt, array, triggered alarm input channel list, subType:object-->
    <alarmInNo>
      <!--req, int, triggered alarm input channel No.-->0
    </alarmInNo>
  </AlarmInNoList>
  <ruleID>
    <!--opt, string, rule ID, desc:if this node is not configured, the search result will not be restricted by any rule-->test
  </ruleID>
  <roomIndex>
    <!--opt, int, interrogation room No., desc:it starts from 1-->0
  </roomIndex>
  <StreamIDParam>
    <!--opt, object, stream ID parameters-->
    <streamID>
      <!--req, string, stream ID, range:[0,72]-->test
    </streamID>
    <channel>
      <!--opt, int, channel No., range:[1,256]-->1
    </channel>
    <cmdType>
      <!--opt, int, trigger type, range:[0,255], desc:connect NVR with cloud storage to use it-->0
    </cmdType>
    <backupVolumeNum>
      <!--opt, int, archive No., range:[0,255], desc:applicable to CVR-->0
    </backupVolumeNum>
    <archiveLabel>
      <!--opt, string, archive No.-->test
    </archiveLabel>
  </StreamIDParam>
  <PosAlarm>
    <!--opt, object, pos video-->
    <caseSensitive>
      <!--req, bool, whether it is case-sensitive-->true
    </caseSensitive>
    <combinateMode>
      <!--req, enum, keyword combination, subType:string, desc:"or", "and"-->or
    </combinateMode>
    <KeyWordList>
      <!--opt, array, keyword list, subType:object, desc:supports up to three key words-->
      <keyWord>
        <!--req, string, keyword, range:[0,128]-->test
      </keyWord>
    </KeyWordList>
  </PosAlarm>
  <TrialParam>
    <!--opt, object, search conditions of trail information-->
    <caseNo>
      <!--opt, string, case No., range:[0,56]-->test
    </caseNo>
    <caseName>
      <!--opt, string, case name, range:[0,56]-->test
    </caseName>
```

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```
    <litigant1>
      <!--opt, string, party 1, range:[0,32]-->test
    </litigant1>
    <litigant2>
      <!--opt, string, party 2, range:[0,32]-->test
    </litigant2>
    <chiefJudge>
      <!--opt, string, jury, range:[0,32]-->test
    </chiefJudge>
    <caseType>
      <!--opt, enum, case type, subType:int, desc:0 (all), 1 (penal case), 2 (civil case)-->0
    </caseType>
  </TrialParam>
  <ConferenceParam>
    <!--opt, object, video conference parameters-->
    <conferenceID>
      <!--opt, string, conference ID, range:[1,64]-->test
    </conferenceID>
  </ConferenceParam>
  <ATMParam>
    <!--opt, object, ATM search-->
    <transactionType>
      <!--opt, enum, transaction type, subType:int, desc:0 (all), 1(search), 2(withdrawal), 3 (deposit), 4(change password), 5(transfer), 6(search with no
card), 7(deposit with no card), 8 (cash swallowed), 9(card swallowed) 10(undefined)-->0
    </transactionType>
    <transationAmount>
      <!--opt, int, amount, step:1, desc:unit: yuan-->0
    </transationAmount>
  </ATMParam>
  <labelSearch>
    <!--opt, object, search by tag type list of video file, desc:if the node is not returned, it means search for all videos. Related URI:
/ISAPI/ContentMgmt/record/control/remark?format=json&security=<security>&iv=<iv>-->
    <labelSearchType>
      <!--opt, enum, type of searching by video file tag, subType:string, desc:"recordLabelTypeList", "allLabel", "noLabel"-->recordLabelTypeList
    </labelSearchType>
    <labelTypeList>
      <!--opt, object, by tag type list of video file, dep:and,{$.CMSearchDescription.recordLabelSearch.recordLabelSearchType,eq,recordLabelTypeList}-->
      <labelTypes>
        <!--opt, object, tag type of video file-->
        <labelType>
          <!--opt, enum, tag type of video file, subType:string, desc:"importance" (important), "unimportance" (not important)-->importance
        </labelType>
      </labelTypes>
    </labelTypeList>
  </labelSearch>
  <dockStationAudioFile>
    <!--opt, object, dock station audio file-->
    <analysisState>
      <!--ro, opt, enum, analysis status, subType:string, desc:"complete" (completed), "running", "notStarted" (not started)-->complete
    </analysisState>
    <keyWordList>
      <!--opt, array, dock station audio file label, subType:object, range:[0,100]-->
      <keyWord>
        <!--opt, string, label, range:[0,256]-->test
      </keyWord>
    </keyWordList>
  </dockStationAudioFile>
  <objectsThrownDetectionSearch>
    <!--opt, object, object thrown from building search condition-->
    <alarmLevel>
      <!--opt, int, alarm priority, range:[1,6], desc:1 (low), 2 (medium), 3 (high), 4 (low and medium), 5 (low and high), 6 (medium and high). If this node
is empty, alarms of all levels will be displayed-->3
    </alarmLevel>
    <lowFloorID>
      <!--opt, int, low floor, range:[-100,100]-->0
    </lowFloorID>
    <highFloorID>
      <!--opt, int, high floor, range:[-100,100]-->0
    </highFloorID>
  </objectsThrownDetectionSearch>
  <alarmResultSearch>
    <!--opt, object, face picture comparison search condition-->
    <faceContrastType>
      <!--req, enum, face picture comparison type, subType:string, desc:"all", "faceContrast" (face picture comparison succeeded), "whiteFaceContrast"
(allowlist face picture comparison), "faceContrastFailure" (face picture comparison failed)-->faceContrast
    </faceContrastType>
  </alarmResultSearch>
  <radarID>
    <!--opt, int, radar channel No.-->1
  </radarID>
  <synthesisType>
    <!--opt, enum, subType:string-->all
  </synthesisType>
</CMSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search results, attr:version{opt, string, protocolVersion}-->
  <searchID>
```

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```
    <!--ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searches, the search history will be saved in the memory to speed up next search-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search status, subType:string, desc:true (succeeded), false (failed)-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK", "NO MATCHES", "MORE"-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of returned records-->500
  </numOfMatches>
  <matchList>
    <!--ro, opt, array, result list, subType:object-->
    <searchMatchItem>
      <!--ro, opt, object, results-->
      <sourceID>
        <!--ro, req, string, source ID, desc:source ID-->test
      </sourceID>
      <trackID>
        <!--ro, opt, int, channel stream ID, desc:"XX01" (main stream videos of XX channel), "XX03" (search for XX channel pictures), 1 by default-->101
      </trackID>
      <timeSpan>
        <!--ro, opt, object, time range-->
        <startTime>
          <!--ro, req, datetime, video start time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:10:00Z
        </startTime>
        <endTime>
          <!--ro, req, datetime, video end time, desc:ISO8601 time (TZ format, UTC)-->1970-01-01T00:20:00Z
        </endTime>
      </timeSpan>
      <mediaSegmentDescriptor>
        <!--ro, opt, object, data descrition-->
        <contentType>
          <!--ro, req, string, file type, desc:corresponding to conditions-->test
        </contentType>
        <codecType>
          <!--ro, req, string, video encoding type-->test
        </codecType>
        <rateType>
          <!--ro, opt, string, video bit rate-->test
        </rateType>
        <playbackURI>
          <!--ro, opt, string, video URL,
desc:"name=" is followed by the video file name, and "size=" is followed by the size of the video file. The "starttime" and "endtime" are UTC time, but as the time returned by front-end devices is UTC time, while the time returned by DVR/NVR devices is the devices' local time, there are different mechanisms to deal with the time:
1. For DVR/NVR devices that return the local time:
    1) When the request URL does not include "timeType=STD" (/ISAPI/ContentMgmt/search), then the returned playbackURI will not include "timeType=STD". The "starttime" and "endtime" will refer to the devices' local time.
    2) When the request URL includes "timeType=STD" (/ISAPI/ContentMgmt/search&timeType=STD), and the devices support parsing "timeType" ("isSupportTimeTypeSTD" is true in GET /ISAPI/System/capabilities), then the returned playbackURI will include "timeType=STD". The "starttime" and "endtime" will refer to the UTC time in ISO 8601 format.
2. For IPC devices that return UTC time, whether or not the request URL includes "timeType=STD", the devices do not need to parse "timeType". The returned playbackURI will not include "timeType=STD", and the "starttime" and "endtime" will refer to the UTC time in ISO 8601 format.
3. To play back the video files, synchronize the playbackURI to the video playback API: /ISAPI/Streaming/tracks/<trackStreamID>?starttime=<startTime>&endtime=<endTimeime>&name=<fileName>&size=<fileSize>&timeType=<timeType>.-->rtsp://10.65.130.168/Streaming/tracks/201/?starttime=20190213T091134Z&amp;endtime=20190213T092116Z&amp;name=01010000093000100&amp;size=732335288;timeType=STD
        </playbackURI>
        <lockStatus>
          <!--ro, opt, enum, whether video file is locked, subType:string, desc:"lock", "unlock"-->lock
        </lockStatus>
        <remainLockTime>
          <!--ro, opt, int, remaining lock time, unit:s, desc:only returned when "lockStatus" is "lock". If 0 is returned, it indicates that it is locked permanently-->1
        </remainLockTime>
        <name>
          <!--ro, opt, string, file name-->test
        </name>
        <size>
          <!--ro, opt, int, file size-->0
        </size>
        <mediaID>
          <!--ro, opt, int, file ID, desc:unique identifier of file-->0
        </mediaID>
        <remark>
          <!--ro, opt, string, file remarks-->test
        </remark>
        <GPSInfo>
          <!--ro, opt, string, file GPS, desc:required, if no GPS information, a blank is returned-->test
        </GPSInfo>
        <recorderCode>
          <!--ro, opt, string, video recorder No.-->test
        </recorderCode>
        <uploadState>
          <!--ro, opt, enum, upload status, subType:string, desc:"uploaded", "nonUploaded", "uploadFailed", "uploading"-->uploaded
        </uploadState>
        <uploadTime>
          <!--ro, opt, datetime, video uploading time, desc:ISO8601 time (TD format, local time and time difference)-->1970-01-01T00:00:00+08:00
        </uploadTime>
        <policeCode>
          <!--ro, opt, string, police No., desc:sensitive information is encrypted-->test
        </policeCode>
```

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```
        <policeName>
          <!--ro, opt, string, police name, desc:sensitive information is encrypted-->test
        </policeName>
        <shootingTime>
          <!--ro, opt, datetime, time, desc:ISO8601 time (TD format, local time and time difference)-->1970-01-01T00:00:00+08:00
        </shootingTime>
        <FileTime>
          <!--ro, opt, int, the file is too long, unit:min-->1
        </FileTime>
        <remarks>
          <!--ro, opt, string, file remarks-->test
        </remarks>
        <cloudStorageType>
          <!--ro, opt, enum, storage server type, subType:string, desc:0 (cloud storage), 1 (CVR), 2 (pstor)-->0
        </cloudStorageType>
        <fileUrl>
          <!--ro, opt, string, file URL-->test
        </fileUrl>
        <thumbnailUrl>
          <!--ro, opt, string, thumbnail URL of picture or video-->test
        </thumbnailUrl>
        <encryption>
          <!--ro, opt, object, encryption-->
          <enabled>
            <!--ro, opt, bool, whether to encrypt the file-->true
          </enabled>
          <encryptionkey>
            <!--ro, opt, string, encryption key, desc:sensitive information is encrypted-->test
          </encryptionkey>
          <encryptionLevel>
            <!--ro, opt, enum, encryption level, subType:int, desc:1 (AES128), 2 (AES256)-->1
          </encryptionLevel>
        </encryption>
        <videoTargetTypeList>
          <!--ro, opt, object, list of target types, desc:this node is only applicable for HEOP protocol of intercom products. Several target types are
included in the time range. Every type is for a stream. Playback is according to video of corresponding time-->
          <videoTargetType>
            <!--ro, opt, enum, list of target types, subType:string, desc:this node is only applicable for HEOP protocol of intercom products. Several
target types are included in the time range. Every type is for a stream. Playback is according to video of corresponding time. "local", "peer"-->local
          </videoTargetType>
        </videoTargetTypeList>
        <TargetRegionList size="8">
          <!--ro, opt, array, search for the target region list of the picture, subType:object, attr:size{req, int}, desc:related URL:
/ISAPI/ContentMgmt/capabilities  targetRegionEventPicSearch-->
          <TargetRegion>
            <!--ro, opt, object, target area-->
            <regionID>
              <!--ro, req, string, detection area ID-->1
            </regionID>
            <RegionCoordinatesList>
              <!--ro, req, array, list of point coordinates, subType:object, range:[3,10], desc:rectangle-->
              <RegionCoordinates>
                <!--ro, opt, object, point coordinates, desc:the origin is the lower-left corner of the screen-->
                <positionX>
                  <!--ro, req, float, X-coordinate, range:[0,1]-->1
                </positionX>
                <positionY>
                  <!--ro, req, float, Y-coordinate, range:[0,1]-->1
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </TargetRegion>
        </TargetRegionList>
        <objectsThrownDetectionResult>
          <!--ro, opt, object, object thrown from building search result-->
          <alarmLevel>
            <!--ro, opt, int, alarm level, desc:it is categorized into three levels according to the target track and the actual event credibility. The
higher the level, the greater the credibility of the alarm being triggered by an actual object thrown from building event.-->0
          </alarmLevel>
          <floorID>
            <!--ro, opt, int, floor No., range:[-100,100]-->0
          </floorID>
        </objectsThrownDetectionResult>
        <synthesisType>
          <!--ro, opt, enum, subType:string-->auto
        </synthesisType>
        <bodyCameraShortSN>
          <!--ro, opt, string, range:[0,9]-->P20151543
        </bodyCameraShortSN>
        <shipsDetection>
          <!--ro, opt, object-->
          <shipName>
            <!--ro, opt, string, range:[1,64]-->浙嘉兴资03181
          </shipName>
          <shipsSpeed>
            <!--ro, opt, float, range:[0.0,300.0], unit:km/h-->23.4
          </shipsSpeed>
          <shipsLength>
            <!--ro, opt, float, range:[0.000,500.000], unit:m-->50.000
          </shipsLength>
          <shipLocation>
            <!--ro, opt, object-->
            <longitudeType>
              <!--ro, opt, enum, subType:string-->E
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```
            </longitudeType>
            <latitudeType>
              <!--ro, opt, enum, subType:string-->S
            </latitudeType>
            <longitude>
              <!--ro, opt, object-->
              <degree>
                <!--ro, opt, int, range:[0,180]-->0
              </degree>
              <minute>
                <!--ro, opt, int, range:[0,60]-->0
              </minute>
              <sec>
                <!--ro, opt, float, range:[0,60.000000]-->0.000
              </sec>
            </longitude>
            <latitude>
              <!--ro, opt, object-->
              <degree>
                <!--ro, opt, int, range:[0,180]-->0
              </degree>
              <minute>
                <!--ro, opt, int, range:[0,60]-->0
              </minute>
              <sec>
                <!--ro, opt, float, range:[0,60.000000]-->0.000
              </sec>
            </latitude>
          </shipLocation>
          <abnormalType>
            <!--ro, opt, enum, subType:int-->shipCourseDeviation
          </abnormalType>
          <shipChannelID>
            <!--ro, opt, int, range:[1,4]-->4
          </shipChannelID>
        </shipsDetection>
      </mediaSegmentDescriptor>
      <metadataMatches>
        <!--ro, opt, object, metadata match information-->
        <metadataDescriptor>
          <!--ro, req, string, metadata description, desc:eg: recordType.meta.com/MOTION. The supported video file type by a specific device can be
returned by opt in RacmCap->recordSearchType of /ISAPI/ContentMgmt/capabilities-->test
        </metadataDescriptor>
      </metadataMatches>
      <face>
        <!--ro, opt, object, face search result-->
        <name>
          <!--ro, opt, string, name-->test
        </name>
        <gender>
          <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
        </gender>
        <certificateType>
          <!--ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (the
other certificate)-->officerID
        </certificateType>
        <certificateNumber>
          <!--ro, opt, string, certificate No.-->test
        </certificateNumber>
        <currTemperature>
          <!--ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃-->36.5
        </currTemperature>
        <mask>
          <!--ro, opt, enum, whether the person wears a mask, subType:string, desc:"unknown", "no", "yes"-->unknown
        </mask>
        <phoneNumber>
          <!--ro, opt, string, phone number, range:[1,64], desc:it is valid when event type is face picture comparison-->test
        </phoneNumber>
        <FDID>
          <!--ro, opt, string, face picture library ID, range:[1,64], desc:it is valid when event type is face picture comparison-->test
        </FDID>
        <FDLibName>
          <!--ro, opt, string, face picture library name, range:[1,64], desc:it is valid when event type is face picture comparison-->test
        </FDLibName>
        <FDPicURL>
          <!--ro, opt, string, face picture library picture URL, desc:it is valid when event type is face picture comparison-->test
        </FDPicURL>
        <similarity>
          <!--ro, opt, float, comparison similarity, range:[0,1], step:0.001, desc:it is valid when event type is face picture comparison-->test
        </similarity>
      </face>
      <AlarmInNoList>
        <!--ro, opt, array, triggered alarm input channel list, subType:object-->
        <alarmInNo>
          <!--ro, req, int, triggered alarm input channel No.-->0
        </alarmInNo>
      </AlarmInNoList>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:if this node is not returned, the search result is not restricted by any rule-->test
      </ruleID>
      <roomIndex>
        <!--ro, opt, int, interrogation room No., desc:it starts from 1-->0
      </roomIndex>
```

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```
<StreamIDParam>
  <!--ro, opt, object, stream ID parameters-->
  <streamID>
    <!--ro, req, string, stream ID, range:[0,72]-->test
  </streamID>
  <cmdType>
    <!--ro, opt, int, trigger type, range:[0,255], desc:connect NVR with cloud storage to use it-->0
  </cmdType>
  <backupVolumeNum>
    <!--ro, opt, int, archive No., range:[0,255], desc:applicable to CVR-->0
  </backupVolumeNum>
  <archiveLabel>
    <!--ro, opt, string, archive No.-->test
  </archiveLabel>
</StreamIDParam>
<PosAlarm>
  <!--ro, opt, object, pos video-->
  <caseSensitive>
    <!--ro, req, bool, whether it is case-sensitive-->true
  </caseSensitive>
  <combinateMode>
    <!--ro, req, enum, keyword combination, subType:string, desc:"or", "and"-->or
  </combinateMode>
  <KeyWordList>
    <!--ro, opt, array, keyword list, subType:object, desc:supports a maximum of three words-->
    <keyWord>
      <!--ro, req, string, keyword, range:[0,128]-->test
    </keyWord>
  </KeyWordList>
</PosAlarm>
<TrialParam>
  <!--ro, opt, object, search conditions of trail information-->
  <caseNo>
    <!--ro, opt, string, case No., range:[0,56]-->test
  </caseNo>
  <caseName>
    <!--ro, opt, string, case name, range:[0,56]-->test
  </caseName>
  <litigant1>
    <!--ro, opt, string, party 1, range:[0,32]-->test
  </litigant1>
  <litigant2>
    <!--ro, opt, string, party 2, range:[0,32]-->test
  </litigant2>
  <chiefJudge>
    <!--ro, opt, string, jury, range:[0,32]-->test
  </chiefJudge>
  <caseType>
    <!--ro, opt, enum, case type, subType:int, desc:0 (all), 1 (penal case), 2 (civil case)-->0
  </caseType>
</TrialParam>
<plateNo>
  <!--ro, opt, string, license plate number, range:[1,64], desc:noPlate (vehicle without license plate), unknown (no license plate recognized),
XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string)-->蓝浙AX2P12
</plateNo>
<ConferenceParam>
  <!--ro, opt, object, video conference parameters-->
  <conferenceName>
    <!--ro, opt, string, conference name, range:[1,64], desc:from the "conferenceName" of POST /ISAPI/VCS/conferences-->test
  </conferenceName>
</ConferenceParam>
<ATMParam>
  <!--ro, opt, object, ATM search-->
  <transactionType>
    <!--ro, opt, enum, transaction type, subType:int, desc:0 (all), 1(search), 2(withdrawal), 3 (deposit), 4 (change password), 5(transfer), 6 (search
with no card), 7 (deposit with no card), 8 (cash swallowed), 9 (card swallowed) 10 (undefined)-->0
  </transactionType>
  <transationAmount>
    <!--ro, opt, int, amount, step:1, desc:unit: yuan-->0
  </transationAmount>
</ATMParam>
<labelTypeList>
  <!--ro, opt, object, tag type list of video file, desc:URL for configuring video tags: /ISAPI/ContentMgmt/record/control/remark?
format=json&security=<security>&iv=<iv>-->
  <labelTypes>
    <!--ro, opt, object, tag type of video file-->
    <labelType>
      <!--ro, opt, enum, tag type of video file, subType:string, desc:"importance", "unimportance"-->importance
    </labelType>
  </labelTypes>
</labelTypeList>
<isInFalseAlarmMaterialLib>
  <!--ro, opt, bool, whether imported to false alarm library, desc:not imported if this node is not returned or false is returned-->true
</isInFalseAlarmMaterialLib>
<vegetationSpectrumDetection>
  <!--ro, opt, object-->
  <vegetationIndex>
    <!--ro, opt, enum, subType:string-->NDVI
  </vegetationIndex>
  <vegetationIndexInitData>
    <!--ro, opt, enum, subType:string-->NDVI
  </vegetationIndexInitData>
  <reflectivityType>
    <!--ro, opt, enum, subType:string-->blue
```

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            <!--ro, opt, enum, subType:string-->blue
          </reflectivityType>
          <isVisibleImage>
            <!--ro, opt, bool-->true
          </isVisibleImage>
        </vegetationSpectrumDetection>
      </searchMatchItem>
      <dockStationAudioFile>
        <!--ro, opt, object, dock station audio file-->
        <analysisState>
          <!--ro, opt, enum, analysis status, subType:string, desc:"complete" (completed), "running", "notStarted" (not started)-->complete
        </analysisState>
        <keyWordList>
          <!--ro, opt, array, dock station audio file label, subType:object, range:[0,100]-->
          <keyWord>
            <!--ro, opt, string, label, range:[0,256]-->test
          </keyWord>
        </keyWordList>
      </dockStationAudioFile>
      <pictureType>
        <!--ro, opt, enum, subType:string-->licensePlatePicture
      </pictureType>
    </matchList>
    <totalMatches>
      <!--ro, opt, int, total number of matched results-->12
    </totalMatches>
  </CMSearchResult>
```

16.3.5.8 Get the capability of searching for videos by target type

Request URL

GET /ISAPI/ContentMgmt/SearchByTargetType/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "SearchDescriptionCap": {
    /*ro, req, object, search description capability*/
        "@size":  1024,
        /*ro, req, int, supported maximum number of search conditions*/
        "channelID": {
        /*ro, req, object, the range of the channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value of channel No.*/
            "@max":  256
            /*ro, req, int, the maximum value of channel No.*/
        },
        "targetTypes": {
        /*ro, req, object, the search target type supported by the device*/
            "@opt": ["human", "vehicle"]
            /*ro, req, array, options, subType:string, desc:"human", "vehicle"*/
        }
    },
    "SearchResultCap": {
    /*ro, opt, object, the capability of results searching*/
        "@size":  1024,
        /*ro, req, int, supported maximum number of search results*/
        "channelID": {
        /*ro, req, object, the range of the channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value of channel No.*/
            "@max":  256
            /*ro, req, int, the maximum value of channel No.*/
        },
        "targetTypes": {
        /*ro, req, object, the search target type supported by the device*/
            "@opt": ["human", "vehicle"]
            /*ro, req, array, options, subType:string, desc:"human", "vehicle"*/
        }
    }
}
```

16.3.5.9 Search for videos by target type

Request URL

POST /ISAPI/ContentMgmt/SearchByTargetType?format=json

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Query Parameter

None

Request Message

```
{
    "SearchDescription": {
    /*opt, object, search description*/
        "searchID":  "test",
        /*req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searches, the search history will be saved in the memory to speed up next search*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list*/
        "maxResults":  100,
        /*req, int, the maximum number of search results this time*/
        "SearchCondList": [
        /*req, array, list of search conditions, subType:object*/
            {
                "channelID":  1,
                /*req, int, channel No.*/
                "targetTypes": ["human", "vehicle"],
                /*req, array, search target type, subType:string, desc:"human", "vehicle"*/
                "searchTimeList": [
                /*opt, array, list of search period, subType:object, desc:if this node is not applied, it indicates that the search will not be performed by
time*/
                    {
                        "searchTime": {
                        /*req, object, search time period*/
                            "startTime":  "1970-01-01T08:00:00+08:00",
                            /*req, datetime, start time of the search, desc:ISO8601 time (TD format, local time and time difference)*/
                            "endTime":  "1970-01-01T00:09:00+08:00"
                            /*req, datetime, end time of the search, desc:ISO8601 time (TD format, local time and time difference)*/
                        }
                    }
                ]
            }
        ]
    }
}
```

Response Message

```
{
    "SearchResult": {
    /*ro, opt, object, search result*/
        "searchID":  "test",
        /*ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during
two searching, the search history will be saved in the memory to speed up next searching*/
        "responseStatus":  true,
        /*ro, req, bool, status of search results*/
        "numOfMatches":  100,
        /*ro, opt, int, number of results returned this time*/
        "totalMatches":  200,
        /*ro, opt, int, total number of matched results*/
        "responseStatusStrg":  "OK",
        /*ro, opt, enum, search status, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data)*/
        "matchList": [
        /*ro, req, array, the information list of returned results, subType:object*/
            {
                "channelID":  1,
                /*ro, req, int, channel No.*/
                "targetTypes": ["human", "vehicle"],
                /*ro, req, array, search target type, subType:string, desc:"human", "vehicle"; multiple elements in the array indicates multi-type, e.g.,
human and vehicle*/
                "RecordInfoList": [
                /*ro, req, array, list of video information, subType:object*/
                    {
                        "RecordTime": {
                        /*ro, req, object, video time range*/
                            "startTime":  "1970-01-01T08:00:00+08:00",
                            /*ro, req, datetime, video start time, desc:ISO8601 time (TD format, local time and time difference)*/
                            "endTime":  "1970-01-01T09:00:00+08:00"
                            /*ro, req, datetime, video end time, desc:ISO8601 time (TD format, local time and time difference)*/
                        }
                    }
                ]
            }
        ]
    }
}
```

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16.3.5.10 Search for video files by VCA event

Request URL

POST /ISAPI/ContentMgmt/SmartSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SmartSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, search condition, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of records for a single search (display on each page)-->1
  </maxResults>
  <trackID>
    <!--req, int, channel No. and stream type, desc:channel No. * 100 + stream type (-1: main stream, -2: sub-stream)-->1
  </trackID>
  <startTime>
    <!--req, datetime, start time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T00:00:00Z
  </startTime>
  <endTime>
    <!--req, datetime, end time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T01:00:00Z
  </endTime>
  <type opt="motionDetection,traversePlaneDetection,FieldDetection,faceDetection">
    <!--req, enum, type for search, subType:string, attr:opt{req, string}, desc:;"motionDetection" (motion detection); "traversePlaneDetection" (line crossing detection); "FieldDetection" (intrusion detection); "faceDetection" (facial detection)-->motionDetection
  </type>
  <MotionDetection>
    <!--opt, object, motion detection parameter, dep:and,{$.SmartSearchDescription.type,eq,motionDetection}-->
    <Grid>
      <!--opt, object, grid area-->
      <rowGranularity>
        <!--req, float, row-->0.000
      </rowGranularity>
      <columnGranularity>
        <!--req, float, column-->0.000
      </columnGranularity>
    </Grid>
    <MotionDetectionLayout>
      <!--opt, object, motion detection layout-->
      <layout>
        <!--opt, object, layout-->
        <gridMap>
          <!--opt, string, grid image, desc:hexadecimal, the image is divided in to N × N grids,each grid can be enabled (value: 1) or disabled (value 0) motion detection,each four horizontal grids consist of a hexadecimal number,e.g.,the corresponding hexadecimal number of "0011" is 3-->0x3
        </gridMap>
      </layout>
    </MotionDetectionLayout>
    <sensitivity min="1" max="100">
      <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </MotionDetection>
  <TraversePlane>
    <!--opt, object, line crossing parameters, dep:and,{$.SmartSearchDescription.type,eq,traversePlaneDetection}-->
    <TraversePlaneParam>
      <!--opt, object, line crossing-->
      <LineList>
        <!--opt, array, line list, subType:object-->
        <Line>
          <!--opt, object, line-->
          <StartPoint>
            <!--opt, object, start point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </StartPoint>
          <EndPoint>
            <!--opt, object, end point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </EndPoint>
        </Line>
      </LineList>
      <crossDirection opt="bothDirection,leftToRight,rightToLeft">
        <!--opt, enum, line crossing direction, subType:string, attr:opt{req, string}, desc:; "bothDirection" (bidirectional); "leftToRight" (from left to right); "rightToLeft" (from right to left)-->bothDirection
```

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        </crossDirection>
        <sensitivity min="1" max="100">
          <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
        </sensitivity>
        <planeHeight>
          <!--opt, int, plane height-->1
        </planeHeight>
        <detectionTarget opt="all,human,vehicle">
          <!--req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
        </detectionTarget>
      </TraversePlaneParam>
      <dwPreTime>
        <!--opt, int, pre-record-->1
      </dwPreTime>
      <dwDelayTime>
        <!--opt, int, post-record-->1
      </dwDelayTime>
      <PTZPos>
        <!--opt, object, PTZ coordinate-->
        <P>
          <!--opt, int, panning adjustment-->1
        </P>
        <T>
          <!--opt, int, tilting adjustment-->1
        </T>
        <Z>
          <!--opt, int, zooming adjustment-->1
        </Z>
      </PTZPos>
      <advancedType opt="humanFeature,objectFeature">
        <!--opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
      </advancedType>
      <advance>
        <!--opt, object, advanced parameters-->
        <HumanFeature>
          <!--opt, object, human body feature, dep:and,{$.SmartSearchDescription.TraversePlane.advancedType,eq,humanFeature}-->
          <age min="0" max="100">
            <!--opt, int, age, attr:min{req, int},max{req, int}-->1
          </age>
          <sex opt="male,female">
            <!--opt, enum, name, subType:string, attr:opt{req, string}, desc:; "male"; "female"-->male
          </sex>
          <eyeGlass>
            <!--opt, bool, whether the person wears glasses-->true
          </eyeGlass>
          <ageGroup>
            <!--opt, enum, age group, subType:string, desc:refer to the dictionary for details; "any" (no limitation, and used for search)-->any
          </ageGroup>
          <ageDeviation>
            <!--opt, int, age tolerance, range:[0,100]-->0
          </ageDeviation>
        </HumanFeature>
        <ObjectFeature>
          <!--opt, object, target features, dep:and,{$.SmartSearchDescription.TraversePlane.advancedType,eq,objectFeature}-->
          <colorRatel>
            <!--opt, int, color ratio-->1
          </colorRatel>
          <R>
            <!--opt, int, red-->1
          </R>
          <G>
            <!--opt, int, green-->1
          </G>
          <B>
            <!--opt, int, blue-->1
          </B>
        </ObjectFeature>
      </advance>
    </TraversePlane>
    <FieldDetection>
      <!--opt, object, intrusion detection, dep:and,{$.SmartSearchDescription.type,eq,FieldDetection}-->
      <normalizedScreenSize>
        <!--opt, object, normalized coordinates-->
        <normalizedScreenWidth>
          <!--opt, int, normalized width-->1
        </normalizedScreenWidth>
        <normalizedScreenHeight>
          <!--opt, int, normalized height-->1
        </normalizedScreenHeight>
      </normalizedScreenSize>
      <param>
        <!--opt, object, parameters-->
        <region>
          <!--opt, object, detection region-->
          <pointList>
            <!--opt, array, point list, subType:object, range:[0,4], desc:rectangle-->
            <point>
              <!--opt, object, point, desc:the origin is the upper-left corner of the screen-->
              <x>
                <!--req, int, X-coordinate, range:[0,1000]-->0
              </x>
              <y>
                <!--req, int, Y-coordinate, range:[0,1000]-->0
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```
            </y>
          </point>
        </pointList>
      </region>
    </param>
    <duration min="1" max="100">
      <!--opt, int, duration, attr:min{req, int},max{req, int}-->1
    </duration>
    <sensitivity min="1" max="100">
      <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <rate min="1" max="100">
      <!--opt, int, speed, attr:min{req, int},max{req, int}-->1
    </rate>
    <detectionTarget opt="all,human,vehicle">
      <!--req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
    </detectionTarget>
    <dwPreTime>
      <!--opt, int, pre-record-->1
    </dwPreTime>
    <dwDelayTime>
      <!--opt, int, Post-record-->1
    </dwDelayTime>
    <PTZPos>
      <!--opt, object, PTZ coordinate-->
      <P>
        <!--opt, int, panning adjustment-->1
      </P>
      <T>
        <!--opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
    </advancedType>
    <advance>
      <!--opt, object, advanced parameters-->
      <HumanFeature>
        <!--opt, object, human body feature, dep:and,{$.SmartSearchDescription.FieldDetection.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
          <!--opt, enum, name, subType:string, attr:opt{req, string}, desc:; "male"; "female"-->male
        </sex>
        <eyeGlass>
          <!--opt, bool, whether the person wears glasses-->true
        </eyeGlass>
        <ageGroup>
          <!--opt, enum, age group, subType:string, desc:refer to the dictionary for details; "any" (no limitation, and used for search)-->any
        </ageGroup>
        <ageDeviation>
          <!--opt, int, age tolerance, range:[0,100]-->0
        </ageDeviation>
      </HumanFeature>
      <ObjectFeature>
        <!--opt, object, target features, dep:and,{$.SmartSearchDescription.FieldDetection.advancedType,eq,objectFeature}-->
        <colorRatel>
          <!--opt, int, color ratio-->1
        </colorRatel>
        <R>
          <!--opt, int, red-->1
        </R>
        <G>
          <!--opt, int, green-->1
        </G>
        <B>
          <!--opt, int, blue-->1
        </B>
      </ObjectFeature>
    </advance>
  </FieldDetection>
  <FaceDetection>
    <!--opt, object, face detection, dep:and,{$.SmartSearchDescription.type,eq,faceDetection}-->
    <param>
      <!--opt, object, parameters-->
      <region>
        <!--opt, object, detection region-->
        <pointList>
          <!--opt, array, point list, subType:object, range:[0,4], desc:rectangle-->
          <point>
            <!--opt, object, point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </point>
```

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DETEKTOR

```
        </pointList>
      </region>
    </param>
    <duration min="1" max="100">
      <!--opt, int, duration, attr:min{req, int},max{req, int}-->1
    </duration>
    <sensitivity min="1" max="100">
      <!--opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <rate min="1" max="100">
      <!--opt, int, speed, attr:min{req, int},max{req, int}-->1
    </rate>
    <detectionTarget opt="all,human,vehicle">
      <!--req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
    </detectionTarget>
    <dwPreTime>
      <!--opt, int, pre-record-->1
    </dwPreTime>
    <dwDelayTime>
      <!--opt, int, post-record-->1
    </dwDelayTime>
    <PTZPos>
      <!--opt, object, PTZ coordinates-->
      <P>
        <!--opt, int, panning adjustment-->1
      </P>
      <T>
        <!--opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
    </advancedType>
    <advance>
      <!--opt, object, advanced parameters-->
      <HumanFeature>
        <!--opt, object, human body feature, dep:and,{$.SmartSearchDescription.FaceDetection.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
          <!--opt, enum, name, subType:string, attr:opt{req, string}, desc:; "male"; "female"-->male
        </sex>
        <eyeGlass>
          <!--opt, bool, whether the person wears glasses-->true
        </eyeGlass>
        <ageGroup>
          <!--opt, enum, age group, subType:string, desc:refer to the dictionary for details; "any" (no limitation, and used for search)-->any
        </ageGroup>
        <ageDeviation>
          <!--opt, int, age tolerance, range:[0,100]-->0
        </ageDeviation>
      </HumanFeature>
      <ObjectFeature>
        <!--opt, object, target features, dep:and,{$.SmartSearchDescription.FaceDetection.advancedType,eq,objectFeature}-->
        <colorRatel>
          <!--opt, int, color ratio-->1
        </colorRatel>
        <R>
          <!--opt, int, red-->1
        </R>
        <G>
          <!--opt, int, green-->1
        </G>
        <B>
          <!--opt, int, blue-->1
        </B>
      </ObjectFeature>
    </advance>
  </FaceDetection>
</SmartSearchDescription>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SmartSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search result, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search record ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one
during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:; “true” (succeeded); “false” (failed)-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, search status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched
data).-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <matchList>
    <!--ro, opt, array, matched result list, subType:object-->
    <searchMatchItem>
      <!--ro, opt, object, matched results-->
      <id>
        <!--ro, req, int, index No.-->1
      </id>
      <timeSpan>
        <!--ro, opt, object, time period-->
        <startTime>
          <!--ro, req, datetime, start time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T00:10:00Z
        </startTime>
        <endTime>
          <!--ro, req, datetime, end time, desc:ISO8601 time (TZ format: UTC time)-->1970-01-01T01:20:00Z
        </endTime>
      </timeSpan>
    </searchMatchItem>
  </matchList>
</SmartSearchResult>
```

16.3.5.11 Get capability of searching videos by VCA event

Request URL

GET /ISAPI/ContentMgmt/SmartSearch/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SmartSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search for videos by VCA event, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--ro, req, int, the start position of the search result in the result list-->1
  </searchResultPosition>
  <maxResults>
    <!--ro, req, int, the maximum number of search results this time-->1
  </maxResults>
  <trackID>
    <!--ro, req, int, channel No. and stream type, desc:channel ID,e.g.,101-main stream if first channel-->1
  </trackID>
  <startTime>
    <!--ro, req, datetime, start time-->1970-01-01T00:00:00+08:00
  </startTime>
  <endTime>
    <!--ro, req, datetime, end time-->1970-01-01T00:00:00+08:00
  </endTime>
  <type opt="motionDetection,traversePlaneDetection,FieldDetection,faceDetection">
    <!--ro, req, enum, type for search, subType:string, attr:opt{req, string}, desc:;”motionDetection” (motion detection); “traversePlaneDetection” (line
crossing detection); “FieldDetection” (intrusion detection); “faceDetection” (facial detection)-->motionDetection
  </type>
  <MotionDetection>
    <!--ro, opt, object, motion detection parameters, dep:and,{$.SmartSearchDescription.type,eq,motionDetection}-->
    <Grid>
```

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```
      <!--ro, opt, object, grid area-->
      <rowGranularity>
        <!--ro, req, float, row-->0.000
      </rowGranularity>
      <columnGranularity>
        <!--ro, req, float, column-->0.000
      </columnGranularity>
    </Grid>
    <MotionDetectionLayout>
      <!--ro, opt, object, motion detection layout-->
      <layout>
        <!--ro, opt, object, layout-->
        <gridMap>
          <!--ro, opt, string, grid image, desc:hexadecimal, the image is divided in to N × N grids,each grid can be enabled (value: 1) or disabled (value
0) motion detection,each four horizontal grids consist of a hexadecimal number,e.g.,the corresponding hexadecimal number of "0011" is 3-->0x3
        </gridMap>
      </layout>
    </MotionDetectionLayout>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </MotionDetection>
  <TraversePlane>
    <!--ro, opt, object, it is valid when type is "traversePlaneDetection", dep:and,{$.SmartSearchDescription.type,eq,traversePlaneDetection}-->
    <TraversePlaneParam>
      <!--ro, opt, object, line crossing-->
      <LineList>
        <!--ro, opt, array, line list, subType:object-->
        <Line>
          <!--ro, opt, object, line-->
          <StartPoint>
            <!--ro, opt, object, start point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </StartPoint>
          <EndPoint>
            <!--ro, opt, object, end point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </EndPoint>
        </Line>
      </LineList>
      <crossDirection opt="bothDirection,leftToRight,rightToLeft">
        <!--ro, opt, enum, crossing direction, subType:string, attr:opt{req, string}, desc:crossing direction-->bothDirection
      </crossDirection>
      <sensitivity min="1" max="100">
        <!--ro, opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
      </sensitivity>
      <planeHeight>
        <!--ro, opt, int, plane height-->1
      </planeHeight>
      <detectionTarget opt="all,human,vehicle">
        <!--ro, req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
      </detectionTarget>
    </TraversePlaneParam>
    <dwPreTime>
      <!--ro, opt, int, pre-recording time-->1
    </dwPreTime>
    <dwDelayTime>
      <!--ro, opt, int, post-recording time-->1
    </dwDelayTime>
    <PTZPos>
      <!--ro, opt, object, PTZ coordinates-->
      <P>
        <!--ro, opt, int, panning adjustment-->1
      </P>
      <T>
        <!--ro, opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--ro, opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--ro, opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
    </advancedType>
    <advance>
      <!--ro, opt, object, it is valid when type is "objectFeature"-->
      <HumanFeature>
        <!--ro, opt, object, human body feature, dep:and,{$.SmartSearchDescription.TraversePlane.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--ro, opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
          <!--ro, opt, enum, name, subType:string, attr:opt{req, string}, desc:; "male"; "female"-->male
```

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DETEKTOR

```
        </sex>
        <eyeGlass>
          <!--ro, opt, bool, whether the person wears glasses-->true
        </eyeGlass>
      </HumanFeature>
      <ObjectFeature>
        <!--ro, opt, object, target features, dep:and,{$.SmartSearchDescription.TraversePlane.advancedType,eq,objectFeature}-->
        <colorRatel>
          <!--ro, opt, int, color ratio-->1
        </colorRatel>
        <R>
          <!--ro, opt, int, red-->1
        </R>
        <G>
          <!--ro, opt, int, green-->1
        </G>
        <B>
          <!--ro, opt, int, blue-->1
        </B>
      </ObjectFeature>
    </advance>
  </TraversePlane>
  <FieldDetection>
    <!--ro, opt, object, intrusion detection, dep:and,{$.SmartSearchDescription.type,eq,FieldDetection}-->
    <normalizedScreenSize>
      <!--ro, opt, object, normalized coordinates-->
      <normalizedScreenWidth>
        <!--ro, opt, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, opt, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <param>
      <!--ro, opt, object, parameters-->
      <region>
        <!--ro, opt, object, area-->
        <pointList>
          <!--ro, opt, array, point list, subType:object, range:[0,4], desc:rectangle-->
          <point>
            <!--ro, opt, object, point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </point>
        </pointList>
      </region>
    </param>
    <duration min="1" max="100">
      <!--ro, opt, int, duration, attr:min{req, int},max{req, int}-->1
    </duration>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <rate min="1" max="100">
      <!--ro, opt, int, speed, attr:min{req, int},max{req, int}-->1
    </rate>
    <detectionTarget opt="all,human,vehicle">
      <!--ro, req, enum, detection target, subType:string, attr:opt{req, string}, desc:; "all" (humans and vehicles); "human"; "vehicle"-->all
    </detectionTarget>
    <dwPreTime>
      <!--ro, opt, int, pre-recording time-->1
    </dwPreTime>
    <dwDelayTime>
      <!--ro, opt, int, post-recording time-->1
    </dwDelayTime>
    <PTZPos>
      <!--ro, opt, object, PTZ coordinates-->
      <P>
        <!--ro, opt, int, panning adjustment-->1
      </P>
      <T>
        <!--ro, opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--ro, opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--ro, opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; "humanFeature" (human body feature); "objectFeature" (object
feature)-->humanFeature
    </advancedType>
    <advance>
      <!--ro, opt, object, it is valid when type is "objectFeature"-->
      <HumanFeature>
        <!--ro, opt, object, human body feature, dep:and,{$.SmartSearchDescription.FieldDetection.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--ro, opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
```

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DETEKTOR

```
          <!--ro, opt, enum, name, subType:string, attr:opt{req, string}, desc:; “male”; “female”-->male
        </sex>
        <eyeGlass>
          <!--ro, opt, bool, whether the person wears glasses-->true
        </eyeGlass>
      </HumanFeature>
      <ObjectFeature>
        <!--ro, opt, object, target features, dep:and,{$.SmartSearchDescription.FieldDetection.advancedType,eq,objectFeature}-->
        <colorRatel>
          <!--ro, opt, int, color ratio-->1
        </colorRatel>
        <R>
          <!--ro, opt, int, red-->1
        </R>
        <G>
          <!--ro, opt, int, green-->1
        </G>
        <B>
          <!--ro, opt, int, blue-->1
        </B>
      </ObjectFeature>
    </advance>
  </FieldDetection>
  <FaceDetection>
    <!--ro, opt, object, face detection, dep:and,{$.SmartSearchDescription.type,eq,faceDetection}-->
    <param>
      <!--ro, opt, object, parameters-->
      <region>
        <!--ro, opt, object, area-->
        <pointList>
          <!--ro, opt, array, point list, subType:object, range:[0,4], desc:rectangle-->
          <point>
            <!--ro, opt, object, point, desc:the origin is the upper-left corner of the screen-->
            <x>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </x>
            <y>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </y>
          </point>
        </pointList>
      </region>
    </param>
    <duration min="1" max="100">
      <!--ro, opt, int, duration, attr:min{req, int},max{req, int}-->1
    </duration>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <rate min="1" max="100">
      <!--ro, opt, int, speed, attr:min{req, int},max{req, int}-->1
    </rate>
    <detectionTarget opt="all,human,vehicle">
      <!--ro, req, enum, detection target, subType:string, attr:opt{req, string}, desc:; ”all” (humans and vehicles); “human”; “vehicle”-->all
    </detectionTarget>
    <dwPreTime>
      <!--ro, opt, int, pre-recording time-->1
    </dwPreTime>
    <dwDelayTime>
      <!--ro, opt, int, post-recording time-->1
    </dwDelayTime>
    <PTZPos>
      <!--ro, opt, object, PTZ coordinates-->
      <P>
        <!--ro, opt, int, panning adjustment-->1
      </P>
      <T>
        <!--ro, opt, int, tilting adjustment-->1
      </T>
      <Z>
        <!--ro, opt, int, zooming adjustment-->1
      </Z>
    </PTZPos>
    <advancedType opt="humanFeature,objectFeature">
      <!--ro, opt, enum, advanced parameter type, subType:string, attr:opt{req, string}, desc:; “humanFeature” (human body feature); “objectFeature” (object feature)-->humanFeature
    </advancedType>
    <advance>
      <!--ro, opt, object, it is valid when type is "objectFeature"-->
      <HumanFeature>
        <!--ro, opt, object, human body feature, dep:and,{$.SmartSearchDescription.FaceDetection.advancedType,eq,humanFeature}-->
        <age min="0" max="100">
          <!--ro, opt, int, age, attr:min{req, int},max{req, int}-->1
        </age>
        <sex opt="male,female">
          <!--ro, opt, enum, name, subType:string, attr:opt{req, string}, desc:; “male”; “female”-->male
        </sex>
        <eyeGlass>
          <!--ro, opt, bool, whether the person wears glasses-->true
        </eyeGlass>
      </HumanFeature>
      <ObjectFeature>
        <!--ro, opt, object, target features, dep:and,{$.SmartSearchDescription.FaceDetection.advancedType,eq,objectFeature}-->
        <colorRatel>
```

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DETEKTOR

```
            <colorRatel>
              <!--ro, opt, int, color ratio-->1
            </colorRatel>
            <R>
              <!--ro, opt, int, red-->1
            </R>
            <G>
              <!--ro, opt, int, green-->1
            </G>
            <B>
              <!--ro, opt, int, blue-->1
            </B>
          </ObjectFeature>
        </advance>
      </FaceDetection>
    </SmartSearchDescription>
```

16.3.5.12 Get the capability of searching for recording start and end time

Request URL

GET /ISAPI/ContentMgmt/time/search/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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DETEKTOR

```
{
    "TimeSearchCond": {
    /*ro, opt, object*/
        "@size":  1024,
        /*ro, req, int, the maximum number of search conditions that can be set*/
        "channelID": {
        /*ro, req, object, channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the maximum value*/
        },
        "streamType": {
        /*ro, opt, object, stream type*/
            "@opt": ["main", "sub", "adaptive"]
            /*ro, req, array, options, subType:string, desc:“main”, “sub”, “adaptive”*/
        }
    },
    "TimeSearchResult": {
    /*ro, opt, object*/
        "@size":  1024,
        /*ro, req, int, the maximum number of search results that can be returned*/
        "channelID": {
        /*ro, req, object, channel No.*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the maximum value*/
        },
        "streamType": {
        /*ro, opt, object, stream type*/
            "@opt": ["main", "sub", "adaptive"]
            /*ro, req, array, options, subType:string, desc:“main”, “sub”, “adaptive”*/
        },
        "hasRecord": {
        /*ro, req, object, whether the video exists (startTime and endTime are valid only when the value of hasRecord is "true")*/
            "@opt": [true, false]
            /*ro, req, array, options, subType:bool*/
        },
        "startTime": {
        /*ro, opt, object, recording start time (e.g.,2010-01-01T00:00:00+08:00, this field is required when the value of hasRecord is "true")*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  64
            /*ro, req, int, the maximum value*/
        },
        "endTime": {
        /*ro, opt, object, recording end time (e.g.,2010-01-01T00:00:00+08:00, this field is required when the value of hasRecord is "true")*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  64
            /*ro, req, int, the maximum value*/
        }
    }
}
```

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DETEKTOR

16.3.5.13 Search for the recording start and end time

Request URL

POST /ISAPI/ContentMgmt/time/search?format=json

Query Parameter

None

Request Message

```
{
    "TimeSearchCond": {
    /*opt, object*/
        "channelID":  1,
        /*req, int, channel No.*/
        "streamType":  "main"
        /*req, enum, stream type, subType:string, desc:"main" (main stream), "sub" (sub-stream), "adaptive" (stream self-adaption)*/
    }
}
```

Response Message

```
{
    "TimeSearchResult": {
    /*ro, opt, object*/
        "channelID":  1,
        /*ro, req, int, channel No.*/
        "streamType":  "main",
        /*ro, req, enum, stream type, subType:string, desc:"main" (main stream), "sub" (sub-stream), "adaptive" (stream self-adaption)*/
        "hasRecord":  true,
        /*ro, req, bool, whether video exists*/
        "startTime":  "1970-01-01T00:00:00+08:00",
        /*ro, opt, datetime, recording start time, dep:and,{$.TimeSearchResult.hasRecord,eq,true}*/
        "endTime":  "1970-01-01T00:00:00+08:00"
        /*ro, opt, datetime, recording end time, dep:and,{$.TimeSearchResult.hasRecord,eq,true}*/
    }
}
```

16.4 PTZ

16.4.1 Preset Management

16.4.1.1 Delete all patrols by channel.

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, read-only,describe the error reason in detail, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.4.1.2 Set parameters of all patrols for a specified channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, list of patrol parameters, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--opt, object, patrol parameters-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <id>
      <!--req, string, ID-->test
    </id>
    <patrolName>
      <!--req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--req, array, list of patrol results, subType:object-->
      <PatrolSequence>
        <!--req, object, patrol result-->
        <presetID>
          <!--req, string, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--req, string, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--req, string, delay-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.4.1.3 Get the parameters of all patrols of a specific channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/patrols

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrolList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PTZPatrol>
    <!--ro, opt, object, patrol parameters-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <patrolName>
      <!--ro, req, string, patrol name-->test
    </patrolName>
    <resumeType>
      <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
    </resumeType>
    <PatrolSequenceList>
      <!--ro, req, array, subType:object-->
      <PatrolSequence>
        <!--ro, req, object, list-->
        <presetID>
          <!--ro, req, string, preset ID-->1
        </presetID>
        <seqSpeed>
          <!--ro, req, string, patrol speed-->1
        </seqSpeed>
        <delay>
          <!--ro, req, int, delay, unit:ms-->1
        </delay>
      </PatrolSequence>
    </PatrolSequenceList>
  </PTZPatrol>
</PTZPatrolList>
```

16.4.1.4 Get patrol parameters of a specified channel

Request URL

GET /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <patrolName>
    <!--ro, req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--ro, opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--ro, req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--ro, req, object, patrol result-->
      <presetID>
        <!--ro, req, string, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--ro, req, string, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--ro, req, int, delay, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

16.4.1.5 Delete a patrol by channel

Request URL

DELETE /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

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16.4.1.6 Set path parameters of a single patrol for a specified channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/patrols/<patrolID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
patrolID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZPatrol xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, patrol parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <id>
    <!--req, string, ID-->test
  </id>
  <patrolName>
    <!--req, string, patrol name-->test
  </patrolName>
  <resumeType>
    <!--opt, enum, type, subType:string, desc:"relative", "absolute"-->relative
  </resumeType>
  <PatrolSequenceList>
    <!--req, array, patrol result list, subType:object-->
    <PatrolSequence>
      <!--req, object, patrol result-->
      <presetID>
        <!--req, string, preset ID-->1
      </presetID>
      <seqSpeed>
        <!--req, string, patrol speed-->1
      </seqSpeed>
      <delay>
        <!--req, int, delay, unit:ms-->1
      </delay>
    </PatrolSequence>
  </PatrolSequenceList>
</PTZPatrol>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.4.2 PTZ Control

16.4.2.1 Set the PTZ absolute positioning data of a specific channel

Request URL

PUT /ISAPI/PTZCtrl/channels/<channelID>/absolute

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PTZData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, PTZ absolute positioning data, attr:version{req, string, protocolVersion}-->
  <AbsoluteHigh>
    <!--opt, object, absolute height-->
    <elevation>
      <!--opt, int, altitude, range:[-900,2700]-->1
    </elevation>
    <azimuth>
      <!--opt, int, panning angle, range:[0,3600]-->1
    </azimuth>
    <absoluteZoom>
      <!--opt, int, absolute zoom, range:[0,1000]-->1
    </absoluteZoom>
  </AbsoluteHigh>
</PTZData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.5 People Counting

16.5.1 Real-Time People Counting

16.5.1.1 Get the people counting parameters of one video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Counting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
```

```
    <!--ro, req, int, normalized width-->1000
   </normalizedScreenWidth>
   <normalizedScreenHeight>
    <!--ro, req, int, normalized height-->1000
   </normalizedScreenHeight>
  </normalizedScreenSize>
  <MountingConfiguration>
   <!--ro, opt, object, installation parameters-->
   <viewingAngle>
    <!--ro, req, enum, "vertical,tilt", subType:string, desc:"tilt", "vertical"-->vertical
   </viewingAngle>
   <mountHeight>
    <!--ro, opt, int, unit: cm, range:[1,1000], unit:cm-->300
   </mountHeight>
   <horizontalDistance>
    <!--ro, opt, int, unit: cm, range:[1,1000], unit:cm-->30
   </horizontalDistance>
   <focalLength>
    <!--ro, opt, int, range:[1,1000], unit:mm-->1
   </focalLength>
  </MountingConfiguration>
  <OverlayConfiguration>
   <!--ro, opt, object, counting OSD parameters-->
   <enabled>
    <!--ro, req, bool, whether to enable the function-->true
   </enabled>
   <positionX>
    <!--ro, req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
   </positionX>
   <positionY>
    <!--ro, req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
   </positionY>
   <OSDType>
    <!--ro, opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->none
   </OSDType>
   <child>
    <!--ro, opt, bool, whether to enable children counting for OSD-->true
   </child>
   <OSDList>
    <!--ro, opt, array, subType:object, range:[0,9]-->
    <OSDType>
     <!--ro, opt, enum, subType:string-->enter
    </OSDType>
   </OSDList>
  </OverlayConfiguration>
  <Demarcation>
   <!--ro, opt, object-->
   <enabled>
    <!--ro, req, bool, whether to enable the function-->true
   </enabled>
   <DemarcationRegionList>
    <!--ro, req, array, subType:object-->
    <DemarcationRegion>
     <!--ro, opt, object-->
     <id>
      <!--ro, req, int-->1
     </id>
     <RegionCoordinatesList>
      <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
       <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
       <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
       </positionX>
       <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
       </positionY>
      </RegionCoordinates>
     </RegionCoordinatesList>
    </DemarcationRegion>
   </DemarcationRegionList>
   <DemarcationLine>
    <!--ro, req, object-->
    <StartPoint>
     <!--ro, req, object, start point, desc:the origin is the lower-left corner of the screen-->
     <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->1
     </positionX>
     <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
     </positionY>
    </StartPoint>
    <EndPoint>
     <!--ro, req, object, end point, desc:the origin is the lower-left corner of the screen-->
     <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->1
     </positionX>
     <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
     </positionY>
    </EndPoint>
   </DemarcationLine>
  </Demarcation>
```

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```
<CountingRegionType>
  <!--ro, req, enum, detection type, subType:string, desc:read-only,detection type: "region,line"-->region
</CountingRegionType>
<detectionPartType>
  <!--ro, opt, enum, subType:string-->headAndShoulders
</detectionPartType>
<CountingRegionList>
  <!--ro, opt, array, detection area list, subType:object-->
  <CountingRegion>
    <!--ro, opt, object, detection area-->
    <id>
      <!--ro, req, int, detection area ID-->1
    </id>
    <direction>
      <!--ro, opt, object-->
      <startPoint>
        <!--ro, req, object-->
        <positionX>
          <!--ro, req, int, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->1
        </positionY>
      </startPoint>
      <endPoint>
        <!--ro, req, object-->
        <positionX>
          <!--ro, req, int, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->1
        </positionY>
      </endPoint>
    </direction>
    <sensitivityLevel>
      <!--ro, opt, int, target detection sensitivity, range:[1,100]-->50
    </sensitivityLevel>
    <spaceGenerationSpeed>
      <!--ro, opt, int, range:[1,100]-->50
    </spaceGenerationSpeed>
    <timeGenerationSpeed>
      <!--ro, opt, int, range:[1,100]-->50
    </timeGenerationSpeed>
    <countingSpeed>
      <!--ro, opt, int, range:[1,100]-->50
    </countingSpeed>
    <detectionType>
      <!--ro, opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
    </detectionType>
    <objectSizeCorrection>
      <!--ro, opt, int, range:[1,100]-->50
    </objectSizeCorrection>
    <RegionCoordinatesList>
      <!--ro, opt, array, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <RuleRegionCoordinatesList>
      <!--ro, opt, array, rule frame, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RuleRegionCoordinatesList>
    <RuleRegionLimit>
      <!--ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->1
      </height>
      <width>
        <!--ro, req, int, width, range:[0,1000]-->1
      </width>
    </RuleRegionLimit>
    <enabled>
      <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
    </enabled>
```

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```
      </enabled>
      <regionColor>
        <!--ro, opt, enum, subType:string-->blue
      </regionColor>
      <TrajectoryCountFilter>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <residenceTime>
          <!--ro, opt, int, range:[1,60], unit:s-->10
        </residenceTime>
      </TrajectoryCountFilter>
    </CountingRegion>
  </CountingRegionList>
  <CountingLineItemList>
    <!--ro, opt, array, list of detection lines, subType:object-->
    <CountingLineItem>
      <!--ro, opt, object, detection line-->
      <id>
        <!--ro, req, int, detection line ID-->1
      </id>
      <direction>
        <!--ro, req, object, detection line direction-->
        <startPoint>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--ro, req, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <sensitivityLevel>
        <!--ro, opt, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
      </sensitivityLevel>
      <spaceGenerationSpeed>
        <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
      </spaceGenerationSpeed>
      <timeGenerationSpeed>
        <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
      </timeGenerationSpeed>
      <countingSpeed>
        <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
      </countingSpeed>
      <detectionType>
        <!--ro, opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
      </detectionType>
      <objectSizeCorrection>
        <!--ro, opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
      </objectSizeCorrection>
      <LineCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <Coordinates>
          <!--ro, req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </Coordinates>
      </LineCoordinatesList>
      <polyline>
        <!--ro, opt, bool, this node should be set to "true" when displaying polyline; otherwise,this node does not exist, desc:this node should be set to
"true" when displaying polyline; otherwise, this node does not exist-->true
      </polyline>
    </CountingLineItem>
  </CountingLineItemList>
  <dataUploadCycle>
    <!--ro, opt, int, interval for uploading data, unit:min-->1
  </dataUploadCycle>
  <SECUploadEnabled>
    <!--ro, opt, bool, uploading mechanism for each second-->true
  </SECUploadEnabled>
  <InterferenceSuppression>
    <!--ro, opt, object-->
    <shadow>
      <!--ro, opt, bool-->true
    </shadow>
    <loitering>
      <!--ro, opt, bool-->true
    </loitering>
    <cart>
```

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```
      <!--ro, opt, bool-->true
    </cart>
  </InterferenceSuppression>
  <EmailReport>
    <!--ro, opt, object-->
    <DayReport>
      <!--ro, opt, bool, daily report-->true
    </DayReport>
    <WeekReport>
      <!--ro, opt, bool, weekly report-->true
    </WeekReport>
    <MonthReport>
      <!--ro, opt, bool, monthly report-->true
    </MonthReport>
    <YearReport>
      <!--ro, opt, bool, yearly report-->true
    </YearReport>
    <reportFormat>
      <!--ro, opt, string, email report format, desc:"excel" (default), "csv", "txt", "xml"-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--ro, opt, object-->
    <calibrateType>
      <!--ro, opt, enum, calibration type, subType:string, desc:"automatic,manual,no",calibration types,including Auto,Manual,and None-->automatic
    </calibrateType>
    <SetupParam>
      <!--ro, req, object, installation parameters-->
      <height>
        <!--ro, opt, float, height, range:[200.0,500.0], unit:cm, desc:the default value is 300-->300.0
      </height>
      <tiltAngle>
        <!--ro, opt, float, angle of pitch, range:[-90.0,90.0], unit:°, desc:the default value is 0-->0.0
      </tiltAngle>
      <heelAngle>
        <!--ro, opt, float, angle of inclination, range:[0.0,180.0], unit:°, desc:the default value is 0-->0.0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--ro, req, object, counting area (in red frame)-->
      <RegionCoordinatesList>
        <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
      <!--ro, opt, object, it is valid when calibrateType is "automaticCalib"-->
      <CalibRegion>
        <!--ro, opt, object, calibration area (in green frame)-->
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </CalibRegion>
    </AutomaticCalib>
  </CountingCalibrate>
  <HeightFilterOverlay>
    <!--ro, opt, object, height filter-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <heightFilter>
      <!--ro, opt, int, height filter, range:[0,200], unit:cm-->0
    </heightFilter>
  </HeightFilterOverlay>
  <countingType>
    <!--ro, opt, enum, method to trigger counting, subType:string, desc:“none”, “alarmInputTrigger”, “videoTrigger"-->none
  </countingType>
  <signalType>
    <!--ro, opt, enum, signal type, subType:string, desc:"level”, “pulses"-->level
  </signalType>
  <RS485TransmissionEnabled>
    <!--ro, opt, bool, "false,true"-->true
  </RS485TransmissionEnabled>
  <dailyResetTime>
    <!--ro, opt, time, e.g.,"22:00:00+08:00", desc:e.g.,"22:00:00+08:00"-->22:00:00+08:00
  </dailyResetTime>
```

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```
<streamOverlayRuleInfos>
  <!--ro, opt, bool, whether to enable overlaying rule information in stream-->true
</streamOverlayRuleInfos>
<detectThreshold>
  <!--ro, opt, int, global detection threshold, range:[0,100], desc:the default value is 90-->90
</detectThreshold>
<aidedTrackEnabled>
  <!--ro, opt, bool, whether to enable auxiliary tracking, desc:it is enabled by default-->true
</aidedTrackEnabled>
<TargetRegionCoordinatesList>
  <!--ro, opt, array, target frame, subType:object, range:[0,4], desc:only supports square-->
  <RegionCoordinates>
    <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
    </positionY>
  </RegionCoordinates>
</TargetRegionCoordinatesList>
<ChildFilter>
  <!--ro, opt, object, counting children-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <heightThreshold>
    <!--ro, opt, int, height threshold of children,the counting will start when the height of children is higher than configured height threshold,unit:
cm,value range: [100,200], range:[100,200], unit:cm, desc:height threshold of children,the counting will start when the height of children is higher than
configured height threshold,unit: cm,value range: [100,200]-->150
  </heightThreshold>
</ChildFilter>
<MisinfoFilter>
  <!--ro, opt, object, false alarm filtering-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <threshold>
    <!--ro, opt, int, threshold of false alarm filtering, range:[0,100], desc:the value ranges from 0 to 100 and the default value is 15. Larger threshold
value corresponds to higher precision-->15
  </threshold>
  <timesLimit>
    <!--ro, opt, int, judge times of false alarm, range:[1,5], desc:value range: [1,5], default value: 3, more attempts corresponds to higher precision--
>3
  </timesLimit>
</MisinfoFilter>
<detectionMode>
  <!--ro, opt, enum, target detection mode, subType:string, desc:detection mode: single-detect based on tracking algorithm,double-detect based on depth
map,double_single-detect based on depth map mainly and tracking algorithm secondarily,single_double-detect based on tracking algorithm mainly and depth map
secondarily-->single
</detectionMode>
<TrajectoryCountFilter>
  <!--ro, opt, object, pattern counting filtering-->
  <enabled>
    <!--ro, req, bool, whether to enable the function, desc:whether to enable the function-->true
  </enabled>
  <movementDisplacement>
    <!--ro, opt, int, displacement range, range:[0,200], unit:cm-->40
  </movementDisplacement>
  <residenceTime>
    <!--ro, opt, float, dwell time duration, range:[0.0,10.0], unit:s-->0.1
  </residenceTime>
</TrajectoryCountFilter>
<RegionsDirectionList>
  <!--ro, opt, array, counting direction, subType:object-->
  <RegionsDirection>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->1
    </id>
    <StartRegionCoordinatesList>
      <!--ro, req, array, start region, subType:object, range:[3,10], desc:start region-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </StartRegionCoordinatesList>
    <EndRegionCoordinatesList>
      <!--ro, req, array, end region, subType:object, range:[3,10], desc:end region-->
      <RegionCoordinates>
        <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,10]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,10]-->1
        </positionY>
      </RegionCoordinates>
    </EndRegionCoordinatesList>
```

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```
    </RegionsDirection>
  </RegionsDirectionList>
  <maintenanceModeEnabled>
    <!--ro, opt, bool, whether to enable maintenance mode-->true
  </maintenanceModeEnabled>
  <OverlayConfigurationList>
    <!--ro, opt, array, subType:object-->
    <OverlayConfiguration>
      <!--ro, opt, object, counting OSD parameters-->
      <id>
        <!--ro, req, int-->1
      </id>
      <name>
        <!--ro, opt, string-->test
      </name>
      <enabled>
        <!--ro, req, bool, whether to enable OSD overlay, desc:whether to enable OSD overlay-->true
      </enabled>
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
      </positionY>
      <OSDType>
        <!--ro, opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->enter
      </OSDType>
      <OSDList>
        <!--ro, opt, array, subType:object, range:[0,9]-->
        <OSDType>
          <!--ro, opt, enum, subType:string-->enter
        </OSDType>
      </OSDList>
    </OverlayConfiguration>
  </OverlayConfigurationList>
  <analysisMode>
    <!--ro, opt, enum, analysis mode, subType:string, desc:"realTimeMode"-real time, "autoSwitchMode"-auto switch-->realTimeMode
  </analysisMode>
  <countingSceneMode>
    <!--ro, opt, enum, subType:string-->entrance
  </countingSceneMode>
  <flowDirectionAnalysis>
    <!--ro, opt, bool-->true
  </flowDirectionAnalysis>
  <dataUploadType>
    <!--ro, opt, object-->
    <enterEnabled>
      <!--ro, opt, bool-->true
    </enterEnabled>
    <leaveEnabled>
      <!--ro, opt, bool-->true
    </leaveEnabled>
  </dataUploadType>
  <InterPupileRegion>
    <!--ro, opt, object-->
    <positionX>
      <!--ro, req, int, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, range:[0,1000]-->50
    </width>
    <height>
      <!--ro, req, int, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--ro, opt, object-->
    <positionX>
      <!--ro, req, int, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, range:[0,1000]-->50
    </width>
    <height>
      <!--ro, req, int, range:[0,1000]-->50
    </height>
  </MaxInterPupilRegion>
  <SizeFilter>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <filterMultiple>
      <!--ro, opt, int, range:[4,6]-->0
    </filterMultiple>
  </SizeFilter>
  <expertModeEnabled>
```

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```
    <!--ro, opt, bool-->false
  </expertModeEnabled>
  <debugEnabled>
    <!--ro, opt, bool-->false
  </debugEnabled>
  <congestion>
    <!--ro, opt, object-->
    <minPeopleNum>
      <!--ro, opt, int, range:[0,999]-->100
    </minPeopleNum>
    <maxFlowRate>
      <!--ro, opt, int, range:[0,9999]-->1000
    </maxFlowRate>
  </congestion>
</Counting>
```

16.5.1.2 Set the people counting parameters for a specific channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Counting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <MountingConfiguration>
    <!--opt, object, installation parameters-->
    <viewingAngle>
      <!--req, enum, "vertical,tilt", subType:string, desc:"tilt", "vertical"-->vertical
    </viewingAngle>
    <mountHeight>
      <!--opt, int, unit: cm, range:[1,1000], unit:cm-->300
    </mountHeight>
    <horizontalDistance>
      <!--opt, int, unit: cm, range:[1,1000], unit:cm-->30
    </horizontalDistance>
    <focalLength>
      <!--opt, int, range:[1,1000], unit:mm-->1
    </focalLength>
  </MountingConfiguration>
  <OverlayConfiguration>
    <!--opt, object, counting OSD parameters-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <OSDType>
      <!--opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->none
    </OSDType>
    <child>
      <!--opt, bool, whether to enable children counting for OSD-->true
    </child>
  </OverlayConfiguration>
  <Demarcation>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <DemarcationRegionList>
      <!--req, array, subType:object-->
      <DemarcationRegion>
```

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```
        <!--opt, object-->
        <id>
          <!--req, int-->1
        </id>
        <RegionCoordinatesList>
          <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </DemarcationRegion>
    </DemarcationRegionList>
    <DemarcationLine>
      <!--req, object-->
      <StartPoint>
        <!--req, object, start point, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </StartPoint>
      <EndPoint>
        <!--req, object, end point, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </EndPoint>
    </DemarcationLine>
  </Demarcation>
  <CountingRegionType>
    <!--req, enum, detection type, subType:string, desc:"region", "line"-->region
  </CountingRegionType>
  <CountingRegionList>
    <!--opt, array, detection area list, subType:object-->
    <CountingRegion>
      <!--opt, object, detection area-->
      <id>
        <!--req, int, detection area ID-->1
      </id>
      <direction>
        <!--req, object-->
        <startPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <sensitivityLevel>
        <!--opt, int, target detection sensitivity, range:[1,100]-->50
      </sensitivityLevel>
      <spaceGenerationSpeed>
        <!--opt, int, range:[1,100]-->50
      </spaceGenerationSpeed>
      <timeGenerationSpeed>
        <!--opt, int, range:[1,100]-->50
      </timeGenerationSpeed>
      <countingSpeed>
        <!--opt, int, range:[1,100]-->50
      </countingSpeed>
      <detectionType>
        <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
      </detectionType>
      <objectSizeCorrection>
        <!--opt, int, range:[1,100]-->50
      </objectSizeCorrection>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
```

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```
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <RuleRegionCoordinatesList>
        <!--opt, array, rule frame, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RuleRegionCoordinatesList>
      <RuleRegionLimit>
        <!--opt, object, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
        <height>
          <!--req, int, height, range:[0,1000]-->1
        </height>
        <width>
          <!--req, int, width, range:[0,1000]-->1
        </width>
      </RuleRegionLimit>
      <enabled>
        <!--opt, bool-->true
      </enabled>
    </CountingRegion>
  </CountingRegionList>
  <CountingLineItemList>
    <!--opt, array, list of detection lines, subType:object-->
    <CountingLineItem>
      <!--opt, object, detection line-->
      <id>
        <!--req, int, detection line ID-->1
      </id>
      <direction>
        <!--req, object, detection line direction-->
        <startPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </startPoint>
        <endPoint>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </endPoint>
      </direction>
      <sensitivityLevel>
        <!--opt, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
      </sensitivityLevel>
      <spaceGenerationSpeed>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </spaceGenerationSpeed>
      <timeGenerationSpeed>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </timeGenerationSpeed>
      <countingSpeed>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </countingSpeed>
      <detectionType>
        <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
      </detectionType>
      <objectSizeCorrection>
        <!--opt, int, range:[1,100], desc:the default value is 50-->50
      </objectSizeCorrection>
      <LineCoordinatesList>
        <!--opt, array, subType:object, range:[0,2]-->
        <Coordinates>
          <!--req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
```

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```
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </Coordinates>
      </LineCoordinatesList>
      <polyline>
        <!--opt, bool, this node should be set to "true" when displaying polyline; otherwise,this node does not exist, desc:this node should be set to
"true" when displaying polyline; otherwise, this node does not exist-->true
      </polyline>
    </CountingLineItem>
  </CountingLineItemList>
  <dataUploadCycle>
    <!--opt, int, interval for uploading data, unit:min-->1
  </dataUploadCycle>
  <SECUploadEnabled>
    <!--opt, bool, uploading data for each second-->true
  </SECUploadEnabled>
  <InterferenceSuppression>
    <!--opt, object-->
    <shadow>
      <!--opt, bool-->true
    </shadow>
    <loitering>
      <!--opt, bool-->true
    </loitering>
    <cart>
      <!--opt, bool-->true
    </cart>
  </InterferenceSuppression>
  <EmailReport>
    <!--opt, object-->
    <DayReport>
      <!--opt, bool, daily report-->true
    </DayReport>
    <WeekReport>
      <!--opt, bool, weekly report-->true
    </WeekReport>
    <MonthReport>
      <!--opt, bool, monthly report-->true
    </MonthReport>
    <YearReport>
      <!--opt, bool, yearly report-->true
    </YearReport>
    <reportFormat>
      <!--opt, string, email report format, desc:"excel" (default), "csv", "txt", "xml"-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--opt, object-->
    <calibrateType>
      <!--opt, enum, calibration type, subType:string, desc:"automatic,manual,no",calibration types,including Auto,Manual,and None-->automatic
    </calibrateType>
    <SetupParam>
      <!--req, object, installation parameters-->
      <height>
        <!--opt, float, height, range:[200.0,500.0], unit:cm, desc:the default value is 300-->300.0
      </height>
      <tiltAngle>
        <!--opt, float, angle of pitch, range:[-90.0,90.0], unit:°, desc:the default value is 0-->0.0
      </tiltAngle>
      <heelAngle>
        <!--opt, float, angle of inclination, range:[0.0,180.0], unit:°, desc:the default value is 0-->0.0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--req, object, counting area (in red frame)-->
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
      <!--opt, object, auto calibration-->
      <CalibRegion>
        <!--opt, object, calibration area (in green frame)-->
        <RegionCoordinatesList>
          <!--opt, array, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--req, object, coordinates of the area, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
```

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```
        </RegionCoordinatesList>
      </CalibRegion>
    </AutomaticCalib>
  </CountingCalibrate>
  <HeightFilterOverlay>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
    </enabled>
    <heightFilter>
      <!--opt, int, filter height, range:[0,200], unit:cm-->0
    </heightFilter>
  </HeightFilterOverlay>
  <countingType>
    <!--opt, enum, method to trigger counting, subType:string, desc:method to trigger counting: "none,alarmInputTrigger,videoTrigger"-->none
  </countingType>
  <signalType>
    <!--opt, enum, type of the signal, subType:string, desc:signal type: "level,pulses",this node is valid when <countingType> is set to "videoTrigger" or
"alarmInputTrigger"-->level
  </signalType>
  <RS485TransmissionEnabled>
    <!--opt, bool, "false,true"-->true
  </RS485TransmissionEnabled>
  <dailyResetTime>
    <!--opt, time, e.g.,"22:00:00+08:00", desc:e.g.,"22:00:00+08:00"-->22:00:00+08:00
  </dailyResetTime>
  <streamOverlayRuleInfos>
    <!--opt, bool, whether to enable overlaying rule information in stream-->true
  </streamOverlayRuleInfos>
  <detectThreshold>
    <!--opt, int, global detection threshold, range:[0,100], desc:the default value is 90-->90
  </detectThreshold>
  <aidedTrackEnabled>
    <!--opt, bool, whether to enable auxiliary tracking, desc:by default, it is enabled-->true
  </aidedTrackEnabled>
  <TargetRegionCoordinatesList>
    <!--opt, array, target frame, subType:object, range:[0,4], desc:only supports square-->
    <RegionCoordinates>
      <!--req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </TargetRegionCoordinatesList>
  <ChildFilter>
    <!--opt, object, counting children-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <heightThreshold>
      <!--opt, int, threshold of children's height, range:[100,200], unit:cm, desc:the counting will start when the height of children is higher than the
configured height threshold-->150
    </heightThreshold>
  </ChildFilter>
  <MisinfoFilter>
    <!--opt, object, whether to filter false alarms-->
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <threshold>
      <!--opt, int, threshold of false alarm filtering, range:[0,100], desc:the value ranges from 0 to 100 and the default value is 15. Larger threshold
value corresponds to higher precision-->15
    </threshold>
    <timesLimit>
      <!--opt, int, limit of times for checking false alarms, range:[1,5], desc:the value ranges from 1 to 5 and the default value is 3. Larger value
corresponds to higher precision-->3
    </timesLimit>
  </MisinfoFilter>
  <detectionMode>
    <!--opt, enum, target detection mode, subType:string, desc:"single" (detect based on tracking algorithm), "double" (detect based on depth map),
"double_single" (detect based on depth map mainly and tracking algorithm secondarily), "single_double" (detect based on tracking algorithm mainly and depth
map secondarily)-->single
  </detectionMode>
  <TrajectoryCountFilter>
    <!--opt, object, track counting filtering-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <movementDisplacement>
      <!--opt, int, displacement range, range:[0,200], unit:cm-->40
    </movementDisplacement>
    <residenceTime>
      <!--opt, float, dwell time duration, range:[0.0,10.0], unit:s-->0.1
    </residenceTime>
  </TrajectoryCountFilter>
  <RegionsDirectionList>
    <!--opt, array, counting direction, subType:object-->
    <RegionsDirection>
      <!--opt, object, area direction-->
      <id>
        <!--req, int, detection area ID-->1
```

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```
        </id>
        <StartRegionCoordinatesList>
          <!--req, array, start area, subType:object, range:[3,10], desc:start region-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </StartRegionCoordinatesList>
        <EndRegionCoordinatesList>
          <!--req, array, end area, subType:object, range:[3,10], desc:end region-->
          <RegionCoordinates>
            <!--req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,10]-->1
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,10]-->1
            </positionY>
          </RegionCoordinates>
        </EndRegionCoordinatesList>
      </RegionsDirection>
    </RegionsDirectionList>
    <maintenanceModeEnabled>
      <!--opt, bool, whether to enable maintenance mode-->true
    </maintenanceModeEnabled>
    <OverlayConfigurationList>
      <!--opt, array, subType:object-->
      <OverlayConfiguration>
        <!--opt, object, counting OSD parameters-->
        <id>
          <!--req, int-->1
        </id>
        <name>
          <!--opt, string-->test
        </name>
        <enabled>
          <!--req, bool, whether to enable OSD overlay-->true
        </enabled>
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000], desc:the origin is the lower-left corner of the screen-->1
        </positionY>
        <OSDType>
          <!--opt, enum, OSD type, subType:string, desc:it is valid when enabled is "true","enter,leave,entreLeave,peoplePassing"-->enter
        </OSDType>
      </OverlayConfiguration>
    </OverlayConfigurationList>
    <analysisMode>
      <!--opt, enum, analysis mode, subType:string, desc:"realTimeMode"-real time, "autoSwitchMode"-auto switch-->realTimeMode
    </analysisMode>
    <countingSceneMode>
      <!--opt, enum, subType:string-->entrance
    </countingSceneMode>
    <flowDirectionAnalysis>
      <!--opt, bool-->true
    </flowDirectionAnalysis>
    <dataUploadType>
      <!--opt, object-->
      <enterEnabled>
        <!--opt, bool-->true
      </enterEnabled>
      <leaveEnabled>
        <!--opt, bool-->true
      </leaveEnabled>
    </dataUploadType>
  </Counting>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.5.1.3 Get capability of getting people counting statistics

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/counting/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CountingCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <CountingRegionType opt="line,region,lineAndRegion">
    <!--ro, opt, string, area type, attr:opt{opt, string}, desc:"line"-->line
  </CountingRegionType>
  <dataUploadCycle opt="1,5,10,15,20,30,60">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </dataUploadCycle>
  <isSupportSECUpload>
    <!--ro, opt, bool-->true
  </isSupportSECUpload>
  <isSupportRecommendValue>
    <!--ro, opt, bool, "true,false"-->true
  </isSupportRecommendValue>
  <isSupportFlashRemoveCouting>
    <!--ro, opt, bool, "true"-->true
  </isSupportFlashRemoveCouting>
  <OSDType opt="enter,leave,enterLeave,
none,enterPeoplePassing,pepolePassing,enterDuplicatePeople,enterLeaveDuplicatePeople,enterLeaveDuplicatePassByShop,enterLeaveDuplicatePassing" def="none">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{opt, string}-->enter
  </OSDType>
  <OverlayConfiguration>
    <!--ro, opt, object-->
    <OSDList size="9">
      <!--ro, opt, object, attr:size{req, int}-->
      <OSDType opt="enter,leave,none">
        <!--ro, opt, string, attr:opt{req, string}-->enter
      </OSDType>
    </OSDList>
  </OverlayConfiguration>
  <InterferenceSuppression>
    <!--ro, opt, object-->
    <shadow opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </shadow>
    <loitering opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </loitering>
    <cart opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </cart>
  </InterferenceSuppression>
  <EmailReport>
    <!--ro, opt, object-->
    <DayReport opt="true,false">
```

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      <!--ro, opt, bool, daily report, attr:opt{req, bool}-->true
    </DayReport>
    <WeekReport opt="true,false">
      <!--ro, opt, bool, weekly report, attr:opt{req, bool}-->true
    </WeekReport>
    <MonthReport opt="true,false">
      <!--ro, opt, bool, monthly report, attr:opt{req, bool}-->true
    </MonthReport>
    <YearReport opt="true,false">
      <!--ro, opt, bool, yearly report, attr:opt{req, bool}-->true
    </YearReport>
    <reportFormat opt="excel,csv,txt,xml">
      <!--ro, opt, string, email report format, attr:opt{req, string}-->excel
    </reportFormat>
  </EmailReport>
  <CountingCalibrate>
    <!--ro, opt, object-->
    <calibrateType opt="automatic,manual,no,smart">
      <!--ro, opt, enum, calibration type, subType:string, attr:opt{req, string}, desc:calibration type-->automatic
    </calibrateType>
    <SetupParam>
      <!--ro, req, object-->
      <height min="200.0" max="500.0">
        <!--ro, opt, float, height, range:[200.0,500.0], unit:cm, attr:min{req, float},max{req, float}-->200.0
      </height>
      <tiltAngle min="-90.0" max="90.0">
        <!--ro, opt, float, range:[-90,90], unit:°, attr:min{req, float},max{req, float}-->-90
      </tiltAngle>
      <heelAngle min="0.0" max="180.0">
        <!--ro, opt, float, range:[0,180], unit:°, attr:min{req, float},max{req, float}-->0
      </heelAngle>
    </SetupParam>
    <CountingArea>
      <!--ro, req, object, counting area (red frame)-->
      <RegionCoordinatesList size="4">
        <!--ro, opt, array, rectangle, subType:object, range:[0,4], attr:size{req, string}, desc:rectangle-->
        <RegionCoordinates>
          <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </CountingArea>
    <AutomaticCalib>
      <!--ro, opt, object, automatic calibration-->
      <CalibRegion>
        <!--ro, opt, object, calibration area (green frame)-->
        <RegionCoordinatesList size="4">
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], attr:size{req, string}, desc:rectangle-->
          <RegionCoordinates>
            <!--ro, req, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </CalibRegion>
    </AutomaticCalib>
  </CountingCalibrate>
  <HeightFilterOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false" default="false">
      <!--ro, opt, bool, attr:opt{req, bool},default{req, bool}-->true
    </enabled>
    <heightFilter min="0" max="250" default="120">
      <!--ro, opt, int, range:[0,250], unit:cm, attr:min{req, int},max{req, int},default{req, int}-->0
    </heightFilter>
  </HeightFilterOverlay>
  <isSupportCalibrate opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportCalibrate>
  <isSupportPosInfoOverlay opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportPosInfoOverlay>
  <countingType opt="none,alarmInputTrigger,videoTrigger">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->none
  </countingType>
  <alarmInputTrigger_signalType opt="level,pulses">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->level
  </alarmInputTrigger_signalType>
  <videoTrigger_RS485Transmission opt="true,false">
    <!--ro, opt, bool, attr:opt{req, bool}-->true
  </videoTrigger_RS485Transmission>
  <isSupportCountingStatus>
    <!--ro, opt, bool-->true
  </isSupportCountingStatus>
  <isSupportTriggerPeopleCountingDataSearch>
```

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```
    <!--ro, opt, bool-->true
  </isSupportTriggerPeopleCountingDataSearch>
  <isSupportDailyResetTime>
    <!--ro, opt, bool, whether it supports resetting and clearing by schedule-->true
  </isSupportDailyResetTime>
  <Polyline>
    <!--ro, opt, object, whether it supports polyline-->
    <maxPointNumber>
      <!--ro, req, int-->1
    </maxPointNumber>
  </Polyline>
  <isSupportBusDetectionArea>
    <!--ro, opt, bool-->true
  </isSupportBusDetectionArea>
  <isSupportStreamOverlayRuleInfos>
    <!--ro, opt, bool, whether it supports overlaying intelligent information in stream-->true
  </isSupportStreamOverlayRuleInfos>
  <isSupportMobileDescribeChange>
    <!--ro, opt, bool, this node is for marking the editing of the displayed parameter name of people counting camera-->true
  </isSupportMobileDescribeChange>
  <isSupportAdvanceConfiguration>
    <!--ro, opt, bool, this node is for marking whether the camera supports advanced configuration-->true
  </isSupportAdvanceConfiguration>
  <isSupportSearchDoorStates>
    <!--ro, opt, bool, this node if for supporting the detection of door status triggered by signal of vehicle-mounted device-->true
  </isSupportSearchDoorStates>
  <isSupportRuleRegion>
    <!--ro, opt, bool, whether it supports setting counting area-->true
  </isSupportRuleRegion>
  <notSupportStatisticType>
    <!--ro, opt, enum, this node is valid when <isSupportTriggerPeopleCountingDataSearch> is set to "true", subType:string, desc:this node is valid when
<isSupportTriggerPeopleCountingDataSearch> is set to "true"-->enternum
  </notSupportStatisticType>
  <detectThreshold min="0" max="100" def="90">
    <!--ro, opt, int, detection threshold, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->90
  </detectThreshold>
  <aidedTrackEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable auxiliary tracking, attr:opt{req, bool},def{req, string}-->true
  </aidedTrackEnabled>
  <SceneLimitCapList size="2">
    <!--ro, opt, array, capability list of scene limit, subType:object, attr:size{req, string}-->
    <SceneLimitCap>
      <!--ro, opt, object, capability of a scene limit-->
      <minTargetWidth>
        <!--ro, opt, int, normalized minimum width of target-->1
      </minTargetWidth>
      <maxTargetWidth>
        <!--ro, opt, int, normalized maximum width of target-->1
      </maxTargetWidth>
      <maxRuleRegionArea>
        <!--ro, opt, int, normalized maximum area of rule frame-->1
      </maxRuleRegionArea>
    </SceneLimitCap>
  </SceneLimitCapList>
  <targetRegionEdgesNum min="4" max="4">
    <!--ro, opt, int, capability of edge quantity of target region frame, attr:min{req, string},max{req, int}, desc:capability of frame edge quantity of
targetregion-->4
  </targetRegionEdgesNum>
  <ruleRegionEdgesNum min="4" max="10">
    <!--ro, opt, int, capability of edge quantity of rule frame, range:[4,10], attr:min{req, int},max{req, int}-->4
  </ruleRegionEdgesNum>
  <ChildFilter>
    <!--ro, opt, object, children counting-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <heightThreshold min="100" max="200" def="150">
      <!--ro, opt, int, threshold of the height of children, range:[100,200], unit:cm, attr:min{req, int},max{req, int},def{req, int}-->150
    </heightThreshold>
  </ChildFilter>
  <MisinfoFilter>
    <!--ro, opt, object, false alarm filtering-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <threshold min="0" max="100" def="15">
      <!--ro, opt, int, filtering threshold, range:[0,100], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the threshold value, the higher
the precision-->15
    </threshold>
    <timesLimit min="1" max="5" def="3">
      <!--ro, opt, int, prevention attempts of false alarm, range:[1,5], attr:min{req, int},max{req, int},def{req, int}, desc:the larger the number of
attempts, the higher the precision-->3
    </timesLimit>
  </MisinfoFilter>
  <isSupportReverseAlarm>
    <!--ro, opt, bool, whether it supports setting reverse counting alarm-->true
  </isSupportReverseAlarm>
  <detectionMode opt="single,double,double_single,single_double" def="double_single">
    <!--ro, opt, enum, detection mode, subType:string, attr:opt{req, string},def{req, string}, desc:"single" (detect based on tracking algorithm), "double"
(detect based on depth map), "double_single" (detect based on depth map mainly and tracking algorithm secondarily), "single_double" (detect based on
tracking algorithm mainly and depth map secondarily)-->double_single
  </detectionMode>
  <TrajectoryCountFilter>
```

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    <!--ro, opt, object, pattern counting filtering-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <movementDisplacement min="0" max="200" def="40">
      <!--ro, opt, int, motion displacement range, range:[0,200], attr:min{req, int},max{req, int},def{req, int}-->40
    </movementDisplacement>
    <residenceTime min="0.0" max="10.0" def="0.1">
      <!--ro, opt, float, dwell time duration, range:[0.0,10.0], unit:s, attr:min{req, float, unit:s},max{req, float, unit:s},def{req, float, unit:s}-->0.1
    </residenceTime>
  </TrajectoryCountFilter>
  <isSupportParameterExport>
    <!--ro, opt, bool, whether it supports one-touch exporting people counting parameters-->true
  </isSupportParameterExport>
  <MultiRegions min="1" max="8">
    <!--ro, opt, int, number of counting regions, attr:min{req, int},max{req, int}, desc:this node will not be returned if people counting in multiple
regions is not supported-->1
  </MultiRegions>
  <isSupportRegionsDirection>
    <!--ro, opt, bool, whether it supports configuring counting direction-->true
  </isSupportRegionsDirection>
  <maintenanceModeEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable maintenance mode, attr:opt{req, bool}-->true
  </maintenanceModeEnabled>
  <MultiOverlayConfiguration>
    <!--ro, opt, object-->
    <id min="1" max="8">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </id>
    <name min="1" max="20">
      <!--ro, opt, string, range:[1,20], attr:min{req, string},max{opt, string}-->1
    </name>
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->false
    </enabled>
    <positionX min="0" max="1000">
      <!--ro, req, int, X-coordinate, range:[0,1000], attr:min{req, int},max{req, int}, desc:the origin is the lower-left corner of the screen-->1
    </positionX>
    <positionY min="0" max="1000">
      <!--ro, req, int, Y-coordinate, range:[0,1000], attr:min{opt, string},max{opt, string}, desc:the origin is the lower-left corner of the screen-->1
    </positionY>
    <OSDType opt="enter,leave,enterLeave,enterLeavePassing">
      <!--ro, opt, enum, OSD type, subType:string, attr:opt{req, string}, desc:OSD type-->enter
    </OSDType>
    <OSDList size="9">
      <!--ro, opt, object, attr:size{req, int}-->
      <OSDType opt="enter,leave,none">
        <!--ro, opt, string, attr:opt{req, string}-->enter
      </OSDType>
    </OSDList>
  </MultiOverlayConfiguration>
  <RuleRegionLimit>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width-->1
    </width>
    <height>
      <!--ro, req, int, height-->1
    </height>
  </RuleRegionLimit>
  <isSupportMultiRegionsEnabled>
    <!--ro, opt, bool-->true
  </isSupportMultiRegionsEnabled>
  <analysisMode opt="realTimeMode,autoSwitchMode">
    <!--ro, opt, string, analysis mode, attr:opt{req, string}, desc:"realTimeMode" (real time), "autoSwitchMode" (auto switch). The engine also has its
analysis mode (refer to the <analysisMode> node in /ISAPI/Intelligent/analysisEngines/<ID>). The device will be added to the engine with the same analysis
mode via the load balancing method. If the engine is not configured correspondingly, this node will be failed to be returned, and the error code of
taskNotMatchEngine will be returned-->test
  </analysisMode>
  <countingSceneMode opt="entrance,exit">
    <!--ro, opt, string, attr:opt{req, string}-->entrance
  </countingSceneMode>
  <flowDirectionAnalysis opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->false
  </flowDirectionAnalysis>
  <isSupportRemoveDuplicate>
    <!--ro, opt, bool-->true
  </isSupportRemoveDuplicate>
  <isSupportCaptureOverlap>
    <!--ro, opt, bool-->true
  </isSupportCaptureOverlap>
  <dataUploadType>
    <!--ro, opt, object-->
    <enterEnabled opt="true,false" def="true">
      <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
    </enterEnabled>
    <leaveEnabled opt="true,false" def="true">
      <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
    </leaveEnabled>
  </dataUploadType>
  <InterPupileRegion>
    <!--ro, opt, object-->
    <positionX>
      <!--ro, req, int, range:[0,1000]-->50
```

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```
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,1000]-->50
      </positionY>
      <width min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, range:[0,1000]-->50
      </height>
    </InterPupileRegion>
    <MaxInterPupilRegion>
      <!--ro, opt, object-->
      <positionX>
        <!--ro, req, int, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,1000]-->50
      </positionY>
      <width min="50" max="50">
        <!--ro, req, int, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <filterMultiple min="4" max="6">
        <!--ro, opt, int, range:[4,6], attr:min{req, int},max{req, int}-->0
      </filterMultiple>
    </SizeFilter>
    <expertModeEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, bool}-->false
    </expertModeEnabled>
    <CountingRegionList>
      <!--ro, opt, object-->
      <CountingRegion>
        <!--ro, opt, object-->
        <regionColor opt="blue,red" def="yellow">
          <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->blue
        </regionColor>
        <TrajectoryCountFilter>
          <!--ro, opt, object-->
          <enabled opt="true,false" def="false">
            <!--ro, req, bool, attr:opt{req, string},def{req, bool}-->true
          </enabled>
          <residenceTime min="1" max="60" def="10">
            <!--ro, opt, int, range:[1,60], unit:s, attr:min{req, int},max{req, int},def{req, int}-->10
          </residenceTime>
        </TrajectoryCountFilter>
      </CountingRegion>
    </CountingRegionList>
    <advanceParams>
      <!--ro, opt, object-->
      <algorithmBrightnessThreshold>
        <!--ro, opt, int-->1
      </algorithmBrightnessThreshold>
      <faceExposurEnable opt="true,false" def="false">
        <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
      </faceExposurEnable>
      <referenceBright min="0" max="100" def="50">
        <!--ro, opt, int, range:[0,100], dep:and,{$.CountingCap.advanceParams.faceExposurEnable,eq,true}, attr:min{opt, string},max{opt, string},def{req,
int}-->50
      </referenceBright>
      <minContinueTime min="1" max="60" def="10">
        <!--ro, opt, int, range:[1,60], attr:min{opt, string},max{opt, string},def{req, int}-->10
      </minContinueTime>
      <faceFilterTime min="0" max="100" def="0">
        <!--ro, opt, int, range:[0,100], attr:min{opt, string},max{opt, string},def{req, int}-->0
      </faceFilterTime>
      <algorithmSceneType opt="indoor,outdoor" def="indoor">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->indoor
      </algorithmSceneType>
    </advanceParams>
    <debugEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </debugEnabled>
    <congestion>
      <!--ro, opt, object-->
      <minPeopleNum min="0" max="999" def="100">
        <!--ro, opt, int, range:[0,999], attr:min{req, int},max{req, int},def{req, int}-->100
      </minPeopleNum>
      <maxFlowRate min="0" max="9999" def="1000">
        <!--ro, opt, int, range:[0,9999], attr:min{req, int},max{req, int},def{req, int}-->1000
      </maxFlowRate>
    </congestion>
    <detectionPartType opt="headAndShoulders,feet">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->headAndShoulders
    </detectionPartType>
```

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```
  <enabled opt="true,false" def="true">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </enabled>
</CountingCap>
```

16.5.1.4 Set the parameters of a specific video input channel of the device

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/lineItem/<lineRuleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
lineRuleID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<CountingLineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, int, detection line ID-->1
  </id>
  <direction>
    <!--req, object, detection line direction-->
    <StartPoint>
      <!--req, object, start point-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </StartPoint>
    <EndPoint>
      <!--req, object, end point-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </EndPoint>
  </direction>
  <sensitivityLevel>
    <!--req, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
  </sensitivityLevel>
  <spaceGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </spaceGenerationSpeed>
  <timeGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </timeGenerationSpeed>
  <countingSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </countingSpeed>
  <detectionType>
    <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
  </detectionType>
  <objectSizeCorrection>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </objectSizeCorrection>
  <LineCoordinatesList>
    <!--opt, array, subType:object-->
    <Coordinates>
      <!--req, object, coordinates-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </Coordinates>
  </LineCoordinatesList>
  <polyline>
    <!--opt, bool-->true
  </polyline>
</CountingLineItem>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.5.1.5 Set the parameters of a specific video input channel of the device

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<CountingRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, area ID-->1
  </id>
  <Direction>
    <!--req, object, direction-->
    <StartPoint>
      <!--req, object, start point, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </StartPoint>
    <EndPoint>
      <!--req, object, end point, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </EndPoint>
  </Direction>
  <sensitivityLevel>
    <!--req, int, target detection sensitivity, range:[1,100], desc:the default value is 50-->50
  </sensitivityLevel>
  <spaceGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </spaceGenerationSpeed>
  <timeGenerationSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </timeGenerationSpeed>
  <countingSpeed>
    <!--opt, int, the default value is 50, range:[1,100], desc:the default value is 50-->50
  </countingSpeed>
  <detectionType>
    <!--opt, enum, detection type, subType:string, desc:"auto", "head", "shoulder"-->auto
  </detectionType>
  <objectSizeCorrection>
    <!--opt, int, range:[1,100]-->50
  </objectSizeCorrection>
  <RegionCoordinatesList>
    <!--req, array, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <RuleRegionCoordinatesList>
    <!--opt, array, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--req, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RuleRegionCoordinatesList>
  <RuleRegionLimit>
    <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->1
    </positionY>
    <height>
      <!--req, int, height, range:[0,1000]-->1
    </height>
    <width>
      <!--req, int, width, range:[0,1000]-->1
    </width>
  </RuleRegionLimit>
  <enabled>
    <!--opt, bool, whether to enable the function, desc:whether to enable the function-->true
  </enabled>
</CountingRegion>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.5.1.6 Reset the people counting statistics

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/counting/resetCount

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.5.1.7 Search people counting report data

Request URL

POST /ISAPI/System/Video/inputs/channels/<channelID>/counting/search

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<CountingStatisticsDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, search people counting report data, attr:version{opt, string, protocolVersion}-->
  <statisticType>
    <!--req, enum, data type, subType:string, desc:"enternum,exitnum,peoplePassing,all"-->enternum
  </statisticType>
  <reportType>
    <!--req, enum, report type, subType:string, desc:"daily,weekly,monthly,yearly"-->daily
  </reportType>
  <timeSpanList>
    <!--opt, array, subType:object-->
    <timeSpan>
      <!--opt, object, time range-->
      <startTime>
        <!--req, datetime, start time-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--req, datetime, end time-->1970-01-01T00:00:00+08:00
      </endTime>
    </timeSpan>
  </timeSpanList>
  <triggerPeopleCountingData>
    <!--opt, bool, whether to enable people counting-->true
  </triggerPeopleCountingData>
  <PagingSearchCond>
    <!--opt, object, this node is valid when <triggerPeopleCountingData> is set to true-->
    <searchID>
      <!--req, string, search ID, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two
searching, the search history will be saved in the memory to speed up next searching-->test
    </searchID>
    <searchResultPosition>
      <!--req, int-->1
    </searchResultPosition>
    <maxResults>
      <!--req, int-->1
    </maxResults>
  </PagingSearchCond>
  <MinTimeInterval>
    <!--opt, enum, minimum time interval, subType:string, desc:minimum time interval: quarter,half,hour,day,month,week-->quarter
  </MinTimeInterval>
  <child>
    <!--opt, bool, whether supports counting children only, desc:true (yes), false (counting both adults and children)-->true
  </child>
  <FaceAttribute>
    <!--opt, object, facial features, desc:facial features-->
    <faceExpression>
      <!--opt, enum, facial expression, subType:string, desc:facial expression-->unknown
    </faceExpression>
    <gender>
      <!--opt, enum, gender, subType:string, desc:gender-->unknown
    </gender>
    <mask>
      <!--opt, enum, whether the person wears a mask, subType:string, desc:whether the person wears a mask-->unknown
    </mask>
    <ageGroup>
      <!--opt, enum, age group, subType:string, desc:age group-->unknown
    </ageGroup>
    <glass>
      <!--opt, enum, subType:string-->yes
    </glass>
  </FaceAttribute>
  <regionID>
    <!--opt, string, region ID, desc:region ID-->1
  </regionID>
  <HumanBodyAttribute>
    <!--opt, object, human body attributes, desc:human body attributes-->
    <things>
      <!--opt, enum, "unknown", "yes", "no", subType:string, desc:"unknown", "yes", "no"-->unknown
    </things>
  </HumanBodyAttribute>
  <statisticTypeList>
    <!--opt, array, subType:object, range:[1,13]-->
    <statisticType>
      <!--opt, enum, subType:string-->enternum
    </statisticType>
  </statisticTypeList>
</CountingStatisticsDescription>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<CountingStatisticsResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search people counting report data, attr:version{opt, string, protocolVersion}-->
  <responseStatus>
    <!--ro, req, bool-->true
  </responseStatus>
  <responseStatusStrg>
```

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```
    <!--ro, req, string, "MORE"-more items to be searched,"OK"-search completed-->test
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int-->1
  </numOfMatches>
  <matchList>
    <!--ro, opt, array, subType:object-->
    <matchElement>
      <!--ro, opt, object-->
      <timeSpan>
        <!--ro, opt, object, time range-->
        <startTime>
          <!--ro, req, datetime, start time-->1970-01-01T00:00:00+08:00
        </startTime>
        <endTime>
          <!--ro, req, datetime, end time-->1970-01-01T00:00:00+08:00
        </endTime>
      </timeSpan>
      <enterCount>
        <!--ro, opt, int-->1
      </enterCount>
      <exitCount>
        <!--ro, opt, int-->1
      </exitCount>
      <peoplePassingCount>
        <!--ro, opt, int-->1
      </peoplePassingCount>
      <duplicatePeopleCount>
        <!--ro, opt, int-->1
      </duplicatePeopleCount>
      <duplicateEnter>
        <!--ro, opt, int-->1
      </duplicateEnter>
      <duplicateExit>
        <!--ro, opt, int-->1
      </duplicateExit>
      <duplicatePass>
        <!--ro, opt, int-->1
      </duplicatePass>
      <FaceExpressionList>
        <!--ro, opt, array, subType:object-->
        <FaceExpression>
          <!--ro, opt, object, facial expression-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </FaceExpression>
      </FaceExpressionList>
      <GenderList>
        <!--ro, opt, array, subType:object-->
        <Gender>
          <!--ro, opt, object, gender-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Gender>
      </GenderList>
      <MaskList>
        <!--ro, opt, array, subType:object-->
        <Mask>
          <!--ro, opt, object, whether the person wears a mask-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Mask>
      </MaskList>
      <AgeGroupList>
        <!--ro, opt, array, subType:object-->
        <AgeGroup>
          <!--ro, opt, object, age group-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </AgeGroup>
      </AgeGroupList>
      <GlassList>
        <!--ro, opt, array, subType:object-->
        <Glass>
          <!--ro, opt, object-->
          <value>
            <!--ro, req, enum, subType:string-->unknown
          </value>
```

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```
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Glass>
      </GlassList>
      <RegionList>
        <!--ro, opt, array, subType:object-->
        <Region>
          <!--ro, opt, object, region-->
          <id>
            <!--ro, req, int-->1
          </id>
          <enterCount>
            <!--ro, opt, int-->1
          </enterCount>
          <exitCount>
            <!--ro, opt, int-->1
          </exitCount>
          <passingCount>
            <!--ro, opt, int-->1
          </passingCount>
          <duplicateEnter>
            <!--ro, opt, int-->1
          </duplicateEnter>
          <duplicateExit>
            <!--ro, opt, int-->1
          </duplicateExit>
        </Region>
      </RegionList>
      <ThingsList>
        <!--ro, opt, array, subType:object-->
        <Things>
          <!--ro, opt, object-->
          <value>
            <!--ro, req, enum, "unknown", "yes", "no", subType:string, desc:"unknown", "yes", "no"-->unknown
          </value>
          <count>
            <!--ro, req, int-->1
          </count>
        </Things>
      </ThingsList>
      <enterRate>
        <!--ro, opt, float, range:[0.00,100.00], step:0.01-->1
      </enterRate>
    </matchElement>
  </matchList>
</CountingStatisticsResult>
```

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16.5.1.8 Get the capability of people counting data replenishment

Request URL

GET /ISAPI/System/Video/inputs/channels/counting/collection/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "CollectionDescriptionCap": {
    /*ro, opt, object*/
        "timeRange":  15,
        /*ro, req, int, supported time range for people counting replenishment*/
        "timeSpanMaxNum":  1,
        /*ro, req, int, the maximum number of time buckets supported for one time search integer*/
        "resultMaxNum":  100,
        /*ro, req, int, the maximum number of items can be searched*/
        "channelMaxNum":  1
        /*ro, req, int, the maximum number of channels supported for one time search*/
    }
}
```

16.5.1.9 Search for people counting data replenishment

Request URL

POST /ISAPI/System/Video/inputs/channels/counting/collection?format=json

Query Parameter

None

Request Message

```
{
    "CollectionDescription": {
    /*opt, object, search for people counting data replenishment*/
        "searchID":  "test",
        /*req, string, search ID, desc:this node is used to check whether searches are the same one. Due to device performance limitation, you can search asynchronously by repeatedly applying the search conditions and getting the search progress (the searchID should be the same)*/
        "searchResultPosition":  0,
        /*req, int, initial position of search result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search*/
        "maxResults":  100,
        /*req, int, the maximum number of search results this time*/
        "timeSpanList": [
        /*opt, array, time period list, subType:object*/
            {
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*req, string, start time*/
                "endTime":  "2004-05-03T17:30:08+08:00"
                /*req, string, end time*/
            }
        ],
        "channels": [1, 2, 3]
        /*opt, array, array, subType:int, desc:sub type in the array is integer32, linked channel No., if the channel information is not specified, it indicates searching for all channels*/
    }
}
```

Response Message

```
{
    "CollectionResult": {
    /*ro, opt, object, search for people counting data replenishment*/
        "responseStatusStrg":  "OK",
        /*ro, req, string, search status description, desc:search status description OK- Searching ended,NO MATCHES-No matched data found,MORE-Search again for more results,the max. length is 32,{dep if errcode == 1 && errMsg == ok}*/
        "numOfMatches":  1,
        /*ro, req, int, returned number of results for current search*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results*/
        "targets": [
        /*ro, opt, array, statistics results, subType:object*/
            {
                "startTime":  "2017-08-25T11:34:59+08:00",
                /*ro, req, string, start time of statistics*/
                "endTime":  "2017-08-25T11:34:59+08:00",
                /*ro, req, string, end time of statistics*/
                "channel":  1,
                /*ro, req, int, channel No.*/
                "enter":  34,
                /*ro, req, int, the number of people entered*/
                "exit":  39,
                /*ro, req, int, the number of people exited*/
                "pass":  0
                /*ro, req, int, the number of people passed by*/
            }
        ],
        "collectionStatus":  "syncing"
        /*ro, opt, enum, the status of people counting data replenishment, subType:string, desc:syncing indicates that the people counting data synchronization is not completed yet; the platform will not parse the targets statistics result and will continue searching until the "finished" is returned if this node does not exist, it indicates that the replenishment result will be returned and the platform will directly parse the targets no matter the synchronization is completed or not; "syncing" (synchronizing the IPC data), "finished" (synchronization completed)*/
    }
}
```

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16.5.1.10 People counting event

EventType:PeopleCounting

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people counting event details, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, opt, string, IPv4 address-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 Address-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, device's communication port No.-->1
  </portNo>
  <protocol>
```

```
    <!--ro, opt, enum, transmission communication protocol, subType:string, desc:"HTTP, HTTPS, EHome". The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->test
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No.-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, event occurred time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, req, int, event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"PeopleCounting" (people counting)-->PeopleCounting
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:"active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, PUID, range:[1,64], desc:required in Ehome alarm, corresponding to deviceID in /ISAPI/System/Network/Ehome-->test
  </deviceID>
  <peopleCounting>
    <!--ro, opt, object, people counting-->
    <statisticalMethods>
      <!--ro, opt, enum, people counting method, subType:string, desc:"realTime" (real-time people counting), "timeRange" (scheduled people counting), "signalTrigger" (alarm input triggered people counting)-->realTime
    </statisticalMethods>
    <RealTime>
      <!--ro, opt, object, real-time, desc:it is required when statisticalMethods is "realTime"-->
      <time>
        <!--ro, req, datetime, people counting time-->1970-01-01T00:00:00+08:00
      </time>
    </RealTime>
    <TimeRange>
      <!--ro, opt, object, time period for real-time people counting, desc:it is required when statisticalMethods is "timeRange" or "signalTrigger".-->
      <startTime>
        <!--ro, req, datetime, start time of people counting-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--ro, req, datetime, end time of people counting-->1970-01-01T00:00:00+08:00
      </endTime>
    </TimeRange>
    <enter>
      <!--ro, opt, int, number of entered people-->1
    </enter>
    <exit>
      <!--ro, opt, int, number of exited people-->1
    </exit>
    <pass>
      <!--ro, opt, int, passers-by number, desc:passers-by number-->1
    </pass>
    <duplicatePeople>
      <!--ro, opt, int, number of duplicated people, desc:number of duplicated people-->1
    </duplicatePeople>
    <FlowStatisticsList>
      <!--ro, opt, array, flow direction counting list, subType:object-->
      <FlowStatistics>
        <!--ro, opt, object, flow direction counting, desc:for deduplicated people counting statistics, the flow direction analysis is only valid in the following circumstances: the rule area is a quadrilateral, the counting broken line is formed with four points, and the counting line has only two points that overlap with the rule area line-->
        <countingLineID>
          <!--ro, opt, int, counting line ID-->1
        </countingLineID>
        <pass>
          <!--ro, opt, int, passers-by number, desc:people who enter the counting area but do not cross the counting line-->1
        </pass>
        <movingDirectionList>
          <!--ro, opt, array, people moving direction list, subType:object, desc:count the moving direction of people who enter the counting area and cross the counting line-->
          <MovingDirection>
            <!--ro, opt, object, people moving direction-->
            <countingLineSegmentID>
              <!--ro, opt, int, area ID, range:[1,3], desc:the four points of the broken line are marked as 1, 2, 3, and 4 respectively in counterclockwise direction from the counting direction. Draw two lines parallel to the counting direction respectively starting from point 2 and point 3 to intersect the counting rule area. The formed three areas are marked as 1, 2, and 3 in counterclockwise direction from the counting direction-->1
            </countingLineSegmentID>
            <count>
              <!--ro, opt, int, the number of people-->0
```

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```
            <!--ro, opt, int, the number of people-->0
          </count>
        </MovingDirection>
      </movingDirectionList>
    </FlowStatistics>
  </FlowStatisticsList>
  <countingSceneMode>
    <!--ro, opt, enum, people counting scene mode, subType:string, desc:"entrance", "exit". The default mode is entrance-->entrance
  </countingSceneMode>
  <deDuplicateEnter>
    <!--ro, opt, int, deduplicated number of entered people-->0
  </deDuplicateEnter>
  <deDuplicateExit>
    <!--ro, opt, int, deduplicated number of exited people-->0
  </deDuplicateExit>
  <faceLibDuplicatePeopleList>
    <!--ro, opt, array, list of duplicated people in the face picture library, subType:object, desc:count the number of duplicated people in every face
picture library-->
    <faceLibDuplicatePeople>
      <!--ro, opt, object, number of duplicated people in the face picture library-->
      <FDID>
        <!--ro, req, string, face picture library ID, desc:check details of the library with /ISAPI/Intelligent/FDLib/search?format=json-->1
      </FDID>
      <name>
        <!--ro, req, string, face picture library name-->11
      </name>
      <duplicate>
        <!--ro, opt, int, number of duplicated people in the library-->1
      </duplicate>
    </faceLibDuplicatePeople>
  </faceLibDuplicatePeopleList>
</peopleCounting>
<childCounting>
  <!--ro, opt, object, children counting-->
  <enter>
    <!--ro, opt, int, number of entered people-->1
  </enter>
  <exit>
    <!--ro, opt, int, number of exited people-->1
  </exit>
</childCounting>
<FaceExpressionList>
  <!--ro, opt, array, facial expression statistics, subType:object-->
  <FaceExpression>
    <!--ro, opt, object, facial expression-->
    <value>
      <!--ro, req, enum, facial attributes, subType:string, desc:"unknown", "poker-faced" (no facial expression), "happy", "surprised", "disgusted",
"sad", "angry", "contemptuous", "panic", "all"-->unknown
    </value>
    <count>
      <!--ro, req, int, number of people corresponding to the face expression-->1
    </count>
  </FaceExpression>
</FaceExpressionList>
<GenderList>
  <!--ro, opt, array, gender statistics, subType:object-->
  <Gender>
    <!--ro, opt, object, gender-->
    <value>
      <!--ro, req, enum, options, subType:string, desc:"unknown", "female", "male", "all"-->unknown
    </value>
    <count>
      <!--ro, req, int, number of people corresponding to the gender-->1
    </count>
  </Gender>
</GenderList>
<MaskList>
  <!--ro, opt, array, mask statistics, subType:object-->
  <Mask>
    <!--ro, opt, object, mask wearing status-->
    <value>
      <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing a mask), "no" (not wearing a mask)-->unknown
    </value>
    <count>
      <!--ro, req, int, number of people corresponding to the masks wearing status-->1
    </count>
  </Mask>
</MaskList>
<AgeGroupList>
  <!--ro, opt, array, age group statistics, subType:object-->
  <AgeGroup>
    <!--ro, opt, object, age group-->
    <value>
      <!--ro, req, enum, options, subType:string, desc:"unknown", "child", "young", "middle", "old", "infant", "kid", "teenager", "prime", "middleAged"
(the middle age), "all"-->unknown
    </value>
    <count>
      <!--ro, req, int, number of people corresponding to the age group-->1
    </count>
  </AgeGroup>
</AgeGroupList>
<GlassList>
  <!--ro, opt, array, glasses statistics, subType:object-->
  <Glass>
```

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```
      <!--ro, opt, object, glasses wearing status-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing glasses), "no" (not wearing glasses), "sunglasses" (wearing sunglasses),
"all"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people corresponding to the glass wearing attribute-->1
      </count>
    </Glass>
  </GlassList>
  <RegionList>
    <!--ro, opt, array, people counting information in the area, subType:object-->
    <Region>
      <!--ro, opt, object, area-->
      <id>
        <!--ro, req, int, area ID-->1
      </id>
      <name>
        <!--ro, opt, string, area name-->test
      </name>
      <enter>
        <!--ro, opt, int, number of entered people-->1
      </enter>
      <exit>
        <!--ro, opt, int, number of exited people-->1
      </exit>
      <totalEnter>
        <!--ro, opt, int, total number of entered people, desc:before resetting on the current day, the number of people will accumulate the number will be
initialized only after manual resetting on the current day or auto resetting on the next day (if the device supports duplicated people counting, the number
will be the deduplicated number of people of the current day)-->1
      </totalEnter>
      <totalExit>
        <!--ro, opt, int, total number of exited people, desc:before resetting on the current day, the number of people will accumulate the number will be
initialized only after manual resetting on the current day or auto resetting on the next day (if the device supports duplicated people counting, the number
will be the deduplicated number of people of the current day)-->1
      </totalExit>
      <totalPass>
        <!--ro, opt, int-->1
      </totalPass>
    </Region>
  </RegionList>
  <targetAttrs>
    <!--ro, opt, string, target attributes, range:[1,1024],
desc:transparent transmission field, which corresponds to targetAttrs in applying task, format:
    <![CDATA[{
    /*opt, transparent transmission field, the following three fileds are recommended*/
    "deviceId": "1",
    /*opt, device ID, string, maximum length 64*/
    "deviceChannel": 1,
    /*opt, device channel, integer32*/
    "deviceName": "shebei"
    /*opt, device name, string, maximum length 128*/
    }]]>, example: <![CDATA[{ "deviceId": "1", "deviceChannel": 1, "deviceName": "shebei"}]]>-->test
  </targetAttrs>
  <targetInfo>
    <!--ro, opt, object, target information, desc:corresponding to targetInfo in heat analysis task /ISAPI/SDT/Management/HeatAnalysis/video/AddTask?
format=json. Either targetInfo or targetAttrs is required-->
    <cameraInfo>
      <!--ro, opt, string, camera information-->test
      <cameraIndexCode>
        <!--ro, req, string, camera ID-->test
      </cameraIndexCode>
      <cameraName>
        <!--ro, opt, string, camera name-->test
      </cameraName>
    </cameraInfo>
    <mapFileID>
      <!--ro, opt, string, map file ID-->test
    </mapFileID>
    <storeID>
      <!--ro, opt, string, store ID-->test
    </storeID>
  </targetInfo>
  <customInfo>
    <!--ro, opt, string, custom information-->test
  </customInfo>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle), 2-"hightPerformanceVehicle" (high-performance
vehicle), 3-"personStructModel" (person structural modeling), 4-"vehicleAndPersonStructModel" (person and vehicle structural modeling), 5-"videoFaceDetect"
(video face detection), 6-"videoFaceDetectModel" (video face detection and modeling), 7-"smart" (face detection + human body detection + behavior
detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
```

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```
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime", "historyvideo", "localvideo"-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to
128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <DetectionRegionList>
    <!--ro, opt, array, detection area, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity level, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <HatList>
    <!--ro, opt, array, wearing hat statistics, subType:object-->
    <Hat>
      <!--ro, opt, object, whether the person wears a hat-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes" (wearing a hat), "no" (not wearing a hat), "helmet" (wearing a helmet)-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people wearing hats-->1
      </count>
    </Hat>
  </HatList>
  <ThingsList>
    <!--ro, opt, array, carrying objects statistics, subType:object-->
    <Things>
      <!--ro, opt, object, carrying objects-->
      <value>
        <!--ro, req, enum, options, subType:string, desc:"unknown", "yes", "no"-->unknown
      </value>
      <count>
        <!--ro, req, int, number of people carrying objects-->1
      </count>
    </Things>
  </ThingsList>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers-->true
  </isDataRetransmission>
```

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```
</EventNotificationAlert>
```

16.6 Abnormal Event Detection

16.6.1 Fast Moving Detection

16.6.1.1 Get the arming schedule of fast moving detection for all channels

Request URL

GET /ISAPI/Event/schedules/rapidMoves

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RapidMoveScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->rapidMove
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RapidMoveScheduleList>
```

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16.6.1.2 Set the arming schedule of fast moving detection for all channels

Request URL

PUT /ISAPI/Event/schedules/rapidMoves

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RapidMoveScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, subType:string-->rapidMove
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RapidMoveScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.1.3 Set the arming schedule of fast moving detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/rapidMoves/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, string-->rapidMove
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.1.4 Get the arming schedule of fast moving detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/rapidMoves/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->rapidMove
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.6.1.5 Set the linkage parameters of fast moving detection

Request URL

PUT /ISAPI/Event/triggers/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--req, enum, subType:string-->rapidMove
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.1.6 Get the linkage parameters of fast moving detection

Request URL

GET /ISAPI/Event/triggers/rapidMove-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->rapidMove
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

16.6.1.7 Fast moving

EventType:rapidMove

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, transmitted parameters, attr:version{req, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
```

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```
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"rapidMove" (fast moving detection)-->rapidMove
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->0
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
```

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```
    <targetAttrs>
      <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
    </targetAttrs>
    <TaskInfo>
      <!--ro, opt, object, task information submitted by data center products-->
      <taskID>
        <!--ro, req, string, task ID, range:[0,64]-->test
      </taskID>
      <algorithmType>
        <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
      </algorithmType>
      <taskName>
        <!--ro, opt, string, task name-->test
      </taskName>
      <algorithmID>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmID>
      <monitorPointID>
        <!--ro, opt, string, camera No.-->test
      </monitorPointID>
      <monitorPointName>
        <!--ro, opt, string, camera name-->test
      </monitorPointName>
      <streamType>
        <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
      </streamType>
      <ruleID>
        <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
      </ruleID>
      <ruleCustomName>
        <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
      </ruleCustomName>
      <algorithmInfo>
        <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
        <algorithmId>
          <!--ro, opt, string, algorithm package ID-->test
        </algorithmId>
        <algoVendorId>
          <!--ro, opt, string, manufacturer ID-->test
        </algoVendorId>
        <algoVersionId>
          <!--ro, opt, string, algorithm version No.-->test
        </algoVersionId>
        <algoChipType>
          <!--ro, opt, string, chip type-->test
        </algoChipType>
        <algoModelVersion>
          <!--ro, opt, string, model version-->test
        </algoModelVersion>
      </algorithmInfo>
    </TaskInfo>
    <timeStamp>
      <!--ro, opt, datetime, timestamp of the frame, which triggers the alarm-->1970-01-01T00:00:00+08:00
    </timeStamp>
    <isDataRetransmission>
      <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
    </isDataRetransmission>
    <ruleID>
      <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
    </ruleID>
    <ruleName>
      <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
    </ruleName>
  </EventNotificationAlert>
```

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16.6.2 Loitering Detection

16.6.2.1 Set arming schedules of loitering detection of all channels

Request URL

PUT /ISAPI/Event/schedules/loiterings

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule-->
    <id>
      <!--req, string, ID, desc:ID-->loitering-1
    </id>
    <eventType>
      <!--opt, string, event type-->loitering
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1: Monday; 2: Tuesday; 3: Wednesday; 4: Thursday; 5: Friday; 6: Saturday; 7: Sunday;-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming schedule, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.2.2 Get arming schedules of the loitering detection of all channels

Request URL

GET /ISAPI/Event/schedules/loiterings

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID-->loitering-1
    </id>
    <eventType>
      <!--ro, opt, string, event type-->loitering
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1: Monday), 2: Tuesday, 3: Wednesday, 4: Thursday, 5: Friday, 6: Saturday, 7: Sunday-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LoiteringScheduleList>
```

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16.6.2.3 Set arming schedule of loitering detection of a channel

Request URL

PUT /ISAPI/Event/schedules/loiterings/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID, desc:ID-->test
  </id>
  <eventType>
    <!--opt, enum, event type, subType:string-->loitering
  </eventType>
  <videoInputChannelID>
    <!--opt, string, video input channel ID-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, arming schedule list, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, day of the week, subType:int, desc:1(Monday); 2(Tuesday); 3(Wednesday); 4(Thursday); 5(Friday); 6(Saturday); 7(Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object, holiday arming schedule-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.6.2.4 Get the arming schedule of loitering detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/loiterings/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->loitering
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.6.2.5 Get the linkage parameters of loitering detection

Request URL

GET /ISAPI/Event/triggers/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->loitering
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.6.2.6 Set the linkage parameters of loitering detection

Request URL

PUT /ISAPI/Event/triggers/loitering-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--req, enum, subType:string-->loitering
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.2.7 Set loitering detection parameters of all channels

Request URL

PUT /ISAPI/Smart/loitering

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Loitering>
    <!--opt, object, loitering detection parameters configuration of a channel-->
    <id>
      <!--req, string, channel No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <LoiteringRegionList>
      <!--opt, array, information list of loitering detection areas, subType:object-->
      <LoiteringRegion>
        <!--opt, object, loitering detection area-->
        <id>
          <!--req, string-->1
        </id>
        <sensitivityLevel>
          <!--opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--opt, int, time threshold-->1
        </timeThreshold>
        <RegionCoordinatesList>
          <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </LoiteringRegion>
    </LoiteringRegionList>
    <mutexAbility opt="PDC">
      <!--opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
  </Loitering>
</LoiteringList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.2.8 Get loitering detection parameters of all channels

Request URL

GET /ISAPI/Smart/loitering

Query Parameter

None

Request Message

None

Response Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Loitering>
    <!--ro, opt, object, loitering detection parameters configuration of a channel-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <LoiteringRegionList>
      <!--ro, opt, array, information list of loitering detection areas, subType:object-->
      <LoiteringRegion>
        <!--ro, opt, object, loitering detection area-->
        <id>
          <!--ro, req, string-->1
        </id>
        <sensitivityLevel>
          <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--ro, opt, int, time threshold-->1
        </timeThreshold>
        <RegionCoordinatesList>
          <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </LoiteringRegion>
    </LoiteringRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
  </Loitering>
</LoiteringList>
```

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16.6.2.9 Get loitering detection parameters of a channel

Request URL

GET /ISAPI/Smart/loitering/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Loitering xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, loitering detection parameters configuration of a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LoiteringRegionList>
    <!--ro, opt, array, information list of loitering detection areas, subType:object-->
    <LoiteringRegion>
      <!--ro, opt, object, loitering detection area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--ro, opt, int, time threshold-->1
      </timeThreshold>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </LoiteringRegion>
  </LoiteringRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, ro,"PDC", attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool-->true
  </isSupportMultiScene>
</Loitering>
```

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16.6.2.10 Get the loitering detection capability

Request URL

GET /ISAPI/Smart/loitering/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Loitering xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the loitering detection capability, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, req, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LoiteringRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LoiteringRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <sensitivityLevel min="1" max="100">
        <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
      </sensitivityLevel>
      <timeThreshold min="1" max="10" def="1">
        <!--ro, req, int, unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
      </timeThreshold>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </LoiteringRegion>
  </LoiteringRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool-->true
  </isSupportMultiScene>
  <isSupportOverlapParams>
    <!--ro, opt, bool-->true
  </isSupportOverlapParams>
</Loitering>
```

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16.6.2.11 Batch get parameters of multiple areas

Request URL

GET /ISAPI/Smart/loitering/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, information list of loitering detection areas, subType:object, attr:version{req, string, protocolVersion}-->
  <LoiteringRegion>
    <!--ro, opt, object, loitering detection area-->
    <id>
      <!--ro, req, string, detection area ID-->1
    </id>
    <sensitivityLevel>
      <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--ro, opt, int, time threshold-->1
    </timeThreshold>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LoiteringRegion>
</LoiteringRegionList>
```

16.6.2.12 Batch set parameters of multiple areas

Request URL

PUT /ISAPI/Smart/loitering/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, information list of loitering detection areas, subType:object, attr:version{req, string, protocolVersion}-->
  <LoiteringRegion>
    <!--opt, object, loitering detection area-->
    <id>
      <!--req, string, area ID-->1
    </id>
    <sensitivityLevel>
      <!--opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--opt, int, time threshold-->1
    </timeThreshold>
    <RegionCoordinatesList>
      <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LoiteringRegion>
</LoiteringRegionList>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.2.13 Set parameters of a single area

Request URL

PUT /ISAPI/Smart/loitering/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, loitering detection area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, area ID-->1
  </id>
  <sensitivityLevel>
    <!--opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--opt, int, time threshold-->1
  </timeThreshold>
  <RegionCoordinatesList>
    <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</LoiteringRegion>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:detailed description of the error code-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.2.14 Get loitering detection parameters of a single area

Request URL

GET /ISAPI/Smart/loitering/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, loitering detection area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, area ID-->1
  </id>
  <sensitivityLevel>
    <!--ro, opt, int, sensitivity, range:[0,100], desc:sensitivity-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--ro, opt, int, time threshold-->1
  </timeThreshold>
  <RegionCoordinatesList>
    <!--ro, opt, array, list of area coordinates, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</LoiteringRegion>
```

16.6.2.15 Get configuration capability of a loitering detection region by channel

Request URL

GET /ISAPI/Smart/loitering/<channelID>/regions/<regionID>/capabilities

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index No.-->1
  </id>
  <sensitivityLevel min="1" max="100">
    <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:default value: 50-->1
  </sensitivityLevel>
  <timeThreshold min="0" max="600" def="0">
    <!--ro, opt, int, time threshold, which is between 0 and 600, range:[0,600], attr:min{req, int},max{req, int},def{req, int}-->1
  </timeThreshold>
  <RegionCoordinatesList>
    <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</LoiteringRegion>
```

16.6.2.16 Get the capability of multiple loitering detection areas

Request URL

GET /ISAPI/Smart/loitering/<channelID>/regions/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LoiteringRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LoiteringRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, area ID-->1
    </id>
    <sensitivityLevel min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:the default value is 50-->1
    </sensitivityLevel>
    <timeThreshold min="0" max="600" def="0">
      <!--ro, opt, int, time threshold, range:[0,600], attr:min{req, int},max{req, int},def{req, int}-->1
    </timeThreshold>
    <RegionCoordinatesList>
      <!--ro, opt, array, list of area coordinates, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LoiteringRegion>
</LoiteringRegionList>
```

16.6.2.17 Loitering detection

EventType:loitering

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"loitering" (loitering detection)-->loitering
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
```

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        <!--ro, req, string, detection area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human"-->human
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
```

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```
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

16.6.3 Object Removal Detection

16.6.3.1 Get the arming schedule of object removal detection for all channels

Request URL

GET /ISAPI/Event/schedules/attendedBaggages

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--ro, opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

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16.6.3.2 Set the arming schedule of object removal detection for all channels

Request URL

PUT /ISAPI/Event/schedules/attendedBaggages

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<AttendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->attendedBaggage-1
    </id>
    <eventType>
      <!--opt, string-->attendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</AttendBaggageScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.3.3 Set the arming schedule of object removal detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/attendedBaggages/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->attendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.3.4 Get the arming schedule of object removal detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/attendedBaggages/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->attendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.6.3.5 Get the linkage parameters of object removal detection

Request URL

GET /ISAPI/Event/triggers/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->attendedBaggage
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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DETEKTOR

16.6.3.6 Set the linkage parameters of object removal detection

Request URL

PUT /ISAPI/Event/triggers/attendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->attendedBaggage-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->attendedBaggage
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.3.7 Object removal

EventType:attendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, object removal alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"attendedBaggage" (object removal)-->attendedBaggage
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
```

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```
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
```

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DESKTOP

```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

16.6.4 People Gathering Detection

16.6.4.1 Get the arming time parameters of people gathering for all channels

Request URL

GET /ISAPI/Event/schedules/groups

Query Parameter

None

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people gathering arming time configuration list, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, people gathering arming time configuration-->
    <id>
      <!--ro, req, string, index, desc:the format is event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string, desc:;group:group-->group
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-
->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming time-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</GroupScheduleList>
```

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DETEKTOR

16.6.4.2 Set the arming time parameters of people gathering for all channels

Request URL

PUT /ISAPI/Event/schedules/groups

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, people gathering arming time configuration list, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, people gathering arming time configuration-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string, desc:;group:group-->group
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start Time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</GroupScheduleList>
```

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DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.4.3 Get the arming schedule parameters of people gathering for a specified channel

Request URL

GET /ISAPI/Event/schedules/groups/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming time, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--ro, opt, enum, event type, subType:string, desc:;group:group-->group
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, arming time list, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object, arming time-->
      <dayOfWeek>
        <!--ro, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)--
>1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object, time range-->
        <beginTime>
          <!--ro, req, time, Start Time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object, holiday arming time-->
    <TimeBlock>
      <!--ro, opt, object, arming time-->
      <TimeRange>
        <!--ro, req, object, time range-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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DETEKTOR

16.6.4.4 Set the arming time parameters of people gathering for all channels

Request URL

PUT /ISAPI/Event/schedules/groups/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, people gathering arming time configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:;group:group-->group
  </eventType>
  <videoInputChannelID>
    <!--opt, string, video input channel ID-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, arming time list, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <dayOfWeek>
        <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object, time range-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object, holiday arming time-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time range-->
        <beginTime>
          <!--req, time, start time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.6.4.5 Set the linkage parameters of people gathering

Request URL

PUT /ISAPI/Event/triggers/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, people gathering linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--req, enum, event type, subType:string, desc:;group:group-->group
  </eventType>
  <eventDescription>
    <!--opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, video input channel ID, desc:if <eventType> is
"VMD,videoloss,tamperdetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage"-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string, wireless ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--opt, object, linkage action-->
      <id>
        <!--req, string, index-->test
      </id>
      <notificationMethod>
        <!--req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP" (uploading
via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading to the
center), "LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio)-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, linkage cycle, subType:string, desc:“beginning”, “beginningandend”, “recurring”-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object, PTZ linkage-->
        <ptzChannelID>
          <!--req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--opt, int, linked No., desc:depend on the content of <actionName>, it can be preset No., patrol No., and pattern No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.4.6 Get linkage parameters of people gathering

Request URL

GET /ISAPI/Event/triggers/group-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, people gathering linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, desc:the format is event type-channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:group:group-->group
  </eventType>
  <eventDescription>
    <!--ro, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID, desc:if <eventType> is
"VMD,videoloss,tamperdetection,regionEntrance,regionExiting,loitering,group,rapidMove,parking,unattendedBaggage,attendedBaggage"-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string, wireless ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object, linkage action-->
      <id>
        <!--ro, req, string, index-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP"
(uploading via HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading
to the center), "LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio)-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, linkage cycle, subType:string, desc:"beginning", "beginningandend", "recurring"-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object, PTZ linkage-->
        <ptzChannelID>
          <!--ro, req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int, linked No., desc:depend on the content of <actionName>, it can be preset No., patrol No., and pattern No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--ro, opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.6.4.7 Get parameters of people gathering for a specified channel

Request URL

PUT /ISAPI/Smart/group/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters configuration of people gathering for a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList>
    <!--opt, array, people gathering area list, subType:object-->
    <GroupRegion>
      <!--opt, object, people gathering area-->
      <id>
        <!--req, string, area ID-->1
      </id>
      <objectOccupation>
        <!--opt, int, percentage-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
</Group>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.6.4.8 Get parameters of people gathering for a specified channel

Request URL

GET /ISAPI/Smart/group/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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DETEKTOR

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters configuration of people gathering for a channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList>
    <!--ro, opt, array, people gathering area list, subType:object-->
    <GroupRegion>
      <!--ro, opt, object, people gathering area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <objectOccupation>
        <!--ro, opt, int, percentage-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
    <headCount>
      <!--ro, opt, int-->5
    </headCount>
    <duration>
      <!--ro, opt, int-->10
    </duration>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
  <shieldedRegionList>
    <!--ro, opt, array, subType:object-->
    <GroupRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </shieldedRegionList>
  <sensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->1
  </sensitivityLevel>
</Group>
```

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16.6.4.9 Get the capability of people gathering

Request URL

GET /ISAPI/Smart/group/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<Group xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, get the capability of people gathering, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, req, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <GroupRegionList size="4">
    <!--ro, opt, object, people gathering area list, attr:size{req, int}-->
    <GroupRegion>
      <!--ro, opt, object, people gathering area-->
      <id>
        <!--ro, req, string, area ID-->1
      </id>
      <objectOccupation min="1" max="100">
        <!--ro, req, int, percentage, range:[1,100], attr:min{req, int},max{req, int}-->1
      </objectOccupation>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
    <headCount min="2" max="20">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->5
    </headCount>
    <duration min="1" max="60">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->10
    </duration>
  </GroupRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, mutual exclusion ability, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool, whether supports multiple scene detection-->true
  </isSupportMultiScene>
  <isSupportOverlapParams>
    <!--ro, opt, bool, whether supports the parameters for overlaying intelligent information of people gathering, desc:GET/PUT
/ISAPI/Smart/group/channels/<channelID>/GroupOverlapParams?format=json-->true
  </isSupportOverlapParams>
  <shieldedRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <GroupRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </GroupRegion>
  </shieldedRegionList>
  <sensitivityLevel min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
</Group>
```

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16.6.4.10 Set parameters of people gathering for all areas

Request URL

PUT /ISAPI/Smart/group/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, batch configure the parameters of people gathering area, subType:object, attr:version{req, string, protocolVersion}-->
  <GroupRegion>
    <!--opt, object, configure the parameter of people gathering area-->
    <id>
      <!--req, string, area ID-->1
    </id>
    <objectOccupation>
      <!--opt, int, percentage-->1
    </objectOccupation>
    <RegionCoordinatesList>
      <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GroupRegion>
</GroupRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.6.4.11 Get the parameters of people gathering in all areas

Request URL

GET /ISAPI/Smart/group/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, batch set the parameters of people gathering area, subType:object, attr:version{req, string, protocolVersion}-->
  <GroupRegion>
    <!--ro, opt, object, set a specified parameter of people gathering area-->
    <id>
      <!--ro, req, string, area ID-->1
    </id>
    <objectOccupation>
      <!--ro, opt, int, percentage-->1
    </objectOccupation>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GroupRegion>
</GroupRegionList>
```

16.6.4.12 Set parameters of people gathering for a specified area

Request URL

PUT /ISAPI/Smart/group/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configure the parameter of people gathering area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, area ID-->1
  </id>
  <objectOccupation>
    <!--opt, int, percentage-->1
  </objectOccupation>
  <RegionCoordinatesList>
    <!--opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</GroupRegion>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.4.13 Get parameters of a specified area for people gathering

Request URL

GET /ISAPI/Smart/group/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, a specified parameter of people gathering area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, area ID-->1
  </id>
  <objectOccupation>
    <!--ro, opt, int, percentage-->1
  </objectOccupation>
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</GroupRegion>
```

16.6.4.14 Get the capability of configuring parameters of a specified area for people gathering

Request URL

GET /ISAPI/Smart/group/<channelID>/regions/<regionID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

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Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people gathering area, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, area ID-->1
  </id>
  <objectOccupation min="1" max="100">
    <!--ro, opt, int, percentage, range:[1,100], attr:min{req, int},max{req, int}-->1
  </objectOccupation>
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</GroupRegion>
```

16.6.4.15 Get the capability of configuring parameters of a specified area for people gathering

Request URL

GET /ISAPI/Smart/group/<channelID>/regions/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<GroupRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, people gathering area list, attr:version{req, string, protocolVersion}-->
  <GroupRegion>
    <!--ro, opt, object, people gathering area-->
    <id>
      <!--ro, req, string, area ID-->1
    </id>
    <objectOccupation min="1" max="100">
      <!--ro, opt, int, percentage, range:[1,100], attr:min{req, int},max{req, int}-->1
    </objectOccupation>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </GroupRegion>
</GroupRegionList>
```

16.6.4.16 People gathering

EventType:group

```
<?xml version="1.0" encoding="UTF-8"?>
```

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```
<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, string, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->test
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"group" (people gathering)-->group
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <objectOccupation>
        <!--ro, opt, int, percentage, range:[1,100]-->1
      </objectOccupation>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture
contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
```

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```
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest
authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/
"deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance
vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect"
(video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing
properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)--
>realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message
indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is
uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
group	[Message content]	application/xml	--	--	--
.value	[Binary picture data]	image/jpeg	随机值	pic.jpeg	--
video	[Video data]	video/mp4	随机值	video.mp4	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

16.6.5 Unattended Baggage Detection

16.6.5.1 Set the arming schedule of unattended baggage detection for all channels

Request URL

PUT /ISAPI/Event/schedules/unattendedBaggages

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<UnattendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->unattendedBaggage-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->unattendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object, range:[0,8]-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</UnattendBaggageScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.5.2 Get the arming schedule of unattended baggage detection for all channels

Request URL

GET /ISAPI/Event/schedules/unattendedBaggages

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<UnattendBaggageScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->unattendedBaggage-1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->unattendedBaggage
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object, range:[0,8]-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</UnattendBaggageScheduleList>
```

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16.6.5.3 Get the arming schedule of unattended baggage detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/unattendedBaggages/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->unattendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object, range:[0,8]-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.6.5.4 Set the arming schedule of unattended baggage detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/unattendedBaggages/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, enum, subType:string-->unattendedBaggage
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, range:[0,56], attr:size{req, int, range:[0,8]}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object, range:[0,8]-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.5.5 Get the linkage parameters of unattended baggage detection

Request URL

GET /ISAPI/Event/triggers/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->unattendedBaggage
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.6.5.6 Set the linkage parameters of unattended baggage detection

Request URL

PUT /ISAPI/Event/triggers/unattendedBaggage-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->unattendedBaggage-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->unattendedBaggage
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.6.5.7 Unattended baggage detection

EventType:unattendedBaggage

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, unattended baggage alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, event triggering frequency, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"unattendedBaggage" (unattended baggage)-->unattendedBaggage
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, target area, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
```

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```
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"others"-->others
      </detectionTarget>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification contents-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
```

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```
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm, desc:ISO 8601 format, e.g., 2019-12-30T15:00:00.23+08:00-->test
  </timeStamp>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
</EventNotificationAlert>
```

16.7 Face Capture

16.7.1 AcuSearch (Face)

16.7.1.1 Get the capability set of searching by picture in captured picture library

Request URL

GET /ISAPI/SDT/Face/searchByPic/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, string, search start time*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, string, search end time*/
    "channelNum":  100,
    /*ro, req, int, number of cameras can be selected*/
    "resultMaxNum":  500,
    /*ro, req, int, maximum results can be searched*/
    "modelThreshold": [
    /*ro, opt, array, maximum and minimum threshold for comparison, subType:object*/
        {
            "min":  0,
            /*ro, opt, int, the minimum value*/
            "max":  100
            /*ro, req, int, the maximum value*/
        }
    ],
    "faceURL": {
    /*ro, opt, object, length of picture URL*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  100
        /*ro, opt, int, the maximum value*/
    },
    "targetModelData": {
    /*ro, opt, object, length of target model data*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  100
        /*ro, opt, int, the maximum value*/
    },
    "sortType":  "similarity,time",
    /*ro, opt, string, sorting type, desc:“similarity” (by similarity), “time” (by time)*/
    "URLType":  "cloudURL",
    /*ro, opt, string, URL type of the returned picture after search, desc:this node is available for devices supporting local URL (such as NVR) and by
default the device will return the local URL*/
    "maxResultsLocal": {
    /*ro, req, object, the maximum records of current search*/
        "min":  0,
        /*ro, opt, int, the minimum value*/
        "max":  1000
        /*ro, opt, int, the maximum value*/
    },
    "isSupportAsync":  true,
    /*ro, opt, bool, whether the device supports asynchronously searching face by picture*/
}
```

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16.7.1.2 Get task progress of using picture to search picture

Request URL

GET /ISAPI/SDT/Face/searchByPic/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID": "ffc98d57-c156-40a8-ae64-84cb98dcee10",
    /*ro, req, string, task ID of using picture to search picture*/
    "progress": 100,
    /*ro, req, int, task progress of using picture to search picture*/
    "analysisModelNum": 100
    /*ro, req, int, total number of models to be compared*/
}
```

16.7.1.3 Search for the results of asynchronously using picture to search picture

Request URL

POST /ISAPI/SDT/Face/searchByPic/result?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

```
{
    "searchResultPosition": 0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults": 100
    /*req, int, the maximum number of search results this time*/
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "activeNumMax",
    /*ro, req, enum, sub status code, subType:string, desc:"ok" (succeeded). It is required when an error occurred. "activeNumMax" (the max. number of
device allowed for License access is reached), "licenseExpired" (License expired)*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "taskID": "test",
    /*ro, req, string, task ID of using picture to search picture*/
    "responseStatusStrg": "OK",
    /*ro, req, string, searching status description*/
    "numOfMatches": 1,
    /*ro, req, int, number of matched results returned this time*/
    "totalMatches": 1,
    /*ro, req, int, total number of matched results by this search*/
    "targets": [
    /*ro, opt, array, list of matched records, subType:object*/
        {
            "id": 1,
            /*ro, req, int, No.*/
            "captureTime": "2004-05-03T17:30:08+08:00",
            /*ro, opt, string, capture time*/
            "isSummerTime": false,
            /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
            "deviceType": "Blade",
            /*ro, req, enum, device type, subType:string, desc:"Blade" (vehicle analysis server), "BladePS" (security vehicle analysis server), "CVR","DM"
(full analysis server), "DMbehavior" (abnormal event detection device), "DVR","DVS", "DockStation" (dock station), "FA" (real-time face picture comparison
and alarm server), "FD" (capture detection server), "FS" (face picture static data server), "HAWK" (video structural analysis server), "HMCP" (model
comparison server), "HybirdNVR" (hybrid DVR), "IPCamera" (network camera), "IPDome" (network speed dome), "IPZoom" (integrated camera), "InfoReleaseSys"
(information release system), "InfoTerminal" (information release terminal), "MCU", "NVR", "PHA" (security control panel), "PURE" (pure analysis server),
"RFID", "SipServer" (SIP server), "Switch", "lightFace" (lightweight facial recognition server)*/
            "deviceId": "test",
            /*ro, opt, string, camera ID*/
            "captureSite": "3#大门处",
            /*ro, opt, string, capture site (camera name)*/
```

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```
            "monitorId":  "1",
            /*ro, opt, string, stream ID (encoding stream ID in CVR). This node will not be returned if the device does not support this function*/
            "ageGroup":  "any",
            /*ro, req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"*/
            "gender":  "any",
            /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
            "glasses":  "no",
            /*ro, req, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses), "unknown",
"yes"*/
            "smile":  "no",
            /*ro, req, enum, whether the person wears a smile, subType:string, desc:smile or not: "unknown", "yes", "no"*/
            "picUrl":  "xxx",
            /*ro, opt, string, background picture URL*/
            "subpicUrl":  "xxx",
            /*ro, req, string, face picture thumbnail URL*/
            "rect": {
            /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                "height":  1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "face_pose": {
            /*ro, opt, object, face angle*/
                "pitch":  0.514,
                /*ro, opt, float, face pitch angle, face pitching up is regarded as positive*/
                "yaw":  0.2909,
                /*ro, opt, float, face yaw angle, face yawing left is regarded as positive*/
                "roll":  0.3669
                /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive*/
            },
            "face_rect": {
            /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                "height":  1.000,
                /*ro, opt, float, height, range:[0.000,1.000]*/
                "width":  1.000,
                /*ro, opt, float, width, range:[0.000,1.000]*/
                "x":  0.000,
                /*ro, opt, float, x-coordinate, range:[0.000,1.000]*/
                "y":  0.000
                /*ro, opt, float, y-coordinate, range:[0.000,1.000]*/
            },
            "similarity":  1.00,
            /*ro, opt, float, similarity*/
            "errInfo": {
            /*ro, opt, object, error description, desc:object, error information. This node is valid when the value of the node subpicUrl is empty*/
                "errMsg":  "test",
                /*ro, req, string, error description, desc:the reason when the value of the node subpicUrl is empty*/
                "errType":  0
                /*ro, req, int, error type, desc:error type, which is the reason type when the value of the node subpicUrl is empty*/
            },
            "channelID":  1,
            /*ro, opt, int, channel No.*/
            "URLType":  "cloudURL",
            /*ro, opt, string, type of returned picture URL, desc:type of returned picture URL*/
            "faceExpression":  "angry",
            /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced",
"sad", "surprised", "unknown"*/
            "mask":  "no",
            /*ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
            "linkFaceBodyId":  "XXX",
            /*ro, opt, string, linkage ID between face and human body picture*/
            "monitorPointID":  "test",
            /*ro, opt, string, camera ID*/
            "thermometryUnit":  "celsius",
            /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
            "currTemperature":  38.0,
            /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
            "temperatureCondition":  "normal"
            /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
        }
    ]
}
```

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16.7.1.4 Submit the task of asynchronously searching by picture

Request URL

POST /ISAPI/SDT/Face/searchByPic?format=json

Query Parameter

None

Request Message

```
{
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, start time of search*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*opt, string, end time of search*/
    "choiceChannel": [
    /*opt, array, list of selected cameras, subType:object*/
        {
            "channelID":  "test",
            /*opt, string, camera ID*/
            "channelName":  "channel1"
            /*opt, string, camera name, range:[0,64]*/
        }
    ],
    "similarityMin":  0.3,
    /*req, float, minimum threshold for comparison, range:[0.00,1.00], desc:minimum threshold for comparison*/
    "similarityMax":  1.0,
    /*req, float, maximum threshold for comparison, range:[0.00,1.00], desc:maximum threshold for comparison*/
    "age":  23,
    /*opt, int, age*/
    "ageGroup":  "any",
    /*req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime", "teenager",
"unknown", "young"*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"*/
    "glasses":  "no",
    /*req, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses), "unknown", "yes"*/
    "smile":  "no",
    /*req, enum, whether the person is smiling, subType:string, desc:"no", "unknown", "yes"*/
    "mask":  "no",
    /*req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
    "modelMaxNum":  100,
    /*req, int, maximum number of model comparison results (the maximum number of returned results), desc:maximum number of model comparison results (the
maximum number of returned results)*/
    "targetModelData":  "test",
    /*req, string, target model data*/
    "sortType":  "similarity",
    /*opt, enum, sorting type, subType:string, desc:"similarity" (sort by similarity), "time" (sort by time)*/
    "URLType":  "cloudURL",
    /*opt, string, type of returned picture URL, desc:the URL type of searched picture*/
    "pictureMerge":  false,
    /*opt, bool, whether to merge multiple pictures of one person, desc:the value of this node is false by default*/
    "thermometryUnit":  "celsius",
    /*opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
    "currTemperature":  38.0,
    /*opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
    "temperatureCondition":  "normal"
    /*opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceInfo	[Message content]	application/json	--	--	--
faceImage	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "ffc98d57-c156-40a8-ae64-84cb98dcee10"
    /*ro, req, string, task ID of searching by picture*/
}
```

16.7.1.5 Get the task information

Request URL

GET /ISAPI/SDT/Face/searchByPic?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskInfo": [
    /*ro, opt, array, task information, subType:object*/
        {
            "taskID":  "abc",
            /*ro, req, string, task ID of searching by picture, range:[0,64]*/
            "taskCreateTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, task creation time*/
            "timeUsed":  100,
            /*ro, opt, int, task execution duration, unit:s, desc:unit: second*/
            "progress":  100,
            /*ro, opt, int, task progress*/
            "TaskDetails": {
            /*ro, opt, object, task details*/
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, string, start time of search*/
                "endTime":  "2004-05-03T17:30:08+08:00",
                /*ro, opt, string, end time of search*/
                "choiceChannel": [
                /*ro, opt, array, list of selected cameras, subType:object*/
                    {
                        "channelID":  "test",
                        /*ro, opt, string, camera ID*/
                        "channelName":  "channel1"
                        /*ro, opt, string, camera name*/
                    }
                ],
                "similarityMin":  0.3,
                /*ro, req, float, minimum threshold for comparison, range:[0.00,1.00], desc:it should be accurate to two decimal place, the default value is
0*/
                "similarityMax":  1.0,
                /*ro, opt, float, maximum threshold for comparison, range:[0.00,1.00], desc:it should be accurate to two decimal place, the default value is
0*/
                "age":  23,
                /*ro, opt, int, age*/
                "ageGroup":  "any",
                /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"*/
                "gender":  "any",
                /*ro, opt, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
                "glasses":  "no",
                /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no" (not wear glasses), "sunglasses" (wears sunglasses),
"unknown"*/
                "smile":  "no",
                /*ro, opt, enum, whether the person wears a smile, subType:string, desc:"no", "unknown", "yes"*/
                "mask":  "no",
                /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"*/
                "modelMaxNum":  100,
                /*ro, opt, int, maximum number of model comparison results, desc:the maximum number of returned results*/
                "targetModelData":  "test",
                /*ro, opt, string, target model data*/
                "sortType":  "similarity",
                /*ro, opt, enum, sorting type, subType:string, desc:"similarity" (sort by similarity), "time" (sort by time)*/
                "URLType":  "cloudURL",
                /*ro, opt, string, type of returned picture URL, desc:the URL type of searched picture*/
                "pictureMerge":  false,
                /*ro, opt, bool, whether to merge multiple pictures of one person, desc:the value of this node is false by default*/
                "thermometryUnit":  "celsius",
                /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                "currTemperature":  38.0,
                /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃*/
                "temperatureCondition":  "normal"
                /*ro, opt, enum, face temperature status, subType:string, desc:"normal", "abnormal", "all"*/
            }
        }
    ]
}
```

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16.7.1.6 Delete the task of searching picture by picture asynchronously

Request URL

DELETE /ISAPI/SDT/Face/searchByPic?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

16.7.2 Face Capture

16.7.2.1 Get the arming schedule of face capture

Request URL

GET /ISAPI/Event/schedules/faceSnap

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming schedule of face capture for all channels, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule of face capture for a specified channel-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <eventType>
      <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpLoad",
"HFP", "IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->ADAS
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel No.-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming time list, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--ro, opt, object, arming advanced configurations-->
          <vehicleDetectSceneID>
            <!--ro, opt, int, vehicle detection scene ID-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--ro, opt, int, scene ID-->1
          </sceneID>
          <SceneList>
            <!--ro, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with "sceneID"-->
            <Scene>
              <!--ro, opt, object, multiple scene list-->
              <sceneID>
                <!--ro, req, string, scene ID-->1
              </sceneID>
              <durationTime>
                <!--ro, req, int, stayed time of a single scene-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapScheduleList>
```

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16.7.2.2 Set the arming schedule of face capture

Request URL

PUT /ISAPI/Event/schedules/faceSnap

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, array, arming schedule of face capture for all channels, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule of face capture for a specified channel-->
    <id>
      <!--wo, req, string, channel No.-->test
    </id>
    <eventType>
      <!--wo, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpLoad",
"HFP", "IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fieldetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
    </eventType>
    <videoInputChannelID>
      <!--wo, opt, string, video input channel No.-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--wo, req, array, arming time list, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--wo, opt, object, arming slots-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:days of the week-->1
        </dayOfWeek>
        <TimeRange>
          <!--wo, req, object, time range-->
          <beginTime>
            <!--wo, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--wo, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--wo, opt, object, arming advanced configurations-->
          <vehicleDetectSceneID>
            <!--wo, opt, int, vehicle detection scene-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--wo, opt, int, scene ID-->1
          </sceneID>
          <SceneList>
            <!--wo, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with "sceneID"-->
            <Scene>
              <!--wo, opt, object, multiple scene list-->
              <sceneID>
                <!--wo, req, string, scene ID-->1
              </sceneID>
              <durationTime>
                <!--wo, req, int, stayed time of a single scene-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--wo, opt, array, holiday arming time list, subType:object-->
      <TimeBlock>
        <!--wo, opt, object, arming time-->
        <TimeRange>
          <!--wo, req, object, time range-->
          <beginTime>
            <!--wo, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--wo, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.2.3 Get the arming schedule of face capture for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, arming schedule of face capture for a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpload", "HFP",
"IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel No.-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--ro, opt, object-->
        <vehicleDetectSceneID>
          <!--ro, opt, int-->1
        </vehicleDetectSceneID>
        <sceneID>
          <!--ro, opt, int-->1
        </sceneID>
        <SceneList>
          <!--ro, opt, array, subType:object-->
          <Scene>
            <!--ro, opt, object-->
            <sceneID>
              <!--ro, req, string-->1
            </sceneID>
            <durationTime>
              <!--ro, req, int-->1
            </durationTime>
          </Scene>
        </SceneList>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
  <backToSceneTime>
    <!--ro, opt, int, range:[5,100], step:1, unit:s-->15
  </backToSceneTime>
</Schedule>
```

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16.7.2.4 Set the arming schedule of face capture for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceSnap/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, arming schedule of face capture for a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--wo, req, string, channel No.-->test
  </id>
  <eventType>
    <!--wo, req, enum, event type, subType:string, desc:“ADAS”, “ADASAlarm”, AID, ANPR, AccessControllerEvent, CDsStatus, DBD, GPSUpload, HFP, IO, IOTD,
LES, LFPD, PALMismatch, PIR, PeopleCounting, PeopleNumChange, Standup, TMA, TMPAM, VMD, abnormalAcceleration, abnormalDriving, advReachHeight, alarmResult,
attendance, attendedBaggage, audioAbnormal, audioexception, behaviorResult, blindSpotDetection, cardMatch, changedStatus, collision, containerDetection,
crowdSituationAnalysis, databaseException, defocus, diskUnformat, diskerror, diskfull, driverConditionMonitor, emergencyAlarm, faceCapture,
faceSnapModeling, facedetection, failDown, faultAlarm, fielddetection, fireDetection, fireEscapeDetection, flowOverrun, framesPeopleCounting, getUp, group,
hdBadBlock, hdImpact, heatmap, highHDTemperature, highTempAlarm, hotSpare, illaccess, ipcTransferAbnorma, ipconflict, keyPersonGetUp, leavePosition,
linedetection , listSyncException, loitering, lowHDTemperature, mixedTargetDetection, modelError, nicbroken, nodeOffline, nonPoliceIntrusio, overSpeed,
overtimeTarry, parking, peopleNumChange, peopleNumCounting, personAbnormalAlarm:, personDensityDetection, personQueueCounting, personQueueDetection,
personQueueRealTime, personQueueTime, playCellphone, pocException, poe, policeAbsent, radarAlarm, radarFieldDetection, radarlineDetection,
radarPerimeterRule, radarTargetDetection, radarVideoDetection, raidException, rapidMove, reachHeight, recordCycleAbnormal, recordException, regionEntrance,
regionExiting, retention, rollover, running, safetyHelmetDetection, scenechangedetection, sensorAlarm, severeHDFailure, shelteralarm, shipsDetection,
sitQuietly, smokeAndFireDetection, smokeDetection, softIO, spacingChange, sysStorFull, takingElevatorDetection, targetCapture, temperature, thermometry,
thirdPartyException, toiletTarry, tollCodeInfo, tossing, unattendedBaggage, vehicleMatchResult, vehicleRcogResult, versionAbnormal, videoException,
videoloss, violationAlarm, violentMotion, yardTarry, faceSnap-->faceSnap
  </eventType>
  <videoInputChannelID>
    <!--wo, opt, string, video input channel No.-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--wo, req, array, arming time list, subType:object, attr:size{req, int}-->
    <TimeBlock>
      <!--wo, opt, object, arming time-->
      <dayOfWeek>
        <!--wo, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)--
>1
      </dayOfWeek>
      <TimeRange>
        <!--wo, req, object, time range-->
        <beginTime>
          <!--wo, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--wo, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--wo, opt, object, arming advanced configurations-->
        <vehicleDetectSceneID>
          <!--wo, opt, int, vehicle detection scene-->1
        </vehicleDetectSceneID>
        <sceneID>
          <!--wo, opt, int, scene ID-->1
        </sceneID>
        <SceneList>
          <!--wo, opt, array, switching rules of multiple scenes, subType:object, desc:mutually exclusive with “sceneID”-->
          <Scene>
            <!--wo, opt, object, multiple scene list-->
            <sceneID>
              <!--wo, req, string, scene ID-->1
            </sceneID>
            <durationTime>
              <!--req, int, stayed time of a single scene-->1
            </durationTime>
          </Scene>
        </SceneList>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, holiday arming time list, subType:object-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time range-->
        <beginTime>
          <!--req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.2.5 Set the arming schedule of face modeling for all channels

Request URL

PUT /ISAPI/Event/schedules/faceSnapModeling

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapModelingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object-->
    <id>
      <!--req, string-->faceSnapModeling-1
    </id>
    <eventType>
      <!--opt, enum, subType:string-->faceSnapModeling
    </eventType>
    <inputIOPortID>
      <!--opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--opt, object-->
          <vehicleDetectSceneID>
            <!--opt, int-->0
          </vehicleDetectSceneID>
          <sceneID>
            <!--opt, int-->0
          </sceneID>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapModelingScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.2.6 Get the arming schedule of face modeling for all channels

Request URL

GET /ISAPI/Event/schedules/faceSnapModeling

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceSnapModelingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, req, object-->
    <id>
      <!--ro, req, string-->faceSnapModeling-1
    </id>
    <eventType>
      <!--ro, opt, enum, subType:string-->faceSnapModeling
    </eventType>
    <inputIOPortID>
      <!--ro, opt, string-->test
    </inputIOPortID>
    <outputIOPortID>
      <!--ro, opt, string-->test
    </outputIOPortID>
    <videoInputChannelID>
      <!--ro, opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object-->
        <dayOfWeek>
          <!--ro, opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--ro, opt, object-->
          <vehicleDetectSceneID>
            <!--ro, opt, int-->0
          </vehicleDetectSceneID>
          <sceneID>
            <!--ro, opt, int-->0
          </sceneID>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object-->
        <TimeRange>
          <!--ro, req, object-->
          <beginTime>
            <!--ro, req, time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceSnapModelingScheduleList>
```

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16.7.2.7 Set the arming schedule of face modeling for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceSnapModeling/faceSnapModeling-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceSnapModeling-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->faceSnapModeling
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--opt, object-->
        <vehicleDetectSceneID>
          <!--opt, int-->0
        </vehicleDetectSceneID>
        <sceneID>
          <!--opt, int-->0
        </sceneID>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.2.8 Get the arming schedule of face modeling for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceSnapModeling/faceSnapModeling-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceSnapModeling-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->faceSnapModeling
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
      <ScheduleProperity>
        <!--ro, opt, object-->
        <vehicleDetectSceneID>
          <!--ro, opt, int-->0
        </vehicleDetectSceneID>
        <sceneID>
          <!--ro, opt, int-->0
        </sceneID>
      </ScheduleProperity>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.7.2.9 Delete the linkage of face capture

Request URL

DELETE /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.7.2.10 Set the linkage parameters of face capture

Request URL

PUT /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, req, object, Configure linkage actions:, attr:version{req, string, protocolVersion}-->
  <id>
    <!--wo, req, string, channel No.-->test
  </id>
  <eventType>
    <!--wo, req, enum, event type, subType:string, desc:“ADAS”, “ADASAlarm”, “AID”, “ANPR”, “AccessControllerEvent”, “CDsStatus”, “DBD”, “GPSUpload”, “HFP”,
“IO”, “IOTD”, “LES”, ‘LFPD“, ”PALMismatch“, ”PIR“, ”PeopleCounting“, ”PeopleNumChange”, “Standup”, “TMA”, “TMPAM”, “VMD”, “abnormalAcceleration”,
“abnormalDriving”, “advReachHeight”, “alarmResult”, “attendance”, “attendedBaggage”, “audioAbnormal”, “audioexception”, “behaviorResult”,
“blindSpotDetection”, “cardMatch”, “changedStatus”, “collision”, “containerDetection”, “crowdSituationAnalysis”, “databaseException”, “defocus”,
“diskUnformat”, “diskerror”, “diskfull”, “driverConditionMonitor”, “emergencyAlarm”, “faceCapture”, “faceSnapModeling”, “facedetection”, “failDown”,
“faultAlarm”, “fielddetection”, “fireDetection”, “fireEscapeDetection”, “flowOverrun”, “framesPeopleCounting”, “getUp”, “group”, “hdBadBlock”, “hdImpact”,
“heatmap”, “highHDTemperature”, “highTempAlarm”, “hotSpare”, “illaccess”, “ipcTransferAbnorma”, “ipconflict”, “keyPersonGetUp”, “leavePosition”,
“linedetection”, “listSyncException”, “loitering”, “LowHDTemperature”, "mixedTargetDetection”, “modelError”, “nicbroken”, “nodeOffline”,
“nonPoliceIntrusio”, “overSpeed”, “overtimeTarry”, “parking”, “peopleNumChange”, “peopleNumCounting”, “personAbnormalAlarm”, “personDensityDetection”,
“personQueueCounting”, “personQueueDetection”, “personQueueRealTime”, “personQueueTime”, “playCellphone”, “pocException”, “poe”, “policeAbsent”,
“radarAlarm”, “radarFieldDetection”, “radarLineDetection”, “radarPerimeterRule”, “radarTargetDetection”, “radarVideoDetection”, “raidException”,
“rapidMove”, “reachHeight”, “recordCycleAbnormal”, “recordException”, “regionEntrance”, “regionExiting”, “retention”, “rollover”, “running”,
“safetyHelmetDetection”, “scenechangedetection”, “sensorAlarm”, “severeHDFailure”, “shelteralarm”, “shipsDetection”, “sitQuietly”, “smokeAndFireDetection”,
“smokeDetection”, “softIO”, “spacingChange”, “sysStorFull”, “takingElevatorDetection”, “targetCapture”, “temperature”, “thermometry”, “thirdPartyException”,
“toiletTarry”, “tollCodeInfo”, “tossing”, “unattendedBaggage”, “vehicleMatchResult”, “vehicleRcogResult”, “versionAbnormal”, “videoException”, “videoloss”,
“violationAlarm”, “violentMotion”, “yardTarry”, “faceSnap”-->faceSnap
  </eventType>
  <eventDescription>
    <!--wo, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--wo, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--wo, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--wo, opt, string, video input channel ID, desc:only valid when event types are VMD, videoloss, shelteralarm, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, thermal, zoneTriggeredAlarm, criticalAlerts, businessCounselingLinkage-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--wo, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--wo, opt, int, event capture interval, step:1, unit:s-->1
```

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```
    </intervalBetweenEvents>
    <WLSensorID>
      <!--wo, opt, string, wireless ID-->test
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--wo, opt, array, configuration list of event linkage, subType:object-->
      <EventTriggerNotification>
        <!--wo, opt, object, list-->
        <id>
          <!--wo, req, string, linkage configuration ID-->test
        </id>
        <notificationMethod>
          <!--wo, req, enum, linkage action, subType:string, desc:"FTP" (uploading via FTP), "HTTP" (uploading via HTTP), "IM" (instant messaging), "IO" (I/O
output), "lightAudioAlarm" (light and audio alarm), "audio" (audio), "beep" (buzzer), "center" (uploading to the center), "cloud" (cloud storage), "email",
"focus" (focus), "monitorAlarm" (alarm on monitor), "ptz" (PTZ), "record" (recording), "syslog" (system log), "trace" (tracking), "whiteLight" (white
light)-->FTP
        </notificationMethod>
        <notificationRecurrence>
          <!--wo, opt, enum, linkage cycle, subType:string, desc:“beginning”, “beginningandend”, “recurring”-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--wo, opt, int, linkage time interval, step:1, unit:ms-->1
        </notificationInterval>
        <videoInputID>
          <!--wo, opt, string, video input ID-->test
        </videoInputID>
        <dynVideoInputID>
          <!--wo, opt, string, dynamic video input ID-->test
        </dynVideoInputID>
        <ptzAction>
          <!--wo, opt, object, PTZ linkage action-->
          <ptzChannelID>
            <!--wo, req, string, PTZ linked channel No.-->test
          </ptzChannelID>
          <actionName>
            <!--wo, req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
          </actionName>
          <actionNum>
            <!--wo, opt, int, linkage value-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--wo, opt, object, white light linkage-->
          <whiteLightDurationTime>
            <!--wo, req, int, light flashing duration, range:[1,60], step:1, unit:s-->1
          </whiteLightDurationTime>
        </WhiteLightAction>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.2.11 Get the linkage parameters of face capture

Request URL

GET /ISAPI/Event/triggers/faceSnap-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Device channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, linkage configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ADAS", "ADASAlarm", "AID", "ANPR", "AccessControllerEvent", "CDsStatus", "DBD", "GPSUpload", "HFP",
"IO", "IOTD", "LES", 'LFPD", "PALMismatch", "PIR", "PeopleCounting", "PeopleNumChange", "Standup", "TMA", "TMPAM", "VMD", "abnormalAcceleration",
"abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult",
"blindSpotDetection", "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus",
"diskUnformat", "diskerror", "diskfull", "driverConditionMonitor", "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown",
"faultAlarm", "fielddetection", "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp", "group", "hdBadBlock", "hdImpact",
"heatmap", "highHDTemperature", "highTempAlarm", "hotSpare", "illaccess", "ipcTransferAbnorma", "ipconflict", "keyPersonGetUp", "leavePosition",
"linedetection", "listSyncException", "loitering", "LowHDTemperature", "mixedTargetDetection", "modelError", "nicbroken", "nodeOffline",
"nonPoliceIntrusio", "overSpeed", "overtimeTarry", "parking", "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm", "personDensityDetection",
"personQueueCounting", "personQueueDetection", "personQueueRealTime", "personQueueTime", "playCellphone", "pocException", "poe", "policeAbsent",
"radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule", "radarTargetDetection", "radarVideoDetection", "raidException",
"rapidMove", "reachHeight", "recordCycleAbnormal", "recordException", "regionEntrance", "regionExiting", "retention", "rollover", "running",
"safetyHelmetDetection", "scenechangedetection", "sensorAlarm", "severeHDFailure", "shelteralarm", "shipsDetection", "sitQuietly", "smokeAndFireDetection",
"smokeDetection", "softIO", "spacingChange", "sysStorFull", "takingElevatorDetection", "targetCapture", "temperature", "thermometry", "thirdPartyException",
"toiletTarry", "tollCodeInfo", "tossing", "unattendedBaggage", "vehicleMatchResult", "vehicleRcogResult", "versionAbnormal", "videoException", "videoloss",
"violationAlarm", "violentMotion", "yardTarry", "faceSnap"-->faceSnap
  </eventType>
  <eventDescription>
    <!--ro, opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string, I/O input port ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string, dynamic I/O input port ID-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, video input channel ID, desc:only valid when event types are VMD, videoloss, shelteralarm, regionEntrance, regionExiting,
loitering, group, rapidMove, parking, unattendedBaggage, attendedBaggage, thermal, zoneTriggeredAlarm, criticalAlerts, businessCounselingLinkage-->test
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string, dynamic video input channel ID-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, event capture interval, step:1, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string, wireless ID-->test
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, step:1, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->test
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->test
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->test
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, req, int, range:[1,60], step:1, unit:s-->1
        </whiteLightDurationTime>
```

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```
      </WhiteLightAction>
      <childDevID>
        <!--ro, opt, string-->1
      </childDevID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

16.7.2.12 Set the face capture rule parameters of a device

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--req, enum, whether to enable the function, subType:string, desc:“true”, “false”-->true
  </enabled>
  <FaceDetectionRegion>
    <!--opt, object, detection area-->
    <RegionCoordinatesList>
      <!--opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </FaceDetectionRegion>
  <InterPupileRegion>
    <!--opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
  <MaxInterPupilRegion>
    <!--opt, object, the maximum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--req, int, pupil distance, range:[0,1000]-->50
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->50
    </height>
  </MaxInterPupilRegion>
  <SizeFilter>
    <!--opt, object, size filter-->
    <enabled>
      <!--req, bool, whether to enable size filter-->true
    </enabled>
    <ratio>
      <!--opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
    </ratio>
    <MaxTargetSize>
      <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000], desc:X-coordinate-->0
      </positionX>
```

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```
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--req, int, height, range:[0,1000]-->0
        </height>
      </MaxTargetSize>
      <MinTargetSize>
        <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--req, int, height, range:[0,1000]-->0
        </height>
      </MinTargetSize>
    </SizeFilter>
    <detectionType>
      <!--opt, string, face type for detection, desc:"abnormalFace" (abnormal face),"normalFace" (normal face),"multiFace" (multiple faces),"sunglassesFace"
(face,on which wearing sunglasses),"callFace" (face of a person who is making a phone call). Multiple values are separated by commas--
>abnormalFace,normalFace
    </detectionType>
    <autoROIEnable>
      <!--opt, bool, whether to enable face automatic ROI, desc:the value of this node is false by default-->false
    </autoROIEnable>
  </FaceRule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.7.2.13 Get the capability of configuring parameters of face capture rules.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--ro, req, object, capability of configuring parameters of face capture rules, attr:version{opt, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, enum, whether to enable, subType:string, attr:opt{opt, string}, desc:whether to enable-->true
  </enabled>
  <FaceDetectionRegion>
```

```
    <FaceDetectionRegion>
      <!--ro, opt, object, detection area-->
      <RegionCoordinatesList>
        <!--ro, opt, array, coordinate list of the area, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, req, object, coordinates of the area, desc:The origin is the lower-left corner of the screen.-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </FaceDetectionRegion>
    <InterPupileRegion>
      <!--ro, opt, object, the rectangle frame on the minimum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width min="0" max="1000">
        <!--ro, req, int, pupillary distance, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </InterPupileRegion>
    <MaxInterPupilRegion>
      <!--ro, opt, object, the maximum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width min="50" max="50">
        <!--ro, req, int, pupillary distance, range:[0,1000], attr:min{opt, string},max{opt, string}-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--ro, opt, object, size filter-->
      <enabled opt="true,false" def="false">
        <!--ro, req, bool, whether to enable size filter, attr:opt{opt, string},def{opt, string}-->true
      </enabled>
      <ratio min="4.0" max="6.0" def="5.0">
        <!--ro, opt, float, ratio of the size filter of the head and shoulders to the minimum pupillary distance, range:[4.0,6.0], attr:min{opt,
string},max{opt, string},def{opt, string}-->5.0
      </ratio>
      <MaxTargetSize>
        <!--ro, opt, object, target frame on the largest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX min="0" max="1000">
          <!--ro, req, int, X-coordinate, attr:min{req, int},max{req, int}-->0
        </positionX>
        <positionY min="0" max="1000">
          <!--ro, req, int, Y-coordinate, attr:min{req, int},max{req, int}-->0
        </positionY>
        <width min="0" max="1000">
          <!--ro, req, int, width, attr:min{req, int},max{req, int}-->0
        </width>
        <height min="0" max="1000">
          <!--ro, req, int, height, attr:min{req, int},max{req, int}-->0
        </height>
      </MaxTargetSize>
      <MinTargetSize>
        <!--ro, opt, object, target frame on the smallest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX min="0" max="1000">
          <!--ro, req, int, X-coordinate, attr:min{req, int},max{req, int}-->0
        </positionX>
        <positionY min="0" max="1000">
          <!--ro, req, int, Y-coordinate, attr:min{req, int},max{req, int}-->0
        </positionY>
        <width min="0" max="1000">
          <!--ro, req, int, width, attr:min{req, int},max{req, int}-->0
        </width>
        <height min="0" max="1000">
          <!--ro, req, int, height, attr:min{req, int},max{req, int}-->0
        </height>
      </MinTargetSize>
    </SizeFilter>
    <isSupportTrigger>
      <!--ro, opt, bool, whether it supports trigger mode, desc:Related URI: /ISAPI/Intelligent/channels/<ID>/faceRule/trigger?format=json-->true
    </isSupportTrigger>
    <detectionType opt="abnormalFace,normalFace,multiFace,sunglassesFace,callFace" def="normalFace,multiFace,sunglassesFace,callFace">
      <!--ro, opt, string, face type for detection, attr:opt{req, string},def{req, string}, desc:abnormalFace (abnormal face), normalFace (normal face),
multiFace (multiple faces), sunglassesFace (face with sunglasses), callFace (face of a person who is making a phone call, which is used to detect faces in
ATM mode). Multiple types are separated by commas.-->test
    </detectionType>
    <defualt opt="true">
```

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```
    <!--ro, opt, bool, whether it supports getting the default parameters, attr:opt{req, string}, desc:If the value of this node is true, it indicates that
the query parameter default can be added to the URI GET /ISAPI/Intelligent/channels/<ID>/faceRule for getting the default parameters, i.e., GET
/ISAPI/Intelligent/channels/<ID>/faceRule?defualt=true.-->true
  </defualt>
  <autoROIEnable opt="true,false">
    <!--ro, opt, bool, whether to enable automatic face ROI, attr:opt{req, string}, desc:By default it is false.-->false
  </autoROIEnable>
  <MEF>
    <!--ro, opt, object, mobile electric fence, desc:Related URI: /ISAPI/Intelligent/channels/<ID>/faceRule/MEF?format=json-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable uploading the electric fence data linked with the event, attr:opt{req, string}-->true
    </enabled>
  </MEF>
  <isSupportFaceSnapPosOverlayParams>
    <!--ro, opt, bool-->true
  </isSupportFaceSnapPosOverlayParams>
  <backToSceneTime min="5" max="100">
    <!--ro, opt, int, range:[5,100], step:1, unit:s, attr:min{req, int},max{req, int}-->15
  </backToSceneTime>
  <minInterPupileDistance min="13" max="666">
    <!--ro, opt, int, range:[13,666], step:1, unit:px, attr:min{req, int},max{req, int}-->13
  </minInterPupileDistance>
  <isSupportSearchFaceSnapAlarm>
    <!--ro, opt, bool-->true
  </isSupportSearchFaceSnapAlarm>
  <isSupportFaceApproachParams>
    <!--ro, opt, bool-->true
  </isSupportFaceApproachParams>
  <isSupportChannelPrivacyMaskList>
    <!--ro, opt, bool-->true
  </isSupportChannelPrivacyMaskList>
</FaceRule>
```

16.7.2.14 Set the scene mode parameters of multiple scenes for face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/multiScene/sceneMode?format=json&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
mode	string	--

Request Message

```
{
    "FaceSnapMultiSceneMode": {
    /*req, object*/
        "sceneMode":  "normal"
        /*req, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.7.2.15 Get the trigger mode parameters capability of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule/trigger/capabilities?format=json

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "TriggerCap": {
    /*ro, req, object*/
        "type": {
        /*ro, opt, object*/
            "@opt": ["IO", "video"]
            /*ro, opt, array, subType:string*/
        },
        "IOSignalType": {
        /*ro, opt, object*/
            "@opt": ["levelSignal", "pulseSignal"]
            /*ro, opt, array, subType:string*/
        },
        "doorStatus": {
        /*ro, opt, object*/
            "@opt": ["open", "close", "N_A"]
            /*ro, opt, array, subType:string*/
        }
    }
}
```

16.7.2.16 Set the trigger mode parameters of face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceRule/trigger?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "Trigger": {
    /*req, object*/
        "type":  "video",
        /*req, enum, subType:string*/
        "IOSignalType":  "levelSignal"
        /*req, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.7.2.17 Get the trigger mode parameters of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule/trigger?format=json

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "Trigger": {
    /*ro, req, object*/
        "type":  "video",
        /*ro, req, enum, subType:string*/
        "IOSignalType":  "levelSignal",
        /*ro, req, enum, subType:string*/
        "doorStatus":  "open"
        /*ro, req, enum, subType:string*/
    }
}
```

16.7.2.18 Get the face capture rule parameters of a device

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceRule?defualt=<defualt>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
defualt	string	Get default parameters via FaceRule.defualt in /ISAPI/Intelligent/channels//faceRule/capabilities.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceRule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face capture rule capability, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--ro, req, enum, whether to enable, subType:string, desc:“true”, “false”-->true
  </enabled>
  <FaceDetectionRegion>
    <!--ro, opt, object, detection area-->
    <RegionCoordinatesList>
      <!--ro, opt, array, list of region coordinates, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, req, object, area point coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->50
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </FaceDetectionRegion>
  <InterPupileRegion>
    <!--ro, opt, object, rectangle frame of minimum pupil distance, desc:the origin is the lower-left corner of the screen-->
    <positionX>
      <!--ro, req, int, X-coordinate, range:[0,1000]-->50
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
    </positionY>
    <width>
      <!--ro, req, int, pupillary distance, range:[0,1000]-->50
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->50
    </height>
  </InterPupileRegion>
```

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```
    <MaxInterPupilRegion>
      <!--ro, opt, object, the maximum pupillary distance, desc:The origin is the lower-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->50
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
      </positionY>
      <width>
        <!--ro, req, int, pupillary distance, range:[0,1000]-->50
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->50
      </height>
    </MaxInterPupilRegion>
    <SizeFilter>
      <!--ro, opt, object, size filter-->
      <enabled>
        <!--ro, req, bool, whether to enable size filter-->true
      </enabled>
      <ratio>
        <!--ro, opt, float, ratio of head-shoulder size filter to minimum pupil distance, range:[4.0,6.0]-->5.0
      </ratio>
      <MaxTargetSize>
        <!--ro, opt, object, target frame on the largest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000], desc:Value range: [0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->0
        </height>
      </MaxTargetSize>
      <MinTargetSize>
        <!--ro, opt, object, target frame on the smallest face, desc:The origin is the lower-left corner of the screen.-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
        <width>
          <!--ro, req, int, width, range:[0,1000]-->0
        </width>
        <height>
          <!--ro, req, int, height, range:[0,1000]-->0
        </height>
      </MinTargetSize>
    </SizeFilter>
    <detectionType>
      <!--ro, opt, string, face type for detection, desc:abnormalFace (abnormal face), normalFace (normal face), multiFace (multiple faces), sunglassesFace
(face with sunglasses), callFace (face of a person who is making a phone call, which is used to detect faces in ATM mode). This node supports to be set to
multiple values and they should be separated by commas.-->abnormalFace,normalFace
    </detectionType>
    <autoROIEnable>
      <!--ro, opt, bool, whether to enable automatic face ROI, desc:By default it is false.-->false
    </autoROIEnable>
    <minInterPupileDistance>
      <!--ro, opt, int, range:[13,666], step:1, unit:px-->58
    </minInterPupileDistance>
  </FaceRule>
```

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16.7.2.19 Get the capability of configuring advanced parameters of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/advancedParam/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
}
```

16.7.2.20 Set the advanced parameters of face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/advancedParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "AdvancedParam": {
    /*wo, req, object*/
        "tarGenerateRate": 3,
        /*wo, opt, int, range:[1,5]*/
        "sensitivity": 3,
        /*wo, opt, int, range:[1,5]*/
        "snapMode": "fast",
        /*wo, req, enum, subType:string*/
        "quicksnapThreshold": 70,
        /*wo, opt, int, range:[0,100], unit:s*/
        "quicksnapMaxtime": 5,
        /*wo, opt, int, range:[0,300], step:1, unit:s*/
        "quicksnapCount": 1,
        /*wo, opt, int, range:[1,100]*/
        "snapTime": 1,
        /*wo, opt, int, range:[1,3]*/
        "snapThreshold": 20,
        /*wo, opt, int, range:[0,100]*/
        "snapInterval": 1,
        /*wo, opt, int*/
        "faceExposurEnable": true,
        /*wo, opt, bool*/
        "referenceBright": 50,
        /*wo, opt, int, range:[0,100]*/
        "minContinueTime": 5,
        /*wo, opt, int, range:[0,60], step:1, unit:min*/
        "faceFilterTime": 0,
        /*wo, opt, int, range:[0,100000], step:1, unit:ms*/
        "faceAttributEnable": true,
        /*wo, opt, bool*/
        "facesnapFilterEnable": false,
        /*wo, opt, bool*/
        "swingAngleLeft": 10,
        /*wo, opt, int, range:[0,70], step:1, unit:°*/
        "swingAngleRight": 1,
        /*wo, opt, int, range:[0,70], step:1, unit:°*/
        "tiltAngleUp": 10,
        /*wo, opt, int, range:[0,50], step:1, unit:°*/
        "tiltAngleDown": 10,
        /*wo, opt, int, range:[0,50], step:1, unit:°*/
        "instantLightMidEnable": true,
        /*wo, opt, bool*/
        "fixedLightMidEnable": true,
        /*wo, opt, bool*/
        "severeEnable": true,
        /*wo, opt, bool*/
        "OSDEnable": true,
        /*wo, opt, bool*/
        "RemoveDuplicate": {
        /*wo, opt, object, dep:and,{$.AdvancedParam.snapMode,eq,best}*/
            "enabled": true,
            /*wo, opt, bool*/
            "threshold": 88,
            /*wo, opt, int*/
            "faceScore": 20,
            /*wo, opt, int*/
            "updateTime": 60
            /*wo, opt, int*/
        },
        "quickContinueSnapEnable": false
        /*wo, opt, bool, dep:and,{$.AdvancedParam.snapMode,eq,fast}*/
    }
}
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.7.2.21 Get the capability of configuring parameters for capturing face pictures and overlaying information on the captured face pictures.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
}
```

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16.7.2.22 Get the parameters of capturing face pictures and overlaying information on the captured face pictures.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "OverlapPic": {
    /*ro, opt, object, parameters of overlaying information on the picture*/
        "AddIntelInfo": {
        /*ro, opt, object, overlay the VCA information*/
            "streamWithIntelInfo":  true,
            /*ro, opt, bool, overlay the VCA information on the stream*/
            "alarmWithTargetInfo":  false
            /*ro, opt, bool, overlay the target frame on the alarm picture*/
        },
        "TargetPicParam": {
        /*ro, opt, object, target picture parameters*/
            "targetPicMode":  "custom",
            /*ro, opt, enum, target picture mode, subType:string, desc:"custom", "head" (head shot), "headShoulders" (head and shoulders picture), "human"
(full-body shot)*/
            "targetPicWidth":  3.0,
            /*ro, opt, float, width*/
            "headHeight":  2.0,
            /*ro, opt, float, head height*/
            "bodyHeight":  7.0,
            /*ro, opt, float, body height*/
            "TargetPicHeight": {
            /*ro, opt, object, target picture with fixed pixels*/
                "enable":  true,
```

```
                /*ro, req, bool, whether to enable*/
                "height":  100
                /*ro, req, int, height*/
            },
            "FaceBeautification": {
            /*ro, opt, object, face beautification*/
                "enable":  true,
                /*ro, req, bool, whether to enable*/
                "level":  100
                /*ro, req, int, beautification level*/
            },
            "faceSharpeningLevel":  50,
            /*ro, opt, int, range:[0,100]*/
            "faceEnhancement":  true,
            /*ro, opt, bool, whether to enable face enhancement*/
            "faceEnhancementLevel":  50,
            /*ro, opt, int, range:[0,100]*/
            "faceCloseupRatio": {
            /*ro, opt, object*/
                "upRatio":  20,
                /*ro, opt, int*/
                "downRatio":  20,
                /*ro, opt, int*/
                "leftRatio":  20,
                /*ro, opt, int*/
                "rightRatio":  20
                /*ro, opt, int*/
            },
            "humanEyeLighteningEnabled":  false,
            /*ro, opt, bool*/
            "colorReproduction":  true
            /*ro, opt, bool*/
        },
        "AlarmPicParam": {
        /*ro, opt, object, alarm picture parameters*/
            "backgroundPicUpload":  true,
            /*ro, opt, bool, whether to upload the background picture*/
            "picQuality":  "general",
            /*ro, opt, enum, picture quality, subType:string, desc:"best", "better", "general"*/
            "PicSize": {
            /*ro, req, object, picture size*/
                "width":  1920,
                /*ro, req, int, width*/
                "height":  1080
                /*ro, req, int, height*/
            },
            "facePictureUpload":  true
            /*ro, opt, bool, whether to upload the close-up face picture (head and shoulders picture), desc:If this field does not exist, the close-up face
picture will be uploaded by default.*/
        },
        "AlarmOsdParam": [
        /*ro, opt, array, parameters of overlaying OSD on the alarm picture, subType:object*/
            {
                "osdIndex":  "cameraNo",
                /*ro, opt, enum, overlaid items, subType:string, desc:"cameraNo" (channel No.), "positionInfo" (position information), "snapTimeOsd"
(capture time), "facingPupil" (pupillary distance)*/
                "osdValue":  "test",
                /*ro, opt, string, overlaid content, desc:If the value of this field is empty, it indicates that the content needs to be overlaid by DSP.*/
                "enabled":  true
                /*ro, opt, bool, whether to enable overlaying the item*/
            }
        ],
        "faceNoiseReduction":  true,
        /*ro, opt, bool*/
        "faceMosaic":  false,
        /*ro, opt, bool*/
        "backgroundCompress": {
        /*ro, opt, object*/
            "enabled":  true,
            /*ro, req, bool*/
            "backgroundCompressLevel":  "middle"
            /*ro, opt, enum, subType:string*/
        }
    }
}
```

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16.7.2.23 Set the overlay and capture parameters for face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/overlapPic?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "OverlapPic": {
    /*opt, object, picture overlay*/
        "AddIntelInfo": {
        /*opt, object, overlay intelligent information*/
            "streamWithIntelInfo":  true,
            /*opt, bool, overlay intelligent information with the stream*/
            "alarmWithTargetInfo":  false
            /*opt, bool, overlay the alarm picture with target information*/
        },
        "TargetPicParam": {
        /*opt, object, target picture parameters*/
            "targetPicMode":  "custom",
            /*ro, opt, enum, target picture mode, subType:string, desc:"custom", "head" (head shot), "headShoulders" (half-body shot), "human" (full-body
shot)*/
            "targetPicWidth":  3.0,
            /*opt, float, target picture width*/
            "headHeight":  2.0,
            /*opt, float, head height*/
            "bodyHeight":  7.0,
            /*opt, float, body height*/
            "TargetPicHeight": {
            /*opt, object, target picture fixed pixels*/
                "enable":  true,
                /*req, bool, whether to enable*/
                "height":  100
                /*req, int, height*/
            },
            "FaceBeautification": {
            /*opt, object, face picture beauty mode*/
                "enable":  true,
                /*req, bool, whether to enable*/
                "level":  100
                /*req, int, beauty level*/
            },
            "faceSharpeningLevel":  50,
            /*opt, int, range:[0,100]*/
            "faceEnhancement":  true,
            /*opt, bool, whether to enable face enhancement*/
            "faceEnhancementLevel":  50,
            /*opt, int, range:[0,100]*/
            "faceCloseupRatio": {
            /*ro, opt, object*/
                "upRatio":  20,
                /*ro, opt, int*/
                "downRatio":  20,
                /*ro, opt, int*/
                "leftRatio":  20,
                /*ro, opt, int*/
                "rightRatio":  20
                /*ro, opt, int*/
            },
            "humanEyeLighteningEnabled":  false,
            /*opt, bool*/
            "colorReproduction":  true
            /*opt, bool*/
        },
        "AlarmPicParam": {
        /*opt, object, alarm picture parameters*/
            "backgroundPicUpload":  true,
            /*opt, bool, whether to upload the background picture*/
            "picQuality":  "general",
            /*opt, enum, picture quality, subType:string, desc:"best", "better", "general"*/
            "PicSize": {
            /*req, object, picture size*/
                "width":  1920,
                /*req, int, width*/
                "height":  1080
                /*req, int, height*/
            },
            "facePictureUpload":  true
            /*opt, bool*/
        },
        "AlarmOsdParam": [
        /*opt, array, text overlay parameters on captured pictures, subType:object*/
            {
                "osdIndex":  "cameraNo",
                /*opt, enum, overlaid item, subType:string, desc:"cameraNo" (channel number), positionInfo (position information), "snapTimeOsd" (capture
time)*/
                "osdValue":  "test",
                /*opt, string, overlay text, desc:when this field does not exists, it indicates DSP overlay*/
                "enabled":  true
                /*opt, bool, whether to enable text overlay*/
            }
        ],
        "faceNoiseReduction":  true,
        /*opt, bool*/
        "faceMosaic":  false,
        /*opt, bool*/
        "backgroundCompress": {
```

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```
        /*opt, object*/
            "enabled":  true,
            /*req, bool*/
            "backgroundCompressLevel":  "middle"
            /*opt, enum, subType:string*/
        }
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

16.7.2.24 Get the face picture privacy mask configuration capability of a specified channel

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "enabled": {
    /*ro, req, object, whether to enable*/
        "@opt": [true, false],
        /*ro, req, array, range, subType:bool*/
        "@def":  false
        /*ro, opt, bool, default value*/
    },
    "privacyMaskLevel": {
    /*ro, opt, object*/
        "@opt": ["high", "mid", "low", "auto"]
        /*ro, opt, array, subType:string*/
    },
    "modeType": {
    /*ro, opt, object*/
        "@opt": [1, 2, 3]
        /*ro, opt, array, subType:int*/
    },
    "nonMosaicFaceProperty": {
    /*ro, opt, object*/
        "@size":  7,
        /*ro, opt, int*/
        "@opt": ["glass", "mask", "hat", "glassAndMask", "glassAndHat", "maskAndHat", "glassAndMaskAdnHat"]
        /*ro, opt, array, subType:string*/
    },
    "nonMosaicFaceLib": {
    /*ro, req, object*/
        "@opt": [true, false],
        /*ro, req, array, subType:bool*/
        "@def":  false
        /*ro, opt, bool*/
    }
}
```

16.7.2.25 Set the privacy mask parameters of a specified channel

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Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "enabled":  true
    /*req, bool, whether to enable*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

16.7.2.26 Get the capability of privacy mask parameters of a specific channel

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/privacyMask?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "enabled":  true,
    /*ro, req, bool, whether to enable, desc:whether to enable*/
    "privacyMaskLevel":  "high",
    /*ro, opt, enum, subType:string*/
    "modeType":  1,
    /*ro, opt, enum, subType:int, dep:or,{$.enabled,eq,true}*/
    "nonMosaicFaceProperty": ["glass", "mask", "hat", "glassAndMask", "glassAndHat", "maskAndHat", "glassAndMaskAdnHat"],
    /*ro, opt, array, subType:string*/
    "nonMosaicFaceLib":  true
    /*ro, opt, bool*/
}
```

16.7.2.27 Restore the face capture algorithm to defaults

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/restoreAlgParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

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Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.7.2.28 Get the capability of configuring shielded area of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/shieldRegion/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ShieldRegionCap": {
    /*ro, opt, object*/
        "@size":  8,
        /*ro, req, int*/
        "enable": {
        /*ro, req, object*/
            "@opt": [true, false]
            /*ro, req, array, subType:bool*/
        },
        "PointF": {
        /*ro, opt, object*/
            "@minSize":  0,
            /*ro, req, int*/
            "@maxSize":  4,
            /*ro, req, int*/
            "x": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@def":  0.001
                /*ro, req, float, range:[0.000,1.000]*/
            },
            "y": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@def":  0.001
                /*ro, req, float, range:[0.000,1.000]*/
            }
        }
    }
}
```

16.7.2.29 Get the shielded area parameters of face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceSnap/shieldRegion?format=json

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Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ShieldRegion": [
    /*ro, opt, array, subType:object*/
        {
            "ruleID":  1,
            /*ro, req, int*/
            "enable":  true,
            /*ro, req, bool*/
            "PointF": [
            /*ro, opt, array, subType:object, range:[3,10]*/
                {
                    "x":  0.120,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "y":  0.120
                    /*ro, req, float, range:[0.000,1.000]*/
                }
            ]
        }
    ]
}
```

16.7.2.30 Set the shielded area parameters of face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceSnap/shieldRegion?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "ShieldRegion": [
    /*opt, array, subType:object*/
        {
            "ruleID":  1,
            /*req, int*/
            "enable":  true,
            /*req, bool*/
            "PointF": [
            /*opt, array, subType:object, range:[3,10]*/
                {
                    "x":  0.120,
                    /*req, float, range:[0.000,1.000]*/
                    "y":  0.120
                    /*req, float, range:[0.000,1.000]*/
                }
            ]
        }
]
```

Response Message

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```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.7.2.31 Get the VCA resource parameters of face capture and recognition channel

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/intelliResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <BehaviorInfo>
    <!--ro, opt, object-->
    <IntelliImage>
      <!--ro, opt, object-->
      <isUpload>
        <!--ro, req, enum, subType:string-->true
      </isUpload>
      <imgQuality>
        <!--ro, opt, enum, subType:string-->best
      </imgQuality>
      <imgResolutionWidth>
        <!--ro, req, int-->1
      </imgResolutionWidth>
      <imgResolutionHeight>
        <!--ro, req, int-->1
      </imgResolutionHeight>
      <isImgTargetOverlap>
        <!--ro, req, bool-->true
      </isImgTargetOverlap>
      <isImgRuleOverlap>
        <!--ro, req, bool-->true
      </isImgRuleOverlap>
      <isImgSizeOverlap>
        <!--ro, opt, bool-->bool
      </isImgSizeOverlap>
    </IntelliImage>
    <VideoOverlapInfo>
      <!--ro, opt, object-->
      <isOverlapIntelli>
        <!--ro, req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--ro, req, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--ro, req, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <IntelliAnalysisEnable>
      <!--ro, req, bool-->true
    </IntelliAnalysisEnable>
    <sceneType>
      <!--ro, opt, enum, subType:string-->perimeterProtection
    </sceneType>
    <DisplayTrajectory>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <durationLimit>
```

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```
        <!--ro, opt, enum, subType:string-->5s
      </durationLimit>
    </DisplayTrajectory>
    <targetMarkingColorEnabled>
      <!--ro, req, bool-->true
    </targetMarkingColorEnabled>
    <picRecordEnable>
      <!--ro, opt, bool-->true
    </picRecordEnable>
    <autoTrack>
      <!--ro, opt, bool-->false
    </autoTrack>
    <NotificationPTZRecheck>
      <!--ro, opt, bool-->false
    </NotificationPTZRecheck>
  </BehaviorInfo>
  <FaceCaptureInfo>
    <!--ro, opt, object-->
    <imgQuality>
      <!--ro, opt, enum, subType:string-->best
    </imgQuality>
    <isImgTargetOverlap>
      <!--ro, req, bool-->true
    </isImgTargetOverlap>
    <VideoOverlapInfo>
      <!--ro, opt, object-->
      <isOverlapIntelli>
        <!--ro, req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--ro, opt, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--ro, opt, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <backgroundUpload>
      <!--ro, opt, bool-->true
    </backgroundUpload>
    <imgResolutionWidth>
      <!--ro, req, int-->1
    </imgResolutionWidth>
    <imgResolutionHeight>
      <!--ro, req, int-->1
    </imgResolutionHeight>
    <TargetImgInfo>
      <!--ro, opt, object-->
      <targetImgType>
        <!--ro, opt, enum, subType:string-->custom
      </targetImgType>
      <TargetImgSize>
        <!--ro, opt, object-->
        <widthRatio>
          <!--ro, req, float-->0.000
        </widthRatio>
        <upperPartHeightRatio>
          <!--ro, req, float-->0.000
        </upperPartHeightRatio>
        <underPartHeightRatio>
          <!--ro, req, float-->0.000
        </underPartHeightRatio>
      </TargetImgSize>
      <fixedPixelEnable>
        <!--ro, opt, bool-->true
      </fixedPixelEnable>
      <heightPixel>
        <!--ro, opt, int-->1
      </heightPixel>
      <FaceBeautification>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, opt, bool-->true
        </enabled>
        <level>
          <!--ro, opt, int, range:[0,100]-->1
        </level>
      </FaceBeautification>
      <faceEnhancement>
        <!--ro, opt, bool-->true
      </faceEnhancement>
    </TargetImgInfo>
    <backgroundIntervalUpload>
      <!--ro, opt, int, range:[0,65534], unit:s-->1
    </backgroundIntervalUpload>
    <picRecordEnable>
      <!--ro, opt, bool-->true
    </picRecordEnable>
  </FaceCaptureInfo>
  <AlgVersionInfo>
    <!--ro, opt, object-->
    <AlgItem>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, string-->test
```

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```
      </id>
      <algName>
        <!--ro, opt, string-->test
      </algName>
    </AlgItem>
  </AlgVersionInfo>
  <JudiciaryInfo>
    <!--ro, opt, object-->
    <IntelliImage>
      <!--ro, req, object-->
      <isUpload>
        <!--ro, req, enum, subType:string-->true
      </isUpload>
      <imgQuality>
        <!--ro, opt, enum, subType:string-->best
      </imgQuality>
      <imgResolutionWidth>
        <!--ro, req, int-->1
      </imgResolutionWidth>
      <imgResolutionHeight>
        <!--ro, req, int-->1
      </imgResolutionHeight>
    </IntelliImage>
    <VideoOverlapInfo>
      <!--ro, opt, object-->
      <isOverlapIntelli>
        <!--ro, req, bool-->true
      </isOverlapIntelli>
    </VideoOverlapInfo>
  </JudiciaryInfo>
  <QRCodeUrl>
    <!--ro, opt, object-->
    <vcaVideoQRCodeUrl>
      <!--ro, opt, string, range:[0,256]-->http://127.0.0.1
    </vcaVideoQRCodeUrl>
  </QRCodeUrl>
</IntelliResource>
```

16.7.2.32 Set the intelligent resource parameters of face capture channel

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/intelliResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <BehaviorInfo>
    <!--opt, object-->
    <IntelliImage>
      <!--opt, object-->
      <isUpload>
        <!--req, enum, subType:string-->true
      </isUpload>
      <imgQuality>
        <!--opt, enum, subType:string-->best
      </imgQuality>
      <imgResolutionWidth>
        <!--opt, int-->1
      </imgResolutionWidth>
      <imgResolutionHeight>
        <!--opt, int-->1
      </imgResolutionHeight>
      <isImgTargetOverlap>
        <!--req, bool-->true
      </isImgTargetOverlap>
      <isImgRuleOverlap>
        <!--req, bool-->true
      </isImgRuleOverlap>
      <isImgSizeOverlap>
        <!--opt, bool-->bool
      </isImgSizeOverlap>
    </IntelliImage>
    <VideoOverlapInfo>
      <!--opt, object-->
      <isOverlapIntelli>
        <!--req, bool-->true
      </isOverlapIntelli>
```

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```
      <isOverlapTarget>
        <!--req, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--req, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <IntelliAnalysisEnable>
      <!--req, bool-->true
    </IntelliAnalysisEnable>
    <DisplayTrajectory>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <durationLimit>
        <!--opt, enum, subType:string-->5s
      </durationLimit>
    </DisplayTrajectory>
    <targetMarkingColorEnabled>
      <!--req, bool-->true
    </targetMarkingColorEnabled>
    <picRecordEnable>
      <!--opt, bool-->true
    </picRecordEnable>
  </BehaviorInfo>
  <FaceCaptureInfo>
    <!--opt, object-->
    <imgQuality>
      <!--opt, enum, subType:string-->best
    </imgQuality>
    <isImgTargetOverlap>
      <!--req, bool-->true
    </isImgTargetOverlap>
    <VideoOverlapInfo>
      <!--opt, object-->
      <isOverlapIntelli>
        <!--req, bool-->true
      </isOverlapIntelli>
      <isOverlapTarget>
        <!--opt, bool-->true
      </isOverlapTarget>
      <isOverlapRule>
        <!--opt, bool-->true
      </isOverlapRule>
    </VideoOverlapInfo>
    <backgroundUpload>
      <!--opt, bool-->true
    </backgroundUpload>
    <imgResolutionWidth>
      <!--req, int-->1
    </imgResolutionWidth>
    <imgResolutionHeight>
      <!--req, int-->1
    </imgResolutionHeight>
    <TargetImgInfo>
      <!--opt, object-->
      <targetImgType>
        <!--opt, enum, subType:string-->custom
      </targetImgType>
      <TargetImgSize>
        <!--opt, object-->
        <widthRatio>
          <!--req, float-->0.000
        </widthRatio>
        <upperPartHeightRatio>
          <!--req, float-->0.000
        </upperPartHeightRatio>
        <underPartHeightRatio>
          <!--req, float-->0.000
        </underPartHeightRatio>
      </TargetImgSize>
      <fixedPixelEnable>
        <!--opt, bool-->true
      </fixedPixelEnable>
      <heightPixel>
        <!--opt, int-->1
      </heightPixel>
      <FaceBeautification>
        <!--opt, object-->
        <enabled>
          <!--opt, bool-->true
        </enabled>
        <level>
          <!--opt, int, range:[0,100]-->1
        </level>
      </FaceBeautification>
      <faceEnhancement>
        <!--opt, bool-->true
      </faceEnhancement>
    </TargetImgInfo>
    <backgroundIntervalUpload>
      <!--opt, int, range:[0,65534], unit:s-->1
```

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DETEKTOR

```
      </backgroundIntervalUpload>
      <picRecordEnable>
        <!--opt, bool-->true
      </picRecordEnable>
    </FaceCaptureInfo>
    <AlgVersionInfo>
      <!--opt, object-->
      <AlgItem>
        <!--opt, object-->
        <id>
          <!--opt, string, index-->test
        </id>
        <algName>
          <!--opt, string-->test
        </algName>
      </AlgItem>
    </AlgVersionInfo>
    <JudiciaryInfo>
      <!--opt, object-->
      <IntelliImage>
        <!--req, object-->
        <isUpload>
          <!--req, enum, subType:string-->true
        </isUpload>
        <imgQuality>
          <!--opt, enum, subType:string-->best
        </imgQuality>
        <imgResolutionWidth>
          <!--req, int-->1
        </imgResolutionWidth>
        <imgResolutionHeight>
          <!--req, int-->1
        </imgResolutionHeight>
      </IntelliImage>
      <VideoOverlapInfo>
        <!--opt, object-->
        <isOverlapIntelli>
          <!--req, bool-->true
        </isOverlapIntelli>
      </VideoOverlapInfo>
    </JudiciaryInfo>
    <QRCodeUrl>
      <!--opt, object-->
      <vcaVideoQRCodeUrl>
        <!--opt, string, range:[0,256]-->http://127.0.0.1
      </vcaVideoQRCodeUrl>
    </QRCodeUrl>
  </IntelliResource>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.7.2.33 Get the configuration capability of intelligent

resources by channel **Request URL**

GET /ISAPI/Intelligent/channels/<channelID>/intelliResource/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<IntelliResourceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
</IntelliResourceCap>
```

16.7.2.34 Set the VCA type parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/intelliType

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliType xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <type>
    <!--req, enum, subType:string-->behavior
  </type>
</IntelliType>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.7.2.35 Get the VCA type parameters

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/intelliType

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IntelliType xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <type>
    <!--ro, req, enum, subType:string-->behavior
  </type>
</IntelliType>
```

16.7.2.36 Get the configuration capability of manual face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/manualFaceSnap/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ManualFaceSnapCap": {
    /*ro, req, object*/
        "enabled": {
        /*ro, req, object*/
            "@opt":  "true,false",
            /*ro, req, string*/
            "#text":  true
            /*ro, opt, bool*/
        }
    }
}
```

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DEFEKTOR

16.7.2.37 Get the parameters of manual face capture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/manualFaceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "ManualFaceSnap": {
    /*ro, req, object*/
        "enabled":  true
        /*ro, req, bool*/
    }
}
```

16.7.2.38 Set the parameters of manual face capture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/manualFaceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "ManualFaceSnap": {
    /*req, object*/
        "enabled":  true
        /*req, bool*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.7.2.39 Get the face modeling configuration capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/modeling/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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DETEKTOR

```
{
    "ModelingCap": {
    /*ro, opt, object*/
        "FaceSnapModeling": {
        /*ro, req, object*/
            "enabled":  "true,false",
            /*ro, req, string*/
            "QuickModeling": {
            /*ro, opt, object*/
                "enabled":  "true,false",
                /*ro, opt, string*/
                "threshold": {
                /*ro, opt, object*/
                    "@min":  20,
                    /*ro, req, int*/
                    "@max":  80,
                    /*ro, req, int*/
                    "@def":  60
                    /*ro, req, int*/
                },
                "snapTime": {
                /*ro, opt, object*/
                    "@min":  0.0,
                    /*ro, req, float*/
                    "@max":  300.0,
                    /*ro, req, float*/
                    "@def":  2.0
                    /*ro, req, float*/
                }
            },
            "LivenessDetection": {
            /*ro, req, object*/
                "enabled":  "true,false"
                /*ro, req, string*/
            },
            "alarmStorageEnable":  "true,false",
            /*ro, opt, string*/
            "mixedTargetDetectionWithFaceSnapModeling":  "true,false"
            /*ro, opt, string*/
        },
        "uploadPictureEnabled":  "true,false"
        /*ro, opt, string*/
    }
}
```

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DETEKTOR

16.7.2.40 Get the face modeling parameters

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/modeling?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "Modeling": {
    /*ro, opt, object*/
        "FaceSnapModeling": {
        /*ro, req, object*/
            "enabled":  true,
            /*ro, opt, bool*/
            "QuickModeling": {
            /*ro, req, object*/
                "enabled":  true,
                /*ro, opt, bool*/
                "threshold":  60,
                /*ro, opt, int*/
                "snapTime":  2.0
                /*ro, opt, float, unit:s*/
            },
            "LivenessDetection": {
            /*ro, req, object*/
                "enabled":  true
                /*ro, opt, bool*/
            },
            "alarmStorageEnable":  true,
            /*ro, opt, bool*/
            "mixedTargetDetectionWithFaceSnapModeling":  true
            /*ro, opt, bool*/
        },
        "uploadPictureEnabled":  false
        /*ro, opt, bool*/
    }
}
```

16.7.2.41 Set the face modeling parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/modeling?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "Modeling": {
    /*opt, object*/
        "FaceSnapModeling": {
        /*req, object*/
            "enabled":  true,
            /*opt, bool*/
            "QuickModeling": {
            /*req, object*/
                "enabled":  true,
                /*opt, bool*/
                "threshold":  60,
                /*opt, int*/
                "snapTime":  2.0
                /*opt, float, unit:s*/
            },
            "LivenessDetection": {
            /*req, object*/
                "enabled":  true
                /*opt, bool*/
            },
            "alarmStorageEnable":  true,
            /*opt, bool*/
            "mixedTargetDetectionWithFaceSnapModeling":  true
            /*opt, bool*/
        },
        "uploadPictureEnabled":  false
        /*opt, bool*/
    }
}
```

Response Message

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DETEKTOR

```
{
    "statusCode": 1,
    /*ro, opt, int*/
    "statusString": "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode": "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode": 1,
    /*ro, opt, int*/
    "errorMsg": "ok"
    /*ro, opt, string*/
}
```

16.7.2.42 Set the overlay parameters of captured face picture

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/picInfoOverlap/faceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "PicInfoOverlap": {
    /*req, object*/
        "enabled": true,
        /*req, bool*/
        "OverlapItem": [
        /*req, array, subType:object*/
            {
                "overlapItemId": "1",
                /*req, string*/
                "itemType": "cameraNo"
                /*req, enum, subType:string*/
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int*/
    "statusString": "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode": "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode": 1,
    /*ro, opt, int*/
    "errorMsg": "ok"
    /*ro, opt, string*/
}
```

16.7.2.43 Get the overlay parameters of captured face picture

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/picInfoOverlap/faceSnap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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DETEKTOR

```
{
    "PicInfoOverlap": {
    /*ro, req, object*/
        "enabled":  true,
        /*ro, req, bool*/
        "OverlapItem": [
        /*ro, req, array, subType:object*/
            {
                "overlapItemId":  "1",
                /*ro, req, string*/
                "itemType":  "cameraNo"
                /*ro, req, enum, subType:string*/
            }
        ]
    }
}
```

16.7.2.44 Clear the face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib/<FDID>/picture?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
identityKey	string	--
FDType	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.2.45 Face capture and recognition event

EventType:faceCapture

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
```

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```
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount": 1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType": "faceCapture",
    /*ro, req, string, event type*/
    "eventState": "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)*/
    "eventDescription": "faceCapture",
    /*ro, req, string, event description, desc:face capture data upload*/
    "channelName": "ABC",
    /*ro, req, string, channel name (camera name)*/
    "deviceID": "test0123",
    /*ro, opt, string, device ID (PUID)*/
    "isDataRetransmission": true,
    /*ro, opt, bool, whether the data is replenished*/
    "faceCapture": [
    /*ro, opt, array, face capture and recognition results, one result can contain multiple pictures and one picture can contains multiple faces,
subType:object*/
        {
            "errorCode": 1,
            /*ro, opt, int, error code*/
            "errorMsg": "ok",
            /*ro, opt, string, error code description*/
            "taskid": "temp_dfasdffsadfdsadada",
            /*ro, opt, string, task ID*/
            "faceTime": "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, capture time*/
            "traceUuid": "sddddddddff",
            /*ro, opt, string, target ID, the same person in multiple capture results corresponds to same target ID*/
            "traceIdx": 1,
            /*ro, opt, int, No. of picture with highest comparison similarity*/
            "targetAttrs": {
            /*ro, opt, object, face attributes*/
                "deviceId": "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID*/
                "deviceChannel": 1,
                /*ro, opt, int, device channel No.*/
                "deviceName": "shebei",
                /*ro, opt, string, device name*/
                "faceTime": "2004-05-03T17:30:08+08:00",
                /*ro, opt, datetime, capture time*/
                "bkgUrl": "test",
                /*ro, opt, string, background picture URL*/
                "thermalURL": "test",
                /*ro, opt, string, thermal picture URL*/
                "contentID": "backgroundImage",
                /*ro, opt, string, background picture (binary)*/
                "pId": "F4F665D6A18E41308CE9934DCDDD1111"
                /*ro, opt, string*/
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object*/
                {
                    "faceId": 1,
                    /*ro, opt, int, face ID*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                        "height": 1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width": 1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x": 0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y": 0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range": 5,
                        /*ro, opt, int, age deviation*/
                        "value": 23,
                        /*ro, opt, int, age*/
                        "ageGroup": "old"
                        /*ro, opt, string, age group*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence": 99.9999,
                        /*ro, opt, float, confidence*/
                        "value": "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence": 99.945,
                        /*ro, opt, float, confidence*/
                        "value": "yes"
                        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes"*/
                    },
                    "mask": {
                    /*ro, opt, object, whether the person wears a mask*/
                        "confidence": 99.945,
                        /*ro, opt, float, confidence*/
```

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```
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person is smiling*/
                        "confidence":  99.945,
                        /*ro, opt, float, confidence*/
                        "value":  "no"
                        /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                    },
                    "faceExpression": {
                    /*ro, opt, object, facial expression*/
                        "value":  "unknown"
                        /*ro, opt, enum, facial expression, subType:string, desc:"unknown", "surprised", "panic", "disgusted", "happy", "sad", "angry",
"poker-faced"*/
                    },
                    "hat": {
                    /*ro, opt, object, whether the person wears a hat*/
                        "value":  "no"
                        /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                    },
                    "URL":  "xxxxxxx",
                    /*ro, opt, string, face picture thumbnail URL*/
                    "contentID":  "faceImage",
                    /*ro, opt, string, face picture thumbnail (binary)*/
                    "stayDuration":  20,
                    /*ro, opt, int, dwell time duration in the image, unit:ms*/
                    "faceScore":  30,
                    /*ro, req, int, face score (face picture quality), range:[0,100]*/
                    "captureEndMark":  true,
                    /*ro, req, bool, end flag of capture, desc:true (ended), false (not ended)*/
                    "targetID":  "xxxxxxx",
                    /*ro, opt, string, face and human body linkage ID*/
                    "FacePictureRect": {
                    /*ro, opt, object, rectangle frame for human face, desc:the origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, req, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, req, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "blockingState":  "noBlocking",
                    /*ro, opt, enum, target face shielding status, subType:string, desc:"noBlocking" (unshielded), "instantaneousInsignificantBlocking"
(lightly shielded), "continuousInsignificantBlocking" (keep shielding), "significantBlocking" (seriously shielded), "OSD", "unknown"*/
                    "swingAngle":  -90,
                    /*ro, opt, int, panning angle, range:[-90,90]*/
                    "tiltAngle":  -90,
                    /*ro, opt, int, tilting angle, range:[-90,90]*/
                    "pupilDistance":  10,
                    /*ro, opt, int, pupil distance of target face, desc:the maximum value is 10 bytes, and the maximum value is (resolution width/1.6),
unit: pixel*/
                    "livenessDetectionStatus":  "liveFace",
                    /*ro, opt, enum, face anti-spoofing detection status, subType:string, desc:"liveFace" (live face), "notLiveFace" (the face is not real),
"detectionFailed" (detection failed), "notEnabled" (disabled).*/
                    "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
                    /*ro, opt, string*/
                    "enterTime":  "2004-05-03T17:30:08+08:00",
                    /*ro, opt, datetime*/
                    "exitTime":  "2004-05-03T17:30:08+08:00",
                    /*ro, opt, datetime*/
                    "perClassify":  0,
                    /*ro, opt, int, face data presorting*/
                    "faceSnapThermometryEnabled":  true,
                    /*ro, opt, bool, whether to enable uploading captured face picture with temperature information*/
                    "currTemperature":  36.8,
                    /*ro, opt, float, face temperature*/
                    "isAbnomalTemperature":  true,
                    /*ro, opt, bool, whether the temperature is abnormal or not: true, false*/
                    "thermometryUnit":  "celsius",
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature":  36.7,
                    /*ro, opt, float, temperature threshold to trigger alarm*/
                    "interiorPosition":  "driver"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "uid":  "20190520103922026OOKb4s3iQPzLignc1sRUbFT0hR039Y0YzuLt0VLfl99yZi",
            /*ro, opt, string, the unique ID of event uploading*/
            "OperatorInfo": {
            /*ro, opt, object, operator information, desc:operator information*/
                "personUUID":  "test",
                /*ro, opt, string, range:[1,32]*/
                "personName":  "test"
                /*ro, opt, string, range:[1,64]*/
            },
            "MEFData": [
            /*ro, opt, array, subType:object*/
                {
                    "time":  "2004-05-03T17:30:08.000+08:00",
                    /*ro, req, datetime, it should be accurate to the millisecond, desc:it should be accurate to the millisecond*/
```


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```
                    "IMSI":  "460001357924680",
                    /*ro, req, string, range:[1,15]*/
                    "bandIdx":  11
                    /*ro, opt, enum, subType:int*/
                }
            ],
            "isNeedVerification":  true
            /*ro, opt, bool*/
        }
    ],
    "GPS": {
    /*ro, opt, object*/
        "divisionEW":  "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude":  100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS":  "N",
        /*ro, req, enum, "S" (Southern Hemisphere), "N" (North Hemisphere), subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)*/
        "latitude":  100
        /*ro, req, int, latitude, desc:latitude*/
    },
    "URLCertificationType":  "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (no authentication, it is used for the cloud protocol), "digest" (digest
authentication, it is used for local picture URL returned by NVR or DVR)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceCapture	[Message content]	application/json	--	--	--
..	[Binary picture data]	image/jpeg	backgroundImage	backgroundImage.jpeg	--
..	[Binary picture data]	image/jpeg	faceImage	faceImage.jpeg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

16.7.3 Face Modeling and Face Picture Comparison

16.7.3.1 Get the alarm information of captured pictures

Request URL

GET /ISAPI/Intelligent/FDLib/alarmInfo?format=json&listType=<listType>

Query Parameter

Parameter Name	Parameter Type	Description
listType	enum	--

Request Message

None

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Response Message


```
{
    "alarmInfo": [
    /*ro, opt, array, supports alarm recognition results output of (multiple/single) person(s), subType:object, desc:supports alarm recognition results output of (multiple/single) person(s); select the capture of maximum similarity*/
        {
            "errorCode": 1,
            /*ro, opt, int, error code*/
            "errorMsg": "ok",
            /*ro, opt, string, error description*/
            "image": "http",
            /*ro, opt, string, the picture URL when submitting the task*/
            "traceUuid": "sddddddddff",
            /*ro, opt, string, trace*/
            "traceIdx": 1,
            /*ro, opt, int, picture index*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId": "1",
                /*ro, opt, string, device ID*/
                "monitorId": "1",
                /*ro, opt, string, stream ID (encoding stream ID in CVR). This node will not be returned if the device does not support this function*/
                "deviceChannel": 1,
                /*ro, opt, int, device Channel No.*/
                "deviceName": "shebei",
                /*ro, opt, string, device name*/
                "faceTime": "2016-09-10 00:00:00.000",
                /*ro, opt, string, time of face picture comparison*/
                "rect": {
                /*ro, opt, object, background picture frame information of face, desc:the origin is the upper-left corner of the screen*/
                    "height": 1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width": 1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x": 0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y": 0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "bkgUrl": "http"
                /*ro, opt, string, picture URL*/
            },
            "faces": [
            /*ro, opt, array, face, subType:object*/
                {
                    "faceId": 1,
                    /*ro, opt, int, face ID*/
                    "eyesPosition": {
                    /*ro, opt, object, eyesPosition*/
                        "leftEye": {
                        /*ro, opt, object, left eye, desc:the origin is the upper-left corner of the screen*/
                            "x": 0.000,
                            /*ro, opt, float, x-coordinate of top-left corner, range:[0.000,1.000]*/
                            "y": 0.000
                            /*ro, opt, float, y-coordinate of top-left corner, range:[0.000,1.000]*/
                        },
                        "rightEye": {
                        /*ro, opt, object, right eye, desc:the origin is the upper-left corner of the screen*/
                            "x": 0.000,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                            "y": 0.000
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    },
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:the origin is the upper-left corner of the screen*/
                        "height": 1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width": 1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x": 0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y": 0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "facePose": {
                    /*ro, opt, object, submit the task*/
                        "pitch": 0.9980,
                        /*ro, opt, float, face pitch angle, face pitching up is regarded as positive*/
                        "roll": 0.6230,
                        /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive*/
                        "yaw": 0.8760
                        /*ro, opt, float, face yaw angle, face yawing left is regarded as positive*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range": 5,
                        /*ro, opt, int, age deviation*/
                        "value": 23,
                        /*ro, opt, int, age*/
                        "ageGroup": "old"
```


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```
                    /*ro, opt, enum, child, subType:string, desc:"unknown", "child", "young", "middle", "old"*/
                },
                "gender": {
                /*ro, opt, object, gender*/
                    "confidence":  99.99990,
                    /*ro, opt, float, confidence*/
                    "value":  "male "
                    /*ro, opt, string, gender*/
                },
                "glass": {
                /*ro, opt, object, whether the person wears glasses*/
                    "confidence":  99.94499999999999,
                    /*ro, opt, float, confidence*/
                    "value":  "yes"
                    /*ro, opt, string, whether the person wears glasses*/
                },
                "smile": {
                /*ro, opt, object, whether the person is smiling*/
                    "confidence":  99.99990,
                    /*ro, opt, float, confidence*/
                    "value":  "yes"
                    /*ro, opt, string, whether the person is smiling*/
                },
                "identify": [
                /*ro, opt, array, recognition result of the group, subType:object*/
                    {
                        "relationId":  9887,
                        /*ro, opt, int, linkage relationship of arming*/
                        "maxsimilarity":  0.780,
                        /*ro, opt, float, maximum similarity*/
                        "similarityRange":  1,
                        /*ro, opt, int, similarity range*/
                        "candidate": [
                        /*ro, opt, array, candidate, subType:object*/
                            {
                                "alarmId":  1,
                                /*ro, opt, int, alarm ID*/
                                "blacklist_id":  "2323",
                                /*ro, opt, string, alarm bloccklist library ID*/
                                "human_data": [
                                /*ro, opt, array, human body data, subType:object*/
                                    {
                                        "bkg_picurl":  "http",
                                        /*ro, opt, string, Background picture URL.*/
                                        "face_id":  "123445",
                                        /*ro, opt, string, face ID*/
                                        "face_picurl":  "http",
                                        /*ro, opt, string, face thumbnail URL*/
                                        "face_rect": {
                                        /*ro, opt, object, front face, desc:the origin is the upper-left corner of the screen*/
                                            "height":  1.000,
                                            /*ro, opt, float, height, range:[0.000,1.000]*/
                                            "width":  1.000,
                                            /*ro, opt, float, width, range:[0.000,1.000]*/
                                            "x":  0.000,
                                            /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                                            "y":  0.000
                                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                                        },
                                        "similarity":  0.780
                                        /*ro, opt, float, person*/
                                    }
                                ],
                                "human_id":  "123",
                                /*ro, opt, string, blocklist person ID*/
                                "reserve_field": {
                                /*ro, opt, object, reserved area*/
                                    "name":  "张三",
                                    /*ro, opt, string, name*/
                                    "gender":  "male",
                                    /*ro, opt, string, gender*/
                                    "bornTime":  "2004-05-03",
                                    /*ro, opt, string, date of birth*/
                                    "city":  "130100",
                                    /*ro, opt, string, city*/
                                    "certificateType":  "officerID",
                                    /*ro, opt, string, certificate type*/
                                    "certificateNumber":  "test",
                                    /*ro, opt, string, certificate No.*/
                                    "caseInfo":  "0",
                                    /*ro, opt, string, case information*/
                                    "tag":  "aa,bb,cc"
                                    /*ro, opt, string, custom tag*/
                                },
                                "similarity":  0.780,
                                /*ro, opt, float, the maximum similarity of a same person in all pictures*/
                                "listType":  "blackList"
                                /*ro, opt, enum, list type, subType:string, desc:the maximum length is 20; "blackList" (blocklist), "whiteList"
(allowlist)*/
                            }
                        ]
                    }
                ]
```

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```
                }
            ],
            "alarmRecordID":  "test",
            /*ro, opt, string, alarm record ID*/
            "alarmAckState":  "acknowledged",
            /*ro, opt, enum, alarm acknowledgment status, subType:string, desc:"acknowledged", "pending" (to be acknowledged. This value can be used when the user cannot acknowledge the alarm when it is triggered but wants to prioritize acknowledging it later), "notAcknowledged" (not acknowledgment), "all"*/
            "alarmCustomInfo":  "test"
            /*ro, opt, string, custom information*/
        }
    ]
}
```

16.7.4 Face Picture Comparison

16.7.4.1 Get the arming schedule of all face comparison channels

Request URL

GET /ISAPI/Event/schedules/faceContrast

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, arming schedule list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, req, object, arming schedule-->
    <id>
      <!--ro, req, string-->faceContrast-1
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->faceContrast
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming schedule list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedule-->
        <dayOfWeek>
          <!--ro, opt, enum, day of the week, subType:int, desc:1: Monday; 2: Tuesday; 3: Wednesday; 4: Thursday; 5: Friday; 6: Saturday; 7: Sunday;-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming schedule, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming schedule-->
        <TimeRange>
          <!--ro, req, object, time period-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceContrastScheduleList>
```

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16.7.4.2 Set the arming schedule of multiple face picture comparison alarms

Request URL

PUT /ISAPI/Event/schedules/faceContrast

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, schedule information list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--req, object, schedule information-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->faceContrast-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->faceContrast
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time period-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FaceContrastScheduleList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.7.4.3 Get the arming schedule of face picture comparison for a specified channel

Request URL

GET /ISAPI/Event/schedules/faceContrast/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrast-1
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->faceContrast
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.7.4.4 Set the arming schedule of face picture comparison for a specified channel

Request URL

PUT /ISAPI/Event/schedules/faceContrast/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrast-1
  </id>
  <eventType>
    <!--opt, enum, subType:string-->faceContrast
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.7.4.5 Set the arming schedule parameters of a specified face picture library by channel

Request URL

PUT /ISAPI/Event/schedules/faceLib/<channelID>/<FDID>?FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
FDID	string	--
FDType	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, arming schedule information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->faceLib-1
  </id>
  <eventType>
    <!--opt, enum, event type, subType:string-->faceLib
  </eventType>
  <videoInputChannelID>
    <!--opt, string, video input channel ID-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <dayOfWeek>
        <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, holiday arming time, subType:object-->
    <TimeBlock>
      <!--opt, object, arming time-->
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.4.6 Set the linkage parameters of face picture comparison

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Request URL

PUT /ISAPI/Event/triggers/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrast-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->faceContrast
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.4.7 Get the linkage parameters of face picture comparison

Request URL

GET /ISAPI/Event/triggers/faceContrast-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrast-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrast
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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DETEKTOR

16.7.4.8 Set the linkage parameters of event when face picture comparison failed

Request URL

PUT /ISAPI/Event/triggers/faceContrastFailure-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrastFailure-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->faceContrastFailure
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.4.9 Get the linkage parameters of event when face picture comparison failed

Request URL

GET /ISAPI/Event/triggers/faceContrastFailure-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrastFailure-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrastFailure
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoLoss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--ro, opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--ro, opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--ro, opt, int, unit:ms-->1
```

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```
      </notificationInterval>
      <videoInputID>
        <!--ro, opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--ro, opt, object-->
        <ptzChannelID>
          <!--ro, req, string-->1
        </ptzChannelID>
        <actionName>
          <!--ro, req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--ro, opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--ro, opt, object-->
        <whiteLightDurationTime>
          <!--ro, opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <AudioAction>
        <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
        <audioID>
          <!--ro, opt, int-->1
        </audioID>
        <alarmTimes>
          <!--ro, opt, int, range:[0,255]-->1
        </alarmTimes>
        <childDevID>
          <!--ro, opt, string, range:[9,32]-->3434645611
        </childDevID>
        <audioContent>
          <!--ro, opt, string, range:[0,256]-->test
        </audioContent>
      </AudioAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

16.7.4.10 Get the linkage parameters of event when face picture comparison completed

Request URL

GET /ISAPI/Event/triggers/faceContrastSuccess-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->faceContrastSuccess-1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->faceContrastSuccess
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--ro, opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--ro, opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
```

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```
    <dynVideoInputChannelID>
      <!--ro, opt, string-->1
    </dynVideoInputChannelID>
    <intervalBetweenEvents>
      <!--ro, opt, int, unit:s-->1
    </intervalBetweenEvents>
    <WLSensorID>
      <!--ro, opt, string-->1
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--ro, opt, array, subType:object-->
      <EventTriggerNotification>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->test
        </id>
        <notificationMethod>
          <!--ro, req, enum, subType:string-->FTP
        </notificationMethod>
        <notificationRecurrence>
          <!--ro, opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--ro, opt, int, unit:ms-->1
        </notificationInterval>
        <videoInputID>
          <!--ro, opt, string-->1
        </videoInputID>
        <dynVideoInputID>
          <!--ro, opt, string-->1
        </dynVideoInputID>
        <ptzAction>
          <!--ro, opt, object-->
          <ptzChannelID>
            <!--ro, req, string-->1
          </ptzChannelID>
          <actionName>
            <!--ro, req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--ro, opt, int-->1
          </actionNum>
        </ptzAction>
        <WhiteLightAction>
          <!--ro, opt, object-->
          <whiteLightDurationTime>
            <!--ro, opt, int, range:[1,60], unit:s-->1
          </whiteLightDurationTime>
        </WhiteLightAction>
        <AudioAction>
          <!--ro, opt, object, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,audio}-->
          <audioID>
            <!--ro, opt, int-->1
          </audioID>
          <alarmTimes>
            <!--ro, opt, int, range:[0,255]-->1
          </alarmTimes>
          <childDevID>
            <!--ro, opt, string, range:[9,32]-->3434645611
          </childDevID>
          <audioContent>
            <!--ro, opt, string, range:[0,256]-->test
          </audioContent>
        </AudioAction>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

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16.7.4.11 Set the linkage parameters of event when face picture comparison completed

Request URL

PUT /ISAPI/Event/triggers/faceContrastSuccess-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->faceContrastSuccess-1
  </id>
  <eventType>
    <!--req, enum, subType:string-->faceContrastSuccess
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string, dep:and,{$.EventTrigger.eventType,eq,VMD},{$.EventTrigger.eventType,eq,videoloss},{$.EventTrigger.eventType,eq,tamperdetection},
{$.EventTrigger.eventType,eq,regionEntrance},{$.EventTrigger.eventType,eq,regionExiting},{$.EventTrigger.eventType,eq,loitering},
{$.EventTrigger.eventType,eq,group},{$.EventTrigger.eventType,eq,rapidMove},{$.EventTrigger.eventType,eq,parking},
{$.EventTrigger.eventType,eq,unattendedBaggage},{$.EventTrigger.eventType,eq,attendedBaggage}-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.4.12 Set the linkage action parameters of a specified face picture library

Request URL

PUT /ISAPI/Event/triggers/faceLib-<channelID>/<FDID>?FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
FDID	string	--
FDType	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, linkage action parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, desc:the format is event type-channel No.-->faceLib-1
  </id>
  <eventType>
    <!--req, enum, event type, subType:string-->faceLib
  </eventType>
  <eventDescription>
    <!--opt, string, event description-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string, I/O input channel ID-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string, dynamic I/O input channel ID-->1
  </dynInputIOPortID>
  <intervalBetweenEvents>
    <!--opt, int, event capture interval, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string, wireless sensor ID-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, linkage action list, subType:object-->
    <EventTriggerNotification>
      <!--opt, object, linkage action-->
      <id>
        <!--req, string, index-->test
      </id>
      <notificationMethod>
        <!--req, enum, linkage action, subType:string, desc:"FTP" (uploading via FTP), "HTTP" (uploading via HTTP), "IM" (instant messaging), "IO" (I/O
output), "LightAudioAlarm" (light and audio alarm), "audio" (audio), "beep" (buzzer), "center" (uploading to the center), "cloud" (cloud storage), "email",
"focus" (focus), "monitorAlarm" (alarm on monitor), "ptz" (PTZ), "record" (recording), "syslog" (system log), "trace" (tracking), "whiteLight" (white
light)-->FTP
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, linkage time interval, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string, I/O output port ID-->1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string, dynamic I/O output port ID-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string, video channel ID-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string, dynamic video input channel ID-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object, PTZ linkage, desc:it is valid when the value of notificationMethod is "PTZ"-->
        <ptzChannelID>
          <!--req, string, PTZ linked channel No.-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
        </actionName>
        <actionNum>
          <!--opt, int, operation No., desc:operation No.-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object, white light linkage, desc:it is valid when the value of notificationMethod is "whiteLight"-->
        <whiteLightDurationTime>
          <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.4.13 Set the face picture comparison parameters

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/faceContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceContrastList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrast>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enable>
      <!--opt, bool-->true
    </enable>
    <faceContrastType>
      <!--req, enum, face picture comparison type, subType:string-->faceContrast
    </faceContrastType>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->1
    </thresholdValue>
    <FDLibList>
      <!--opt, array, subType:object-->
      <FDLib>
        <!--opt, object, face picture library-->
        <id>
          <!--req, int-->1
        </id>
        <FDID>
          <!--req, string, a unique ID of face picture library-->test
        </FDID>
        <thresholdValue>
          <!--opt, int, range:[0,100]-->1
        </thresholdValue>
        <customFaceLibID>
          <!--opt, string, custom face picture library ID-->test
        </customFaceLibID>
      </FDLib>
    </FDLibList>
    <faceSnapDataUpload>
      <!--opt, bool, upload captured face image, desc:true by default-->true
    </faceSnapDataUpload>
    <AttendanceSaveEnable>
      <!--opt, bool, desc:true by default-->true
    </AttendanceSaveEnable>
    <contrastFailureAlarmUpload>
      <!--opt, bool, desc:true by default-->true
    </contrastFailureAlarmUpload>
    <alarmOutType>
      <!--opt, enum, subType:string-->off
    </alarmOutType>
    <plusesTime>
      <!--opt, int, unit:ms-->1
    </plusesTime>
    <contrastSuccessAlarmUpload>
```

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```
    <!--opt, bool-->true
  </contrastSuccessAlarmUpload>
  <LivenessDetection>
    <!--opt, object-->
    <enable>
      <!--req, bool-->true
    </enable>
  </LivenessDetection>
  <voiceEnabled>
    <!--opt, bool-->true
  </voiceEnabled>
  <faceSnapDataSave>
    <!--opt, bool-->true
  </faceSnapDataSave>
  <QuickContrast>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <snapTime>
      <!--req, float, unit:s-->0.000
    </snapTime>
    <threshold>
      <!--opt, int-->1
    </threshold>
    <quickConfigMode>
      <!--opt, enum, subType:string-->custom
    </quickConfigMode>
    <Custom>
      <!--opt, object-->
      <timeOutMode>
        <!--opt, enum, subType:string-->infinite
      </timeOutMode>
      <timeOut>
        <!--opt, int, range:[1,99], unit:s-->1
      </timeOut>
      <duplicateContrastMode>
        <!--opt, enum, subType:string-->success
      </duplicateContrastMode>
      <duplicateContrastTimes>
        <!--opt, int, range:[1,99]-->1
      </duplicateContrastTimes>
    </Custom>
  </QuickContrast>
  <mixedTargetDetectionWithFaceContrast>
    <!--opt, bool-->true
  </mixedTargetDetectionWithFaceContrast>
  <brokenNetHttp>
    <!--opt, bool-->true
  </brokenNetHttp>
  <relationId>
    <!--opt, string, arming ID-->test
  </relationId>
  <alarmStorageEnable>
    <!--opt, bool-->true
  </alarmStorageEnable>
  <realTimeContrast>
    <!--opt, bool-->true
  </realTimeContrast>
  <analysisTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </analysisTime>
  <contrastSuccessAlarmout>
    <!--opt, bool, desc:true by default-->true
  </contrastSuccessAlarmout>
  <contrastFailureAlarmout>
    <!--opt, bool, desc:true by default-->true
  </contrastFailureAlarmout>
  <removeDuplicateEnabled>
    <!--opt, bool-->true
  </removeDuplicateEnabled>
  <removeDuplicateTime>
    <!--opt, float, unit:s-->0.000
  </removeDuplicateTime>
  <repetitionTime>
    <!--opt, int-->1
  </repetitionTime>
  <topOverlayNameEnabled>
    <!--opt, bool-->true
  </topOverlayNameEnabled>
  <personListShowEnabled>
    <!--opt, bool-->true
  </personListShowEnabled>
  <refreshListTime>
    <!--opt, int, unit:s-->1
  </refreshListTime>
  <personCountShowEnabled>
    <!--opt, bool-->true
  </personCountShowEnabled>
  <FaceROI>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
```

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DETEKTOR

```
      </enabled>
      <level>
        <!--opt, int-->1
      </level>
    </FaceROI>
    <uploadResultCount>
      <!--opt, int-->1
    </uploadResultCount>
    <normalTemperatureAlarmout>
      <!--opt, bool-->true
    </normalTemperatureAlarmout>
    <abnormalTemperatureAlarmout>
      <!--opt, bool-->true
    </abnormalTemperatureAlarmout>
    <matchType>
      <!--opt, enum, subType:int-->0
    </matchType>
    <alarmSourceType>
      <!--opt, enum, subType:string-->timing
    </alarmSourceType>
    <timingInterval>
      <!--opt, enum, subType:int, dep:and,{$.FaceContrastList[*].FaceContrast.alarmSourceType,eq,timing}-->5
    </timingInterval>
  </FaceContrast>
</FaceContrastList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.7.4.14 Get the face picture comparison parameters.

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<FaceContrastList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FaceContrast>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int, ID-->1
    </id>
    <enable>
      <!--ro, opt, bool-->true
    </enable>
    <faceContrastType>
      <!--ro, req, enum, subType:string-->faceContrast
    </faceContrastType>
    <thresholdValue>
      <!--ro, opt, int, Detection Threshold, range:[0,100], desc:Detection Threshold-->1
    </thresholdValue>
    <FDLibList>
      <!--ro, opt, array, subType:object-->
    </FDLibList>
    <contrastFailureAlarmUpload>
      <!--ro, opt, bool, true by default, desc:true by default-->true
    </contrastFailureAlarmUpload>
    <QuickContrast>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <snapTime>
        <!--ro, req, float, unit:s-->0.000
      </snapTime>
      <threshold>
        <!--ro, opt, int-->1
      </threshold>
      <quickConfigMode>
        <!--ro, opt, enum, subType:string-->custom
      </quickConfigMode>
      <Custom>
        <!--ro, opt, object-->
        <timeOutMode>
          <!--ro, opt, enum, subType:string-->infinite
        </timeOutMode>
        <duplicateContrastMode>
          <!--ro, opt, enum, subType:string-->success
        </duplicateContrastMode>
      </Custom>
    </QuickContrast>
    <mixedTargetDetectionWithFaceContrast>
      <!--ro, opt, bool-->true
    </mixedTargetDetectionWithFaceContrast>
    <contrastSuccessAlarmout>
      <!--ro, opt, bool, true by default, desc:true by default-->true
    </contrastSuccessAlarmout>
    <contrastFailureAlarmout>
      <!--ro, opt, bool, true by default, desc:true by default-->true
    </contrastFailureAlarmout>
  </FaceContrast>
</FaceContrastList>
```

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16.7.4.15 Get the face picture comparison capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/faceContrast/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<FaceContrastCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <enable opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enable>
  <faceContrastType opt="faceContrast,blacklistFaceContrast,whitelistFaceContrast">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->faceContrast
```

```
</faceContrastType>
<thresholdValue min="1" max="10">
  <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
</thresholdValue>
<SupportFDLibList opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</SupportFDLibList>
<contrastFailureAlarmUpload opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</contrastFailureAlarmUpload>
<plusesTime min="100" max="900">
  <!--ro, opt, int, unit:ms, attr:min{req, int},max{req, int}-->1
</plusesTime>
<isSupportSingleThresholdValue>
  <!--ro, opt, bool-->true
</isSupportSingleThresholdValue>
<contrastSuccessAlarmUpload opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</contrastSuccessAlarmUpload>
<isSupportFaceScore>
  <!--ro, opt, bool-->true
</isSupportFaceScore>
<isSupportCustomFaceLibID>
  <!--ro, opt, bool-->true
</isSupportCustomFaceLibID>
<QuickContrast>
  <!--ro, opt, object-->
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <threshold min="20" max="80" def="60">
    <!--ro, opt, int, attr:min{req, int},max{req, int},def{req, int}-->1
  </threshold>
  <snapTime min="0.0" max="300.0" def="2.0">
    <!--ro, opt, float, unit:s, attr:min{req, int},max{req, int},def{req, int}-->0.000
  </snapTime>
  <quickConfigMode opt="custom,faceSignIn,faceRecognition">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->custom
  </quickConfigMode>
  <CustomCap>
    <!--ro, opt, object-->
    <timeOutMode opt="infinite,finite" def="finite">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->infinite
    </timeOutMode>
    <timeOut min="1" max="99" default="1">
      <!--ro, opt, int, range:[1,99], unit:s, attr:min{req, int},max{req, int},default{opt, int}-->1
    </timeOut>
    <duplicateContrastMode opt="success,infinite,finite" def="success">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->success
    </duplicateContrastMode>
    <duplicateContrastTimes min="1" max="99" default="1">
      <!--ro, opt, int, range:[1,99], attr:min{req, int},max{req, int},default{opt, int}-->1
    </duplicateContrastTimes>
  </CustomCap>
  <FaceSignInCap>
    <!--ro, opt, object-->
    <timeOutMode opt="infinite,finite" def="finite">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->infinite
    </timeOutMode>
    <duplicateContrastMode opt="success,infinite,finite" def="success">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->success
    </duplicateContrastMode>
  </FaceSignInCap>
  <FaceRecognitionCap>
    <!--ro, opt, object-->
    <timeOutMode opt="infinite,finite" def="finite">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->infinite
    </timeOutMode>
    <duplicateContrastMode opt="success,infinite,finite" def="success">
      <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->success
    </duplicateContrastMode>
  </FaceRecognitionCap>
</QuickContrast>
<mixedTargetDetectionWithFaceContrast>
  <!--ro, opt, bool-->true
</mixedTargetDetectionWithFaceContrast>
<brokenNetHttp opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</brokenNetHttp>
<relationId min="1" max="64">
  <!--ro, opt, string, range:[1,64], attr:min{req, int},max{req, int}-->test
</relationId>
<FaceContrast>
  <!--ro, opt, object-->
  <linkageType opt="faceContrast,faceContrastSuccess,faceContrastFailure,faceContrastNonRealTime">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </linkageType>
</FaceContrast>
<contrastSuccessAlarmout opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</contrastSuccessAlarmout>
<contrastFailureAlarmout opt="true,false">
  <!--ro, opt, bool, attr:opt{req, string}-->true
</contrastFailureAlarmout>
```

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```
    <supportFDLibNum min="1" max="10">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </supportFDLibNum>
    <supportFDLibType opt="ordinary,private,stranger">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->ordinary
    </supportFDLibType>
  </FaceContrastCap>
```

16.7.4.16 Set the prompt when face picture comparison failure

Request URL

PUT /ISAPI/Intelligent/FDLib/faceContrastPrompt/channels/<channelID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "prompt":  "test"
    /*req, string, range:[0,16]*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

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16.7.4.17 Get the prompt of face picture comparison failure

Request URL

GET /ISAPI/Intelligent/FDLib/faceContrastPrompt/channels/<channelID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "prompt":  "test"
    /*ro, req, string, range:[0,16]*/
}
```

16.7.4.18 Search for face picture comparison results of a single channel

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string-->test
  </searchID>
  <searchResultPosition>
    <!--req, int-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string-->test
  </monitorName>
  <similarity>
    <!--opt, float, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, subType:string-->unlimit
  </eventType>
  <FaceModeList>
    <!--opt, array, subType:object-->
    <FaceMode>
      <!--opt, object-->
      <ModeInfo>
        <!--opt, object-->
        <similarity>
          <!--opt, float, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string-->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <PictureInfoList>
    <!--opt, array, subType:object-->
    <PictureInfo>
      <!--opt, object-->
      <FDID>
        <!--req, string-->test
      </FDID>
      <PID>
        <!--req, string-->test
      </PID>
    </PictureInfo>
  </PictureInfoList>
  <totalSearchResults>
    <!--opt, int-->1
  </totalSearchResults>
  <FDIDList>
    <!--opt, array, subType:object-->
    <FDID>
      <!--opt, string-->test
    </FDID>
  </FDIDList>
  <sortord>
    <!--opt, enum, subType:string-->similarity
  </sortord>
  <sex>
    <!--opt, enum, subType:string-->male
  </sex>
  <glass>
    <!--opt, enum, subType:string-->yes
  </glass>
  <smile>
    <!--opt, enum, subType:string-->yes
  </smile>
  <high_risk_group>
    <!--opt, enum, subType:string-->yes
  </high_risk_group>
  <importLib>
    <!--opt, enum, subType:string-->yes
```

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```
  </importLib>
  <CustomFaceLibIDList>
    <!--opt, array, subType:object-->
    <customFaceLibID>
      <!--opt, string-->test
    </customFaceLibID>
  </CustomFaceLibIDList>
</FCSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, subType:string-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, subType:object-->
    <MatchElement>
      <!--ro, opt, object-->
      <snapPicURL>
        <!--ro, opt, string-->test
      </snapPicURL>
      <snapTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </snapTime>
      <facePicURL>
        <!--ro, opt, string-->test
      </facePicURL>
      <FaceMatchInfoList>
        <!--ro, opt, array, subType:object-->
        <FaceMatchInfo>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, req, string-->test
          </FDID>
          <FDname>
            <!--ro, opt, string-->test
          </FDname>
          <thresholdValue>
            <!--ro, opt, int, range:[0,100]-->1
          </thresholdValue>
          <bornTime>
            <!--ro, opt, date-->1970-01-01+08:00
          </bornTime>
          <name>
            <!--ro, opt, string-->test
          </name>
          <sex>
            <!--ro, opt, enum, subType:string-->male
          </sex>
          <province>
            <!--ro, req, string-->0
          </province>
          <city>
            <!--ro, opt, string-->11
          </city>
          <certificateType>
            <!--ro, req, enum, subType:string-->ID
          </certificateType>
          <certificateNumber>
            <!--ro, opt, string, range:[8,32]-->test
          </certificateNumber>
          <picURL>
            <!--ro, req, string-->test
          </picURL>
          <PID>
            <!--ro, req, string-->test
          </PID>
          <PersonInfoExtendList>
            <!--ro, opt, array, subType:object-->
            <PersonInfoExtend>
              <!--ro, opt, object-->
```

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```
            <id>
              <!--ro, req, int-->1
            </id>
            <enable>
              <!--ro, req, bool-->true
            </enable>
            <name>
              <!--ro, opt, string-->test
            </name>
            <value>
              <!--ro, opt, string-->test
            </value>
          </PersonInfoExtend>
        </PersonInfoExtendList>
        <phoneNumber>
          <!--ro, opt, string-->test
        </phoneNumber>
        <caseInfo>
          <!--ro, opt, string-->test
        </caseInfo>
        <similarity>
          <!--ro, opt, float, range:[0.0,100.0]-->0.0
        </similarity>
        <modelStatus>
          <!--ro, opt, enum, subType:string-->modeling
        </modelStatus>
        <ModelingStatus>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, opt, string-->test
          </FDID>
          <PID>
            <!--ro, opt, string-->test
          </PID>
          <name>
            <!--ro, req, string-->test
          </name>
          <status>
            <!--ro, req, enum, subType:string-->abnormal
          </status>
          <reason>
            <!--ro, req, enum, subType:string-->analysisFailed
          </reason>
          <customHumanID>
            <!--ro, opt, string-->123456789012345678901234567890123456789012
          </customHumanID>
          <customFaceLibID>
            <!--ro, opt, string-->123456789012345678901234567890123456789012
          </customFaceLibID>
          <modeData>
            <!--ro, opt, string-->test
          </modeData>
        </ModelingStatus>
      </FaceMatchInfo>
    </FaceMatchInfoList>
    <monitorName>
      <!--ro, opt, string-->test
    </monitorName>
    <glass>
      <!--ro, req, enum, subType:string-->any
    </glass>
    <smile>
      <!--ro, req, enum, subType:string-->no
    </smile>
    <high_risk_group>
      <!--ro, opt, enum, subType:string-->yes
    </high_risk_group>
    <similarity>
      <!--ro, opt, float, range:[0.0,100.0]-->0.0
    </similarity>
    <Rect>
      <!--ro, opt, object-->
      <height>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </height>
      <width>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </width>
      <x>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </x>
      <y>
        <!--ro, req, float, range:[0.000,1.000]-->0.000
      </y>
    </Rect>
    <channelID>
      <!--ro, opt, int-->1
    </channelID>
    <name>
      <!--ro, opt, string-->test
    </name>
    <sex>
      <!--ro, opt, enum, subType:string-->male
    </sex>
```

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```
      <ageGroup>
        <!--ro, req, enum, subType:string-->any
      </ageGroup>
      <mask>
        <!--ro, req, enum, subType:string-->no
      </mask>
      <imageQuality>
        <!--ro, opt, int-->1
      </imageQuality>
      <faceExpression>
        <!--ro, req, enum, subType:string-->angry
      </faceExpression>
      <contrastStatus>
        <!--ro, opt, bool-->true
      </contrastStatus>
      <age>
        <!--ro, opt, int, range:[1,255]-->1
      </age>
      <targetID>
        <!--ro, opt, string, range:[1,48]-->test
      </targetID>
      <MEFData>
        <!--ro, opt, array, subType:object-->
        <MEF>
          <!--ro, opt, object-->
          <time>
            <!--ro, req, datetime-->2004-05-03T17:30:08+08:00.000
          </time>
          <IMSI>
            <!--ro, req, string, range:[1,15]-->test
          </IMSI>
        </MEF>
      </MEFData>
    </MatchElement>
  </MatchList>
</FCSearchResult>
```

16.7.4.19 Search for the face picture comparison alarm information

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FCSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time, desc:the maximum number of search results this time-->1
  </maxResults>
  <snapStartTime>
    <!--opt, datetime, capture start time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapStartTime>
  <snapEndTime>
    <!--opt, datetime, capture end time with ISO8601 time format-->1970-01-01T00:00:00+08:00
  </snapEndTime>
  <faceMatchInfoEnable>
    <!--opt, bool, whether to only upload data with matched information-->true
  </faceMatchInfoEnable>
  <monitorName>
    <!--opt, string, camera name-->test
  </monitorName>
  <similarity>
    <!--opt, float, similarity, range:[0.0,100.0]-->0.0
  </similarity>
  <eventType>
    <!--opt, enum, event type, subType:string, desc:event type-->unlimit
  </eventType>
  <FaceModeList>
    <!--opt, array, face picture information list, subType:object-->
    <FaceMode>
      <!--opt, object, face picture information-->
      <ModeInfo>
        <!--opt, object, model information-->
        <similarity>
```

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```
        <!--opt, float, similarity, range:[0.0,100.0]-->0.0
      </similarity>
      <modeData>
        <!--opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method-
->test
      </modeData>
    </ModeInfo>
  </FaceMode>
 </FaceModeList>
 <PictureInfoList>
  <!--opt, array, list of the picture information, subType:object, desc:list of the picture information-->
  <PictureInfo>
   <!--opt, object, picture information-->
   <FDID>
    <!--req, string-->test
   </FDID>
   <PID>
    <!--req, string-->test
   </PID>
  </PictureInfo>
 </PictureInfoList>
 <ChannelList>
  <!--opt, array, search channel list, subType:object-->
  <Channel>
   <!--opt, object-->
   <channelID>
    <!--req, int, channel No.-->1
   </channelID>
  </Channel>
 </ChannelList>
 <totalSearchResults>
  <!--opt, int, the total number of results in a single search, desc:the total number of results in a single search-->1
 </totalSearchResults>
 <FDIDList>
  <!--opt, array, face picture list, subType:object-->
  <FDID>
   <!--opt, string, face picture library ID-->test
  </FDID>
 </FDIDList>
 <sortord>
  <!--opt, enum, subType:string-->similarity
 </sortord>
 <sex>
  <!--opt, enum, gender, subType:string, desc:gender-->male
 </sex>
 <glass>
  <!--opt, enum, whether the person wears glasses, subType:string, desc:whether the person wears glasses-->yes
 </glass>
 <smile>
  <!--opt, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->yes
 </smile>
 <high_risk_group>
  <!--opt, enum, subType:string-->yes
 </high_risk_group>
 <importLib>
  <!--opt, enum, subType:string-->yes
 </importLib>
 <CustomFaceLibIDList>
  <!--opt, array, custom list of face picture library ID, subType:object-->
  <customFaceLibID>
   <!--opt, string, custom face picture library ID-->test
  </customFaceLibID>
 </CustomFaceLibIDList>
 <thermometryUnit>
  <!--opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
 </thermometryUnit>
 <currTemperature>
  <!--opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
 </currTemperature>
 <temperatureCondition>
  <!--opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
 </temperatureCondition>
 <temperatureValue>
  <!--opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchDescription.temperatureCondition,eq,above}-->37.2
 </temperatureValue>
 <humanIDList>
  <!--opt, array, subType:object-->
  <humanID>
   <!--req, string-->test
  </humanID>
 </humanIDList>
 <collectEnabled>
  <!--opt, enum, subType:string-->yes
 </collectEnabled>
</FCSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
```

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DETEKTOR

```
<FCSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search results, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching completed)-->MORE
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <progress>
    <!--ro, opt, int, range:[0,100]-->1
  </progress>
  <MatchList>
    <!--ro, opt, array, matched results list, subType:object-->
    <MatchElement>
      <!--ro, opt, object, matched results-->
      <snapPicURL>
        <!--ro, opt, string, URL of captured background picture-->test
      </snapPicURL>
      <snapTime>
        <!--ro, opt, datetime, capture time-->1970-01-01T00:00:00+08:00
      </snapTime>
      <facePicURL>
        <!--ro, opt, string, face sub-picture URL-->test
      </facePicURL>
      <FaceMatchInfoList>
        <!--ro, opt, array, subType:object-->
        <FaceMatchInfo>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, req, string-->test
          </FDID>
          <FDname>
            <!--ro, opt, string, face picture library name-->test
          </FDname>
          <thresholdValue>
            <!--ro, opt, int, detection threshold, range:[0,100], desc:detection threshold-->1
          </thresholdValue>
          <bornTime>
            <!--ro, opt, date, date of birth-->1970-01-01+08:00
          </bornTime>
          <name>
            <!--ro, opt, string, name-->test
          </name>
          <sex>
            <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
          </sex>
          <province>
            <!--ro, req, string-->0
          </province>
          <city>
            <!--ro, opt, string, city, desc:see the data dictionary for details-->11
          </city>
          <certificateType>
            <!--ro, req, enum, certificate type, subType:string, desc:certificate type-->ID
          </certificateType>
          <certificateNumber>
            <!--ro, opt, string, certificate No., range:[8,32], desc:certificate No.-->test
          </certificateNumber>
          <picURL>
            <!--ro, req, string-->test
          </picURL>
          <PID>
            <!--ro, req, string, picture ID-->test
          </PID>
          <PersonInfoExtendList>
            <!--ro, opt, array, person extended information list, subType:object-->
            <PersonInfoExtend>
              <!--ro, opt, object, person extended information-->
              <id>
                <!--ro, req, int, serial No. of person custom extension information-->1
              </id>
              <enable>
                <!--ro, req, bool, enable person extended information or not-->true
              </enable>
              <name>
                <!--ro, opt, string, person tag extended information name-->test
              </name>
              <value>
                <!--ro, opt, string, person tag extended information-->test
              </value>
            </PersonInfoExtend>
          </PersonInfoExtendList>
          <phoneNumber>
```

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```
            <!--ro, opt, string, phone No.-->test
          </phoneNumber>
          <caseInfo>
            <!--ro, opt, string, case description-->test
          </caseInfo>
          <similarity>
            <!--ro, opt, float, similarity, range:[0.0,100.0]-->0.0
          </similarity>
          <modelStatus>
            <!--ro, opt, enum, subType:string-->modeling
          </modelStatus>
          <ModelingStatus>
            <!--ro, opt, object-->
            <FDID>
              <!--ro, opt, string, face picture library ID-->test
            </FDID>
            <PID>
              <!--ro, opt, string, picture ID-->test
            </PID>
            <name>
              <!--ro, req, string, picture name-->test
            </name>
            <status>
              <!--ro, req, enum, subType:string-->abnormal
            </status>
            <reason>
              <!--ro, req, enum, subType:string-->analysisFailed
            </reason>
            <customHumanID>
              <!--ro, opt, string-->123456789012345678901234567890l2
            </customHumanID>
            <customFaceLibID>
              <!--ro, opt, string, custom face picture library ID, desc:custom face picture library ID-->123456789012345678901234567890l2
            </customFaceLibID>
            <modeData>
              <!--ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method-->test
            </modeData>
          </ModelingStatus>
        </FaceMatchInfo>
      </FaceMatchInfoList>
      <monitorName>
        <!--ro, opt, string, camera information-->test
      </monitorName>
      <glass>
        <!--ro, req, enum, whether the person wears glasses, subType:string, desc:"any" (unlimited condition which is used in search), "no" (not wear
glasses), "sunglasses" (wears sunglasses), "unknown"-->any
      </glass>
      <smile>
        <!--ro, req, enum, whether the person wears a smile, subType:string, desc:whether the person wears a smile-->no
      </smile>
      <high_risk_group>
        <!--ro, opt, enum, subType:string-->yes
      </high_risk_group>
      <similarity>
        <!--ro, opt, float, similarity, range:[0.0,100.0], desc:similarity-->0.0
      </similarity>
      <Rect>
        <!--ro, opt, object-->
        <height>
          <!--ro, req, float, width, range:[0.000,1.000]-->0.000
        </height>
        <width>
          <!--ro, req, float, height, range:[0.000,1.000]-->0.000
        </width>
        <x>
          <!--ro, req, float, X-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
        </x>
        <y>
          <!--ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the reference origin is the upper left corner of image-->0.000
        </y>
      </Rect>
      <channelID>
        <!--ro, opt, int, channel No., desc:channel No.-->1
      </channelID>
      <name>
        <!--ro, opt, string-->test
      </name>
      <sex>
        <!--ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
      </sex>
      <ageGroup>
        <!--ro, req, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old", "prime",
"teenager", "unknown", "young"-->any
      </ageGroup>
      <mask>
        <!--ro, req, enum, whether the person wears a mask, subType:string, desc:"no", "unknown", "yes"-->no
      </mask>
      <imageQuality>
        <!--ro, opt, int, picture quality-->1
      </imageQuality>
      <faceExpression>
        <!--ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-faced", "sad",
```

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```
"surprised", "unknown"-->angry
      </faceExpression>
      <contrastStatus>
        <!--ro, opt, bool, comparison result-->true
      </contrastStatus>
      <age>
        <!--ro, opt, int, age, range:[1,255]-->1
      </age>
      <targetID>
        <!--ro, opt, string, target ID, range:[1,48]-->test
      </targetID>
      <thermometryUnit>
        <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
      </thermometryUnit>
      <currTemperature>
        <!--ro, opt, float, face temperature, range:[-20.0,105.0], unit:℃, desc:face temperature-->36.7
      </currTemperature>
      <temperatureCondition>
        <!--ro, opt, enum, face temperature status, subType:string, desc:face temperature status-->normal
      </temperatureCondition>
      <temperatureValue>
        <!--ro, opt, float, temperature value, range:[-20,105], unit:℃, dep:and,{$.FCSearchResult.MatchList[*].MatchElement.temperatureCondition,eq,above}-
->37.2
      </temperatureValue>
      <MEFData>
        <!--ro, opt, array, subType:object-->
        <MEF>
          <!--ro, opt, object, electronic fence data-->
          <time>
            <!--ro, req, datetime-->2004-05-03T17:30:08+08:00.000
          </time>
          <IMSI>
            <!--ro, req, string, range:[1,15]-->test
          </IMSI>
        </MEF>
      </MEFData>
      <alarmTemperature>
        <!--ro, opt, float, alarm temperature threshold, range:[-20.0,60.0], unit:℃-->37.6
      </alarmTemperature>
      <collectEnabled>
        <!--ro, opt, bool-->false
      </collectEnabled>
    </MatchElement>
  </MatchList>
</FCSearchResult>
```

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16.7.4.20 Get the capabilities of configuring prompt for face picture comparison

Request URL

GET /ISAPI/Intelligent/FDLib/prompt/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "faceContrastPromptLen": {
    /*ro, opt, object*/
        "@min":  "0",
        /*ro, req, string, range:[0,16]*/
        "@max":  "16"
        /*ro, req, string, range:[0,16]*/
    },
    "whiteListFaceContrastPromptLen": {
    /*ro, opt, object*/
        "@min":  "0",
        /*ro, req, string, range:[0,16]*/
        "@max":  "16"
        /*ro, req, string, range:[0,16]*/
    }
}
```

16.7.4.21 Face picture comparison event

EventType:alarmResult

```
{
    "ipAddress":  "172.6.64.7",
```

```
    /*ro, req, string, IPv4 address of the device that triggers the alarm, range:[1,32]*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm, range:[1,128]*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm, range:[1,65535]*/
    "protocol":  "HTTP",
    /*ro, opt, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address, range:[1,32]*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, req, int, times that the same alarm has been uploaded*/
    "eventType":  "alarmResult",
    /*ro, req, enum, event type, subType:string, desc:"alarmResult" (face picture comparison)*/
    "eventState":  "active",
    /*ro, req, enum, continuous event status, subType:string, desc:active (valid event); inactive (invalid event)*/
    "eventDescription":  "alarmResult",
    /*ro, req, string, event description, range:[1,128], desc:event description*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name), range:[1,128]*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID (PUID), range:[1,64], desc:this field must be returned when ISAPI event information is transmitted via ISUP*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:if the real-time detection data upload fails due to the network condition, the data will be
uploaded again after the device recovers*/
    "alarmResult": [
    /*ro, opt, array, face picture comparison results, subType:object, desc:face picture comparison results*/
        {
            "errorCode":  1,
            /*ro, opt, int, when analyzing face picture failed, the corresponding error code will be returned*/
            "errorMsg":  "ok",
            /*ro, opt, string, when analyzing face picture failed, the corresponding error information will be returned, desc:when analyzing face picture
failed, the corresponding error information will be returned*/
            "image":  " http://10.7.52.174:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/011_163957_23006&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, picture URL*/
            "modelData":  "test",
            /*ro, opt, string, modeling data of background picture, which is encoded by Base64, range:[1,2048]*/
            "traceUuid":  "sddddddddff",
            /*ro, opt, string, target ID, desc:one person corresponds to one target ID in multiple alarms (related URI:
/ISAPI/SDT/Management/Task/facePicture); the maximum size*/
            "traceIdx":  1,
            /*ro, opt, int, picture index, desc:one person can correspond to multiple pictures*/
            "targetAttrs": {
            /*ro, opt, object, target attributes*/
                "deviceId":  "ee3c4000-a22d-11b4-8334-5803fb77894c",
                /*ro, opt, string, device UUID, range:[1,64]*/
                "deviceChannel":  1,
                /*ro, opt, int, device channel No.*/
                "deviceName":  "shebei",
                /*ro, opt, string, device name*/
                "faceTime":  "2016-09-10 00:00:00.000",
                /*ro, opt, string, capture time, range:[1,64]*/
                "rect": {
                /*ro, opt, object, background picture frame information of face, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "contentID":  "backgroundImage",
                /*ro, opt, string, contentID of the picture data binary data, range:[1,128], desc:the bkgURL cannot be returned when this field is
returned*/
                "bkgUrl":  "http://absddsdfsdfsdk.jpg",
                /*ro, opt, string, background picture URL*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, captured background picture ID, range:[1,32], desc:captured background picture ID*/
                "ThermalImage": {
                /*ro, opt, object, thermal imaging picture information*/
                    "resourcesContentType":  "binary",
                    /*ro, opt, string, resource transmission type, desc:"url" (transmitted via URL), "binary" (transmitted in binary format). When the
returned alarm message contains additional resource (e.g., pictures), this field should be returned*/
                    "resourcesContent":  "test"
                    /*ro, opt, string, resource ID, desc:when the returned alarm message contains additional resources (e.g., pictures), this field should
be returned. When the value of resourcesContentType is "binary", it must be the same as Content-ID of the resource. When the value of resourcesContentType
is "url", its value is the resource URL*/
                }
            },
            "faces": [
            /*ro, opt, array, face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on,
subType:object, desc:face picture information, which contains the recognition results, face attributes, modeling, matched list, and so on*/
                {
                    "faceId":  1,
                    /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
```

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```
                    /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                    "faceRect": {
                    /*ro, opt, object, face thumbnail frame, desc:it is used for modeling. The origin is the upper-left corner of the screen*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "faceMark": {
                    /*ro, opt, object, face features in the picture*/
                        "leftEye": {
                        /*ro, opt, object, left eye coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightEye": {
                        /*ro, opt, object, right eye's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "noseTip": {
                        /*ro, opt, object, nose tip's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "leftMouth": {
                        /*ro, opt, object, left mouth corner's coordinates, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        },
                        "rightMouth": {
                        /*ro, opt, object, right mouth corner's position, desc:the origin is the upper-left corner of the screen*/
                            "x":  0.13,
                            /*ro, opt, float, X-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                            "y":  0.87
                            /*ro, opt, float, Y-coordinate, range:[0.000,1.000], desc:normalized coordinates, it is accurate to three decimal places*/
                        }
                    },
                    "recommendFaceRect": {
                    /*ro, opt, object, recommended face frame coordinates, desc:recommended face frame coordinates*/
                        "height":  1.000,
                        /*ro, opt, float, height, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, opt, float, width, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                    },
                    "facePose": {
                    /*ro, opt, object, face angle*/
                        "pitch":  0.998,
                        /*ro, opt, float, face pitch angle, face pitching up is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                        "yaw":  0.876,
                        /*ro, opt, float, face yaw angle, face yawing left is regarded as positive, range:[-90.000,90.000], desc:this field does not exists
if it is invalid. It is accurate to three decimal places*/
                        "roll":  0.623
                        /*ro, opt, float, face roll angle, face rolling clockwise is regarded as positive, range:[-90.000,90.000], desc:this field does not
exists if it is invalid. It is accurate to three decimal places*/
                    },
                    "age": {
                    /*ro, opt, object, age*/
                        "range":  5,
                        /*ro, opt, int, age deviation; if the age deviation already exists, the age is required*/
                        "value":  23,
                        /*ro, opt, int, age*/
                        "ageGroup":  "any"
                        /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle age), "old",
"prime", "teenager", "unknown", "young"*/
                    },
                    "gender": {
                    /*ro, opt, object, gender*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "male"
                        /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
                    },
                    "glass": {
                    /*ro, opt, object, whether the person wears glasses*/
                        "confidence":  99.94499999999999,
                        /*ro, opt, float, confidence*/
```


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```
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes", "sunglasses"*/
                    },
                    "smile": {
                    /*ro, opt, object, whether the person is smiling*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
                    },
                    "mask": {
                    /*ro, opt, object, whether the person wears a mask*/
                        "confidence":  99.94499999999999,
                        /*ro, opt, float, confidence*/
                        "value":  "yes"
                        /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes"*/
                    },
                    "faceExpression": {
                    /*ro, opt, object, facial expression*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "angry"
                        /*ro, req, enum, facial expression, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-
faced", "sad", "surprised", "unknown"*/
                    },
                    "hat": {
                    /*ro, opt, object, whether the person wears a hat*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  "unknown"
                        /*ro, opt, enum, whether the person wears a hat, subType:string, desc:"no", "yes", "unknown"*/
                    },
                    "score": {
                    /*ro, opt, object, face score (face picture quality)*/
                        "confidence":  99.99990,
                        /*ro, opt, float, confidence*/
                        "value":  65
                        /*ro, opt, int, score value, range:[1,100]*/
                    },
                    "modeldata":  "test",
                    /*ro, opt, string, face model data (encoded by base64)*/
                    "AbsoluteHigh": {
                    /*ro, opt, object, PTZ position*/
                        "elevation":  -900,
                        /*ro, opt, int, tilting angle, range:[-900,2700]*/
                        "azimuth":  22,
                        /*ro, opt, int, azimuth, range:[0,3600]*/
                        "absoluteZoom":  1
                        /*ro, opt, int, zooming ratio, range:[1,1000]*/
                    },
                    "identify": [
                    /*ro, opt, array, arming linkage information, subType:object*/
                        {
                            "relationId":  "9887",
                            /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                            "maxsimilarity":  0.780,
                            /*ro, opt, float, the maximum similarity of face comparison results, range:[0.000,1.000], step:0.001*/
                            "similarityRange":  1,
                            /*ro, opt, enum, similarity range, subType:int, desc:0 (smaller than the minimum value), 1 (between the minimum value and
maximum value), 2 (larger than the maximum value). If the similarity is smaller than the minimum value, the person will be marked*/
                            "candidate": [
                            /*ro, opt, array, matched face picture information, subType:object*/
                                {
                                    "alarmId":  1,
                                    /*ro, opt, int, alarm ID, desc:it cannot be duplicated of one arming linkage*/
                                    "blacklist_id":  "2323",
                                    /*ro, opt, string, face picture library ID, range:[1,64], desc:it corresponds to the FDID of face picture library. For
cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                    "human_data": [
                                    /*ro, opt, array, person information of matched face pictures, subType:object*/
                                        {
                                            "face_id":  "123445",
                                            /*ro, opt, string, face ID, range:[1,32]*/
                                            "face_picurl":  "http://127.0.0.1:6120/face_picurl",
                                            /*ro, opt, string, face picture URL*/
                                            "pId":  "F4F665D6A18E41308CE9934DCDDD766F",
                                            /*ro, opt, string, face thumbnail ID, range:[1,64], desc:face thumbnail ID*/
                                            "face_rect": {
                                            /*ro, opt, object, face thumbnail coordinates, desc:the origin is the upper-left corner of the screen*/
                                                "height":  1.000,
                                                /*ro, opt, float, height, range:[0.000,1.000]*/
                                                "width":  1.000,
                                                /*ro, opt, float, width, range:[0.000,1.000]*/
                                                "x":  0.000,
                                                /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                                                "y":  0.000
                                                /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                                            },
                                            "similarity":  0.997000,
                                            /*ro, opt, float, face picture similarity*/
                                            "isNoSaveFDPicture":  true,
                                            /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (do not save); if this node
is not returned, it indicates that saving the picture*/
```


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```
                                        "contentID":  "faceLibImage",
                                        /*ro, opt, string, picture in the face picture library, desc:picture in the face picture library*/
                                        "bkg_picurl":  "http://10639103:80/FDLib?
FDID=529FB9EAC2A04041A603E18F300CD3A5&pId=F4F665D6A18E41308CE9934DCDDD766F&size=22332"
                                        /*ro, opt, string, background picture URL*/
                                    }
                                ],
                                "human_id":  "123",
                                /*ro, opt, string, person ID, desc:person ID in the list, which corresponds to FPID in the face picture library*/
                                "reserve_field": {
                                /*ro, opt, object, reserved (transmitted data)*/
                                    "name":  "张三",
                                    /*ro, req, string, name, range:[1,256]*/
                                    "gender":  "any",
                                    /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male",
"female", "unknown"*/
                                    "bornTime":  "2004-05-03",
                                    /*ro, req, string, date of birth*/
                                    "city":  "130100",
                                    /*ro, opt, string, city code of person birth, range:[1,32], desc:see the data dictionary for details*/
                                    "certificateType":  "officerID",
                                    /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport"
(passport), "other" (the other certificate)*/
                                    "certificateNumber":  "test",
                                    /*ro, opt, string, certificate No., range:[0,32]*/
                                    "ageGroup":  "any",
                                    /*ro, opt, enum, age group, subType:string, desc:"any", "child", "infant", "kid", "middle", "middleAged" (the middle
age), "old", "prime", "teenager", "unknown", "young"*/
                                    "phoneNumber":  "test",
                                    /*ro, opt, string, phone number, range:[1,64]*/
                                    "cityName":  "石家庄市",
                                    /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field
"city")*/
                                    "provinceName":  "河北省"
                                    /*ro, opt, string, province, range:[0,32], desc:province*/
                                },
                                "similarity":  0.780,
                                /*ro, opt, float, range:[0.000,1.000]*/
                                "listType":  "blackList",
                                /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                "extendData": [
                                /*ro, opt, array, person extended information, subType:object*/
                                    {
                                        "extendID":  1,
                                        /*ro, opt, int, person extended information ID, desc:it starts from 1*/
                                        "enable":  true,
                                        /*ro, opt, bool, enable person extended information or not*/
                                        "name":  "test",
                                        /*ro, opt, string, name of person tag extended information, range:[1,96]*/
                                        "value":  "test"
                                        /*ro, opt, string, person tag extended information, range:[0,64]*/
                                    }
                                ],
                                "FDLibName":  "faceLib1",
                                /*ro, opt, string, face picture library name, desc:face picture library name*/
                                "FDLibThreshold":  50,
                                /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                "customFaceLibID":  "test",
                                /*ro, opt, string, custom face picture library ID*/
                                "customHumanID":  "test"
                                /*ro, opt, string, custom person ID*/
                            }
                        ],
                        "alarmRecordID":  "test",
                        /*ro, opt, string, alarm record ID*/
                        "libAttribute":  "general"
                        /*ro, opt, enum, library type, subType:string, desc:"general" (normal library), "blackList" (blocklist library), "VIP" (VIP
library), "passerby" (allowlist library)*/
                    }
                ],
                "contentID":  "faceImage",
                /*ro, opt, string, background picture ID, desc:background picture ID*/
                "URL":  "test",
                /*ro, opt, string, face picture thumbnail URL, range:[1,256], desc:face picture thumbnail URL*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, picture ID in face picture library, range:[1,32]*/
                "stayDuration":  20,
                /*ro, opt, int, dwell time duration in the image, unit:ms*/
                "prompt":  "test",
                /*ro, opt, string, prompt contents of face picture comparison, range:[1,16]*/
                "contrastStatus":  true,
                /*ro, opt, bool, comparison result*/
                "realTimeContrast":  true,
                /*ro, opt, bool, whether to enable real-time face picture comparison*/
                "remainReportNum":  0,
                /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target, desc:the remaining times of uploading the
comparison alarm of the current target*/
                "UID":  "test",
                /*ro, opt, string*/
                "UUPID":  "test",
                /*ro, opt, string, background picture ID of face picture library, range:[1,64], desc:it consists of device serial No., imported time,
and random digits*/
                "frontMatchType":  1,
                /*ro, opt, enum, face picture library type, subType:int, desc:1 (cluster library), 0 (normal library)*/
```


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```
                    "isImportFD":  true,
                    /*ro, opt, bool, whether to import the captured face data to the library, desc:true (yes), false (no). When face picture comparison
failed, the face picture quality meets the requirements, and is imported to the library, true will be returned. When face picture comparison failed, and the
face picture quality does not meet the requirements, false will be returned*/
                    "faceSnapThermometryEnabled":  true,
                    /*ro, opt, bool, whether to enable temperature screening when capturing face pictures*/
                    "currTemperature":  36.7,
                    /*ro, opt, float, face temperature, range:[-20.0,150.0], unit:℃, desc:it should be used with thermometryUnit together*/
                    "isAbnomalTemperature":  true,
                    /*ro, opt, bool, whether the temperature is abnormal, desc:true (yes), false (no)*/
                    "thermometryUnit":  "celsius",
                    /*ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"*/
                    "alarmTemperature":  36.7,
                    /*ro, opt, float, range:[-20.0,150.0], unit:℃*/
                    "facePicQuality":  "low"
                    /*ro, opt, enum, subType:string*/
                }
            ],
            "alarmRecordID":  "test",
            /*ro, opt, string, alarm record ID*/
            "uid":  "test",
            /*ro, opt, string, the unique ID of event uploading, range:[1,64], desc:it can be composed of time (accurate to the millisecond) and random
digits*/
            "remainReportNum":  0,
            /*ro, opt, int, the remaining times of uploading the comparison alarm of the current target*/
            "algorithmInfo": {
            /*ro, opt, object, algorithm version information, desc:this field is required for secondary analysis*/
                "algorithmId":  "test",
                /*ro, opt, string, algorithm package ID*/
                "algoVendorId":  "test",
                /*ro, opt, string, manufacturer ID*/
                "algoVersionId":  "test",
                /*ro, opt, string, algorithm version No.*/
                "algoChipType":  "test",
                /*ro, opt, string, chip type*/
                "algoModelVersion":  "test"
                /*ro, opt, string, model version*/
            }
        }
    ],
    "URLCertificationType":  "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "GPS": {
    /*ro, opt, object, device GPS information*/
        "divisionEW":  "E",
        /*ro, req, enum, "E" (Eastern Hemisphere), W" (West Hemisphere), subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)*/
        "longitude":  100,
        /*ro, req, int, longitude, desc:longitude*/
        "divisionNS":  "N",
        /*ro, req, enum, subType:string*/
        "latitude":  100
        /*ro, req, int, latitude, desc:latitude*/
    }
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
alarmResult	[Message content]	application/json	--	--	--
..value	[Binary picture data]	image/jpeg	backgroundImage	backgroundImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
faceLibImage	[Binary picture data]	image/jpeg	faceLibImage	faceLibImage.jpg	--
...value	[Binary picture data]	image/jpeg			--
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

16.7.5 Face Picture Library

16.7.5.1 Get the parameters of person custom information

Request URL

GET /ISAPI/Intelligent/faceContrast/personInfoExtend

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, int-->1
    </id>
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <name>
      <!--ro, opt, string-->test
    </name>
    <value>
      <!--ro, opt, string-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

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16.7.5.2 Set the parameters of person custom information

Request URL

PUT /ISAPI/Intelligent/faceContrast/personInfoExtend

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--opt, object-->
    <id>
      <!--req, int-->1
    </id>
    <enable>
      <!--req, bool-->true
    </enable>
    <name>
      <!--opt, string-->test
    </name>
    <value>
      <!--opt, string-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.7.5.3 Get the capability of person custom information

Request URL

GET /ISAPI/Intelligent/faceContrast/personInfoExtend/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PersonInfoExtendList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <PersonInfoExtend>
    <!--ro, opt, object-->
    <id min="1" max="10">
      <!--ro, req, int, attr:min{req, int},max{req, int}-->1
    </id>
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <name min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </name>
    <value min="1" max="10">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </value>
  </PersonInfoExtend>
</PersonInfoExtendList>
```

16.7.5.4 Delete all face picture libraries

Request URL

DELETE /ISAPI/Intelligent/FDLib

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.5.5 Import the face pictures to a specific face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/<FDID>/data?checkCode=<checkCode>&FDType=<FDType>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
checkCode	string	Verification code, and the maximum length is 16. When "security" is 1, it indicates that "checkCode" in the URL is encrypted in AES128 CBC mode; when "security" is 2, it indicates that "checkCode" in the URL is encrypted in AES256 CBC mode.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
mode	string	When importing a face library, an error "faceLibraryNumError" occurs to prompt the user whether to overwrite. If the overwrite import mode is selected, and the face library has reached its maximum capacity, the subsequently imported faces will overwrite the earliest imported face information.

Request Message

Binary Data

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.7.5.6 Update the specified face picture of a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>/imageData?identityKey=<identityKey>&FDType=<FDType>&modeling=<modeling>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	face picture library ID
pictureID	string	picture ID
identityKey	string	The maximum length is 64. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
modeling	enum	whether it is synchronous modeling true (synchronous modeling), false or not applied (asynchronous modeling).

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ImageData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture parameters, attr:version{req, string, protocolVersion}-->
  <picType>
    <!--req, enum, picture type, subType:string, desc:"binary" (binary picture in form format), "url" (picture transmitted in URL)-->binary
  </picType>
  <picUrl>
    <!--opt, picurl, picture URL-->http://127.0.0.1
  </picUrl>
  <Rect>
    <!--opt, object, target frame, desc:the origin is the upper-left corner of the screen. When there are multiple face targets in one picture, you need to
specify the modeling target-->
    <height>
      <!--req, float, height, range:[0.000,1.000]-->0.000
    </height>
    <width>
      <!--req, float, width, range:[0.000,1.000]-->0.000
    </width>
    <x>
      <!--req, float, X-coordinate, range:[0.000,1.000]-->0.000
    </x>
    <y>
      <!--req, float, Y-coordinate, range:[0.000,1.000]-->0.000
    </y>
  </Rect>
</ImageData>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
ImageData	[Message content]	application/xml	--	--	--
faceImage	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

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- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

16.7.5.7 Delete the frequency information of a face picture in

the face picture library **Request URL**

DELETE /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>/occurrenceData/<indexID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
pictureID	string	--
indexID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when statusCode is not 1*/
}
```

16.7.5.8 Get frequency information of all face pictures in the face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>/occurrenceData?format=json

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
pictureID	string	--

Request Message

None

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Response Message

```
{
    "OccurrenceDataList": [
    /*ro, req, array, list of face picture frequency information, subType:object*/
        {
            "id":  1,
            /*ro, req, int, serial No., range:[1,4294967295]*/
            "channel":  1,
            /*ro, req, int, device channel No., range:[1,4294967295]*/
            "snapTime":  "1970-01-01T00:00:00+08:00",
            /*ro, req, datetime, picture captured time*/
            "bkgUrl":  "test",
            /*ro, opt, string, background picture URL, range:[1,256]*/
            "faceId":  1,
            /*ro, opt, int, face ID, range:[1,4294967295], desc:the face ID cannot be duplicated in the same picture*/
            "age": {
            /*ro, opt, object, age*/
                "range":  5,
                /*ro, opt, int, age deviation, range:[0,255], desc:if the age deviation already exists, the age is required*/
                "value":  23,
                /*ro, opt, int, age, range:[0,255]*/
                "ageGroup":  "old"
                /*ro, opt, enum, age group, subType:string, desc:"unknown","child", "young", "middle", "old"*/
            },
            "gender": {
            /*ro, opt, object, gender*/
                "confidence":  99.99990,
                /*ro, opt, float, confidence, range:[0.000000,100.000000]*/
                "value":  "male"
                /*ro, opt, enum, gender, subType:string, desc:"male", "female", "unknown"*/
            },
            "glass": {
            /*ro, opt, object, whether the person wears glasses*/
                "confidence":  99.94490,
                /*ro, opt, float, confidence, range:[0.000000,100.000000]*/
                "value":  "yes"
                /*ro, opt, enum, whether the person wears glasses, subType:string, desc:"no", "yes"*/
            },
            "smile": {
            /*ro, opt, object, whether the person is smiling*/
                "confidence":  99.99990,
                /*ro, opt, float, confidence, range:[0.000000,100.000000]*/
                "value":  "yes"
                /*ro, opt, enum, whether the person is smiling, subType:string, desc:"no", "yes"*/
            },
            "mask": {
            /*ro, opt, object, whether the person wears a mask*/
                "confidence":  99.99990,
                /*ro, opt, float, confidence, range:[0.000000,100.000000]*/
                "value":  "yes"
                /*ro, opt, enum, whether the person wears a mask, subType:string, desc:"no", "yes", "unknown"*/
            },
            "faceExpression": {
            /*ro, opt, object, face expression*/
                "confidence":  99.99990,
                /*ro, opt, float, confidence, range:[0.000000,100.000000]*/
                "value":  "unknown"
                /*ro, opt, enum, face expression type, subType:string, desc:"angry", "contemptuous", "disgusted", "unknown", "happy", "panic", "poker-
faced", "sad", "surprised", "unknown"*/
            },
            "URL":  "test"
            /*ro, opt, string, face thumbnail URL, range:[1,256]*/
        }
    ]
}
```

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16.7.5.9 Get the information of the person linked with the face picture.

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
pictureID	string	Picture ID
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<FaceAppendData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <bornTime>
    <!--ro, opt, string-->test
  </bornTime>
  <name>
    <!--ro, req, string, range:[1,32]-->test
  </name>
  <sex>
    <!--ro, opt, enum, gender, subType:string, desc:male (male), female (female)-->male
  </sex>
  <province>
    <!--ro, opt, string-->11
  </province>
  <city>
    <!--ro, opt, string-->11
  </city>
  <certificateType>
    <!--ro, opt, enum, ID type, subType:string, desc:officerID (officer ID), ID (ID card), passportID (passport), other (other)-->officerID
  </certificateType>
  <certificateNumber>
    <!--ro, opt, string, ID No., range:[8,32], desc:When the value of certificateType is officerID, ID, or other, the size of the value of certificateNumber
can be between 8 bytes and 18 bytes. When the value of certificateType is passportID, the size of the value of certificateType can be between 8 bytes and 32
bytes.-->test
  </certificateNumber>
  <PersonInfoExtendList>
    <!--ro, opt, array, subType:object-->
    <PersonInfoExtend>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int, person extension information No., desc:The No. starts from 1 and increases by 1 subsequently.-->1
      </id>
      <enable>
        <!--ro, req, bool-->true
      </enable>
      <name>
        <!--ro, opt, string-->test
      </name>
      <value>
        <!--ro, opt, string-->test
      </value>
    </PersonInfoExtend>
  </PersonInfoExtendList>
  <phoneNumber>
    <!--ro, opt, string, range:[1,64]-->test
  </phoneNumber>
  <RegionCoordinatesList>
    <!--ro, opt, array, face target, subType:object, range:[0,4], desc:This node determines a rectangle frame to guide data modeling and processing in later
searching.-->
    <RegionCoordinates>
      <!--ro, opt, object, coordinates of the rectangle frame, desc:The origin is the upper-left corner of the screen.-->
      <positionX>
        <!--ro, req, int, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <customHumanId>
    <!--ro, opt, string, range:[1,64]-->test
  </customHumanId>
  <faceDescribe>
    <!--ro, opt, string-->test
  </faceDescribe>
  <IMSIList>
    <!--ro, opt, array, IMSI list, subType:object, desc:If the face picture library is an allowed library, it must be linked to at least one IMSI number and
up to 5 ones.-->
    <IMSI>
      <!--ro, opt, string, range:[1,15]-->123456789012345
    </IMSI>
  </IMSIList>
  <plateNo>
    <!--ro, opt, string, range:[0,32]-->test
  </plateNo>
</FaceAppendData>
```

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16.7.5.10 Set the face picture related information

Request URL

PUT /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
pictureID	string	--
identityKey	string	--
FDType	string	--

Request Message

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DETEKTOR

```xml
<?xml version="1.0" encoding="UTF-8"?>

<FaceAppendData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, face picture related information, attr:version{req, string, protocolVersion}-->
  <bornTime>
    <!--opt, string, date of birth-->test
  </bornTime>
  <name>
    <!--req, string, name, range:[1,32]-->test
  </name>
  <sex>
    <!--opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
  </sex>
  <province>
    <!--opt, string, province, desc:see the data dictionary for details-->11
  </province>
  <city>
    <!--opt, string, city, desc:see the data dictionary for details-->11
  </city>
  <certificateType>
    <!--opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (other
certificate)-->officerID
  </certificateType>
  <certificateNumber>
    <!--opt, string, certificate No., range:[8,32], desc:certificate No.-->test
  </certificateNumber>
  <PersonInfoExtendList>
    <!--opt, array, custom extension information of person, subType:object-->
    <PersonInfoExtend>
      <!--opt, object-->
      <id>
        <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
      </id>
      <enable>
        <!--req, bool-->true
      </enable>
      <name>
        <!--opt, string, name of person tag extended information-->test
      </name>
      <value>
        <!--opt, string, person tag extended information-->test
      </value>
    </PersonInfoExtend>
  </PersonInfoExtendList>
  <phoneNumber>
    <!--opt, string, range:[1,64]-->test
  </phoneNumber>
  <RegionCoordinatesList>
    <!--opt, array, face picture target, subType:object, range:[0,4], desc:face picture target-->
    <RegionCoordinates>
      <!--opt, object, coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <customHumanId>
    <!--opt, string, custom person ID, range:[1,64]-->test
  </customHumanId>
  <faceDescribe>
    <!--opt, string-->test
  </faceDescribe>
  <IMSIList>
    <!--opt, array, subType:object-->
    <IMSI>
      <!--opt, string, range:[1,15]-->123456789012345
    </IMSI>
  </IMSIList>
  <plateNo>
    <!--opt, string, range:[0,32]-->test
  </plateNo>
</FaceAppendData>
```

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DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.5.11 Delete the linked information of a picture in the face picture

library **Request URL**

DELETE /ISAPI/Intelligent/FDLib/<FDID>/picture/<pictureID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
pictureID	string	Picture ID
identityKey	string	The maximum length is 64. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.7.5.12 Get the number of remaining face pictures in a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>/picture/surplusCapacity?FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SurplusCapacity xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, undefined, attr:version{req, string, protocolVersion}-->
  <FDPicturesNumber>
    <!--ro, opt, int, supported number of remaining face pictures in the face picture library-->1
  </FDPicturesNumber>
</SurplusCapacity>
```

16.7.5.13 Set the basic parameters of a face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/<FDID>/updateHumanId?format=json&FDType=<FDType>

Query Parameter

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Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

```
{
    "UpdateHumanID": [
    /*opt, array, subType:object*/
        {
            "index":  1,
            /*req, int*/
            "PID":  "test",
            /*req, string*/
            "customHumanID":  "test"
            /*req, string*/
        }
    ]
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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16.7.5.14 Delete a face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib/<FDID>?FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.7.5.15 Set the basic parameters of a face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/<FDID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID.
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for 8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, face picture library basic information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, index-->1
  </id>
  <FDID>
    <!--opt, string, face picture library ID-->test
  </FDID>
  <name>
    <!--opt, string, face picture library name-->test
  </name>
  <thresholdValue>
    <!--opt, int, detection threshold, range:[0,100]-->1
  </thresholdValue>
  <customInfo>
    <!--opt, string, custom information-->test
  </customInfo>
  <customFaceLibID>
```

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DEFEKTOR

```
    <!--opt, string, ID of custom human body picture library-->test
  </customFaceLibID>
  <faceLibType>
    <!--opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger library),
"allowable" (allowlist library), "privacyMask" (privacy mask library)-->ordinary
  </faceLibType>
  <autoUpdata>
    <!--opt, bool, whether to enable auto update of background picture in face picture library, desc:if autoUpdata in the capability set only supports
"true", the auto update feature is enabled by default; if not, it is disabled-->true
  </autoUpdata>
  <qualityThreshold>
    <!--opt, int, picture quality threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </qualityThreshold>
  <importFDThreshold>
    <!--opt, int, put-in-storage similarity threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </importFDThreshold>
  <Age>
    <!--opt, object, configuration capability of face picture library clustering aging parameters, desc:this node is valid when <autoUpdata> is "true"-->
    <judgementCycleType>
      <!--req, enum, evaluation cycle type of clustering data aging in face picture library, subType:string, desc:"day" (evaluate at 0 every day), "week"
(evaluate at 0 every Sunday), "month" (default; evaluate at 0 on the first day of every month), "duration" (evaluate by duration), "quarter" (evaluate at 0
on the first day of every quarter), "halfYear" (evaluate at 0 on the first day of every six months)-->day
    </judgementCycleType>
    <statisticalCycleType>
      <!--opt, enum, statistics cycle type of clustering matches in face picture library, subType:string, desc:"day" (daily), "week" (weekly), "month"
(monthly)-->day
    </statisticalCycleType>
    <statisticalCycle>
      <!--opt, int, statistics cycle range of clustering matches in face picture library, range:[0,99999], desc:this node is based on statisticalCycleType--
>1
    </statisticalCycle>
    <minFMCount>
      <!--opt, int, min. number of face picture matches within statistics cycle, range:[0,99999], desc:the default value is 30-->1
    </minFMCount>
    <FMCountEnabled>
      <!--opt, bool, whether to enable restriction on total number of clustering matches in face picture library-->true
    </FMCountEnabled>
    <FMCount>
      <!--opt, int, range of total face picture matches, range:[0,99999]-->100
    </FMCount>
    <backgroundPicThreshold>
      <!--opt, int, background picture replacement similarity threshold, range:[1,100]-->90
    </backgroundPicThreshold>
    <judgementCycle>
      <!--opt, int, evaluation cycle range of clustering data aging in face picture library, range:[1,365], unit:d, dep:and,
{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
    </judgementCycle>
    <agingExecDataTime>
      <!--opt, time, clustering library aging execution time-->00:00:00+08:00
    </agingExecDataTime>
    <automaticClearEnabled>
      <!--opt, bool, whether to enable auto clear-up when the clustering library is full-->true
    </automaticClearEnabled>
  </Age>
  <FaceCounting>
    <!--opt, object, face picture statistics-->
    <enabled>
      <!--req, bool, whether to enable face-picture people counting deduplication-->true
    </enabled>
    <uploadType>
      <!--req, enum, reporting mode after face-picture people counting deduplication is enabled, subType:string, desc:"out_repetition" (report outside
deduplication time), "noUpload" (do not report)-->out_repetition
    </uploadType>
    <channelList>
      <!--opt, object, video channel list, desc:this node is used on normal cameras accessed to NVR/CVR to specify which video channels should use the face
picture library to perform people counting-->
      <channelID>
        <!--req, int, video channel No., range:[1,256]-->true
      </channelID>
    </channelList>
  </FaceCounting>
  <faceMonitor>
    <!--opt, bool, whether it is the face picture library linked with intelligent face picture arming-->true
  </faceMonitor>
  <monitorFaceNum>
    <!--opt, bool, max. face pictures supported by face picture arming-->true
  </monitorFaceNum>
  <customNoConfig>
    <!--opt, bool, this node is invalid for Web client, desc:it is only valid in person arming mode-->true
  </customNoConfig>
</FDLibBaseCfg>
```

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

16.7.5.16 Get the basic parameters of a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/<FDID>?identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID.
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face picture library basic information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, index-->1
  </id>
  <FDID>
    <!--ro, opt, string, face picture library ID-->test
  </FDID>
  <name>
    <!--ro, opt, string, face picture library name-->test
  </name>
  <thresholdValue>
    <!--ro, opt, int, detection threshold, range:[0,100]-->1
  </thresholdValue>
  <customInfo>
    <!--ro, opt, string, custom information-->test
  </customInfo>
  <customFaceLibID>
    <!--ro, opt, string, custom face picture library ID-->test
```

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HIKVISION

```
  </customFaceLibID>
  <faceLibType>
    <!--ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger library (clustering library)), "allowable" (allowlist library), "privacyMask" (privacy mask library)-->ordinary
  </faceLibType>
  <autoUpdata>
    <!--ro, opt, bool, whether to enable auto update of background picture in face picture library, desc:if autoUpdata in the capability set only supports "true", the auto update feature is enabled by default; if not, it is disabled.-->true
  </autoUpdata>
  <qualityThreshold>
    <!--ro, opt, int, picture quality threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </qualityThreshold>
  <importFDThreshold>
    <!--ro, opt, int, put-in-storage similarity threshold, range:[0,100], desc:this node is valid when <autoUpdata> is "true"-->1
  </importFDThreshold>
  <Age>
    <!--ro, opt, object, evaluation cycle, desc:this node is valid when <autoUpdata> is "true"-->
    <judgementCycleType>
      <!--ro, req, enum, evaluation cycle type, subType:string, desc:"day" (evaluate at 0 o'clock every day), "week" (evaluate at 0 o'clock every Sunday), "month" (default; evaluate at 0 o'clock on the first day of every month), "duration" (evaluate by duration), "quarter" (evaluate at 0 o'clock on the first day of every quarter), "halfYear" (evaluate at 0 o'clock on the first day of every six months)-->day
    </judgementCycleType>
    <statisticalCycleType>
      <!--ro, opt, enum, statistics cycle type of face picture matches, subType:string, desc:"day" (daily), "week" (weekly), "month" (monthly)-->day
    </statisticalCycleType>
    <statisticalCycle>
      <!--ro, opt, int, statistics cycle of face picture matches, range:[0,99999]-->1
    </statisticalCycle>
    <minFMCount>
      <!--ro, opt, int, min. number of face picture matches within statistics cycle, range:[0,99999], desc:the default value is 30-->1
    </minFMCount>
    <FMCountEnabled>
      <!--ro, opt, bool, whether to enable total number of face picture matches-->true
    </FMCountEnabled>
    <FMCount>
      <!--ro, opt, int, total number of face picture matches, range:[0,99999]-->100
    </FMCount>
    <backgroundPicThreshold>
      <!--ro, opt, int, background picture replacement similarity threshold, range:[1,100]-->90
    </backgroundPicThreshold>
    <judgementCycle>
      <!--ro, opt, int, evaluation cycle of clustering data aging in face picture library, range:[1,365], unit:d, dep:and, {$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
    </judgementCycle>
    <agingExecDataTime>
      <!--ro, opt, time, clustering library aging execution time-->00:00:00+08:00
    </agingExecDataTime>
    <automaticClearEnabled>
      <!--ro, opt, bool, whether to enable auto clear-up when the clustering library is full-->true
    </automaticClearEnabled>
  </Age>
  <FaceCounting>
    <!--ro, opt, object, face picture statistics-->
    <enabled>
      <!--ro, req, bool, whether to enable face-picture people counting deduplication-->true
    </enabled>
    <uploadType>
      <!--ro, req, enum, reporting mode after face-picture people counting deduplication is enabled, subType:string, desc:"out_repetition" (report outside deduplication time), "noUpload" (do not report)-->out_repetition
    </uploadType>
    <channelList>
      <!--ro, opt, object, video channel list, desc:this node is used on normal cameras accessed to NVR/CVR to specify which video channels should use the face picture library to perform people counting-->
      <channelID>
        <!--ro, req, int, video channel No., range:[1,256]-->true
      </channelID>
    </channelList>
  </FaceCounting>
  <faceMonitor>
    <!--ro, opt, bool, whether it is the face picture library linked with intelligent face picture arming-->true
  </faceMonitor>
  <monitorFaceNum>
    <!--ro, opt, bool, max. face pictures supported by face picture arming-->true
  </monitorFaceNum>
  <customNoConfig>
    <!--ro, opt, bool, this node is not valid for Web client, desc:it is only valid in person arming mode-->true
  </customNoConfig>
</FDLibBaseCfg>
```

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DETEKTOR

16.7.5.17 Get the capability of asynchronously importing base picture data to face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/asyncImportDatas/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "AsyncImportDatas": {
    /*ro, req, object*/
        "customFaceLibID": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, req, int, range:[1,64]*/
            "@max":  64
            /*ro, req, int, range:[1,64]*/
        },
        "maxTaskNum":  10,
        /*ro, req, int, range:[1,10]*/
        "taskID": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, req, int, range:[1,64]*/
            "@max":  1
            /*ro, req, int, range:[1,64]*/
        },
        "URL": {
        /*ro, req, object*/
            "@min":  16,
            /*ro, req, int*/
            "@max":  255
            /*ro, req, int*/
        },
        "type": {
        /*ro, req, object*/
            "@opt": [0]
            /*ro, opt, array, subType:int*/
        },
        "URLCertificationType": {
        /*ro, req, object*/
            "@opt": ["CEOSSToken", "AWS2_0"]
            /*ro, req, array, subType:string*/
        },
        "accessKey": {
        /*ro, opt, object*/
            "@min":  32,
            /*ro, opt, int, range:[32,256]*/
            "@max":  256
            /*ro, opt, int, range:[32,256]*/
        },
        "secretKey": {
        /*ro, opt, object*/
            "@min":  32,
            /*ro, opt, int, range:[32,256]*/
            "@max":  256
            /*ro, opt, int, range:[32,256]*/
        },
        "token": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int, range:[1,64]*/
            "@max":  64
            /*ro, opt, int, range:[1,64]*/
        },
        "isSupportPrivateFaceLib":  true,
        /*ro, opt, bool*/
        "isSupportEncryption":  true
        /*ro, opt, bool*/
    }
}
```

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DETEKTOR

16.7.5.18 Get the status search capability of asynchronous import of base picture data

Request URL

GET /ISAPI/Intelligent/FDLib/asyncImportDatas/statusSearch/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "StatusSearch": {
    /*ro, req, object*/
        "searchResultPosition": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, opt, int, range:[1,10000]*/
            "@max":  10000
            /*ro, req, int, range:[1,10000]*/
        },
        "maxResults": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, req, int, range:[1,10000]*/
            "@max":  10000
            /*ro, req, int, range:[1,10000]*/
        },
        "modelMaxNum": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int, range:[1,10000]*/
            "@max":  10000
            /*ro, req, int, range:[1,10000]*/
        },
        "customFaceLibID": {
        /*ro, req, object*/
            "@min":  1,
            /*ro, req, int, range:[1,64]*/
            "@max":  64
            /*ro, req, int, range:[1,64]*/
        },
        "importDatasStatus": {
        /*ro, req, object*/
            "@opt":  "downloadFailed"
            /*ro, req, enum, subType:string*/
        },
        "taskID": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, range:[0,64]*/
            "@max":  64
            /*ro, req, int, range:[0,64]*/
        }
    }
}
```

16.7.5.19 Asynchronously import the data to the face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/asyncImportDatas?format=json&identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--

Request Message

```
{
    "AsyncImportDatas": {
    /*req, object, data object, which is imported asynchronously*/
        "customFaceLibID":  "1234567890",
        /*req, string, ID of custom face picture library, range:[1,64]*/
        "taskID":  "123abcABC",
        /*req, string, task ID, range:[1,64], desc:the platform can import multiple data files via multiple tasks, which are distinguished by taskID. Only
the following characters are supported: "0-9" "A-Z" "a-z" and "-"*/
        "URL":  "http://10.21.84.28:6011/pic?4299E9285F34B167FECCDF6B3FB286F7",
        /*req, string, data file URL*/
        "type":  0,
        /*opt, enum, data file type, subType:int, desc:0-XML (default)*/
        "URLCertificationType":  "AWS2_0",
        /*req, enum, authentication mode of picture URL, subType:string, desc:"CEOSSToken" (OSS authentication), "AWS2_0" (Cloud)*/
        "accessKey":  "Vl7UVrl995wmKQ7SLJin1aKcUp81Y44ll6aZ9fh9251S4cSr6",
        /*opt, string, cloud storage access key, dep:and,{$.AsyncImportDatas.URLCertificationType,eq,AWS2_0}, desc:it is valid when the value of node
URLCertificationType is "AWS2_0"*/
        "secretKey":  "FY0sb288x7J244V5Z1Wdn18wOyo6JKRg82iK5056563Z4A767H",
        /*opt, string, cloud storage encryption key, dep:and,{$.AsyncImportDatas.URLCertificationType,eq,AWS2_0}, desc:it is valid when the value of node
URLCertificationType is "AWS2_0"*/
        "token":  "1Wdn18wOyo6JKRg82iK5056563Z4A767HcaZ68KQ5X8NpJ"
        /*opt, string, OSS authentication token, dep:and,{$.AsyncImportDatas.URLCertificationType,eq,CEOSSToken}*/
    }
}
```

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Response Message

```
{
    "requestURL":  "/ISAPI/Intelligent/FDLib/asyncImportDatas?format=json",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

16.7.5.20 Get the status of asynchronously importing base picture data to face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/asyncImportDatasTasks?format=json&identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	If the URL does not contain the identityKey parameter, it indicates searching for tasks of the normal custom library; if the identityKey parameter is contained, it indicates searching for tasks related to the private library only.

Request Message

None

Response Message

```
{
    "AsyncImportDatasTasks": [
    /*ro, opt, array, subType:object*/
        {
            "customFaceLibID":  "test",
            /*ro, req, string, range:[1,64]*/
            "taskID":  "test",
            /*ro, req, string, range:[1,64]*/
            "URL":  "test",
            /*ro, req, string, range:[1,255]*/
            "type":  0,
            /*ro, opt, enum, subType:int*/
            "URLCertificationType":  "AWS2_0",
            /*ro, req, enum, subType:string*/
            "status":  0,
            /*ro, req, enum, subType:int*/
            "state":  "URLError",
            /*ro, opt, enum, subType:string*/
            "progress":  1
            /*ro, req, int, range:[1,100], step:1*/
        }
    ]
}
```

16.7.5.21 Get face picture library capability

Request URL

GET /ISAPI/Intelligent/FDLib/capabilities

Query Parameter

None

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>
<FDLibCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, face picture library capability object, attr:version{opt, string, protocolVersion}-->
  <CreateFDLibList size="15">
    <!--ro, req, array, face picture library capability object array, subType:object, attr:size{opt, string}-->
    <CreateFDLib>
      <!--ro, req, object, face picture library capability object-->
      <id max="15" min="1">
        <!--ro, req, int, number range capability of face picture library, range:[1,15], attr:max{opt, string},min{opt, string}-->10
      </id>
      <name max="64" min="1">
        <!--ro, opt, string, name length of face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
      </name>
      <thresholdValue max="100" min="0" def="60">
        <!--ro, opt, int, detection threshold range capability of face picture library, range:[0,100], attr:max{opt, string},min{opt, string},def{req, int},
desc:the higher the threshold, the lower the detection accuracy-->50
      </thresholdValue>
      <customInfo max="128" min="1">
        <!--ro, opt, string, custom description length in the face picture library, range:[1,128], attr:max{opt, string},min{opt, string}-->128
      </customInfo>
      <faceDataMax max="150000" min="1">
        <!--ro, opt, int, the maximum number of human face base picture allowed in the face picture library, range:[1,150000], attr:max{opt,
string},min{opt, string}-->30000
      </faceDataMax>
      <customFaceLibID max="64" min="1">
        <!--ro, opt, string, the custom ID length in the face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
      </customFaceLibID>
      <indentityKey max="64" min="0">
        <!--ro, opt, string, description length of the secret key in the private face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-
->64
      </indentityKey>
    </CreateFDLib>
  </CreateFDLibList>
  <FaceAppendData>
    <!--ro, opt, object, the object capability of background picture data extension of face picture library-->
    <bornTime>
      <!--ro, opt, date, birth date of the person in the face picture-->1970-01-01+08:00
    </bornTime>
    <name max="64" min="1">
      <!--ro, opt, string, the description length of person name, range:[1,64], attr:max{opt, string},min{opt, string}-->64
    </name>
    <sex opt="male,female">
      <!--ro, opt, enum, gender, subType:string, attr:opt{opt, string}, desc:male; female;-->male
    </sex>
    <province>
      <!--ro, opt, string, province of birth-->01
    </province>
    <city>
      <!--ro, opt, string, city-->01
    </city>
    <certificateType opt="officerID,ID,passportID,other">
      <!--ro, opt, enum, certificate type capability, subType:string, attr:opt{opt, string}, desc:"officerID"-officer ID,"ID"-identify card,"passportID"-
passport,"other"-other certificate-->officerID
    </certificateType>
    <certificateNumber max="32" min="1">
      <!--ro, opt, string, capability of certificate No., range:[1,128], attr:max{opt, string},min{opt, string}-->32
    </certificateNumber>
    <PersonInfoExtendList size="128">
      <!--ro, opt, array, object array of base picture's extended information in the face picture library, subType:object, attr:size{opt, string}, desc:if
the node is returned, it indicates that the device supports. Corresponding URL: /ISAPI/Intelligent/FDLib/pictureUpload-->
      <PersonInfoExtend>
        <!--ro, opt, object, the capability of base picture's extended information in the face picture library-->
        <id max="15000" min="1">
          <!--ro, req, int, No. range of extended information, range:[1,15000], attr:max{opt, string},min{opt, string}-->30000
        </id>
        <enable opt="true,false">
          <!--ro, req, bool, whether to enable the extended information, attr:opt{opt, string}-->true
        </enable>
        <name max="128" min="1">
          <!--ro, opt, string, the name length of extended information, range:[1,128], attr:max{opt, string},min{opt, string}-->128
        </name>
        <value max="256" min="1">
          <!--ro, opt, string, the content length of extended information, range:[1,256], attr:max{opt, string},min{opt, string}-->256
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
    <isSupportPhoneNumber>
      <!--ro, opt, bool, whether the device supports telephone number capability-->true
    </isSupportPhoneNumber>
    <customHumanID max="128" min="1">
      <!--ro, opt, string, the capability of the person ID of custom face picture, range:[1,128], attr:max{opt, string},min{opt, string}-->64
    </customHumanID>
  </FaceAppendData>
  <FDSearchDescription>
    <!--ro, opt, object, the search capability of face picture library-->
    <searchID max="63" min="1">
      <!--ro, opt, string, search ID, attr:max{opt, string},min{opt, string}, desc:it is used to confirm the upper-level platform or system. If the platform
or the system is the same one during two searches, the search history will be saved in the memory to speed up next search; min/max will be returned
according to the device actual capability, if not returned: min=1 max=63 (default)-->63
    </searchID>
    <searchResultPosition>
      <!--ro, opt, int, start position, desc:if this node is not returned, confirm <isSupportStandardSearch> exists and the node "true" is returned; it will
start from 0 by default; if not returned, and it will start from 1 by default-->1
    </searchResultPosition>
```

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```
    <maxResults>
      <!--ro, opt, int, the number of search results this time, desc:the number of search results in a single search is between 1 and 500; if exceeding the max number of search results supported by the device, the device will return the max number of results in a single search by default; if this node is not returned, the number of search results is between 1 and 500 by default-->30
    </maxResults>
    <FDID>
      <!--ro, opt, string, face picture library ID-->1
    </FDID>
    <startTime>
      <!--ro, opt, datetime, the birth start time of the persons in the face picture library-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, the birth end time of the persons in the face picture library-->1970-01-01T00:00:00+08:00
    </endTime>
    <name max="64" min="1">
      <!--ro, opt, string, person name, attr:max{opt, string},min{opt, string}-->64
    </name>
    <sex opt="male,female">
      <!--ro, opt, enum, gender, subType:string, attr:opt{opt, string}, desc:male; female;-->male
    </sex>
    <province>
      <!--ro, opt, string, province of birth-->01
    </province>
    <city>
      <!--ro, opt, string, city-->01
    </city>
    <certificateType opt="officerID,ID,passportID,other">
      <!--ro, opt, enum, certificate type capability, subType:string, attr:opt{opt, string}, desc:"officerID" (officer ID), "ID" (identify card), "passportID" (passport), "other" (other certificate)-->officerID
    </certificateType>
    <certificateNumber max="32" min="1">
      <!--ro, opt, string, capability of certificate No. length, range:[1,128], attr:max{opt, string},min{opt, string}-->32
    </certificateNumber>
    <FaceModeCap>
      <!--ro, opt, object, the object capability of searching picture by picture-->
      <FaceMode>
        <!--ro, opt, object, face model-->
        <similarity max="100.0" min="0.0">
          <!--ro, opt, float, similarity, attr:max{opt, string},min{opt, string}-->50.0
        </similarity>
      </FaceMode>
      <maxResultNum>
        <!--ro, opt, int, the maximum number of returned results-->30
      </maxResultNum>
    </FaceModeCap>
    <modelStatus opt="modeling,unmodeled,modelingFailed">
      <!--ro, opt, enum, the modeling status of the background picture in the face picture library, subType:string, attr:opt{opt, string}, desc:"modeling" (modeled), "unmodeled" (not modeled), "modelingFailed" (modeling failed)-->modeling
    </modelStatus>
    <modelingStatus opt="success,failed,none">
      <!--ro, opt, enum, the modeling result of the background picture in the face picture library, subType:string, attr:opt{opt, string}, desc:"success" (modeling succeeded), "failed" (modeling failed), "none" (not modeled)-->success
    </modelingStatus>
    <customFaceLibID max="64" min="0">
      <!--ro, opt, string, the custom ID length in the face picture library, range:[1,64], attr:max{opt, string},min{opt, string}-->64
    </customFaceLibID>
    <indentityKey max="64" min="0">
      <!--ro, opt, string, the private key length capability of the private library, attr:max{opt, string},min{opt, string}-->64
    </indentityKey>
    <faceScore min="-1" max="100">
      <!--ro, opt, int, face score (face picture quality), range:[-1,100], attr:min{opt, string},max{opt, string}, desc:valid range: [0-100]; -1 (rating unknown, i.e., the face picture is not rated)-->-1
    </faceScore>
    <isSupportFDIDList>
      <!--ro, opt, bool, whether the device supports searching by picture in the multiple libraries, desc:corresponding to the node FDIDList in /ISAPI/Intelligent/FDLib/FDSearch-->true
    </isSupportFDIDList>
    <pictureStatus opt="normal,abnormal">
      <!--ro, opt, string, face picture status, attr:opt{req, string}, desc:abnormalFace (abnormal face), normalFace (normal face)-->test
    </pictureStatus>
  </FDSearchDescription>
  <isSuportAnalysisFace>
    <!--ro, opt, bool, whether the device supports face analysis-->true
  </isSuportAnalysisFace>
  <isSuportFDLibImpEmp opt="true,false">
    <!--ro, opt, bool, whether to support the import and export of the face picture library, attr:opt{opt, string}-->true
  </isSuportFDLibImpEmp>
  <isSuportFCSearch opt="true,false">
    <!--ro, opt, bool, whether supports searching captured face picture library, attr:opt{opt, string}-->true
  </isSuportFCSearch>
  <isSupportFDLibEachImport>
    <!--ro, opt, bool, the background picture import data of the face picture library-->true
  </isSupportFDLibEachImport>
  <FCSearchDescription>
    <!--ro, opt, object, the object capability of searching captured face picture library-->
    <searchID max="63" min="1">
      <!--ro, opt, string, search ID, attr:max{opt, string},min{opt, string}-->63
    </searchID>
    <searchResultPosition>
      <!--ro, opt, int, start position; this node is not returned-->1
    </searchResultPosition>
    <maxResults>
      <!--ro, opt, int, the number of search results this time-->30
```

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```
    </maxResults>
    <snapStartTime>
      <!--ro, opt, datetime, ISO 8601 time-->1970-01-01T00:00:00+08:00
    </snapStartTime>
    <snapEndTime>
      <!--ro, opt, datetime, ISO 8601 time-->1970-01-01T00:00:00+08:00
    </snapEndTime>
    <eventType opt="unlimit,faceContrast,whiteFaceContrast,faceCapture,faceContrastFailure">
      <!--ro, opt, enum, type of face capture event, subType:string, attr:opt{opt, string}, desc:"unlimit", "faceContrast" (face picture comparison),
"whiteFaceContrast" (allowlist face picture comparison (strangers alarm)), "faceCapture" (face capture), "faceContrastFailure" (face picture comparison
failed)-->unlimit
    </eventType>
    <FaceModeCap>
      <!--ro, opt, object, the object capability of searching picture by picture-->
      <FaceMode>
        <!--ro, opt, object, face model-->
        <similarity max="100.0" min="0.0">
          <!--ro, opt, float, similarity, range:[0.0,100.0], attr:max{opt, string},min{opt, string}-->50.0
        </similarity>
      </FaceMode>
      <maxResultNum>
        <!--ro, opt, int, the maximum number of returned results, desc:due to its large storage capacity, the search time for face capture library can be
long; consequently, in order to improve search efficiency, the number of samples and results of the search is limited-->30
      </maxResultNum>
    </FaceModeCap>
    <PictureSearchCap>
      <!--ro, opt, object, this node is returned when searching for pictures by ID is supported-->
      <maxSearchPictureNum max="1" min="1">
        <!--ro, opt, int, the maximum number of pictures allowed to be searched for, attr:max{opt, string},min{opt, string}-->50
      </maxSearchPictureNum>
    </PictureSearchCap>
    <isSupportMultiChannelSearch>
      <!--ro, opt, bool, whether the device supports multichannel search-->true
    </isSupportMultiChannelSearch>
    <isSupportTotalSearchResult>
      <!--ro, opt, bool, whether the device supports the device result limit for each search-->true
    </isSupportTotalSearchResult>
  </FCSearchDescription>
  <isSupportFCSearchDataPackage>
    <!--ro, opt, bool, whether supports packaging the search results of face comparison alarms-->true
  </isSupportFCSearchDataPackage>
  <maxFCSearchDataPackageNum>
    <!--ro, opt, int, the maximum number of face search results package allowed-->50
  </maxFCSearchDataPackageNum>
  <isSupportFDExecuteControl>
    <!--ro, opt, bool, whether the device supports creating relation between face picture libraries and cameras-->true
  </isSupportFDExecuteControl>
  <isSupportManualModeling>
    <!--ro, opt, bool, whether the device supports modeling control-->true
  </isSupportManualModeling>
  <isSupportSingleManualModeling>
    <!--ro, opt, bool, whether the device supports single modeling-->true
  </isSupportSingleManualModeling>
  <isSupportRangeManualModeling>
    <!--ro, opt, bool, whether the device supports all modeling-->true
  </isSupportRangeManualModeling>
  <isSupportSingleManualModelingSearch>
    <!--ro, opt, bool, whether the device supports searching for single modeling-->true
  </isSupportSingleManualModelingSearch>
  <isSupportRangeManualModelingSearch>
    <!--ro, opt, bool, whether the device supports all modeling control-->true
  </isSupportRangeManualModelingSearch>
  <isSupportModelingStatusSearch>
    <!--ro, opt, bool, whether the device supports searching for the condition for modeling status-->true
  </isSupportModelingStatusSearch>
  <Reboot>
    <!--ro, opt, object, the specific parameters of prompt for device reboot to take effect-->
    <thresholdValue>
      <!--ro, opt, bool, after editing the threshold for the face picture library, you need to reboot the device to take effect-->true
    </thresholdValue>
    <delFaceLib>
      <!--ro, opt, bool, after deleting the face picture library, you need to reboot the device to take effect-->true
    </delFaceLib>
  </Reboot>
  <isSupportCoverImportpictures>
    <!--ro, opt, bool, whether to support importing and overwriting the existing pictures-->true
  </isSupportCoverImportpictures>
  <isSupportCoverImportFaceLib>
    <!--ro, opt, bool, whether to support importing and overwriting the existing face picture library-->true
  </isSupportCoverImportFaceLib>
  <isSupportConcurrentImportpictures>
    <!--ro, opt, bool, whether to support concurrent import of pictures-->true
  </isSupportConcurrentImportpictures>
  <isSupportSurplusCapacity>
    <!--ro, opt, bool, whether to support searching for the number of remaining face pictures of specified face picture library-->true
  </isSupportSurplusCapacity>
  <isSupportFDlibProgress>
    <!--ro, opt, bool, whether to support searching for the import and output status of the face picture library-->true
  </isSupportFDlibProgress>
  <isSupportFDlibPhotoWall>
    <!--ro, opt, bool, whether to support the picture display capability-->true
  </isSupportFDlibPhotoWall>
  <isSupportStandardSearch>
    <!--ro, opt, bool, this node is used for the inconsistency between the implementation of some devices and documentation, desc:due to the inconsistency
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    <!--ro, opt, bool, this node is used for the inconsistency between the implementation of some devices and documentation, desc:due to the inconsistency between the implementation of some devices and documentation, the modification is as follows: the responseStatus returns true, capitalize the first letters of matchList and matchElement, in order to keep consistent with the documentation; the face picture library search (/ISAPI/Intelligent/FDLib/FDSearch), the face comparison search (/ISAPI/Intelligent/FDLib/FCSearch), the face modeling status search and searchResultPosition, all starts from 0-->true
  </isSupportStandardSearch>
  <isSupportFaceDataExport>
    <!--ro, opt, bool, it indicates whether to support exporting the data of face picture library-->true
  </isSupportFaceDataExport>
  <isSupportNewlyPictureUpload>
    <!--ro, opt, bool, whether to support importing pictures into the face picture library, desc:it is used for the device to be compatible with large-sized picture import-->true
  </isSupportNewlyPictureUpload>
  <isSupportFCSearchNormal>
    <!--ro, opt, bool, whether to support searching for all face picture data, desc:it is used for searching for face picture result via ISAPI protocols; if the search condition is not configured, by default, it will search all and get pictures via the URI: NET_DVR_SmartSearchPicture-->true
  </isSupportFCSearchNormal>
  <isSupportPrompt>
    <!--ro, opt, bool, whether to support custom prompt, desc:if the device supports, corresponding URL: /ISAPI/Intelligent/FDLib/FCSearch/capabilities?format=json; the specific function supported by the device can be judged via the URL-->true
  </isSupportPrompt>
  <isSupportCustomHumanID>
    <!--ro, opt, bool, whether to support custom person ID for developer-->true
  </isSupportCustomHumanID>
  <isSupportCustomFaceLibID>
    <!--ro, opt, bool, whether to support custom face picture library ID for developer-->true
  </isSupportCustomFaceLibID>
  <isSupportFCSearchJsonFormat>
    <!--ro, opt, bool, whether to support searching for the face comparison alarm information in json format, desc:if the device supports, corresponding URL: /ISAPI/Intelligent/FDLib/FCSearch/capabilities?format=json; the function supported by the device can be judged via the URL-->true
  </isSupportFCSearchJsonFormat>
  <isSupportFCSearchDataPackageJsonFormat>
    <!--ro, opt, bool, whether to support packing for the face comparison alarm information in json format, desc:if the device supports, corresponding URL: POST /ISAPI/Intelligent/FDLib/FCSearch/dataPackage?format=json-->true
  </isSupportFCSearchDataPackageJsonFormat>
  <isSupportFDCapacity>
    <!--ro, opt, bool, whether to support getting the capacity of face picture library-->true
  </isSupportFDCapacity>
  <SupportUploadPictureType opt="binary,url">
    <!--ro, opt, enum, the supported types of the pictures to be imported, subType:string, attr:opt{opt, string}, desc:"binary", "url"; if it is not returned, it is binary by default-->binary
  </SupportUploadPictureType>
  <isSupportNoSaveUploadPicture>
    <!--ro, opt, bool, whether to support not uploading the original pictures-->true
  </isSupportNoSaveUploadPicture>
  <isSupportUploadModeling>
    <!--ro, opt, bool, whether to support directly uploading the picture modeling data-->true
  </isSupportUploadModeling>
  <importImageSize max="300" min="1">
    <!--ro, opt, int, the size of supported JPG/JPEG pictures, unit:KB, attr:max{opt, string},min{opt, string}-->30
  </importImageSize>
  <FDModelingStatusSearchDescription>
    <!--ro, opt, object, task ID-->
    <searchID max="63" min="1">
      <!--ro, opt, string, search ID, attr:max{opt, string, range:[0,64]},min{opt, string}, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searches, the search history will be saved in the memory to speed up next search; min/max will be returned according to the device actual capability, if not returned: min=1 max=63 (default)-->searchIDtest
    </searchID>
    <searchResultPosition>
      <!--ro, opt, int, start position; this node is not returned, desc:this node is not returned, confirm <isSupportStandardSearch> exists and the node "true" is returned, it will start from 0 by default; if not returned, it will start from 1 by default-->30
    </searchResultPosition>
    <maxResults>
      <!--ro, opt, int, the number of search results this time, desc:the number of search results in a single search is between 1 and 500; if exceeding the max number of search results supported by the device, the device will return the max number of results in a single search by default; if this node is not returned, the number of search results is between 1 and 500 by default-->30
    </maxResults>
    <FDIDList>
      <!--ro, opt, string, ID of face picture library supported by the device to search for face modeling data-->test
    </FDIDList>
    <taskID max="64" min="1">
      <!--ro, opt, string, taskID, attr:max{opt, string},min{opt, string}, desc:the taskID has priority, corresponding URL (/ISAPI/Intelligent/FDLib/manualModelingRangeTask or /ISAPI/Intelligent/FDLib/manualModeling?FDID=&&PID=), this node will be returned-->64
    </taskID>
  </FDModelingStatusSearchDescription>
  <StrangerLibFaceAppendData>
    <!--ro, opt, object, facial picture attributes supported by stranger face picture library-->
  </StrangerLibFaceAppendData>
  <StrangerLibFDSearchDescription>
    <!--ro, opt, object, stranger face picture library-->
    <FDID>
      <!--ro, opt, string, the unique ID of face picture library-->1
    </FDID>
    <occurrences max="30" min="1">
      <!--ro, opt, int, person frequency range, attr:max{opt, string},min{opt, string}-->30
    </occurrences>
    <occurrencesSearchType opt="greaterThanOrEqual,lessThan,equal">
      <!--ro, opt, enum, frequency search method, subType:string, attr:opt{opt, string}, desc:"greaterThanOrEqual", "lessThan", "equal". This node will be returned only when the node occurrences is returned-->greaterThanOrEqual
    </occurrencesSearchType>
  </StrangerLibFDSearchDescription>
  <isSupportOccurrencesData>
    <!--ro, opt, bool, whether to support directly getting the picture frequency information, desc:corresponding URL: /ISAPI/Intelligent/FDLib/<FDID>/picture/<PID>/occurrenceData?format=json-->true
  </isSupportOccurrencesData>
```

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```
  <isSupportFaceLibFormat>
    <!--ro, opt, bool, whether to support formating face picture library, desc:corresponding URL: /ISAPI/Intelligent/FDLib/format?format=json-->true
  </isSupportFaceLibFormat>
  <isSupportSurplusCapacityALL>
    <!--ro, opt, bool, whether to support searching for the capacity of all face picture libraries, desc:corresponding URL:
/ISAPI/Intelligent/FDLib/surplusCapacity?format=json-->true
  </isSupportSurplusCapacityALL>
  <AsyncImportCap>
    <!--ro, opt, object, the size range of a single data file-->
    <singleFileMaxFaceNum max="100">
      <!--ro, opt, int, the maximum number of face pictures of a single file, attr:max{opt, string}-->100
    </singleFileMaxFaceNum>
    <singleFileMaxSize max="100">
      <!--ro, opt, int, the size range of a single data file, unit:KB, attr:max{opt, int}, desc:unit:kb-->100
    </singleFileMaxSize>
  </AsyncImportCap>
  <isSupportAsyncImportData>
    <!--ro, opt, bool, whether to support asynchronously importing the data to the face picture library, desc:corresponding URL:
/ISAPI/Intelligent/FDLib/asyncImportDatas/capabilities?format=json-->true
  </isSupportAsyncImportData>
  <isSupportTaskStatusSearch>
    <!--ro, opt, bool, whether to support searching for the status of asynchronous import task in the face picture library, desc:corresponding URL:
/ISAPI/Intelligent/FDLib/asyncImportDatasTasks?format=json-->true
  </isSupportTaskStatusSearch>
  <SyncDeletePicNum max="10">
    <!--ro, opt, int, number of face pictures allowed in a single synchronous deleting, attr:max{opt, int}-->1
  </SyncDeletePicNum>
  <isSupportPICCertification>
    <!--ro, opt, bool, whether to support cloud storage, desc:corresponding URL: /ISAPI/Intelligent/FDLib/PICCertification?format=json-->true
  </isSupportPICCertification>
  <isSupportFaceScore>
    <!--ro, opt, bool, whether to support searching for the rating details of face pictures, desc:corresponding URL:
/ISAPI/Intelligent/FDLib/<FDID>/picture/<PID>/faceScore?format=json-->true
  </isSupportFaceScore>
</FDLibCap>
```

16.7.5.22 Get the progress of packaging alarm information

Request URL

GET /ISAPI/Intelligent/FDLib/FCSearch/dataPackage/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID, and the length range is 1 to 64 characters.

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL, range:[1,256]*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "progress":  22,
    /*ro, opt, int, packaging progress, range:[0,100]*/
    "URL":  "test"
    /*ro, opt, string, storage URL generated by device after packaging, range:[1,128], desc:you can download the data files via the URL*/
}
```

16.7.5.23 Start packaging and exporting the searched information about face picture comparison alarm

Request URL

POST /ISAPI/Intelligent/FDLib/FCSearch/dataPackage?format=json

Query Parameter

None

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Request Message

```
{
    "searchID":  "test",
    /*opt, string, search ID, desc:it is unchanged when the search condition remains the same*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, range:[0,4294967295], desc:in a single search, if you cannot get all the records
in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResultNumber":  100,
    /*req, int, the maximum records to be found, range:[1,4294967295]*/
    "maxResults":  1,
    /*req, int, the maximum results per search, range:[1,100]*/
    "FDID":  "aa,bb",
    /*req, string, face picture library ID, range:[0,63], desc:"-1" (all libraries). If there are multiple libraries, you can separate them by commas*/
    "cameraID":  "aa,bb",
    /*opt, string, camera ID (channel ID), range:[0,4095], desc:"-1" (all cameras). If there are multiple cameras, you can separate each one by commas*/
    "snapStartTime":  "1970-01-01T00:00:00Z",
    /*opt, datetime, capture start time*/
    "snapEndTime":  "1970-01-01T01:00:00Z",
    /*opt, datetime, capture end time*/
    "maxSimilarity":  0.99,
    /*opt, float, the maximum similarity, range:[0.00,1.00]*/
    "minSimilarity":  0.99,
    /*opt, float, the minimum similarity, range:[0.00,1.00]*/
    "sortType":  "similarity",
    /*opt, enum, sorting type, subType:string, desc:"similarity" (by similarity), "time" (by time), "attendanceNumber" (by check-in counts)*/
    "Attendance": {
    /*opt, object, attendance information*/
        "Schedules": [
        /*opt, array, attendance time, subType:object*/
            {
                "startTime":  "00:00:00+08:00",
                /*opt, time, start time*/
                "endTime":  "00:00:00+08:00",
                /*opt, time, end time*/
                "dayOfWeek": [1, 2]
                /*opt, array, days of the week, subType:int, dep:and,{$.Attendance.type,eq,attendance}, desc:from Monday to Sunday, multiple selections are
supported*/
            }
        ],
        "MatchFace": [
        /*opt, array, matched face information, subType:object*/
            {
                "FDID":  "test",
                /*opt, string, face picture library ID, range:[0,63]*/
                "PID":  "test"
                /*opt, string, picture ID, range:[0,64]*/
            }
        ],
        "type":  "check-in"
        /*opt, enum, mode switch, subType:string, desc:"attendance", "check-in"*/
    },
    "listType":  "blackList"
    /*opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL, range:[1,256]*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID":  "test"
    /*ro, opt, string, packaging task ID, range:[0,32], desc:it can be used for packaging progress search*/
}
```

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16.7.5.24 Search face picture data

Request URL

POST /ISAPI/Intelligent/FDLib/FDSearch?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, initial position of search result list, desc:if it cannot get all records in one time searching,you can search the records followed specified position for next search. For video intercom devices,this field can only be set to 0 as the picture will be returned along with the message-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, maximum number of records for single searching-->1
  </maxResults>
  <FDID>
    <!--opt, string, face picture library ID-->test
  </FDID>
  <startTime>
    <!--opt, date, start time-->1970-01-01+08:00
  </startTime>
  <endTime>
    <!--opt, date, end time-->1970-01-01+08:00
  </endTime>
  <name>
    <!--opt, string, name, range:[1,32], desc:name-->test
  </name>
  <sex>
    <!--opt, enum, gender, subType:string, desc:gender-->male
  </sex>
  <province>
    <!--opt, string-->11
  </province>
  <city>
    <!--opt, string, city, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes-->11
  </city>
  <certificateType>
    <!--opt, enum, certificate type, subType:string, desc:the maximum size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify card,passport,other-->officerID
  </certificateType>
  <certificateNumber>
    <!--opt, string, certificate No., range:[8,32], desc:certificate No.,the maximum size is 32 bytes-->test
  </certificateNumber>
  <phoneNumber>
    <!--opt, string, range:[1,64]-->test
  </phoneNumber>
  <FaceModeList>
    <!--opt, array, subType:object-->
    <FaceMode>
      <!--opt, object-->
      <ModeInfo>
        <!--opt, object-->
        <similarity>
          <!--opt, float, similarity, range:[0.0,100.0]-->0.0
        </similarity>
        <modeData>
          <!--opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method-->test
        </modeData>
      </ModeInfo>
    </FaceMode>
  </FaceModeList>
  <modelStatus>
    <!--opt, enum, subType:string-->modeling
  </modelStatus>
  <modelingStatus>
    <!--opt, enum, subType:string-->success
  </modelingStatus>
  <customFaceLibID>
    <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
  </customFaceLibID>
  <sortord>
```

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```
    <!--opt, enum, subType:string-->time
  </sortord>
  <order>
    <!--opt, enum, subType:string-->DESC
  </order>
  <StrangerLibFaceAppendData>
    <!--opt, object-->
    <PersonInfoExtendList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <PersonInfoExtend>
        <!--opt, object, person extended information-->
        <id>
          <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
        </id>
        <enable>
          <!--req, bool, enable person extended information or not-->true
        </enable>
        <name>
          <!--opt, string, person tag extended information name, desc:person tag extended information name-->test
        </name>
        <value>
          <!--opt, string, person tag extended information, desc:person tag extended information-->test
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
  </StrangerLibFaceAppendData>
  <OccurrencesInfo>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <occurrences>
      <!--opt, int-->1
    </occurrences>
    <occurrencesSearchType>
      <!--opt, enum, subType:string-->greaterThanOrEqual
    </occurrencesSearchType>
  </OccurrencesInfo>
  <faceScore>
    <!--opt, int, face score (face picture quality), range:[-1,100], desc:face score (face picture quality)-->1
  </faceScore>
  <FDIDList>
    <!--opt, object-->
    <FDID>
      <!--req, string, range:[1,64]-->test
    </FDID>
  </FDIDList>
  <faceScoreMax>
    <!--opt, int, range:[-1,100]-->0
  </faceScoreMax>
  <pictureStatus>
    <!--opt, enum, subType:string-->normal
  </pictureStatus>
  <importStartTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </importStartTime>
  <importEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </importEndTime>
  <faceContrastSuccessStartTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </faceContrastSuccessStartTime>
  <faceContrastSuccessEndTime>
    <!--opt, datetime-->1970-01-01T00:00:00+08:00
  </faceContrastSuccessEndTime>
  <userType>
    <!--opt, enum, subType:string-->normal
  </userType>
  <associatedPlateType>
    <!--opt, enum, subType:string-->greenChannel
  </associatedPlateType>
</FDSearchDescription>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, search result, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, search ID, desc:it is used to check whether the current search requester is the same as the previous one. If they are the same, the search record will be stored in the device to speed up the next search-->test
  </searchID>
  <responseStatus>
    <!--ro, req, enum, search result, subType:string, desc:search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data).-->OK
  </responseStatusStrg>
```

```
    <numOfMatches>
      <!--ro, req, int, number of results returned this time-->1
    </numOfMatches>
    <totalMatches>
      <!--ro, req, int, total number of matched results-->1
    </totalMatches>
    <progress>
      <!--ro, opt, int, range:[0,100]-->1
    </progress>
    <MatchList>
      <!--ro, opt, array, searched matched data list, subType:object-->
      <MatchElement>
        <!--ro, opt, object-->
        <FDID>
          <!--ro, opt, string, face picture library ID-->test
        </FDID>
        <thresholdValue>
          <!--ro, opt, int, range:[0,100]-->1
        </thresholdValue>
        <bornTime>
          <!--ro, opt, date, birthday of person in the face picture-->1970-01-01+08:00
        </bornTime>
        <name>
          <!--ro, req, string, name, range:[1,32], desc:name-->test
        </name>
        <sex>
          <!--ro, opt, enum, gender, subType:string, desc:male; female;-->male
        </sex>
        <province>
          <!--ro, opt, string-->11
        </province>
        <city>
          <!--ro, opt, string, city, desc:city code of birth for the person in the face picture,the maximum size is 32 bytes-->11
        </city>
        <certificateType>
          <!--ro, opt, enum, certificate type, subType:string, desc:the max. size is 10 bytes,certificate type: "officerID"-officer ID,"ID"-identify
card,passport,other-->officerID
        </certificateType>
        <certificateNumber>
          <!--ro, opt, string, certificate No., range:[8,32], desc:certificate No.,the max. size is 32 bytes-->test
        </certificateNumber>
        <picURL>
          <!--ro, req, string-->test
        </picURL>
        <PID>
          <!--ro, req, string, picture ID-->test
        </PID>
        <PersonInfoExtendList>
          <!--ro, opt, array, subType:object-->
          <PersonInfoExtend>
            <!--ro, opt, object, person extended information-->
            <id>
              <!--ro, req, int, serial No. of person custom extension information-->1
            </id>
            <enable>
              <!--ro, req, bool, enable person extended information or not-->true
            </enable>
            <name>
              <!--ro, opt, string, person tag extended information name, desc:person tag extended information name-->test
            </name>
            <value>
              <!--ro, opt, string, person tag extended information, desc:person tag extended information-->test
            </value>
          </PersonInfoExtend>
        </PersonInfoExtendList>
        <similarity>
          <!--ro, opt, float, similarity, range:[0.0,100.0]-->0.0
        </similarity>
        <ModelingStatus>
          <!--ro, opt, object-->
          <FDID>
            <!--ro, opt, string, face picture library ID-->test
          </FDID>
          <PID>
            <!--ro, opt, string, picture ID-->test
          </PID>
          <name>
            <!--ro, req, string, picture name-->test
          </name>
          <status>
            <!--ro, req, enum, subType:string-->success
          </status>
          <reason>
            <!--ro, opt, enum, subType:string-->noface
          </reason>
          <customHumanID>
            <!--ro, opt, string-->123456789012345678901234567890l2
          </customHumanID>
          <customFaceLibID>
            <!--ro, opt, string-->123456789012345678901234567890l2
          </customFaceLibID>
          <modeData>
            <!--ro, opt, string, target modeling data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64
method-->test
```

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```
        </modeData>
      </ModelingStatus>
      <isNoSaveFDPicture>
        <!--ro, opt, bool-->true
      </isNoSaveFDPicture>
      <humanId>
        <!--ro, opt, string-->test
      </humanId>
      <modelStatus>
        <!--ro, opt, enum, subType:string-->modeling
      </modelStatus>
      <FMCount>
        <!--ro, opt, int-->1
      </FMCount>
      <importTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </importTime>
      <UUPID>
        <!--ro, opt, string-->test
      </UUPID>
      <occurrences>
        <!--ro, opt, int-->1
      </occurrences>
      <IMSIList>
        <!--ro, opt, array, subType:object-->
        <IMSIItem>
          <!--ro, opt, object-->
          <IMSI>
            <!--ro, opt, string-->123456789012345
          </IMSI>
          <count>
            <!--ro, opt, int-->10
          </count>
          <lastFMTime>
            <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
          </lastFMTime>
        </IMSIItem>
      </IMSIList>
      <lastFMTime>
        <!--ro, opt, datetime-->1970-01-01T00:00:00+08:00
      </lastFMTime>
      <faceScore>
        <!--ro, opt, int, range:[-1,100]-->1
      </faceScore>
      <pictureStatus>
        <!--ro, opt, enum, subType:string-->normal
      </pictureStatus>
      <userType>
        <!--ro, opt, enum, subType:string-->normal
      </userType>
      <associatedPlateType>
        <!--ro, opt, enum, subType:string-->greenChannel
      </associatedPlateType>
      <contrastStatus>
        <!--ro, opt, enum, subType:string-->contrast
      </contrastStatus>
    </MatchElement>
  </MatchList>
</FDSearchResult>
```

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16.7.5.25 Format all face picture libraries (including normal and private libraries)

Request URL

PUT /ISAPI/Intelligent/FDLib/format?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16.7.5.26 Specify a face picture library to start model the face pictures in a batch

Request URL

GET /ISAPI/Intelligent/FDLib/manualModeling?range=<range>&FDID=<FDID>&FDType=<FDType>&identityKey=<identityKey>&pictureSaveMode=<pictureSaveMode>

Query Parameter

Parameter Name	Parameter Type	Description
range	string	Control data range, and when the range is unmodeled, it indicates that unmodeled image data in the face database should be uniformly modeled; when the range is all, it indicates that all images stored in the device (regardless of whether they have modeling data) need to be remodeled, for example, after a device upgrade (algorithm library upgrade), the modeling data needs to be updated.
FDID	string	Face picture library ID
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
identityKey	string	When importing a face library, an error "faceLibraryNumError" occurs to prompt the user whether to overwrite. If the overwrite import mode is selected, and the face library has reached its maximum capacity, the subsequently imported faces will overwrite the earliest imported face information.
pictureSaveMode	string	When pictureSaveMode is noSave, it indicates that the uploaded original image will not be saved. When isSupportNoSaveUploadPicture is returned in /ISAPI/Intelligent/channels//faceContrast/capabilities, it represents the operation of not saving the uploaded original image. After the device modeling is successful, the uploaded original image will be deleted.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <id>
    <!--ro, opt, string, task ID, desc:task ID-->test
  </id>
</ResponseStatus>
```

16.7.5.27 Search control results of discrete modeling

Request URL

POST /ISAPI/Intelligent/FDLib/manualModelingRangeTask?identityKey=<identityKey>&pictureSaveMode=<pictureSaveMode>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
pictureSaveMode	string	When pictureSaveMode is noSave, it indicates that the uploaded original image will not be saved. When isSupportNoSaveUploadPicture is returned in /ISAPI/Intelligent/channels//faceContrast/capabilities, it represents the operation of not saving the uploaded original image. After the device modeling is successful, the uploaded original image will be deleted.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingRangeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <ModelingRange>
    <!--opt, object-->
    <FDID>
      <!--req, string, face picture library ID-->test
    </FDID>
    <PID>
      <!--req, string, picture ID-->test
    </PID>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
    </customFaceLibID>
  </ModelingRange>
</ModelingRangeList>
```

Response Message

dracrun@gmail.com DEFEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.7.5.28 Get the modeling status of a face picture in a specific face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/modelingStatus?FDID=<FDID>&PID=<pictureID>&identityKey=<identityKey>&FDType=<FDType>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	Face picture library ID
pictureID	string	Picture ID
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <FDID>
    <!--ro, opt, string-->test
  </FDID>
  <PID>
    <!--ro, opt, string-->test
  </PID>
  <name>
    <!--ro, req, string-->test
  </name>
  <status>
    <!--ro, req, enum, subType:string-->success
  </status>
  <reason>
    <!--ro, opt, enum, subType:string-->noface
  </reason>
  <customHumanID>
    <!--ro, opt, string-->123456789012345678901234567890123456
  </customHumanID>
  <customFaceLibID>
    <!--ro, opt, string-->123456789012345678901234567890123456
  </customFaceLibID>
  <modeData>
    <!--ro, opt, string-->test
  </modeData>
</ModelingStatus>
```

16.7.5.29 Search for face pictures by modeling status

Request URL

POST /ISAPI/Intelligent/FDLib/modelingStatus?status=<status>&identityKey=<identityKey>&FDType=<FDType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
status	string	Modeling status: "success" (search for successfully modeled data), "failed" (search for failed modeling data), "none" (search for unmodeled data).
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.
FDType	string	Face picture library type. When FDType=custom is not applied, the FDID in this URL is the same as returned by URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries. When FDType=custom is applied, the FDID in this URL is the same as applied in the request of URL (POST /ISAPI/Intelligent/FDLib) for creating face picture libraries.
terminalNo	string	Terminal No., which is used for information release terminal and can be obtained from the TerminalInfo->id field in the response message of POST /ISAPI/Publish/TerminalMgr/terminalSearch.

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FDModelingStatusSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, undefined, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <searchResultPosition>
    <!--req, int, the start position of the search result in the result list-->1
  </searchResultPosition>
  <maxResults>
    <!--req, int, the maximum number of search results this time-->1
  </maxResults>
  <CondList>
    <!--opt, array, subType:object-->
    <Cond>
      <!--opt, object-->
      <FDID>
        <!--opt, string-->test
      </FDID>
    </Cond>
  </CondList>
  <taskID>
    <!--opt, string, task ID, desc:taskID and CondList cannot exist at the same time, and taskID is in prior-->test
  </taskID>
</FDModelingStatusSearchDescription>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDModelingStatusSearchResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, undefined, attr:version{req, string, protocolVersion}-->
  <searchID>
    <!--ro, req, string, unique ID of search records, desc:it is used to confirm the upper-level platform or system. If the platform or the system is the same one during two searching, the search history will be saved in the memory to speed up next searching-->test
  </searchID>
  <responseStatus>
    <!--ro, req, bool, search result-->true
  </responseStatus>
  <responseStatusStrg>
    <!--ro, req, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched data)-->OK
  </responseStatusStrg>
  <numOfMatches>
    <!--ro, req, int, number of results returned this time-->1
  </numOfMatches>
  <totalMatches>
    <!--ro, req, int, total number of matched results-->1
  </totalMatches>
  <ModelingStatusList>
    <!--ro, opt, array, subType:object-->
    <ModelingStatus>
      <!--ro, opt, object-->
      <FDID>
        <!--ro, opt, string-->test
      </FDID>
      <PID>
        <!--ro, opt, string-->test
      </PID>
      <name>
        <!--ro, req, string-->test
      </name>
      <status>
        <!--ro, req, enum, subType:string-->success
      </status>
      <reason>
        <!--ro, opt, enum, subType:string-->noface
      </reason>
      <customHumanID>
        <!--ro, opt, string-->123456789012345678901234567890 12
      </customHumanID>
      <customFaceLibID>
        <!--ro, opt, string-->123456789012345678901234567890 12
      </customFaceLibID>
      <modeData>
        <!--ro, opt, string-->test
      </modeData>
    </ModelingStatus>
  </ModelingStatusList>
</FDModelingStatusSearchResult>
```

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16.7.5.30 Search for all modeling statuses of the device

Request URL

POST /ISAPI/Intelligent/FDLib/modelingStatusRange?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64. When "security" is 1, it indicates that the node "identityKey" in the URL are encrypted in AES128 CBC mode; when "security" is 2, it indicates that the node "identityKey" in the URL are encrypted in AES256 CBC mode. The node "identityKey" stands for the identity key of the operation platform. The device can verify the operator's identity by checking if this parameter exists in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingRangeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <ModelingRange>
    <!--opt, object-->
    <FDID>
      <!--req, string-->test
    </FDID>
    <PID>
      <!--req, string-->test
    </PID>
    <customFaceLibID>
      <!--opt, string-->test
    </customFaceLibID>
  </ModelingRange>
</ModelingRangeList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ModelingStatusList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <ModelingStatus>
    <!--ro, opt, object-->
    <FDID>
      <!--ro, opt, string-->test
    </FDID>
    <PID>
      <!--ro, opt, string-->test
    </PID>
    <name>
      <!--ro, req, string-->test
    </name>
    <status>
      <!--ro, req, enum, subType:string-->success
    </status>
    <reason>
      <!--ro, opt, enum, subType:string-->noface
    </reason>
    <customHumanID>
      <!--ro, opt, string-->123456789012345678901234567890123456789012
    </customHumanID>
    <customFaceLibID>
      <!--ro, opt, string-->123456789012345678901234567890123456789012
    </customFaceLibID>
    <modeData>
      <!--ro, opt, string-->test
    </modeData>
  </ModelingStatus>
</ModelingStatusList>
```

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16.7.5.31 Import face pictures and the additional information to a specific face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/pictureUpload?identityKey=<identityKey>&type=<type>&dataType=<dataType>&pictureSaveMode=<pictureSaveMode>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--
type	string	--
dataType	string	--
pictureSaveMode	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PictureUploadData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, picture information, attr:version{req, string, protocolVersion}-->
  <FDID>
    <!--req, string, a unique ID of face comparison picture library-->test
  </FDID>
  <customFaceLibID>
    <!--opt, string, custom face picture library ID, desc:custom face picture library ID-->test
  </customFaceLibID>
  <FaceAppendData>
    <!--opt, object, face picture information-->
    <bornTime>
      <!--opt, string, date of birth-->test
    </bornTime>
    <name>
      <!--req, string, name, range:[1,32]-->test
    </name>
    <sex>
      <!--opt, enum, gender, subType:string, desc:"male", "female", "unknown"-->male
    </sex>
    <province>
      <!--opt, string, province, desc:see the data dictionary for details-->11
    </province>
    <city>
      <!--opt, string, city, desc:see the data dictionary for details-->11
    </city>
    <certificateType>
      <!--opt, enum, certificate type, subType:string, desc:"officerID"-officer ID,"ID"-identify card,"passportID"-passport,"other"-other certificate--
>officerID
    </certificateType>
    <certificateNumber>
      <!--opt, string, certificate No., range:[8,32], desc:certificate No.-->test
    </certificateNumber>
    <PersonInfoExtendList>
      <!--opt, array, custom extension information of person, subType:object-->
      <PersonInfoExtend>
        <!--opt, object, person information-->
        <id>
          <!--req, int, serial No. of person custom extension information, desc:serial No. of person custom extension information-->1
        </id>
        <enable>
          <!--req, bool, enable person extended information or not-->true
        </enable>
        <name>
          <!--opt, string, person tag extended information name-->test
        </name>
        <value>
          <!--opt, string, person tag extended information-->test
        </value>
      </PersonInfoExtend>
    </PersonInfoExtendList>
    <caseInfo>
      <!--opt, string, case information, range:[1,64]-->test
    </caseInfo>
    <customHumanID>
      <!--opt, string, custom person ID, range:[1,64]-->test
    </customHumanID>
    <phoneNumber>
      <!--opt, string, phone number, range:[1,64]-->test
    </phoneNumber>
    <RegionCoordinatesList>
      <!--opt, array, face picture target, subType:object, range:[0,4], desc:face picture target-->
      <RegionCoordinates>
        <!--opt, object, coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <modeldata>
```

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DETEKTOR

```
      <!--opt, string, modeling data, desc:modeling data-->test
    </modeldata>
    <IMSIList>
      <!--opt, array, subType:object-->
      <IMSI>
        <!--opt, string, range:[1,15]-->123456789012345
      </IMSI>
    </IMSIList>
    <userType>
      <!--opt, enum, subType:string-->normal
    </userType>
    <associatedPlateType>
      <!--opt, enum, subType:string-->greenChannel
    </associatedPlateType>
  </FaceAppendData>
  <picURL>
    <!--opt, string, picture URL-->test
  </picURL>
</PictureUploadData>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
PictureUploadData	[Message content]	application/xml	--	--	--
importImage	[Binary picture data]	image/jpeg		file.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MaskInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, information, attr:version{req, string, protocolVersion}, desc:information-->
  <PID>
    <!--ro, opt, string, picture ID, desc:picture ID-->test
  </PID>
  <picURL>
    <!--ro, opt, picurl, picture URL generated by device after importing pictures, desc:picture URL generated by device after importing pictures--
>http://127.0.0.1
  </picURL>
  <isNoSavePDPicture>
    <!--ro, opt, bool-->true
  </isNoSavePDPicture>
  <FaceGradingDetails>
    <!--ro, opt, object, face score (face picture quality), desc:face score (face picture quality)-->
    <featureConfidence>
      <!--ro, opt, float, [0,1], range:[0.000,1.000], desc:[0,1]-->0.000
    </featureConfidence>
    <faceRect>
      <!--ro, opt, enum, rectangle frame for human face, subType:int, desc:rectangle frame for human face-->1
    </faceRect>
    <pupillaryDistance>
      <!--ro, opt, int, pupil distance, desc:pupil distance-->1
    </pupillaryDistance>
    <colorConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </colorConfidence>
    <grayScale>
      <!--ro, opt, int, grayscale value, range:[0,255]-->1
    </grayScale>
    <grayMean>
      <!--ro, opt, int, range:[0,255]-->1
    </grayMean>
    <grayVariance>
      <!--ro, opt, int, range:[0,128]-->1
    </grayVariance>
    <clearityScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </clearityScore>
    <pitchAngle>
      <!--ro, opt, float, range:[-90.000,90.000]-->0.000
    </pitchAngle>
    <yawAngle>
      <!--ro, opt, float, range:[-90.000,90.000]-->0.000
    </yawAngle>
    <rollAngle>
      <!--ro, opt, float, face roll angle, range:[-90.000,90.000], desc:face roll angle-->0.000
    </rollAngle>
    <poseConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </poseConfidence>
    <frontalScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </frontalScore>
    <visibleScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </visibleScore>
    <totalScore>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </totalScore>
    <detectConfidence>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </detectConfidence>
    <invisibleClass>
      <!--ro, opt, array, subType:object, range:[1,8]-->
    </invisibleClass>
    <faceClass>
      <!--ro, opt, enum, face picture type, subType:int, desc:face picture type-->0
    </faceClass>
    <illuminationClass>
      <!--ro, opt, enum, subType:int-->0
    </illuminationClass>
    <leftEyeOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </leftEyeOpen>
    <rightEyeOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </rightEyeOpen>
    <mouthOpen>
      <!--ro, opt, float, range:[0.000,1.000]-->0.000
    </mouthOpen>
    <pictureStatus>
      <!--ro, opt, enum, subType:string-->normal
    </pictureStatus>
  </FaceGradingDetails>
</MaskInfo>
```

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16.7.5.32 Get the capability of configuring face picture pre-classification parameters.

Request URL

GET /ISAPI/Intelligent/FDLib/presort/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "presortInfo": {
    /*ro, req, object, face picture pre-classification parameters*/
        "presortSpeedupTimes":  "1,20,50"
        /*ro, req, string, value range of times of speeding up pre-classification*/
    }
}
```

16.7.5.33 Set presorting parameters of human face pictures

Request URL

PUT /ISAPI/Intelligent/FDLib/presort?format=json

Query Parameter

None

Request Message

```
{
    "presortInfo": {
    /*req, object*/
        "presortSpeedupTimes":  1
        /*opt, enum, 1 (1 days), 15 (15 days), 50 (50 days), subType:int, desc:1 (1 days), 15 (15 days), 50 (50 days)*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16.7.5.34 Get presorting parameters of face pictures

Request URL

GET /ISAPI/Intelligent/FDLib/presort?format=json

Query Parameter

None

Request Message

None

Response Message

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```
{
    "presortInfo": {
    /*ro, req, object*/
        "presortSpeedupTimes":  1
        /*ro, opt, enum, subType:int*/
    }
}
```

16.7.5.35 Get the capacity of face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/surplusCapacity?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "SurplusCapacity": {
    /*ro, req, object*/
        "FDPicturesNumber":  100,
        /*ro, req, int*/
        "capacity":  100
        /*ro, req, int*/
    }
}
```

16.7.5.36 Set the face picture library parameters

Request URL

PUT /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, face picture library information list, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibBaseCfg>
    <!--opt, object, face picture library information-->
    <id>
      <!--opt, int, range:[1,15]-->1
    </id>
    <FDID>
      <!--opt, string, unique ID of face picture library-->test
    </FDID>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, additional information of face picture library, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <faceLibType>
      <!--ro, opt, enum, face picture library type, subType:string, desc:"ordinary" (normal library), "private" (private library), "stranger" (stranger
library), "allowable" (allowlist library), "privacyMask" (privacy mask library)-->allowable
    </faceLibType>
    <autoUpdata>
      <!--opt, bool-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object-->
      <judgementCycleType>
        <!--req, enum, subType:string-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--ro, req, enum, subType:string-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, range:[0,99999]-->1
      </FMCount>
      <backgoundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgoundPicThreshold>
      <judgementCycle>
        <!--opt, int, range:[1,365], unit:d, dep:and,{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
    </Age>
    <FaceCounting>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, subType:string-->noUpload
      </uploadType>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, int-->1
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool-->true
    </customNoConfig>
    <uploadID>
      <!--opt, string, range:[1,128], dep:and,{$.FDLibBaseCfglist.FDLibBaseCfg.autoUpdata,eq,true}-->test
    </uploadID>
  </FDLibBaseCfg>
</FDLibBaseCfgList>
```

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DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.7.5.37 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64 characters. This parameter stands for the identity key of the operation platform. The device can know the operator's identity by checking this parameter in the operation URL (only available for iVMS-8200 and Hik-Central platform). The face picture library managed by this key can only be operated on remotely-managed platform.

Request Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<CreateFDLibList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, create the list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <CreateFDLib>
    <!--opt, object, create a face picture library-->
    <id>
      <!--opt, int, sub item No. in the list; it increases from 1, range:[1,15]-->1
    </id>
    <name>
      <!--req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--opt, string, custom information, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <autoUpdata>
      <!--opt, bool, whether to enable auto update of background picture in face picture library-->true
    </autoUpdata>
    <qualityThreshold>
      <!--opt, int, picture quality threshold, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--opt, int, inbound similarity threshold, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--opt, object, the aggregate aging parameters configuration of face picture library; it is valid when autoUpdata is true-->
      <judgementCycleType>
        <!--req, enum, the time type for aging judgement cycle of the aggregate data, subType:string, desc:"day", "week", "month", "duration" (judge
according to actual duration)-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--req, enum, statistics cycle type of clustering matches in face picture library, subType:string, desc:"day", "week", "month"-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--req, int, statistics cycle of face picture matches, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--req, int, min. number of face picture matches within statistics cycle, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--opt, bool, whether to enable total number of face picture matches-->true
      </FMCountEnabled>
      <FMCount>
        <!--opt, int, total number of face picture matches, range:[0,99999]-->1
      </FMCount>
      <backgroundPicThreshold>
        <!--opt, int, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--opt, int, evaluation cycle range of clustering data aging in face picture library, range:[1,365], unit:d, dep:and,
{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->60
      </judgementCycle>
      <agingExecDataTime>
        <!--opt, time-->00:00:00+08:00
      </agingExecDataTime>
      <automaticClearEnabled>
        <!--opt, bool-->true
      </automaticClearEnabled>
    </Age>
    <FaceCounting>
      <!--opt, object, the configuration for customer traffic (face) deduplication of face picture comparison-->
      <enabled>
        <!--req, bool, whether to enable customer traffic (face) deduplication-->true
      </enabled>
      <uploadType>
        <!--req, enum, uploading mode after the customer traffic (face) deduplication is enabled, subType:string, desc:"out_repetition" (uploading outside
the deduplication period); "noUpload" (not uploading outside the deduplication period)-->noUpload
      </uploadType>
    </FaceCounting>
    <faceLibType>
      <!--opt, enum, face picture library type, subType:string, desc:if this node is not returned, a normal library is created by default; "ordinary"
(normal library); "privacyMask" (privacy mask library)-->ordinary
    </faceLibType>
  </CreateFDLib>
</CreateFDLibList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, create the result list of face picture library, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibInfo>
    <!--ro, opt, object, face picture library information-->
    <id>
      <!--ro, req, int, sub item No. in the list; it increases from 1 and corresponds to the id of CreateFDLibList-->1
    </id>
    <name>
      <!--ro, req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <FDID>
      <!--ro, req, string, returned face picture library ID when it created, the library ID of the same type is unique, range:[1,64]-->test
    </FDID>
    <statusCode>
      <!--ro, opt, enum, status code; it is returned when creating a face picture library failed, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3
(Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->1
    </statusCode>
    <subStatusCode>
      <!--ro, opt, string, sub status code; it is returned when creating a face picture library failed-->existSameNameFacelib
    </subStatusCode>
  </FDLibInfo>
</FDLibInfoList>
```

16.7.5.38 Get the information, including library ID, library type, name, and custom information, of all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib?identityKey=<identityKey>

Query Parameter

Parameter Name	Parameter Type	Description
identityKey	string	The maximum length is 64 characters.

Request Message

None

Response Message

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DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<FDLibBaseCfgList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FDLibBaseCfg>
    <!--ro, opt, object, face picture library information-->
    <id>
      <!--ro, opt, int, range:[1,15]-->1
    </id>
    <FDID>
      <!--ro, opt, string, face picture library ID-->test
    </FDID>
    <name>
      <!--ro, req, string, face picture library name, range:[1,64]-->faceLib1
    </name>
    <thresholdValue>
      <!--ro, opt, int, detection threshold, range:[0,100]-->50
    </thresholdValue>
    <customInfo>
      <!--ro, opt, string, custom information, range:[1,128]-->test
    </customInfo>
    <customFaceLibID>
      <!--ro, opt, string, custom face picture library ID, range:[1,64]-->test
    </customFaceLibID>
    <faceLibType>
      <!--ro, opt, enum, face picture library type, subType:string, desc:face picture library type-->ordinary
    </faceLibType>
    <autoUpdata>
      <!--ro, opt, bool-->true
    </autoUpdata>
    <qualityThreshold>
      <!--ro, opt, int, range:[0,100]-->50
    </qualityThreshold>
    <importFDThreshold>
      <!--ro, opt, int, range:[0,100]-->40
    </importFDThreshold>
    <Age>
      <!--ro, opt, object-->
      <judgementCycleType>
        <!--ro, req, enum, subType:string-->day
      </judgementCycleType>
      <statisticalCycleType>
        <!--ro, req, enum, "day", "week", "month", subType:string, desc:"day", "week", "month"-->month
      </statisticalCycleType>
      <statisticalCycle>
        <!--ro, req, int, range:[0,99999]-->1
      </statisticalCycle>
      <minFMCount>
        <!--ro, req, int, range:[0,99999]-->1
      </minFMCount>
      <FMCountEnabled>
        <!--ro, opt, bool-->true
      </FMCountEnabled>
      <FMCount>
        <!--ro, opt, int, range:[0,99999]-->1
      </FMCount>
      <backgroundPicThreshold>
        <!--ro, opt, int, range:[1,100]-->90
      </backgroundPicThreshold>
      <judgementCycle>
        <!--ro, opt, int, range:[1,365], unit:d, dep:and,{$.FDLibCap.CreateFDLibList[*].CreateFDLib.Age.judgementCycleType,eq,duration}-->30
      </judgementCycle>
    </Age>
    <FaceCounting>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <uploadType>
        <!--ro, req, enum, subType:string-->noUpload
      </uploadType>
      <faceSimilarity>
        <!--ro, opt, int, range:[0,100]-->40
      </faceSimilarity>
    </FaceCounting>
    <faceMonitor>
      <!--ro, opt, bool-->true
    </faceMonitor>
    <monitorFaceNum>
      <!--ro, opt, int-->1
    </monitorFaceNum>
    <customNoConfig>
      <!--ro, opt, bool-->true
    </customNoConfig>
    <uploadID>
      <!--ro, opt, string, range:[1,128], dep:and,{$.FDLibBaseCfglist.FDLibBaseCfg.autoUpdata,eq,true}-->test
    </uploadID>
  </FDLibBaseCfg>
</FDLibBaseCfgList>
```

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DETEKTOR

16.7.5.39 Export the package of face capture library attributes

Request URL

POST /ISAPI/SDT/Face/captureSearch/dataPackage

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int*/
    "maxResults":  100,
    /*req, int*/
    "startTime":  "2004-05-03T17:30:08+08:00",
    /*opt, datetime*/
    "endTime":  "2004-05-03T17:30:08+08:00",
    /*opt, datetime*/
    "choiceChannel": [
    /*opt, array, subType:object*/
        {
            "channelID":  "test"
            /*opt, string*/
        }
    ],
    "ageGroup":  "any",
    /*opt, enum, subType:string*/
    "gender":  "male",
    /*opt, enum, subType:string*/
    "glasses":  "yes",
    /*opt, enum, subType:string*/
    "smile":  "yes",
    /*opt, enum, subType:string*/
    "isInLibrary":  "yes",
    /*opt, enum, subType:string*/
    "thermometryUnit":  "celsius",
    /*opt, enum, subType:string*/
    "currTemperature":  38.0,
    /*opt, float, range:[-20.0,150.0]*/
    "temperatureCondition":  "normal",
    /*opt, enum, subType:string*/
    "FPID":  " "
    /*opt, string*/
}
```

Response Message

Binary Data

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DETEKTOR

16.7.5.40 Delete the task according to task ID

Request URL

DELETE /ISAPI/SDT/Management/Task/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "test",
    /*ro, opt, string, status description, desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "test",
    /*ro, opt, string, sub status code, desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

16.7.5.41 Pause the task according to task ID

Request URL

GET /ISAPI/SDT/Management/Task/pause/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field must be returned when the value of statusCode is not 1*/
}
```

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16.7.5.42 Obtain the task progress

Request URL

GET /ISAPI/SDT/Management/Task/progress/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	--

Request Message

None

Response Message

```
{
    "progress": {
    /*ro, opt, object, task progress*/
        "value":  22
        /*ro, req, int, value of task searching progress, range:[0,100]*/
    }
}
```

16.7.5.43 Resume the task according to task ID

Request URL

GET /ISAPI/SDT/Management/Task/resume/<taskID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, URI*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:status code*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:which corresponds to subStatusCode, this field is required when statusCode is not 1. The returned value is the
transformed decimal number*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1*/
}
```

16.8 Human Body Recognition

16.8.1 AcuSearch (Human Body)

16.8.1.1 Analyze and model human body pictures

Request URL

POST /ISAPI/Intelligent/humanModel?format=json

Query Parameter

None

Request Message

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```
{
    "URL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
    /*opt, string, picture URL,the picture data will be uploaded in form format if no value is assigned to this node, desc:picture URL,the picture data will
be uploaded in form format if no value is assigned to this node*/
    "picData":  "JXU4RkQ5JXU5MUNDJXU2NjJGJXU4OTgxJXU3RjE2JXU3ODAxJXU3Njg0JXU1MTg1JXU1QkI5JXVGRjAx",
    /*opt, string*/
    "algorithm": [
    /*opt, array, algorithm type, subType:object, desc:algorithm type*/
        {
            "classificationID":  "1",
            /*req, string, algorithm category ID, desc:algorithm category ID*/
            "targetType":  "1",
            /*req, string, target type, desc:target type*/
            "analysisType":  "1,2,3",
            /*opt, string, algorithm analysis type, desc:algorithm analysis type*/
            "analysisSourceType":  "video",
            /*req, string, VCA source type, desc:VCA source type*/
            "version":  "1.0",
            /*opt, string, algorithm version, desc:algorithm version*/
            "vendor":  ""
            /*opt, string, algorithm manufacturer, desc:algorithm manufacturer*/
        }
    ],
    "Rect": {
    /*opt, object, target frame, desc:target frame*/
        "height":  0.107407,
        /*req, float, height, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
        "width":  0.025,
        /*req, float, width, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
        "x":  0.645833,
        /*req, float, X-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
        "y":  0.651852
        /*req, float, Y-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "confidence":  99.99,
    /*ro, opt, float, confidence, range:[0.00,100.00], desc:confidence*/
    "modelData":  "abc"
    /*ro, opt, string, target model data, desc:used for searching human body data*/
}
```

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16.8.1.2 Get the capability of detecting human bodies in the picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "URLLen": {
```

```
    /*ro, req, object, URL length*/
        "min":  0,
        /*ro, req, int, the minimum value*/
        "max":  256
        /*ro, req, int, the maximum value*/
    },
    "Rect": {
    /*ro, req, object, target frame, desc:the origin is the lower-left corner of the screen*/
        "height": {
        /*ro, req, object, height*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, req, object, width*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        },
        "x": {
        /*ro, req, object, x-coordinate*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, Y-coordinate*/
            "min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "max":  1.000
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
        }
    },
    "modelDataLen": {
    /*ro, req, object, data length of the target model*/
        "min":  0,
        /*ro, req, int, the minimum value*/
        "max":  256
        /*ro, req, int, the maximum value*/
    },
    "maxResults": {
    /*ro, req, object, the maximum number of search results this time*/
        "min":  0,
        /*ro, req, int, the minimum value*/
        "max":  1000
        /*ro, req, int, the maximum value*/
    },
    "startTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, start time*/
    "endTime":  "1970-01-01T00:00:00+08:00",
    /*ro, req, datetime, end time*/
    "channelIDLen": {
    /*ro, req, object, camera ID length*/
        "min":  0,
        /*ro, req, int, the minimum value*/
        "max":  64
        /*ro, req, int, the maximum value*/
    },
    "similarity": {
    /*ro, opt, object, similarity*/
        "min":  0.0,
        /*ro, req, float, the minimum value*/
        "max":  1.00
        /*ro, req, float, the maximum value*/
    },
    "taskIDLen": {
    /*ro, req, object, task ID length*/
        "min":  0,
        /*ro, req, int, the minimum value*/
        "max":  64
        /*ro, req, int, the maximum value*/
    },
    "sortType": {
    /*ro, opt, object, sorting type*/
        "@opt": ["similarity", "time"]
        /*ro, req, array, range, subType:string, desc:similarity (by similarity(by default)), time (by time)*/
    },
    "URLType": {
    /*ro, opt, object, the URL type of searched picture*/
        "@opt": ["cloudURL"]
        /*ro, req, array, range, subType:string*/
    },
    "maxCloudURLResults": {
    /*ro, req, object, the maximum number of search results this time, desc:the maximum records of current search. It is valid when the value of URLType is
"cloudURL"*/
        "min":  0,
        /*ro, req, int, the minimum value*/
        "max":  1000
        /*ro, req, int, the maximum value*/
    },
```

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```
}
```

16.8.1.3 Get task progress of searching for human body by picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/searchByPic/progress?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	UUID format

Request Message

None

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded); it is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded); it is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode, this field is required when statusCode is not 1*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "progress": 1,
    /*ro, req, int, progress of searching by picture task, range:[0,100], desc:progress of searching by picture task*/
    "matchePicNums": 100
    /*ro, opt, int, range:[0,10000]*/
}
```

16.8.1.4 Search the result of human body search

Request URL

POST /ISAPI/Intelligent/humanRecognition/searchByPic/result?format=json

Query Parameter

None

Request Message

```
{
    "taskID": "5d811723-d165-11ea-9b07-005056c00008",
    /*req, string, task ID, desc:task ID*/
    "searchResultPosition": 0,
    /*req, int, The end position of search result in result list. In a single search,if you cannot get all the records in the result list,you can mark the end position and get the following records after the marked position in the next search., desc:The end position of search result in result list. In a single search,if you cannot get all the records in the result list,you can mark the end position and get the following records after the marked position in the next search.*/
    "maxResults": 100
    /*req, int, the maximum number of records that can be searched in this time, desc:the maximum number of records that can be searched in this time*/
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "activeNumMax",
    /*ro, opt, enum, sub status code, subType:string, desc:"ok" (succeeded). It is required when an error occurred. "activeNumMax" (the max. number of device allowed for License access is reached), "licenseExpired" (License expired)*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "taskID": "5d811723-d165-11ea-9b07-005056c00008",
    /*ro, req, string, task ID, desc:task ID*/
    "responseStatusStrg": "OK",
    /*ro, opt, enum, searching status, subType:string, desc:searching status: "OK"-searching ended,"NO MATCHES"-no data found,"MORE"-searching,the max. size
```

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```
is 32 bytes. It is valid only when errorCode is 1 and errorMsgis ok*/
    "numOfMatches":  1,
    /*ro, opt, int, returned number of results for current search, desc:returned number of results for current search*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results, desc:total number of matched results*/
    "targetInfo": [
    /*ro, req, array, target information, subType:object, desc:target information*/
        {
            "monitorPointIndexCode":  "abc",
            /*ro, req, string, camera No., desc:camera No.*/
            "monitorPointName":  "abc",
            /*ro, req, string, camera name, desc:camera name*/
            "captureTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, string, capture time, desc:capturing time,e.g.,2004-05-03T17:30:08+08:00*/
            "isSummerTime":  false,
            /*ro, opt, bool, whether the time returned currently is in DST (Daylight Saving Time) system*/
            "streamType":  "localvideo",
            /*ro, opt, enum, streaming type, subType:string, desc:streaming type: "realtime,historyvideo,localvideo"*/
            "targetSize":  "unknown",
            /*ro, opt, enum, target size, subType:string, desc:target size*/
            "direction":  "lowerLeft",
            /*ro, opt, enum, target direction, subType:string, desc:target direction*/
            "speed":  "unknown",
            /*ro, opt, enum, vehicle speed, subType:string, desc:target speed*/
            "targetType":  "human",
            /*ro, opt, enum, target type, subType:string, desc:"human", "twoWheelVehicle" (two wheeler), "threeWheelVehicle" (tricycle)*/
            "gender":  "unknown",
            /*ro, opt, enum, gender, subType:string, desc:gender*/
            "glass":  "unknown",
            /*ro, opt, enum, whether the person wears glasses, subType:string, desc:whether the person wears glasses*/
            "ageGroup":  "unknown",
            /*ro, opt, enum, age group, subType:string, desc:age group*/
            "ride":  "unknown",
            /*ro, opt, enum, whether the person is riding a bike, subType:string, desc:whether the person is riding a bike*/
            "bag":  "unknown",
            /*ro, opt, enum, whether the person is carrying a bag, subType:string, desc:whether the person is with a bag on his/her back*/
            "things":  "unknown",
            /*ro, opt, enum, whether the person is carrying things, subType:string, desc:whether the person is with things on his/her hands*/
            "hat":  "unknown",
            /*ro, opt, enum, whether the person wears a hat, subType:string, desc:whether the person is wearing hat*/
            "jacketType":  "unknown",
            /*ro, opt, enum, top's type, subType:string, desc:jacket type*/
            "trousersType":  "unknown",
            /*ro, opt, enum, trousers type, subType:string, desc:trousers type*/
            "jacketColor":  "unknown",
            /*ro, opt, enum, jacket color, subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue", "yellow", "green",
"brown", "pink", "purple", "deepGray", "cyan", "orange", "mixture"*/
            "trousersColor":  "unknown",
            /*ro, opt, enum, trousers color, subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue", "yellow",
"green", "brown", "pink", "purple", "deepGray", "cyan", "orange", "mixture"*/
            "hairStyle":  "shortHair",
            /*ro, opt, enum, hairstyle, subType:string, desc:hairstyle*/
            "mask":  "unknown",
            /*ro, opt, enum, whether the person wears a mask, subType:string, desc:whether the person is wearing mask or not*/
            "cyclingType":  "unknown",
            /*ro, opt, enum, riding type, subType:string, desc:cycling type*/
            "cyclingPersonNumber":  "unknown",
            /*ro, opt, enum, number of riders, subType:string, desc:number of clcling person*/
            "similarity":  0.53,
            /*ro, opt, float, similarity, desc:float,similarity,value ranges from 0.0 to 1.00*/
            "targetRect": {
            /*ro, opt, object, target frame, desc:target frame*/
                "height":  0.107407,
                /*ro, req, float, height, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
                "width":  0.025,
                /*ro, req, float, width, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
                "x":  0.645833,
                /*ro, req, float, x-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
                "y":  0.651852
                /*ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
            },
            "picUrl":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, req, string, picture URL, desc:picture URL*/
            "subPicUrl":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
            /*ro, opt, string, thumbnail URL, desc:thumbnail picture URL*/
            "URLType":  "cloudURL"
            /*ro, opt, string, the URL type of searched picture. This node is applicable to the device which supports local URL. To get the picture of could
storage,set the value to "cloudURL", desc:the URL type of searched picture. This node is applicable to the device which supports local URL. To get the
picture of could storage,set the value to "cloudURL"*/
        }
    ]
}
```

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16.8.1.5 Get the capability of searching for the task of searching by human body picture

Request URL

GET /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "searchID": {
    /*ro, req, object, search ID, desc:this node is used to check whether searches are the same one; due to device performance limitation, you can apply the search conditions multiple times (the search ID should be the same) to complete one search and get the search progress*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,32]*/
        "@max":  32
        /*ro, opt, int, the maximum value, range:[0,32]*/
    },
    "maxResultNumber": {
    /*ro, opt, object, the maximum number of records that need to be searched*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,4294967295]*/
        "@max":  2000,
        /*ro, opt, int, the maximum value, range:[0,4294967295]*/
        "@def":  0
        /*ro, opt, int, the default value, range:[0,4294967295]*/
    },
    "maxResults": {
    /*ro, req, object, the maximum number of records that can be displayed on each page by this search*/
        "@min":  0,
        /*ro, opt, int, the minimum value, range:[0,100]*/
        "@max":  100
        /*ro, opt, int, the maximum value, range:[1,100]*/
    }
}
```

16.8.1.6 Get the tasks of searching by human body picture

Request URL

POST /ISAPI/Intelligent/humanRecognition/searchByPic/taskInfo?format=json

Query Parameter

None

Request Message

```
{
    "searchID":  "C93A238C-6AD0-0001-4CB8-1D961A901701",
    /*req, string, search ID, range:[0,64]*/
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, range:[0,4294967295], desc:in a single search, if you cannot get all the records in the result list, you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  100
    /*opt, int, the maximum number of search results this time, range:[0,100]*/
}
```

Response Message

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```
{
    "responseStatusStrg":  "MORE",
    /*ro, req, enum, searching status description, subType:string, desc:"MORE" (searching for more data), "NO MATCHES" (no matched data),"OK" (searching
completed)*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time, range:[0,100]*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results, range:[0,4294967295]*/
    "TaskInfo": [
    /*ro, opt, array, task information, subType:object*/
        {
            "taskID":  "abc",
            /*ro, req, string, task ID of searching by picture, range:[0,64]*/
            "taskCreateTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, task creation time*/
            "taskFinishTime":  "2004-05-03T17:30:08+08:00",
            /*ro, opt, datetime, task end time*/
            "timeUsed":  100,
            /*ro, opt, int, task execution duration, unit:s, desc:unit: second*/
            "progress":  100,
            /*ro, opt, int, task progress, range:[0,100]*/
            "TaskDetails": {
            /*ro, opt, object, task details*/
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*ro, req, string, start time, desc:ISO 8601 format*/
                "endTime":  "2004-05-03T17:30:08+08:00",
                /*ro, req, string, end time, desc:ISO 8601 format*/
                "choiceChannel": [
                /*ro, opt, array, camera list, subType:object, desc:which specifies cameras where the human body pictures are captured. If this node is not
configured, it indicates no camera limit*/
                    {
                        "channelID":  "abc",
                        /*ro, opt, string, camera ID, desc:the maximum string size is 32*/
                        "channelName":  "channel1"
                        /*ro, opt, string, camera name, range:[0,64]*/
                    }
                ],
                "similarity":  1,
                /*ro, opt, int, similarity, range:[0.00,1.00], desc:the larger the value is, the more accurate it will be*/
                "modelData":  "abc",
                /*ro, opt, string, target model data, desc:target model data*/
                "taskID":  "5d811723-d165-11ea-9b07-005056c00008",
                /*ro, opt, string, task ID of searching by picture, desc:task ID of searching by picture*/
                "URLType":  "cloudURL",
                /*ro, opt, string, the URL type of searched picture, desc:this node is for devices supporting local URL, and the local URL will be returned
by default*/
                "maxResults":  100
                /*ro, opt, int, the maximum number of search results this time, desc:the search will stop when the number of results reaches the configured
value*/
            }
        }
    ]
}
```


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16.8.1.7 Delete a task of searching by human body picture

Request URL

DELETE /ISAPI/Intelligent/humanRecognition/searchByPic?format=json&taskID=<taskID>

Query Parameter

Parameter Name	Parameter Type	Description
taskID	string	Task ID, which is returned by the URL (POST /ISAPI/Intelligent/humanRecognition/searchByPic?format=json).

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, Status Code, desc:this field is required when the status code is not "1" (success)*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:this field is required when the status code is not "OK" (success)*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:this field is required when the status code is not "ok" (success)*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error message, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

16.8.1.8 Detect human body

Request URL

POST /ISAPI/Intelligent/humanRecognition?format=json

Query Parameter

None

Request Message

```
{
    "URL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile?
id=00000001/temp001/212_14444186_17930&token=7a57a5a7ffffffffc1a0316369671314",
    /*opt, string, picture URL,the picture data will be uploaded in form format if no value is assigned to this node, desc:picture URL,the picture data will
be uploaded in form format if no value is assigned to this node*/
    "picData":  "JXU4RkQ5JXU5MUNDJXU2NjJGJXU4OTgxJXU3RjE2JXU3ODAxJXU3Njg0JXU1MTg1JXU1QkI5JXVGRjAx",
    /*opt, string*/
    "algorithm": [
    /*opt, array, algorithm type, subType:object, desc:algorithm type*/
        {
            "classificationID":  "1",
            /*req, string, algorithm category ID, desc:algorithm category ID*/
            "targetType":  "1",
            /*req, string, target type: 1-face,2-vehicle,3-active targets (human body and vehicle),4-human body, desc:target type: 1-face,2-vehicle,3-active
targets (human body and vehicle),4-human body*/
            "analysisType":  "1,2,3",
            /*opt, string, algorithm analysis type, desc:algorithm analysis type*/
            "analysisSourceType":  "video",
            /*req, string, VCA source type, desc:VCA source type*/
            "version":  "1.0",
            /*opt, string, algorithm version, desc:algorithm version*/
            "vendor":  ""
            /*opt, string, algorithm manufacturer, desc:algorithm manufacturer*/
        }
    ]
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
humanRecognition	[Message content]	application/json	--	--	--
human	[Binary picture data]	image/jpeg	human	human	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the

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transmitted data of form unit is file, and it refers to the file name of form unit body.

- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "Rect": [
    /*ro, opt, array, target frame, subType:object, range:[0.000,1.000], desc:target frame*/
        {
            "height":  0.107407,
            /*ro, req, float, height, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
            "width":  0.025,
            /*ro, req, float, width, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen*/
            "x":  0.645833,
            /*ro, req, float, X-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
            "y":  0.651852
            /*ro, req, float, Y-coordinate, range:[0.000,1.000], desc:the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the
screen*/
        }
    ]
}
```

16.9 Perimeter Protection

16.9.1 Intrusion Detection

16.9.1.1 Set intrusion detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/fieldDetections

Query Parameter

None

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->fieldDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FieldDetectionScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.1.2 Get arming schedules of all channels for intrusion detection

Request URL

GET /ISAPI/Event/schedules/fieldDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, configuration of intrusion detection arming schedules-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->fieldDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedules-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</FieldDetectionScheduleList>
```

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16.9.1.3 Set the linkage triggering parameters of intrusion detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/FieldDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <notificationMethod>
      <!--req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60]-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <AudioAction>
      <!--opt, object-->
      <audioID>
        <!--req, int-->1
      </audioID>
      <alarmTimes>
        <!--req, int, range:[0,9]-->1
      </alarmTimes>
    </AudioAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.1.4 Get the linkage triggering parameters of intrusion detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/FieldDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60]-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <AudioAction>
      <!--ro, opt, object-->
      <audioID>
        <!--ro, req, int-->1
      </audioID>
      <alarmTimes>
        <!--ro, req, int, range:[0,9]-->1
      </alarmTimes>
    </AudioAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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16.9.1.5 Set the linkage parameters of intrusion detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/FieldDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->fielddetection
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <LightAudioByIPAction>
```

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```
        <!--opt, object-->
        <notificationType>
          <!--req, enum, subType:string-->audio
        </notificationType>
        <notificationName>
          <!--opt, string-->test
        </notificationName>
      </LightAudioByIPAction>
      <SupplementLightAlarm>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.1.6 Get the linkage parameters of intrusion detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/FieldDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->fielddetection
  </eventType>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

16.9.1.7 Set intrusion detection parameters of all channels

Request URL

PUT /ISAPI/Smart/FieldDetection

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetection>
    <!--opt, object-->
    <id>
      <!--req, string, channel No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--opt, bool, whether the device supports video dual-VCA-->true
    </intelliBackSearch>
    <startTriggerTime>
      <!--req, int, unit:ms-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--req, int, unit:ms-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--opt, int, minimum pixels of each object-->1
    </minObjectSize>
    <maxObjectSize>
      <!--opt, int, maximum pixels of each object-->1
    </maxObjectSize>
    <FieldDetectionRegionList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <FieldDetectionRegion>
        <!--opt, object, intrusion region-->
        <id>
          <!--req, string-->1
        </id>
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
```

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```
          <sensitivityLevel>
            <!--req, int, sensitivity, range:[0,100]-->1
          </sensitivityLevel>
          <timeThreshold>
            <!--req, int-->1
          </timeThreshold>
          <objectOccupation>
            <!--req, int, percentage-->1
          </objectOccupation>
          <detectionTarget>
            <!--opt, enum, detection target, subType:string, desc:detection target-->all
          </detectionTarget>
          <RegionCoordinatesList>
            <!--opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->1
              </positionX>
              <positionY>
                <!--req, int, y-coordinate, range:[0,1000]-->1
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
          <humanMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (human)-->true
          </humanMisinfoFilterEnabled>
          <vehicleMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
          </vehicleMisinfoFilterEnabled>
          <priority>
            <!--opt, enum, priority, subType:string, desc:priority-->low
          </priority>
          <alarmConfidence>
            <!--opt, enum, subType:string-->low
          </alarmConfidence>
          <recordConfidence>
            <!--opt, enum, subType:string-->low
          </recordConfidence>
        </FieldDetectionRegion>
      </FieldDetectionRegionList>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (human)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
    </FieldDetection>
</FieldDetectionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.1.8 Get intrusion detection parameters of all channels

Request URL

GET /ISAPI/Smart/FieldDetection

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, intrusion detection parameter list, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetection>
    <!--ro, opt, object, intrusion detection parameters-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--ro, opt, bool, whether the device supports video dual-VCA-->true
    </intelliBackSearch>
    <startTriggerTime>
      <!--ro, req, int, unit:ms-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--ro, req, int, unit:ms-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--ro, opt, int, minimum pixels of each object-->1
    </minObjectSize>
    <maxObjectSize>
      <!--ro, opt, int, maximum pixels of each object-->1
    </maxObjectSize>
    <FieldDetectionRegionList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <FieldDetectionRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->1
        </id>
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--ro, req, int, sensitivity-->1
        </sensitivityLevel>
        <timeThreshold>
          <!--ro, req, int, time threshold-->1
        </timeThreshold>
        <objectOccupation>
          <!--ro, req, int, percentage-->1
        </objectOccupation>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <RegionCoordinatesList>
          <!--ro, opt, array, rectangle, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </FieldDetectionRegion>
    </FieldDetectionRegionList>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
```

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```
    </vehicleMisinfoFilterEnabled>
  </FieldDetection>
</FieldDetectionList>
```

16.9.1.9 Get intrusion detection parameters by video input channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, intrusion detection, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--ro, opt, bool, whether the device supports video dual-VCA-->true
  </intelliBackSearch>
  <startTriggerTime>
    <!--ro, req, int, unit: milliseconds, unit:ms-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--ro, req, int, unit: milliseconds, unit:ms-->1
  </endTriggerTime>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--ro, opt, int, minimum pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--ro, opt, int, maximum pixels of each object-->1
  </maxObjectSize>
  <FieldDetectionRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, ID-->1
      </id>
      <enabled>
        <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold>
        <!--ro, req, int, time threshold-->1
      </timeThreshold>
      <objectOccupation>
        <!--ro, req, int, percentage-->1
      </objectOccupation>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
      </detectionTarget>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
```

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```
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--ro, opt, enum, priority, subType:string, desc:priority-->low
      </priority>
      <alarmConfidence>
        <!--ro, opt, enum, subType:string-->low
      </alarmConfidence>
      <recordConfidence>
        <!--ro, opt, enum, subType:string-->low
      </recordConfidence>
      <ruleName>
        <!--ro, opt, string, range:[1,64]-->test
      </ruleName>
      <detectionTargetProperty>
        <!--ro, opt, object-->
        <human>
          <!--ro, opt, object, dep:or,{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,all},
{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,human}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->nonMotor
            </property>
          </propertyList>
        </human>
        <animal>
          <!--ro, opt, object, dep:or,{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,all},
{$.FieldDetection.FieldDetectionRegionList.FieldDetectionRegion.detectionTarget,eq,animal}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->large
            </property>
          </propertyList>
        </animal>
      </detectionTargetProperty>
    </FieldDetectionRegion>
  </FieldDetectionRegionList>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <contAlarmForStaticTargetEnabled>
    <!--ro, opt, bool-->true
  </contAlarmForStaticTargetEnabled>
  <targetAlarmInterval>
    <!--ro, opt, int, range:[1,5], step:1, unit:s, dep:and,{$.FieldDetection.targetAlarmInterval,eq,true}-->1
  </targetAlarmInterval>
  <stackTarget>
    <!--ro, opt, bool-->true
  </stackTarget>
  <targetSensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->50
  </targetSensitivityLevel>
</FieldDetection>
```

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16.9.1.10 Set intrusion detection parameters by channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, intrusion detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
```

```
<enabled>
  <!--req, bool, whether to enable the function-->true
</enabled>
<intelliBackSearch>
  <!--opt, bool, whether the device supports video dual-VCA-->true
</intelliBackSearch>
<startTriggerTime>
  <!--req, int, trigger start time, unit:ms-->1
</startTriggerTime>
<endTriggerTime>
  <!--req, int, trigger end time, unit:ms-->1
</endTriggerTime>
<normalizedScreenSize>
  <!--opt, object, normalized size-->
  <normalizedScreenWidth>
    <!--req, int, normalized width-->1
  </normalizedScreenWidth>
  <normalizedScreenHeight>
    <!--req, int, normalized height-->1
  </normalizedScreenHeight>
</normalizedScreenSize>
<minObjectSize>
  <!--opt, int, minimum pixels of each object-->1
</minObjectSize>
<maxObjectSize>
  <!--opt, int, maximum pixels of each object-->1
</maxObjectSize>
<FieldDetectionRegionList size="4">
  <!--opt, array, intrusion detection region list, subType:object, attr:size{req, int}-->
  <FieldDetectionRegion>
    <!--opt, object, intrusion detection region-->
    <id>
      <!--req, string, region ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--req, int, time threshold-->1
    </timeThreshold>
    <objectOccupation>
      <!--req, int, percentage-->1
    </objectOccupation>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string-->test
    </detectionTarget>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </FieldDetectionRegion>
</FieldDetectionRegionList>
<humanMisinfoFilterEnabled>
  <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
</humanMisinfoFilterEnabled>
<vehicleMisinfoFilterEnabled>
  <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
</vehicleMisinfoFilterEnabled>
</FieldDetection>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.1.11 Get configuration capability of intrusion detection by channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>
<FieldDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <FieldDetectionRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <FieldDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, region ID-->1
      </id>
      <minRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </minRegionCoordinatesNum>
      <maxRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </maxRegionCoordinatesNum>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity-->1
      </sensitivityLevel>
      <timeThreshold min="0" max="60" def="0">
        <!--ro, req, int, time threshold, attr:min{req, int},max{req, int},def{req, int}-->1
      </timeThreshold>
      <detectionTarget opt="all,human,vehicle,animal,others" def="human">
        <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string},def{req, string}, desc:detection target-->all
      </detectionTarget>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
      <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </alarmConfidence>
    </FieldDetectionRegion>
  </FieldDetectionRegionList>
  <isSupportMultiScene opt="true,false">
    <!--ro, req, bool, whether it supports multiple scene detection, attr:opt{req, string}-->true
  </isSupportMultiScene>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool, whether it supports false human body alarm prevention, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned-->true
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--ro, opt, bool, whether it supports false vehicle alarm prevention, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
  </isSupportVehicleMisinfoFilter>
  <isSupportTargetMultiSelect>
    <!--ro, opt, bool, whether it supports selecting multiple detection targets, desc:if this function is supported, this node will be returned and its
value is true; otherwise, this node will not be returned-->true
  </isSupportTargetMultiSelect>
</FieldDetection>
```

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16.9.1.12 Get parameters of all intrusion detection regions of a specific channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <channelEnabled>
    <!--ro, opt, bool-->true
  </channelEnabled>
  <FieldDetectionRegion>
    <!--ro, opt, object, intrusion detection-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--ro, req, int, time threshold-->1
    </timeThreshold>
    <objectOccupation>
      <!--ro, req, int-->1
    </objectOccupation>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <RegionCoordinatesList>
      <!--ro, opt, array, list of area coordinates, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of video recording-->low
    </recordConfidence>
  </FieldDetectionRegion>
</FieldDetectionRegionList>
```

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16.9.1.13 Set parameters of all intrusion detection regions of a specified channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters of all intrusion detection regions, subType:object, attr:version{req, string, protocolVersion}-->
  <FieldDetectionRegion>
    <!--opt, object, intrusion detection regions-->
    <id>
      <!--req, string, No.-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, sensitivity-->1
    </sensitivityLevel>
    <timeThreshold>
      <!--req, int, time threshold-->1
    </timeThreshold>
    <objectOccupation>
      <!--req, int, percentage-->1
    </objectOccupation>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </FieldDetectionRegion>
</FieldDetectionRegionList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.1.14 Get parameters of an intrusion detection region of a specific channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <sensitivityLevel>
    <!--ro, req, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--ro, req, int, time threshold-->1
  </timeThreshold>
  <objectOccupation>
    <!--ro, req, int-->1
  </objectOccupation>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--ro, opt, object, list of area coordinates-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates-->
      <positionX>
        <!--ro, req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification: "low","mediumLow","mediumHigh","high","low"-
->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of recording video: "low","mediumLow","mediumHigh","high","low"-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.FieldDetectionRegion.detectionTarget,eq,all},{$.FieldDetectionRegion.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.FieldDetectionRegion.detectionTarget,eq,all},{$.FieldDetectionRegion.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</FieldDetectionRegion>
```

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16.9.1.15 Set parameters of an intrusion detection region of a specified channel

Request URL

PUT /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of an intrusion detection region of a specified channel, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--req, int, sensitivity-->1
  </sensitivityLevel>
  <timeThreshold>
    <!--req, int, time threshold-->1
  </timeThreshold>
  <objectOccupation>
    <!--req, int, percentage-->1
  </objectOccupation>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--opt, object, region coordinate list-->
    <RegionCoordinates>
      <!--opt, object, region coordinates-->
      <positionX>
        <!--req, int, X-coordinate-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</FieldDetectionRegion>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.1.16 Get configuration capability of intrusion detection by channel

Request URL

GET /ISAPI/Smart/FieldDetection/<channelID>/regions/<regionID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<FieldDetectionRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, detection area ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel min="1" max="100">
    <!--ro, req, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <timeThreshold min="1" max="100" def="1">
    <!--ro, req, int, range:[1,100], attr:min{req, int},max{req, int},def{req, int}-->1
  </timeThreshold>
  <objectOccupation min="1" max="100">
    <!--ro, req, int, range:[1,100], attr:min{req, int},max{req, int}-->1
  </objectOccupation>
  <maxRegionCoordinatesNum>
    <!--ro, req, int, the maximum number of coordinates-->1
  </maxRegionCoordinatesNum>
  <minRegionCoordinatesNum>
    <!--ro, req, int, the minimum number of coordinates-->1
  </minRegionCoordinatesNum>
  <detectionTarget opt="all,human,vehicle,animal,others">
    <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
  </detectionTarget>
  <RegionCoordinatesList>
    <!--ro, opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, false alarm reduction of human body, attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, false alarm reduction of vehicle, attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:priority: "low,middle,high"-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</FieldDetectionRegion>
```

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16.9.1.17 Intrusion detection

EventType:fielddetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
```

```
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"fielddetection" (intrusion detection)-->fielddetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
```

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```
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <targetSpeed>
    <!--ro, opt, int-->0
  </targetSpeed>
```

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```
  <targetDistance>
    <!--ro, opt, int, unit:m-->2
  </targetDistance>
  <triggerType>
    <!--ro, opt, enum, subType:string-->video
  </triggerType>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <nonTargetRate>
    <!--ro, opt, int, range:[1,100], unit:%-->1
  </nonTargetRate>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
fielddetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

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16.9.2 Line Crossing Detection

16.9.2.1 Get the arming schedules of line crossing detection of all channels

Request URL

GET /ISAPI/Event/schedules/lineDetections

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the arming schedules of line crossing detection of all channels, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming schedule information-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->lineDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming time-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LineDetectionScheduleList>
```

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16.9.2.2 Set line crossing detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/lineDetections

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming information-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->lineDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</LineDetectionScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.2.3 Get the arming schedule of line crossing detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/lineDetections/lineDetections_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, enum, subType:string-->lineDetection
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.9.2.4 Set the arming schedule of line crossing detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/lineDetections/lineDetections_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, enum, subType:string-->lineDetection
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.2.5 Set alarm linkage parameters of line crossing detection by channel

Request URL

PUT /ISAPI/Event/triggers/LineDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, linkage action list, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object, linkage action-->
    <id>
      <!--req, string, No.-->test
    </id>
    <notificationMethod>
      <!--req, enum, linkage action, subType:string, desc:"email", "IM" (instant messaging), "IO" (I/O output), "syslog" (system log), "HTTP" (uploading via
HTTP), "FTP" (uploading via FTP), "beep" (buzzer), "ptz" (PTZ), "record" (recording), "monitorAlarm" (alarm on monitor), "center" (uploading to the center),
"LightAudioAlarm" (light and audio alarm), "focus" (focus), "trace" (tracking), "whiteLight" (white light), "audio" (audio), "supplementLight" (supplement
light)-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, linkage time interval, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string, video channel ID-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string, dynamic video input channel ID-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object, PTZ linkage-->
      <ptzChannelID>
        <!--req, string, PTZ linked channel No.-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, PTZ linkage action, subType:string, desc:"preset", "pattern", "patrol"-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object, linkage parameters of the white light, desc:it is valid when the value of notificationMethod is "whiteLight"-->
      <whiteLightDurationTime>
        <!--opt, int, light flashing duration, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, supplement light alarm duration, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.2.6 Get alarm linkage parameters of a line crossing detection channel

Request URL

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GET /ISAPI/Event/triggers/LineDetection-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object, linkage type-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, linkage type, subType:string, desc:linkage type-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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16.9.2.7 Get the linkage parameters of line crossing detection event for a specified channel

Request URL

GET /ISAPI/Event/triggers/LineDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->linedetection
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--ro, opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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16.9.2.8 Set the linkage parameters of line crossing detection event for a specified channel

Request URL

PUT /ISAPI/Event/triggers/LineDetection-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->LineDetection
```

```
    </eventType>
    <eventDescription>
      <!--opt, string-->test
    </eventDescription>
    <inputIOPortID>
      <!--opt, string-->1
    </inputIOPortID>
    <dynInputIOPortID>
      <!--opt, string-->1
    </dynInputIOPortID>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <dynVideoInputChannelID>
      <!--opt, string-->1
    </dynVideoInputChannelID>
    <intervalBetweenEvents>
      <!--opt, int, unit:s-->1
    </intervalBetweenEvents>
    <WLSensorID>
      <!--opt, string-->1
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--opt, array, subType:object-->
      <EventTriggerNotification>
        <!--opt, object-->
        <id>
          <!--req, string-->test
        </id>
        <notificationMethod>
          <!--req, enum, subType:string-->email
        </notificationMethod>
        <notificationRecurrence>
          <!--opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--opt, int, unit:ms-->1
        </notificationInterval>
        <outputIOPortID>
          <!--opt, string-->1
        </outputIOPortID>
        <dynOutputIOPortID>
          <!--opt, string-->1
        </dynOutputIOPortID>
        <videoInputID>
          <!--opt, string-->1
        </videoInputID>
        <dynVideoInputID>
          <!--opt, string-->1
        </dynVideoInputID>
        <ptzAction>
          <!--opt, object-->
          <ptzChannelID>
            <!--req, string-->1
          </ptzChannelID>
          <actionName>
            <!--req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--opt, int-->1
          </actionNum>
          <presetDurationTime>
            <!--opt, int, unit:s-->1
          </presetDurationTime>
        </ptzAction>
        <cellphoneNumber>
          <!--opt, string, range:[1,11]-->test
        </cellphoneNumber>
        <LightAudioByIPAction>
          <!--opt, object-->
          <notificationType>
            <!--req, enum, subType:string-->audio
          </notificationType>
          <notificationName>
            <!--opt, string-->test
          </notificationName>
        </LightAudioByIPAction>
        <SupplementLightAlarm>
          <!--opt, object-->
          <durationTime>
            <!--opt, int, range:[0,90], unit:s-->5
          </durationTime>
        </SupplementLightAlarm>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.2.9 Set line crossing detection parameters of all channels

Request URL

PUT /ISAPI/Smart/LineDetection

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineDetection>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <intelliBackSearch>
      <!--opt, bool-->true
    </intelliBackSearch>
    <duration>
      <!--opt, int-->1
    </duration>
    <startTriggerTime>
      <!--opt, int-->1
    </startTriggerTime>
    <endTriggerTime>
      <!--opt, int-->1
    </endTriggerTime>
    <normalizedScreenSize>
      <!--opt, object-->
      <normalizedScreenWidth>
        <!--req, int-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <minObjectSize>
      <!--opt, int-->1
    </minObjectSize>
    <maxObjectSize>
      <!--opt, int-->1
    </maxObjectSize>
    <LineItemList size="4">
      <!--opt, array, subType:object, attr:size{req, int}-->
      <LineItem>
        <!--opt, object-->
        <id>
          <!--req, string-->1
        </id>
        <enabled>
          <!--req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--req, int, detection sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <directionSensitivity>
          <!--opt, enum, detection direction, subType:string, desc:detection direction-->any
        </directionSensitivity>
        <CoordinatesList>
          <!--opt, array, subType:object, range:[0,2]-->
          <Coordinates>
```

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```
            <Coordinates>
              <!--opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->1
              </positionX>
              <positionY>
                <!--req, int, y-coordinate, range:[0,1000]-->1
              </positionY>
            </Coordinates>
          </CoordinatesList>
          <detectionTarget>
            <!--opt, enum, detection target, subType:string, desc:detection target-->all
          </detectionTarget>
          <humanMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (human body)-->true
          </humanMisinfoFilterEnabled>
          <vehicleMisinfoFilterEnabled>
            <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
          </vehicleMisinfoFilterEnabled>
          <priority>
            <!--opt, enum, priority, subType:string, desc:priority-->low
          </priority>
          <alarmConfidence>
            <!--opt, enum, subType:string-->low
          </alarmConfidence>
          <recordConfidence>
            <!--opt, enum, subType:string-->low
          </recordConfidence>
        </LineItem>
      </LineItemList>
      <recogRuleType>
        <!--opt, enum, subType:string-->vectorMode
      </recogRuleType>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (human body)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
    </LineDetection>
  </LineDetectionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.2.10 Get line crossing detection parameters of all channels

Request URL

GET /ISAPI/Smart/LineDetection

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetectionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineDetection>
    <!--ro, opt, object-->
```

```
      <id>
        <!--ro, req, string, channel No.-->test
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <intelliBackSearch>
        <!--ro, opt, bool-->true
      </intelliBackSearch>
      <duration>
        <!--ro, opt, int, duration-->1
      </duration>
      <startTriggerTime>
        <!--ro, opt, int-->1
      </startTriggerTime>
      <endTriggerTime>
        <!--ro, opt, int-->1
      </endTriggerTime>
      <normalizedScreenSize>
        <!--ro, opt, object-->
        <normalizedScreenWidth>
          <!--ro, req, int-->1
        </normalizedScreenWidth>
        <normalizedScreenHeight>
          <!--ro, req, int-->1
        </normalizedScreenHeight>
      </normalizedScreenSize>
      <minObjectSize>
        <!--ro, opt, int-->1
      </minObjectSize>
      <maxObjectSize>
        <!--ro, opt, int-->1
      </maxObjectSize>
      <LineItemList size="4">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <LineItem>
          <!--ro, opt, object-->
          <id>
            <!--ro, req, string, ID-->1
          </id>
          <enabled>
            <!--ro, req, bool, whether to enable the function-->true
          </enabled>
          <sensitivityLevel>
            <!--ro, req, int, detection sensitivity, range:[0,100]-->1
          </sensitivityLevel>
          <directionSensitivity>
            <!--ro, opt, enum, detection direction, subType:string, desc:detection direction-->any
          </directionSensitivity>
          <CoordinatesList>
            <!--ro, opt, array, rectangle, subType:object, range:[0,2], desc:rectangle-->
            <Coordinates>
              <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--ro, req, int, X-coordinate, range:[0,1000]-->1
              </positionX>
              <positionY>
                <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
              </positionY>
            </Coordinates>
          </CoordinatesList>
          <detectionTarget>
            <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
          </detectionTarget>
          <humanMisinfoFilterEnabled>
            <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
          </humanMisinfoFilterEnabled>
          <vehicleMisinfoFilterEnabled>
            <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
          </vehicleMisinfoFilterEnabled>
          <priority>
            <!--ro, opt, enum, priority, subType:string, desc:priority-->low
          </priority>
          <alarmConfidence>
            <!--ro, opt, enum, subType:string-->low
          </alarmConfidence>
          <recordConfidence>
            <!--ro, opt, enum, subType:string-->low
          </recordConfidence>
        </LineItem>
      </LineItemList>
      <recogRuleType>
        <!--ro, opt, enum, subType:string-->vectorMode
      </recogRuleType>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
      </vehicleMisinfoFilterEnabled>
    </LineDetection>
  </LineDetectionList>
```

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16.9.2.11 Set line crossing detection parameters by channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, line crossing detection parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <intelliBackSearch>
    <!--opt, bool, whether to enable dual-VCA-->true
  </intelliBackSearch>
  <duration>
    <!--opt, int, duration-->1
  </duration>
  <startTriggerTime>
    <!--opt, int, trigger start time-->1
  </startTriggerTime>
  <endTriggerTime>
    <!--opt, int, trigger end time-->1
  </endTriggerTime>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <minObjectSize>
    <!--opt, int, minimum number of pixels of each object-->1
  </minObjectSize>
  <maxObjectSize>
    <!--opt, int, maximum number of pixels of each object-->1
  </maxObjectSize>
  <LineItemList size="4">
    <!--opt, array, subType:object, attr:size{req, int}-->
    <id>
      <!--req, string, No.-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any" (all directions)-
->any
    </directionSensitivity>
    <CoordinatesList>
      <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
```

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```
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </LineItemList>
  <recogRuleType>
    <!--opt, enum, direction recognition mode, subType:string, desc:"vectorMode" (vector mode), "slopeMode" (slope mode)-->vectorMode
  </recogRuleType>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</LineDetection>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.2.12 Get configuration capability of line crossing detection by channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <LineItemList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <LineItem>
      <!--ro, opt, object-->
      <id>
```

```
        <!--ro, req, string-->test
      </id>
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
      </enabled>
      <sensitivityLevel min="1" max="100">
        <!--ro, req, int, detection sensitivity, attr:min{req, int},max{req, int}-->1
      </sensitivityLevel>
      <directionSensitivity opt="left-right,right-left,any">
        <!--ro, opt, enum, detection direction, subType:string, attr:opt{req, string}, desc:detection direction-->any
      </directionSensitivity>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[0,2], desc:line-->
        <RegionCoordinates>
          <!--ro, req, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget opt="all,human,vehicle,animal,others" def="human">
        <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string},def{req, string}, desc:detection target-->all
      </detectionTarget>
      <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </alarmConfidence>
      <recordConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </recordConfidence>
      <ruleName min="1" max="64">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
      </ruleName>
      <detectionTargetProperty>
        <!--ro, opt, object-->
        <human>
          <!--ro, opt, object-->
          <propertyList size="1">
            <!--ro, opt, object, attr:size{req, int}-->
            <property opt="nonMotor">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
            </property>
          </propertyList>
        </human>
        <animal>
          <!--ro, opt, object-->
          <propertyList size="3">
            <!--ro, opt, object, attr:size{req, int}-->
            <property opt="large,small,poultry">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
            </property>
          </propertyList>
        </animal>
      </detectionTargetProperty>
    </LineItem>
  </LineItemList>
  <isSupportPerimeterDetection opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </isSupportPerimeterDetection>
  <isSupportMultiScene opt="true,false">
    <!--ro, opt, bool, whether supports multiple scene detection, attr:opt{req, string}-->true
  </isSupportMultiScene>
  <recogRuleType>
    <!--ro, opt, enum, subType:string-->vectorMode
  </recogRuleType>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool, whether supports false human body alarm prevention, desc:If this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned.-->true
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--ro, opt, bool, whether supports false vehicle alarm prevention, desc:If this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned.-->true
  </isSupportVehicleMisinfoFilter>
  <isSupportTargetMultiSelect>
    <!--ro, opt, bool, whether supports selecting multiple detection targets, desc:If this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned.-->true
  </isSupportTargetMultiSelect>
  <isSupportAllDayUpload>
    <!--ro, opt, bool, whether supports uploading all-day event-->true
  </isSupportAllDayUpload>
  <isSupportOverlapParams>
    <!--ro, opt, bool-->true
  </isSupportOverlapParams>
  <warningSystemEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </warningSystemEnabled>
  <isSupportStackTarget>
    <!--ro, opt, bool-->true
  </isSupportStackTarget>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
```

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```
    </isSupportAllTargetDetectionWebDisplay>
    <targetSensitivityLevel min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
    </targetSensitivityLevel>
  </LineDetection>
```

16.9.2.13 Add detection lines for a specific channel

Request URL

POST /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineItem>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--ro, opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any"-->any
    </directionSensitivity>
    <CoordinatesList>
      <!--ro, opt, array, list of coordinates, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"low", "middle" "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, confidence of alarm notification, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, confidence of video recording, subType:string-->low
    </recordConfidence>
  </LineItem>
</LineItemList>
```

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16.9.2.14 Set parameters of all line crossing detection lines by channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters of all line crossing detection lines, subType:object, attr:version{req, string, protocolVersion}-->
  <LineItem>
    <!--opt, object, parameters of a line crossing detection line-->
    <id>
      <!--req, string, No.-->test
    </id>
    <enabled>
      <!--req, bool, whether to enable the function-->true
    </enabled>
    <sensitivityLevel>
      <!--req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any" (all directions)-
->any
    </directionSensitivity>
    <CoordinatesList>
      <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </LineItem>
</LineItemList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.2.15 Get parameters of all detection lines of a specific channel

Request URL

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GET /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItemList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <LineItem>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivityLevel>
      <!--ro, req, int, detection sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <directionSensitivity>
      <!--ro, opt, enum, detection direction, subType:string, desc:"left-right" (from left to right), "right-left" (from right to left), "any"-->any
    </directionSensitivity>
    <CoordinatesList>
      <!--ro, opt, array, list of coordinates, subType:object, range:[0,2], desc:line-->
      <Coordinates>
        <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </Coordinates>
    </CoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, subType:string-->low
    </recordConfidence>
  </LineItem>
</LineItemList>
```

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16.9.2.16 Delete all detection lines of a specific channel

Request URL

DELETE /ISAPI/Smart/LineDetection/<channelID>/lineItem

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.2.17 Get parameters of a detection line of a specific channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
ruleID	string	Rule ID

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable the function, desc:whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--ro, req, int, detection sensitivity-->1
  </sensitivityLevel>
  <directionSensitivity>
    <!--ro, opt, enum, detection direction, subType:string, desc:"left-right,right-left,any"-->any
  </directionSensitivity>
  <CoordinatesList>
    <!--ro, opt, array, line, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all,human,vehicle,others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm reduction of vehicle-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:“low”, “middle”, “high”-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:"low", "mediumLow", "mediumHigh", "high", "low”-->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of recording video, subType:string, desc:"low", "mediumLow", "mediumHigh", "high", "low”-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.LineItem.detectionTarget,eq,all},{$.LineItem.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.LineItem.detectionTarget,eq,all},{$.LineItem.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</LineItem>
```

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16.9.2.18 Set parameters of a detection line of a specified channel

Request URL

PUT /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
ruleID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of a detection line, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <sensitivityLevel>
    <!--req, int, detection sensitivity-->1
  </sensitivityLevel>
  <directionSensitivity>
    <!--opt, enum, detection direction, subType:string, desc:"Left-right" (from left to right), "right-left" (from right to left), "any" (all directions)--
>any
  </directionSensitivity>
  <CoordinatesList>
    <!--opt, array, coordinates list, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</LineItem>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.2.19 Delete a detection line of a specific channel

Request URL

DELETE /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--
ruleID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.2.20 Get configuration capability of line crossing detection by channel

Request URL

GET /ISAPI/Smart/LineDetection/<channelID>/lineItem/<ruleID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
ruleID	string	Rule ID

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<LineItem xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->test
  </id>
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
  <sensitivityLevel min="1" max="100">
    <!--ro, req, int, sensitivity, attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <directionSensitivity opt="left-right,right-left,any">
    <!--ro, opt, enum, direction, subType:string, attr:opt{req, string}, desc:direction-->any
  </directionSensitivity>
  <CoordinatesList>
    <!--ro, opt, array, line, subType:object, range:[0,2], desc:line-->
    <Coordinates>
      <!--ro, opt, object, coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->1
      </positionY>
    </Coordinates>
  </CoordinatesList>
  <detectionTarget opt="all,human,vehicle,animal,others">
    <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (human body), attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle), attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:priority-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</LineItem>
```

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16.9.2.21 Line crossing detection

EventType:linedetection

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
```

```
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When
ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"linedetection" (Line Crossing Detection)-->linedetection
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
```

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```
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transmitted parameters-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <targetSpeed>
    <!--ro, opt, datetime-->2019-12-30T15:29:23+08:00
  </targetSpeed>
  <targetDistance>
    <!--ro, opt, int, unit:m-->2
  </targetDistance>
  <triggerType>
    <!--ro, opt, enum, subType:string-->video
  </triggerType>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
```

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```
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <applicationScene>
    <!--ro, opt, enum, Application Scene, subType:string-->indoor
  </applicationScene>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
linedetection	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

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16.9.3 Region Entrance Detection

16.9.3.1 Set region entrance detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/regionEntrances

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, string, event type-->regionEntrance
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of a week, subType:int, desc:day of a week-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionEntranceScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.3.2 Get arming schedules of all channels for region entrance detection

Request URL

GET /ISAPI/Event/schedules/regionEntrances

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, configuration of region entrance detection arming schedules-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->1
    </id>
    <eventType>
      <!--ro, opt, string, event type-->regionEntrance
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedule-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionEntranceScheduleList>
```

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16.9.3.3 Set the arming schedule of region entrance detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/regionEntrances/regionEntrance-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--opt, string-->regionEntrance
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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DETEKTOR

16.9.3.4 Get the arming schedule of region entrance detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/regionEntrances/regionEntrance-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, opt, string-->regionEntrance
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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DETEKTOR

16.9.3.5 Get the linkage triggering parameters of region entrance for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionEntrance-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->test
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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DETEKTOR

16.9.3.6 Set the linkage triggering parameters of region entrance for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionEntrance-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <notificationMethod>
      <!--req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.3.7 Get the linkage parameters of region entrance for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionEntrance-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, enum, subType:string-->regionEntrance
  </eventType>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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DETEKTOR

16.9.3.8 Set the linkage parameters of region entrance for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionEntrance-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->regionEntrance
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->1
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->1
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <SupplementLightAlarm>
        <!--opt, object-->
        <durationTime>
          <!--opt, int, range:[0,90], unit:s-->5
        </durationTime>
      </SupplementLightAlarm>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.3.9 Set region entrance detection parameters of all channels

Request URL

PUT /ISAPI/Smart/regionEntrance

Query Parameter

None

Request Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_RegionEntrance for details, attr:version{req, string, protocolVersion}-->
  <RegionEntrance>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionEntranceRegionList>
      <!--opt, object, area list-->
      <RegionEntranceRegion>
        <!--opt, object, intrusion region-->
        <id>
          <!--req, string, detection area ID-->1
        </id>
        <sensitivityLevel>
          <!--opt, int, sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--opt, enum, subType:string-->low
        </recordConfidence>
      </RegionEntranceRegion>
    </RegionEntranceRegionList>
    <mutexAbility opt="PDC">
      <!--opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionEntrance>
</RegionEntranceList>
```

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DETEKTOR

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.3.10 Get region entrance detection parameters of all channels

Request URL

GET /ISAPI/Smart/regionEntrance

Query Parameter

None

Request Message

None

Response Message

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DETEKTOR

```xml
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_RegionEntrance for details, attr:version{req, string, protocolVersion}-->
  <RegionEntrance>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, channel No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionEntranceRegionList>
      <!--ro, opt, array, area list, subType:object-->
      <RegionEntranceRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->1
        </id>
        <sensitivityLevel>
          <!--ro, opt, int, sensitivity, range:[1,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </RegionEntranceRegion>
    </RegionEntranceRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionEntrance>
</RegionEntranceList>
```

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DETEKTOR

16.9.3.11 Set parameters of the region entrance detection rule by channel

Request URL

PUT /ISAPI/Smart/regionEntrance/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntrance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, region entrance detection configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionEntranceRegionList>
    <!--opt, object, region list-->
    <RegionEntranceRegion>
      <!--opt, object, region entrance detection region-->
      <id>
        <!--req, string, region ID-->1
      </id>
      <sensitivityLevel>
        <!--opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--opt, object, region coordinate list-->
        <RegionCoordinates>
          <!--opt, object, region coordinates-->
          <positionX>
            <!--req, int, X-coordinate-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--opt, enum, alarm confidence, subType:string, desc:"Low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--opt, enum, video confidence, subType:string, desc:"Low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
    </RegionEntranceRegion>
  </RegionEntranceRegionList>
  <mutexAbility opt="PDC">
    <!--opt, string, mutually exclusive capability: "PDC" (people counting statistics, which is not supported during region entrance detection),
attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--opt, bool, whether the device supports multiple scenes detection-->true
  </isSupportMultiScene>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</RegionEntrance>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.3.12 Get region entrance detection parameters by video input channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntrance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, read-only, "PDC"-people counting statistics, which is not supported during region entrance detection, attr:version{req, string,
protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionEntranceRegionList>
    <!--ro, opt, object-->
    <RegionEntranceRegion>
      <!--ro, opt, object, Intrusion Region-->
      <id>
        <!--ro, req, string-->1
      </id>
      <sensitivityLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
```

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```
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
        <ruleName>
          <!--ro, opt, string, range:[1,64]-->test
        </ruleName>
        <detectionTargetProperty>
          <!--ro, opt, object-->
          <human>
            <!--ro, opt, object, dep:or,{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,all},
{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,human}-->
            <propertyList>
              <!--ro, opt, array, subType:object-->
              <property>
                <!--ro, opt, enum, subType:string-->nonMotor
              </property>
            </propertyList>
          </human>
          <animal>
            <!--ro, opt, object, dep:or,{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,all},
{$.RegionEntrance.RegionEntranceRegionList.RegionEntranceRegion.detectionTarget,eq,animal}-->
            <propertyList>
              <!--ro, opt, array, subType:object-->
              <property>
                <!--ro, opt, enum, subType:string-->large
              </property>
            </propertyList>
          </animal>
        </detectionTargetProperty>
      </RegionEntranceRegion>
    </RegionEntranceRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple-scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <stackTarget>
      <!--ro, opt, bool-->true
    </stackTarget>
    <targetSensitivityLevel>
      <!--ro, opt, int, range:[0,100]-->50
    </targetSensitivityLevel>
  </RegionEntrance>
```

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16.9.3.13 Get configuration capability of region entrance detection by channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntrance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ro, mutex capability, it indicates that which function is mutually exclusive with the region entrance detection function:
"PDC,videoFrameRate50,videoFrameRate60,19-1920*1080@50fps,20-1920*1080@60fps,scheduled recording", attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
  </id>
```

```
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized screen size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized screen width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized screen height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionEntranceRegionList size="4">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <RegionEntranceRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->1
      </id>
      <sensitivityLevel min="1" max="100">
        <!--ro, req, int, sensitivity, attr:min{req, int},max{req, int}-->1
      </sensitivityLevel>
      <RegionCoordinatesList size="10">
        <!--ro, opt, array, region coordinate list, subType:object, range:[3,10], attr:size{req, int}, desc:region coordinate list-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <maxRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </maxRegionCoordinatesNum>
      <minRegionCoordinatesNum>
        <!--ro, opt, int-->1
      </minRegionCoordinatesNum>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
      </detectionTarget>
      <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </alarmConfidence>
      <recordConfidence opt="low,mediumLow,mediumHigh,high" def="low">
        <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
      </recordConfidence>
      <ruleName min="1" max="64">
        <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
      </ruleName>
      <detectionTargetProperty>
        <!--ro, opt, object-->
        <human>
          <!--ro, opt, object-->
          <propertyList size="1">
            <!--ro, opt, object, attr:size{req, int}-->
            <property opt="nonMotor">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
            </property>
          </propertyList>
        </human>
        <animal>
          <!--ro, opt, object-->
          <propertyList size="3">
            <!--ro, opt, object, attr:size{req, int}-->
            <property opt="large,small,poultry">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
            </property>
          </propertyList>
        </animal>
      </detectionTargetProperty>
    </RegionEntranceRegion>
  </RegionEntranceRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene opt="true,false">
    <!--ro, req, bool, whether supports multiple scenes detection, attr:opt{req, string}-->true
  </isSupportMultiScene>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool, whether supports preventing false human body alarm, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
  </isSupportHumanMisinfoFilter>
  <isSupportVehicleMisinfoFilter>
    <!--ro, opt, bool, whether supports preventing false vehicle alarm, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
  </isSupportVehicleMisinfoFilter>
  <isSupportTargetMultiSelect>
    <!--ro, opt, bool, whether supports selecting multiple detection targets, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned-->true
  </isSupportTargetMultiSelect>
  <isSupportAllDayUpload>
```

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```
    <isSupportAllDayUpload>
      <!--ro, opt, bool, whether supports uploading all-day event-->true
    </isSupportAllDayUpload>
    <isSupportOverlapParams>
      <!--ro, opt, bool-->true
    </isSupportOverlapParams>
    <isSupportStackTarget>
      <!--ro, opt, bool-->true
    </isSupportStackTarget>
    <isSupportAllTargetDetectionWebDisplay>
      <!--ro, opt, bool-->false
    </isSupportAllTargetDetectionWebDisplay>
    <targetSensitivityLevel min="0" max="100">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
    </targetSensitivityLevel>
  </RegionEntrance>
```

16.9.3.14 Get parameters of all region entrance detection regions by channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, refer to the message XML_RegionEntranceRegion for details, attr:version{req, string, protocolVersion}-->
  <RegionEntranceRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, region ID-->1
    </id>
    <sensitivityLevel>
      <!--ro, req, int, sensitivity-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:priority-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, subType:string-->low
    </recordConfidence>
  </RegionEntranceRegion>
</RegionEntranceRegionList>
```

16.9.3.15 Set parameters of all region entrance detection regions by channel

Request URL

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PUT /ISAPI/Smart/regionEntrance/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, region parameters list, attr:version{req, string, protocolVersion}-->
  <RegionEntranceRegion>
    <!--opt, object, region parameters-->
    <id>
      <!--req, string, region ID-->1
    </id>
    <sensitivityLevel>
      <!--req, int, sensitivity-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </RegionEntranceRegion>
</RegionEntranceRegionList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.3.16 Set parameters of a region entrance detection region by channel

Request URL

PUT /ISAPI/Smart/regionEntrance/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, region configuration, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--req, int, sensitivity-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all" is
mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"Low", "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</RegionEntranceRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.3.17 Get parameters of a region entrance detection region of a specific channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>/regions/<regionID>

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region ID

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--ro, req, int, sensitivity-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--ro, opt, array, list of area coordinates, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification: "low","mediumLow","mediumHigh","high","low"-
->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of recording video: "low","mediumLow","mediumHigh","high","low"-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.RegionEntranceRegion.detectionTarget,eq,all},{$.RegionEntranceRegion.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.RegionEntranceRegion.detectionTarget,eq,all},{$.RegionEntranceRegion.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</RegionEntranceRegion>
```

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16.9.3.18 Get the capability of configuring the parameters of all region entrance detection regions of a specific channel

Request URL

GET /ISAPI/Smart/regionEntrance/<channelID>/regions/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <RegionEntranceRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, region ID-->1
    </id>
    <sensitivityLevel min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, object, list of area coordinates-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates-->
        <positionX>
          <!--ro, req, int, X-coordinate-->1
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <maxRegionCoordinatesNum>
      <!--ro, req, int-->1
    </maxRegionCoordinatesNum>
    <minRegionCoordinatesNum>
      <!--ro, req, int-->1
    </minRegionCoordinatesNum>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled opt="true,false">
      <!--ro, opt, bool, whether to enable false alarm filtering (human), attr:opt{req, string}-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled opt="true,false">
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle), attr:opt{req, string}-->true
    </vehicleMisinfoFilterEnabled>
    <priority opt="low,middle,high">
      <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:"low", "middle", "high"-->low
    </priority>
    <isSupportAllTargetDetectionWebDisplay>
      <!--ro, opt, bool-->false
    </isSupportAllTargetDetectionWebDisplay>
  </RegionEntranceRegion>
</RegionEntranceRegionList>
```

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16.9.3.19 Get configuration capability of region entrance detection by channel

Request URL

GET /ISAPI/Smart/regionEntrance/<regionID>/regions/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
regionID	string	Region ID
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionEntranceRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, region ID-->test
  </id>
  <sensitivityLevel min="1" max="100">
    <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--ro, opt, array, region coordinates list, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <maxRegionCoordinatesNum>
    <!--ro, req, int-->1
  </maxRegionCoordinatesNum>
  <minRegionCoordinatesNum>
    <!--ro, req, int-->1
  </minRegionCoordinatesNum>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all,human,vehicle,others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (human body), attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle), attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:priority-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</RegionEntranceRegion>
```

16.9.3.20 Region entrance

EventType:regionEntrance

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is
```

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```
the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"regionEntrance" (region entrance)-->regionEntrance
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area points coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height, range:[0.000,1.000]-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
```

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  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits-->test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:Rule No.-->0
  </ruleID>
  <ruleName>
```

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```
    <!--ro, opt, string, rule name, range:[0,32], desc:Rule Name-->test
  </ruleName>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionEntrance	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

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16.9.4 Region Exiting Detection

16.9.4.1 Get arming schedules of all channels for region exiting detection

Request URL

GET /ISAPI/Event/schedules/regionExitings

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, list of arming schedules, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID, desc:format: event type-channel No.-->test
    </id>
    <eventType>
      <!--ro, opt, string, event type-->regionExiting
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, object, list of arming schedules, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <dayOfWeek>
          <!--ro, opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, object, holiday arming schedule-->
      <TimeBlock>
        <!--ro, opt, object, arming schedules-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionExitingScheduleList>
```

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16.9.4.2 Set region exiting detection arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/regionExitings

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, refer to the message XML_Schedule for details, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, string, event type-->regionExiting
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, object, arming time list, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, day of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, object, holiday arming time-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</RegionExitingScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.4.3 Get the arming schedule of region exiting detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/regionExitings/regionExiting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <eventType>
    <!--ro, opt, string-->regionExiting
  </eventType>
  <videoInputChannelID>
    <!--ro, opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

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16.9.4.4 Set the arming schedule of region exiting detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/regionExitings/regionExiting-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->test
  </id>
  <eventType>
    <!--opt, string-->regionExiting
  </eventType>
  <videoInputChannelID>
    <!--opt, string-->test
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, object, attr:size{req, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->00:00:00
        </beginTime>
        <endTime>
          <!--req, time-->00:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.4.5 Get the linkage triggering parameters of region exiting for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionExiting-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, enum, subType:string-->regionExiting
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--ro, opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--ro, opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--ro, opt, string--> 1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--ro, opt, string--> 1
    </dynOutputIOPortID>
    <videoInputID>
      <!--ro, opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--ro, opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--ro, opt, object-->
      <ptzChannelID>
        <!--ro, req, string-->1
      </ptzChannelID>
      <actionName>
        <!--ro, req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--ro, opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--ro, opt, object-->
      <whiteLightDurationTime>
        <!--ro, opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--ro, opt, object-->
      <durationTime>
        <!--ro, opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
    <customOverHTTPList>
      <!--ro, opt, array, subType:object, dep:and,{$.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,customOverHTTP}-->
      <customOverHTTP>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, int-->1
        </id>
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <hostID>
          <!--ro, opt, int-->1
        </hostID>
        <customInfo>
          <!--ro, opt, string, range:[1,512]-->test
        </customInfo>
      </customOverHTTP>
    </customOverHTTPList>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

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16.9.4.6 Set the linkage triggering parameters of region exiting for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionExiting-<channelID>/notifications

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EventTriggerNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EventTriggerNotification>
    <!--opt, object-->
    <id>
      <!--req, enum, subType:string-->regionExiting
    </id>
    <notificationMethod>
      <!--ro, req, enum, subType:string-->email
    </notificationMethod>
    <notificationRecurrence>
      <!--opt, enum, subType:string-->beginning
    </notificationRecurrence>
    <notificationInterval>
      <!--opt, int, unit:ms-->1
    </notificationInterval>
    <outputIOPortID>
      <!--opt, string-->1
    </outputIOPortID>
    <dynOutputIOPortID>
      <!--opt, string-->1
    </dynOutputIOPortID>
    <videoInputID>
      <!--opt, string-->1
    </videoInputID>
    <dynVideoInputID>
      <!--opt, string-->1
    </dynVideoInputID>
    <ptzAction>
      <!--opt, object-->
      <ptzChannelID>
        <!--req, string-->1
      </ptzChannelID>
      <actionName>
        <!--req, enum, subType:string-->preset
      </actionName>
      <actionNum>
        <!--opt, int-->1
      </actionNum>
    </ptzAction>
    <WhiteLightAction>
      <!--opt, object-->
      <whiteLightDurationTime>
        <!--opt, int, range:[1,60], unit:s-->1
      </whiteLightDurationTime>
    </WhiteLightAction>
    <SupplementLightAlarm>
      <!--opt, object-->
      <durationTime>
        <!--opt, int, range:[0,90], unit:s-->5
      </durationTime>
    </SupplementLightAlarm>
  </EventTriggerNotification>
</EventTriggerNotificationList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.4.7 Set the linkage parameters of region exiting for a specified channel

Request URL

PUT /ISAPI/Event/triggers/regionExiting-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, string-->regionExiting
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string-->test
      </outputIOPortID>
      <dynOutputIOPortID>
        <!--opt, string-->test
      </dynOutputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
        <presetDurationTime>
          <!--opt, int, unit:s-->1
        </presetDurationTime>
      </ptzAction>
      <cellphoneNumber>
        <!--opt, string, range:[0,11]-->test
      </cellphoneNumber>
      <LightAudioByIPAction>
        <!--opt, object-->
        <notificationType>
          <!--req, enum, subType:string-->audio
        </notificationType>
        <notificationName>
          <!--opt, string-->10.16.99.98:8000:光
        </notificationName>
      </LightAudioByIPAction>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.4.8 Get the linkage parameters of region exiting for a specified channel

Request URL

GET /ISAPI/Event/triggers/regionExiting-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupportTriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, string-->regionEntrance
  </eventType>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
    <EventTriggerNotification>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <notificationMethod>
        <!--ro, req, enum, subType:string-->email
      </notificationMethod>
      <dynVideoInputID>
        <!--ro, opt, string-->1
      </dynVideoInputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```

16.9.4.9 Get region exiting detection parameters of all channels

Request URL

GET /ISAPI/Smart/regionExiting

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Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExiting>
    <!--ro, opt, object, region exiting-->
    <id>
      <!--ro, req, string, channel No.-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionExitingRegionList>
      <!--ro, opt, array, subType:object-->
      <RegionExitingRegion>
        <!--ro, opt, object, region exiting-->
        <id>
          <!--ro, req, string, detection area ID-->1
        </id>
        <sensitivityLevel>
          <!--ro, req, int, sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--ro, opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--ro, opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--ro, opt, enum, subType:string-->low
        </recordConfidence>
      </RegionExitingRegion>
    </RegionExitingRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionExiting>
</RegionExitingList>
```

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16.9.4.10 Set region exiting detection parameters of all channels

Request URL

PUT /ISAPI/Smart/regionExiting

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExiting>
    <!--opt, object-->
    <id>
      <!--req, string-->1
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <normalizedScreenSize>
      <!--opt, object, normalized size-->
      <normalizedScreenWidth>
        <!--req, int, normalized width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionExitingRegionList>
      <!--opt, array, subType:object-->
      <RegionExitingRegion>
        <!--opt, object-->
        <id>
          <!--req, string-->1
        </id>
        <sensitivityLevel>
          <!--req, int, sensitivity, range:[0,100]-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--opt, array, region coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <detectionTarget>
          <!--opt, enum, detection target, subType:string, desc:detection target-->all
        </detectionTarget>
        <humanMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (human body)-->true
        </humanMisinfoFilterEnabled>
        <vehicleMisinfoFilterEnabled>
          <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
        </vehicleMisinfoFilterEnabled>
        <priority>
          <!--opt, enum, priority, subType:string, desc:priority-->low
        </priority>
        <alarmConfidence>
          <!--opt, enum, subType:string-->low
        </alarmConfidence>
        <recordConfidence>
          <!--opt, enum, subType:string-->low
        </recordConfidence>
      </RegionExitingRegion>
    </RegionExitingRegionList>
    <mutexAbility opt="PDC">
      <!--opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--opt, bool, whether supports multiple scene detection-->true
    </isSupportMultiScene>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
  </RegionExiting>
</RegionExitingList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.4.11 Set parameters of region exiting detection rule by channel

Request URL

PUT /ISAPI/Smart/regionExiting/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExiting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of region exiting detection rule, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, channel No.-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionExitingRegionList>
    <!--opt, array, region list of region exiting detection, subType:object-->
    <RegionExitingRegion>
      <!--opt, object, region exiting detection region-->
      <id>
        <!--req, string, region ID-->1
      </id>
      <sensitivityLevel>
        <!--req, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--opt, enum, alarm confidence, subType:string, desc:"Low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--opt, enum, video confidence, subType:string, desc:"Low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
    </RegionExitingRegion>
  </RegionExitingRegionList>
  <mutexAbility opt="PDC">
    <!--opt, string, mutually exclusive capability: "PDC" (people counting statistics, which is not supported during region entrance detection),
attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--opt, bool, whether supports multi-scene detection-->true
  </isSupportMultiScene>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
</RegionExiting>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.4.12 Get parameters of region exiting detection rule by channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExiting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, read-only, "PDC"-people counting statistics, which is not supported during region exiting detection, attr:version{req, string,
protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <normalizedScreenSize>
    <!--ro, opt, object, normalized size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <RegionExitingRegionList>
    <!--ro, opt, array, list of region exiting, subType:object-->
    <RegionExitingRegion>
      <!--ro, opt, object, region exiting-->
      <id>
        <!--ro, req, string, region ID-->1
      </id>
      <sensitivityLevel>
        <!--ro, req, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
      </detectionTarget>
      <humanMisinfoFilterEnabled>
```

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```
        <!--ro, opt, bool, whether to enable false alarm filtering of human body detection-->true
      </humanMisinfoFilterEnabled>
      <vehicleMisinfoFilterEnabled>
        <!--ro, opt, bool, whether to enable false alarm filtering of vehicle detection-->true
      </vehicleMisinfoFilterEnabled>
      <priority>
        <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
      </priority>
      <alarmConfidence>
        <!--ro, opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </alarmConfidence>
      <recordConfidence>
        <!--ro, opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
      </recordConfidence>
      <ruleName>
        <!--ro, opt, string, range:[1,64]-->test
      </ruleName>
      <detectionTargetProperty>
        <!--ro, opt, object-->
        <human>
          <!--ro, opt, object, dep:or,{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,all},
{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,human}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->nonMotor
            </property>
          </propertyList>
        </human>
        <animal>
          <!--ro, opt, object, dep:or,{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,all},
{$.RegionExiting.RegionExitingRegionList.RegionExitingRegion.detectionTarget,eq,animal}-->
          <propertyList>
            <!--ro, opt, array, subType:object-->
            <property>
              <!--ro, opt, enum, subType:string-->large
            </property>
          </propertyList>
        </animal>
      </detectionTargetProperty>
    </RegionExitingRegion>
  </RegionExitingRegionList>
  <mutexAbility opt="PDC">
    <!--ro, opt, string, attr:opt{req, string}-->PDC
  </mutexAbility>
  <isSupportMultiScene>
    <!--ro, opt, bool, whether supports multi-scene detection-->true
  </isSupportMultiScene>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <stackTarget>
    <!--ro, opt, bool-->true
  </stackTarget>
  <targetSensitivityLevel>
    <!--ro, opt, int, range:[0,100]-->50
  </targetSensitivityLevel>
</RegionExiting>
```

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16.9.4.13 Get configuration capability of region exiting detection by channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExiting xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, ro, mutex capability, it indicates that which function is mutually exclusive with the region exiting detection:
"PDC,videoFrameRate50,videoFrameRate60,19-1920*1080@50fps,20-1920*1080@60fps,scheduled recording", attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, channel No.-->test
```

```
    </id>
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <normalizedScreenSize>
      <!--ro, opt, object, normalized screen size-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized screen width-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized screen height-->1
      </normalizedScreenHeight>
    </normalizedScreenSize>
    <RegionExitingRegionList size="4">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <RegionExitingRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, req, string-->1
        </id>
        <sensitivityLevel min="1" max="100">
          <!--ro, req, int, sensitivity, attr:min{req, int},max{req, int}-->1
        </sensitivityLevel>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinate list, subType:object, range:[3,10], desc:region coordinate list-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <maxRegionCoordinatesNum>
          <!--ro, opt, int-->1
        </maxRegionCoordinatesNum>
        <minRegionCoordinatesNum>
          <!--ro, opt, int-->1
        </minRegionCoordinatesNum>
        <detectionTarget opt="all,human,vehicle,animal,others">
          <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:detection target-->all
        </detectionTarget>
        <alarmConfidence opt="low,mediumLow,mediumHigh,high" def="low">
          <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
        </alarmConfidence>
        <recordConfidence opt="low,mediumLow,mediumHigh,high" def="low">
          <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->low
        </recordConfidence>
        <ruleName min="1" max="64">
          <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
        </ruleName>
        <detectionTargetProperty>
          <!--ro, opt, object-->
          <human>
            <!--ro, opt, object-->
            <propertyList size="1">
              <!--ro, opt, object, attr:size{req, int}-->
              <property opt="nonMotor">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
              </property>
            </propertyList>
          </human>
          <animal>
            <!--ro, opt, object-->
            <propertyList size="3">
              <!--ro, opt, object, attr:size{req, int}-->
              <property opt="large,small,poultry">
                <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
              </property>
            </propertyList>
          </animal>
        </detectionTargetProperty>
      </RegionExitingRegion>
    </RegionExitingRegionList>
    <mutexAbility opt="PDC">
      <!--ro, opt, string, attr:opt{req, string}-->PDC
    </mutexAbility>
    <isSupportMultiScene>
      <!--ro, req, bool, whether supports multiple scenes.-->true
    </isSupportMultiScene>
    <isSupportHumanMisinfoFilter>
      <!--ro, opt, bool, whether supports preventing false human body alarm, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
    </isSupportHumanMisinfoFilter>
    <isSupportVehicleMisinfoFilter>
      <!--ro, opt, bool, whether supports preventing false vehicle alarm, desc:if this function is supported, this node will be returned and its value is
true; otherwise, this node will not be returned-->true
    </isSupportVehicleMisinfoFilter>
    <isSupportTargetMultiSelect>
      <!--ro, opt, bool, whether supports selecting multiple detection targets, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will not be returned-->true
    </isSupportTargetMultiSelect>
```

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```
  <isSupportAllDayUpload>
    <!--ro, opt, bool, whether supports uploading all-day event-->true
  </isSupportAllDayUpload>
  <isSupportOverlapParams>
    <!--ro, opt, bool-->true
  </isSupportOverlapParams>
  <isSupportStackTarget>
    <!--ro, opt, bool-->true
  </isSupportStackTarget>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <targetSensitivityLevel min="0" max="100">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->50
  </targetSensitivityLevel>
</RegionExiting>
```

16.9.4.14 Set parameters of all region exiting detection regions by channel

Request URL

PUT /ISAPI/Smart/regionExiting/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameter list of all region exiting detection regions, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExitingRegion>
    <!--opt, object, parameters of region exiting detection region-->
    <id>
      <!--req, string, region ID-->1
    </id>
    <sensitivityLevel>
      <!--req, int, sensitivity, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value "all"
is mutually exclusive with "others"-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
    </priority>
    <alarmConfidence>
      <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </alarmConfidence>
    <recordConfidence>
      <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
    </recordConfidence>
  </RegionExitingRegion>
</RegionExitingRegionList>
```

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.9.4.15 Get parameters of all region exiting detection regions by channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>/regions

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <RegionExitingRegion>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string-->1
    </id>
    <sensitivityLevel>
      <!--ro, req, int, range:[0,100]-->1
    </sensitivityLevel>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[3,10]-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->1
        </positionX>
        <positionY>
          <!--ro, req, int, range:[0,1000]-->1
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <detectionTarget>
      <!--ro, opt, enum, detection target, subType:string, desc:detection target-->all
    </detectionTarget>
    <humanMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (human body)-->true
    </humanMisinfoFilterEnabled>
    <vehicleMisinfoFilterEnabled>
      <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
    </vehicleMisinfoFilterEnabled>
    <priority>
      <!--ro, opt, enum, priority, subType:string, desc:priority-->low
    </priority>
    <alarmConfidence>
      <!--ro, opt, enum, subType:string-->low
    </alarmConfidence>
    <recordConfidence>
      <!--ro, opt, enum, subType:string-->low
    </recordConfidence>
  </RegionExitingRegion>
</RegionExitingRegionList>
```

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16.9.4.16 Get parameters of a region exiting detection region of a specific channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--ro, req, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--ro, opt, array, list of area coordinates, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--ro, opt, enum, detection target, subType:string, desc:"all", "human", "vehicle", "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (human)-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--ro, opt, bool, whether to enable false alarm filtering (vehicle)-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--ro, opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--ro, opt, enum, confidence of alarm notification, subType:string, desc:confidence of alarm notification: "low","mediumLow","mediumHigh","high","low"-
->low
  </alarmConfidence>
  <recordConfidence>
    <!--ro, opt, enum, confidence of video recording, subType:string, desc:confidence of recording video: "low","mediumLow","mediumHigh","high","low"-->low
  </recordConfidence>
  <ruleName>
    <!--ro, opt, string, range:[1,64]-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object, dep:or,{$.RegionExitingRegion.detectionTarget,eq,all},{$.RegionExitingRegion.detectionTarget,eq,human}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object, dep:or,{$.RegionExitingRegion.detectionTarget,eq,all},{$.RegionExitingRegion.detectionTarget,eq,animal}-->
      <propertyList>
        <!--ro, opt, array, subType:object-->
        <property>
          <!--ro, opt, enum, subType:string-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</RegionExitingRegion>
```

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16.9.4.17 Set parameters of a region exiting detection region by channel

Request URL

PUT /ISAPI/Smart/regionExiting/<channelID>/regions/<regionID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
regionID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of a region exiting detection region, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, region ID-->1
  </id>
  <sensitivityLevel>
    <!--req, int, sensitivity, range:[0,100]-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--opt, array, region coordinate list, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget>
    <!--opt, enum, detection target type, subType:string, desc:"all", "human", "vehicle", "others" (targets except human bodies and vehicles). The value
"all" is mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of human body detection-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled>
    <!--opt, bool, whether to enable false alarm filtering of vehicle detection-->true
  </vehicleMisinfoFilterEnabled>
  <priority>
    <!--opt, enum, priority, subType:string, desc:"low", "middle", "high"-->low
  </priority>
  <alarmConfidence>
    <!--opt, enum, alarm confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </alarmConfidence>
  <recordConfidence>
    <!--opt, enum, video confidence, subType:string, desc:"low" (default value), "mediumLow", "mediumHigh", "high"-->low
  </recordConfidence>
</RegionExitingRegion>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.9.4.18 Get capability of configuring parameters of a region exiting detection region by channel

Request URL

GET /ISAPI/Smart/regionExiting/<channelID>/regions/<regionID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
regionID	string	Region No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionExitingRegion xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, region ID-->test
  </id>
  <sensitivityLevel min="1" max="100">
    <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
  </sensitivityLevel>
  <RegionCoordinatesList>
    <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
    <RegionCoordinates>
      <!--ro, opt, object, region coordinates, desc:the origin is the lower-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
  <detectionTarget opt="all,human,vehicle,animal,others">
    <!--ro, opt, enum, detection target, subType:string, attr:opt{req, string}, desc:"all", "human", "vehicle", "others" (targets except human bodies and
vehicles). The value "all" is mutually exclusive with "others"-->all
  </detectionTarget>
  <humanMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </humanMisinfoFilterEnabled>
  <vehicleMisinfoFilterEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable false alarm filtering of vehicle detection, attr:opt{req, string}-->true
  </vehicleMisinfoFilterEnabled>
  <priority opt="low,middle,high">
    <!--ro, opt, enum, priority, subType:string, attr:opt{req, string}, desc:"Low", "middle", "high"-->low
  </priority>
  <isSupportAllTargetDetectionWebDisplay>
    <!--ro, opt, bool-->false
  </isSupportAllTargetDetectionWebDisplay>
  <ruleName min="1" max="64">
    <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
  </ruleName>
  <detectionTargetProperty>
    <!--ro, opt, object-->
    <human>
      <!--ro, opt, object-->
      <propertyList size="1">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="nonMotor">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->nonMotor
        </property>
      </propertyList>
    </human>
    <animal>
      <!--ro, opt, object-->
      <propertyList size="3">
        <!--ro, opt, object, attr:size{req, int}-->
        <property opt="large,small,poultry">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->large
        </property>
      </propertyList>
    </animal>
  </detectionTargetProperty>
</RegionExitingRegion>
```

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16.9.4.19 Region exiting

EventType:regionExiting

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, event alarm, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome".
The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP--
>HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"regionExiting" (region exiting)-->regionExiting
  </eventType>
  <eventState>
    <!--ro, req, enum, event status, subType:string, desc:for durative event: "active" (valid), "inactive" (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, string, event description-->test
  </eventDescription>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <DetectionRegionList>
    <!--ro, opt, array, detection area list, subType:object-->
    <DetectionRegionEntry>
      <!--ro, opt, object, detection area-->
      <regionID>
        <!--ro, req, string, area ID-->test
      </regionID>
      <sensitivityLevel>
        <!--ro, opt, int, sensitivity, range:[0,100]-->1
      </sensitivityLevel>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--ro, opt, object, area point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->50
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->50
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <detectionTarget>
        <!--ro, opt, enum, target type, subType:string, desc:"human", "vehicle", "others"-->human
      </detectionTarget>
      <TargetRect>
        <!--ro, opt, object, target area, desc:the origin is the upper-left corner of the screen-->
        <X>
          <!--ro, req, float, target X-coordinate, range:[0.000,1.000]-->0.000
        </X>
        <Y>
          <!--ro, req, float, target Y-coordinate, range:[0.000,1.000]-->0.000
        </Y>
        <width>
          <!--ro, req, float, target width, range:[0.000,1.000]-->1.000
        </width>
        <height>
          <!--ro, req, float, target height-->1.000
        </height>
      </TargetRect>
      <sceneID>
        <!--ro, opt, int, scene ID-->1
      </sceneID>
```

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```
    </DetectionRegionEntry>
  </DetectionRegionList>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,64]-->test
  </channelName>
  <Extensions>
    <!--ro, opt, object, extended information-->
    <serialNumber>
      <!--ro, opt, string, serial No.-->test
    </serialNumber>
    <eventPush>
      <!--ro, opt, string, event notification-->test
    </eventPush>
  </Extensions>
  <detectionPictureTransType>
    <!--ro, opt, enum, picture transmission mode, subType:string, desc:"url", "binary", "base64"-->url
  </detectionPictureTransType>
  <detectionPicturesNumber>
    <!--ro, opt, int, number of pictures contained in the alarm received in arming mode, desc:this field will not be returned when there is no picture contained in the alarm-->1
  </detectionPicturesNumber>
  <bkgUrl>
    <!--ro, opt, string, background picture URL, desc:background picture URL-->test
  </bkgUrl>
  <URLCertificationType>
    <!--ro, opt, enum, picture URL authentication method, subType:string, desc:authentication mode of picture URL: "no"-no authentication, "digest"-digest authentication (it is used for local picture URL returned by NVR or DVR)-->no
  </URLCertificationType>
  <pId>
    <!--ro, opt, string, picture ID, desc:the field value is the picture Content-ID when the value of detectionPictureTransType is "binary"-->test
  </pId>
  <visibleLightURL>
    <!--ro, opt, string, visible light picture URL-->test
  </visibleLightURL>
  <thermalURL>
    <!--ro, opt, string, thermal imaging picture URL-->test
  </thermalURL>
  <targetAttrs>
    <!--ro, opt, object, transparent transmission field-->{ /*可选,透传字段,推荐使用以下三个字段*/ "deviceId": "1", /*可选, 设备ID, string类型, 最大长度为64*/ "deviceChannel": 1, /*可选, 设备通道号, integer32类型*/ "deviceName": "shebei", /*可选, 设备名称, string类型, 最大长度为128*/ }
  </targetAttrs>
  <TaskInfo>
    <!--ro, opt, object, task information submitted by data center products-->
    <taskID>
      <!--ro, req, string, task ID, range:[0,64]-->test
    </taskID>
    <algorithmType>
      <!--ro, opt, enum, algorithm type, subType:int, desc:1-"hightQualityVehicle" (high-quality vehicle),2-"hightPerformanceVehicle" (high-performance vehicle),3-"personStructModel" (person structural modeling),4-"vehicleAndPersonStructModel" (person and vehicle structural modeling),5-"videoFaceDetect" (video face detection),6-"videoFaceDetectModel" (video face detection and modeling),7-"smart" (face detection (output frame without recognizing properties)+human body detection (output properties and frame)+behavior detection)-->1
    </algorithmType>
    <taskName>
      <!--ro, opt, string, task name-->test
    </taskName>
    <algorithmID>
      <!--ro, opt, string, algorithm package ID-->test
    </algorithmID>
    <monitorPointID>
      <!--ro, opt, string, camera No.-->test
    </monitorPointID>
    <monitorPointName>
      <!--ro, opt, string, camera name-->test
    </monitorPointName>
    <streamType>
      <!--ro, opt, enum, stream type, subType:string, desc:"realtime" (real-time stream), "historyvideo" (history stream), "localvideo" (local stream)-->realtime
    </streamType>
    <ruleID>
      <!--ro, opt, string, rule ID, desc:it is used to distinguish different tasks with the same name, and the ruleID in the event uploading message indicates that the event is related to this task-->test
    </ruleID>
    <ruleCustomName>
      <!--ro, req, string, user-defined event name, range:[0,128], desc:it is defined by the caller, and it is included in the message when the alarm is uploaded. Up to 128 characters are allowed-->test
    </ruleCustomName>
    <algorithmInfo>
      <!--ro, opt, object, algorithm version information, desc:this field is required for secondary analysis-->
      <algorithmId>
        <!--ro, opt, string, algorithm package ID-->test
      </algorithmId>
      <algoVendorId>
        <!--ro, opt, string, manufacturer ID-->test
      </algoVendorId>
      <algoVersionId>
        <!--ro, opt, string, algorithm version No.-->test
      </algoVersionId>
      <algoChipType>
        <!--ro, opt, string, chip type-->test
      </algoChipType>
      <algoModelVersion>
        <!--ro, opt, string, model version-->test
      </algoModelVersion>
```

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```
    </algorithmInfo>
  </TaskInfo>
  <timeStamp>
    <!--ro, opt, string, timestamp of the frame, which triggers the alarm-->test
  </timeStamp>
  <uid>
    <!--ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits--
>test
  </uid>
  <isDataRetransmission>
    <!--ro, opt, bool, whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded
again after the device recovers.)-->true
  </isDataRetransmission>
  <ruleID>
    <!--ro, opt, int, rule No., range:[0,7], desc:rule No.-->0
  </ruleID>
  <ruleName>
    <!--ro, opt, string, rule name, range:[0,32], desc:rule name-->test
  </ruleName>
  <visibleLightAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </visibleLightAbsoluteHigh>
  <thermalAbsoluteHigh>
    <!--ro, opt, object-->
    <elevation>
      <!--ro, opt, float, range:[-90.00,270.00], step:0.001-->0.00
    </elevation>
    <azimuth>
      <!--ro, opt, float, range:[0,360.00], step:0.001-->0.00
    </azimuth>
    <absoluteZoom>
      <!--ro, opt, float, zooming ratio, range:[0,10000.00]-->0.00
    </absoluteZoom>
    <focus>
      <!--ro, opt, int, range of the focus, range:[1,100000], step:1-->1
    </focus>
  </thermalAbsoluteHigh>
  <deviceLocation>
    <!--ro, opt, object-->
    <longitudeType>
      <!--ro, req, enum, subType:string-->W
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, subType:string, desc:"S"-south, "N"-north-->S
    </latitudeType>
    <longitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, degree(s), range:[0,180]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </longitude>
    <latitude>
      <!--ro, req, object-->
      <degree>
        <!--ro, req, int, range:[0,90]-->1
      </degree>
      <minute>
        <!--ro, req, float, minute(s), range:[0,60]-->0.0
      </minute>
      <sec>
        <!--ro, req, float, range:[0,60]-->0.0
      </sec>
    </latitude>
  </deviceLocation>
  <laserRanging>
    <!--ro, opt, int, range:[1,10000], step:1, unit:m-->1
  </laserRanging>
</EventNotificationAlert>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
regionExiting	[Message content]	application/xml	--	--	--
visibleLightImage	[Binary picture data]	image/jpeg	visibleLightImage	visibleLightImage.jpg	--
thermalImage	[Binary picture data]	image/jpeg	thermalImage	thermalImage.jpg	--
targetImage	[Binary picture data]	image/jpeg	targetImage	targetImage.jpg	--
POSInfo	[Stream data]	application/octet-stream	POSInfo	POSInfo.dat	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

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16.10 Multi-Target-Type Detection

16.10.1 Global Target Detection

16.10.1.1 Get the global target detection capability

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/humanRecognition/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HumanRecognition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->test
  </id>
  <normalizedScreenSize>
    <!--ro, opt, object-->
    <normalizedScreenWidth>
      <!--ro, req, int-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
```

```
      <!--ro, req, int-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <HumanRecognitionRegionList size="8">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <HumanRecognitionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string-->test
      </id>
      <regionName min="1" max="10">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->test
      </regionName>
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <LineCoordinatesList size="2">
        <!--ro, opt, array, subType:object, range:[0,2], attr:size{req, int}-->
        <LineCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </LineCoordinates>
      </LineCoordinatesList>
    </HumanRecognitionRegion>
  </HumanRecognitionRegionList>
  <isSupportHumanMisinfoFilter>
    <!--ro, opt, bool-->true
  </isSupportHumanMisinfoFilter>
  <isSupportPicRecognition>
    <!--ro, opt, bool-->true
  </isSupportPicRecognition>
  <HumanRecognitionModeSearchCap>
    <!--ro, opt, object-->
    <searchTargetsNumMax>
      <!--ro, req, int-->5
    </searchTargetsNumMax>
    <HumanMode>
      <!--ro, opt, object-->
      <age_group opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
      </age_group>
      <gender opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
      </gender>
      <jacet_color opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
      </jacet_color>
      <glass opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->no
      </glass>
      <bag opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->yes
      </bag>
      <ride opt="test">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
      </ride>
      <similarity min="0" max="100">
        <!--ro, opt, float, range:[0.0,100.0], attr:min{req, int},max{req, int}-->0.000
      </similarity>
    </HumanMode>
    <eventType opt="unlimit,humanRecognition">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unlimit
    </eventType>
  </HumanRecognitionModeSearchCap>
  <analysisParamList>
    <!--ro, opt, array, subType:object-->
    <analysisParam>
      <!--ro, opt, object-->
      <analysisMode opt="normal,groupBehavior,individualBehavior">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
      </analysisMode>
      <captureInterval min="5" max="3600" def="12">
        <!--ro, opt, int, range:[5,3600], unit:s, attr:min{req, int},max{req, int},def{req, int}-->12
      </captureInterval>
      <maxRegionNum>
        <!--ro, opt, int-->1
```

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```
        </maxRegionNum>
        <cruiseLinkageCapture opt="true,false">
          <!--ro, opt, bool, attr:opt{req, string}-->true
        </cruiseLinkageCapture>
        <analysisControl opt="auto,non-auto">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->auto
        </analysisControl>
        <timeInterval min="1" max="60" def="12">
          <!--ro, opt, int, range:[0,60], unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
        </timeInterval>
      </analysisParam>
    </analysisParamList>
    <isSupportConfidenceConfig>
      <!--ro, opt, bool-->true
    </isSupportConfidenceConfig>
    <InterPupileRegion>
      <!--ro, opt, object-->
      <positionX min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </positionX>
      <positionY min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </positionY>
      <width min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </width>
      <height min="0" max="1000">
        <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->1
      </height>
    </InterPupileRegion>
    <sid min="1" max="8">
      <!--ro, opt, string, attr:min{req, int},max{req, int}-->test
    </sid>
    <sceneDurationTime min="1" max="86400" def="5">
      <!--ro, opt, int, range:[1,86400], unit:s, attr:min{req, int},max{req, int},def{req, int}-->1
    </sceneDurationTime>
    <targetCutout>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, opt, bool-->true
      </enabled>
      <size>
        <!--ro, req, int-->1
      </size>
      <zoom>
        <!--ro, req, int-->1
      </zoom>
    </targetCutout>
    <isSupportpicOverlay>
      <!--ro, opt, bool-->true
    </isSupportpicOverlay>
    <isSupportRealTimeShow>
      <!--ro, opt, bool-->true
    </isSupportRealTimeShow>
    <enabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </enabled>
  </HumanRecognition>
```

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16.10.2 Multi-Target-Type Detection

16.10.2.1 Set arming schedules of multiple multi-target-type detections of all channels

Request URL

PUT /ISAPI/Event/schedules/mixedTargetDetection

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<MixedTargetDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, undefined, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string, ID, desc:ID-->mixedTargetDetection-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->mixedTargetDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming the task by day each week-->
        <dayOfWeek>
          <!--opt, enum, subType:int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--opt, object-->
          <vehicleDetectSceneID>
            <!--opt, int-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--opt, int-->1
          </sceneID>
          <SceneList>
            <!--opt, array, subType:object-->
            <Scene>
              <!--opt, object, scene information-->
              <sceneID>
                <!--req, string, range:[1,16]-->1
              </sceneID>
              <durationTime>
                <!--req, int, duration-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object, arming the task by day each week-->
        <TimeRange>
          <!--req, object-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
    <parkTime>
      <!--opt, int, unit:s-->1
    </parkTime>
  </Schedule>
</MixedTargetDetectionScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.10.2.2 Get arming schedules of multiple multi-target-type detections of all channels

Request URL

GET /ISAPI/Event/schedules/mixedTargetDetection

Query Parameter

None

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<MixedTargetDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, refer to the message XML_Schedule for details, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, the message XML_Schedule for details-->
    <id>
      <!--ro, req, string, ID-->mixedTargetDetection-1
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->test
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <dayOfWeek>
          <!--ro, opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-
->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
        <ScheduleProperity>
          <!--ro, opt, object-->
          <vehicleDetectSceneID>
            <!--ro, opt, int-->1
          </vehicleDetectSceneID>
          <sceneID>
            <!--ro, opt, int-->1
          </sceneID>
          <SceneList>
            <!--ro, opt, array, subType:object-->
            <Scene>
              <!--ro, opt, object, scene information-->
              <sceneID>
                <!--ro, req, string, range:[1,16]-->1
              </sceneID>
              <durationTime>
                <!--ro, req, int, duration-->1
              </durationTime>
            </Scene>
          </SceneList>
        </ScheduleProperity>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, arming time-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
    <parkTime>
      <!--ro, opt, int, unit:s-->1
    </parkTime>
  </Schedule>
</MixedTargetDetectionScheduleList>
```

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16.10.2.3 Get the captured picture information of multi-target-detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/alarmInfo?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "MixedTargetDetectionResult": {
    /*ro, opt, object*/
        "ResultList": [
        /*ro, req, array, subType:object*/
            {
                "captureTime":  "2004-05-03T17:30:08.500+08:00",
                /*ro, opt, date*/
                "faceImageURL":  "test",
                /*ro, opt, picurl*/
                "humanImageURL":  "test",
                /*ro, opt, picurl*/
                "targetID":  1,
                /*ro, opt, int*/
                "Face": {
                /*ro, opt, object*/
                    "Rect": {
                    /*ro, req, object*/
                        "height":  1.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    },
                    "Property": [
                    /*ro, opt, array, subType:object*/
                        {
                            "description":  "age",
                            /*ro, opt, enum, subType:string*/
                            "value":  "23"
                            /*ro, opt, string*/
                        }
                    ],
                    "faceImageURL":  "test",
                    /*ro, opt, string*/
                    "faceBackgroundImageURL":  "test"
                    /*ro, opt, string*/
                },
                "Human": {
                /*ro, opt, object*/
                    "Rect": {
                    /*ro, req, object*/
                        "height":  1.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "y":  0.000
                        /*ro, req, float, range:[0.000,1.000]*/
                    },
                    "Property": [
                    /*ro, opt, array, subType:object*/
                        {
                            "description":  "jacketColor",
                            /*ro, opt, enum, subType:string*/
                            "value":  "23"
                            /*ro, opt, string*/
                        }
                    ],
                    "modeldata":  "test",
                    /*ro, opt, string*/
                    "humanImageURL":  "test",
                    /*ro, opt, string*/
                    "humanBackgroundImageURL":  "test"
                    /*ro, opt, string*/
                },
                "NonMotor": {
                /*ro, opt, object*/
                    "Rect": {
                    /*ro, req, object*/
                        "height":  1.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "width":  1.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "x":  0.000,
                        /*ro, req, float, range:[0.000,1.000]*/
```

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```
                    /*ro, req, float, range:[0.000,1.000]*/
                    "y": 0.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, subType:object*/
                    {
                        "description": "nonMotorType",
                        /*ro, opt, enum, subType:string*/
                        "value": "23"
                        /*ro, opt, string*/
                    }
                ],
                "nonMotorImageURL": "test"
                /*ro, opt, string*/
            },
            "Vehicle": {
            /*ro, opt, object*/
                "VehicleRect": {
                /*ro, req, object*/
                    "height": 1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "width": 1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "x": 0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "y": 0.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "PlateRect": {
                /*ro, req, object*/
                    "height": 1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "width": 1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "x": 0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "y": 0.000
                    /*ro, req, float, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, subType:object*/
                    {
                        "description": "plateNo",
                        /*ro, opt, enum, subType:string*/
                        "value": "23"
                        /*ro, opt, string*/
                    }
                ],
                "vehicleImageURL": "test",
                /*ro, opt, string*/
                "plateImageURL": "test",
                /*ro, opt, string*/
                "vehicleBackgroundImageURL": "test"
                /*ro, opt, string*/
            },
            "AbsoluteHigh": {
            /*ro, opt, object*/
                "elevation": 11,
                /*ro, opt, int*/
                "azimuth": 22,
                /*ro, opt, int*/
                "absoluteZoom": 33
                /*ro, opt, int*/
            },
            "sid": "123",
            /*ro, opt, string*/
            "sceneName": "场景1"
            /*ro, opt, string*/
        }
    ]
  }
}
```


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16.10.2.4 Get advanced configuration capability of multi-target-type detection algorithm library

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/algParam/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetAlgParamCap": {
    /*ro, req, object, algorithm configuration capability of multi-target-type detection*/
        "enabled":  "true,false",
        /*ro, req, string, whether to enable algorithm configuration*/
        "snapMode": {
        /*ro, opt, object, Capture Mode*/
            "@opt":  "best, quick",
            /*ro, opt, string*/
            "@def":  "best"
            /*ro, opt, string, the default value*/
        },
        "bestSnapThreshold": {
        /*ro, opt, object, picture number threshold of optimal capture: 0 to 20 frames; it is valid only when snapMode is "best"*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  100,
            /*ro, opt, int, maximum value*/
            "@def":  20
            /*ro, req, int, the default value*/
        },
        "exposureEnabled":  "true,false",
        /*ro, opt, string, whether to enable face picture exposure*/
        "brightRef": {
        /*ro, opt, object, reference brightness*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  100,
            /*ro, opt, int, maximum value*/
            "@def":  50
            /*ro, opt, int, the default value*/
        },
        "expDurationTime": {
        /*ro, opt, object, shortest exposure time*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  3600,
            /*ro, opt, int, maximum value*/
            "@def":  60
            /*ro, req, int, the default value*/
        },
        "faceFilteringTime": {
        /*ro, opt, object, Face Filtering Time*/
            "@min":  0,
            /*ro, opt, int, minimum value*/
            "@max":  100,
            /*ro, opt, int, maximum value*/
            "@def":  5
            /*ro, req, int, the default value*/
        },
        "UploadDataType": {
        /*ro, opt, object*/
            "faceEnabled":  "true,false",
            /*ro, opt, string*/
            "humanEnabled":  "true,false",
            /*ro, opt, string*/
        },
        "RemoveDuplicateCap": {
        /*ro, opt, object, remove duplicates of captured face pictures; it takes effect when the value of snapMode is "best"*/
            "enabled":  "true,false",
            /*ro, req, string, whether to enable algorithm configuration*/
            "threshold": {
            /*ro, opt, object, threshold of removing duplicates, when the picture similarity is larger than the value, the picture is duplicated*/
                "@min":  70,
                /*ro, opt, int, minimum value*/
                "@max":  100,
                /*ro, opt, int, maximum value*/
                "@def":  88
                /*ro, req, int, the default value*/
            },
            "faceScore": {
            /*ro, opt, object, the face grading threshold for removing duplicates. Only when the face grading is larger than the configured threshold, the
face picture will be compared for removing duplicates, in case that low quality face pictures influence the comparison effect*/
                "@min":  20,
                /*ro, opt, int, minimum value*/
                "@max":  60,
                /*ro, opt, int, maximum value*/
                "@def":  20
                /*ro, opt, int, the default value*/
            },
            "updateTime": {
            /*ro, opt, object, face modal data updating time interval (the time interval between the modal data being added into the library and being
deleted)*/
                "@min":  0,
                /*ro, opt, int, minimum value*/
                "@max":  300,
                /*ro, opt, int, maximum value*/
                "@def":  60
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```
                /*ro, opt, int, the default value*/
            }
        },
    }
}
```

16.10.2.5 Get advanced parameters of multi-target-type detection algorithm library

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/algParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetAlgParam": {
    /*ro, req, object, algorithm configuration capability of multi-target-type detection*/
        "enabled": true,
        /*ro, req, bool, whether to enable algorithm configuration*/
        "targetSpeed": 3,
        /*ro, opt, int, target speed*/
        "sensitiveLevel": 3,
        /*ro, opt, int, sensitivity level*/
        "snapMode": "best",
        /*ro, opt, enum, capture mode, subType:string, desc:"best" (optimal capture), "quick" (fast capture)*/
        "bestSnapInterval": 1,
        /*ro, opt, int, it is valid only snapMode is "best", range:[1,255], desc:it is valid only snapMode is "best"*/
        "bestSnapThreshold": 20,
        /*ro, opt, int, picture number threshold of optimal capture, range:[0,100], desc:it is valid only snapMode is "best"*/
        "quickSnapThreshold": 20,
        /*ro, opt, int, picture number threshold of fast capture, range:[20,80], desc:it is valid only snapMode is "quick"*/
        "exposureEnabled": true,
        /*ro, opt, bool, whether to enable face picture exposure*/
        "brightRef": 50,
        /*ro, opt, int, reference brightness, range:[0,100]*/
        "expDurationTime": 60,
        /*ro, req, int, shortest exposure time, range:[0,3600], unit:s*/
        "faceFilteringTime": 5,
        /*ro, req, int, face filtering time, range:[0,100], unit:s*/
        "bestSnapNum": 1,
        /*ro, opt, int, range:[0,10]*/
        "longestSnapTime": 1,
        /*ro, opt, int, longest capture duration, range:[1,10], unit:s, desc:it is valid when snapMode is "quick"*/
        "UploadDataType": {
        /*ro, opt, object*/
            "faceEnabled": true,
            /*ro, opt, bool*/
            "humanEnabled": true,
            /*ro, opt, bool*/
            "nonMotorEnabled": true,
            /*ro, opt, bool*/
            "vehiceEnabled": true
            /*ro, opt, bool*/
        },
        "RemoveDuplicate": {
        /*ro, req, object, remove duplicates of captured face pictures, desc:it is valid when snapMode is "best"*/
            "enabled": true,
            /*ro, req, bool, whether to enable algorithm configuration*/
            "threshold": 88,
            /*ro, req, int, threshold of removing duplicates, desc:when the picture similarity is larger than the value, the picture is duplicated*/
            "faceScore": 20,
            /*ro, opt, int, the face grading threshold for removing duplicate, desc:only when the face grading is larger than the configured threshold, the face picture will be compared for removing duplicates, in case that low quality face pictures influence the comparison effect*/
            "updateTime": 60
            /*ro, opt, int, duplicates removing library update time, range:[0,300], desc:duplicates removing library update time*/
        },
        "uploadWithoutLicense": true,
        /*ro, opt, bool*/
        "panoramaTargetRemoveDuplicate": false,
        /*ro, opt, bool*/
        "miscatchFilterEnabled": false,
        /*ro, opt, bool*/
        "capturingPeopleEnabled": true,
        /*ro, opt, bool*/
        "lineCrossingCaptureSensitivity": 50,
        /*ro, opt, int, range:[0,100]*/
        "relatedMixedTargetUidTime": 30
```

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```
        "relatedMixedTargetDuration": 30,
        /*ro, opt, int, range:[5,30], step:1, unit:s*/
        "nonMotorPlateCustomText": "test",
        /*ro, opt, string, range:[0,16]*/
        "filterDataType": {
        /*ro, opt, object*/
            "retrogradeEnabled": true,
            /*ro, opt, bool*/
            "mannedEnabled": true,
            /*ro, opt, bool*/
            "unproopedHelmetEnabled": true,
            /*ro, opt, bool*/
            "unblockedBicycleEnabled": true,
            /*ro, opt, bool*/
            "blockedBicycleEnabled": true
            /*ro, opt, bool*/
        },
        "nonMotorOverspeedThreshoid": 30,
        /*ro, opt, int, range:[10,50], step:1, unit:km/h*/
        "vehicleOverspeedThreshoid": 40,
        /*ro, opt, int, range:[30,80], step:1, unit:km/h*/
        "vehicleDisplacementFilterEnabled": true,
        /*ro, opt, bool*/
        "vehicleDisplacementFilterThreshold": 30,
        /*ro, opt, int, range:[1,100]*/
        "framesCaptureEnabled": false,
        /*ro, opt, bool*/
        "facePosture": "normal",
        /*ro, opt, enum, subType:string*/
        "snapAlgLevel": 1
        /*ro, opt, int, range:[1,5]*/
    }
}
```

16.10.2.6 Set advanced parameters of multi-target-type detection algorithm library

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/algParam?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "MixedTargetAlgParam": {
    /*req, object, algorithm configuration capability of multi-target-type detection*/
        "enabled": true,
        /*req, bool, whether to enable algorithm configuration*/
        "targetSpeed": 3,
        /*opt, int, target generation speed, range:[1,5]*/
        "sensitiveLevel": 3,
        /*opt, int, sensitivity level, range:[1,5]*/
        "snapMode": "best",
        /*opt, enum, capture mode, subType:string, desc:"best" (optimal capture), "quick" (fast capture)*/
        "bestSnapInterval": 1,
        /*opt, int, it is valid when the snapMode is "best", range:[1,255]*/
        "bestSnapThreshold": 20,
        /*opt, int, picture number threshold of optimal capture: 0 to 20 frames, it is valid when snapMode is "best", range:[0,100]*/
        "quickSnapThreshold": 50,
        /*opt, int, picture number threshold of fast capture: 20 to 80 frames, it is valid when snapMode is "quick", range:[20,80]*/
        "exposureEnabled": true,
        /*opt, bool, whether to enable face picture exposure*/
        "brightRef": 50,
        /*opt, int, reference brightness, range:[0,100]*/
        "expDurationTime": 60,
        /*opt, int, minimum exposure time, range:[0,3600]*/
        "faceFilteringTime": 5,
        /*opt, int, face filtering time, range:[0,100]*/
        "bestSnapNum": 1,
        /*opt, int, optimal capture times, range:[0,10]*/
        "longestSnapTime": 5,
        /*opt, int, maximum capture duration: 0 to 10 seconds, it is valid when snapMode is "quick", range:[1,10]*/
        "UploadDataType": {
        /*opt, object*/
            "faceEnabled": true,
            /*opt, bool*/
            "humanEnabled": true,
            /*opt, bool*/
            "nonMotorEnabled": true,
            /*opt, bool*/
            "vehiceEnabled": true
            /*opt, bool*/
        }
```

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```
        },
        "RemoveDuplicate": {
        /*opt, object, remove duplicates of captured face pictures, it is valid when snapMode is "best"*/
            "enabled": true,
            /*req, bool, whether to enable algorithm configuration*/
            "threshold": 88,
            /*opt, int, threshold of removing duplicates, when the picture similarity exceeds the threshold, the picture is duplicated, range:[70,100]*/
            "faceScore": 20,
            /*opt, int, the face grading threshold for removing duplicates, only when the face grading exceeds the configured threshold, the face picture
will be compared for removing duplicates, in case that low quality face pictures leads to less-effective comparison, range:[20,60]*/
            "updateTime": 60
            /*opt, int, range:[0,300]*/
        },
        "uploadWithoutLicense": true,
        /*opt, bool*/
        "panoramaTargetRemoveDuplicate": true,
        /*opt, bool*/
        "miscatchFilterEnabled": true,
        /*opt, bool*/
        "capturingPeopleEnabled": true,
        /*opt, bool*/
        "lineCrossingCaptureSensitivity": 50,
        /*opt, int, range:[0,100]*/
        "relatedMixedTargetUidTime": 30,
        /*opt, int, range:[5,30], step:1, unit:s*/
        "nonMotorPlateCustomText": "test",
        /*opt, string, range:[0,16]*/
        "filterDataType": {
        /*opt, object*/
            "retrogradeEnabled": true,
            /*opt, bool*/
            "mannedEnabled": true,
            /*opt, bool*/
            "unproopedHelmetEnabled": true,
            /*opt, bool*/
            "unblockedBicycleEnabled": true,
            /*opt, bool*/
            "blockedBicycleEnabled": true
            /*opt, bool*/
        },
        "nonMotorOverspeedThreshoid": 30,
        /*opt, int, range:[10,50], step:1, unit:km/h*/
        "vehicleOverspeedThreshoid": 40,
        /*opt, int, range:[30,80], step:1, unit:km/h*/
        "vehicleDisplacementFilterEnabled": true,
        /*opt, bool*/
        "vehicleDisplacementFilterThreshold": 30,
        /*opt, int, range:[1,100]*/
        "framesCaptureEnabled": false,
        /*opt, bool*/
        "facePosture": "normal",
        /*opt, enum, subType:string*/
        "snapAlgLevel": 1
        /*opt, int, range:[1,5]*/
    }
}
```

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Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode": "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx": "0x00000000"
    /*ro, opt, string*/
}
```

16.10.2.7 Get the rule parameters capability of frequently appeared person detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetDetectionCap": {
    /*ro, opt, object*/
        "enabled":  "true,false",
        /*ro, req, string, whether to enable multi-target-type detection*/
        "TypesDetectionCap": [
        /*ro, opt, array, subType:object*/
            {
                "ruleMode":  "human,face,vehice,nonMotor",
                /*ro, opt, string, rule mode*/
                "enabled":  "true,false",
                /*ro, opt, string, whether to enable the function*/
                "Property": [
                /*ro, opt, array, subType:object*/
                    {
                        "description":  "playMobilePhone",
                        /*ro, opt, string, attributes description*/
                        "enabled":  "true,false"
                        /*ro, opt, string, whether to enable multi-target-type detection*/
                    }
                ]
            }
        ],
        "RuleInfoCap": {
        /*ro, opt, object, rule information, desc:rule information,currently,only one rule is supported*/
            "maxSize":  1,
            /*ro, req, int, capability of maximum queue management rules*/
            "ruleID": {
            /*ro, req, object, rule ID*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "ruleMode": {
            /*ro, opt, object, rule mode, desc:rule type: face + human body + vehicle + non-motor vehicle,human body + vehicle + non-motor vehicle*/
                "@opt":  "face_human_vehice_nonMotor,human_vehice_nonMotor,face_human",
                /*ro, req, string, options*/
                "@def":  "face_human_vehice_nonMotor"
                /*ro, req, string, the default value*/
            },
            "RegionCap": {
            /*ro, req, object, region that configured with rule, desc:polygon with 3 to 10 edges*/
                "minSize":  3,
                /*ro, req, int, capability of minimum edges of rule region*/
                "maxSize":  10,
                /*ro, req, int, capability of maximum queue management rules*/
                "x": {
                /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                }
            },
            "InterPupileRegionCap": {
            /*ro, req, object, minimum rectangle region to detect pupil distance*/
                "x": {
                /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object, float,Y-coordinate,the value is between 0.000 and 1, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000
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            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "height": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        }
    },
    "MaxInterPupilRegionCap": {
    /*ro, opt, object*/
        "x": {
        /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "width": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        },
        "height": {
        /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
            "@min":  0.000,
            /*ro, req, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000,
            /*ro, req, float, the maximum value, range:[0.000,1.000]*/
            "#text":  0.120
            /*ro, opt, float, range:[0.000,1.000]*/
        }
    },
    "LaneLineInfoCap": {
    /*ro, opt, object, lane information capability, desc:it is valid when ruleMode is "face_human_vehice"*/
        "LaneListCap": {
        /*ro, req, object, lane list capability*/
            "maxSize":  4,
            /*ro, req, int, capability of maximum left lane lines*/
            "LeftLaneLineCap": {
            /*ro, req, object, left lane line capability*/
                "maxSize":  2,
                /*ro, req, int, capability of maximum queue management rules*/
                "x": {
                /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                }
            }
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        },
        "RightLaneLineCap": {
        /*ro, req, object, right lane line capability*/
            "maxSize":  2,
            /*ro, req, int, capability of maximum queue management rules*/
            "x": {
            /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            },
            "y": {
            /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                "@min":  0.000,
                /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            }
        }
    },
    "alarmDisplayMode": {
    /*ro, opt, object*/
        "@opt":  "face_human_vehice_nonMotor,human_vehice_nonMotor,face_human",
        /*ro, opt, string*/
        "@def":  "face_human_vehice_nonMotor"
        /*ro, opt, string*/
    },
    "retrogradeDetectionList": {
    /*ro, opt, object*/
        "@size":  2,
        /*ro, opt, int*/
        "retrogradeDetectionLine": {
        /*ro, opt, object*/
            "@size":  2,
            /*ro, opt, int*/
            "x": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            },
            "y": {
            /*ro, req, object*/
                "@min":  0.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "@max":  1.000,
                /*ro, req, float, range:[0.000,1.000]*/
                "#text":  0.120
                /*ro, opt, float, range:[0.000,1.000]*/
            }
        },
        "retrogradeDirectionLine": {
        /*ro, opt, object*/
            "startPoint": {
            /*ro, req, object*/
                "x": {
                /*ro, req, object*/
                    "@min":  0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object*/
                    "@min":  0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                }
            },
            "stopPoint": {
            /*ro, req, object*/
                "x": {
                /*ro, req, object*/
                    "@min":  0.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
```

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```
                    "y": {
                    /*ro, req, object*/
                        "@min":  0.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "@max":  1.000,
                        /*ro, req, float, range:[0.000,1.000]*/
                        "#text":  0.120
                        /*ro, opt, float, range:[0.000,1.000]*/
                    }
                }
            }
        }
    },
    "KeyRegionInfo": {
    /*ro, opt, object, key region information*/
        "enabled":  "true,false",
        /*ro, req, string, whether to enable the key region*/
        "RegionList": {
        /*ro, opt, object, key region array*/
            "maxSize":  8,
            /*ro, req, int, capability of maximum queue management rules*/
            "id": {
            /*ro, req, object, region ID*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  8
                /*ro, req, int, the maximum value*/
            },
            "Region": {
            /*ro, req, object, key region frame (rectangle)*/
                "x": {
                /*ro, req, object, X-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "y": {
                /*ro, req, object, Y-coordinate, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "width": {
                /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                },
                "height": {
                /*ro, req, object, rectangle width, desc:the origin is the lower-left corner of the screen*/
                    "@min":  0.000,
                    /*ro, req, float, the minimum value, range:[0.000,1.000]*/
                    "@max":  1.000,
                    /*ro, req, float, the maximum value, range:[0.000,1.000]*/
                    "#text":  0.120
                    /*ro, opt, float, range:[0.000,1.000]*/
                }
            }
        }
    },
    "MultiScene": {
    /*ro, opt, object, multi-scene information*/
        "maxSceneNum":  8,
        /*ro, req, int, maximum number of scenes*/
        "sid": {
        /*ro, req, object, scene ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  8
            /*ro, req, int, the maximum value*/
        },
        "sceneName": {
        /*ro, opt, object, scene name*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the minimum value*/
        },
        "parkTime": {
        /*ro, opt, object*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  256
            /*ro, req, int, the maximum value*/
        }
```

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```
        },
        "isSupportCapturePicture":  true,
        /*ro, opt, bool, whether supports setting capture parameters for multi-target-type detection, desc:whether supports setting capture parameters for
multi-target-type detection,if not supports,this node will not be returned*/
        "isSupportCapturePicOverlap":  true,
        /*ro, opt, bool, whether supports setting parameters of overlaying information on captured pictures, desc:whether supports setting parameters of
overlaying information on captured pictures,if not supports,this node will not be returned*/
        "isSupportAlgParam":  true,
        /*ro, opt, bool, whether supports setting advanced algorithm library parameters for multi-target-type detection, desc:whether supports setting
advanced algorithm library parameters for multi-target-type detection,if not supports,this node will not be returned*/
        "isSupportAlarmInfoGet":  true,
        /*ro, opt, bool, whether supports getting captured multi-target-type information, desc:whether supports getting captured multi-target-type
information,"true"-support,"false"-not support*/
        "province": {
        /*ro, opt, object, reserved, desc:reserved*/
            "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,255",
            /*ro, opt, string, options*/
            "@def":  33
            /*ro, opt, int, the default value*/
        },
        "ruleModeChangedPrompt": {
        /*ro, opt, object*/
            "opt":  "prompt1"
            /*ro, req, string, options, desc:options*/
        },
        "MultiSceneMode": {
        /*ro, opt, object*/
            "sceneMode": {
            /*ro, opt, object*/
                "@opt":  "normal,expert",
                /*ro, req, string, options*/
                "@def":  "normal"
                /*ro, req, string, the default value*/
            },
            "normalSceneNum": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  1
                /*ro, req, int, the maximum value*/
            },
            "expertSceneNum": {
            /*ro, opt, object*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  8
                /*ro, req, int, the maximum value*/
            }
        },
        "isSupportOverlay":  true,
        /*ro, opt, bool, whether it supports OSD overlay of mixed targets, desc:whether it supports OSD overlay of mixed targets,"true" will be returned for
supporting; Otherwise,this node will not be returned*/
        "pictureQuality": {
        /*ro, opt, object, picture quality*/
            "@opt":  "high,middle,low",
            /*ro, req, string*/
            "@def":  "middle"
            /*ro, req, string, the default value*/
        },
        "isSupportBinaryPicUp":  true,
        /*ro, opt, bool*/
        "convertBinToBmpEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, req, array, range, subType:bool*/
        },
        "MEF": {
        /*ro, opt, object*/
            "enabled":  "true,false",
            /*ro, req, string, whether to enable multi-target-type detection, desc:whether to enable multi-target-type detection*/
            "offsetTimes": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, unit:min, desc:the minimum value*/
                "@max":  10,
                /*ro, req, int, maximum value, unit:min*/
                "@def":  1
                /*ro, req, int, the default value, unit:min*/
            },
            "leftOffsetTimes": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, unit:s, desc:the minimum value*/
                "@max":  300,
                /*ro, req, int, maximum value, unit:s*/
                "@def":  60
                /*ro, req, int, the default value, unit:s*/
            },
            "rightOffsetTimes": {
            /*ro, opt, object*/
                "@min":  0,
                /*ro, req, int, the minimum value, unit:s, desc:the minimum value*/
                "@max":  30,
```

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```
                /*ro, req, int, maximum value, unit:s*/
                "@def":  30
                /*ro, req, int, the default value, unit:s*/
            }
        },
        "plateInfo": {
        /*ro, opt, object*/
            "region": {
            /*ro, opt, object*/
                "@opt": ["AFandAM"],
                /*ro, opt, array, subType:string*/
                "@def":  "APAC"
                /*ro, opt, string*/
            },
            "CRIndex": {
            /*ro, opt, object*/
                "@opt": ["0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "21", "22",
"23", "25", "26", "28", "29", "30", "31", "33", "34", "35", "36", "37", "38", "39", "40", "42", "43", "44", "45", "46", "47", "49", "51", "53", "55", "59",
"60", "62", "65", "68", "70", "71", "72", "73", "76", "77", "84", "87", "89", "90", "94", "95", "91", "104", "107", "108", "110", "114", "115", "130",
"137", "139", "140", "144", "145", "146", "151", "156", "157", "169", "174", "175", "177", "180", "181", "188", "206", "210", "215", "216", "217", "218",
"219", "220", "221", "222", "227", "228", "256"]
                /*ro, opt, array, subType:string*/
            },
            "relatedRegionAndCRList": [
            /*ro, opt, array, subType:object*/
                {
                    "region":  "AFandAM",
                    /*ro, req, enum, subType:string*/
                    "CRIndex": {
                    /*ro, opt, object*/
                        "@opt": ["0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12", "13", "14", "15", "16", "17", "18", "19", "20", "21",
"22", "23", "25", "26", "28", "29", "30", "31", "33", "34", "35", "36", "37", "38", "39", "40", "42", "43", "44", "45", "46", "47", "49", "51", "53", "55",
"59", "60", "62", "65", "68", "70", "71", "72", "73", "76", "77", "84", "87", "89", "90", "94", "95", "91", "104", "107", "108", "110", "114", "115", "130",
"137", "139", "140", "144", "145", "146", "151", "156", "157", "169", "174", "175", "177", "180", "181", "188", "206", "210", "215", "216", "217", "218",
"219", "220", "221", "222", "227", "228", "256"]
                        /*ro, opt, array, subType:string*/
                    }
                }
            ]
        },
        "nonMotorizedViolationUpload": {
        /*ro, opt, object*/
            "postEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            },
            "illegalEnabled": {
            /*ro, opt, object*/
                "@opt": [true, false]
                /*ro, req, array, subType:bool*/
            }
        }
    },
    "isSupportLineCrossingCapture":  true,
    /*ro, opt, bool*/
    "isSupportOneClickZoom":  true
    /*ro, opt, bool*/
}
```


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16.10.2.8 Get configuration capability of overlaying information on multi-target-type detection pictures

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capturePicOverlap/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "MixedTargetCapturePicOverlapCap": {
    /*req, object, configuration capability of overlaying information on multi-target-type detection pictures*/
        "overlapItemNum": {
        /*opt, object*/
            "@max":  1
            /*req, int, the maximum value*/
        },
        "overlapItemId": {
        /*opt, object*/
            "@min":  1,
            /*opt, int, the minimum value*/
            "@max":  "3"
            /*req, string, the maximum value*/
        },
        "itemType": {
        /*opt, object, attribute details of different overlay element types*/
            "@opt":
"cameraNo,captureTime,positionInfo,motionDirection,jacketColor,trousersColor,jacketType,trousersType,bag,things,hat,plateNo,vehicleLogo,vehicleType,GPS,enterLeaveTime,hatStyle,cyclingType"
            /*opt, string, attribute details of different overlay element types*/
        },
        "targetAttribute": {
        /*opt, object*/
            "@opt":
"motionDirection,jacketColor,trousersColor,jacketType,trousersType,bag,things,hat,plateNo,vehicleLogo,vehicleType,enterLeaveTime,hatStyle,cyclingType"
            /*opt, string, attribute details of different overlay element types*/
        }
    }
}
```

Response Message

```
{
    "MixedTargetCapturePicOverlapCap": {
    /*ro, req, object*/
        "overlapItemNum": {
        /*ro, req, object, the maximum number of elements can be overlaid*/
            "@max":  1
            /*ro, req, int, the maximum value*/
        },
        "overlapItemId": {
        /*ro, req, object, overlaid element ID*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  3
            /*ro, req, int, the maximum value*/
        },
        "itemType": {
        /*ro, req, object, overlay element types*/
            "@opt":  "cameraNo"
            /*ro, req, string, "cameraNo"-camera No., "captureTime"-capture time, "positionInfo"-camera information, "motionDirection"-motion direction, "jacketColor"-Tops' color, "trousersColor"-bottoms' color, "jacketType"-Tops' type, "trousersType"-bottoms' type, "bag"-backpack, "things"-handbag, "hat"-wearing hat, "plateNo"-license plate number, "vehicleLogo"-vehicle brand, "vehicleType"-vehicle type, "GPS"-longitude and latitude information, "enterLeaveTime"-entering and leaving time, desc:"cameraNo"-camera No., "captureTime"-capture time, "positionInfo"-camera information, "motionDirection"-motion direction, "jacketColor"-Tops' color, "trousersColor"-bottoms' color, "jacketType"-Tops' type, "trousersType"-bottoms' type, "bag"-backpack, "things"-handbag, "hat"-wearing hat, "plateNo"-license plate number, "vehicleLogo"-vehicle brand, "vehicleType"-vehicle type, "GPS"-longitude and latitude information, "enterLeaveTime"-entering and leaving time*/
        },
        "targetAttribute": {
        /*ro, opt, object, attribute details of different overlay element types*/
            "@opt":  "motionDirection"
            /*ro, req, string, attribute details of different overlay element types, desc:attribute details of different overlay element types*/
        }
    }
}
```

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16.10.2.9 Get the captured picture overlay parameters of multi-target-detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capturePicOverlap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetCapturePicOverlap": {
    /*ro, req, object*/
        "OverlapItemList": [
        /*ro, req, array, subType:object, range:[0,22]*/
            {
                "overlapItem": {
                /*ro, req, object*/
                    "overlapItemId":  1,
                    /*ro, req, int*/
                    "itemType":  "cameraNo"
                    /*ro, req, enum, subType:string*/
                }
            }
        ]
    }
}
```

16.10.2.10 Set the captured picture overlay parameters of multi-target-detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/capturePicOverlap?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetCapturePicOverlap": {
    /*req, object*/
        "OverlapItemList": [
        /*req, array, subType:object*/
            {
                "overlapItem": {
                /*req, object*/
                    "overlapItemId":  1,
                    /*req, int*/
                    "itemType":  "cameraNo"
                    /*req, enum, subType:string*/
                }
            }
        ]
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.10.2.11 Set the parameters of multiple scene mode switching of multi-target-detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/multiScene/sceneMode?format=json

Query Parameter

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Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetMultiSceneMode": {
    /*req, object*/
        "sceneMode":  "normal"
        /*req, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

16.10.2.12 Get the parameters of multiple scene mode switching of multi-target-detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/multiScene/sceneMode?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetMultiSceneMode": {
    /*ro, req, object*/
        "sceneMode":  "normal"
        /*ro, req, enum, subType:string*/
    }
}
```

16.10.2.13 Get OSD overlay configuration capability for detecting mixed targets

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

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```
{
    "MixedTargetOverlayCap": {
    /*ro, req, object, OSD overlay configuration capability for mixed targets*/
        "enabled": {
        /*ro, opt, object, whether to enable OSD overlay capability for detecting mixed targets; OSD calculation data will be cleared if this node is
reset*/
            "@opt":  "true,false",
            /*ro, opt, string, options*/
            "@def":  false
            /*ro, opt, bool, the default value*/
        },
        "x": {
        /*ro, opt, object, x-coordinate; the value ranges from 0.000 to 1; the upper-left corner is set as the origin point*/
            "@min":  0.000,
            /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum value, range:[0.000,1.000]*/
        },
        "y": {
        /*ro, opt, object, y-coordinate; the value ranges from 0.000 t0 1; the upper-left corner is set as the origin point*/
            "@min":  0.000,
            /*ro, opt, float, the minimum value, range:[0.000,1.000]*/
            "@max":  1.000
            /*ro, opt, float, the maximum value, range:[0.000,1.000]*/
        },
        "fontSize": {
        /*ro, opt, object, font size of overlaid information*/
            "@opt": ["16*16", "32*32", "48*48", "64*64", "96*96", "adaptive"]
            /*ro, opt, array, "adaptive"-size adaptive (by default), subType:string*/
        },
        "targetDynamicMaskEnabled": {
        /*ro, opt, object*/
            "@opt": [true, false]
            /*ro, opt, array, subType:bool*/
        },
        "targetDynamicMaskLevel": {
        /*ro, opt, object*/
            "@opt": ["high", "low", "medium"]
            /*ro, opt, array, subType:string*/
        },
        "isSupportOSDReset":  true,
        /*ro, opt, bool, if "true" is returned, it indicates that manually clearing OSD calculation data of mixed targets detection is supported (the URI is
available: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection/overlay/OSDReset?format=json*/
        "isSupportOSDAutoReset":  true
        /*ro, opt, bool, if "true" is returned, it indicates that auto clearing OSD calculation data of mixed targets detection is supported (the URI is
available: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection/overlay/OSDAutoReset?format=json*/
    }
}
```

16.10.2.14 Clear OSD calculation data of detecting mixed targets manually

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay/OSDReset?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

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16.10.2.15 Set OSD overlay configuration parameters for detecting mixed targets

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "MixedTargetOverlay": {
    /*req, object, OSD overlay configuration parameters for detecting mixed targets*/
        "enabled": false,
        /*opt, bool, whether to enable OSD overlay capability for detecting mixed targets*/
        "x": 0.120,
        /*opt, float, x-coordinate, the value ranges from 0.000 to 1, the upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "y": 0.120,
        /*opt, float, y-coordinate, the value ranges from 0.000 to 1, the upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "fontSize": "adaptive",
        /*opt, enum, font size of overlaid information, subType:string, desc:"16*16", "32*32", "48*48", "64*64", "96*96", "adaptive" (size adaptive, by
default)*/
        "targetDynamicMaskEnabled": true,
        /*opt, bool*/
        "targetDynamicMaskLevel": "high"
        /*opt, enum, subType:string, dep:or,{$.MixedTargetOverlay.targetDynamicMaskEnabled,eq,true}*/
    }
}
```

Response Message

```
{
    "statusCode": 1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString": "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode": "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode": 1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode": "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx": "0x00000000"
    /*ro, opt, string*/
}
```

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16.10.2.16 Get OSD overlay configuration parameters for detecting mixed targets

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection/overlay?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetOverlay": {
    /*ro, req, object, OSD overlay configuration parameters for detecting mixed targets*/
        "enabled":  false,
        /*ro, opt, bool, whether to enable OSD overlay capability for detecting mixed targets. By default, it is set to false*/
        "x":  0.120,
        /*ro, opt, float, X-coordinate, the value ranges from 0.000 to 1. The upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "y":  0.120,
        /*ro, opt, float, Y-coordinate, the value ranges from 0.000 to 1. The upper-left corner is set as the origin point, range:[0.000,1.000]*/
        "fontSize":  "adaptive",
        /*ro, opt, enum, font size of overlaid information, subType:string, desc:"16*16", "32*32", "48*48", "64*64", "96*96", "adaptive" (size adaptive, by
default)*/
        "targetDynamicMaskEnabled":  true,
        /*ro, opt, bool*/
        "targetDynamicMaskLevel":  "high"
        /*ro, opt, enum, subType:string, dep:or,{$.MixedTargetOverlay.targetDynamicMaskEnabled,eq,true}*/
    }
}
```

16.10.2.17 Set single-scene mode parameters of multi-target-type detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MixedTargetDetection": {
    /*opt, object, rule parameters of multi-target-type detection (by default, the first scene)*/
        "enabled":  true,
        /*req, bool, whether to enable globally*/
        "TypesDetection": [
        /*opt, array, subType:object*/
            {
                "ruleMode":  "human",
                /*opt, enum, rule type, subType:string, desc:human: human body; face; vehice; nonMotor: non-motor vehicle;*/
                "enabled":  true,
                /*opt, bool, whether to enable*/
                "Property": [
                /*opt, array, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*opt, enum, subType:string*/
                        "enabled":  false
                        /*opt, bool, whether to enable*/
                    }
                ]
            }
        ],
        "RuleInfo": [
        /*opt, array, rule information, subType:object*/
            {
                "ruleID":  1,
                /*req, int, rule ID*/
                "ruleMode":  "face_human_vehice_nonMotor",
                /*req, enum, rule type, subType:string, desc:rule type: face + human body + vehicle + non-motor vehicle,human body + vehicle + non-motor
vehicle*/
                "Region": [
                /*req, array, region configured with rules, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x":  0.120,
                        /*req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.120
                        /*req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ],
                "InterPupileRegion": {
                /*req, object, minimum rectangle region to detect pupil distance, desc:The origin is the lower-left corner of the screen.*/
                    "x":  0.120,
                    /*req, float, X-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*req, float, Y-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*req, float, rectangle width, range:[0.000,1.000]*/
                    "height":  0.120
                    /*req, float, rectangle height, range:[0.000,1.000]*/
                },
                "MaxInterPupilRegion": {
                /*opt, object, maximum rectangle region to detect pupil distance, desc:the origin is the lower-left corner of the screen.*/
```

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```
                /* opt, object, maximum rectangle region to detect pupil distance, desc:the origin is the lower-left corner of the screen. */
                    "x":  0.120,
                    /*req, float, X-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*req, float, Y-coordinate of the point in the top left corner, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*req, float, rectangle width, range:[0.000,1.000]*/
                    "height":  0.120
                    /*req, float, rectangle height, range:[0.000,1.000]*/
                },
                "LaneInfo": {
                /*opt, object, lane information, desc:lane information,it is valid when ruleMode is "face_human_vehice"*/
                    "LaneList": [
                    /*req, array, lane list, subType:object*/
                        {
                            "LeftLaneLine": [
                            /*req, array, left lane line capability, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen.*/
                                {
                                    "x":  0.120,
                                    /*req, float, Y-coordinate, range:[0.000,1.000]*/
                                    "y":  0.120
                                    /*req, float, Y-coordinate, range:[0.000,1.000]*/
                                }
                            ]
                        }
                    ],
                    "RightLaneLine": [
                    /*req, array, right lane line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen.*/
                        {
                            "x":  0.120,
                            /*req, float, Y-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                }
            }
        ],
        "KeyRegionInfo": {
        /*opt, object, key region information*/
            "enabled":  true,
            /*req, bool, whether to enable key region*/
            "RegionList": [
            /*opt, array, key region array, subType:object*/
                {
                    "id":  1,
                    /*req, int, detection area ID*/
                    "Region": [
                    /*req, array, key region, subType:object, desc:The origin is the lower-left corner of the screen.*/
                        {
                            "x":  0.120,
                            /*req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120,
                            /*req, float, Y-coordinate, range:[0.000,1.000]*/
                            "width":  0.120,
                            /*req, float, rectangle width, range:[0.000,1.000]*/
                            "height":  0.120
                            /*req, float, rectangle height, range:[0.000,1.000]*/
                        }
                    ]
                }
            ]
        },
        "province":  1,
        /*opt, enum, reserved, subType:int, desc:reserved*/
        "pictureQuality":  "high",
        /*opt, enum, picture quality, subType:string*/
        "isSupportBinaryPicUp":  true,
        /*opt, bool*/
        "convertBinToBmpEnabled":  true,
        /*opt, bool*/
        "MEF": {
        /*opt, object*/
            "enabled":  true,
            /*req, bool, whether to enable multi-target-type detection*/
            "offsetTimes":  1,
            /*opt, int, range:[0,10], unit:min*/
            "leftOffsetTimes":  60,
            /*opt, int, range:[0,300], unit:s*/
            "rightOffsetTimes":  30
            /*opt, int, range:[0,30], unit:s*/
        }
    }
}
```

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Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

16.10.2.18 Get single-scene mode parameters of multi-target-type detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/mixedTargetDetection?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
{
    "MixedTargetDetection": {
    /*ro, opt, object*/
        "enabled":  true,
        /*ro, req, bool, whether to enable multi-target-type detection*/
        "TypesDetection": [
        /*ro, opt, array, subType:object*/
            {
                "ruleMode":  "human",
                /*ro, opt, enum, rule type, subType:string, desc:face + human body + vehicle + non-motor vehicle, human body + vehicle + non-motor vehicle*/
                "enabled":  true,
                /*ro, opt, bool, whether to enable multi-target-type detection*/
                "Property": [
                /*ro, opt, array, Attribute Information, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*ro, opt, enum, subType:string*/
                        "enabled":  false
                        /*ro, opt, bool, whether to enable multi-target-type detection*/
                    }
                ]
            }
        ],
        "RuleInfo": [
        /*ro, opt, array, rule information, subType:object*/
            {
                "ruleID":  1,
                /*ro, req, int, rule ID*/
                "ruleMode":  "face_human_vehice_nonMotor",
                /*ro, opt, enum, rule type, subType:string, desc:rule type: face + human body + vehicle + non-motor vehicle,human body + vehicle + non-motor
vehicle*/
                "Region": [
                /*ro, req, array, rule area, subType:object, range:[3,10], desc:the origin is the lower-left corner of the screen*/
                    {
                        "x":  0.120,
                        /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                        "y":  0.120
                        /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    }
                ],
                "InterPupileRegion": {
                /*ro, req, object, minimum rectangle region to detect pupil distance, desc:the origin is the lower-left corner of the screen*/
                    "x":  0.120,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                    "height":  0.120
                    /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                },
                "MaxInterPupilRegion": {
                /*ro, opt, object, the origin is the lower-left corner of the screen, desc:the origin is the lower-left corner of the screen*/
                    "x":  0.120,
```

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```
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.120,
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                    "width":  0.120,
                    /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                    "height":  0.120
                    /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                },
                "LaneInfo": {
                /*ro, opt, object, lane information, desc:it is valid when ruleMode is "face_human_vehice"*/
                    "LaneList": [
                    /*ro, req, array, lane list, subType:object*/
                        {
                            "LeftLaneLine": [
                            /*ro, req, array, left lane line capability, subType:object, range:[0,2], desc:the origin is the lower-left corner of the
screen*/
                                {
                                    "x":  0.120,
                                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                                    "y":  0.120
                                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                                }
                            ]
                        }
                    ],
                    "RightLaneLine": [
                    /*ro, req, array, right lane line, subType:object, range:[0,2], desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                            "y":  0.120
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                        }
                    ]
                },
                "retrogradeDetectionList": [
                /*ro, opt, array, subType:object, range:[0,2]*/
                    {
                        "retrogradeDetectionLine": [
                        /*ro, opt, array, subType:object, range:[2,2]*/
                            {
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            }
                        ],
                        "retrogradeDirectionLine": {
                        /*ro, opt, object*/
                            "startPoint": {
                            /*ro, req, object*/
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            },
                            "stopPoint": {
                            /*ro, req, object*/
                                "x":  0.120,
                                /*ro, req, float, range:[0.000,1.000]*/
                                "y":  0.120
                                /*ro, req, float, range:[0.000,1.000]*/
                            }
                        }
                    }
                ],
                "nonMotorOverspeedThreshoid":  30,
                /*ro, opt, int, range:[10,50], step:1, unit:km/h*/
                "vehicleOverspeedThreshoid":  40
                /*ro, opt, int, range:[30,80], step:1, unit:km/h*/
            }
        ],
        "KeyRegionInfo": {
        /*ro, opt, object, key region information*/
            "enabled":  true,
            /*ro, req, bool, whether to enable multi-target-type detection*/
            "RegionList": [
            /*ro, opt, array, key region array, at least two key regions are required, subType:object*/
                {
                    "id":  1,
                    /*ro, req, int, detection area ID*/
                    "Region": [
                    /*ro, req, array, Key Area, subType:object, desc:the origin is the lower-left corner of the screen*/
                        {
                            "x":  0.120,
                            /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                            "y":  0.120,
                            /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                            "width":  0.120,
                            /*ro, req, float, rectangle height, range:[0.000,1.000]*/
                            "height":  0.120
                            /*ro, req, float, rectangle width, range:[0.000,1.000]*/
                        }
                    ]
```

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```
                }
            ]
        },
        "province": 1,
        /*ro, opt, enum, reserved, subType:int, desc:reserved*/
        "pictureQuality": "high",
        /*ro, opt, enum, picture quality, subType:string, desc:picture quality*/
        "isSupportBinaryPicUp": true,
        /*ro, opt, bool*/
        "convertBinToBmpEnabled": true,
        /*ro, opt, bool*/
        "MEF": {
        /*ro, opt, object*/
            "enabled": true,
            /*ro, req, bool, whether to enable multi-target-type detection*/
            "offsetTimes": 1,
            /*ro, opt, int, range:[0,10], unit:min*/
            "leftOffsetTimes": 60,
            /*ro, opt, int, range:[0,300], unit:s*/
            "rightOffsetTimes": 30
            /*ro, opt, int, range:[0,30], unit:s*/
        },
        "plateInfo": {
        /*ro, opt, object*/
            "region": "AFandAM",
            /*ro, opt, enum, subType:string*/
            "CRIndex": "0"
            /*ro, opt, enum, subType:string*/
        },
        "nonMotorizedViolationUpload": {
        /*ro, opt, object*/
            "postEnabled": true,
            /*ro, opt, bool*/
            "illegalEnabled": true
            /*ro, opt, bool*/
        }
    }
}
```

16.10.2.19 Set the shielded area parameters of multi-target-detection event

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/Shield/mixedTargetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shield xmlns="http://www.isapi.org/ver20/XMLSchema" version="1.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, enum, subType:string-->true
  </enabled>
  <RegionList>
    <!--opt, array, subType:object-->
    <Region>
      <!--opt, object-->
      <id>
        <!--req, string-->1
      </id>
      <enabled>
        <!--req, enum, subType:string-->true
      </enabled>
      <RegionCoordinatesList>
        <!--opt, array, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object-->
          <positionX>
            <!--req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </Region>
  </RegionList>
</Shield>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

16.10.2.20 Multi-target-type detection

EventType:mixedTargetDetection

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "dateTimeMillisecond":  999,
    /*ro, opt, int, the milliseconds of the alarm trigger time, range:[0,999], desc:it should be used together with field dateTime*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:the event triggering frequency. The pulse event is the event which is triggered continuously according to the detection frequency of the device, e.g., the motion detection. The instant event is the event triggered when object recognition occurs, and the recognition of each object will trigger the event once, e.g., the facial recognition. When the pulse events are triggered, this node is used to tell whether the events are triggered by the same source. For example, if it is motion detection, the alarms will be uploaded continuously according to the device detection frequency; if it changes to another source, then the counting will start again*/
    "eventType":  "mixedTargetDetection",
    /*ro, req, enum, event type, subType:string, desc:"mixedTargetDetection" (multi-target-type detection event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Mixed target detection",
    /*ro, req, enum, event description, subType:string, desc:"Mixed target detection" (multi-target-type detection event)*/
    "channelName":  "ABC",
    /*ro, opt, string, channel name (camera name), desc:it can be configured via URI (/ISAPI/Streaming/channels/<ID>)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID, desc:it can be configured via URI (/ISAPI/System/Network/Ehome)*/
    "isDataRetransmission":  true,
    /*ro, opt, bool, whether the data is replenished, desc:whether it is a resending data (if the real-time detection data upload fails due to the network condition, the data will be uploaded again after the device recovers.)*/
    "customChanID":  "test",
    /*ro, opt, string, custom camera channel No., desc:the maximum string size is 64. The value of this node is that of the node <customChanID> configured by /ISAPI/ContentMgmt/InputProxy/channels/<ID>*/
    "faceBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"faceBackgroundImage" (face background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanBackgroundImage" (human body background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "humanImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"humanImage" (human body thumbnail)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
```

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```
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "nonMotorBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"nonMotorBackgroundImage" (background picture of the non-motor vehicle)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleBackgroundImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleBackgroundImage" (vehicle background picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "vehicleImageResolution": {
    /*ro, opt, object, captured picture resolution, desc:"vehicleImage" (vehicle captured picture)*/
        "height":  1,
        /*ro, req, int, resolution (height), range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, resolution (width), range:[1,65535], unit:px*/
    },
    "CaptureResult": [
    /*ro, opt, array, captured results of detected target, subType:object*/
        {
            "targetID":  1,
            /*ro, req, int, captured target ID*/
            "Face": {
            /*ro, opt, object, face attribute*/
                "Rect": {
                /*ro, opt, object, face thumbnail coordinates, desc:rectangular frame coordinates of the face thumbnail, which is based on the background
picture (faceBackgroundImage). The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, face attribute details, subType:object*/
                    {
                        "description":  "age",
                        /*ro, opt, enum, attributes description, subType:string, desc:"age" (age), "ageGroup" (age group), "gender" (gender), "glass"
(whether the person wears glasses), "smile" (whether the person wears a smile), "mask" (whether the person wears a mask), "faceExpression" (face
expression), "hat" (whether the person wears a hat), "score" (face score)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share
one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "faceBackgroundImage",
                /*ro, opt, enum, "faceBackgroundImage" (background picture of the face), subType:string, desc:"faceBackgroundImage" (face background
picture), "humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle
license plate thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and
non-motor vehicle are detected, these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "FacePictureRect": {
                /*ro, opt, object, coordinates of face picture containing head and shoulder, desc:it is based on the face background picture matting and
equals to the face thumbnail uploaded by the device. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, face thumbnail (contains head and shoulder) ID, which indicates the uniqueness of the picture data, range:[0,32],
desc:both "contentID" and "pId" are available for the data transmission format Content-Type: multipart/form-data, and implement the relation between picture
data and structured data of corresponding contents. "contentID1" and "contentID2" correspond to "contentID", "pId1" and "pId2" correspond to "name" and
"filename". The advantage of "pId" is that it has better compatibility, it can match data via "name" and indicate the file name and format via "filename".
Both "contentID" and "pId" need to be returned, while "faceImageURL" and "faceBackgroundImageURL" are not required*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, face background picture ID*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "faceBackgroundImageURL":  "test",
                /*ro, opt, string, face background picture URL, desc:face background picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, face capture time*/
                "snapTimeMillisecond":  999
                /*ro, opt, int, the milliseconds of face capture time, range:[0,999], desc:it should be used together with the field snapTime*/
            },
            "Human": {
            /*ro, opt, object, human body attributes*/
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            /*ro, opt, object, human body attributes*/
                "Rect": {
                /*ro, req, object, target frame (based on the background picture), desc:coordinates of the human body target frame, which is based on the background picture matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "jacketColor",
                        /*ro, opt, enum, attributes description, subType:string, desc:"jacketColor" (top color), "trousersColor" (bottoms color), "ageGroup" (age group), "gender", "glass" (whether the person wears glasses), "ride" (whether the person is riding a bike), "bag" (whether the person is carrying a bag), "things"(whether the person is carrying things), "hat" (whether the person wears a hat), "trousersType" (bottoms type), "mask" (whether the person wears a mask), "hairStyle" (hairstyle), "cyclingPersonNumber" (number of riders), "direction" (target direction), "score" (human body grading), "hatStyle" (hat style), "cyclingType" (riding type), "jacketType" (top type)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "modeldata":  "test",
                /*ro, opt, string, human body model data (encoded by base64)*/
                "contentID1":  "humanImage",
                /*ro, opt, enum, human body thumbnail, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage" (non-motor vehicle thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "humanBackgroundImage",
                /*ro, opt, enum, human body background picture, subType:string, desc:"faceBackgroundImage" (face background picture), "humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture), "plateImage" (vehicle license plate thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, human body thumbnail ID, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one thumbnail. The value of pId1 is same as that of pId1 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle thumbnail*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId2 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanImageURL":  "test",
                /*ro, opt, string, human body thumbnail URL, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanImageURL is same as that of nonMotorImageURL, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle thumbnail*/
                "humanBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the human body, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, human body capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, milliseconds of human body capture time, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "humanMergeImage",
                /*ro, opt, string, "humanMergeImage" (picture of human body merging details)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of the picture of human body merging details, range:[0,32]*/
                "contentID4":  "humanBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of human body, subType:string, desc:"humanBackgroundImage2" (2nd background picture of human body)*/
                "contentID5":  "humanBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of human body, subType:string, desc:"humanBackgroundImage3" (3rd background picture of human body)*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 2nd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId4 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, ID of 3rd background picture of human body, range:[0,32], desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one background picture. The value of pId5 is same as that of pId2 in nonMotor, i.e., human body and non-motor vehicle share the binary picture data of one non-motor vehicle background picture*/
                "humanBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of 2nd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "humanBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of 3rd human body background picture, desc:when properties of both human body and non-motor vehicle of a target are detected at the same, these properties can share one picture. The value of humanBackgroundImageURL is same as that of nonMotorBackgroundImageURL*/
                "HumanImageRect": {
                /*ro, opt, object, target rectangle frame based on the human body thumbnail, desc:the coordinates of the human body target frame, which is based on the human body thumbnail matting. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "smallImageRectInBackground": {
                /*ro, opt, object, rectangle frame of human body thumbnail (based on the background picture), desc:the rectangle frame is the area of the small human body picture based on humanBackgroundImage. The matted picture is the same as humanImage. The origin is the upper-left corner of the screen*/
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                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of
mixedTargetDetection (multi-target-type detection event). when a device uploads human body properties, other related targets attributes will be linked, and
the client software can get the information about targets around the captured human body*/
                "contentID6":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test"
                /*ro, opt, string, close-up picture URL*/
            },
            "NonMotor": {
            /*ro, opt, object, non-motor vehicle attributes*/
                "Rect": {
                /*ro, req, object, frame information of non-motor vehicle thumbnail, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "nonMotorType",
                        /*ro, opt, enum, attributes description, subType:string, desc:"detectType" (device type), "nonMotorType" (non-motor vehicle type),
"plateColor" (license plate color of non-motor vehicle), "plateNo" (license plate number of non-motor vehicle), "directionIndex" (vehicle driving
direction), "laneNo" (lane number), "RFID" (linked RFID), "EPlateChipNo" (chip number of electronic license plate), "EPlateCardNo" (card number of
electronic license plate), "EPlateListenAuthCode" (licensing authority code of electronic license plate), "EPlateNo" (electronic license plate number),
"EPlateManufacturingDate" (data of production of electronic license plate), "EPlateInspectionValidity" (inspection validity of electronic license plate),
"EPlateMandaScrapPeriod" (compulsory scrapping period of electronic license plate), "EPlatePassengersCapacity" (authorized passenger capacity),
"EPlateCarryingCapacity" (tractive tonnage), "EPlateVehicleColor" (non-motor vehicle color), "EPlateNonMotorType" (non-motor vehicle type),
"EPlateNonMotorPlateType" (license plate type of non-motor vehicle), "EPlateColor" (license plate color of non-motor vehicle), "EPlateUseType" (function
properties of non-motor vehicle), "EPlateTriggerTime" (detection trigger time), "retrograde" (wrong-way driving of non-motor vehicle), "manned" (carrying
passengers on non-motor vehicle), "hatStyle" (hat style)*/
                        "value":  "23"
                        /*ro, opt, string, it depends on the value of description, desc:the returned value type of this field is string*/
                    }
                ],
                "contentID1":  "nonMotorImage",
                /*ro, opt, enum, face picture ID, subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "nonMotorImage"
(non-motor vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body
and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "nonMotorBackgroundImage",
                /*ro, opt, enum, background picture of the non-motor vehicle, subType:string, desc:"faceBackgroundImage" (face background picture),
"humanBackgroundImage" (human body background picture), "nonMotorBackgroundImage" (non-motor vehicle background picture). When there are multiple properties
for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one
picture and "nonMotorBackgroundImage" will be returned*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, ID of the non-motor vehicle thumbnail, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, background picture ID of the non-motor vehicle, range:[0,32]*/
                "nonMotorImageURL":  "test",
                /*ro, opt, string, URL of the non-motor vehicle thumbnail*/
                "nonMotorBackgroundImageURL":  "test",
                /*ro, opt, string, background picture URL of the non-motor vehicle*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, non-motor vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "contentID3":  "nonMotorBackgroundImage2",
                /*ro, opt, enum, 2nd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage2" (2nd background picture of
the non-motor vehicle)*/
                "contentID4":  "nonMotorBackgroundImage3",
                /*ro, opt, enum, 3rd background picture of the non-motor vehicle, subType:string, desc:"nonMotorBackgroundImage3" (3rd background picture of
the non-motor vehicle)*/
                "contentID5":  "faceImage",
                /*ro, opt, enum, face thumbnail, subType:string, desc:"faceImage" (face thumbnail)*/
                "contentID6":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD2223",
                /*ro, opt, string, 2nd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD2224",
                /*ro, opt, string, 3rd background picture ID of the non-motor vehicle, range:[0,32]*/
                "pId5":  "F4F665D6A18E41308CE9934DCDDD2225",
                /*ro, opt, string, face thumbnail ID, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD2226",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "nonMotorBackgroundImage2URL":  "test",
                /*ro, opt, string, 2nd background picture URL of the non-motor vehicle*/
                "nonMotorBackgroundImage3URL":  "test",
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                /*ro, opt, string, 3rd background picture URL of the non-motor vehicle*/
                "faceImageURL":  "test",
                /*ro, opt, string, face thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the first picture*/
                "relatedMixedTargetUid": ["test", "1"],
                /*ro, opt, array, UID array of linked multi-target-type detection events, subType:string, desc:corresponding to the "uid" of mixedTargetDetection (multi-target-type detection event). when a device uploads non-motor vehicle properties, other related targets attributes will be linked, and the client software can get the information about targets around the captured non-motor vehicle*/
                "contentID7":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId7":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test",
                /*ro, opt, string, close-up picture URL*/
                "nonMotorSpeed":  23.4
                /*ro, opt, float, driving speed of non-motor vehicle, range:[0,1024], unit:km/h*/
            },
            "Vehicle": {
            /*ro, opt, object, vehicle attributes*/
                "VehicleRect": {
                /*ro, req, object, target frame information of vehicle picture (based on background picture), desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "PlateRect": {
                /*ro, req, object, frame information of license plate picture, desc:the origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "Property": [
                /*ro, opt, array, attributes data, subType:object*/
                    {
                        "description":  "plateNo",
                        /*ro, opt, enum, attributes description, subType:string, desc:"plateNo" (license plate No.), "plateType" (license plate type), "plateColor" (license plate color), "vehicleType"(vehicle type), "vehicleColor" (vehicle color), "vehicleLogo" (vehicle brand), "vehicleSublogo" (vehicle sub-brand), "vehicleSize" (vehicle size)*/
                        "value":  "23",
                        /*ro, opt, string, it depends on the value of description, desc:it depends on the value of description*/
                        "confidence":  99.00
                        /*ro, opt, float, confidence, range:[0.00,100.00]*/
                    }
                ],
                "contentID1":  "vehicleImage",
                /*ro, opt, enum, vehicle thumbnail (on the basis of the target frame, a certain proportion is expanded), subType:string, desc:"faceImage" (face thumbnail), "humanImage" (human body thumbnail), "vehicleImage" (vehicle thumbnail). When there are multiple properties for the detected target, multiple properties can share one picture, e.g., if human body and non-motor vehicle are detected, these two properties share one picture and "nonMotorImage" will be returned*/
                "contentID2":  "plateImage",
                /*ro, opt, enum, license plate thumbnail, subType:string, desc:"plateImage" (license plate thumbnail)*/
                "contentID3":  "vehicleBackgroundImage",
                /*ro, opt, enum, vehicleBackgroundImage, subType:string, desc:"vehicleBackgroundImage" (vehicle background picture 1)*/
                "contentID4":  "vehicleBackgroundImage2",
                /*ro, opt, enum, vehicleBackgroundImage2, subType:string, desc:"vehicleBackgroundImage2" (vehicle background picture 2)*/
                "contentID5":  "vehicleBackgroundImage3",
                /*ro, opt, enum, vehicleBackgroundImage3, subType:string, desc:"vehicleBackgroundImage3" (vehicle background picture 3)*/
                "contentID6":  "vehicleBackgroundImage4",
                /*ro, opt, enum, vehicleBackgroundImage4, subType:string, desc:"vehicleBackgroundImage4" (vehicle background picture 4)*/
                "contentID8":  "vehicleBackgroundImage5",
                /*ro, opt, enum, vehicleBackgroundImage5, subType:string, desc:"vehicleBackgroundImage5" (vehicle background picture 5)*/
                "contentID9":  "pilotPicture",
                /*ro, opt, enum, pilotPicture, subType:string, desc:"pilotPicture" (driver's picture)*/
                "contentID10":  "copilotPicture",
                /*ro, opt, enum, copilotPicture, subType:string, desc:"copilotPicture" (co-driver's picture)*/
                "pId1":  "F4F665D6A18E41308CE9934DCDDD1111",
                /*ro, opt, string, vehicle thumbnail ID, range:[0,32]*/
                "pId2":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                "pId3":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, vehicle background picture ID, range:[0,32]*/
                "pId4":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 2nd vehicle background picture, range:[0,32]*/
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                "pId5":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 3rd vehicle background picture, range:[0,32]*/
                "pId6":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 4th vehicle background picture, range:[0,32]*/
                "pId8":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of the 5th vehicle background picture, range:[0,32]*/
                "pId9":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, driver's picture, range:[0,32]*/
                "pId10":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, co-driver's picture, range:[0,32]*/
                "vehicleImageURL":  "test",
                /*ro, opt, string, vehicle thumbnail URL*/
                "plateImageURL":  "test",
                /*ro, opt, string, license plate thumbnail URL*/
                "vehicleBackgroundImageURL":  "test",
                /*ro, opt, string, vehicle background picture URL*/
                "vehicleBackgroundImage2URL":  "test",
                /*ro, opt, string, URL of the 2nd vehicle background picture*/
                "vehicleBackgroundImage3URL":  "test",
                /*ro, opt, string, URL of the 3rd vehicle background picture*/
                "vehicleBackgroundImage4URL":  "test",
                /*ro, opt, string, URL of the 4th vehicle background picture*/
                "vehicleBackgroundImage5URL":  "test",
                /*ro, opt, string, URL of the 5th vehicle background picture*/
                "pilotPictureURL":  "test",
                /*ro, opt, string, driver's picture URL*/
                "copilotPictureURL":  "test",
                /*ro, opt, string, co-driver's picture URL*/
                "snapTime":  "2015-06-04T19:25:16-07:00",
                /*ro, opt, datetime, vehicle capture time*/
                "snapTimeMillisecond":  999,
                /*ro, opt, int, it should be used together with the field snapTime, range:[0,999], desc:it should be used together with the field snapTime*/
                "humanMergeImageURL":  "test",
                /*ro, opt, string, URL of the picture of human body merging details*/
                "illegalCode":  "12313",
                /*ro, opt, string, violation code, range:[0,64]*/
                "illegalName":  "test",
                /*ro, opt, string, violation name, range:[0,128]*/
                "illegalDescription":  "test",
                /*ro, opt, string, violation description, range:[0,256]*/
                "dwIllegalTime":  100,
                /*ro, opt, int, duration of the violation, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the first picture*/
                "VehicleImageRect": {
                /*ro, opt, object, rectangle target frame in the vehicle thumbnail, desc:the frame is the target area based on vehicleImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "smallImageRectInBackground": {
                /*ro, opt, object, vehicle thumbnail frame (based on the background picture), desc:the rectangle frame of vehicle thumbnail is based on vehicleBackgroundImage. The matted picture equals to vehicleImage. The origin is the upper-left corner of the screen*/
                    "height":  1.000,
                    /*ro, req, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, req, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, req, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, req, float, Y-coordinate, range:[0.000,1.000]*/
                },
                "MultiPlateInfo": [
                /*ro, opt, array, information about multiple license plates, subType:object*/
                    {
                        "MultiPlateRect": {
                        /*ro, opt, object, license plate thumbnail frame, desc:based on the vehicle background picture*/
                            "height":  1.000,
                            /*ro, req, float, height, range:[0.000,1.000]*/
                            "width":  1.000,
                            /*ro, req, float, width, range:[0.000,1.000]*/
                            "x":  0.000,
                            /*ro, req, float, x-coordinate, range:[0.000,1.000]*/
                            "y":  0.000
                            /*ro, req, float, y-coordinate, range:[0.000,1.000]*/
                        },
                        "plateNo":  "noPlate",
                        /*ro, req, enum, license plate number, subType:string, desc:"noPlate" (vehicle without license plate), "unknown" (no license plate recognized), XXXXXXXX (license plate number. For motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string*/
                        "plateColor":  "black",
                        /*ro, req, enum, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow), "orange", "other" (other color), "red", "unknown", "white", "yellow"*/
                        "confidence":  100.0,
                        /*ro, opt, float, confidence, range:[0,100.0]*/
                        "province":  0,
                        /*ro, req, enum, province index, subType:int*/
                        "contentID7":  "HongkongPlateImage",
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                        /*ro, opt, enum, license plate thumbnail, subType:string, desc:license plate thumbnail*/
                        "pId7":  "F4F665D6A18E41308CE9934DCDDD3333",
                        /*ro, opt, string, license plate thumbnail ID, range:[0,32]*/
                        "multiplateImageURL":  "test"
                        /*ro, opt, string, thumbnail URL of multiple license plates*/
                    }
                ],
                "carDriveDirect":  "forward",
                /*ro, opt, enum, vehicle driving direction, subType:string, desc:"back" (backward), "eastWest" (from east to west), "forward" (forward),
"northSouth" (from north to south), "northeastToSouthwest" (from northeast to southwest), "northwestToSoutheast" (from northwest to southeast), "other"
(other), "southNorth" (from south to north), "southeastToNorthwest" (from southeast to northwest), "southwestToNortheast" (from southwest to northeas),
"unknown" (unknown), "westEast" (from west to east)*/
                "contentID11":  "featurePicture",
                /*ro, opt, enum, close-up picture, subType:string, desc:"featurePicture" (close-up picture)*/
                "pId11":  "F4F665D6A18E41308CE9934DCDDD2222",
                /*ro, opt, string, close-up picture ID, range:[0,32]*/
                "featurePictureURL":  "test",
                /*ro, opt, string, close-up picture URL*/
                "vehicleSpeed":  23.4
                /*ro, opt, float, vehicle driving speed, range:[0,1024], unit:km/h*/
            },
            "AbsoluteHigh": {
            /*ro, opt, object, PTZ position information*/
                "elevation":  11,
                /*ro, opt, int, tilting angle, desc:tilting angle*/
                "azimuth":  22,
                /*ro, opt, int, azimuth, desc:panning angle*/
                "absoluteZoom":  33
                /*ro, opt, int, zooming ratio, desc:zooming rate,value range: 1 to 1000*/
            },
            "GPS": {
            /*ro, opt, object, device GPS information*/
                "longitude":  -180.000000,
                /*ro, req, float, longitude*/
                "latitude":  -90.000000
                /*ro, req, float, latitude*/
            },
            "FaceContrastResult": [
            /*ro, opt, array, face picture comparison result, one result can contains comparison results of multiple persons, the captured picture with
highest comparison similarity is adopted, subType:object*/
                {
                    "errorCode":  1,
                    /*ro, opt, int, error code, desc:error information returned when analyzing face picture failed*/
                    "errorMsg":  "ok",
                    /*ro, opt, string, the modeling data of base picture in the face picture library, desc:when analyzing face picture failed, the
corresponding error information will be returned*/
                    "modelData":  "test",
                    /*ro, opt, string, the modeling data of base picture in the face picture library, range:[0,2048], desc:base64 encoded*/
                    "faces": [
                    /*ro, opt, array, matched face information, subType:object, desc:the information includes detection results, properties, modeling data,
and matched list. One picture can contains multiple faces*/
                        {
                            "faceId":  1,
                            /*ro, opt, int, face ID, desc:the face ID cannot be duplicated in the same picture*/
                            "identify": [
                            /*ro, opt, array, features, subType:object*/
                                {
                                    "relationId":  "9887",
                                    /*ro, opt, string, arming linkage ID, range:[0,64], desc:arming linkage ID*/
                                    "maxsimilarity":  0.204000,
                                    /*ro, opt, float, the maximum similarity, range:[0.000000,1.000000], desc:the maximum similarity in the matched results
the similarity is between 0 and 1, and it corrects to two decimals*/
                                    "candidate": [
                                    /*ro, opt, array, person information list, subType:object*/
                                        {
                                            "blacklist_id":  "2323",
                                            /*ro, opt, string, alarm list ID, range:[0,64], desc:it corresponds to the FDID of face picture library. For
cameras, only the default FDID will be returned, the custom face picture library ID will be returned in customFaceLibID*/
                                            "human_data": [
                                            /*ro, opt, array, face picture information list, subType:object*/
                                                {
                                                    "face_id":  "123445",
                                                    /*ro, opt, string, face ID*/
                                                    "contentID":  "faceLibImage",
                                                    /*ro, opt, string, picture in the face picture library*/
                                                    "similarity":  0.204000,
                                                    /*ro, opt, float, face picture similarity, range:[0.000000,1.000000]*/
                                                    "isNoSaveFDPicture":  true,
                                                    /*ro, opt, bool, whether to not save the picture of face picture library, desc:true (not save). If the
function of not saving original pictures during picture importing or modeling is enabled, true should be returned, otherwise, this field will not be
returned*/
                                                    "pId":  "F4F665D6A18E41308CE9934DCDDD766F",
                                                    /*ro, opt, string, picture ID in face picture library, range:[0,32], desc:picture ID in face picture
library*/
                                                    "face_picurl":  "http://127.0.0.1:6120/face_picurl"
                                                    /*ro, opt, string, face thumbnail URL*/
                                                }
                                            ],
                                            "human_id":  "123",
                                            /*ro, opt, string, corresponds to the FPID in the face picture library, range:[0,64], desc:person ID in the
list, which corresponds to FPID in the face picture library*/
                                            "reserve_field": {
                                            /*ro, req, object, transmitted data*/
                                                "name":  "张三"
```


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```
                                        "name":  "张三",
                                        /*ro, opt, string, name, range:[0,96]*/
                                        "gender":  "any",
                                        /*ro, req, enum, gender, subType:string, desc:"any" (unlimited condition), "unknown", "male", "female"*/
                                        "bornTime":  "1970-01-01+08:00",
                                        /*ro, opt, date, date of birth*/
                                        "city":  "130100",
                                        /*ro, opt, string, city code of person birth, desc:see the data dictionary for details*/
                                        "certificateType":  "officerID",
                                        /*ro, opt, enum, certificate type, subType:string, desc:"officerID" (officer ID), "ID" (identify card), "passport" (passport), "other" (other certificate)*/
                                        "certificateNumber":  "test",
                                        /*ro, opt, string, certificate No., range:[0,32]*/
                                        "ageGroup":  "unknown",
                                        /*ro, opt, enum, age group, subType:string, desc:“any”, “child”, “infant”, “kid”, “middle”, “middleAged” (the middle age), “old”, “prime”, “teenager”, “unknown”, “young”*/
                                        "phoneNumber":  "test",
                                        /*ro, opt, string, phone number, range:[0,64]*/
                                        "cityName":  "石家庄市",
                                        /*ro, opt, string, city (corresponding to the field "city"), range:[0,32], desc:city (corresponding to the field "city)*/
                                        "provinceName":  "河北省"
                                        /*ro, opt, string, province, range:[0,32], desc:province*/
                                    },
                                    "similarity":  0.780,
                                    /*ro, opt, float, the maximum similarity, range:[0.000,1.000], desc:the maximum similarity of all matched pictures for the person*/
                                    "listType":  "blackList",
                                    /*ro, opt, enum, list type, subType:string, desc:"blacklist" (blocklist), "whitelist" (allowlist)*/
                                    "extendData": [
                                    /*ro, opt, array, person extended information, subType:object*/
                                        {
                                            "extendID":  1,
                                            /*ro, opt, int, ID of the extended person information*/
                                            "enable":  true,
                                            /*ro, opt, bool, whether to enable the person extended information*/
                                            "name":  "test",
                                            /*ro, opt, string, name of person tag extended information, range:[0,96]*/
                                            "value":  "test"
                                            /*ro, opt, string, person tag extended information, range:[0,64]*/
                                        }
                                    ],
                                    "FDLibName":  "faceLib1",
                                    /*ro, opt, string, detection threshold of face picture library*/
                                    "FDLibThreshold":  50,
                                    /*ro, opt, int, detection threshold of face picture library, range:[0,100]*/
                                    "customFaceLibID":  "test",
                                    /*ro, opt, string, custom face picture library ID*/
                                    "customHumanID":  "test"
                                    /*ro, opt, string, custom person ID*/
                                }
                            ]
                        }
                    ],
                    "stayDuration":  20
                    /*ro, opt, int, dwell time duration in the image, unit:ms*/
                }
            ]
        }
    ],
    "ModelingResult": {
    /*ro, opt, object, modeling result of captured face picture*/
        "modelingStatus":  "succeeded",
        /*ro, req, enum, modeling status, subType:string, desc:"succeeded" (modeling succeeded), "resourceInsuf" (insufficient engine capacity), "faceDetectingFail" (face detection failed), "faceModelingFail" (face modeling failed), "faceGradingFail" (face quality grading), "featureExtractFail" (extracting features failed), "featureAnalyzeFail" (features analysis failed), "otherErrors" (other error of the engine), "DSPInternalError" (DSP internal error)*/
        "facePicQuality":  "lower",
        /*ro, opt, enum, quality level of face thumbnail, subType:string, desc:"lower", "medium", "high"*/
        "stayDuration":  20,
        /*ro, opt, int, dwell time duration in the image, unit:ms*/
        "modeldata":  "test"
        /*ro, opt, string, model data (encoded by base64)*/
    },
    "sid":  "123",
    /*ro, opt, string, scene ID*/
    "sceneName":  "场景1",
    /*ro, opt, string, scene name*/
    "uid":  "test",
    /*ro, opt, string, the unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits*/
    "behaviorLinkageUid":  "test",
    /*ro, opt, string, which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes, range:[0,64], desc:which can consists of time (accurate to millisecond) and random digits,the maximum value size is 64 bytes*/
    "position": [500, 500],
    /*ro, opt, array, offset position coordinates of target center point, subType:int, desc:an X-coordinate is positive if it is to the right of the Y-axis and negative if it is to the left of the Y-axis. A Y-coordinate is positive if it is above of the X-axis and negative values are not supported. The origin is the center point (the center point of all lane lines) of the horizontal line where the camera is located and which is perpendicular to the lane line*/
    "laneNo":  1,
    /*ro, opt, int, number of the lane where the target is located*/
    "speed":  23.4,
    /*ro, opt, float, vehicle speed*/
    "targetType":  "human",
```


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```
            /*ro, opt, enum, vehicle type, subType:string, desc:"vehicle", "human", "nonMotor" (non-motor vehicles)*/
            "horizonSpeed":  33.4,
            /*ro, opt, float, horizontal speed of target*/
            "NormalizationPosition": [500, 500],
            /*ro, opt, array, normalized coordinates, subType:int, desc:normalized position coordinates,the value of x-coordinate is between 0 and 1000,and
the value of y-coordinate is between -500 and 1500,e.g.,[500,500]*/
            "targetMaxY":  340,
            /*ro, opt, int, the maximum Y-coordinate of target, desc:the value should be smaller than 1500*/
            "TargetGPSInfo": [
            /*ro, opt, array, GPS information of the target, subType:object*/
                {
                    "longitude":  -180.000000,
                    /*ro, req, float, longitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                    "latitude":  -90.000000
                    /*ro, req, float, latitude information, corrects to 6 decimal places, and the value range is [-180.000000,180.000000]*/
                }
            ],
            "perimeterLinkageUid":  "test",
            /*ro, opt, string, UID of linked perimeter events uploading, desc:64-byte unique identifier, which can be composed of time (accurate to the
millisecond) and random digits*/
            "MEFData": [
            /*ro, opt, array, mobile electronic fence data corresponding to the target, subType:object, desc:mobile electronic fence data of multiple users
is supported (the data needs to be encrypted)*/
                {
                    "time":  "2004-05-03T17:30:08+08:00",
                    /*ro, req, datetime, electronic fence data acquired time*/
                    "IMSI":  "460001357924680",
                    /*ro, req, string, SIM card number (international mobile subscriber identity), range:[1,15]*/
                    "IMEI":  "359355041886388",
                    /*ro, opt, string, international mobile equipment identity, range:[0,17]*/
                    "bandIdx":  11
                    /*ro, opt, enum, frequency band type used by the current user, subType:int, desc:0 (China Unicom B3), 1 (China Telecom B3), 2 (China
Mobile B39), 3 (China Mobile B38), 4 (China Mobile B40), 5 (China Mobile B41), 6 (China Unicom B1), 7 (China Telecom B1), 8 (China Mobile 2G), 9 (China
Unicom 2G), 10 (China Unicom WCDMA), 11 (China Telecom CDMA), 12 (China Telecom CDMA2000), 13 (China Mobile TD), 19 (China Mobile B1), 20 (China Mobile B3),
21 (China Mobile B8)*/
                }
            ]
        }
    ],
    "captureResultTargetNum":  10,
    /*ro, opt, int, number of uploaded alarm targets*/
    "URLCertificationType":  "digest",
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no"-no authentication, "digest"-digest authentication (it is used for local
picture URL returned by NVR or DVR)*/
    "triggerSourceType":  "linkedTrackingCapture"
    /*ro, opt, enum, alarm source type, subType:string, desc:"ruleTrigger" (rule parameters triggered alarm), "linkedTrackingCapture" (alarm triggered by
linked tracking capture). If this field is not returned or the returned value is "ruleTrigger", it indicates that the alarm is triggered by multi-target-
type detection rules (corresponding URL for rule configurations: /ISAPI/Intelligent/channels/<ID>/mixedTargetDetection?format=json). If the returned value
is "linkedTrackingCapture", it indicates that the alarm is triggered by linked tracking capture (corresponding URL for configurations:
/ISAPI/MasterSlaveTracking/linkedTracking?format=json)*/
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name
mixedTargetDetection	[Message content]	application/json	--	--
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg
..value	[Binary picture data]	image/jpeg	faceBackgroundImage	faceBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	humanImage	humanImage.jpg
..value	[Binary picture data]	image/jpeg	humanBackgroundImage	humanBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	humanMergeImage	humanMergeImage.jpg

..value	[Binary picture data]	image/jpeg	humanBackgroundImage2	humanBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	humanBackgroundImage3	humanBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	nonMotorImage	nonMotorImage.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage	nonMotorBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage2	nonMotorBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	nonMotorBackgroundImage3	nonMotorBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	faceImage	faceImage.jpg
..value	[Binary picture data]	image/jpeg	plateImage	plateImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleImage	vehicleImage.jpg
..value	[Binary picture data]	image/jpeg	plateImage	plateImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage	vehicleBackgroundImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage2	vehicleBackgroundImage2.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage3	vehicleBackgroundImage3.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage4	vehicleBackgroundImage4.jpg
..value	[Binary picture data]	image/jpeg	HongkongPlateImage	HongkongPlateImage.jpg
	[Binary			

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..value	[Binary picture data]	image/jpeg	MacaoPlateImage	MacaoPlateImage.jpg
.....value	[Binary picture data]	image/jpeg	faceLibImage	faceLibImage.jpg
..value	[Binary picture data]	image/jpeg	vehicleBackgroundImage5	vehicleBackgroundImage.jpg5
..value	[Binary picture data]	image/jpeg	pilotPicture	pilotPicture.jpg
..value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	featurePicture	featurePicture
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg
..value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg

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Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.

- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

16.11 Algorithm Scheduling

16.11.1 Scheduling and Managing Local Smart Resources

16.11.1.1 Get parameters of intelligent resources switch

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/VCAResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VCAResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, intelligent resources, attr:version{req, string, protocolVersion}-->
  <type>
    <!--ro, req, enum, “basicBehavior”, subType:string, desc:“basicBehavior”-->basicBehavior
  </type>
</VCAResource>
```

16.11.1.2 Set parameters of intelligent resources switch

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/VCAResource

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VCAResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, VCA Resource, attr:version{req, string, protocolVersion}-->
  <type>
    <!--req, enum, type, subType:string,
desc:"basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFaceDetection,smart264AndHeatMap,smartIntelligentMonitor,smartTrafficDataCollection,roadDetection,humanRecognition,perimeterCapture,vehicleDetection,HVTDetection,mixedTargetDetection,trackingCaptureMode,nonTrackingCaptureMode,close,faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,safetyHelmet,faceCounting,personArming,AIOpenPlatform"-->basicBehavior
  </type>
</VCAResource>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
VCAResource	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

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16.11.1.3 Get capability of switching intelligent resources

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/VCAResource/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VCAResource xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, intelligent resources, attr:version{req, string, protocolVersion}-->
  <type
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFac
eDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetection,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition,
perimeterCapture, vehicleDetection,HVTDetection,mixedTargetDetection,     trackingCaptureMode,nonTrackingCaptureMode,close,
faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,AIOpenPlatform,mixedTargetFaceHuman,mixedT
argetAll,safetyHelmet,faceContrast,none,faceMonitor,vehicleMonitor,faceCounting,heelPDC,personDensity,behaviorMixedTargetAll,operationMonitor,fielddetection
,personArming,smokeDetectAlarm,smokeDetection,personDensityQueueLeavePosition,bannerDetection,ATMSurround,
ATMPanel,ATMSafetyCabin,intelligentTraffic,heatmap,teacherBehaviorAndFaceContrast,mixedTargetAllCapture,riverwayGaugeReading,MixedTargetAllAuxiliaryCapture,
radarHearNurse,smokeDetectionAndFlameCombustion,PackingSpaceRecognition" def="basicBehavior">
    <!--ro, req, enum, resource type, subType:string, attr:opt{req, string},def{opt, string}, desc:resource type-->basicBehavior
  </type>
  <PromptList>
    <!--ro, opt, array, prompt for switching intelligent resources, subType:object-->
    <Prompt>
      <!--ro, opt, object, list-->
      <resourceType opt="roadDetection">
        <!--ro, opt, string, currently,only "roadDetection" is supported, attr:opt{req, string}, desc:currently, only "roadDetection" is supported-->test
      </resourceType>
      <resourcePrompt opt="smartCode">
        <!--ro, opt, string, prompt contents, attr:opt{req, string}, desc:prompt contents-->test
      </resourcePrompt>
    </Prompt>
  </PromptList>
  <isSupportAPPWorking>
    <!--ro, opt, bool-->true
  </isSupportAPPWorking>
  <isSupportTest>
    <!--ro, opt, bool-->true
  </isSupportTest>
  <VCAResourcePrompt opt="streamingCodingPrompt">
    <!--ro, opt, string, attr:opt{req, string}-->streamingCodingPrompt
  </VCAResourcePrompt>
  <sceneRecommendationTypeList>
    <!--ro, opt, array, subType:object-->
    <sceneRecommendationType>
      <!--ro, opt, object-->
      <scene opt="Y,L,one,concentricBayonet">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->Y
      </scene>
      <recommendationType
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFac
eDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetection,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition,
perimeterCapture, vehicleDetection,HVTDetection,mixedTargetDetection,     trackingCaptureMode,nonTrackingCaptureMode,close,
faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,AIOpenPlatform,mixedTargetFaceHuman,mixedT
argetAll,safetyHelmet,faceContrast,none,faceMonitor,vehicleMonitor,faceCounting,heelPDC,personDensity,behaviorMixedTargetAll,operationMonitor,fielddetection
,personArming,smokeDetectAlarm,smokeDetection,personDensityQueueLeavePosition,bannerDetection,ATMSurround,
ATMPanel,ATMSafetyCabin,intelligentTraffic,heatmap" def="basicBehavior">
        <!--ro, req, enum, subType:string, attr:opt{req, string},def{opt, string}-->basicBehavior
      </recommendationType>
    </sceneRecommendationType>
  </sceneRecommendationTypeList>
  <synergyRecommendationTypeList>
    <!--ro, opt, array, subType:object-->
    <synergyRecommendationType>
      <!--ro, opt, object-->
      <synergyMode opt="associationDeDuplication,trackAssociation,targetAssociation">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->associationDeDuplication
      </synergyMode>
      <priority min="1" max="3">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
      </priority>
      <recommendationType
opt="basicBehavior,fullBehavior,facesnapBehavior,facesnap,TFS,smartVehicleDetection,smartHVTDetection,smart,judicial,smart264AndRoadDetection,smart264AndFac
eDetection,smart264AndHeatMap,smartVehicleIllegalParkingDetection,smartIntelligentMonitor,smartTrafficDataCollection, roadDetection,humanRecognition,
perimeterCapture, vehicleDetection,HVTDetection,mixedTargetDetection,     trackingCaptureMode,nonTrackingCaptureMode,close,
faceHumanModelingContrast,cityManagement,teacherBehavior,12MPLiveView,personQueueDetection,verticalPeopleCounting,AIOpenPlatform,mixedTargetFaceHuman,mixedT
argetAll,safetyHelmet,faceContrast,none,faceMonitor,vehicleMonitor,faceCounting,heelPDC,personDensity,behaviorMixedTargetAll,operationMonitor,fielddetection
,personArming,smokeDetectAlarm,smokeDetection,personDensityQueueLeavePosition,bannerDetection,ATMSurround,
ATMPanel,ATMSafetyCabin,intelligentTraffic,heatmap" def="basicBehavior">
        <!--ro, req, enum, subType:string, attr:opt{req, string},def{opt, string}-->basicBehavior
      </recommendationType>
    </synergyRecommendationType>
  </synergyRecommendationTypeList>
</VCAResource>
```

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16.11.1.4 Get the functional mutex configuration parameters

Request URL

GET /ISAPI/Thermal/channels/<channelID>/thermIntell

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ThermIntell xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, channel No.-->1
  </id>
  <intellType>
    <!--ro, req, enum, each two types are mutually exclusive,"thermometryAndSmart"-temperature measurement+abnormal event detection
(default),"shipsDetection"-ship detection,"fireDetection"-fire source detection,"pip"-picture in picture function,"faceThermometry"-temperature
screening,"thermometryAndSmokeFireDetection"-temperature measurement+smoke and fire source detection,"thermometryAndFireDetection"-temperature
measurement+fire source detection,"basicBehavior"-abnormal event detection,"thermometry"-temperature measurement, subType:string,
desc:"thermometryAndSmart"-temperature measurement+abnormal event detection (default), "shipsDetection"-ship detection, "fireDetection"-fire source
detection, "pip"-picture in picture function, "faceThermometry"-temperature screening, "thermometryAndSmokeFireDetection"-temperature measurement+smoke and
fire source detection, "thermometryAndFireDetection"-temperature measurement+fire source detection, "basicBehavior"-abnormal event detection, "thermometry"-
temperature measurement-->thermometryAndSmart
  </intellType>
</ThermIntell>
```

16.11.1.5 Set the smart functions parameters of thermography

Request URL

PUT /ISAPI/Thermal/channels/<channelID>/thermIntell

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ThermIntell xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, smart functions parameters of thermography, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, channel No.-->1
  </id>
  <intellType>
    <!--req, enum, VCA type (each two types are mutually exclusive), subType:string, desc:"thermometryAndSmart"-temperature measurement + abnormal event
detection (default), "shipsDetection"-ship detection, "fireDetection"-fire source detection, "pip"-picture in picture function, "faceThermometry"-
temperature screening, "thermometryAndSmokeFireDetection"-temperature measurement + smoke and fire source detection, "thermometryAndFireDetection"-
temperature measurement + fire source detection, "basicBehavior"-abnormal event detection, "thermometry"-temperature measurement-->thermometryAndSmart
  </intellType>
</ThermIntell>
```

Response Message

drackum@gmail.com
DETEKTOR

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

16.11.1.6 Get the configuration capability of smart functions of thermography

Request URL

GET /ISAPI/Thermal/channels/<channelID>/thermIntell/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ThermIntell xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, channel No.-->1
  </id>
  <intellType
opt="thermometryAndSmart,shipsDetection,fireDetection,pip,faceThermometry,thermometryAndSmokeFireDetection,thermometryAndFireDetection,thermometryAndOpenPla
tForm,basicBehavior,thermometry,thermometryAndFireDetectionAndAIOpenPlatForm,thermometryAndAIOpenPlatForm,thermometryAndPeopleNumCounting,thermometryAndHeal
thTracking,healthTracking,FireDetectionAndAIOpenPlatFormAndHEOP,perimeterDetection" def="thermometryAndSmart">
    <!--ro, req, string, smart function type, attr:opt{req, string},def{req, string}, desc:"thermometryAndSmart"-temperature measurement+abnormal event
detection (default), "shipsDetection"-ship detection, "fireDetection"-fire source detection, "pip"-picture in picture function, "faceThermometry"-
temperature screening, "thermometryAndSmokeFireDetection"-temperature measurement+smoke and fire source detection, "thermometryAndFireDetection"-temperature
measurement+fire source detection, "basicBehavior"-abnormal event detection, "thermometry"-temperature measurement,
"thermometryAndFireDetectionAndAIOpenPlatForm"-temperature measurement+fire source detection+AI open platform-->thermometryAndSmart
  </intellType>
</ThermIntell>
```

16.12 VCA

16.12.1 Face Picture Library Management

16.12.1.1 Get face picture library capability

Request URL

GET /ISAPI/Intelligent/FDLib/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

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DETEKTOR

```
{
    "requestURL": "test",
    /*ro, opt, string, request URL*/
    "statusCode": 1,
    /*ro, req, int, status code*/
    "statusString": "test",
    /*ro, req, string, status description*/
    "subStatusCode": "test",
    /*ro, req, string, sub status code*/
    "errorCode": 1,
    /*ro, opt, int, error code, this field is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg": "ok",
    /*ro, opt, string, detailed error description, this field is required when the value of statusCode is not 1*/
    "FDNameMaxLen": 64,
    /*ro, req, int, maximum length of face picture library name*/
    "customInfoMaxLen": 192,
    /*ro, req, int, maximum length of custom information*/
    "FDMaxNum": 3,
    /*ro, req, int, maximum number of face picture libraries*/
    "FDRecordDataMaxNum": 12345,
    /*ro, req, int, maximum face records supported by face picture library*/
    "supportFDFunction": "post,delete,put,get,setUp",
    /*ro, req, string, the supported operations on face picture library, desc:"post" (create), "delete" (delete), "put" (edit), "get" (search), "setUp"
(set)*/
    "isSuportFDSearch": true,
    /*ro, req, bool, whether supports searching in face picture library, desc:whether supports searching in face picture library*/
    "isDisplayCaptureNum": {
    /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, opt, array, subType:bool*/
    },
    "isSupportFDSearchDataPackage": true,
    /*ro, req, bool, whether supports packaging the found data in the face picture library*/
    "isSuportFSsearchByPic": true,
    /*ro, req, bool, whether supports searching by picture in the face picture library, desc:whether supports searching by picture in the face picture
library*/
    "isSuportFSsearchByPicGenerate": true,
    /*ro, req, bool, whether supports exporting results of using picture to search picture from the face picture library*/
    "isSuportFDSearchDuplicate": true,
    /*ro, req, bool, whether supports duplication checking*/
    "isSuportFDSearchDuplicateGenerate": true,
    /*ro, req, bool, whether supports exporting the duplication checking results*/
    "isSuportFCSearch": true,
    /*ro, req, bool, whether supports searching face picture comparison alarms*/
    "isSupportFCSearchDataPackage": true,
    /*ro, req, bool, whether supports packaging the search results of face picture comparison alarms*/
    "isSupportFDExecuteControl": true,
    /*ro, req, bool, whether supports creating relation between face picture libraries and cameras*/
    "generateMaxNum": 1234,
    /*ro, opt, int, maximum face records can be exported from face picture library*/
    "faceLibType": "blackFD,staticFD,infraredFD",
    /*ro, opt, string, face picture library type*/
    "modelMaxNum": 1000,
    /*ro, opt, int, the maximum number of search results*/
    "isSupportModelData": true,
    /*ro, opt, bool, whether it supports applying model data*/
    "isSuportFDLibArmingType": true,
    /*ro, opt, bool, whether it supports face picture library arming type*/
    "isSuportFDLibSearch": true,
    /*ro, opt, bool, whether it supports searching face picture library*/
    "FDArmingRecordDataMaxNum": 12345,
    /*ro, opt, int, the supported maximum number of face records in the face picture arming library*/
    "isSupportControlPersonRecordByHumanId": true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by humanId*/
    "isSupportControlPersonRecordByRowKey": true,
    /*ro, opt, bool, whether it supports modifying and deleting the face record by rowKey*/
    "isSupportFaceLibRebuildCfg": true,
    /*ro, opt, bool, whether it supports recreating face picture library information and configuration*/
    "isSupportFDMove": true,
    /*ro, opt, bool, whether it supports moving face data in the face picture library in a batch, desc:the related URI is
/ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json*/
    "faceURLLen": 1024,
    /*ro, opt, int, the maximum size of the face picture URL, desc:if this node is not returned, the default size of the face picture URL supported by the
device is 256 bytes; otherwise, the device should support that the value of this node is greater than or equal to 256*/
    "isSupportArmingLibCfg": true,
    /*ro, opt, bool, whether it supports configuring parameters of the armed face picture library, desc:related URI:
/ISAPI/Intelligent/FDLib/armingLibCfg/capabilities?format=json*/
    "isSupportModelTransformation": true,
    /*ro, opt, bool, whether it supports converting face picture models in the face picture list library, desc:related URI:
/ISAPI/Intelligent/FDLib/model/transformation/capabilities?format=json*/
    "featurePointTypeList": ["face", "leftEye", "rightEye", "leftMouthCorner", "rightMouthCorner", "nose"],
    /*ro, opt, array, feature point types of face pictures supported by the device, subType:string*/
    "isSupportPresort": true,
    /*ro, opt, bool, whether it supports configuring pre-categorization, desc:related URI: /ISAPI/Intelligent/FDLib/presort/capabilities?format=json*/
    "libAttribute": {
    /*ro, opt, object, library attribute type, desc:"general" (normal library), "blockList" (blocklist library), "VIP" (VIP library), "passerby" (allowlist
library, which cannot be deleted)*/
        "@opt": ["general", "blackList", "VIP", "passerby"]
        /*ro, opt, array, options, subType:string*/
    },
    "faceType": {
    /*ro, opt, object, face picture type*/
```


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DETEKTOR

```
    /*ro, opt, object, face picture type*/
        "@opt": ["normalFace", "patrolFace", "hijackFace", "superFace"]
        /*ro, opt, array, options, subType:string*/
    },
    "saveFacePic": {
    /*ro, opt, object, whether to save the registered face picture*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "leaderPermission": {
    /*ro, opt, object, first authentication permission*/
        "@size":  4,
        /*ro, opt, int, the maximum number of array members, range:[1,4]*/
        "@min":  1,
        /*ro, opt, int, member value range, range:[1,4]*/
        "@max":  4
        /*ro, opt, int, member value range, range:[1,4]*/
    },
    "facePicFormat": {
    /*ro, opt, object*/
        "@opt": ["jpg", "png", "bmp"]
        /*ro, req, array, subType:string*/
    }
}
```

16.12.1.2 Search face picture library information by arming mode

Request URL

POST /ISAPI/Intelligent/FDLib/search?format=json

Query Parameter

None

Request Message

```
{
    "searchResultPosition":  0,
    /*req, int, the start position of the search result in the result list, desc:in a single search, if you cannot get all the records in the result list,
you can mark the end position and get the following records after the marked position in the next search*/
    "maxResults":  0,
    /*req, int, the maximum number of search results this time*/
    "libArmingType":  "armingLib"
    /*opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
}
```

Response Message

```
{
    "responseStatusStrg":  "OK",
    /*ro, opt, enum, searching status description, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO MATCH" (no matched
data).*/
    "numOfMatches":  1,
    /*ro, opt, int, number of results returned this time*/
    "totalMatches":  1,
    /*ro, opt, int, total number of matched results*/
    "FDLib": [
    /*ro, opt, array, face picture library information, subType:object*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID*/
            "faceLibType":  "blackFD",
            /*ro, opt, enum, face picture library type, desc:"blackFD" (list library), "staticFD" (static library)*/
            "name":  "test",
            /*ro, opt, string, name of the face picture library, range:[0,48]*/
            "customInfo":  "test",
            /*ro, opt, string, custom information, range:[0,192]*/
            "libArmingType":  "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:"armingLib" (armed library (default)), "nonArmingLib" (not armed library)*/
            "libAttribute":  "blackList"
            /*ro, opt, enum, library attribute type, subType:string, desc:"blackList" (blocklist), "VIP" (VIP library), "passerby" (passer-by library, which
cannot be deleted)*/
        }
    ]
}
```

16.12.1.3 Delete all face picture libraries

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json

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Query Parameter

None

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16.12.1.4 Create a face picture library

Request URL

POST /ISAPI/Intelligent/FDLib?format=json

Query Parameter

None

Request Message

```
{
    "faceLibType":  "blackFD,staticFD",
    /*req, string, face picture library type, desc:face picture library type*/
    "name":  "test",
    /*req, string, name of the face picture library, range:[0,48], desc:name of the face picture library*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192], desc:custom information*/
    "libArmingType":  "armingLib",
    /*opt, enum, subType:string*/
    "libAttribute":  "blackList",
    /*opt, enum, subType:string*/
    "FDID":  "test"
    /*opt, string, face picture library ID, desc:face picture library ID*/
}
```

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, req, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test"
    /*ro, opt, string, face picture library ID, desc:face picture library ID*/
}
```

16.12.1.5 Delete a specified face picture library

Request URL

DELETE /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

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DETEKTOR

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

16.12.1.6 Edit the information of a specific face picture library information, including name and custom information

Request URL

PUT /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	--
terminalNo	string	--

Request Message

```
{
    "name":  "test",
    /*opt, string, face picture library name, range:[0,48]*/
    "customInfo":  "test",
    /*opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*opt, enum, arming type of the list library, subType:string, desc:“armingLib” (armed face picture library), “nonArmingLib” (not armed face picture
library). The default value is "armingLib”*/
    "libAttribute":  "blackList"
    /*opt, enum, library type, subType:string, desc:“blackList” (blocklist library), “VIP” (VIP library), “passerby” (passerby library). The passerby
library cannot be deleted*/
}
```

Response Message

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DETEKTOR

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

16.12.1.7 Get the information, including library ID, library type, name, and custom information, of all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library), "infraredFD" (IR face picture library)
terminalNo	string	Terminal No. It is an integer starting from 1.

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code, desc:status code*/
    "statusString":  "OK",
    /*ro, req, string, status description, range:[1,64], desc:status description*/
    "subStatusCode":  "ok",
    /*ro, req, string, sub status code, range:[1,64], desc:sub status code*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:error description*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:"blackFD" (list library), "staticFD" (static library)*/
    "name":  "test",
    /*ro, opt, string, face picture library name, range:[1,48]*/
    "customInfo":  "test",
    /*ro, opt, string, custom information, range:[0,192]*/
    "libArmingType":  "armingLib",
    /*ro, opt, enum, library arming type, subType:string, desc:library arming type*/
    "libAttribute":  "blackList",
    /*ro, opt, enum, subType:string*/
    "personnelFileEnabled":  true
    /*ro, opt, bool*/
}
```

16.12.2 Face Picture Library Record Management

16.12.2.1 Get the total number of face records in all face picture libraries

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json

Query Parameter

None

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DETEKTOR

Request Message

None

Response Message

```
{
    "requestURL":  "http://10.7.52.31:8080/kms/services/rest/dataInfoService/downloadFile",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, req, string, see the description of this node and above nodes in the message of JSON_ResponseStatus, desc:this node is required when the value of
statusCode is not 1*/
    "FDRecordDataInfo": [
    /*ro, opt, array, information of face records in face picture library, subType:object, desc:this node is valid when errorCode is 1 and errorMsg is
"ok"*/
        {
            "FDID":  "test",
            /*ro, opt, string, face picture library ID, desc:the maximum size is 63 bytes*/
            "faceLibType":  "blackFD",
            /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type "blackFD" list library,"staticFD" static library, the
maximum size is 32 bytes*/
            "name":  "test",
            /*ro, opt, string, face picture library name, desc:the maximum size is 48 bytes*/
            "recordDataNumber":  123,
            /*ro, opt, int, number of records*/
            "libArmingType":  "armingLib",
            /*ro, opt, enum, arming type, subType:string, desc:arming type*/
            "libAttribute":  "general",
            /*ro, opt, enum, subType:string*/
            "personnelFileEnabled":  true
            /*ro, opt, bool*/
        }
    ],
    "totalRecordDataNumber":  500000,
    /*ro, opt, int, total number of records, desc:total number of records*/
    "FDCapacity": {
    /*ro, opt, object*/
        "total":  0.000,
        /*ro, req, float*/
        "use":  0.000,
        /*ro, req, float*/
        "remain":  0.000,
        /*ro, req, float, remaining space, desc:remaining space*/
        "maxRecordDataNumber":  0,
        /*ro, opt, int*/
        "useRecordDataNumber":  0,
        /*ro, opt, int*/
        "remainRecordDataNumber":  0
        /*ro, opt, int*/
    }
}
```

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DETEKTOR

16.12.2.2 Get the total number of face records in a face picture library

Request URL

GET /ISAPI/Intelligent/FDLib/Count?format=json&FDID=<FDID>&faceLibType=<faceLibType>&terminalNo=<terminalNo>

Query Parameter

Parameter Name	Parameter Type	Description
FDID	string	--
faceLibType	string	"blackFD" (list library), "staticFD" (static library), "infraredFD" (IR face picture library)
terminalNo	string	--

Request Message

None

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
    "FDID":  "test",
    /*ro, opt, string, face picture library ID, range:[1,64]*/
    "faceLibType":  "blackFD",
    /*ro, opt, enum, face picture library type, subType:string, desc:face picture library type*/
    "name":  "test",
    /*ro, opt, string, name of the face picture library, desc:the maximum length is 48*/
    "recordDataNumber":  123
    /*ro, opt, int, total number of records*/
}
```

16.12.2.3 Add a face record to the face picture library

Request URL

POST /ISAPI/Intelligent/FDLib/FaceDataRecord?format=json

Query Parameter

None

Request Message

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DETEKTOR

```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture by URL, desc:select this field or modelData*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type*/
    "FDID":  "test",
    /*req, string, face picture library ID, range:[1,64]*/
    "FPID":  " test",
    /*opt, string, face record ID, range:[1,1000], desc:face record ID,it can be generated by device or inputted. If it is inputted,it should be the unique ID with the combination of letters and digits,and the maximum length is 63 bytes; if it is generated by the device automatically,it is the same as the employee No. (person ID)*/
    "name":  "张三",
    /*req, string, name of person in the face picture, range:[1,1000], desc:the maximum size is 96 bytes*/
    "gender":  "male",
    /*opt, enum, gender of person in the face picture, subType:string, desc:“male”, “female”, “unknown”; the maximum size is 32 bytes*/
    "bornTime":  "2004-05-03",
    /*req, string, birthday of person in the face picture, desc:the maximum size is 20 bytes*/
    "city":  "130100",
    /*opt, string, city code of birth for the person in the face picture, desc:the maximum size is 32 bytes*/
    "certificateType":  "ID",
    /*opt, enum, certificate type, subType:string, desc:"ID" (ID card), "officerID" (officer ID), "other" (other type), "LawyerCertificate" (lawyer qualification), "birthCertificate", "householdRegister", "judgeCertificate", "studentIDCard", "workCard" (work permit),  "policeID", "foreignPermanentResidentIDCard", "passport", "pressCard", "prosecutorCard", "temporaryIDCard", "temporaryResidencePermit"*/
    "certificateNumber":  "test",
    /*opt, string, certificate No., range:[1,64], desc:the maximum size is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, desc:the maximum size is 192 bytes; it is valid when faceLibType is "blackFD”*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag, desc:up to 4 tags, which are separated by commas. The maximum size is 195 bytes. It is valid when faceLibType is "blackFD”*/
    "address":  "test",
    /*opt, string, person address, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD”*/
    "customInfo":  "test",
    /*opt, string, custom information, desc:the maximum size is 192 bytes. It is valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target modeling data, desc:target model data,non-modeled binary data needs to be encrypted by Base64 during transmission*/
    "transfer":  true,
    /*opt, bool, whether to enable transfer, desc:whether to enable transfer*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal” (operate by terminal)*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point, desc:object coordinates of the feature point*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:this node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:this node is required when featurePointType is "face"*/
            }
        }
    ],
    "cutOutRect": {
    /*opt, object, picture cropping frame, desc:the origin is the upper-left corner of the screen, which is used for the target frame in matting*/
        "x":  0.000,
        /*req, float, x-coordinate, range:[0,1], desc:normalize coordinate*/
        "y":  0.000,
        /*req, float, y-Coordinate, range:[0,1], desc:normalize coordinate*/
        "width":  0.000,
        /*req, float, width, range:[0,1]*/
        "height":  0.000
        /*req, float, height, range:[0,1]*/
    },
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress person face), "superFace" (super person face)*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:if the array is empty, it indicates clearing first authentication permission of the person; if not, it indicates that the person have first authentication permission for the door 1, 2, 3, and 4*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		facePic.jpg	--

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Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "requestURL":  "test",
    /*ro, opt, string, request URL*/
    "statusCode":  1,
    /*ro, req, int, status code*/
    "statusString":  "test",
    /*ro, req, string, status description*/
    "subStatusCode":  "test",
    /*ro, req, string, sub status code*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:this field is required when the value of statusCode is not 1*/
    "errorMsg":  "ok",
    /*ro, opt, string, error description, desc:see the description of this node and above nodes in the message of JSON_ResponseStatus*/
    "FPID":  "test",
    /*ro, opt, string, face record ID, range:[1,64], desc:face record ID returned when the face record is added,it is unique,and the maximum size is 63
bytes. This node is valid when errorCode is "1" and errorMsg is "ok"*/
    "rowKey":  "test"
    /*ro, opt, string, rowKey of face record, desc:rowKey of face record is returned after face data is added. It is unique and in string format. The
maximum length is 64 bytes.*/
}
```

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16.12.2.4 Edit face records in the face picture library in a batch

Request URL

PUT /ISAPI/Intelligent/FDLib/FDModify?format=json

Query Parameter

None

Request Message

```
{
    "faceURL":  "test",
    /*opt, string, picture storage URL inputted when uploading the face picture by URL, the maximum length is 256 bytes*/
    "faceLibType":  "blackFD",
    /*req, enum, face picture library type, subType:string, desc:face picture library type: "blackFD"-list library,"staticFD"-static library,the maximum
length is 32 bytes*/
    "FDID":  "test",
    /*req, string, face picture library ID, desc:face picture library ID,the maximum length is 63 bytes,multiple face picture libraries should be separated
by commas*/
    "FPID":  " ",
    /*opt, string, face record ID*/
    "name":  "test",
    /*opt, string, name*/
    "gender":  "any",
    /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used in search), "male", "female", "unknown"*/
    "bornTime":  "2004-05-03",
    /*req, time, date of birth of the person in the face picture in ISO8601 time format, the maximum length is 20 bytes, desc:date of birth of the person in
the face picture in ISO8601 time format,the maximum length is 20 bytes*/
    "city":  "130100",
    /*opt, string, code of the city of birth for the person in the face picture, the maximum length is 32 bytes, desc:code of the city of birth for the
person in the face picture,the maximum length is 32 bytes*/
    "certificateType":  "ID",
    /*req, enum, certificate type, subType:string*/
    "certificateNumber":  "test",
    /*opt, string, ID No., the maximum length is 32 bytes*/
    "caseInfo":  "test",
    /*opt, string, case information, it is valid when faceLibType is "blackFD", range:[0,192]*/
    "tag":  "aa,bb,cc,dd",
    /*opt, string, custom tag*/
    "address":  "test",
    /*opt, string, person address, it is valid when faceLibType is "staticFD"*/
    "customInfo":  "test",
    /*opt, string, custom information, it is valid when faceLibType is "staticFD"*/
    "modelData":  "test",
    /*opt, string, target model data, non-modeled binary data needs to be encrypted by base64 during transmission*/
    "operateType":  "byTerminal",
    /*opt, enum, operation type, subType:string, desc:"byTerminal"-by terminal*/
    "terminalNoList": [1],
    /*opt, array, terminal ID list, subType:int, desc:this node is required when operation type is "byTerminal"; currently, only one terminal is supported*/
    "PicFeaturePoints": [
    /*opt, array, feature points to be applied. If the device only supports three types of feature points, when the platform applies more than three types
of feature points,the device will not return error information, subType:object*/
        {
            "featurePointType":  "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of
mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, point coordinates*/
                "x":  1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized X-coordinate which is between 0 and 1000*/
                "y":  1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized Y-coordinate which is between 0 and 1000*/
                "width":  1,
                /*opt, int, width, range:[0,1000], desc:width which is between 0 and 1000. This node is required when featurePointType is "face"*/
                "height":  1
                /*opt, int, height, range:[0,1000], desc:height which is between 0 and 1000. This node is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType":  "normalFace",
    /*opt, enum, face picture type, subType:string*/
    "saveFacePic":  true,
    /*opt, bool, whether to save face pictures*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, subType:int, range:[1,4]*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		faceImage.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

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```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error details, desc:this field is required when statusCode is not 1*/
}
```

16.12.2.5 Get the capability of moving face data in the face picture library in a batch

Request URL

GET /ISAPI/Intelligent/FDLib/FDMove/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "FDMove": [
    /*ro, opt, array, subType:object*/
        {
            "id": {
            /*ro, opt, object, number of data that can be moved each time*/
                "@min":  0,
                /*ro, opt, int*/
                "@max":  0
                /*ro, opt, int, the maximum value*/
            },
            "FDSrc": {
            /*ro, opt, object, source face picture library from which the face record will be moved out*/
                "FDID": {
                /*ro, opt, object, face picture library ID*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                },
                "FPID": {
                /*ro, opt, object, face record ID*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                },
                "humanId": {
                /*ro, opt, object, person ID*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int, the maximum value*/
                }
            }
```

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```
},
"FDDst": {
/*ro, opt, object, destination face picture library which the face record will be moved to*/
    "FDID": {
    /*ro, opt, object, face picture library ID*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "FPID": {
    /*ro, opt, object, face record ID*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "humanId": {
    /*ro, opt, object, person ID*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "name": {
    /*ro, opt, object, name of the person in the face picture*/
        "@min":  0,
        /*ro, opt, int, the minimum value*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "gender": {
    /*ro, opt, object, gender of the person in the face picture*/
        "@min":  0,
        /*ro, opt, int, the minimum value*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "bornTime": {
    /*ro, opt, object, birth date of the person in the face picture*/
        "@min":  0,
        /*ro, opt, int, the minimum value*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "city": {
    /*ro, opt, object, city code of birth of the person in the face picture*/
        "@min":  0,
        /*ro, opt, int, the minimum value*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "certificateType": {
    /*ro, opt, object*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int*/
    },
    "certificateNumber": {
    /*ro, opt, object, certificate No.*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "caseInfo": {
    /*ro, opt, object, remarks*/
        "@min":  0,
        /*ro, opt, int, the minimum value*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "tag": {
    /*ro, opt, object, custom tag*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "address": {
    /*ro, opt, object, person address*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
    "customInfo": {
    /*ro, opt, object, custom information*/
        "@min":  0,
        /*ro, opt, int*/
        "@max":  0
        /*ro, opt, int, the maximum value*/
    },
```

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```
                "rowKey": {
                /*ro, opt, object, main key of face picture library*/
                    "@min":  0,
                    /*ro, opt, int*/
                    "@max":  0
                    /*ro, opt, int*/
                },
                "transfer": {
                /*ro, opt, object, whether to transfer*/
                    "@opt": [true, false]
                    /*ro, opt, array, subType:bool*/
                }
            }
        }
    ]
}
```

16.12.2.6 Move face data in the face picture library in a batch

Request URL

POST /ISAPI/Intelligent/FDLib/FDMove?format=json

Query Parameter

None

Request Message

```
{
    "FDMove": [
    /*req, array, subType:object*/
        {
            "id":  1,
            /*req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results*/
            "FDSrc": {
            /*opt, object, face records to be moved*/
                "FDID":  "test",
                /*req, string, face picture library ID*/
                "FPID":  "test",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX"
                /*opt, string, person ID*/
            },
            "FDDst": {
            /*opt, object, destination face picture library (face records will be moved to this face picture library)*/
                "FDID":  "test",
                /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
                "FPID":  " ",
                /*req, string, face record ID*/
                "humanId":  "humanIDXXX",
                /*opt, string, person ID*/
                "name":  "test",
                /*opt, string, name*/
                "gender":  "any",
                /*req, enum, gender, subType:string, desc:"any" (unlimited condition which is used for search), "male", "female", "unknown"*/
                "bornTime":  "2004-05-03",
                /*req, string, birth date of the person in the face picture, range:[0,20]*/
                "city":  "130100",
                /*opt, string, city code, desc:city of birth*/
                "certificateType":  "ID",
                /*req, enum, certificate type, subType:string, desc:certificate type*/
                "certificateNumber":  "test",
                /*opt, string, certificate No.*/
                "caseInfo":  "test",
                /*opt, string, remarks*/
                "tag":  "aa,bb,cc,dd",
                /*opt, string, custom tag*/
                "address":  "test",
                /*opt, string, person address*/
                "customInfo":  "test",
                /*opt, string, custom information*/
                "rowKey ":  "test",
                /*opt, string, key of the face record database,the maximum size is 64 bytes. Searching by rowKey can be efficient*/
                "transfer":  true
                /*opt, bool, whether to enable transferring, desc:whether to enable transferring*/
            }
        }
    ]
}
```

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DETEKTOR

Response Message

```
{
    "FDMoveResult": [
    /*ro, req, array, result of batch moving, subType:object*/
        {
            "id": 1,
            /*ro, req, int, serial No. of the element in the array,it starts from 1 and is used to match returned results, desc:serial No. of the element in the array,it starts from 1 and is used to match returned results*/
            "errorCode": 1,
            /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
            "errorMsg": "ok"
            /*ro, req, string, error description, desc:this node is required when the value of statusCode is not 1*/
        }
    ]
}
```

16.12.2.7 Set the face picture data in the face picture library

Request URL

PUT /ISAPI/Intelligent/FDLib/FDSetUp?format=json

Query Parameter

None

Request Message

```
{
    "faceURL": "http",
    /*opt, string, picture storage URL, which is inputted when uploading the face picture by URL*/
    "faceLibType": "blackFD",
    /*req, enum, face picture library type, subType:string, desc:"blackFD" (list library), "staticFD" (static library), the maximum length is 32 bytes*/
    "FDID": "test",
    /*req, string, face picture library ID, desc:the maximum length is 63 bytes. Multiple libraries are separated by commas*/
    "FPID": " ",
    /*opt, string, face record ID, desc:it can be generated by the device or inputted. If it is inputted, it should be the unique ID with the combination of letters and digits, and the maximum length is 63 bytes; if it is generated by the device automatically, it is the same as the employee No. (person ID)*/
    "deleteFP": true,
    /*opt, bool, whether to delete the face record, desc:true (yes). This field required when the face record needs to be deleted; for adding or editing the face record, it should not be configured*/
    "modelData": "test",
    /*opt, string, target model data, desc:during the process of transmission, the non-modeling binary data will be encrypted with Base64 method*/
    "PicFeaturePoints": [
    /*opt, array, feature picture coordinates, subType:object, desc:if the device only supports three types of feature points, when the platform applies more than three types of feature points, the device will not return error information*/
        {
            "featurePointType": "face",
            /*req, enum, feature point type, subType:string, desc:"face", "leftEye" (left eye), "rightEye" (right eye), "leftMouthCorner" (left corner of mouth), "rightMouthCorner" (right corner of mouth), "nose"*/
            "coordinatePoint": {
            /*opt, object, coordinates of the feature point*/
                "x": 1,
                /*req, int, X-coordinate, range:[0,1000], desc:normalized coordinates*/
                "y": 1,
                /*req, int, Y-coordinate, range:[0,1000], desc:normalized coordinates*/
                "width": 1,
                /*opt, int, width, range:[0,1000], desc:it is required when featurePointType is "face"*/
                "height": 1
                /*opt, int, height, range:[0,1000], desc:it is required when featurePointType is "face"*/
            }
        }
    ],
    "faceType": "normalFace",
    /*opt, enum, face picture type, subType:string, desc:"normalFace" (normal face, default value), "patrolFace" (patrol person face), "hijackFace" (duress person face), "superFace" (super person face)*/
    "saveFacePic": true,
    /*opt, bool, whether to save the registered face picture*/
    "leaderPermission": [1, 2, 3, 4]
    /*opt, array, first authentication permission, subType:int, range:[1,4], desc:first authentication permission*/
}
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
faceURL	[Message content]	application/json	--	--	--
img	[Binary picture data]	image/jpeg		faceImage.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, req, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, req, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

16.12.3 Hard Hat Detection

16.12.3.1 Get arming schedules for hard hat detection of all channels

Request URL

GET /ISAPI/Event/schedules/safetyHelmetDetection

Query Parameter

None

Request Message

None

Response Message

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```xml
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, arming schedules for hard hat detection of all channels, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--ro, opt, object, arming settings-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <eventType>
      <!--ro, opt, enum, event type, subType:string-->safetyHelmetDetection
    </eventType>
    <videoInputChannelID>
      <!--ro, opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--ro, req, array, list of time periods, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--ro, opt, object, time-->
        <dayOfWeek>
          <!--ro, opt, int-->1
        </dayOfWeek>
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--ro, opt, array, subType:object-->
      <TimeBlock>
        <!--ro, opt, object, time range-->
        <TimeRange>
          <!--ro, req, object, time range-->
          <beginTime>
            <!--ro, req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SafetyHelmetDetectionScheduleList>
```

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16.12.3.2 Set arming schedules for hard hat detection of all channels

Request URL

PUT /ISAPI/Event/schedules/safetyHelmetDetection

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the structure XML_Schedule, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object-->
    <id>
      <!--req, string-->test
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->safetyHelmetDetection
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->test
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, subType:object, attr:size{req, int}-->
      <TimeBlock>
        <!--opt, object-->
        <dayOfWeek>
          <!--opt, int-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--req, time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, subType:object-->
      <TimeBlock>
        <!--opt, object-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->00:00:00+08:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->00:00:00+08:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</SafetyHelmetDetectionScheduleList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

16.12.3.3 Get the parameters of hard hat detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, hard hat detection parameters, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object, normalized value-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[1,1000]-->1000
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height, range:[1,1000]-->1000
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled>
    <!--ro, opt, bool, whether to enable the rule-->true
  </enabled>
  <SafetyHelmetDetectionRegionList>
    <!--ro, opt, array, subType:object-->
    <SafetyHelmetDetectionRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, string, region ID-->test
      </id>
      <RegionCoordinatesList>
        <!--ro, opt, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget>
    <!--ro, req, bool, whether to overlay target frame on the alarm picture-->true
  </stackTarget>
  <stackRule>
    <!--ro, req, bool, whether to overlay rule frame on the alarm picture-->true
  </stackRule>
  <alarmLinkageDuration>
    <!--ro, req, enum, alarm linkage duration, subType:string, desc:alarm linkage duration-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--ro, opt, object-->
    <Resolution>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, picture ID-->test
      </id>
      <name>
        <!--ro, req, enum, pixels, subType:string, desc:pixels-->1080p
      </name>
      <value>
        <!--ro, req, enum, resolution, subType:string, desc:resolution-->1920*1080
      </value>
    </Resolution>
    <quality>
      <!--ro, req, enum, picture quality, subType:string, desc:picture quality-->best
    </quality>
    <alarmUpload>
      <!--ro, req, bool, whether to upload alarm picture-->true
    </alarmUpload>
    <localStorage>
      <!--ro, opt, bool, whether to store pictures locally-->true
    </localStorage>
  </ImageParam>
  <videoOverlay>
    <!--ro, opt, object, video overlay parameters-->
    <stackTarget>
      <!--ro, req, bool, whether to overlay alarm target frame-->true
    </stackTarget>
    <stackRule>
      <!--ro, req, bool, whether to overlay alarm rule frame-->true
    </stackRule>
  </videoOverlay>
  <wearHelmetEnabled>
    <!--ro, opt, bool-->true
  </wearHelmetEnabled>
  <analysisMode>
    <!--ro, opt, enum, analysis mode, subType:string, desc:the engine also has its analysis mode (refer to the <analysisMode> node in
/ISAPI/Intelligent/analysisEngines/<ID>). The device will be added to the engine with the same analysis mode via the load balancing method. If the engine is
not configured correspondingly this node will be failed to e returned and the error code of taskNotMatchEngine will be returned: "realTimeMode"-real-time
```

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```
not configured correspondingly,this node will be jutted to e returned,and the error code of taskNotMatchEngine will be returned, "realTimeMode -real time
mode,"autoSwitchMode"-auto-switch mode-->realTimeMode
  </analysisMode>
  <helmetFilterEnabled>
    <!--ro, opt, bool-->false
  </helmetFilterEnabled>
  <durationCapEnabled>
    <!--ro, opt, bool-->false
  </durationCapEnabled>
  <durationCapInterval>
    <!--ro, opt, int, range:[0,3600]-->30
  </durationCapInterval>
  <shieldRegionList>
    <!--ro, opt, array, subType:object-->
    <shieldRegion>
      <!--ro, opt, object-->
      <id>
        <!--ro, opt, string-->test
      </id>
      <regionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[3,10]-->
        <regionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->1
          </positionY>
        </regionCoordinates>
      </regionCoordinatesList>
    </shieldRegion>
  </shieldRegionList>
  <sensitivity>
    <!--ro, opt, int, range:[0,100]-->1
  </sensitivity>
</SafetyHelmetDetection>
```

16.12.3.4 Set the parameters of hard hat detection

Request URL

PUT /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetection xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, hard hat detection, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--opt, object-->
    <normalizedScreenWidth>
      <!--req, int, normalized width, range:[1,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height, range:[1,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled>
    <!--opt, bool, whether to enable the hard hat detection rules-->true
  </enabled>
  <SafetyHelmetDetectionRegionList>
    <!--opt, array, subType:object-->
    <SafetyHelmetDetectionRegion>
      <!--opt, object-->
      <id>
        <!--opt, string, detection area ID-->test
      </id>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin (0, 0) is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget>
    <!--req, bool, whether to overlay the alarm target frame-->true
  </stackTarget>
  <stackRule>
    <!--req, bool, whether to overlay the alarm rule frame-->true
  </stackRule>
  <alarmLinkageDuration>
    <!--req, enum, alarm linkage duration time, subType:string, desc:alarm linkage duration time-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--opt, object, image parameters-->
    <Resolution>
      <!--opt, object-->
      <id>
        <!--req, string, picture ID-->test
      </id>
      <name>
        <!--req, enum, subType:string-->1080p
      </name>
      <value>
        <!--req, enum, resolution, subType:string, desc:resolution-->1920*1080
      </value>
    </Resolution>
    <quality>
      <!--req, enum, picture quality, subType:string, desc:picture quality-->best
    </quality>
    <alarmUpload>
      <!--req, bool-->true
    </alarmUpload>
    <localStorage>
      <!--opt, bool-->true
    </localStorage>
  </ImageParam>
  <videoOverlay>
    <!--opt, object-->
    <stackTarget>
      <!--req, bool, whether to overlay the target frame-->true
    </stackTarget>
    <stackRule>
      <!--req, bool, whether to overlay the rule frame-->true
    </stackRule>
  </videoOverlay>
  <wearHelmetEnabled>
    <!--opt, bool-->true
  </wearHelmetEnabled>
  <analysisMode>
    <!--opt, enum, analysis mode, subType:string, desc:analysis mode-->realTimeMode
  </analysisMode>
</SafetyHelmetDetection>
```

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Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
SafetyHelmetDetection	[Message content]	application/xml	--	--	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
</ResponseStatus>
```

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16.12.3.5 Get the configuration capability of hard hat detection

Request URL

GET /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SafetyHelmetDetectionCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, opt, object, ro,normalized value-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width, range:[1,1000]-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
```

```
      <!--ro, req, int, normalized height, range:[1,1000]-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <SafetyHelmetDetectionRegionList size="8">
    <!--ro, opt, array, detection area list, subType:object, attr:size{req, int}-->
    <SafetyHelmetDetectionRegion>
      <!--ro, opt, object, detection area-->
      <id>
        <!--ro, opt, string-->test
      </id>
      <RegionCoordinatesList size="10">
        <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:area coordinates-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->1
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SafetyHelmetDetectionRegion>
  </SafetyHelmetDetectionRegionList>
  <stackTarget opt="true,false">
    <!--ro, req, bool, whether to overlay alarm target frame, attr:opt{req, string}-->true
  </stackTarget>
  <stackRule opt="true,false">
    <!--ro, req, bool, whether to overlay alarm rule frame, attr:opt{req, string}-->true
  </stackRule>
  <alarmLinkageDuration opt="5,10,30,60,120,300,600">
    <!--ro, req, enum, alarm linkage duration, subType:int, attr:opt{req, string}, desc:alarm linkage duration-->5
  </alarmLinkageDuration>
  <ImageParam>
    <!--ro, opt, object, picture parameters-->
    <ResolutionList size="3">
      <!--ro, opt, array, resolution, subType:object, attr:size{req, int}-->
      <Resolution>
        <!--ro, opt, object, resolution-->
        <id>
          <!--ro, req, string, ID-->test
        </id>
        <name>
          <!--ro, req, enum, resolution name, subType:string, desc:resolution name-->1080p
        </name>
        <value>
          <!--ro, req, enum, subType:string-->1920*1080
        </value>
      </Resolution>
    </ResolutionList>
    <quality opt="best,good,general">
      <!--ro, req, enum, picture quality, subType:string, attr:opt{req, string}, desc:"best","good","general"-normal-->best
    </quality>
    <alarmUpload opt="true,false">
      <!--ro, req, bool, whether to upload alarm picture, attr:opt{req, string}-->true
    </alarmUpload>
    <localStorage opt="true,false">
      <!--ro, opt, bool, whether to store pictures locally, attr:opt{req, string}-->true
    </localStorage>
  </ImageParam>
  <AdvanceConfiguration>
    <!--ro, opt, object-->
    <headSensitivity min="1" max="5" def="3">
      <!--ro, opt, int, head detection sensitivity, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates more sensitive detection-->1
    </headSensitivity>
    <humanSensitivity min="1" max="5" def="3">
      <!--ro, opt, int, human body detection sensitivity, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates more sensitive
detection-->1
    </humanSensitivity>
    <alarmAnalysisFrame min="1" max="50" def="12">
      <!--ro, opt, int, number of frames required for alarm analysis, attr:min{req, int},max{req, int},def{req, int}, desc:larger value indicates that more
frames are needed to be analyzed and more slowly the alarm will be triggered-->1
    </alarmAnalysisFrame>
    <FDLibList size="10">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <FDLib>
        <!--ro, opt, object, face picture library-->
        <id min="1" max="10">
          <!--ro, req, int, No., range:[1,10], attr:min{req, int},max{req, int}-->1
        </id>
        <FDID min="1" max="10">
          <!--ro, req, string, attr:min{req, int},max{req, int}-->test
        </FDID>
        <thresholdValue min="1" max="10">
          <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
        </thresholdValue>
        <customFaceLibID min="1" max="10">
          <!--ro, opt, string, custom face picture library ID, attr:min{req, int},max{req, int}, desc:custom face picture library ID-->test
        </customFaceLibID>
      </FDLib>
    </FDLibList>
```

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```
    </AdvanceConfiguration>
    <SearchDescriptionCap>
      <!--ro, opt, object, ability of searching for safety helmet event, desc:ability of searching for safety helmet event-->
      <timeSpanMaxNum>
        <!--ro, req, enum, the maximum number of time periods in which search can be conducted simultaneously, subType:string, desc:the maximum number of time
  periods in which search can be conducted simultaneously-->all
      </timeSpanMaxNum>
      <resultMaxNum>
        <!--ro, req, int, the maximum number of items that can be searched for at a time-->1
      </resultMaxNum>
      <faceContrastResult opt="all,success,failed">
        <!--ro, req, string, comparison result, attr:opt{req, string}-->test
      </faceContrastResult>
      <FDLibList size="10">
        <!--ro, opt, array, subType:object, attr:size{req, int}-->
        <FDLib>
          <!--ro, opt, object, face picture library-->
          <FDID min="1" max="10">
            <!--ro, req, string, attr:min{req, int},max{req, int}-->test
          </FDID>
        </FDLib>
      </FDLibList>
      <name min="1" max="10">
        <!--ro, opt, string, (search condition) name, attr:min{req, int},max{req, int}, desc:720p,D1"-->test
      </name>
    </SearchDescriptionCap>
    <videoOverlay>
      <!--ro, opt, object, video overlay parameters-->
      <stackTarget opt="true,false">
        <!--ro, req, bool, whether to overlay alarm target frame, attr:opt{req, string}-->true
      </stackTarget>
      <stackRule opt="true,false">
        <!--ro, req, bool, whether to overlay alarm rule frame, attr:opt{req, string}-->true
      </stackRule>
    </videoOverlay>
    <wearHelmetEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </wearHelmetEnabled>
    <analysisMode opt="realTimeMode,autoSwitchMode">
      <!--ro, opt, string, attr:opt{req, string}-->test
    </analysisMode>
    <helmetFilterEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </helmetFilterEnabled>
    <durationCapEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->false
    </durationCapEnabled>
    <durationCapInterval min="0" max="3600">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->30
    </durationCapInterval>
    <shieldRegionList size="8">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <shieldRegion>
        <!--ro, opt, object-->
        <id>
          <!--ro, opt, string-->test
        </id>
        <regionCoordinatesList size="10">
          <!--ro, opt, array, subType:object, range:[3,10], attr:size{req, int}-->
          <regionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->1
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->1
            </positionY>
          </regionCoordinates>
        </regionCoordinatesList>
      </shieldRegion>
    </shieldRegionList>
    <sensitivity min="0" max="100">
      <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </SafetyHelmetDetectionCap>
```

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16.12.3.6 Search the hard hat detection events

Request URL

POST /ISAPI/Intelligent/channels/<channelID>/safetyHelmetDetection/search?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "SearchDescription": {
    /*opt, object, search condition*/
        "searchID":  "test",
        /*req, string, search ID, desc:this node is used to check whether searches are the same one. Due to device performance limitation, you can apply the
search conditions multiple times (the search ID should be the same) to complete one search and get the search progress*/
        "searchResultPosition":  0,
        /*req, int, the start position of the search result in the result list*/
        "maxResultNumber":  10000,
        /*opt, int, maximum number of results that can be searched*/
        "maxResults":  100,
        /*req, int, the maximum number of search results this time*/
        "timeSpanList": [
        /*opt, array, list of time periods, subType:object*/
            {
                "startTime":  "2004-05-03T17:30:08+08:00",
                /*req, string, start time*/
                "endTime":  "2004-05-03T17:30:08+08:00"
                /*req, string, end time*/
            }
        ],
        "faceContrastResult":  "all",
        /*req, enum, comparison result, subType:string, desc:"all" (including succeeded and failed), "success", "failed"*/
        "FDLibList": [
        /*opt, array, list of face picture libraries, subType:object*/
            {
                "FDLib": {
                /*opt, object, face picture library*/
                    "FDID":  "test"
                    /*opt, string, a unique ID of face picture library*/
                }
            }
        ],
        "name":  "test",
        /*opt, string, person name, desc:this node is valid when FDLibList exists*/
        "channelID": [1, 2]
        /*opt, array, channel No., subType:int, desc:multiple channel No.s are allowed*/
    }
}
```

Response Message

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```
{
    "SearchResult": {
    /*ro, req, object, search result*/
        "responseStatusStrg":  "OK",
        /*ro, req, enum, search status description: "OK"-search ended,"MORE"-searching,"NO_MATCHES"-no data found,the maximum size is 32 bytes,{dep if
errcode == 1 && errMsg == ok}, subType:string, desc:"OK" (searching completed), "MORE" (searching for more data), "NO_MATCHES" (no matched data)*/
        "numOfMatches":  1,
        /*ro, req, int, number of results returned this time, desc:number of records matched in this search,{dep if errcode == 1 && errMsg == ok}*/
        "totalMatches":  1,
        /*ro, req, int, total number of matched results, desc:total number of records matched,{dep if errcode == 1 && errMsg == ok}*/
        "targets": [
        /*ro, opt, array, search result, subType:object*/
            {
                "time":  "2017-08-25T11:34:59+08:00",
                /*ro, req, string, statistics time*/
                "channel":  1,
                /*ro, req, int, channel No.*/
                "Target": {
                /*ro, req, object, target information*/
                    "Rect": [
                    /*ro, opt, array, target frame, subType:object, range:[0,1], desc:the origin is the upper-left corner of the screen*/
                        {
                            "targetID":  1,
                            /*ro, opt, int, target ID*/
                            "height":  1.000,
                            /*ro, req, float, height, range:[0.000,1.000]*/
                            "width":  1.000,
                            /*ro, req, float, width, range:[0.000,1.000]*/
                            "x":  0.000,
                            /*ro, req, float, X-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
                            "y":  0.000
                            /*ro, req, float, Y-coordinate of the upper-left corner of the frame, range:[0.000,1.000]*/
                        }
                    ],
                    "face": [
                    /*ro, opt, array, face picture information, this node is returned when face picture comparison is linked to hard hat detection,
subType:object*/
                        {
                            "FDID":  "1123",
                            /*ro, opt, string, face picture library ID, desc:the maximum length is 63*/
                            "FPID":  "test",
                            /*ro, opt, string, face record ID, desc:face record ID,if it is inputted,it should be the unique ID consisting of letters and
digits,and the maximum length is 63 bytes*/
                            "name":  "test",
                            /*ro, opt, string, name*/
                            "caputureFacePicURL":  "test",
                            /*ro, req, string, captured face picture URL, desc:the maximum size is 256, the captured picture will be returned no matter the
face picture comparison is successful or not*/
                            "libFacePicURL":  "test"
                            /*ro, opt, string, URL of picture, which matches the captured face picture, desc:URL of the matched picture in the library,the
maximum size is 256 bytes*/
                        }
                    ]
                },
                "backgroundImageURL":  "test"
                /*ro, opt, string, background picture URL*/
            }
        ]
    }
}
```

16.12.3.7 Hard hat detection results

EventType:safetyHelmetDetection

```
{
    "ipAddress":  "172.6.64.7",
    /*ro, req, string, IPv4 address of the device that triggers the alarm*/
    "ipv6Address":  "1080:0:0:0:8:800:200C:417A",
    /*ro, opt, string, IPv6 address of the device that triggers the alarm*/
    "portNo":  80,
    /*ro, opt, int, communication port No. of the device that triggers the alarm*/
    "protocol":  "HTTP",
    /*ro, opt, enum, transmission communication protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The
value should be "HTTP" when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP*/
    "macAddress":  "01:17:24:45:D9:F4",
    /*ro, opt, string, MAC address*/
    "channelID":  1,
    /*ro, opt, int, channel No. of the device that triggers the alarm, desc:when ISAPI protocol is transmitted via HCNetSDK, the channel No. is the video
channel No. of private protocol. When ISAPI protocol is transmitted via EZ protocol, the channel No. is the video channel No. of EZ protocol. When ISAPI
protocol is transmitted via ISUP, the channel No. is the video channel No. of ISUP*/
    "releatedChannelList": [1, 2, 3],
    /*ro, opt, array, list of alarm related channels, which are of the same camera with channelID, subType:int, desc:this parameter is used for live view or
playback on the platform*/
    "dateTime":  "2004-05-03T17:30:08+08:00",
    /*ro, req, datetime, alarm trigger time*/
    "activePostCount":  1,
    /*ro, opt, int, times that the same alarm has been uploaded, desc:times that the same alarm has been uploaded*/
```

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```
    "eventType":  "safetyHelmetDetection",
    /*ro, req, string, event type, desc:"safetyHelmetDetection" (hard hat detection event)*/
    "eventState":  "active",
    /*ro, req, enum, event status, subType:string, desc:for durative event: active (valid event or event starts), inactive (invalid event or the event
ends). For the heartbeat, the field value indicates the heartbeat data, and it is uploaded every 10 seconds*/
    "eventDescription":  "Safety Helmet Detection",
    /*ro, req, string, event description, desc:Hard Hat Detection*/
    "channelName":  "ABC",
    /*ro, req, string, channel name (camera name)*/
    "deviceID":  "test0123",
    /*ro, opt, string, device ID which is also the PUID, desc:this field must be returned when ISAPI event information is transmitted via ISUP*/
    "safetyHelmetDetectionBackgroundImageResolution": {
    /*ro, opt, object*/
        "height":  1,
        /*ro, req, int, range:[1,65535], unit:px*/
        "width":  1
        /*ro, req, int, range:[1,65535], unit:px*/
    },
    "Target": [
    /*ro, opt, array, target, subType:object, desc:multiple targets can be output after being recognized once*/
        {
            "recognitionType":  "safetyHelmet",
            /*ro, req, enum, recognition type, subType:string, desc:"safetyHelmet" (hard hat detection)*/
            "TargetInfo": {
            /*ro, opt, object, target information*/
                "recognition":  "safetyHelmet",
                /*ro, req, enum, recognition type, subType:string, desc:"safetyHelmet" (hard hat detection)*/
                "dataSource":  "timing"
                /*ro, req, enum, detection data source, subType:string, desc:"timing" (scheduled), "trigger" (when exceeded the configured level limit)*/
            },
            "rect": [
            /*ro, opt, array, rectangle frame of the hard hat, subType:object, desc:only one element will be returned. The origin is the upper-left corner
of the screen*/
                {
                    "height":  1.000,
                    /*ro, opt, float, height, range:[0.000,1.000]*/
                    "width":  1.000,
                    /*ro, opt, float, width, range:[0.000,1.000]*/
                    "x":  0.000,
                    /*ro, opt, float, X-coordinate, range:[0.000,1.000]*/
                    "y":  0.000
                    /*ro, opt, float, Y-coordinate, range:[0.000,1.000]*/
                }
            ],
            "contentID":  "safetyHelmet1_image",
            /*ro, opt, string, thumbnail content ID, range:[0,256]*/
            "pId":  "F4F665D6A18E41308CE9934DCDDD2222",
            /*ro, opt, string, thumbnail ID of hard hat detection alarm, range:[0,64]*/
            "uid":  "test",
            /*ro, opt, string, unique ID of event uploading, range:[0,64], desc:it can be composed of time (accurate to the millisecond) and random digits*/
            "safetyHelmetURL":  "test",
            /*ro, opt, string, thumbnail URL of hard hat detection alarm, range:[0,256]*/
            "faceContrastResult":  "success",
            /*ro, opt, enum, face picture comparison result, subType:string, desc:this field is returned only when the linkage between hard hat detection
and face picture (face comparison linkage) has been configured. "success" (comparison succeeded), "failed" (comparison failed)*/
            "Face": {
            /*ro, opt, object, face picture information*/
                "FDID":  "1123",
                /*ro, req, string, face picture library ID, range:[0,63]*/
                "FPID":  "test",
                /*ro, opt, string, face record ID, range:[0,63], desc:unique ID consisting of letters and digits*/
                "contentID":  "faceLibPic",
                /*ro, opt, string, contentID of picture, which matches the captured face picture, range:[0,256]*/
                "pId":  "F4F665D6A18E41308CE9934DCDDD3333",
                /*ro, opt, string, ID of picture, which matches the captured face picture, range:[0,64]*/
                "libFacePicURL":  "test"
                /*ro, opt, string, URL of picture, which matches the captured face picture, range:[0,256]*/
            },
            "FaceList": [
            /*ro, opt, array, face picture information list, subType:object, desc:face picture information list*/
                {
                    "FDID":  "1123",
                    /*ro, req, string, face picture library ID, range:[0,63]*/
                    "FPID":  "test",
                    /*ro, opt, string, face record ID, range:[0,63], desc:unique ID consisting of letters and digits*/
                    "contentID":  "faceLibPic",
                    /*ro, opt, string, contentID of picture, which matches the captured face picture, range:[0,256]*/
                    "pId":  "F4F665D6A18E41308CE9934DCDDD3333",
                    /*ro, opt, string, ID of picture, which matches the captured face picture, range:[0,64]*/
                    "libFacePicURL":  "test",
                    /*ro, opt, string, URL of picture, which matches the captured face picture, range:[0,256]*/
                    "name":  "test",
                    /*ro, opt, string, person name, range:[1,32], desc:the sensitive information should be encrypted*/
                    "certificateType":  "passportID",
                    /*ro, opt, enum, certificate type, subType:string, desc:the sensitive information should be encrypted. "officerID" (officer ID), "ID"
(ID card), "passportID" (passport), "other" (other type)*/
                    "certificateNumber":  "test"
                    /*ro, opt, string, certificate number, range:[8,32], desc:the sensitive information should be encrypted*/
                }
            ],
            "status":  "wear",
            /*ro, opt, enum, whether the person wears a hard hat, subType:string, desc:"unwear" (not wear, default value), "wear" (wear)*/
            "safetyHelmetColour":  "yellow",
            /*ro, opt, enum, subType:string, dep:and {$.Target[*].status,eq,wear}*/
```

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```
            /*ro, opt, enum, subType:string, dep:and,{$.Target[*].status,eq,wear}*/
            "reflectiveVestStatus":  "wear",
            /*ro, opt, enum, subType:string*/
            "presetID":  0
            /*ro, opt, int*/
        }
    ],
    "contentID":  "background_image",
    /*ro, opt, string, thumbnail content ID, range:[0,256]*/
    "pId":  "F4F665D6A18E41308CE9934DCDDD1111",
    /*ro, opt, string, thumbnail ID of hard hat detection alarm, range:[0,64]*/
    "imageURL":  "test",
    /*ro, opt, string, background picture URL, range:[0,256]*/
    "URLCertificationType":  "digest"
    /*ro, opt, enum, picture URL authentication method, subType:string, desc:"no" (it is for the cloud storage protocol), "digest" (digest authentication,
it is for the device local storage)*/
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
safetyHelmetDetection	[Message content]	application/json	--	--	--
.value	[Binary picture data]	image/jpeg	safetyHelmet1_image	safetyHelmet1_image.jpg	--
.value	[Binary picture data]	image/jpeg	faceLibPic	faceLibPic.jpg	--
.value	[Binary picture data]	image/jpeg	background_image	background_image.jpg	--

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Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

16.13 Audio (General)

16.13.1 Audio Recording

16.13.1.1 Get monthly audio information of an audio channel

Request URL

POST /ISAPI/ContentMgmt/recordAudio/<audioID>/searchDailyDistribution?format=json

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

```
{
    "year":  2021,
    /*req, int, year*/
    "month":  1
    /*req, int, month, range:[1,12]*/
}
```

Response Message

```
{
    "AudioList": [
    /*ro, opt, array, monthly calendar list, subType:object*/
        {
            "dayID":  1,
            /*ro, req, int, day, range:[1,31]*/
            "record":  true
            /*ro, req, bool, whether there is audio*/
        }
    ]
}
```

16.13.1.2 Get the capability of audio recording schedule parameters

Request URL

GET /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--

Request Message

None

Response Message

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```
{
    "AudioRecordPlanCap": {
    /*opt, object, get the capability of audio recording schedule parameters, desc:related URL: GET/PUT
/ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json*/
        "enable": {
        /*req, object, whether to enable recording schedule range*/
            "@opt": [true, false],
            /*opt, array, whether to enable recording schedule enumeration list, subType:bool*/
            "@def":  true
            /*opt, bool, default value*/
        },
        "mode": {
        /*opt, object, recording type range*/
            "@opt": ["timing"],
            /*opt, array, recording type enumeration list, subType:string*/
            "@def":  "timing"
            /*opt, string, default value*/
        },
        "weekTimeList": {
        /*opt, object, week time*/
            "@size":  7,
            /*opt, int, the maximum number allowed*/
            "@minSize":  1,
            /*opt, int, the minimum number allowed*/
            "dayOfWeek": {
            /*opt, object, day of a week range*/
                "@opt": [1, 2, 3, 4, 5, 6, 7],
                /*opt, array, day of a week enumeration list, subType:int*/
                "@def":  1
                /*opt, int, default value*/
            },
            "timeRange": {
            /*req, object, time period*/
                "@size":  8,
                /*opt, int, the maximum number allowed*/
                "@minSize":  1,
                /*opt, int, the minimum number allowed*/
                "startTime":  "undefined",
                /*req, string, start time value example*/
                "endTime":  "08:00:00"
                /*req, string, end time value example*/
            }
        },
        "holidayTimeList": {
        /*opt, object, holiday time*/
            "@size":  32,
            /*opt, int, the maximum number allowed*/
            "startTime":  "undefined",
            /*req, string, start time value example, desc:format: 00:00*/
            "endTime":  "undefined"
            /*req, string, end time value example, desc:format: 00:00*/
        },
        "preRecordTimeSeconds": {
        /*opt, object, pre-record time range*/
            "@def":  5,
            /*opt, int, default value*/
            "@min":  0,
            /*opt, int, the minimum value*/
            "@max":  10
            /*opt, int, the maximum value*/
        },
        "postRecordTimeSeconds": {
        /*opt, object, post-record time range*/
            "@def":  5,
            /*opt, int, default value*/
            "@min":  0,
            /*opt, int, the minimum value*/
            "@max":  10
            /*opt, int, the maximum value*/
        }
    }
}
```

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16.13.1.3 Get audio recording schedule

Request URL

GET /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	audio channel

Request Message

None

Response Message

```
{
    "enable":  true,
    /*req, bool, whether to enable audio recording schedule*/
    "mode":  "timing",
    /*opt, enum, audio recording type, subType:string, desc:"timing" (scheduled audio recording)*/
    "weekTimeList": [
    /*opt, array, week time list, subType:object*/
        {
            "dayOfWeek":  1,
            /*opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
            "timeRange": [
            /*req, array, time period, subType:object*/
                {
                    "startTime":  "undefined",
                    /*req, string, start time*/
                    "endTime":  "08:00:00"
                    /*req, string, end time, range:[0,64]*/
                }
            ]
        }
    ],
    "holidayTimeList": [
    /*opt, array, holiday time list, subType:object*/
        {
            "startTime":  "undefined",
            /*req, string, start time, desc:format: 00:00*/
            "endTime":  "undefined"
            /*req, string, end time, desc:format: 00:00*/
        }
    ],
    "preRecordTimeSeconds":  5,
    /*opt, int, pre-record, range:[0,10], unit:s*/
    "postRecordTimeSeconds":  5
    /*opt, int, post-record, range:[0,10], unit:s*/
}
```

16.13.1.4 Set audio recording schedule

Request URL

PUT /ISAPI/System/RecordAudio/channels/<audioID>/AudioRecordPlan?format=json

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	audio channel

Request Message

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```
{
    "enable":  true,
    /*req, bool, whether to enable audio recording schedule*/
    "mode":  "timing",
    /*opt, enum, audio recording type, subType:string, desc:"timing" (scheduled audio recording)*/
    "weekTimeList": [
    /*opt, array, week time list, subType:object*/
        {
            "dayOfWeek":  1,
            /*opt, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
            "timeRange": [
            /*req, array, time period, subType:object*/
                {
                    "startTime":  "undefined",
                    /*req, string, start time*/
                    "endTime":  "08:00:00"
                    /*req, string, end time, range:[0,64]*/
                }
            ]
        }
    ],
    "holidayTimeList": [
    /*opt, array, holiday time list, subType:object*/
        {
            "startTime":  "undefined",
            /*req, string, start time, desc:format: 00:00*/
            "endTime":  "undefined"
            /*req, string, end time, desc:format: 00:00*/
        }
    ],
    "preRecordTimeSeconds":  5,
    /*opt, int, pre-record, range:[0,10], unit:s*/
    "postRecordTimeSeconds":  5
    /*opt, int, post-record, range:[0,10], unit:s*/
}
```

Response Message

```
{
    "statusCode":  1,
    /*opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

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16.14 Two-Way Audio

16.14.1 Two-Way Audio

16.14.1.1 Get the parameters of all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, configuration of the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType>
      <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM",
"MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--ro, opt, int, input volume, range:[1,100]-->100
    </speakerVolume>
    <microphoneVolume>
      <!--ro, opt, int, output volume, range:[1,100]-->100
    </microphoneVolume>
    <noisereduce>
      <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--ro, opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-
adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled>
        <!--ro, req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--ro, opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--ro, opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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16.14.1.2 Set audio parameters for all two-way audio channels

Request URL

PUT /ISAPI/System/TwoWayAudio/channels

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, parameters configuration of the two-way audio channel, subType:object, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--opt, object, configuration of the two-way audio channel-->
    <id>
      <!--req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--req, bool, whether to enable the function or not-->true
    </enabled>
    <audioCompressionType>
      <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType>
      <!--req, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume>
      <!--opt, int, input volume-->100
    </speakerVolume>
    <microphoneVolume>
      <!--opt, int, output volume-->100
    </microphoneVolume>
    <noisereduce>
      <!--opt, bool, whether to enable the environmental noise filter or not-->true
    </noisereduce>
    <audioBitRate>
      <!--opt, int, audio frame rate, unit:kbs-->100
    </audioBitRate>
    <audioInputType>
      <!--req, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--opt, object, linked video channel-->
      <enabled>
        <!--req, bool, whether to enable the linked video input channel or not-->true
      </enabled>
      <videoInputChannelList>
        <!--req, array, list of linked video channels, subType:object-->
        <videoInputChannelID>
          <!--opt, string, linked video channel-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate>
      <!--opt, float, audio sampling rate, unit:kHz-->48.00
    </audioSamplingRate>
    <lineOutForbidden>
      <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
    </lineOutForbidden>
    <muteDuringPanTilt>
      <!--opt, bool, whether to mute during motion-->true
    </muteDuringPanTilt>
    <audioOutputType>
      <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <matrixPickUp>
      <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
    </matrixPickUp>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.14.1.3 Receive two-way audio data

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--
audioSessionID	string	--

Request Message

Binary Data

Response Message

Binary Data

16.14.1.4 Send two-way audio data

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<audioID>/audioData?sessionId=<audioSessionID>&type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
audioID	string	--
audioSessionID	string	Two-way audio session ID
type	enum	If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub-device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub-devices, with multiple childDevID values configured in /ISAPI/System/TwoWayAudio/channels//open.

Request Message

Binary Data

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

16.14.1.5 Set audio parameters of a specified two-way audio channel

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	--

Request Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--opt, object, linked video channel-->
    <enabled>
      <!--req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the value
is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the value
is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
</TwoWayAudioChannel>
```

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Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

16.14.1.6 Get the capability of configuring audio parameters for a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/capabilities?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, capability of configuring audios for the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus,G.722.1">
    <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711uLaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,G.722.1">
    <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711uLaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume min="0" max="100">
    <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
  </speakerVolume>
  <microphoneVolume min="0" max="100">
    <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
  </microphoneVolume>
  <noisereduce opt="true,false">
    <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
  </noisereduce>
  <audioBitRate min="1" max="100">
    <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, int},max{opt, int}-->100
  </audioBitRate>
  <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp,doubleMic,networkPickUp">
    <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID min="1" max="2048">
        <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate min="1.00" max="48.00">
    <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
  </audioSamplingRate>
  <lineOutForbidden opt="true,false">
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
  </lineOutForbidden>
  <muteDuringPanTilt opt="true,false">
    <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
  </muteDuringPanTilt>
  <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker,HDMI" def="LineOut">
    <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <isSupportEchoCancellation>
    <!--ro, opt, bool, whether the device supports echo cancellation-->true
  </isSupportEchoCancellation>
  <timeOut min="10" max="600">
    <!--ro, opt, int, timeout during two-way audio, attr:min{req, int},max{req, int}, desc:when the Web Client initiates the two-way audio actively towards
the radar device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some
noise. When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client
stops speaking or not.-->300
  </timeOut>
  <matrixPickUp opt="true,false">
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it is
valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
  </matrixPickUp>
  <isSupportRestore>
    <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
  </isSupportRestore>
  <networkPickUpStatus opt="online,offline">
    <!--ro, opt, enum, network pick-up status, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}, attr:opt{opt, string},
desc:network status of the network pickup linked with the audio channel: "online", "offline"-->online
  </networkPickUpStatus>
  <noisereduceType opt="normal,smart">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

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16.14.1.7 Stop two-way audio

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/close?sessionId=<audioSessionID>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-Way Audio Channel ID
audioSessionID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, Custom Error Information Description, range:[0,1024], desc:Custom detailed error information returned by device application, used
for quick positioning and diagnosis.-->badXmlFormat
  </description>
</ResponseStatus>
```

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16.14.1.8 Start two-way audio

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/open?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
type	enum	If this filed does not exist, it indicates two-way audio via channel, and the audioID is the channel ID. When the type is childDevID, it indicates two-way audio with a single sub device, and the audioID is the gateway intercom resource index (obtained from /ISAPI/System/DynamicResource/SearchResources?format=json); when the type is childDevList, it indicates broadcasting to multiple sub devices, with multiple childDevID values specified in the message, and the audioID in the URL is invalid.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudio xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters for this two-way audio, attr:version{req, string, protocolVersion}-->
  <audioLevel>
    <!--opt, int, audio level, range:[0,15], desc:range: [0,15]-->0
  </audioLevel>
  <microphoneVolume>
    <!--opt, int, output volume-->100
  </microphoneVolume>
  <audioCompressionType>
    <!--opt, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioSamplingRate>
    <!--opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <isBroadcast>
    <!--opt, bool-->true
  </isBroadcast>
  <broadcastType>
    <!--opt, enum, subType:string-->alarm
  </broadcastType>
  <childList>
    <!--opt, array, subType:object, range:[0,8], dep:and,{$.TwoWayAudio.isBroadCast,eq,true}-->
    <childDevID>
      <!--opt, string, range:[9,32]-->test
    </childDevID>
    <voiceTalkIndex>
      <!--opt, int, range:[1,256]-->test
    </voiceTalkIndex>
  </childList>
</TwoWayAudio>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioSession xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, result returned when starting two-way audio succeeds, attr:version{opt, string, protocolVersion}-->
  <sessionId>
    <!--ro, req, string, two-way audio session ID-->1
  </sessionId>
</TwoWayAudioSession>
```

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16.14.1.9 Restore audio parameters of the two-way audio channel

Request URL

PUT /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```

16.14.1.10 Get the audio parameters of a specified two-way audio channel

Request URL

GET /ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>?audioInputType=<audioInputType>

Query Parameter

Parameter Name	Parameter Type	Description
twoWayAudioChannelID	string	Two-way audio channel No.
audioInputType	string	Audio input type, corresponding to the "audioInputType" in the message. This parameter is mainly used to obtain the parameters corresponding to the specified audio input type.

Request Message

None

Response Message

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```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, configuration of the two-way audio channel, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, audio channel ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable or not-->true
  </enabled>
  <audioCompressionType>
    <!--ro, req, enum, encoding type of the audio output, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM", "MP2L2", "Opus", "G.722.1"-->G.711alaw
  </audioCompressionType>
  <audioInboundCompressionType>
    <!--ro, opt, enum, encoding type of the audio input, subType:string, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a", "G.729b", "PCM", "MP3",
"AC3", "AAC", "ADPCM"-->G.711alaw
  </audioInboundCompressionType>
  <speakerVolume>
    <!--ro, opt, int, input volume, range:[1,100]-->100
  </speakerVolume>
  <microphoneVolume>
    <!--ro, opt, int, output volume, range:[1,100]-->100
  </microphoneVolume>
  <noisereduce>
    <!--ro, opt, bool, whether to enable the environmental noise filter or not-->true
  </noisereduce>
  <audioBitRate>
    <!--ro, opt, int, audio frame rate, unit:kbs-->100
  </audioBitRate>
  <audioInputType>
    <!--ro, opt, enum, audio input type, subType:string, desc:"MicIn" (microphone-level input), "LineIn" (line-level input), "selfAdaptive" (self-adaptive),
"wirelessPickUp" (wireless audio pickup)-->MicIn
  </audioInputType>
  <associateVideoInputs>
    <!--ro, opt, object, linked video channel-->
    <enabled>
      <!--ro, req, bool, whether to enable the linked video input channel or not-->true
    </enabled>
    <videoInputChannelList>
      <!--ro, req, array, list of linked video channels, subType:object-->
      <videoInputChannelID>
        <!--ro, opt, string, linked video channel-->1
      </videoInputChannelID>
    </videoInputChannelList>
  </associateVideoInputs>
  <audioSamplingRate>
    <!--ro, opt, float, audio sampling rate, unit:kHz-->48.00
  </audioSamplingRate>
  <lineOutForbidden>
    <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported-->true
  </lineOutForbidden>
  <micInForbidden>
    <!--ro, opt, bool, whether the audio input is not supported. If this node is not returned or its value is "false", audio input is supported;  if the
value is "true", audio input is not supported-->true
  </micInForbidden>
  <muteDuringPanTilt>
    <!--ro, opt, bool, whether to mute during motion-->true
  </muteDuringPanTilt>
  <audioOutputType>
    <!--ro, opt, enum, audio output type, subType:string, desc:"Close" (output closed), "LineOut" (line-level output), "Speaker" (speaker output),
"selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
  </audioOutputType>
  <timeOut>
    <!--ro, opt, int, timeout during two-way audio, range:[10,600], unit:s, desc:when the Web Client initiates the two-way audio actively towards the radar
device, the noise data will be transmitted to the radar device although the person on the Web Client does not speak, because there is always some noise.
When this node is configured, the device will stop the two-way audio after a specific time period, regardless of whether the person on the Web Client stops
speaking or not.-->10
  </timeOut>
  <matrixPickUp>
    <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node is valid only when the audio input type is "Linein")-->true
  </matrixPickUp>
  <networkPickUpStatus>
    <!--ro, opt, enum, subType:string, dep:and,{$.TwoWayAudioChannel.audioInputType,eq,networkPickUp}-->online
  </networkPickUpStatus>
  <noisereduceType>
    <!--ro, opt, enum, subType:string-->normal
  </noisereduceType>
</TwoWayAudioChannel>
```

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16.14.1.11 Get the capability of configuring parameters for all two-way audio channels

Request URL

GET /ISAPI/System/TwoWayAudio/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TwoWayAudioChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capability of configuring audios for all two-way audio channels, attr:version{opt, string, protocolVersion}-->
  <TwoWayAudioChannel>
    <!--ro, opt, object, capability of configuring audios for the two-way audio channel-->
    <id>
      <!--ro, req, string, audio channel ID-->1
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable or not-->true
    </enabled>
    <audioCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM,MP2L2,Opus">
      <!--ro, req, enum, encoding type of the audio output, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729",
"G.729a", "G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM", "MP2L2", "Opus"-->G.711alaw
    </audioCompressionType>
    <audioInboundCompressionType opt="G.711alaw,G.711ulaw,G.726,G.729,G.729a,G.729b,PCM,MP3,AC3,AAC,ADPCM">
      <!--ro, opt, enum, encoding type of the audio input, subType:string, attr:opt{opt, string}, desc:"G.711alaw", "G.711ulaw", "G.726", "G.729", "G.729a",
"G.729b", "PCM", "MP3", "AC3", "AAC", "ADPCM"-->G.711alaw
    </audioInboundCompressionType>
    <speakerVolume min="0" max="100">
      <!--ro, opt, int, input volume, attr:min{opt, int},max{opt, int}-->100
    </speakerVolume>
    <microphoneVolume min="0" max="100">
      <!--ro, opt, int, output volume, attr:min{opt, int},max{opt, int}-->100
    </microphoneVolume>
    <noisereduce opt="true,false">
      <!--ro, opt, bool, whether to enable the environmental noise filter or not, attr:opt{opt, string}-->true
    </noisereduce>
    <audioBitRate min="1" max="100">
      <!--ro, opt, int, audio frame rate, unit:kbs, attr:min{opt, string},max{opt, string}-->100
    </audioBitRate>
    <audioInputType opt="MicIn,LineIn,selfAdaptive,wirelessPickUp">
      <!--ro, opt, enum, audio input type, subType:string, attr:opt{opt, string}, desc:"MicIn" (microphone-level input), "LineIn" (line-level input),
"selfAdaptive" (self-adaptive), "wirelessPickUp" (wireless audio pickup)-->MicIn
    </audioInputType>
    <associateVideoInputs>
      <!--ro, opt, object, linked video channel-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable the linked video input channel or not, attr:opt{opt, string}-->true
      </enabled>
      <videoInputChannelList>
        <!--ro, req, array, list of linked video channels, subType:object-->
        <videoInputChannelID min="1" max="2048">
          <!--ro, opt, string, linked video channel, attr:min{opt, string},max{opt, string}-->1
        </videoInputChannelID>
      </videoInputChannelList>
    </associateVideoInputs>
    <audioSamplingRate min="1.00" max="48.00">
      <!--ro, opt, float, audio sampling rate, unit:kHz, attr:min{opt, float},max{opt, float}-->48.00
    </audioSamplingRate>
    <lineOutForbidden opt="true,false">
      <!--ro, opt, bool, whether the audio output is not supported. If this node is not returned or its value is "false", audio output is supported; if the
value is "true", audio output is not supported, attr:opt{opt, string}-->true
    </lineOutForbidden>
    <muteDuringPanTilt opt="true,false">
      <!--ro, opt, bool, whether to mute during motion, attr:opt{opt, string}-->true
    </muteDuringPanTilt>
    <audioOutputType opt="Close,LineOut,Speaker,selfAdaptive,LineOut_Speaker" def="LineOut">
      <!--ro, opt, enum, audio output type, subType:string, attr:opt{opt, string},def{opt, string}, desc:"Close" (output closed), "LineOut" (line-level
output), "Speaker" (speaker output), "selfAdaptive" (self-adaptive), "LineOut_Speaker" (both line-level output and speaker output)-->LineOut
    </audioOutputType>
    <isSupportEchoCancellation>
      <!--ro, opt, bool, whether the device supports echo cancellation-->true
    </isSupportEchoCancellation>
    <matrixPickUp opt="true,false">
      <!--ro, opt, bool, whether to use the wired matrix pickup or not (this node will not be returned if the wired matrix pickup is not supported, and it
is valid only when the audio input type is "Linein"), attr:opt{opt, string}-->false
    </matrixPickUp>
    <isSupportRestore>
      <!--ro, opt, bool, whether the device supports restoring audio parameters of the two-way audio channel, desc:corresponding API:
/ISAPI/System/TwoWayAudio/channels/<twoWayAudioChannelID>/restore?format=json-->true
    </isSupportRestore>
  </TwoWayAudioChannel>
</TwoWayAudioChannelList>
```

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17 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on https://tpp.hikvision.com and visit our Training Center: https://tpp.hikvision.com/tpp/Training. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.

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